Heinrich Weber

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(Textbook of Algebra)

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LEHRBUCH

DER

ALGEBRA.

[Textbook of Algebra]

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[TEXTBOOK OF ALGEBRA]

BY

HEINRICH WEBER,

PROFESSOR OF MATHEMATICS AT THE UNIVERSITY OF GÖTTINGEN.

IN TWO VOLUMES.

FIRST VOLUME.

WITH 28 FIGURES PRINTED IN THE TEXT.

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PREFACE.

In view of the development that algebra has undergone in recent decades, a comprehensive presentation and linking of the various theoretical considerations and manifold applications may well be useful, even after the textbook of Serret, which was excellent for its time.

For years I have cherished the plan of such an undertaking, which indeed appeared vast and far-reaching and required a good deal of preparatory work. Only after I had several times traversed the field as a whole in university lectures, and had treated individual parts in greater detail, did I resolve to set about the execution of the work, of which the first volume now lies before the reader in completed form.

It was my intention to provide a textbook which, without presupposing much prior knowledge, would introduce the reader to modern algebra and also lead him on to the higher and more difficult parts, in which interest in the subject first becomes truly alive. At the same time, the necessary tools, the elementary as well as the higher ones, were to be derived from the course of the development itself, in order to make the presentation as independent of other textbooks as possible.

There are two things that have become of quite special importance for the most recent development of algebra; these are, on the one hand, group theory, which is gaining ever greater dominance and whose ordering and clarifying influence can be felt everywhere, and then the intervention of number theory.

Although algebra in part goes beyond the theory of numbers, and reaches over into other fields, e.g. the theory of functions, or, in its applications, into geometry as well, the science of numbers nevertheless always remains the foremost example for all algebraic considerations, and the questions of number theory that stand today in the foreground of interest are predominantly of an algebraic nature. This marked out the path that I had to follow in my work.

The extensive material has been divided into two volumes. The first volume contains the elementary part of algebra, which, by a traditional expression, may be called calculation with letters, then the rules for the numerical computation of the roots of equations, and the beginnings of Galois's theory.

The second volume, which will, I hope, follow the first within a short time, is to present the general theory of finite groups, the theory of linear substitution groups, and applications to various individual problems, and is to conclude with the theory of algebraic numbers, where an attempt is made to unite the various points of view from which this theory has hitherto been considered.

Finally, the second volume is to provide an alphabetical index to both volumes.

As cannot be otherwise in a discipline that is in the course of rapid development and on which work is being done from the most diverse quarters, in algebra, too, the use of language and the notation are very varied and frequently not in agreement. This makes a unified presentation, as well as penetration into the various works, very difficult.

I have therefore endeavoured to adhere uniformly to a mode of expression that is as appropriate as possible, and in doing so I have frequently taken counsel with colleagues. I may express the hope of having thereby contributed to the consolidation of a uniform terminology.

The references to the literature and the historical notes given in the book make no claim whatsoever to completeness, although I have endeavoured

to mention, as far as possible, the most important sources and literary aids at the appropriate places.

It is a pleasant duty for me to express my thanks here to many a friend and colleague who has taken a lively and active interest in the progress of my work. These thanks go first of all to my friend Dedekind for his faithful help with the proofreading. And even though he has exercised no influence on the plan and the execution of my work, I should nevertheless not like to leave unmentioned something from many years ago. Through a set of notes of a lecture course on higher algebra, in particular on the theory of Galois, which he gave in Göttingen in the winter of 1857/58, I gained an even livelier interest in this theory. Until then it had probably never been presented at our universities with such completeness.

I must also make mention here of the manifold stimulation and instruction that I owe to my friend and colleague F. Klein. He has followed the progress of the work with the liveliest interest, and his expert advice, always most readily given, has been of great influence on many parts of the book.

I cannot name here all the colleagues and friends who, as the reader will find in the relevant places, have supported me with their advice. But I must still make mention here of Messrs. E. Hess in Marburg, Fr. Meyer in Clausthal and R. Fricke in Braunschweig. Through their expert and careful execution of the laborious correction of the proof sheets, they have furthered the accuracy and correctness of the text.

Finally, my thanks go to the publishing house, which, by readily accommodating my wishes and by its care in printing and production, has contributed substantially to the success of the whole.

Göttingen, November 1894.

The Author.

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INTRODUCTION.

We presuppose, in our considerations, the natural numbers $1, 2, 3, \ldots$ and the rules according to which calculations are made with these numbers as known and given. The fundamental arithmetical operations, the so-called four species, are addition, multiplication, to which, as repetition, exponentiation belongs, subtraction, and division. The first two are called the direct operations of calculation; they are distinguished by the fact that in the realm of the natural numbers they can be carried out without restriction. The first of the indirect or inverse operations, subtraction, can be carried out only when the minuend is greater than the subtrahend.

The problem of division can be understood in two ways. In the first elementary interpretation one asks how often the divisor is contained in the dividend. A number is not contained in a smaller one. But if the dividend is equal to or greater than the divisor, then the answer to the question gives a quotient and, in most cases, a remainder which is smaller than the divisor. If no remainder is left, one says that the division comes out exactly, or that the dividend is divisible by the divisor, or that the divisor is a factor or divisor of the number which forms the dividend.

This problem, which already presents itself at the first stages of the art of calculation, leads to a more deeply lying distinction among numbers, which is the foundation of all number theory.

Since a factor can never be greater than the number of which it is a factor, every number has only a finite number of factors. Every number is divisible by 1 and by itself, and a number which has no other divisor is called a prime-

number. The number 1 itself is customarily not designated a prime number, for reasons of expediency. If two numbers are divisible by a third, then the sum and the difference of the first two are also divisible by the third; and if one number is divisible by a second and this second by a third, then the first is also divisible by the third.

Two numbers always have the common divisor 1. If they have no other common divisor, as for example the pairs of numbers 5 and 7, or 21 and 38, then the two numbers are called relative prime numbers, or also numbers prime to one another.

Among the common divisors of any two given numbers one will be the greatest, and it is a very important problem to find this greatest common divisor. This is accomplished by a procedure known under the name algorithm of the greatest common divisor, which already occurs in Euclid.[1]

Let a, a_1 be the two numbers; take the larger of them, say a, as dividend and the smaller a_1 as divisor, and determine the quotient q_1 and, if the division does not come out exactly, the remainder a_2, so that

$$a = q_1 a_1 + a_2,$$

and $a_2 < a_1$. Every common divisor of a and a_1 is then also a common divisor of a_1 and a_2, and conversely. If one proceeds with a_1, a_2 just as with a, a_1, and puts, when the division does not come out exactly,

$$a_1 = q_2 a_2 + a_3,$$

then again $a_3 < a_2$, and every common divisor of a_1 and a_2 is also a common divisor of a_2 and a_3, and conversely. If one continues dividing in this way, then, since the numbers $a, a_1, a_2, a_3, \ldots$ continually decrease, the division must necessarily come out exactly after a finite number of steps; that is, at last a pair of equations

$$a_{\nu-2} = q_{\nu-1} a_{\nu-1} + a_\nu, \qquad a_{\nu-1} = q_\nu a_\nu$$

must occur. Then a_ν is a common divisor of all preceding a's, hence also of a and a_1, and every common divisor of a and a_1 is a divisor of a_ν. Thus a_ν is the greatest common divisor of a and a_1,

[1] Elements, Book VII, II, vol. II of Heiberg's edition.

and at the same time we have arrived at the theorem that every common divisor of two numbers must divide their greatest common divisor.

If a and a_1 are relatively prime numbers, then the last divisor is $a_\nu = 1$. Multiplying, under this assumption, the preceding equations by an arbitrary number b, it follows that b is the greatest common divisor of ab and $a_1 b$, and hence that every common divisor of ab and a_1, or of a and $a_1 b$, must be a divisor of b. Thus if a_1 is relatively prime to a and to b, it is also relatively prime to ab; and from the special assumption that a_1 is a prime number it follows that a product can be divisible by a prime number only when at least one of the factors is divisible by it.

Thus, if the product ab is divisible by a_1 and a_1 is relatively prime to a, then b must be divisible by a_1.

Let a, b be any two numbers with greatest common divisor d, and let $a = da'$, $b = db'$. Then a' and b' are relatively prime to one another. Every number m which is at once a multiple of a and of b has the form $m = am' = da'm'$, and in this $a'm'$ must be a multiple of b'; therefore m' too must be a multiple of b'. That is, every number which is at once a multiple of a and of b is divisible by $a'b'd$. This number $a'b'd$, which itself is a common multiple of a and b, is therefore called the least common multiple of a and b.

If a number is divisible by two relatively prime numbers, then it is also divisible by their product; and if, therefore, a number m is divisible by several numbers of which each two are relatively prime to one another, then it is also divisible by the product of all these numbers.

On this rests the proof of the important theorem that a number m can always be represented, and in only one way, as a product of prime numbers. For since a divisor cannot be greater than the dividend, m can certainly be divisible by only a finite number of prime numbers. If a is one of these prime numbers and a^α the highest power of a which divides m, then α too is a definite number. Let likewise $b^\beta, c^\gamma, \ldots$ be the highest powers of the remaining prime numbers occurring in m, namely $b, c, \ldots$, by which m can be divided. Then, since any two of the numbers $a^\alpha, b^\beta, c^\gamma, \ldots$ are relatively prime, m must also be divisible by the product $a^\alpha b^\beta c^\gamma \cdots$; and m must be equal to this product, since otherwise another

prime number, or a higher power of one of the prime numbers $a, b, c, \ldots$, would have to divide m.

We now give an overview of the extensions of the concept of number which are necessary in mathematics and which were gradually introduced.

By a manifold or set, or by the abbreviated sign $\mathfrak{M}$, we understand a system of objects or elements of any kind, which is so delimited and complete in itself that, for every arbitrary object, it is completely determined whether it belongs to the system or not, regardless of whether we are able, in each particular case, actually to make the decision or not.

A set is called ordered when, of any two different elements, one that is fully determined in itself always counts as the greater, and in such a way that from $a > b$, $b > c$ one always has $a > c$. If $a > b$ and $b > c$, or $a > b > c$, we say that b lies between a and c.

The natural numbers form an ordered set; between two successive elements of it there lies no further element. Such a manifold is called discrete. An ordered set with the property that between any two elements further elements can always be found is called dense. One can form a dense set by combining the natural numbers in pairs and regarding these pairs as elements of a set. These pairs shall be called fractions and denoted by $m : n$ or m/n, and two such fractions $m : n$ and $m' : n'$ are set equal when $mn' = nm'$. If all fractions equal to one another are combined into one element, one obtains a manifold which is ordered once one further stipulates that $m : n$ is greater than $m' : n'$ when $mn' > nm'$. That this manifold is dense is seen as follows: if $\mu = m : n$, $\mu' = m' : n'$ are two fractions and $\mu > \mu'$, then, with h arbitrary, one can put

$$\mu = \frac{hmn'}{hnn'}, \qquad \mu' = \frac{hm'n}{hnn'}$$

and therein $hmn' > hm'n$. One can always take h so large that further numbers lie between hmn' and $hm'n$, and if p is such a number, then $p : hnn'$ lies between μ and μ'.

The points of a straight line can also be regarded as an ordered set if by greater and smaller one understands some indication of position, for instance farther right and farther left, or higher and lower.

A division of an ordered set $\mathfrak{M}$ into two parts A, B of such a kind that every element a of A is smaller than every element b of B is called a cut in $\mathfrak{M}$ and is suitably denoted by (A, B). Such a cut arises when one selects an element μ of $\mathfrak{M}$, assigns all smaller elements to A, all larger elements to B, and assigns μ itself at will to A or to B. Strictly speaking, according as one does the one or the other, two cuts arise, but we shall always regard them as equal. If, in a cut (A, B), either A contains a greatest element or B a least element μ, then we say that μ produces the cut (A, B). But it may also occur that neither A has a greatest nor B a least element.

If every cut in a dense set is produced by a definite element μ, then the set is called continuous.

Continuity as well as density are properties which, by the nature of the matter, are inaccessible to our sense perception; consequently they also cannot be demonstrated with rigor in things of the external world, in spatial magnitudes, intervals of time, or masses, however much they may seem to lie in the essence of our intuition. But pure conceptual systems can very well be constructed to which density without continuity, or also density and continuity, belong[1].

The rational fractions furnish an example of a dense set. This manifold, which we shall denote by $\mathfrak{R}$,

is not continuous. For take any rational fraction $\mu = m : n$, in which m and n have no common divisor and are not both square numbers. Then μ is not the square of a rational fraction. For if $\mu = p^2 : q^2$, then $mq^2 = np^2$ would follow, and from this $m = p^2$, $n = q^2$, contrary to the assumption. Therefore if we form a cut (A, B) in the manifold $\mathfrak{R}$ by assigning to A every element a of $\mathfrak{R}$ whose square is smaller than μ, and to B every element b whose square is greater than μ, then neither A contains a greatest element nor B a least element, and the cut (A, B) is not produced by an element of $\mathfrak{R}$.

For suppose that $a = p : q$ is any element of A, so that $p^2 : q^2 < m : n$, or $np^2 < mq^2$. Let y be any natural number and choose another natural number x so that $x > y$ and $x(mq^2 - np^2) > ny(2p+1)$. Then

$$x^2(mq^2 - np^2) > nxy(2p + 1) > n(2pxy + y^2),$$

and hence also

$$mq^2x^2 > n(px + y)^2.$$

Thus, if we put $a' = (px + y) : qx$, then $a' > a$ and $a'^2 < \mu$; hence a' too is contained in A. In exactly the same way one can show that there is no least element in B.

The manifold $\mathfrak{R}$ can, however, serve as our starting point for constructing a continuous set. The totality of all cuts in $\mathfrak{R}$ is certainly a manifold, which may be denoted by $\mathfrak{S}$. Consider two different elements of it, $\alpha = (A, B)$ and $\alpha' = (A', B')$. Then either A is a part of A' or A' is a part of A. For if some

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manifolds to which continuity belongs were formed by Weierstrass and G. Cantor.

The definition of continuity, as we here lay it down after Dedekind, is exhaustive in the following respect: a set continuous in this sense, if it also possesses the property of measurability presently to be discussed, cannot be a part of a richer continuous set. I do not know whether this property has already been proved anywhere, and hope to

[1]This was proved by Dedekind, to whom we owe in general the definition of continuity given above. Compare the writings of Dedekind, *Stetigkeit und irrationale Zahlen*, Brunswick 1872, 1892, and *Was sind und was sollen die Zahlen?*, Brunswick 1888, 1893. Other

return to it on another occasion. I remark, however, that such a property can be proved only for measurable sets. A merely ordered set can always, however dense it may be, be regarded as a part of one still denser.

element a belongs to A, then every smaller element of $\mathfrak{R}$ also belongs to A. If A is a part of A', we shall call α smaller than α', and thereby the set $\mathfrak{S}$ has become ordered.

If we regard the cuts produced by rational fractions as equivalent to those rational fractions themselves and call them, briefly, rational cuts, then the set $\mathfrak{S}$ contains the set $\mathfrak{R}$, and $\mathfrak{S}$ is therefore in any case a dense set. But the set $\mathfrak{S}$ is also continuous. For let us denote the cuts by the large German letters $\mathfrak{A}, \mathfrak{B}, \ldots$, and consider any cut in the manifold of cuts, $(\mathfrak{A}, \mathfrak{B})$. We can determine an element of $\mathfrak{S}$, say $\alpha = (A, B)$, by putting into A every rational fraction that produces one of the cuts of $\mathfrak{A}$, and putting into B all the remaining rational fractions, which therefore produce the rational cuts in $\mathfrak{B}$. This cut α in $\mathfrak{R}$ produces the cut $(\mathfrak{A}, \mathfrak{B})$ in $\mathfrak{S}$. This will be proved when it is shown that every element α' of $\mathfrak{S}$ which is smaller than α belongs to $\mathfrak{A}$, and every element β' of $\mathfrak{S}$ which is greater than α belongs to $\mathfrak{B}$.

Let $\alpha' = (A', B')$ and $\alpha' < \alpha$. Then there are rational fractions in A which are not contained in A'. There is therefore a rational μ such that $\alpha' < \mu < \alpha$; this μ produces a cut contained in $\mathfrak{A}$, hence itself belongs to $\mathfrak{A}$. Since $\alpha' < \mu$, α' also belongs to $\mathfrak{A}$. In exactly the same way one shows that every β' greater than α belongs to $\mathfrak{B}$, and the continuity of $\mathfrak{S}$ is thereby proved.

This very abstract way of looking at the matter gives us the assurance that the assumption of a continuous set contains no contradiction, and that such sets at least exist in the realm of thought. Geometry, and also analysis, which has always willingly attached itself to geometrical intuition, long tacitly assumed the existence of continuous sets, for example in the points of a straight line or of any other connected line-segment, as a kind of axiom. The difference between a dense and a continuous set, which lies at the basis of the distinction between commensurable and incommensurable segments, was also not missed by the ancients. [1]

We too shall in what follows not renounce the aid of geometrical intuition, and shall, for example, without hesitation regard the points of a straight line as a continuous set.

An ordered set $\mathfrak{M}$ is called measurable under the following assumptions: addition and multiplication are generally executable in $\mathfrak{M}$, and likewise the subtraction of a smaller element from a greater one; that is, from any two elements a, b (which may also be identical) a new element $a + b$ of $\mathfrak{M}$ can be derived according to a definite rule, in such a way that $a + b$ is greater than a and greater than b, and the familiar laws of addition expressed in the formulae

$$a + b = b + a, \qquad (a + b) + c = a + (b + c)$$

hold. Moreover, for two elements a, c, of which the second is the greater, a third element b can be found such that $a + b = c$, which is also expressed by the sign of subtraction $b = c - a$. Thus two unequal elements of $\mathfrak{M}$ always have a definite difference. From these assumptions it follows that a sum becomes greater when one of its summands becomes greater. The repeated, say m-fold, addition of the same element a is called multiplication, and its result is denoted by ma.

Finally there is added one more assumption, namely that, for every given a, a sufficiently high multiple ma is greater than any other element b given in advance. Among the elements of a measurable set there is therefore no greatest element.

In a dense measurable set there is also no least element. For if a were the least and b an arbitrary element, then between b and $b + a$ no elements could lie; for if $b < c < b + a$, then it would follow from the definition of measurability that $a' = c - b$ would be smaller than a. Conversely it also follows that a measurable set in which no least element occurs is dense. For if a and $a + b$ are any two elements, one need only add to a an element smaller than b in order to obtain an element between a and $a + b$.

[1] Euclid, Elements, Book X.

If a set is continuous, then the assumptions for measurability can be simplified still further, because then subtraction is a consequence of addition. For if a and c are two ele-

ments of a continuous ordered set $\mathfrak{M}$ in which addition exists, and if $c > a$, then one obtains a cut (A, B) in $\mathfrak{M}$ by assigning to A all elements x for which $a + x \leq c$, and to B those for which $a + x > c$. This cut is produced by an element b, for which $a + b = c$ must hold.

The natural numbers form, according to our definition, a measurable set in which a least element, namely 1, occurs. The manifold of rational fractions is likewise measurable when addition and subtraction are explained according to the known rules of fractional calculation. Especially important, and as it were typical for the measurable sets, is the manifold of rectilinear segments, or lengths of a line, which are added simply by laying them end to end. Quantities of material, compared by means of the balance, and intervals of time, measured by the clock, also furnish examples of measurable sets. The manner of measuring does not lie in the nature of the manifold itself, but is put into it by the thinking observer; thus, for example, it would be just as admissible to understand by the sum $a + b$ of two segments a and b the hypotenuse of a right triangle with legs a, b, instead of, as is usually assumed, the segment composed from a and b by laying them end to end.

In order to make the continuous set $\mathfrak{S}$ of cuts in the manifold $\mathfrak{R}$ of rational fractions into a measurable one, observe first the following.

If $\alpha = (A, B)$ is an element in $\mathfrak{S}$ and μ is an arbitrarily given rational fraction, then one can always choose an element a' in A so that $a' + \mu = b'$ is contained in B. For if one chooses two arbitrary elements a, b, then one can determine the natural number m so that $m\mu > b - a$, hence $a + m\mu$ is contained in B. If then h is the smallest integer for which $a + h\mu$ is contained in B, then $a + (h - 1)\mu = a'$ is in A and hence $a' + \mu = b'$ is in B.

We now understand by the sum $(A, B) + (A', B') = (A'', B'')$, or $\alpha + \alpha' = \alpha''$, the cut in $\mathfrak{R}$ which is obtained by assigning a rational fraction a'' to A'' only when an a in A and an a' in A' exist such that $a'' \leq a + a'$. In fact (A'', B'') is a cut; for if a'' is contained in A'', the same holds for every smaller fraction, and there are fractions which are in A'' and fractions which are not in

A'', namely fractions of the forms $a + a'$ and $b + b'$. Further, α'' is greater than α and than α'. For A'' first contains all of A, and if a' is an arbitrary fraction in A', one can choose a in A so that the element $a + a'$ of A'' is contained in B; hence A'' is more comprehensive than A. The cuts produced by rational fractions μ, μ' give, by addition, the cut produced by $\mu + \mu'$.

We now pass to the definition of ratios, which since antiquity have been regarded as the foundation of the theory of numbers, and here follow Euclid first of all. [1]

If one combines the elements of a measurable set $\mathfrak{M}$ into pairs, and regards these pairs themselves as elements, then a new manifold arises. We denote such a pair by $a : b$, or also by a/b; but, when a and b are different elements, we distinguish $a : b$ from $b : a$, and call a the numerator and b the denominator of $a : b$. We shall call these pairs ratios, and shall now order and make measurable this new set.

Assume first that two whole numbers m, n exist such that $na = mb$, as is always the case, for example, when $\mathfrak{M}$ is the system of natural numbers, or when a, b are two commensurable segments. Then, if p, q are two other whole numbers, $qa = pb$ holds if and only if $mq = np$. The pair of numbers p, q is completely determined by this requirement if the condition is added that p, q are to be relatively prime. Then, if h is an arbitrary whole number, $m = hp$, $n = hq$. In this case we call the ratio $a : b$ rational and set it equal to the rational fraction $m : n$ or $p : q$. Accordingly these rational fractions may be regarded as ratios of whole numbers.

All rational ratios equal to one another form a rational number, and the rational numbers form, like the rational fractions, an ordered, dense, and measurable manifold.

[1] Elements, Book V.

The natural numbers themselves are inserted into the manifold of rational numbers if by the natural number m one understands the ratio $m : 1$.

We now return to any measurable manifold $\mathfrak{M}$ and take from it any two elements a and b. If one chooses, which is always possible, two natural numbers m, n so that $na > mb$, then the ratio $a : b$ is called greater than the rational ratio $m : n$, or

$$\frac{a}{b} > \frac{m}{n},$$

and if $m : n > p : q$, then also $a : b > p : q$. Similarly, if $n'a < m'b$, then

$$\frac{a}{b} < \frac{m'}{n'}.$$

If $a : b > m : n$, then one can insert one, and consequently also arbitrarily many, rational numbers $m_1 : n_1$ such that

$$\frac{a}{b} > \frac{m_1}{n_1} > \frac{m}{n}$$

hold. That is, one can insert arbitrarily many rational ratios between $a : b$ and $m : n$. To show this, choose an arbitrary whole number k and determine the whole number h so that $h(na - mb) > kb$, which is always possible. Then

$$\frac{a}{b} > \frac{hm + k}{hn} > \frac{m}{n}.$$

And in exactly the same way, if $n'a < m'b$ and $h'(m'b - n'a) > k'a$, one obtains

$$\frac{a}{b} < \frac{h'm'}{h'n' + k'} < \frac{m'}{n'}.$$

If now $a : b$ and $\alpha : \beta$ are any two ratios, which for brevity we shall also denote by e, ϵ, and whose elements belong to the same or to different manifolds, then two cases are possible: 1) no rational ratio μ lies between e and ϵ, or 2) a rational ratio lies between e and ϵ.

In case 1) the two ratios e and ϵ are called equal to one another. One sees that two ratios which are equal to a third are also equal to one another; for if $e < \mu < \epsilon$, then every other ratio e' is either smaller

than, equal to, or greater than μ. If e' is equal to μ, then rational ratios lie both between e and e' and between e' and ϵ. If, however, e' is smaller than μ, then μ lies between e' and ϵ, and if e' is greater than μ, then μ lies between e and e'; hence e' cannot at the same time be equal to e and equal to ϵ.

In case 2) the ratios e, ϵ are called unequal. Thus either α) $e < \mu < \epsilon$ or β) $e > \mu' > \epsilon$ may hold, and these two cases exclude one another, because from $e < \mu < \epsilon$, $\epsilon < \mu'$ it follows that $\mu < \mu'$, and consequently $e < \mu'$ results.

The relation of magnitude 2α) or 2β) also remains unchanged when e or ϵ is replaced by an element equal to it. For if $\mu < \epsilon$ and $\mu \gtrless \epsilon'$, then a rational ratio lies between ϵ and ϵ', and ϵ, ϵ' are not equal.

In case 2α), e is called smaller than ϵ; in case 2β), e is called greater than ϵ. Between two unequal ratios one can insert an arbitrary number of rational ratios.

If now we collect all ratios equal to one another, we obtain a generic concept which we designate as a number in the general sense of the word. A number is therefore a name or sign for a certain manifold whose elements are precisely the ratios equal to one of them. [1] Under this concept of number are included the rational ratios, and hence also the natural numbers as the ratios $m : 1$; they form the rational numbers. Numbers which do not arise from rational ratios are called irrational numbers. According to what has been developed so far, the numbers form an ordered set, and their order can be fixed if, for each number, any one of the ratios contained under it is chosen as representative.

[1] The natural numbers too can be traced back in a simple and consistent way to the generic concept.

Of two ratios with the same denominator and unequal numerators, that one is greater whose numerator is greater; and

of two ratios with the same numerator and unequal denominators, that one is smaller whose denominator is greater. For if a, a', b are arbitrary elements of a measurable set and $a' > a$, choose first a whole number n such that $na > b$ and $n(a' - a) > b$. Then take m to be the least whole number satisfying $mb > na$; then $na < mb$, but $mb < na'$. For if $mb \geq na' \geq na + n(a' - a)$, then $mb > na + b$; hence, contrary to the supposition, already $(m-1)b > na$. Thus

$$\frac{a}{b} < \frac{m}{n} < \frac{a'}{b},$$

and hence $a : b < a' : b$. In exactly the same way one proves that if $b' > b$, then $a : b > a : b'$. This theorem is also expressed by saying that a ratio increases with the numerator and decreases with increasing denominator.

From this one easily obtains the following theorem. If a, b, c, d are elements of the same measurable set and $a : b = c : d$, then also $a : c = b : d$. For suppose that $a : c < b : d$; then there would have to be two whole numbers m, n such that

$$na < mc, \qquad nb > md.$$

Then, however, by the theorem just proved, $na : nb < mc : md$, and hence $a : b < c : d$, contrary to the supposition.

To this is now attached the following principal theorem. If of four quantities a, b, c, d any three are given arbitrarily from a continuous measurable set, then the fourth can be determined in the same set so that $a : b = c : d$.

The theorem is an immediate consequence of the assumed continuity. For if, for example, an element x of $\mathfrak{M}$ is put into A or into B according as $x : b$ is smaller or greater than $c : d$, then a cut is obtained, which is produced by an element a satisfying $a : b = c : d$. But this theorem also holds in certain non-continuous sets, for example for the rational fractions.

It follows that as representatives of two numbers one can always choose two ratios whose elements belong to the same

manifold and have the same arbitrarily chosen denominator. Addition is then defined so that

$$\frac{a}{c} + \frac{b}{c} = \frac{a+b}{c}$$

holds. This rule contains as a special case the addition of rational fractions. To justify it generally it remains only to show that, if $a : c = a' : c'$ and $b : c = b' : c'$, then also $(a + b) : c = (a' + b') : c'$ must hold. We prove this by showing that, if $a : c = a' : c'$ and $(a + b) : c > (a' + b') : c'$, then also $b : c > b' : c'$ must hold. Let therefore, with m, n two whole numbers,

$$\frac{a+b}{c} > \frac{m}{n} > \frac{a'+b'}{c'},$$

then $n(a + b) > mc$, and hence

$$\frac{b}{c} > \frac{mc - na}{nc}, \qquad \frac{mc' - na'}{nc'} > \frac{b'}{c'}.$$

On the other hand, from the supposition $a : c = a' : c'$ it follows easily that

$$\frac{mc - na}{nc} = \frac{mc' - na'}{nc'}$$

holds, and therefore $b : c > b' : c'$, as was required to be proved.

It is thus shown that the numbers, as we have defined them, also form a measurable set. If a and c are taken from a continuous manifold, then, with fixed c, the ratios $a : c$ also form a continuous set; hence, since continuous sets exist at all, it follows that the numbers too form a continuous set.

If $\alpha, \beta, \gamma, \delta$ are now numbers, then from the proportion $\alpha : \beta = \gamma : \delta$ any one of the four numbers can be determined by the three others given. If one puts $\delta = 1$ and seeks α, one obtains the multiplication $\alpha = \beta\gamma$; the commutativity of the factors is a consequence of the theorem that from $\alpha : \beta = \gamma : \delta$ there follows $\alpha : \gamma = \beta : \delta$. If one seeks γ, one obtains division; and from the definition of addition given above there follows the fundamental formula $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$. The four fundamental operations are therefore executable in the domain of numbers, with the sole restriction that in subtraction the subtrahend must be smaller than the minuend.

The construction of a cut in the sequence of numbers always supplies the proof for the existence of a number satisfying prescribed requirements. Thus one obtains a cut (A, B) by putting into A all and only those numbers whose square is smaller than a prescribed number α; to this cut there corresponds a definite number whose square is equal to α and which is denoted by $\sqrt{\alpha}$. In this way the existence of square roots is proved.

The numerical sequences introduced by G. Cantor for the definition of irrational numbers can also be referred back to cuts. [1]

According to Cantor, by a numerical sequence one is to understand any unbounded system of numbers ordered in a definite way:

$$S = x_1, x_2, x_3, x_4, \ldots$$

having the property that there is a definite number g below which none of the numbers of S sinks, and that, if δ is any chosen number and x_n, x_m are different elements of S, then $x_n - x_m$ or $x_m - x_n$ always remains smaller than δ when m and n have exceeded a sufficiently large number.

There are always numbers which are not exceeded by the numbers of such a sequence; for, once δ has been chosen and n suitably determined, x_m, whatever the value of m, never exceeds the greatest of the numbers $x_1, x_2, \ldots, x_n, x_n + \delta$. One now obtains a cut (A, B) by throwing into B those numbers which, when n has a sufficiently high value, are no longer exceeded by any x_n, and all other numbers, which are therefore exceeded by infinitely many x_n, into A. If this cut is produced by the number α, then, however small ε may be, there are always infinitely many numbers x_n between $\alpha - \varepsilon$ and α; and one may say that this number α, completely determined by S, is produced by the numerical sequence S. According to Cantor, the numerical sequence S is directly the definition of the number α. Of course one and the same

number α can be produced by very different numerical sequences. These numerical sequences, however, are all to be regarded as equal to one another. Among other things, one may take all numbers of S to be rational fractions. Conversely, it is also easily shown that for every given number α one can always give numerical sequences S by which α is produced, so that the totality of the numerical sequences S likewise forms a continuous set.

In the explanation of the fundamental operations an inconvenient restriction has appeared in subtraction, from which we can free ourselves by introducing zero and the negative numbers.

Let x denote any element of the number system defined so far, which we shall now call the system of positive numbers. We take this number system a second time and, to distinguish it, denote in this second system, which is to be called the system of negative numbers, every

[1] Cantor, *Ueber die Ausdehnung eines Satzes aus der Theorie der trigonometrischen Reihen.* Mathematische Annalen, vol. 5 (1872); compare also Heine, *Elemente der Functionenlehre.* Journal für Mathematik, vol. 74 (1872).

element by $-x$. This second system we now order in exactly the reverse way from the first, so that wherever in the system x "greater" stands, in the system $-x$ "smaller" is put, and conversely. Addition and subtraction in $-x$ are explained just as in x, so that $(-x)+(-y) = -(x+y)$, $(-x)-(-y) = -(x-y)$ shall hold.

But we wish to order these two number systems together in such a way that every $-x$ is to be smaller than every x. We thus obtain an ordered set in which there is no greatest and no least element. This system is also in general continuous; only the single cut $(-x, x)$ is produced by no element, and here, where the two systems meet, a violation of continuity is still present. To restore continuity we must therefore add, corresponding to the cut $(-x, x)$, a number zero, or 0, which is precisely defined by this cut. Then we have an ordered continuous set unbounded on both sides, the complete series of real numbers.

In this enlarged number domain we now explain addition generally by putting, by definition:

$$x+(-x)=0, \qquad x+0=x,$$
$$x+(-y)=x-y, \quad \text{if } x>y, \quad =-(y-x), \quad \text{if } y>x.$$

With this definition of addition, if z_1, z_2, z_3 are any three numbers of the whole number domain, the laws

$$z_1+z_2=z_2+z_1, \qquad (z_1+z_2)+z_3=z_1+(z_2+z_3),$$

hold; they are called the commutative and associative laws. The sum of any number of summands is formed by successively combining any two summands into a sum. Subtraction no longer needs separate consideration if z_1-z_2 is defined as $z_1+(-z_2)$ and $-(-z)=z$ is set.

The number line is represented intuitively by points: from a fixed point of a straight line, denoted by 0, the positive numbers are laid off as segments on one side, say the right, and the negative numbers on the other (left) side. The image of the sum of two segments z_1+z_2 is obtained by laying off from the point z_1 the segment of length $\pm z_2$ to the right or left, according as z_2 is positive or negative.

Multiplication and division are introduced into the enlarged number domain by the equations

$$x(-y)=(-x)y=-xy,$$
$$(-x)(-y)=xy, \qquad 0x=0$$

are defined. Division, as the inverse of multiplication, is always possible except when the divisor is zero.

A further extension of the concept of number consists in introducing complex quantities. We combine every two numbers of the complete number line into pairs (x, y), and regard two such pairs (x, y) and (a, b) as equal only when $x=a$, $y=b$. These pairs of numbers form a manifold whose elements cannot be ordered, but with which addition, subtraction, multiplication, and division are to be performed according to the following rules. Let

$$(x,y)+(a,b)=(x+a, y+b),$$
$$(x,y)(a,b)=(xa-yb, xb+ya),$$

and we further stipulate that $(x, 0)=x$, which does not contradict these equations. The pair (x, y) is $=0$ only when x and y are both zero. For abbreviation we denote $(0, 1)$ by i. The equations above then give the consequences:

$$(x,0)(0,1)=(0,x) \quad \text{or} \quad =ix,$$
$$(x,0)+(0,y)=(x,y)=x+yi,$$

$$x + yi + a + bi = (x + a) + (y + b)i,$$

and the inverse of addition:

$$x + yi - (a + bi) = (x - a) + i(y - b),$$

and further multiplication:

$$(x + yi)(a + bi) = xa - yb + i(xb + ya), \qquad i^2 = -1,$$

$$(x + yi)(x - yi) = x^2 + y^2,$$

$$x + yi = (a + bi)\frac{(a - bi)(x + yi)}{a^2 + b^2}$$

or

$$\frac{x + yi}{a + bi} = \frac{ax + by + i(ay - bx)}{a^2 + b^2},$$

which defines division except when $a + bi = 0$.

Numbers of the form ix are called purely imaginary numbers, and i the imaginary unit. The numbers $a + bi$ are called imaginary or complex. The systems of real and purely imaginary numbers are contained among them as special cases.

Thus, in the domain of complex numbers $x + yi$, the fundamental operations can be carried out without restriction (except for division by zero), and calculation with real numbers is a special case.

Following Gauss, one represents the complex numbers $z = x + yi$ geometrically by the points of a plane, laying down a rectangular coordinate system and regarding the point with coordinates x, y as the image of the numerical value z. The points of the x-axis represent the real numbers x in the way discussed above. The points of the y-axis are the images of the purely imaginary numbers yi. The origin is the image of the number 0. The radius vector from the origin to the point z has the numerical value $\varrho = \sqrt{x^2 + y^2}$, which is called the absolute value, or the magnitude, or, according to older terminology, the modulus of the complex number z.

The only number having absolute value 0 is 0. Every positive number occurs as the absolute value of infinitely many complex numbers, and the image points of all numbers with the same absolute value lie on a circle whose center is at the origin.

Two imaginary numbers which differ only by the sign of i, namely $x + yi$ and $x - yi$, are called conjugate imaginary. Their product is the square of the absolute value of either one.

If one of two conjugate imaginary numbers is zero, the other is also zero; hence in every correct numerical equation one may replace i by $-i$ without disturbing its correctness.

We shall also state the frequently used theorem that the absolute value of the sum of two nonzero complex numbers is never greater than the sum of the absolute values of the summands, and is equal to it only when the ratio (the quotient) of the two summands is real and positive. Namely, let

$$z = x + yi, \qquad c = a + bi, \qquad Z = (x + a) + (y + b)i,$$

$$\varrho^2 = x^2 + y^2, \qquad r^2 = a^2 + b^2, \qquad R^2 = (x + a)^2 + (y + b)^2,$$

then

$$(r + \varrho - R)(r + \varrho + R) = (r + \varrho)^2 - R^2 = 2(r\varrho - ax - by);$$

this is certainly positive when $ax + by \leqq 0$. If, however, $ax + by > 0$, then from

$$r^2\varrho^2 - (ax + by)^2 = (ay - bx)^2,$$

it follows that $r\varrho \geqq ax + by$, with equality only when $ay - bx = 0$. Hence $r + \varrho > R$, except in the special case $r + \varrho = R$ when $ay - bx = 0$ and $ax + by > 0$, which proves the assertion.

In the geometric representation this theorem says that in a triangle one side is smaller than the sum of the other two. The theorem has the further consequence that the absolute value of a sum

cannot be smaller than the difference of the absolute values of the summands. For, applying the preceding theorem to the sum $c = Z - z$, one obtains $r \leqq R + \varrho$, or

$$R \geqq r - \varrho.$$

Equality occurs here only when the quotient $z : c$ is real and negative.

It remains to say a word about the most important aid of algebra, literal calculation. This aid is so generally used that the term literal calculation is sometimes employed outright as a synonym for algebra. We presuppose the rules for calculating with such literal expressions. Equations between literal expressions can be of two kinds: either they are so-called identities, that is, the two expressions set equal can be transformed by the rules of calculation until they agree exactly. Such literal equations yield correct numerical equations when the letters are replaced by any real or complex numbers, provided that division by zero is not required. The letters in such equations are often called variables, because one may imagine, without ever reaching a contradiction, that different numerical values are successively substituted for them.

Another kind of equation between literal expressions does not have the character of an identity. It instead contains a requirement imposed on the numbers that may be substituted for the letters without error. Algebra has the task of determining numerical values that satisfy such a requirement, that is, of solving the equation. In these equations the letters are also called "unknowns." Very often one and the same equation contains letters of two kinds: some for which arbitrary numerical values are to be substituted, and others whose numerical value is to be determined.

FIRST BOOK.

THE FOUNDATIONS.

First Section.

Rational Functions.

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§. 1.

Whole Functions.

The next object of consideration is whole rational functions, or briefly whole functions of one variable. By these we understand expressions of the following form:

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n, \tag{1}$$

where n is an integral exponent, the degree of the function $f(x)$. The degree is a natural number. Sometimes, however, it is also useful to speak of whole rational functions of degree 0, by which a constant is understood. x is called the variable, $a_0, a_1, a_2, \ldots, a_{n-1}, a_n$ the coefficients. Both x and $a_0, a_1, \ldots, a_n$ are symbols for indeterminate quantities, with which one proceeds according to the rules of literal calculation, and for which definite numerical values may also be substituted under some circumstances (cf. the Introduction).

If the function $f(x)$ is written in the way shown in (1), we call it ordered according to descending powers of x. The summands may be arranged in any other order; for example, the function may also be ordered according to ascending powers of x:

$$f(x) = a_n + a_{n-1}x + \cdots + a_1x^{n-1} + a_0x^n.$$

By addition (subtraction) and multiplication of whole rational functions there again arise whole rational functions. The rules of literal calculation directly give the laws of formation of these new functions.

In addition, the coefficient of any power x^ν in the sum is equal to the sum of the coefficients of x^ν in the individual summands. The degree of the function that arises is

equal to the highest of the degrees of the summands, and can decrease only in the special case in which the highest degree occurs in several summands and the sum of the coefficients of the highest powers is zero.

In the multiplication of two or more whole rational functions there arises a function whose degree is equal to the sum of the degrees of the factors.

To survey the law of formation of the coefficients for the product, put

$$\begin{aligned} A(x) &= a_0x^m + a_1x^{m-1} + a_2x^{m-2} + \cdots, \\ B(x) &= b_0x^n + b_1x^{n-1} + b_2x^{n-2} + \cdots, \end{aligned} \tag{2}$$

$$A(x)B(x) = C(x) = c_0x^{m+n} + c_1x^{m+n-1} + c_2x^{m+n-2} + \cdots \tag{3}$$

and obtain

$$\begin{aligned} c_0 &= a_0b_0, \\ c_1 &= a_0b_1 + a_1b_0, \\ c_2 &= a_0b_2 + a_1b_1 + a_2b_0, \\ &\cdots, \end{aligned}$$

or, in general notation, if ν denotes one of the numbers $0, 1, 2, \ldots, m+n$:

$$c_\nu = a_0b_\nu + a_1b_{\nu-1} + a_2b_{\nu-2} + \cdots + a_{\nu-1}b_1 + a_\nu b_0, \tag{4}$$

where all a whose index is greater than m, and all b whose index is greater than n, are to be set equal to zero. For c_ν is the coefficient of $x^{m+n-\nu}$, and hence $c_\nu x^{m+n-\nu}$ arises by multiplying all terms of the form $a_\mu x^{m-\mu}$ by all terms of the form $b_{\nu-\mu}x^{n-\nu+\mu}$ and then adding. If a_0 and $b_0 = 1$—that is, if both are equal to one—then also $c_0 = 1$.

If in all factors of such a product the highest power of x has coefficient 1, then in the product too the coefficient of the highest power is $= 1$.

From these rules for calculating with whole functions it follows that the rules of calculation expressed in formulas such as $ab = ba$, $(ab)c = a(bc)$, $(a+b)c = ac + bc$ and the like also hold when a, b, c are whole functions of x, and in the sense that whole functions are to be regarded as equal only when equal powers have equal coefficients.

By whole functions of several variables we understand sums of products of whole, positive powers of the variables $x, y, z \ldots$ with arbitrary coefficients that count as constant. One can think of these

functions as arising

from functions of one variable x when the coefficients $a_0, a_1 \dots a_n$ occurring in them are themselves regarded as whole rational functions of other variables $y, z \dots$. Thus functions of m variables arise from functions of $m-1$ variables. All terms of such a function are of the form $x^r y^s z^t \cdots$, multiplied by a coefficient; and if the function is properly collected, then each combination of the exponents $r, s, t \dots$ occurs only once. Two whole functions so ordered are regarded as equal only when they contain the same products $x^r y^s z^t \cdots$ with the same coefficients.

§. 2.

A Theorem of Gauss.

We shall at once apply the multiplication rule for two whole rational functions in order to prove a theorem of Gauss which will be useful to us later, but which here is to serve as an introduction to the mode of calculation and as an example.[2]

We consider here the case in which the coefficients in the functions $A(x), B(x)$ are whole numbers, so that, by (4), the coefficients of $C(x) = A(x)B(x)$ are also whole numbers.

If all integral coefficients $a_0, a_1, \dots, a_m$ of a function $A(x)$ have no common divisor, then the function $A(x)$ is called an original or primitive function, and the theorem we wish to prove is:

If $A(x)$ and $B(x)$ are primitive functions, then their product $C(x)$ is also a primitive function.

The proof follows almost immediately from looking at formula (4).

For if all coefficients $c_0, c_1, c_2 \dots c_{m+n}$, as given in (4), had a common divisor greater than 1, then there would also have to be at least one prime number which divides all these coefficients. Let p be such a prime number. By the assumption that $A(x), B(x)$ are primitive, this prime can divide neither all $a_0, a_1 \dots a_m$ nor all $b_0, b_1 \dots b_n$.

Suppose now that p divides

$$a_0, a_1 \dots a_{r-1}, \quad \text{but not } a_r,$$

and divides

$$b_0, b_1 \dots b_{s-1}, \quad \text{but not } b_s,$$

Then, by assumption, $r \leqq m$, and may also be zero if p already does not divide a_0. Likewise $s \leqq n$.

If now, by (4), we form c_{r+s} and arrange it in the following way,

$$\begin{aligned} c_{r+s} = {} & a_r b_s + a_{r-1} b_{s+1} + a_{r-2} b_{s+2} + \cdots \\ & + a_{r+1} b_{s-1} + a_{r+2} b_{s-2} + \cdots , \end{aligned} \tag{1}$$

one sees at once that c_{r+s} cannot be divisible by p, as had been assumed; for the first term $a_r b_s$ is not divisible by p, while all other terms, since they are multiplied by one of the coefficients $a_0, a_1 \dots a_{r-1}, b_0, b_1 \dots b_{s-1}$, are divisible by p. Thus the assumption that $C(x)$ is not primitive while $A(x)$ and $B(x)$ are primitive leads to a contradiction.

This theorem can be transferred to functions of several variables. We call a whole rational function of m variables with integral coefficients original or primitive if the coefficients have no common divisor, and we state the theorem that the product of two primitive functions is again a primitive function.

To see its truth, we need only take, in the argument carried out above, the coefficients $a_0, a_1, a_2 \dots$, $b_0, b_1, b_2 \dots$ not as whole numbers but as whole rational functions of $m-1$ variables $y, z \dots$, and call such a function divisible by a prime number p when all its coefficients are divisible by p. If we then suppose that the theorem to be proved has already been proved for functions of $m-1$

[2] Gauss, *Disquisitiones arithmeticae*, Art. 42.

variables, formula (1) gives its correctness for functions of m variables, hence its general validity by complete induction, or by the conclusion from $m-1$ to m.

An imprimitive function is one whose integral coefficients all have a common divisor. We call this divisor the divisor of the function. Then we can also express the theorem just proved as follows:

The divisor of a product of two whole functions is equal to the product of the divisors of the two functions.

For if PA and QB are two whole functions with divisors P and Q, then A and B are primitive functions. Hence $AB = C$ is also a primitive function, and PQ is the divisor of the function $PA \cdot QB = PQC$.

Insofar as it refers to functions of one variable, we can give the theorem the following form, without changing its content essentially; in this form it is especially useful. Let

$$\varphi(x) = x^m + \alpha_1 x^{m-1} + \alpha_2 x^{m-2} + \cdots + \alpha_m,$$
$$\psi(x) = x^n + \beta_1 x^{n-1} + \beta_2 x^{n-2} + \cdots + \beta_n$$

be two whole rational functions in which the highest powers of x have coefficient 1, while the remaining coefficients are rational numbers. Then, in the product

$$\varphi(x)\psi(x) = x^{m+n} + \gamma_1 x^{m+n-1} + \gamma_2 x^{m+n-2} + \cdots + \gamma_{m+n}$$

the coefficients γ cannot all be whole numbers if the coefficients α, β in $\varphi(x)$ and $\psi(x)$ are not all whole numbers.

For let the smallest common denominator of the coefficients α of φ be denoted by a_0, and that of the coefficients β of ψ by b_0. Then $a_0\varphi(x) = A(x)$ and $b_0\psi(x) = B(x)$ are primitive functions of x. If the coefficients γ were whole numbers, the product $a_0 b_0 \varphi(x)\psi(x)$ would have divisor $a_0 b_0$ and, if a_0 and b_0 were not both equal to 1, would therefore not be primitive; but this would contradict the theorem proved above.

§. 3.

Division.

Let, as before,

$$\begin{aligned} A = A(x) &= a_0 x^m + a_1 x^{m-1} + \cdots, \\ B = B(x) &= b_0 x^n + b_1 x^{n-1} + \cdots \end{aligned} \tag{1}$$

be two whole rational functions of x; but now suppose

that $m \geqq n$ and that a_0 and b_0 are different from zero. Then the difference

$$A - \frac{a_0}{b_0} x^{m-n} B \tag{2}$$

is also a whole rational function of x, but its degree is smaller than m, since the highest power in the two terms of the difference has the same coefficient and therefore cancels. We put this difference equal to

$$A' = A'(x) = a_0' x^{m'} + a_1' x^{m'-1} + \cdots, \qquad m' < m. \tag{3}$$

If now m' is still $\geqq n$, then in (2) we can put A' in place of A and repeat the same reasoning.

This gives a chain of equations:

$$\begin{aligned} A - \frac{a_0}{b_0} x^{m-n} B &= A', \\ A' - \frac{a_0'}{b_0} x^{m'-n} B &= A'', \\ A'' - \frac{a_0''}{b_0} x^{m''-n} B &= A''', \\ &\cdots, \end{aligned} \tag{4}$$

and this chain can be continued until the degree of the function that has arisen is smaller than n. Since in the sequence of functions $A, A', A'' \ldots$ the degree of each following function is lowered by at least one unit, the chain of equations (4) consists of at most $m-n+1$ members; but it may also contain fewer members when several powers cancel simultaneously. If we add all equations (4), denote the last of the functions $A, A' \ldots$ by C, and, for abbreviation, set

$$Q = \frac{a_0}{b_0} x^{m-n} + \frac{a_0'}{b_0} x^{m'-n} + \cdots , \tag{5}$$

so that Q also is a whole rational function of x, then there follows

$$A = QB + C. \tag{6}$$

The operation just described, by which the functions Q, C are found from A, B, is called division. A is the dividend, B the divisor, C the remainder, and Q the quotient. The degree of the remainder is always lower than the degree of the divisor.

The coefficients of the functions Q and C are composed from the coefficients a and b by addition, subtraction, multiplication, and division. In the denominator, however, only powers of b_0 occur; hence, if $b_0 = 1$, the coefficients of Q and C are whole functions of the a and b. The highest power of b_0 that can occur in the denominator is the $(m-n+1)$st, since in the chain (4) the factor b_0 is added once in the denominator in each succeeding equation. In special cases, however, the highest power of b_0 in all denominators may be lower.

Take, for example, for A a function of the third degree,

$$f(x) = a_0 x^3 + a_1 x^2 + a_2 x + a_3 \tag{7}$$

and for B the so-called first derivative of $f(x)$, which is of the second degree,

$$f'(x) = 3a_0 x^2 + 2a_1 x + a_2, \tag{8}$$

and hence:

$$Q = \frac{1}{3} x + \frac{a_1}{9a_0}, \tag{9}$$

$$C = \frac{6a_0 a_2 - 2a_1^2}{9a_0} x + \frac{9a_0 a_3 - a_1 a_2}{9a_0}. \tag{10}$$

How the calculation prescribed in equations (4) is expediently arranged may here be assumed as known from the elements. We draw attention to the analogy between this calculation and division in the decimal number system. A function $f(x)$ represents a number written decimally if the coefficients $a_0, a_1, \ldots$ are whole numbers between zero (inclusive) and 10 (exclusive), and $x = 10$ is set. If numbers greater than 10 are also permitted among the coefficients, a number can be represented by $f(x)$ in different ways. This is used in the division procedure to avoid fractional and negative coefficients in the results in the simplest way.

§. 4.

Division by a Linear Function.

We shall apply the general rule for division to one special case. Let the dividend be

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_n \tag{1}$$

arbitrary, while the divisor is of the first degree, or, as one also says, linear. We shall also suppose the coefficient of the first power of x in the divisor to be $= 1$, and therefore take the divisor in the simple form $(x - \alpha)$, where α remains arbitrary. In this case the quotient Q is of degree $(n-1)$, and the remainder C is of degree 0, that is, independent of x. Thus set

$$f(x) = (x - \alpha)Q + C, \tag{2}$$

so that C no longer contains x; and if we set

$$Q = q_0x^{n-1} + q_1x^{n-2} + \cdots + q_{n-2}x + q_{n-1} \tag{3}$$

then there follows from (2):

$$\begin{aligned} f(x) = {} & q_0x^n + q_1x^{n-1} + \cdots + q_{n-2}x^2 + q_{n-1}x + C \\ & - \alpha q_0x^{n-1} - \cdots - \alpha q_{n-3}x^2 - \alpha q_{n-2}x - \alpha q_{n-1}, \end{aligned} \tag{4}$$

and by comparison with (1):

$$\begin{aligned} q_0 &= a_0, \\ q_1 - \alpha q_0 &= a_1, \\ q_2 - \alpha q_1 &= a_2, \\ &\cdots, \\ q_{n-1} - \alpha q_{n-2} &= a_{n-1}, \\ C - \alpha q_{n-1} &= a_n. \end{aligned} \tag{5}$$

From this one obtains

$$\begin{aligned} q_0 &= a_0, \\ q_1 &= a_0\alpha + a_1, \\ q_2 &= a_0\alpha^2 + a_1\alpha + a_2, \\ &\cdots, \\ q_{n-1} &= a_0\alpha^{n-1} + a_1\alpha^{n-2} + \cdots + a_{n-1}, \\ C &= a_0\alpha^n + a_1\alpha^{n-1} + \cdots + a_{n-1}\alpha + a_n = f(\alpha). \end{aligned} \tag{6}$$

C arises from $f(x)$ by setting $x = \alpha$, and therefore can also be denoted by $f(\alpha)$.

Accordingly we also have the formula

$$\frac{f(x) - f(\alpha)}{x - \alpha} = Q(x), \tag{7}$$

where $Q(x)$ is a whole function of degree $n - 1$.

If in expressions (6) we put the sign x in place of the indeterminate quantity α, there arises a sequence of whole rational functions of x, which, if for simplicity we set $a_0 = 1$, we write as follows:

$$\begin{aligned} f_0 &= 1, \\ f_1 &= x + a_1, \\ f_2 &= x^2 + a_1x + a_2, \\ &\cdots, \\ f_{n-1} &= x^{n-1} + a_1x^{n-2} + a_2x^{n-3} + \cdots + a_{n-1}. \end{aligned} \tag{8}$$

These functions $f_0, f_1, \ldots, f_{n-1}$ will later render us good service. For the present we add the following remarks.

According to (8), the powers $1, x, x^2, \ldots, x^{n-1}$ of x can be expressed linearly in terms of the functions $f_0, f_1, \ldots, f_{n-1}$, in such a way that only whole rational combinations of the a occur in the coefficients, for example

$$\begin{aligned} 1 &= f_0, \\ x &= f_1 - a_1f_0, \\ x^2 &= f_2 - a_1f_1 + (a_1^2 - a_2)f_0, \\ &\cdots, \end{aligned}$$

and it follows that every whole rational function of x whose degree is not greater than $n - 1$ can likewise be expressed linearly in terms of $f_0, f_1, \ldots, f_{n-1}$ in the form

$$y_0f_0 + y_1f_1 + \cdots + y_{n-1}f_{n-1}, \tag{9}$$

where the coefficients $y_0, y_1, \ldots, y_{n-1}$ are independent of x.

Thus, if $F(x)$ is any whole rational function of x, then by § 3, regarding $f(x)$ as divisor, one can set

$$F(x) = Qf(x) + y_0 f_0 + y_1 f_1 + \cdots + y_{n-1} f_{n-1} \tag{10}$$

where Q is also a whole rational function of x.

For the recursive calculation of the functions $f_r(x)$, relation (8) gives

$$f_r(x) - x f_{r-1}(x) = a_r. \tag{11}$$

§. 5.

Fractional Functions; Divisibility.

If $F(x)$ and $f(x)$ are two whole rational functions of x, then the fraction

$$\frac{F(x)}{f(x)} \tag{1}$$

is called a fractional rational, or briefly a fractional or rational function of x.

If the degree of the numerator is lower than the degree of the denominator, the function is called a proper fraction; in the opposite case it is called an improper fraction.

According to § 3, the whole rational functions Q and $\varphi(x)$ can be determined so that

$$F(x) = Qf(x) + \varphi(x), \tag{2}$$

and the degree of $\varphi(x)$ is smaller than the degree of $f(x)$; consequently

$$\frac{F(x)}{f(x)} = Q + \frac{\varphi(x)}{f(x)} \tag{3}$$

and hence the theorem:

Every fractional function can be decomposed into the sum of a whole function and a proper fractional function.

If n is the degree of $f(x)$, then the degree of $\varphi(x)$ is at most $n-1$; it may also be lower. In particular, the case may occur in which $\varphi(x)$ vanishes identically.

In this case the function $F(x)$ is said to be divisible by $f(x)$. The function

$$\frac{F(x)}{f(x)}$$

is in this case only apparently fractional; in reality it is equal to the whole function Q.

The formula

$$\frac{x^m - 1}{x - 1} = x^{m-1} + x^{m-2} + x^{m-3} + \cdots + 1$$

gives a simple example.

For divisibility of functions the same laws hold as for divisibility of numbers, in particular the following:

1. If the function $F(x)$ is divisible by the function $f(x)$, and $f(x)$ is divisible by a third function $\varphi(x)$, then $F(x)$ is also divisible by $\varphi(x)$.

For if

$$F = Qf, \qquad f = q\varphi,$$

where Q and q are whole rational functions, then

$$F = Qq\varphi,$$

and since Qq is a whole rational function, F is divisible by φ.

2. If $F(x)$ is divisible by $f(x)$ and Q is any whole rational function, then $QF(x)$ is also divisible by $f(x)$.

3. If $F(x)$ and $f(x)$ are divisible by $\varphi(x)$, then $F(x) \pm f(x)$ is also divisible by $\varphi(x)$, or more generally:

4. If $F_1, F_2 \ldots$ are divisible by $f(x)$ and $Q_1, Q_2 \ldots$ are arbitrary whole rational functions, then

$$Q_1F_1 + Q_2F_2 + \cdots$$

is also divisible by $f(x)$.

The last theorem includes the two preceding ones and is proved simply as follows.

If $F_1, F_2 \ldots$ are divisible by f, then the whole rational functions $\Phi_1, \Phi_2 \ldots$ can be determined so that

$$F_1 = \Phi_1 f, \qquad F_2 = \Phi_2 f, \ldots$$

and consequently

$$Q_1F_1 + Q_2F_2 + \cdots = (Q_1\Phi_1 + Q_2\Phi_2 + \cdots)f.$$

Since $Q_1\Phi_1 + Q_2\Phi_2 + \cdots$ is a whole rational function, the theorem is proved.

5. Every function is divisible by itself.

With respect to divisibility or indivisibility of functions nothing is changed if the functions are multiplied by arbitrary factors independent of x.

A non-zero quantity independent of x may be regarded as a function of degree zero. If we call such a quantity a constant, we may say:

6. Every function is divisible by every constant.

If a function is divisible by another, then the degree of the quotient is equal to the difference between the degree of the dividend and the degree of the divisor. The former therefore cannot be smaller than the latter. If the degrees are equal, the quotient is a constant, and hence:

7. If, of two whole rational functions of the same degree, one is divisible by the other, then they differ from one another only by a constant factor, and the second is also divisible by the first.

8. By § 4, the necessary and sufficient condition that the function $f(x)$ be divisible by the linear function $x - \alpha$ is that $f(\alpha) = 0$.

§. 6.

Greatest Common Divisor.

It is a problem of fundamental importance to decide whether two whole rational functions have, besides the constants, any further common divisor. The solution is found by the algorithm of the greatest common divisor in exactly the same way as the corresponding question concerning divisibility of integers is answered. See the Introduction.

Let

$$f(x) = A, \qquad \varphi(x) = A' \tag{1}$$

be two given functions; the degree of $\varphi(x)$ is supposed to be lower, or at least not higher, than that of $f(x)$.

We divide A by A' and denote the remainder, whose degree is lower than the degree of A', by A'', thus:

$$A = Q'A' + A'';$$

then we divide A' by A'' and denote the remainder by A''', and continue in this way in forming the sequence of functions

$$A, A', A'', A''', \ldots, \tag{2}$$

whose degrees $n, n', n'', n''' \ldots$ always decrease and therefore, after a finite number of divisions, necessarily go down to zero. Let the last remainder of degree zero, and therefore a constant, be $A^{(\nu)}$. Then we have the chain of equations:

$$\begin{aligned} A &= Q'A' + A'', \\ A' &= Q''A'' + A''', \\ &\cdots \\ A^{(\nu-3)} &= Q^{(\nu-2)}A^{(\nu-2)} + A^{(\nu-1)}, \\ A^{(\nu-2)} &= Q^{(\nu-1)}A^{(\nu-1)} + A^{(\nu)}, \end{aligned} \tag{3}$$

where $A^{(\nu)}$ is a constant.

If now A, A' have any common divisor, then by the theorems of the preceding paragraph, as is seen from the equations (3), it is also a divisor of A'', A''', A'''' and so on down to $A^{(\nu)}$. If $A^{(\nu)}$ is a non-zero constant, then no divisor of $f(x)$ and $\varphi(x)$ depending on x can exist. Such functions are called prime to one another, or relatively prime. One also says, taking account only of divisors depending on x, that the functions have no common divisor.

The condition that the two functions have a common divisor is therefore

$$A^{(\nu)} = 0. \tag{4}$$

In this case $A^{(\nu-1)}$ is a divisor of $A^{(\nu-2)}$, as the last equation (3) shows, and by the preceding equation it is a divisor of $A^{(\nu-3)}$, and so on, hence also a divisor of A and A', that is, of f and φ. Conversely, every common divisor of A, A' is also a divisor of $A^{(\nu-1)}$; accordingly $A^{(\nu-1)}$ is called the greatest common divisor of f and φ. The algorithm (3) shows that $A^{(\nu)}$ or $A^{(\nu-1)}$ can be derived from the coefficients of f and φ by the rational operations of arithmetic, and in such a manner that division is always only by the coefficients of the highest powers of x in the functions $A, A', A'', \ldots$ which are not zero.

We shall apply these considerations to two examples.

First let

$$\begin{aligned} A &= a_0x^2 + a_1x + a_2, \\ B &= b_0x^2 + b_1x + b_2 \end{aligned} \tag{5}$$

be two functions of the second degree, so that a_0, b_0 are non-zero.

The first step is to form the equation

$$A = \frac{a_0}{b_0}B + C, \tag{6}$$

where

$$C = c_0x + c_1 \tag{7}$$

and

$$c_0 = \frac{a_1b_0 - b_1a_0}{b_0}, \qquad c_1 = \frac{a_2b_0 - b_2a_0}{b_0}. \tag{8}$$

If $c_0 = 0$, so that C is constant, then A, B have a common factor only if also $c_1 = 0$; and then A is divisible by B, that is, A and B differ only by a constant factor. If however c_0 is different from zero, we continue the calculation by putting

$$B = QC + D$$

where, according to § 4, if there $\alpha = -c_1 : c_0$ is put,

$$D = \frac{b_0 c_1^2 - b_1 c_0 c_1 + b_2 c_0^2}{c_0^2}. \tag{9}$$

If D is non-zero, then A, B have no common divisor. If however $D = 0$, then C is the greatest common factor of A and B.

Substituting the values c_0, c_1 from (8) into (9), and suppressing the denominator $b_0 c_0^2$, one obtains the condition for the existence of a common divisor C of A and B in the form

$$\begin{aligned} a_0^2 b_2^2 + a_2^2 b_0^2 - 2a_0 a_2 b_0 b_2 - a_1 a_2 b_0 b_1 - a_0 a_1 b_1 b_2 \\ + a_0 a_2 b_1^2 + a_1^2 b_0 b_2 = 0 \end{aligned} \tag{10}$$

or

$$(a_0 b_2 - b_0 a_2)^2 + (a_0 b_1 - a_1 b_0)(a_2 b_1 - a_1 b_2) = 0. \tag{11}$$

This condition is also fulfilled when c_0 and c_1 are both zero, and is therefore the necessary and sufficient condition that A, B have a common divisor.

The left hand side of (10) or (11) is called the resultant of the functions A and B.

As a second example we take the one already chosen in § 3. Put

$$\begin{aligned} f(x) &= a_0 x^3 + a_1 x^2 + a_2 x + a_3 = A, \\ f'(x) &= 3a_0 x^2 + 2a_1 x + a_2 = B, \end{aligned} \tag{12}$$

and suppose a_0 different from zero. Then by § 3 we have:

$$A = QB + C, \tag{13}$$

$$C = c_0 x + c_1, \tag{14}$$

where

$$c_0 = \frac{6a_0 a_2 - 2a_1^2}{9a_0}, \qquad c_1 = \frac{9a_0 a_3 - a_1 a_2}{9a_0}. \tag{15}$$

If c_0 is zero, the algorithm is already closed. If c_1 is non-zero, then A and B have no common divisor; but if c_1 is also zero, then B itself is the greatest common divisor of A and B, that is, A is divisible by B. The conditions for this are therefore

$$3a_0 a_2 - a_1^2 = 0, \qquad 9a_0 a_3 - a_1 a_2 = 0. \tag{16}$$

If c_0 is not zero, we go one step further and put

$$B = PC + D, \tag{17}$$

where D becomes constant and has the expression

$$D = \frac{a_2 c_0^2 - 2a_1 c_0 c_1 + 3a_0 c_1^2}{c_0^2}. \tag{18}$$

If this expression is non-zero, then A and B are prime to one another; if it is zero, then A and B have the greatest common divisor C. If for c_0, c_1 one substitutes the values (15), omits the denominator, still removes the non-vanishing factor $9a_0$, and changes the sign, this condition, after a simple calculation, assumes the form

$$a_1^2a_2^2 + 18a_0a_1a_2a_3 - 4a_0a_2^3 - 4a_1^3a_3 - 27a_0^2a_3^2 = 0. \tag{19}$$

It is, as one readily sees either by calculation or from (18), also fulfilled when the conditions (16) hold, and is therefore the necessary and sufficient condition that $f(x)$ and $f'(x)$ have a common divisor. The left hand side of (19), which is a whole rational and homogeneous function of the coefficients of $f(x)$, is called the discriminant of the function $f(x)$.

We derive one more theorem from the algorithm (3), which is often used.

From the first of these equations it follows that

$$A'' = A - Q'A', \tag{20}$$

and if this value of A'' is substituted in the following equation, one obtains

$$A''' = (1 + Q'Q'')A' - Q''A,$$

thus, denoting by p, p' whole rational functions,

$$A''' = pA + p'A'. \tag{21}$$

Substituting the expressions (20), (21) in the third equation (3), one obtains for A'''' again an expression of the form (21), and so one may continue, obtaining finally

$$A^{(\nu)} = PA + P'A', \tag{22}$$

where P, P' are whole rational functions whose coefficients are composed from the coefficients of A and A' by rational operations.

In the formula (22), $A^{(\nu)}$ is a constant. This theorem is especially important in the case where $A^{(\nu)}$ is non-zero, so that A, A' are relatively prime. If in this case we put

$$P = A^{(\nu)}F(x), \qquad P' = A^{(\nu)}\Phi(x),$$

then, after suppressing the factor $A^{(\nu)}$, the theorem can be expressed as follows:

I. If $f(x)$ and $\psi(x)$ are two whole functions without a common divisor, then two other whole functions $F(x)$ and $\Phi(x)$ can be determined which satisfy the equation

$$F(x)f(x) + \Phi(x)\psi(x) = 1 \tag{23}$$

identically.

The theorem can still be generalized. Multiplying (23) by an arbitrary whole function $\chi(x)$ gives

$$F(x)\chi(x)f(x) + \Phi(x)\chi(x)\psi(x) = \chi(x), \tag{24}$$

and by § 3 we can put

$$\Phi(x)\chi(x) = Q(x)f(x) + \varphi(x) \tag{25}$$

in such a way that $Q(x), \varphi(x)$ are whole rational functions of x and the degree of $\varphi(x)$ is lower than that of $f(x)$. Substituting this in (24), and putting in place of $F(x)\chi(x) + Q(x)\psi(x)$ again $F(x)$, we obtain

$$F(x)f(x) + \varphi(x)\psi(x) = \chi(x).$$

We can therefore generalize the preceding theorem as follows:

II. If $f(x), \psi(x), \chi(x)$ are given whole rational functions, and $f(x)$ and $\psi(x)$ are without a common divisor, then the

whole rational functions $F(x), \varphi(x)$ can be so determined, with $\varphi(x)$ of lower degree than $f(x)$, that the equation

$$F(x)f(x) + \varphi(x)\psi(x) = \chi(x) \tag{26}$$

is identically satisfied.

§. 7.

Products of Linear Factors.

According to § 1, by multiplication of whole functions we again obtain such functions of higher degree, and the degree of the product is determined by the sum of the degrees of the individual factors.

If therefore we multiply n linear factors by one another, there arises a function of the nth degree, whose manner of formation we must examine somewhat more closely.

We shall take the linear factors in the simple form $x - \alpha_1$, $x - \alpha_2$, ..., $x - \alpha_n$, and put, since the coefficient of the highest power is (by § 1) = 1,

$$\begin{aligned} f(x) &= (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n) \\ &= x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n. \end{aligned} \tag{1}$$

A slight consideration shows the following law of formation of the coefficients $a_1, a_2 \ldots a_n$:

$-a_1$ is equal to the sum of the α's; a_2 is equal to the sum of the products of every two of the α's; $-a_3$ is the sum of the products of every three of the α's; in general $(-1)^\nu a_\nu$ is the sum of the products of every ν of the quantities α, or, in formulas,

$$\begin{aligned} -a_1 &= \sum \alpha_1, \\ +a_2 &= \sum \alpha_1\alpha_2, \\ &\cdots \\ (-1)^\nu a_\nu &= \sum \alpha_1\alpha_2\cdots\alpha_\nu, \\ &\cdots \\ (-1)^n a_n &= \alpha_1\alpha_2\cdots\alpha_n. \end{aligned} \tag{2}$$

But to see still more clearly the correctness of this law of formation, one uses complete induction.

One first verifies the correctness in the first cases by actually carrying out the multiplication

$$\begin{aligned} (x-\alpha_1)(x-\alpha_2) &= x^2 - (\alpha_1+\alpha_2)x + \alpha_1\alpha_2, \\ (x-\alpha_1)(x-\alpha_2)(x-\alpha_3) &= x^3 - (\alpha_1+\alpha_2+\alpha_3)x^2 \\ &\quad + (\alpha_1\alpha_2 + \alpha_1\alpha_3 + \alpha_2\alpha_3)x - \alpha_1\alpha_2\alpha_3. \end{aligned}$$

If we assume the law of formation proved for $n-1$ factors, and put

$$(x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_{n-1}) = x^{n-1} + a_1'x^{n-2} + \cdots + a_{n-1}',$$

then multiplication by $x - \alpha_n$ gives

$$\begin{aligned} a_1 &= a'_1 - \alpha_n, \\ a_2 &= a'_2 - \alpha_n a'_1, \\ a_3 &= a'_3 - \alpha_n a'_2, \\ &\cdots \\ a_\nu &= a'_\nu - \alpha_n a'_{\nu-1}, \\ &\cdots \\ a_n &= -\alpha_n a'_{n-1}, \end{aligned} \tag{3}$$

and from this the correctness of the formulas (2) is immediately evident.

It is important to determine the number of terms which occur in each of the sums (2). We denote the number of terms occurring in the sum $(-1)^\nu a_\nu$ by $B_\nu^{(n)}$; it is the number of combinations without repetition of n elements to the νth class, that is, of every ν distinct ones among the n elements. To determine it, suppose first that the $B_{\nu-1}^{(n)}$ combinations to the $(\nu-1)$st class have been formed. From each of these combinations, by adding one at a time of the missing $n-\nu+1$ elements, one can derive $n-\nu+1$ combinations to the νth class. In this way however every combination is formed ν times, namely by addition of each of its elements; hence one obtains the relation

$$B_\nu^{(n)} = B_{\nu-1}^{(n)} \frac{n-\nu+1}{\nu} \tag{4}$$

whereas $B_1^{(n)}$ plainly has the value n. Therefore, if one multiplies the equations following from (4),

$$\begin{aligned} B_1^{(n)} &= n, \\ B_2^{(n)} &= B_1^{(n)} \frac{n-1}{2}, \\ &\cdots \\ B_\nu^{(n)} &= B_{\nu-1}^{(n)} \frac{n-\nu+1}{\nu} \end{aligned} \tag{5}$$

it follows that:

$$B_\nu^{(n)} = \frac{n(n-1)(n-2)\cdots(n-\nu+1)}{1\cdot 2\cdot 3\cdots\nu}. \tag{6}$$

In what follows we shall often, when m is an arbitrary positive integer, use the symbol

$$\Pi(m) = 1\cdot 2\cdot 3\cdots m, \qquad \Pi(0) = 1 \tag{7}$$

so that

$$\Pi(m) = m\Pi(m-1). \tag{8}$$

With the help of this symbol the expression for $B_\nu^{(n)}$ can be displayed more conveniently as

$$B_\nu^{(n)} = B_{n-\nu}^{(n)} = \frac{\Pi(n)}{\Pi(\nu)\Pi(n-\nu)}, \tag{9}$$

which, if $B_0^{(n)} = 1$ is put, is also valid for $\nu = 0$ and $\nu = n$, and makes evident the invariance of $B_\nu^{(n)}$ under the interchange of ν with $n-\nu$.

The derivation of the formula (4), which we have just given according to the rules of the theory of combinations, is indeed perfectly correct and clear, but for its exact foundation it requires some

reflection which cannot easily be compressed into a few words. We shall therefore subsequently prove its correctness once more by the method of complete induction.

The formulas (3), namely, give the following recurrence formulas:

$$
\begin{aligned}
B_1^{(n)} &= B_1^{(n-1)} + 1, \\
B_2^{(n)} &= B_2^{(n-1)} + B_1^{(n-1)}, \\
&\cdots \\
B_\nu^{(n)} &= B_\nu^{(n-1)} + B_{\nu-1}^{(n-1)}, \\
&\cdots \\
B_n^{(n)} &= B_{n-1}^{(n-1)}.
\end{aligned}
\tag{10}
$$

Now the formula (6) can be confirmed very easily by counting for the first values of n. If one assumes it correct for $n-1$, then (10) gives

$$B_\nu^{(n)} = \frac{\Pi(n-1)}{\Pi(\nu)\Pi(n-\nu-1)} + \frac{\Pi(n-1)}{\Pi(\nu-1)\Pi(n-\nu)},$$

whence, by (8),

$$B_\nu^{(n)} = \frac{\Pi(n)}{\Pi(\nu)\Pi(n-\nu)},$$

and by this the general validity of formula (9) is proved.

From the meaning of $B_\nu^{(n)}$ it follows, and it is also proved from the formulas (10) by complete induction, that the $B_\nu^{(n)}$ are positive integers.

§. 8.

The Binomial Theorem.

If in formula (1) of the preceding paragraph we set the previously arbitrary quantities $\alpha_1, \alpha_2 \ldots \alpha_n$ equal to one another, we obtain the binomial theorem, which consists in nothing other than arranging the nth power of a binomial $x+y$ according to powers of x. Namely, if we put

$$\alpha_1 = \alpha_2 = \cdots = \alpha_n = -y$$

then formula (2) gives

$$a_\nu = (-1)^\nu \sum \alpha_1\alpha_2\cdots\alpha_\nu = y^\nu B_\nu^{(n)},$$

where, according to the definition of the preceding paragraph, $B_\nu^{(n)}$ means the number of terms of the sum, which all become equal to one another and equal to $(-1)^\nu y^\nu$.

Accordingly one obtains

$$(x+y)^n = x^n + B_1^{(n)}x^{n-1}y + B_2^{(n)}x^{n-2}y^2 + \cdots + B_n^{(n)}y^n, \tag{1}$$

or, written out,

$$
\begin{aligned}
(x+y)^n &= x^n + nx^{n-1}y + \frac{n(n-1)}{1\cdot 2}x^{n-2}y^2 + \cdots + y^n \\
&= \Pi(n)\sum_{\nu=0}^{n}\frac{x^{n-\nu}y^\nu}{\Pi(n-\nu)\Pi(\nu)} = \Pi(n)\sum\frac{x^\alpha y^\beta}{\Pi(\alpha)\Pi(\beta)},
\end{aligned}
$$

the last sum being extended over all non-negative integer values α, β which satisfy the condition $\alpha+\beta = n$.

Accordingly the coefficients $B_\nu^{(n)}$ are called the binomial coefficients.

For reference we put here a small table of the first values of the binomial coefficients:

$$\begin{array}{lccccccc} n=1 & 1 & 1 & & & & & \\ n=2 & 1 & 2 & 1 & & & & \\ n=3 & 1 & 3 & 3 & 1 & & & \\ n=4 & 1 & 4 & 6 & 4 & 1 & & \\ n=5 & 1 & 5 & 10 & 10 & 5 & 1 & \\ n=6 & 1 & 6 & 15 & 20 & 15 & 6 & 1 \\ n=7 & 1 & 7 & 21 & 35 & 35 & 21 & 7 \quad 1 \end{array}$$

Among the various properties of the binomial coefficients we shall derive two, of which we shall afterwards make an interesting application.

From (1), replacing x, y by $1, x$ and taking $n = 0, 1, 2, 3 \ldots$, the formulas

$$\begin{aligned} 1 &= B_0^{(0)}, \\ 1+x &= B_0^{(1)} + B_1^{(1)} x, \\ (1+x)^2 &= B_0^{(2)} + B_1^{(2)} x + B_2^{(2)} x^2, \\ &\cdots \\ (1+x)^n &= B_0^{(n)} + B_1^{(n)} x + B_2^{(n)} x^2 + \cdots + B_n^{(n)} x^n. \end{aligned} \tag{2}$$

are obtained.

We now use the well-known summation formula for the geometric series:

$$1 + (1+x) + (1+x)^2 + \cdots + (1+x)^n = \frac{(1+x)^{n+1} - 1}{x},$$

where, if $(1+x)^{n+1}$ is again expanded by the binomial formula, replacing n by $n+1$ in the last formula (2) and setting $B_0^{(n+1)} = 1$,

$$\frac{(1+x)^{n+1} - 1}{x} = B_1^{(n+1)} + B_2^{(n+1)} x + \cdots + B_{n+1}^{(n+1)} x^n. \tag{3}$$

Comparing this with the sum of the right hand sides of (2), and, according to the rules of § 1, setting equal the coefficients of corresponding powers of x, one obtains

$$\begin{aligned} B_0^{(0)} + B_0^{(1)} + B_0^{(2)} + \cdots + B_0^{(n)} &= B_1^{(n+1)}, \\ B_1^{(1)} + B_1^{(2)} + \cdots + B_1^{(n)} &= B_2^{(n+1)}, \\ &\cdots \\ B_n^{(n)} &= B_{n+1}^{(n+1)}, \end{aligned} \tag{4}$$

or in general

$$B_\nu^{(\nu)} + B_\nu^{(\nu+1)} + \cdots + B_\nu^{(n)} = B_{\nu+1}^{(n+1)}. \tag{5}$$

If however the equations (2) are multiplied in order by

$$B_0^{(n)}, \quad -B_1^{(n)}, \quad +B_2^{(n)}, \ldots, \quad \pm B_n^{(n)}$$

where the upper sign is to be taken for even and the lower for odd n, the sum of the left hand sides gives

$$B_0^{(n)} - B_1^{(n)}(1+x) + B_2^{(n)}(1+x)^2 - \cdots \pm B_n^{(n)}(1+x)^n = [1-(1+x)]^n = (-x)^n$$

and equating the coefficients of equal powers on the right and left hand sides yields the system of formulas:

$$
\begin{aligned}
0 &= B_0^{(0)}B_0^{(n)} - B_0^{(1)}B_1^{(n)} + B_0^{(2)}B_2^{(n)} - \cdots \pm B_0^{(n)}B_n^{(n)},\\
0 &= -B_1^{(1)}B_1^{(n)} + B_1^{(2)}B_2^{(n)} - \cdots \pm B_1^{(n)}B_n^{(n)},\\
&\cdots\\
\pm 1 &= \pm B_n^{(n)}B_n^{(n)},
\end{aligned}
\tag{6}
$$

and this system of formulas can also be written in the reversed order, using the relation $B_\nu^{(n)} = B_{n-\nu}^{(n)}$, as

$$
\begin{aligned}
1 &= B_0^{(n)}B_0^{(n)},\\
0 &= B_1^{(n)}B_0^{(n)} - B_0^{(n-1)}B_1^{(n)},\\
0 &= B_2^{(n)}B_0^{(n)} - B_1^{(n-1)}B_1^{(n)} + B_0^{(n-2)}B_2^{(n)},\\
&\cdots\\
0 &= B_n^{(n)}B_0^{(n)} - B_{n-1}^{(n-1)}B_1^{(n)} + B_{n-2}^{(n-2)}B_2^{(n)} - \cdots \pm B_0^{(0)}B_n^{(n)},
\end{aligned}
\tag{7}
$$

or, in a summarizing notation,

$$
\sum_{\nu=0}^{\mu}(-1)^\nu B_{\mu-\nu}^{(n-\nu)}B_\nu^{(n)} = \begin{cases} 1, & \mu = 0,\\ 0, & \mu = 1, 2, \ldots, n. \end{cases}
\tag{8}
$$

§. 9.

Interpolation.

We shall make an application of the last obtained formulae to a problem from the theory of interpolation.

A whole rational function of degree n is to be determined which, for $n+1$ given values of the variable x, has prescribed values.

Thus the problem is the determination of the $n+1$ coefficients $a_0, a_1, \ldots, a_n$ in

$$
f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n
\tag{1}
$$

from the $n+1$ equations

$$
f(\alpha_0) = A_0, \quad f(\alpha_1) = A_1, \quad \ldots, \quad f(\alpha_n) = A_n,
\tag{2}
$$

when $\alpha_0, \alpha_1, \ldots, \alpha_n$ are the given values of x, and $A_0, A_1, \ldots, A_n$ are the corresponding values of $f(x)$.

Equations (2) form a system of equations of the first degree for the unknowns $a_0, a_1, \ldots, a_n$, which in general can be solved by determinants, as we shall see in the next section. We shall later return to this problem,

but treat it now under the special supposition that the given values of x are the first $n+1$ whole numbers $0, 1, 2, \ldots, n$.

The binomial coefficients $B_\nu^{(n)}$, as they were defined by § 7, (6), keep their meaning even when n is not a whole number, as was previously presupposed, but an arbitrary variable quantity. Then

$$
B_\nu^{(x)} = \frac{x(x-1)\cdots(x-\nu+1)}{1\cdot 2\cdot 3\cdots\nu}
\tag{3}
$$

is a whole rational function of degree ν in x. [3] The coefficient of x^ν is different from zero, and therefore x^ν can also be expressed by $B_\nu^{(x)}$ and by lower powers of x. Thus every whole function $f(x)$ of degree n in x can also be linearly expressed by $B_0^{(x)}, B_1^{(x)}, \ldots, B_n^{(x)}$ in the form

$$f(x) = M_0 B_0^{(x)} + M_1 B_1^{(x)} + \cdots + M_n B_n^{(x)}, \tag{4}$$

where $M_0, M_1, \ldots, M_n$ are constants; and the function $f(x)$ is determined when these constants are determined.

Now, by our supposition, $f(0), f(1), f(2), \ldots, f(n)$ are given. Since, by (3), $B_\nu^{(x)}$ always vanishes when x has one of the values $0, 1, 2, \ldots, \nu - 1$, the following linear equations for the unknowns M are obtained from (4):

$$\begin{aligned} f(0) &= M_0 B_0^{(0)}, \\ f(1) &= M_0 B_0^{(1)} + M_1 B_1^{(1)}, \\ f(2) &= M_0 B_0^{(2)} + M_1 B_1^{(2)} + M_2 B_2^{(2)}, \\ &\cdots \\ f(n) &= M_0 B_0^{(n)} + M_1 B_1^{(n)} + M_2 B_2^{(n)} + \cdots + M_n B_n^{(n)}. \end{aligned} \tag{5}$$

These equations are now to be solved with respect to $M_0, M_1, \ldots, M_n$, which is done very easily with the help of the final equations (8) of the preceding paragraph. The first equation (5) gives directly

$$M_0 = f(0). \tag{6}$$

If the first equation (5) is multiplied by $B_0^{(1)}$, the second by $-B_1^{(1)}$, and they are added, one obtains, by the formula system mentioned (applied to $n = 1$),

$$-M_1 = B_0^{(1)} f(0) - B_1^{(1)} f(1),$$

and so in general, if the first ν equations (5) are multiplied in order by

$$B_0^{(\nu)}, \quad -B_1^{(\nu)}, \quad +B_2^{(\nu)}, \quad \ldots, \quad \pm B_\nu^{(\nu)}$$

and then added,

$$\pm M_\nu = B_0^{(\nu)} f(0) - B_1^{(\nu)} f(1) + B_2^{(\nu)} f(2) - \cdots \pm B_\nu^{(\nu)} f(\nu), \tag{7}$$

whereby, according to (4), the function $F(x)$ is determined and the problem is solved. It is clear that, so long as we make no special supposition about the values $f(0), f(1), \ldots, f(n)$, every arbitrary whole rational function of x can be represented in this form.

§. 10.

Solution of the Interpolation Problem by Differences.

The definition of the whole rational function

$$B_\nu^{(x)} = \frac{x(x-1)\cdots(x-\nu+1)}{1 \cdot 2 \cdot 3 \cdots \nu} \tag{1}$$

gives the relation, already proved earlier for the special case of a positive integer x,

$$B_\nu^{(x+1)} = B_\nu^{(x)} + B_{\nu-1}^{(x)}, \qquad B_0^{(x+1)} = B_0^{(x)} = 1. \tag{2}$$

[3] The meaning of these generalized binomial coefficients for the expansion of powers of a binomial does not belong here, but to analysis.

From this we obtain for the coefficients $M_0, M_1, \ldots, M_n$ a mode of determination which is much more convenient for practical calculation than the application of the formulae of the preceding paragraph.

Namely, by the formulae (4) and (6) of § 9,

$$f(x) = f(0) + M_1 B_1^{(x)} + M_2 B_2^{(x)} + \cdots + M_n B_n^{(x)}, \tag{3}$$

and if in this expression we replace x by $x+1$ and form the difference

$$\Delta_x = f(x+1) - f(x) \tag{4}$$

then, with regard to (2),

$$\Delta_x = M_1 + M_2 B_1^{(x)} + M_3 B_2^{(x)} + \cdots + M_n B_{n-1}^{(x)}, \tag{5}$$

from which follows (since (5) is an equation of the same kind as (3), except that $n-1$ has taken the place of n):

$$M_1 = \Delta_0 = f(1) - f(0).$$

Put

$$\begin{aligned} \Delta_{x+1} - \Delta_x &= \Delta'_x, \\ \Delta'_{x+1} - \Delta'_x &= \Delta''_x, \\ &\cdots \\ \Delta_{x+1}^{(n-2)} - \Delta_x^{(n-2)} &= \Delta_x^{(n-1)}. \end{aligned} \tag{6}$$

then, accordingly,

$$M_0 = f(0), \quad M_1 = \Delta_0, \quad M_2 = \Delta'_0, \quad \ldots, \quad M_n = \Delta_0^{(n-1)},$$

and formula (3) gives

$$f(x) = f(0) + \Delta_0 B_1^{(x)} + \Delta'_0 B_2^{(x)} + \Delta''_0 B_3^{(x)} + \cdots + \Delta_0^{(n-1)} B_n^{(x)}. \tag{7}$$

Likewise the functions $\Delta_x, \Delta'_x, \Delta''_x, \ldots$ can be expressed:

$$\begin{aligned} \Delta_x &= \Delta_0 + \Delta'_0 B_1^{(x)} + \Delta''_0 B_2^{(x)} + \cdots + \Delta_0^{(n-1)} B_{n-1}^{(x)}, \\ \Delta'_x &= \Delta'_0 + \Delta''_0 B_1^{(x)} + \cdots + \Delta_0^{(n-1)} B_{n-2}^{(x)}. \end{aligned} \tag{8}$$

and so on.

The $\Delta_x, \Delta'_x, \Delta''_x, \ldots, \Delta_x^{(n-1)}$ are whole rational functions of degrees $n-1, n-2, \ldots, 0$; the last of them is therefore constant. In order to represent them all, one needs only the values

$$f(0), \quad \Delta_0, \quad \Delta'_0, \quad \Delta''_0, \quad \ldots, \quad \Delta_0^{(n-1)},$$

which one computes most easily by arranging a table. For the case $n = 3$, for example, the table has the form:

$f(0)$			
$f(1)$	Δ_0		
$f(2)$	Δ_1	Δ'_0	
$f(3)$	Δ_2	Δ'_1	Δ''_0

When $f(0), f(1), \ldots$ are given, the table is filled by simple subtractions.

§. 11.

Arithmetic Series of Higher Order.

The foregoing is the foundation for the theory of arithmetic series of higher order.

We denote by

$$u_0, u_1, u_2, u_3, \ldots \tag{1}$$

any infinite series of quantities and by

$$\Delta_0 = u_1 - u_0, \Delta_1 = u_2 - u_1, \Delta_2 = u_3 - u_2, \ldots \tag{2}$$

the series of its differences; and

$$\Delta_0' = \Delta_1 - \Delta_0, \Delta_1' = \Delta_2 - \Delta_1, \ldots \tag{3}$$

the series of its second differences, and so on.

It is clear that the series (1) is completely determined when its first term and the series of its first differences are given; for

$$u_1 = u_0 + \Delta_0, \quad u_2 = u_0 + \Delta_0 + \Delta_1, \quad \ldots,$$

$$u_m = u_0 + \Delta_0 + \Delta_1 + \cdots + \Delta_{m-1}.$$

Likewise the series (1) is completely determined when the first two terms and the series of its second differences are given, and so forth. The series (1) is called an arithmetic series of nth order when the series of its nth differences is constant, and hence the series of the $(n+1)$st differences consists entirely of zeros.

One obtains an arithmetic series of nth order by substituting the numbers $0, 1, 2, 3, \ldots$ for x in a whole function $f(x)$ of degree n. For if one puts

$$\Delta_x = f(x+1) - f(x),$$

$$\Delta_x' = \Delta_{(x+1)} - \Delta_x,$$

$$\ldots\ldots\ldots\ldots$$

$$\Delta_x^{(n-1)} = \Delta_{(x+1)}^{(n-2)} - \Delta_x^{(n-2)},$$

then Δ_x is of degree $(n-1)$, Δ_x' of degree $(n-2)$, and so on with respect to x, and therefore $\Delta_x^{(n-1)}$ is constant.

If now

$$u_0, u_1, u_2, \ldots$$

is an arithmetic series of nth order, then the whole series is completely determined when the n values $u_0, u_1, u_2, \ldots, u_{n-1}$ and, besides these, the constant nth difference are given. But this latter is determined if also the $(n+1)$st term u_n is given. We can therefore state the theorem:

An arithmetic series of nth order is completely determined when its first $n+1$ terms are given.

Since a function $f(x)$ of degree n is likewise completely determined by the arbitrarily given values

$$f(0), f(1), f(2), \ldots, f(n)$$

it follows that from the whole rational functions $f(x)$ of degree n all arithmetic series of nth order are produced when in them the series of natural numbers is substituted for x.

The expression for the general term is then given by formula (7) of the preceding paragraph.

The sums of the first $m+1$ terms of an arithmetic series of nth order,

$$s_m = u_0 + u_1 + \cdots + u_m,$$

form an arithmetic series of $(n+1)$st order, since their first differences

$$s_{m+1} - s_m = u_{m+1}$$

form an arithmetic series of nth order.

Thus, with the help of formula (7) of § 10, the sum s_m can be determined in general when $s_0, s_1, \ldots, s_{n+1}$ are assumed known.

To find the generating function $F(x)$ of s_m, when $f(x)$ is the generating function of u_m, put

$$F(0) = f(0),$$

$$F(1) = f(0) + f(1),$$

$$F(2) = f(0) + f(1) + f(2),$$

$$\ldots\ldots\ldots\ldots$$

and in formula (7), § 10,

$$F(x) = F(0) + D_0 B_1^{(x)} + D_0' B_2^{(x)} + \cdots$$

one has to put

$$\begin{aligned} D_0 &= F(1) - F(0) = f(1), \quad & D_0' &= f(2) - f(1) = \Delta_1, \\ D_1 &= F(2) - F(1) = f(2), \quad & D_1' &= f(3) - f(2) = \Delta_2, \\ D_2 &= F(3) - F(2) = f(3), \quad & D_0'' &= \Delta_2 - \Delta_1 = \Delta_1'. \end{aligned}$$

Thus one obtains

$$F(x) = f(0) + f(1)B_1^{(x)} + \Delta_1 B_2^{(x)} + \Delta_1' B_3^{(x)} + \cdots . \tag{4}$$

For example, take $f(x) = x^2$; then $F(m)$ gives the sum of the first m square numbers. Here

$$\Delta_x = 2x + 1, \qquad \Delta_x' = 2,$$

and therefore

$$F(x) = x + 3\frac{x(x-1)}{1\cdot 2} + 2\frac{x(x-1)(x-2)}{1\cdot 2\cdot 3} = \frac{x(x+1)(2x+1)}{6}.$$

For $f(x) = x^3$ the same calculation gives

$$F(x) = \left(\frac{x(x+1)}{2}\right)^2.$$

Thus the sum of the first m cubes is equal to the square of the mth triangular number.

§. 12.

The Polynomial Theorem.

In § 8 the following form of the binomial theorem was derived:

$$(x+y)^n = \Pi(n) \sum^{\alpha,\beta} \frac{x^\alpha y^\beta}{\Pi(\alpha)\Pi(\beta)}, \tag{1}$$

where the sum extends over all combinations of two numbers α, β, neither of which is negative, and which satisfy the condition

$$\alpha + \beta = n. \tag{2}$$

This form permits, first by induction, a generalization to the nth power of a polynomial:

$$(x + y + z + \cdots)^n = \Pi(n) \sum^{\alpha,\beta,\gamma,\ldots} \frac{x^\alpha y^\beta z^\gamma \cdots}{\Pi(\alpha)\Pi(\beta)\Pi(\gamma)\cdots}, \tag{3}$$

with the determination that $\alpha, \beta, \gamma, \ldots$ run through all positive or zero integer values which satisfy

$$\alpha + \beta + \gamma + \cdots = n. \tag{4}$$

But to prove the correctness of this formula in general, we assume that it is proved when the polynomial contains one term fewer, as is indeed the case when the polynomial contains only two terms.

We then put

$$u = y + z + \cdots \tag{5}$$

and apply formula (1) to $(x + u)$, from which there follows

$$(x + y + z + \cdots)^n = \Pi(n) \sum^{\alpha,\nu} \frac{x^\alpha u^\nu}{\Pi(\alpha)\Pi(\nu)}, \tag{6}$$

with the restriction

$$\alpha + \nu = n. \tag{7}$$

By the assumption, however, it has already been proved that

$$u^\nu = \Pi(\nu) \sum^{\beta,\gamma,\ldots} \frac{y^\beta z^\gamma \cdots}{\Pi(\beta)\Pi(\gamma)\cdots}, \tag{8}$$

where

$$\beta + \gamma + \cdots = \nu. \tag{9}$$

When this is inserted in (6), formula (3) is obtained immediately, and (7) passes over into (4).

The coefficients

$$P^{(n)}_{\alpha,\beta,\gamma,\ldots} = \frac{\Pi(n)}{\Pi(\alpha)\Pi(\beta)\Pi(\gamma)\cdots}, \tag{10}$$

which, according to their meaning, are whole numbers, are called the polynomial coefficients.

For example, for the third power of the trinomial one obtains

$$\begin{aligned}(x + y + z)^3 = {} & x^3 + y^3 + z^3 + 3x^2y + 3xy^2 + 3x^2z + 3xz^2 \\ & + 3y^2z + 3yz^2 + 6xyz.\end{aligned} \tag{11}$$

§. 13.

Derived Functions.

Let

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n \tag{1}$$

be a whole rational function of order n.

If in it we replace x by a binomial $x + y$, then we can apply the binomial theorem to each separate term, and can arrange the result according to descending or according to ascending powers of x or of y. We shall carry out the arrangement according to ascending powers of y. The highest power of y which occurs is the nth, and the coefficient of the zeroth power of y is the function $f(x)$ itself, as one sees by putting $y = 0$. We therefore put, denoting the other coefficients by

$$f'(x), \qquad \frac{f''(x)}{\Pi(2)}, \qquad \frac{f'''(x)}{\Pi(3)}, \ldots :$$

$$f(x+y) = f(x) + yf'(x) + \frac{y^2}{1\cdot 2}f''(x) + \cdots = \sum_{\nu=0}^{n} \frac{y^\nu}{\Pi(\nu)}\, f^{(\nu)}(x). \tag{2}$$

The functions $f'(x), f''(x), f'''(x), \ldots$ are called the first, second, third, ... derivatives or derived functions of $f(x)$. They are whole functions of x, and $f^{(\nu)}(x)$ cannot exceed the degree $n - \nu$, since the sum of the exponents of x and y in no term exceeds the degree n.

The first derivative, which is thus the coefficient of the first power of y in the expansion of $f(x + y)$ according to ascending powers of y, is obtained by applying the binomial theorem to (1):

$$f'(x) = na_0x^{n-1} + (n-1)a_1x^{n-2} + (n-2)a_2x^{n-3} + \cdots . \tag{3}$$

The principal theorem on derived functions follows from (2) if we transform x into $x + z$ or y into $y + z$:

$$f(x+y+z) = \sum_{\nu=0}^{n} \frac{y^\nu}{\Pi(\nu)} f^{(\nu)}(x+z) = \sum_{\nu=0}^{n} \frac{(y+z)^\nu}{\Pi(\nu)} f^{(\nu)}(x). \tag{4}$$

If by $f^{(\nu,\mu)}(x)$ we denote the μth derivative of $f^{(\nu)}(x)$, then by (2)

$$f^{(\nu)}(x+z) = \sum_{\mu=0}^{n-\nu} \frac{z^\mu}{\Pi(\mu)} f^{(\nu,\mu)}(x), \tag{5}$$

and by the binomial theorem

$$\frac{(y+z)^\nu}{\Pi(\nu)} = \sum^{\beta,\gamma} \frac{y^\beta z^\gamma}{\Pi(\beta)\Pi(\gamma)}, \qquad \beta + \gamma = \nu. \tag{6}$$

Substituting this in (4), one obtains

$$\sum_{\nu=0}^{n} \sum_{\mu=0}^{n-\nu} \frac{y^\nu z^\mu}{\Pi(\nu)\Pi(\mu)} f^{(\nu,\mu)}(x) = \sum^{\beta,\gamma} \frac{y^\beta z^\gamma}{\Pi(\beta)\Pi(\gamma)} f^{(\beta+\gamma)}(x). \tag{7}$$

The last sum is to be extended over all non-negative numbers β, γ whose sum does not exceed the degree n of $f(x)$. But the same number-combinations are also run through by the exponents ν, μ on the left-hand side, and comparison of the coefficients of equal powers and products gives (by § 1)

$$f^{(\nu,\mu)}(x) = f^{(\nu+\mu)}(x), \tag{8}$$

and therefore the theorem:

The μth derivative of the νth derivative is the $(\nu+\mu)$th derivative of the original function.

Thus all the higher derivatives are obtained by forming from each preceding one, according to the rule (3), the first derivative:

$$\begin{aligned} f(x) &= a_0x^n + a_1x^{n-1} + a_2x^{n-2} + a_3x^{n-3} + \cdots, \\ f'(x) &= na_0x^{n-1} + (n-1)a_1x^{n-2} + (n-2)a_2x^{n-3} + \cdots, \\ f''(x) &= n(n-1)a_0x^{n-2} + (n-1)(n-2)a_1x^{n-3} + \cdots. \end{aligned} \tag{9}$$

These derivatives assume a somewhat simpler form if one uses another notation, which is often in use and for certain purposes almost indispensable, and which we shall discuss in connection with this.

Because of the indeterminacy of the coefficients $a_0, a_1, \ldots, a_n$, it is clearly no restriction if we write a whole rational function of degree n in the form

$$f(x) = a_0x^n + B_1^{(n)}a_1x^{n-1} + B_2^{(n)}a_2x^{n-2} + \cdots + a_n, \tag{10}$$

or explicitly

$$f(x) = a_0x^n + na_1x^{n-1} + \frac{n(n-1)}{1\cdot 2}a_2x^{n-2} + \cdots. \tag{11}$$

When a function $f(x)$ is represented in this way, we shall say that it is "written with the binomial coefficients."

A greater agreement is already shown by the formulae (9), which then read:

$$\begin{aligned} f(x) &= a_0x^n + na_1x^{n-1} + \frac{n(n-1)}{1\cdot 2}a_2x^{n-2} + \cdots, \\ \frac{1}{n}f'(x) &= a_0x^{n-1} + (n-1)a_1x^{n-2} + \frac{(n-1)(n-2)}{1\cdot 2}a_2x^{n-3} + \cdots, \\ \frac{1}{n(n-1)}f''(x) &= a_0x^{n-2} + (n-2)a_1x^{n-3} + \frac{(n-2)(n-3)}{1\cdot 2}a_2x^{n-4} + \cdots, \end{aligned} \tag{12}$$

wherein the right-hand sides all likewise appear written with the binomial coefficients. We shall later become more acquainted with the usefulness of this notation, but already here we must emphasize that the choice of the one or the other

manner of representation is nevertheless not wholly indifferent, and that the first often deserves preference. Especially in those cases in which the coefficients are numbers and the number-theoretic behavior of these coefficients is at issue, one must not overlook that through the binomial coefficients a numerical element foreign to the matter is introduced. That Gauss in the theory of quadratic forms (in the Disq. ar.) uses the notation with binomial coefficients when he represents the quadratic forms by $ax^2 + 2bxy + cy^2$, and that this notation has come into general use, has led in number theory to an unnecessary and very regrettable complication.

§. 14.

Derivative of a Product.

The derived functions which we have considered here are no others than the differential quotients known from differential calculus; however, here, where only whole rational functions are involved, we have obtained the concept without using infinitesimal calculus, from the coefficients of development of the powers of y in the function $f(x+y)$. If we denote the νth derivative of $f(x)$ by $D_\nu f$, then by (2), § 13:

$$f(x+y) = f(x) + yD_1f + \frac{y^2}{1\cdot 2}D_2f + \frac{y^3}{1\cdot 2\cdot 3}D_3f + \cdots, \tag{1}$$

and from this follow at once the two fundamental rules expressed in the formulae

$$D_\nu(Cf) = CD_\nu f; \tag{2}$$

$$D_\nu(f+\varphi) = D_\nu f + D_\nu\varphi, \tag{3}$$

where C is a constant, and φ a second whole rational function of x.

A generalization of formula (2) gives the representation of the derivatives of the product $f\varphi$. Namely, if after (1) one writes, for abbreviation,

$$\begin{aligned} f(x+y) &= u_0 + yu_1 + y^2u_2 + \cdots + y^nu_n, \\ \varphi(x+y) &= v_0 + yv_1 + y^2v_2 + \cdots + y^mv_m, \end{aligned} \tag{4}$$

so that

$$u_\nu = \frac{D_\nu f}{\Pi(\nu)}, \qquad v_\nu = \frac{D_\nu\varphi}{\Pi(\nu)}, \tag{5}$$

then carrying out the multiplication of the two formulae (4), and seeking in the product the coefficient of y^ν, gives

$$\frac{D_\nu(f\varphi)}{\Pi(\nu)} = u_\nu v_0 + u_{\nu-1}v_1 + u_{\nu-2}v_2 + \cdots + u_0v_\nu, \tag{6}$$

or

$$D_\nu(f\varphi) = \varphi D_\nu f + B_1^{(\nu)}D_{\nu-1}fD_1\varphi + B_2^{(\nu)}D_{\nu-2}fD_2\varphi + \cdots, \tag{7}$$

where $B_1^{(\nu)}, B_2^{(\nu)}, \ldots$ are the binomial coefficients. In a similar way one can, by using the polynomial coefficients, form the derivatives of a product of more than two factors.

We shall form the first derivative, which we now denote by D instead of by D_1, for a product of n factors. For two factors we obtain from (7)

$$D(f\varphi) = \varphi Df + fD\varphi = \varphi f'(x) + f\varphi'(x),$$

and generally, if we denote the factors by $u_1, u_2, \ldots, u_n$ and their derivatives by $u_1', u_2', \ldots, u_n'$,

$$D(u_1u_2\cdots u_n) = u_1'u_2\cdots u_n + u_1u_2'\cdots u_n + \cdots + u_1u_2\cdots u_n', \tag{8}$$

or more briefly,

$$\frac{D(u_1u_2\cdots u_n)}{u_1u_2\cdots u_n} = \sum\frac{Du_\nu}{u_\nu}.$$

Thus if we consider a product of linear factors

$$f(x) = (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n), \tag{9}$$

we obtain, since the first derivatives of $x-\alpha_1$, $x-\alpha_2, \ldots, x-\alpha_n$ are all equal to 1,

$$\begin{aligned} f'(x) = &(x-\alpha_2)(x-\alpha_3)\cdots(x-\alpha_n) \\ &+ (x-\alpha_1)(x-\alpha_3)\cdots(x-\alpha_n) + \cdots \\ &+ (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_{n-1}), \end{aligned} \tag{10}$$

for which one can also write

$$f'(x) = \frac{f(x)}{x-\alpha_1} + \frac{f(x)}{x-\alpha_2} + \cdots + \frac{f(x)}{x-\alpha_n}. \tag{11}$$

A very important result follows from this if x is put equal to one of the values $\alpha_1, \alpha_2, \ldots, \alpha_n$, namely

$$(12)\qquad \begin{aligned} f'(\alpha_1) &= (\alpha_1-\alpha_2)(\alpha_1-\alpha_3)\cdots(\alpha_1-\alpha_n),\\ f'(\alpha_2) &= (\alpha_2-\alpha_1)(\alpha_2-\alpha_3)\cdots(\alpha_2-\alpha_n),\\ &\cdots\\ f'(\alpha_n) &= (\alpha_n-\alpha_1)(\alpha_n-\alpha_2)\cdots(\alpha_n-\alpha_{n-1}). \end{aligned}$$

§. 15.

Whole Functions of Several Variables: Forms.

Up to now we have chiefly considered whole rational functions of one variable; but we cannot always restrict ourselves to this, and indeed we have already used functions of several variables above. By a whole rational function of degree n in several variables $F(x, y, z, \ldots)$ we understand a sum of terms

$$\sum A_{\alpha,\beta,\gamma,\ldots} x^\alpha y^\beta z^\gamma \cdots,$$

where $\alpha, \beta, \gamma, \ldots$ are non-negative integral exponents whose sum $\alpha+\beta+\gamma+\cdots$ does not exceed the value n, and in at least one term actually attains it. Thus the degree is determined by the greatest value assumed by the sum $\alpha+\beta+\gamma+\cdots$.

If the sum of the exponents $\alpha+\beta+\gamma+\cdots$ has the same value in all terms, then the function is called homogeneous.

A fundamental property of homogeneous functions of degree n is this: if all variables are multiplied by the same factor, the result is the same as if the function were multiplied by the nth power; in signs, if t denotes an arbitrary variable,

$$(1)\qquad F(tx, ty, tz, \ldots) = t^n F(x, y, z, \ldots);$$

for if in the product $x^\alpha y^\beta z^\gamma \cdots$ the variables are replaced by $tx, ty, tz, \ldots$, it receives the factor

$$t^{\alpha+\beta+\gamma+\cdots}.$$

If now $\alpha+\beta+\gamma+\cdots$ has in all terms the same value n, the factor t^n can be taken out before the sum. But if the sum $\alpha+\beta+\gamma+\cdots$ has different values in the separate terms, no such common factor can be taken out, at least

not without leaving different powers of t in the separate terms.

By increasing the number of variables one can transform every non-homogeneous function into a homogeneous one of the same degree. If $m-1$ is the number of variables in a non-homogeneous function of degree n, then we put

$$x = \frac{x_1}{x_m}, \qquad y = \frac{x_2}{x_m}, \qquad z = \frac{x_3}{x_m}, \ldots,$$

and obtain in

$$x_m^n F\left(\frac{x_1}{x_m}, \frac{x_2}{x_m}, \frac{x_3}{x_m}, \ldots\right)$$

a whole homogeneous function of degree n in the variables $x_1, x_2, \ldots, x_m$, which we denote by

$$\Phi(x_1, x_2, \ldots, x_m).$$

It is sometimes advisable to write homogeneous functions of several variables with the polynomial coefficients. We therefore put

$$(2)\qquad \Phi(x_1, x_2, \ldots, x_m) = \sum \frac{\Pi(n)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)} A_{\alpha_1,\alpha_2,\ldots,\alpha_m} x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_m^{\alpha_m},$$

where the sum is extended over all non-negative numbers satisfying the condition

$$\alpha_1 + \alpha_2 + \cdots + \alpha_m = n. \tag{3}$$

This notation, without the restriction (3), is also applicable to non-homogeneous functions.

But the homogeneous function can also be represented as

$$\Phi(x_1, x_2, \ldots, x_m) = \sum A_{\nu_1,\nu_2,\ldots,\nu_n} x_{\nu_1} x_{\nu_2} \cdots x_{\nu_n}, \tag{4}$$

where each of the indices $\nu_1, \nu_2, \ldots, \nu_n$, independently of the others, runs through the range of values $1, 2, \ldots, m$. Thus the sum (4) consists of m^n terms, but not all of them are different from one another. The product $x_{\nu_1} x_{\nu_2} \cdots x_{\nu_n}$ remains unchanged if the indices $\nu_1, \nu_2, \ldots, \nu_n$ are permuted among themselves in any way. The number of permutations of n elements is $\Pi(n)$. If among these elements $\alpha_1, \alpha_2, \ldots$ respectively are equal to one another, then the number of permutations is reduced to

$$\frac{\Pi(n)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots},$$

whence it follows that in (4) any product $x_1^{\alpha_1} x_2^{\alpha_2} \cdots$ occurs exactly

$$\frac{\Pi(n)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots}$$

times. If one further fixes that $A_{\nu_1,\nu_2,\ldots,\nu_n}$ shall not change when the indices are permuted in any way, then the notations (2) and (4) prove to be identical if, by collecting equal factors,

$$x_{\nu_1} x_{\nu_2} \cdots x_{\nu_n} = x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_m^{\alpha_m}$$

and

$$A_{\nu_1,\nu_2,\ldots,\nu_n} = A_{\alpha_1,\alpha_2,\ldots,\alpha_m}$$

are put.

If the number of terms that occur in the function Φ [according to (2)] is denoted by (m, n), then, by first counting the terms which have the factor x_1 and then the remaining ones, which form a homogeneous function of order n of the remaining $m - 1$ variables, one finds the recursion formula

$$(m, n) = (m, n-1) + (m-1, n), \tag{5}$$

with whose aid one proves by complete induction the expression

$$(m, n) = \frac{m(m+1)\cdots(m+n-1)}{1 \cdot 2 \cdots n} = \frac{\Pi(m+n-1)}{\Pi(n)\Pi(m-1)} \tag{6}$$

to be correct.

Whole homogeneous functions are also called forms. According to the number of variables one distinguishes unary forms (simple powers), binary, ternary, quaternary forms. It is the binary forms that especially interest us here; their theory is essentially identical with the theory of whole rational functions of one variable. One passes from the binary forms back to these functions by regarding one of the homogeneous variables as constant, for example by giving it the value 1.

§. 16.

The Derivatives of Functions of Several Variables.

In § 13 we defined the derived functions of a whole rational function of one variable. The concept is immediately transferable to functions of several variables, by forming the derivatives with respect to each variable for itself, as though it were the only one. Thus, if one writes for instance as in § 15

$$F(x, y, z, \ldots) = \sum A_{\alpha,\beta,\gamma,\ldots} x^\alpha y^\beta z^\gamma \cdots, \tag{1}$$

one obtains the first derivative with respect to x:

$$F'(x) = \sum \alpha A_{\alpha,\beta,\gamma,\ldots} x^{\alpha-1} y^{\beta} z^{\gamma} \cdots , \tag{2}$$

or with respect to y:

$$F'(y) = \sum \beta A_{\alpha,\beta,\gamma,\ldots} x^{\alpha} y^{\beta-1} z^{\gamma} \cdots \tag{3}$$

and so on. From these functions one can again form, according to the same rules, the derivatives with respect to the different variables, and thus obtains the higher derivatives.

In order to display the results more clearly, let $\Phi(x_1, x_2, \ldots, x_m)$ be a whole rational function of order n in the m variables $x_1, x_2, \ldots, x_m$. We replace these variables by binomials

$$x_1 + \xi_1, \quad x_2 + \xi_2, \ldots, \quad x_m + \xi_m$$

and expand, in each term of the function

$$\Phi(x_1 + \xi_1, x_2 + \xi_2, \ldots, x_m + \xi_m) = \Phi(x + \xi),$$

by carrying out the multiplication

$$(x_1 + \xi_1)^{\mu_1} (x_2 + \xi_2)^{\mu_2} \cdots (x_m + \xi_m)^{\mu_m}$$

according to powers of $\xi_1, \xi_2, \ldots, \xi_m$. If equal powers and products of the variables ξ are collected each into one term, then for $\Phi(x + \xi)$, in the notation (2) of § 15, one obtains a representation which in differential calculus is called the Taylor expansion:

$$\Phi(x + \xi) = \sum \frac{\xi_1^{\alpha_1} \xi_2^{\alpha_2} \cdots \xi_m^{\alpha_m}}{\Pi(\alpha_1)\Pi(\alpha_2) \cdots \Pi(\alpha_m)} D_{\alpha_1,\alpha_2,\ldots,\alpha_m} \Phi. \tag{4}$$

The coefficients, which we denote by

$$D_{\alpha_1,\alpha_2,\ldots,\alpha_m} \Phi,$$

are functions of the variables x and are called, when

$$\alpha_1 + \alpha_2 + \cdots + \alpha_m = \nu,$$

the derivatives of order ν of the function Φ.

They are also written, according to the notation usual in differential calculus, as

$$D_{\alpha_1,\alpha_2,\ldots,\alpha_m} \Phi = \frac{\partial^{\nu} \Phi}{\partial x_1^{\alpha_1} \partial x_2^{\alpha_2} \cdots \partial x_m^{\alpha_m}}. \tag{5}$$

The law of formation of the derivatives can be summarized in the following propositions, where for brevity we omit the indices on the symbol D.

I. If C is a constant, then

$$D(C\Phi) = CD\Phi.$$

II. If Φ and Ψ are any two functions, then

$$D(\Phi + \Psi) = D\Phi + D\Psi.$$

Both follow immediately from (4).

Thus we can easily form the derived functions in general if we know them for the special case in which Φ is a product of powers, that is, if we know

$$D_{\alpha_1,\alpha_2,\ldots,\alpha_m} (x_1^{\mu_1} x_2^{\mu_2} \cdots x_m^{\mu_m}),$$

where the μ are arbitrary non-negative exponents.

But

$$(x_1+\xi_1)^{\mu_1}=\sum^{\alpha_1}\frac{\Pi(\mu_1)}{\Pi(\alpha_1)\Pi(\mu_1-\alpha_1)}\xi_1^{\alpha_1}x_1^{\mu_1-\alpha_1},$$

and therefore

$$\begin{aligned}&(x_1+\xi_1)^{\mu_1}\cdots(x_m+\xi_m)^{\mu_m}\\&=\sum^{\alpha_1\ldots\alpha_m}\frac{\Pi(\mu_1)\cdots\Pi(\mu_m)x_1^{\mu_1-\alpha_1}\cdots x_m^{\mu_m-\alpha_m}}{\Pi(\mu_1-\alpha_1)\cdots\Pi(\mu_m-\alpha_m)\Pi(\alpha_1)\cdots\Pi(\alpha_m)}\xi_1^{\alpha_1}\cdots\xi_m^{\alpha_m}.\end{aligned}\tag{6}$$

Comparison with (4) and (6) therefore gives

$$\begin{aligned}&D_{\alpha_1,\ldots,\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m})\\&=\frac{\Pi(\mu_1)\cdots\Pi(\mu_m)}{\Pi(\mu_1-\alpha_1)\cdots\Pi(\mu_m-\alpha_m)}x_1^{\mu_1-\alpha_1}\cdots x_m^{\mu_m-\alpha_m},\end{aligned}\tag{7}$$

as long as $\alpha_1\leq\mu_1,\ldots,\alpha_m\leq\mu_m$. On the other hand,

$$D_{\alpha_1,\ldots,\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m})=0,\tag{8}$$

as soon as one of the indices α is greater than the corresponding exponent μ.

Now let $\beta_1,\beta_2,\ldots,\beta_m$ denote a second system of indices, and form from the function (7) the derivative $D_{\beta_1,\beta_2,\ldots,\beta_m}$. By a repeated application of the same formulae (7) and (8) one obtains

$$D_{\beta_1,\ldots,\beta_m}D_{\alpha_1,\ldots,\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m})=D_{\beta_1+\alpha_1,\ldots,\beta_m+\alpha_m}(x_1^{\mu_1}\cdots x_m^{\mu_m}),\tag{9}$$

and from this, by II, follows the general validity of the theorem

$$\text{III.}\qquad D_{\beta_1,\ldots,\beta_m}D_{\alpha_1,\ldots,\alpha_m}\Phi=D_{\beta_1+\alpha_1,\ldots,\beta_m+\alpha_m}\Phi,$$

which is a generalization of the theorem (8), § 13; hence the higher derivatives can be formed by continued differentiation of the lower ones.

For the first derivatives of a function Φ,

$$D_{1,0,\ldots,0}\Phi,\quad D_{0,1,\ldots,0}\Phi,\ldots,D_{0,0,\ldots,1}\Phi,$$

we also use the shorter signs

$$\Phi'(x_1),\quad\Phi'(x_2),\ldots,\Phi'(x_m),$$

and likewise for the second derivatives

$$D_{2,0,\ldots,0},\quad D_{1,1,\ldots,0},\ldots$$

the signs

$$\Phi''(x_1,x_1),\quad\Phi''(x_1,x_2)=\Phi''(x_2,x_1),\ldots.$$

This notation too can be generalized and would lead to an expression corresponding to formula (4), § 15.

Because of their frequent use, we mention the formulae for the quadratic forms especially. Let

$$\Phi(x)=\sum a_{i,k}x_ix_k,\tag{10}$$

where i,k independently run through the range of values $1,2,\ldots,m$ and $a_{i,k}=a_{k,i}$. Then

$$\begin{aligned}\frac{1}{2}\Phi'(x_1)&=a_{1,1}x_1+a_{1,2}x_2+\cdots+a_{1,m}x_m,\\\frac{1}{2}\Phi'(x_2)&=a_{2,1}x_1+a_{2,2}x_2+\cdots+a_{2,m}x_m,\\&\cdots\\\frac{1}{2}\Phi'(x_m)&=a_{m,1}x_1+a_{m,2}x_2+\cdots+a_{m,m}x_m,\end{aligned}\tag{11}$$

and we further put

$$\Phi(x,\xi) = \Phi(\xi,x) = \sum \xi_i \Phi'(x_i) = \sum x_i \Phi'(\xi_i). \tag{12}$$

Then

$$\Phi(x+\xi) = \Phi(x) + \Phi(x,\xi) + \Phi(\xi). \tag{13}$$

The function $\Phi(x,\xi)$ is called the polar of Φ. It is linear and homogeneous both with respect to the x and with respect to the ξ.

It can be expressed by

$$\Phi(x,\xi) = 2\sum a_{ik}\xi_i x_k \tag{14}$$

and satisfies the condition

$$\Phi(x,x) = 2\Phi(x). \tag{15}$$

§. 17.

Euler's Theorem on Homogeneous Functions.

From the preceding developments one can easily derive a fundamental theorem on homogeneous functions, discovered by Euler and named after him.

We obtain it most simply from formula (4) of the preceding paragraph if, denoting by t an arbitrary variable, we put

$$\xi_1 = tx_1, \quad \xi_2 = tx_2, \ldots, \quad \xi_m = tx_m \tag{1}$$

and then apply the fundamental formula § 15 (1) for homogeneous functions. We thus obtain first:

$$(1+t)^n \Phi(x) = \sum \frac{(tx_1)^{\alpha_1}(tx_2)^{\alpha_2}\cdots(tx_m)^{\alpha_m}}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)} D_{\alpha_1,\alpha_2,\ldots,\alpha_m}\Phi. \tag{2}$$

If one applies the binomial theorem to the left-hand side of (2) and then equates on both sides the coefficients of equal powers of t, one obtains for every $\nu = 1, 2, \ldots, n$:

$$\frac{\Pi(n)}{\Pi(n-\nu)}\Phi(x_1,x_2,\ldots,x_m) = \sum^{\alpha} \frac{\Pi(\nu)}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)} x_1^{\alpha_1}x_2^{\alpha_2}\cdots x_m^{\alpha_m} D_{\alpha_1,\alpha_2,\ldots,\alpha_m}\Phi, \tag{3}$$

where the sum is extended over all systems of values of the α satisfying the condition

$$\alpha_1 + \alpha_2 + \cdots + \alpha_m = \nu. \tag{4}$$

In this form the theorem to be proved is contained in its full generality. For the special case $\nu = 1$ we obtain the formula

$$n\Phi(x_1,x_2,\ldots,x_m) = x_1\Phi'(x_1) + x_2\Phi'(x_2) + \cdots + x_m\Phi'(x_m), \tag{5}$$

of which formula (15) of the preceding paragraph is a special case, and for $\nu = 2$:

$$n(n-1)\Phi(x_1,x_2,\ldots,x_m) = \sum_{i,k=1}^{m} x_i x_k \Phi''(x_i,x_k), \tag{6}$$

where the sum is to be extended from $i = 1$ to $i = m$ and from $k = 1$ to $k = m$, so that every term with unequal i, k occurs twice in the sum.

If, when the α are subject to the condition (4), we put

$$\Phi_\nu(\xi,x)=\sum\frac{\xi_1^{\alpha_1}\xi_2^{\alpha_2}\cdots\xi_m^{\alpha_m}}{\Pi(\alpha_1)\Pi(\alpha_2)\cdots\Pi(\alpha_m)}D_{\alpha_1,\alpha_2,\ldots,\alpha_m}\Phi, \tag{7}$$

then by (4) of § 16,

$$\Phi(x+\xi)=\Phi(x)+\Phi_1(x,\xi)+\Phi_2(x,\xi)+\cdots+\Phi_n(x,\xi), \tag{8}$$

and since the left-hand side remains unchanged when x and ξ are interchanged, the relation follows:

$$\Phi_{n-\nu}(x,\xi)=\Phi_\nu(\xi,x), \tag{9}$$

and therefore in particular

$$\Phi_n(x,\xi)=\Phi(\xi). \tag{10}$$

The function $\Phi_\nu(x,\xi)$, regarded as a function of x, is called the νth polar of the function Φ for the system of values ξ.

We shall now specialize formula (3) to the case of a binary form ($m=2$). We denote the variables by x,y and put, for abbreviation,

$$\Phi(x,y)=u,\qquad D_{h,\nu-h}\Phi=u_h.$$

From (3) we obtain

$$\frac{\Pi(n)}{\Pi(n-\nu)}u=\sum_{h=0}^{\nu}\frac{\Pi(\nu)}{\Pi(h)\Pi(\nu-h)}u_hx^hy^{\nu-h}, \tag{9}$$

where ν can assume any value not greater than n.

Second Section.
Determinants.

§. 18.
Permutations of n Elements.

We consider a system of n distinct elements of any kind, for example the n digits

$$1,2,3,\ldots,n,$$

whose complex, in this definite arrangement, we shall denote by $\mathfrak{A}$. The elements of $\mathfrak{A}$ can be arranged in different ways, for example

$$2,1,3,\ldots,n.$$

The passage from one arrangement to another is called a permutation.

If we denote by $\Pi(n)$ the number of the different arrangements, which can depend only on the number n of the elements, then first

$$\Pi(1)=1,\qquad \Pi(2)=2.$$

To determine the number in general, we think of an n-th element as added to $n-1$ elements. In every arrangement of the $n-1$ elements the n-th element can now be put in n different places,

namely before the first, between the first and the second, between the second and the third, and so on, finally after the $(n-1)$-st; and all arrangements thus obtained are distinct. Hence

$$\Pi(n) = n\Pi(n-1), \tag{1}$$

from which, by complete induction,

$$\Pi(n) = 1 \cdot 2 \cdot 3 \cdots n \tag{2}$$

is obtained, so that here the sign $\Pi(n)$ has the same meaning as in the first section (§ 7).

Any arrangement of the system $\mathfrak{A}$ will be denoted by $\mathfrak{A}'$, or more fully, if $\alpha_1, \alpha_2, \ldots, \alpha_n$ are the digits $1, 2, \ldots, n$ in some order, by

$$\mathfrak{A}' = \alpha_1, \alpha_2, \ldots, \alpha_n. \tag{3}$$

One can derive any arrangement from any other in very different ways, that is, one can carry out a permutation in different ways. Among the several possibilities, those carried out by so-called transpositions, that is, by successive interchange of only two elements, are of special interest for us. By several transpositions performed successively one can derive every arrangement, for example $\mathfrak{A}'$, from $\mathfrak{A}$. For this purpose one may proceed, for instance, by first interchanging in $\mathfrak{A}$ the element 1 with that which stands in the first place in $\mathfrak{A}'$, that is, with α_1 (unless $\alpha_1 = 1$); then, if α_2 is not already $= 2$, interchanging 2 with α_2, and so on.

Thus, for example, to pass from $(1, 2, 3, 4)$ to $(4, 3, 2, 1)$, one forms the arrangements

$$(1, 2, 3, 4), \qquad (4, 2, 3, 1), \qquad (4, 3, 2, 1).$$

If we briefly denote a transposition by the two interchanged digits, and therefore the interchange of 1 with 2 by $(1, 2)$, then here we have performed successively the transpositions

$$(1, 4), \quad (2, 3).$$

It is to be observed that the passage from one arrangement to a definite other one can be reached in infinitely many different ways by successive transpositions. Thus the arrangement $(1, 2, 3, 4)$ also passes into $(4, 3, 2, 1)$ by the transpositions

$$(1, 2), \quad (1, 3), \quad (2, 4), \quad (1, 2).$$

§. 19.

Permutations of the First and Second Kind.

The $\Pi(n)$ arrangements of n elements can be separated into two kinds according to the following point of view.

From the n elements of our system one can form

$$\frac{n(n-1)}{2}$$

and no more pairs. We now assign to the n elements

$1, 2, \ldots, n$, in a definite way, n real numerical values $a_1, a_2, \ldots, a_n$, and from these numerical values form the $\frac{n(n-1)}{2}$ differences

$$a_1 - a_2, \quad a_1 - a_3, \ldots,$$

where the lower index is to belong to the minuend. The product of differences

$$P = (a_1 - a_2)(a_1 - a_3) \cdots (a_1 - a_n)(a_2 - a_3) \cdots (a_2 - a_n) \cdots (a_{n-1} - a_n) \tag{1}$$

will, when $a_1, a_2, \ldots, a_n$ are mutually distinct numerical values, have a value different from zero, for example a positive one if $a_1 > a_2 > a_3 > \cdots > a_n$ was assumed.

If now we interchange the indices $1, 2, 3, \ldots, n$ in some way, and so pass, say, from $\mathfrak{A}$ to $\mathfrak{A}'$, then P passes into

$$P' = (a_{\alpha_1} - a_{\alpha_2})(a_{\alpha_1} - a_{\alpha_3}) \cdots (a_{\alpha_1} - a_{\alpha_n}) \cdot (a_{\alpha_2} - a_{\alpha_3}) \cdots (a_{\alpha_2} - a_{\alpha_n}) \cdots (a_{\alpha_{n-1}} - a_{\alpha_n}), \tag{2}$$

and, apart from the sign, this product consists of the same factors as P. Thus P' is either equal to P or opposite to P.

I. We count the arrangement $\mathfrak{A}'$, and therefore also the permutation which transforms $\mathfrak{A}$ into $\mathfrak{A}'$, as of the first or of the second kind according as P is equal or opposite to P'; hence $\mathfrak{A}$ itself belongs to the first kind.

II. By a simple transposition (h, k), in which h, k denote any two of the digits $1, 2, \ldots, n$, both P and P' change their sign.

For the factors which contain neither h nor k are not affected by this transposition; furthermore, in P and P' there is the factor $\pm(a_h - a_k)$, and the pairs of factors

$$\pm(a_h - a_\nu)(a_k - a_\nu),$$

where ν runs through the series $1, 2, \ldots, n$ with the exception of h, k. The first factor changes its sign, while the pair of factors remains unchanged under the transposition (h, k). Hence follows:

III. The permutations of the first kind are composed of an even number of transpositions, and those of the second kind of an odd number.

The so-called identical permutation, which leaves $\mathfrak{A}$ unchanged, is therefore also to be counted as of the first kind.

From this one obtains further the conclusion: in however many different ways one may derive $\mathfrak{A}'$ from $\mathfrak{A}$ by transpositions, the number of these transpositions is in all these ways consistently even or odd, according as $\mathfrak{A}'$ belongs to the first or to the second kind.

If in all arrangements

$$\mathfrak{A}, \mathfrak{A}', \mathfrak{A}'', \ldots$$

of the n elements we perform one transposition, say $(1, 2)$, then each of these arrangements passes into a definite other one, say $\mathfrak{A}$ into $\mathfrak{B}$, $\mathfrak{A}'$ into $\mathfrak{B}'$, $\mathfrak{A}''$ into $\mathfrak{B}''$, and so on; and if the same transposition is repeated once more, then $\mathfrak{B}$ again passes into $\mathfrak{A}$, $\mathfrak{B}'$ again into $\mathfrak{A}'$, and so on. Hence the arrangements $\mathfrak{B}, \mathfrak{B}', \mathfrak{B}'', \ldots$ are all distinct from one another, and consequently in their totality agree with the totality of the $\mathfrak{A}$'s. Since, as we have seen above, all difference-products

$$P, P', P'', \ldots$$

which are formed from P with the different arrangements $\mathfrak{A}, \mathfrak{A}', \mathfrak{A}'', \ldots$ change their sign under a transposition, it follows that to each $\mathfrak{A}$ of the first kind there corresponds a $\mathfrak{B}$ of the second kind, and to each $\mathfrak{A}$ of the second kind a $\mathfrak{B}$ of the first kind.

IV. Accordingly the number of arrangements of the first kind is just as large as the number of arrangements of the second kind, namely $\frac{1}{2}\Pi(n)$[4].

For $n = 3$ we have the following six arrangements, of which the first horizontal row forms the first kind:

$$\begin{array}{lll} (1,2,3), & (2,3,1), & (3,1,2), \\ (3,2,1), & (2,1,3), & (1,3,2). \end{array} \tag{3}$$

[4]These theorems have here been obtained from the consideration of the product P, hence from a numerical quantity. How one can arrive at the same results without using such a function will be seen in the fourteenth section.

§. 20.

Determinants.

We now consider a system of n^2 arbitrary quantities, with which the rational operations of arithmetic can be performed. For a simple notation of these quantities we choose one letter with a double index $a_i^{(k)}$, where both i and k are to run through the series of digits $1, 2, 3, \ldots, n$. For better survey we arrange these quantities in a square, so that all a's with the same upper index stand in a horizontal row, and all a's with the same lower index in a vertical row, and denote this square by Δ:

$$\Delta = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & a_3^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & a_3^{(2)} & \cdots & a_n^{(2)} \\ a_1^{(3)} & a_2^{(3)} & a_3^{(3)} & \cdots & a_n^{(3)} \\ \vdots & \vdots & \vdots & & \vdots \\ a_1^{(n)} & a_2^{(n)} & a_3^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}. \tag{1}$$

For brevity one calls the horizontal rows rows, and the vertical rows columns.

But by the square enclosed between vertical lines we shall understand not only the complex of the quantities a, but a definite arithmetical combination of these quantities, which can be calculated as soon as the a's are given numerically, and which we shall now describe.

Form the product of the terms standing in the diagonal running from upper left to lower right:

$$M = a_1^{(1)} a_2^{(2)} a_3^{(3)} \cdots a_n^{(n)}. \tag{2}$$

From this derive $\Pi(n)$ products $M, M', M'', \ldots$ by permuting the lower indices, and give to each product so obtained the positive or negative sign according as the applied permutation belongs to the first or to the second kind; thus, in the notation of the preceding paragraph,

$$M' = \pm a_{\alpha_1}^{(1)} a_{\alpha_2}^{(2)} \cdots a_{\alpha_n}^{(n)}. \tag{3}$$

The sum of these products

$$M + M' + M'' + \cdots = \Sigma M$$

is to be Δ. The quantity Δ is called the determinant of the n^2 elements $a_i^{(k)}$, more precisely, when the distinction is necessary, an n-rowed determinant (also a determinant of the n-th degree or n-th order). The term M of this sum, that is, the product of all elements standing in the diagonal of the square, is called the principal term.

For example, if $n = 2$, we obtain

$$\Delta = a_1^{(1)} a_2^{(2)} - a_2^{(1)} a_1^{(2)}, \tag{4}$$

and for $n = 3$ [according to (3) of the preceding paragraph]:

$$\begin{aligned} \Delta = {} & a_1^{(1)} a_2^{(2)} a_3^{(3)} + a_2^{(1)} a_3^{(2)} a_1^{(3)} + a_3^{(1)} a_1^{(2)} a_2^{(3)} \\ & - a_1^{(1)} a_3^{(2)} a_2^{(3)} - a_2^{(1)} a_1^{(2)} a_3^{(3)} - a_3^{(1)} a_2^{(2)} a_1^{(3)}. \end{aligned} \tag{5}$$

Or in another notation:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc, \tag{6}$$

$$\begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix} = ab'c'' + bc'a'' + ca'b'' - ac'b'' - ba'c'' - cb'a''. \tag{7}$$

The reader is advised to practise the calculation of such determinants on numerical examples.

The notation (1) is in many cases too cumbersome; other, shorter symbols are therefore also in use. Thus Jacobi, writing out only the principal term of the expanded determinant, puts

$$\Delta = \sum \pm a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)}, \tag{8}$$

and Kronecker, still more briefly,

$$\Delta = \left| a_i^{(k)} \right|. \tag{9}$$

Both notations are entirely clear only when the elements are denoted by two indices in the manner presupposed here, and are wholly inapplicable when, for example, the elements are given numerically.

Determinants sometimes occur for which

$$a_i^{(k)} = a_k^{(i)},$$

so that in (1) the elements standing symmetrically with respect to the diagonal of the square are equal to one another. In these cases we shall ordinarily place both indices below (to indicate their equivalence), and therefore put

$$a_{i,k} = a_{k,i}.$$

Such determinants are called symmetric.

§. 21.

Principal Theorems on Determinants.

From the concept of the determinant the first theorems useful for application are easily obtained.

If in the product [§ 20, (3)]

$$M' = \pm a_{\alpha_1}^{(1)} a_{\alpha_2}^{(2)} \cdots a_{\alpha_n}^{(n)} \tag{1}$$

we rearrange the factors, its value is not changed. We can therefore also order the factors in such a way that the lower indices occur in their natural order $1, 2, \ldots, n$. The upper indices will then occur in a certain permuted order; thus M' will receive the form

$$\pm a_1^{(\beta_1)} a_2^{(\beta_2)} \cdots a_n^{(\beta_n)}, \tag{2}$$

where

$$(\beta_1, \beta_2, \ldots, \beta_n) = \mathfrak{B}$$

is, just as $(\alpha_1, \alpha_2, \ldots, \alpha_n) = \mathfrak{A}$, an arrangement of the digits $1, 2, \ldots, n$. One obtains the arrangement $\mathfrak{B}$ by undoing, in the factors of M', the transpositions which led to $\mathfrak{A}$, beginning with the last, in order to recover the original arrangement in the row of the lower indices. The resulting order of the upper indices is then the arrangement $\mathfrak{B}$. It follows that $\mathfrak{B}$ belongs to the first or the second kind according as $\mathfrak{A}$ belongs to the first or the second kind, since both arise through the same number of transpositions. The totality of the $\mathfrak{B}$'s represents, just like the totality of the $\mathfrak{A}$'s, all permutations of the n elements, since two different $\mathfrak{A}$'s can never lead to the same $\mathfrak{B}$. Thus is proved:

I. The determinant Δ can also be formed by permuting the upper indices in the principal term $a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)}$ in all possible ways, assigning to every product so formed the positive or negative sign according as the applied permutation belongs to the first or second kind, and then taking the sum of all these products.

In the representation § 20, (1) of Δ, the rows are characterized by the upper indices, the columns by the lower indices. Hence this theorem may also be expressed as follows:

II. A determinant is not changed when the rows are made columns and the columns are made rows.

If, in all arrangements $\mathfrak{A}, \mathfrak{A}', \mathfrak{A}'', \ldots$ of the n elements, we interchange any two elements with one another, the totality of these arrangements remains unchanged, but every arrangement of the first kind passes into an arrangement of the second kind, and conversely. If, therefore, in the terms $M, M', M'', \ldots$ of which Δ is composed, any two lower indices are interchanged, every term with positive sign passes into another which in Δ bore the negative sign, and conversely; therefore Δ changes its sign. With the help of II follows the theorem:

III. If in Δ two lower or two upper indices are interchanged with one another, the determinant changes only its sign.

Expressed somewhat differently: if two rows or two columns are interchanged with one another, the determinant changes only its sign; and from this more generally:

IV. If in a determinant the rows or the columns are permuted, the absolute value is not changed, and the sign is not changed or passes into the opposite sign according as the applied permutation belongs to the first or second kind.

From III one obtains the following fundamental theorem:

V. If in two rows or in two columns the terms standing in the same place are equal to one another (more briefly, if two rows are equal to one another), then the determinant has the value zero.

For the interchange of the two rows changes the sign by III, but on the other hand, since both rows are identical,

cannot change anything; thus for Δ only the value zero remains.

One only expresses theorem V differently when one says:

VI. One obtains a vanishing determinant if the elements of one row are replaced by the corresponding elements of another row, or, shortly, if one lower or upper index is replaced by another.

§. 22.

Subdeterminants.

In every term of the expanded determinant

$$\sum \pm a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)},$$

whose value we now wish to denote by A, every one of the numbers $1, 2, \ldots, n$ occurs once and only once as a lower index. Thus a certain complex of terms will contain the factor $a_1^{(1)}$, another complex the factor $a_1^{(2)}$, and so on, finally a complex the factor $a_1^{(n)}$; every term of the determinant occurs in one and only one of these complexes.

If, therefore, we denote the first of these complexes by $a_1^{(1)} A_1^{(1)}$, the second by $a_1^{(2)} A_1^{(2)}$, and the last by $a_1^{(n)} A_1^{(n)}$, then the determinant can be represented in the following way:

$$A = a_1^{(1)} A_1^{(1)} + a_1^{(2)} A_1^{(2)} + \cdots + a_1^{(n)} A_1^{(n)}. \tag{1}$$

Instead of the lower index 1, we could equally well single out any other index ν, and hence put

$$A = a_\nu^{(1)} A_\nu^{(1)} + a_\nu^{(2)} A_\nu^{(2)} + \cdots + a_\nu^{(n)} A_\nu^{(n)}. \tag{2}$$

Here the product $a_\nu^{(\mu)} A_\nu^{(\mu)}$ denotes the complex of all terms of the determinant which contain the factor $a_\nu^{(\mu)}$.

Since the same rules hold for upper as for lower indices, the determinant can also be written in the following way:

$$A = a_1^{(\mu)} A_1^{(\mu)} + a_2^{(\mu)} A_2^{(\mu)} + \cdots + a_n^{(\mu)} A_n^{(\mu)}, \tag{3}$$

where μ, too, may denote any one of the indices $1, 2, \ldots, n$.

The quantities $A_\nu^{(\mu)}$ thereby completely defined are called the subdeterminants of the determinant A. To become precisely acquainted with their mode of formation, let us first consider the complex $a_1^{(1)} A_1^{(1)}$. It is obtained by leaving the lower index 1 unchanged in the product

$$a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)}$$

and permuting only the remaining indices $2, 3, \ldots, n$ in all ways, and forming the sum of the terms so obtained with regard to the rule of signs. Hence $A_1^{(1)}$ is the $(n-1)$-rowed determinant

$$A_1^{(1)} = \begin{vmatrix} a_2^{(2)} & a_3^{(2)} & \cdots & a_n^{(2)} \\ a_2^{(3)} & a_3^{(3)} & \cdots & a_n^{(3)} \\ \vdots & \vdots & & \vdots \\ a_2^{(n)} & a_3^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \tag{4}$$

or the determinant obtained from A by deleting, in the square representing A [§ 20, (1)], the first row and the first column.

From this one easily obtains the meaning of $A_\nu^{(\mu)}$. By making $\nu - 1$ interchanges of rows one can bring the ν-th row to the first, and by adding $\mu - 1$ interchanges of columns one can bring the μ-th column to the first; apart from this the remaining rows and columns keep their relative order. The determinant itself has taken the factor $(-1)^{\mu+\nu}$, and has remained unchanged in absolute value. In the arrangement thus changed the element $a_\nu^{(\mu)}$ has taken the place of the element $a_1^{(1)}$, and from this one infers the following rule of formation:

One obtains the subdeterminant $A_\nu^{(\mu)}$ by deleting from the square representing the determinant the row and column which intersect in $a_\nu^{(\mu)}$, and adjoining the factor $(-1)^{\mu+\nu}$.

For example, for the determinant of three rows, one obtains the representation

$$\begin{aligned} \begin{vmatrix} a & b & c \\ a' & b' & c' \\ a'' & b'' & c'' \end{vmatrix} &= a \begin{vmatrix} b' & c' \\ b'' & c'' \end{vmatrix} - b \begin{vmatrix} a' & c' \\ a'' & c'' \end{vmatrix} + c \begin{vmatrix} a' & b' \\ a'' & b'' \end{vmatrix} \\ &= a(b'c'' - c'b'') + b(c'a'' - a'c'') + c(a'b'' - b'a''). \end{aligned} \tag{5}$$

Since the lower index ν does not occur at all in $A_\nu^{(\mu)}$, $A_\nu^{(\mu)}$ is not changed if the lower index ν is replaced by another. But then, by § 21, VI, the determinant vanishes. We therefore obtain from (2) the important relation, in which μ, ν can be any two distinct digits $1, 2, \ldots, n$:

$$0 = a_\mu^{(1)} A_\nu^{(1)} + a_\mu^{(2)} A_\nu^{(2)} + \cdots + a_\mu^{(n)} A_\nu^{(n)}, \tag{6}$$

and likewise from (3)

$$0 = a_1^{(\mu)} A_1^{(\nu)} + a_2^{(\mu)} A_2^{(\nu)} + \cdots + a_n^{(\mu)} A_n^{(\nu)}. \tag{7}$$

For instance from (5), if a, b, c are replaced by a', b', c', there follows

$$a'(b'c'' - c'b'') + b'(c'a'' - a'c'') + c'(a'b'' - b'a'') = 0, \tag{8}$$

a formula of whose correctness one is convinced by the simplest calculation.

If we multiply the relation (6) by an arbitrary factor λ and add it to (2), we obtain the formula

$$\begin{aligned} A = (a_\nu^{(1)} + \lambda a_\mu^{(1)}) A_\nu^{(1)} + (a_\nu^{(2)} + \lambda a_\mu^{(2)}) A_\nu^{(2)} \\ + \cdots + (a_\nu^{(n)} + \lambda a_\mu^{(n)}) A_\nu^{(n)}, \end{aligned} \tag{9}$$

which expresses for us the following theorem:

VII. The determinant does not change its value if to the elements of one row one adds the corresponding elements of another row multiplied by an arbitrary common factor.

The same theorem holds also for the columns. It is often used with advantage for the simplification and numerical calculation of determinants. We add the following theorems, which can be read off at once from the representations (2), (3).

VIII. If all elements of a row or of a column have a common factor, then this factor can be omitted from the row or column and put as a factor before the determinant.

For by (2) it is

$$pa_\nu^{(1)}A_\nu^{(1)} + pa_\nu^{(2)}A_\nu^{(2)} + \cdots + pa_\nu^{(n)}A_\nu^{(n)} = pA.$$

IX. If in a row or in a column all elements but one vanish, then the determinant reduces

to the product of this one element and the corresponding subdeterminant.

For if $a_\nu^{(1)}, a_\nu^{(2)}, \ldots, a_\nu^{(n)}$, with the exception of $a_\nu^{(\mu)}$, vanish, then by (2)

$$A = a_\nu^{(\mu)}A_\nu^{(\mu)}.$$

The value of A is then wholly independent of the $a_1^{(\mu)}, a_2^{(\mu)}, \ldots, a_n^{(\mu)}$ with the exception of $a_\nu^{(\mu)}$.

To make an application of these theorems, we determine the value of the determinant

$$\Delta = \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix},$$

where a, b, c are arbitrary quantities. If we multiply the second column by a and subtract it from the third, then multiply the first by a and subtract it from the second, then by VII

$$\Delta = \begin{vmatrix} 1 & 0 & 0 \\ 1 & b-a & b(b-a) \\ 1 & c-a & c(c-a) \end{vmatrix},$$

and by IX

$$\Delta = \begin{vmatrix} b-a & b(b-a) \\ c-a & c(c-a) \end{vmatrix},$$

and finally by VIII

$$\Delta = (b-a)(c-a)\begin{vmatrix} 1 & b \\ 1 & c \end{vmatrix} = (b-a)(c-a)(c-b). \tag{10}$$

In the same way one can also treat the n-rowed determinant

$$\begin{vmatrix} 1 & a_1 & a_1^2 & \cdots & a_1^{n-1} \\ 1 & a_2 & a_2^2 & \cdots & a_2^{n-1} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & a_n & a_n^2 & \cdots & a_n^{n-1} \end{vmatrix}$$

and finds its value equal to

$$(a_2-a_1)(a_3-a_1)\cdots(a_n-a_1)(a_3-a_2)\cdots(a_n-a_2)\cdots(a_n-a_{n-1}). \tag{11}$$

If one orders the columns in the reverse sequence, then, according as n is even or odd,

$$\frac{n}{2} \quad \text{or} \quad \frac{n-1}{2}$$

interchanges are necessary, so that the determinant thus ordered differs from Δ by the factor

$$(-1)^{n(n-1)/2}.$$

It comes to the same thing if one gives the opposite sign to the

$$\frac{n(n-1)}{2}$$

factors of the product (11). Thus, along with (11), the equality also holds:

$$(12) \qquad \begin{vmatrix} a_1^{n-1} & a_1^{n-2} & \cdots & a_1 & 1 \\ a_2^{n-1} & a_2^{n-2} & \cdots & a_2 & 1 \\ \vdots & \vdots & & \vdots & \vdots \\ a_n^{n-1} & a_n^{n-2} & \cdots & a_n & 1 \end{vmatrix} = \begin{matrix} (a_1-a_2)(a_1-a_3)\cdots(a_1-a_n) \\ (a_2-a_3)\cdots(a_2-a_n) \\ \cdots \\ (a_{n-1}-a_n). \end{matrix}$$

We still mention here a notation for the subdeterminants which is borrowed from differential calculus and is often used with advantage, especially when derivatives are to be formed. If the quantities $a_i^{(k)}$ are regarded as independent variables, then the determinant A is only of the first degree with respect to each of them. The derivative or differential quotient taken with respect to $a_i^{(k)}$ is therefore equal to the coefficient of $a_i^{(k)}$, hence

$$\frac{\partial A}{\partial a_i^{(k)}} = A_i^{(k)}.$$

If therefore, for example, the $a_i^{(k)}$ are functions of a variable t, then A is also a function of t, and by the first rules of differential calculus one obtains the derivative of A with respect to t in the form

$$A'(t) = \sum A_i^{(k)} \frac{\partial a_i^{(k)}}{\partial t},$$

where $\partial a_i^{(k)}/\partial t$ denotes the derivative of $a_i^{(k)}$ with respect to t.

§. 23.

The Subdeterminants in the Wider Sense.

We can now generalize the considerations of the preceding paragraph in the following way. Just as there we started from the problem of seeking out all terms in the developed determinant A which contain the factor $a_1^{(1)}$, so now we seek all terms which contain the factor

$$a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)},$$

where ν may be any number less than n. These terms are obtained from the principal term

$$a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)} a_{\nu+1}^{(\nu+1)} \cdots a_n^{(n)}$$

if, in permuting the lower indices, one leaves $1, 2, \ldots, \nu$ unchanged and permutes only $\nu+1, \ldots, n$ in all possible ways, with observance of the rule of signs. Thus the aggregate of the required terms is

$$(1) \qquad a_1^{(1)} a_2^{(2)} \cdots a_\nu^{(\nu)} \begin{vmatrix} a_{\nu+1}^{(\nu+1)} & \cdots & a_n^{(\nu+1)} \\ \vdots & & \vdots \\ a_{\nu+1}^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}.$$

I. The determinant of $n-\nu$ rows which occurs here as a factor, and which we denote by

$$A^{1,2,\ldots,\nu}_{1,2,\ldots,\nu},$$

arises from A by suppressing the first ν rows and columns.

We shall now generalize this result in the following manner. Choose any ν elements

$$a^{(\beta_1)}_{\alpha_1}, a^{(\beta_2)}_{\alpha_2}, \ldots, a^{(\beta_\nu)}_{\alpha_\nu}$$

so that no two of them occur in the same row or in the same column, that is, so that neither a lower nor an upper index is repeated. We denote by

$$a^{(\beta_1)}_{\alpha_1} a^{(\beta_2)}_{\alpha_2} \cdots a^{(\beta_\nu)}_{\alpha_\nu} A^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu} \tag{2}$$

the aggregate of those terms of the determinant which contain the product of these elements as a factor.

By interchanging rows and columns, whereby at most the sign of the determinant is changed, one can always cause the elements

$$a^{(\beta_1)}_{\alpha_1}, a^{(\beta_2)}_{\alpha_2}, \ldots, a^{(\beta_\nu)}_{\alpha_\nu} \tag{3}$$

to take the place of the elements.

$$a^{(1)}_1, a^{(2)}_2, \ldots, a^{(\nu)}_\nu$$

Then rule I applies to the determination of $A^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}$, and one obtains:

II. Apart from sign, $A^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}$ is obtained as an $(n-\nu)$-rowed determinant by deleting from A all rows and all columns which meet in one of the elements (3), and by leaving the remaining rows and columns in their order.

For the determination of the sign one uses the following rule. Arrange the lower and upper indices $1, 2, \ldots, n$ in the orders

$$\alpha_1, \alpha_2, \ldots, \alpha_\nu, \alpha_{\nu+1}, \ldots, \alpha_n, \tag{4}$$

$$\beta_1, \beta_2, \ldots, \beta_\nu, \beta_{\nu+1}, \ldots, \beta_n, \tag{5}$$

where the missing $\alpha_{\nu+1}, \ldots, \alpha_n$, and likewise $\beta_{\nu+1}, \ldots, \beta_n$, are taken in increasing order.

III. The $(n-\nu)$-rowed determinant described in II receives the positive or negative sign according as the two arrangements (4), (5) of the digits $1, 2, \ldots, n$ both belong to the same kind or to different kinds.

Indeed, the determinant changes its sign through each interchange of two lower or two upper indices. In order to reduce the general case (2) to the special case (1), one has to make just so many transpositions of upper and lower indices that the permutations (4) and (5) both pass into the original arrangement $1, 2, 3, \ldots, n$; the same number of changes of sign has then taken place.

The quantities so defined

$$A^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}$$

are called the ν-th subdeterminants, or subdeterminants of the ν-th order. They are represented by $(n-\nu)$-rowed determinants. With respect to these subdeterminants, III implies the theorem:

IV. The subdeterminant $A^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}$ changes only its sign if two of its lower or two of its upper indices are interchanged; more generally, it remains unaltered in absolute value if the order of the indices $\alpha_1, \alpha_2, \ldots, \alpha_\nu$ is replaced by any other order, and changes sign or not according as this permutation is of the second or of the first kind.

If $\alpha'_1, \alpha'_2, \ldots, \alpha'_\nu$ denotes any ordering of $\alpha_1, \alpha_2, \ldots, \alpha_\nu$, then the determinant A also contains the complex of terms

$$\pm a^{(\beta_1)}_{\alpha'_1} a^{(\beta_2)}_{\alpha'_2} \cdots a^{(\beta_\nu)}_{\alpha'_\nu} A^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}.$$

Collecting all these terms gives the complex

$$A^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu} \Sigma \pm a^{(\beta_1)}_{\alpha_1} a^{(\beta_2)}_{\alpha_2} \cdots a^{(\beta_\nu)}_{\alpha_\nu}. \tag{6}$$

The ν-rowed determinant which occurs in (6),

$$\Sigma \pm a^{(\beta_1)}_{\alpha_1} a^{(\beta_2)}_{\alpha_2} \cdots a^{(\beta_\nu)}_{\alpha_\nu} = \begin{vmatrix} a^{(\beta_1)}_{\alpha_1} & a^{(\beta_1)}_{\alpha_2} & \cdots & a^{(\beta_1)}_{\alpha_\nu} \\ a^{(\beta_2)}_{\alpha_1} & a^{(\beta_2)}_{\alpha_2} & \cdots & a^{(\beta_2)}_{\alpha_\nu} \\ \vdots & \vdots & & \vdots \\ a^{(\beta_\nu)}_{\alpha_1} & a^{(\beta_\nu)}_{\alpha_2} & \cdots & a^{(\beta_\nu)}_{\alpha_\nu} \end{vmatrix},$$

is called the subdeterminant complementary to $A^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}$, and is denoted by $B^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}$. It contains exactly the rows and columns missing in $A^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}$, and,

apart from sign, agrees with the subdeterminant of the $(n-\nu)$-th order

$$A^{\beta_{\nu+1}\ldots\beta_n}_{\alpha_{\nu+1}\ldots\alpha_n}.$$

Thus the complex of terms (6) is denoted by

$$A^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu} B^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}. \tag{7}$$

If now, for $\alpha_1, \alpha_2, \ldots, \alpha_\nu$, one chooses each combination of ν of the digits $1, 2, \ldots, n$, whose number (according to § 7) is $B^{(n)}_\nu$, while $\beta_1, \beta_2, \ldots, \beta_\nu$ is held fixed, one obtains the same number of complexes of the form (7), and every term of the determinant A occurs in one and only one of them.

V. Consequently, by summing all expressions (7), one obtains the determinant A:

$$A = \sum^{\alpha} A^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu} B^{\beta_1,\beta_2,\ldots,\beta_\nu}_{\alpha_1,\alpha_2,\ldots,\alpha_\nu}. \tag{8}$$

Of course one may also keep the combination of the α's fixed and sum with respect to the β's.

This is Laplace's theorem.

Another representation of the determinant A by the first and second subdeterminants is obtained as follows.

Choose in A a row and a column which intersect in an element, say $a^{(\mu)}_\nu$. In every term of A there occurs one element with lower index ν and one element with upper index μ. Thus in A we have first the complex $a^{(\mu)}_\nu A^{(\mu)}_\nu$, and furthermore the various complexes $a^{(i)}_\nu a^{(\mu)}_k A^{i,\mu}_{\nu,k}$, in which i may denote any index different from μ, and k any index different from ν.

VI. We can therefore put:

$$A = a^{(\mu)}_\nu A^{(\mu)}_\nu + \sum^{i,k} a^{(i)}_\nu a^{(\mu)}_k A^{i,\mu}_{\nu,k}, \tag{9}$$

or, according to IV,

$$A = a^{(\mu)}_\nu A^{(\mu)}_\nu - \sum^{i,k} a^{(i)}_\nu a^{(\mu)}_k A^{\mu,i}_{\nu,k}. \tag{10}$$

We further observe concerning this theorem that $A^{\mu,i}_{\nu,k}$ is the first subdeterminant, corresponding to the element $a^{(i)}_k$, of the $(n-1)$-rowed determinant $A^{(\mu)}_\nu$; for $A^{\mu,i}_{\nu,k}$ is the coefficient

of $a^{(\mu)}_\nu a^{(i)}_k$ in the expansion of A, and $A^{(\mu)}_\nu$ is the coefficient of $a^{(\mu)}_\nu$; consequently $A^{\mu,i}_{\nu,k}$ is the coefficient of $a^{(i)}_k$ in the determinant $A^{(\mu)}_\nu$.

According to this theorem, the so-called bordered determinant

$$U = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} & u_1 \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} & u_2 \\ \vdots & \vdots & & \vdots & \vdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} & u_n \\ v_1 & v_2 & \cdots & v_n & q \end{vmatrix} \tag{11}$$

can be expanded with respect to the elements of the last row and column, and one obtains:

$$U = qA - \sum^{i}\sum^{k} u_i v_k A_k^{(i)}. \tag{12}$$

This equation is obtained from (10) by replacing n by $n+1$, and distinguishing the elements of the last row and column by a different notation.

For the higher subdeterminants also, notation by differential quotients is sometimes convenient, so that, for example,

$$A_{\nu,k}^{i,\mu} = \frac{\partial^2 A}{\partial a_\nu^{(i)}\, \partial a_k^{(\mu)}} \tag{13}$$

is put.

§. 24.

Linear Homogeneous Equations.

The principal application of determinants, to which the whole theory owes its origin, is the solution of linear equations.

We shall attack the problem here at once in its most general form, since the special form affords scarcely any simplification and can afterwards easily be derived from the general result.

We consider a system of m equations of the first degree in which n unknowns $x_1, x_2, \ldots, x_n$ occur homogeneously:

$$\begin{aligned} a_1^{(1)}x_1 + a_2^{(1)}x_2 + \cdots + a_n^{(1)}x_n &= 0, \\ a_1^{(2)}x_1 + a_2^{(2)}x_2 + \cdots + a_n^{(2)}x_n &= 0, \\ &\vdots \\ a_1^{(m)}x_1 + a_2^{(m)}x_2 + \cdots + a_n^{(m)}x_n &= 0. \end{aligned} \tag{1}$$

where the coefficients $a_i^{(k)}$ are regarded as given quantities. For the present we shall make no assumption whatever about the numbers m, n, but set ourselves the general task of determining all systems of values of $x_1, x_2, \ldots, x_n$ which satisfy equations (1).

We can give one solution of equations (1) at once: whatever the coefficients $a_i^{(k)}$ may be, they are satisfied when

$$x_1 = 0, \quad x_2 = 0, \quad \ldots, \quad x_n = 0. \tag{2}$$

We may also mention another extreme case: if all the coefficients $a_i^{(k)}$ have the value zero, then equations (1) are satisfied for arbitrary values of $x_1, x_2, \ldots, x_n$.

We precede the general answer to the question by the following remarks. We write the system of coefficients of (1) in the form of a rectangle

$$\begin{matrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(m)} & a_2^{(m)} & \cdots & a_n^{(m)}. \end{matrix} \tag{3}$$

Such a scheme, which by itself has no numerical meaning, is called a matrix, insofar as it is regarded as the source of a larger number of determinants. The determinants arising from the matrix are obtained by deleting arbitrary rows and columns, in an arbitrary number limited only so far that the remaining elements form a square, and by regarding this square as a determinant. Thus one obtains from the matrix one-rowed, two-rowed, and so forth, determinants. The highest determinants are n- or m-rowed, according as n or m is the smaller number (or n-rowed if $n = m$).

We now make the assumption that among the ν-rowed determinants of the matrix at least one is different from zero, while the $(\nu + 1)$-rowed and consequently also the higher determinants, if such exist, shall all vanish. ν may be any number which is not greater than the smaller of the two numbers n and m (or, if $n = m$, does not exceed this common value).

Such a number ν can always be found if we exclude the already disposed of, wholly uninteresting case in which all the coefficients $a_i^{(k)}$ vanish.

Without limiting generality, we may, to simplify the notation, assume that the non-vanishing ν-rowed determinant is

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_\nu^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} \end{vmatrix}. \tag{4}$$

For it is plainly open to us to arrange the system of equations (1) in any order, and furthermore we can choose the notation of the unknowns x so that any ν of them are the first ν. We denote the subdeterminants of A, as before, by $A_i^{(k)}$, where i, k range from 1 to ν.

I. If first $\nu = n$, which presupposes that m is not smaller than n, then equations (1) have no solution other than that contained in equations (2).

For let us take the first n equations of (1):

$$\begin{aligned} a_1^{(1)}x_1 + a_2^{(1)}x_2 + \cdots + a_n^{(1)}x_n &= 0, \\ a_1^{(2)}x_1 + a_2^{(2)}x_2 + \cdots + a_n^{(2)}x_n &= 0, \\ &\vdots \\ a_1^{(n)}x_1 + a_2^{(n)}x_2 + \cdots + a_n^{(n)}x_n &= 0, \end{aligned} \tag{5}$$

and multiply these in succession by $A_\mu^{(1)}, A_\mu^{(2)}, \ldots, A_\mu^{(n)}$, where μ may be any one of the indices $1, 2, \ldots, n$, and add them. Then, since by §. 22, (2) and (6),

$$\sum_{1,\nu}^{i} A_\mu^{(i)} a_\lambda^{(i)} = 0 \quad \text{or} \quad = A$$

holds according as λ is different from μ or not,

$$Ax_\mu = 0,$$

and since by our hypothesis A is different from zero,

$$x_\mu = 0.$$

We single out the most frequently used special case $m = n$, and give the theorem for this case in the following form:

II. If a system of n linear homogeneous equations with n unknowns has a determinant different from zero, then all the unknowns have the value zero; or:

If a system of n linear equations with as many homogeneously occurring unknowns has a solution in which not all the unknowns vanish, then the determinant of the system vanishes.

Here, by the determinant of a system of n linear homogeneous equations with n unknowns is meant the determinant formed from the n^2 coefficients of these equations.

We next consider the case in which ν is smaller than n. Since m must be equal to or greater than ν, we choose the first ν equations of the system (1), and write them thus:

$$\begin{aligned} a_1^{(1)}x_1 + a_2^{(1)}x_2 + \cdots + a_\nu^{(1)}x_\nu &= -a_{\nu+1}^{(1)}x_{\nu+1} - \cdots - a_n^{(1)}x_n, \\ a_1^{(2)}x_1 + a_2^{(2)}x_2 + \cdots + a_\nu^{(2)}x_\nu &= -a_{\nu+1}^{(2)}x_{\nu+1} - \cdots - a_n^{(2)}x_n, \\ &\vdots \\ a_1^{(\nu)}x_1 + a_2^{(\nu)}x_2 + \cdots + a_\nu^{(\nu)}x_\nu &= -a_{\nu+1}^{(\nu)}x_{\nu+1} - \cdots - a_n^{(\nu)}x_n. \end{aligned} \tag{6}$$

We again let μ denote one of the indices $1, 2, \ldots, \nu$, multiply equations (6) in succession by $A_\mu^{(1)}, A_\mu^{(2)}, \ldots, A_\mu^{(\nu)}$, and add them. It follows, as before, using §. 22, (2), (6):

$$Ax_\mu = -C_{\nu+1,\mu}x_{\nu+1} - \cdots - C_{n,\mu}x_n, \tag{7}$$

where, for abbreviation, we have put

$$C_{\lambda,\mu} = \sum_{1,\nu}^{i} a_\lambda^{(i)} A_\mu^{(i)}, \qquad \lambda = \nu+1, \nu+2, \ldots, n. \tag{8}$$

By §. 22, (2), $C_{\lambda,\mu}$ is the determinant obtained from the determinant defined by (4) by replacing the elements of the μth column $a_\mu^{(1)}, a_\mu^{(2)}, \ldots, a_\mu^{(\nu)}$ by $a_\lambda^{(1)}, a_\lambda^{(2)}, \ldots, a_\lambda^{(\nu)}$. By (7), since A is different from zero, $x_1, x_2, \ldots, x_\nu$ are now expressed linearly in terms of $x_{\nu+1}, \ldots, x_n$ and the known quantities.

It remains to show

III. that the expressions (7) satisfy equations (1), whatever values $x_{\nu+1}, \ldots, x_n$ may have; thus $n - \nu$ of the unknowns remain arbitrary, and the other ν depend on them according to (7).

To see the truth of this theorem, we need only substitute the expressions (7) in equations (1). The calculation is greatly simplified by use of a summation sign Σ, to which, as above, we append the summation letters above and the limits below. We can then first write equations (7) as

$$Ax_\mu = -\sum_{h=\nu+1}^{n} \sum_{i=1}^{\nu} a_h^{(i)} A_\mu^{(i)} x_h, \qquad \mu = 1, 2, \ldots, \nu. \tag{9}$$

We multiply by $a_\mu^{(k)}$, where k denotes any one of the digits $1, 2, \ldots, m$, and sum with respect to μ:

$$A\sum_{\mu=1}^{\nu} a_\mu^{(k)} x_\mu = -\sum_{h=\nu+1}^{n} x_h \sum_{i=1}^{\nu}\sum_{\mu=1}^{\nu} a_h^{(i)} a_\mu^{(k)} A_\mu^{(i)}, \qquad k = 1, 2, \ldots, m. \tag{10}$$

To both sides we add the sum $A\sum_{h=\nu+1}^{n} a_h^{(k)} x_h$, and obtain

$$A\sum_{\mu=1}^{n} a_\mu^{(k)} x_\mu = \sum_{h=\nu+1}^{n} x_h \left(A a_h^{(k)} - \sum_{i=1}^{\nu}\sum_{\mu=1}^{\nu} a_h^{(i)} a_\mu^{(k)} A_\mu^{(i)} \right). \tag{11}$$

The factor of x_h in the sum on the right-hand side is, by §. 23, (12), the determinant

$$\begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} & a_h^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_\nu^{(2)} & a_h^{(2)} \\ \vdots & \vdots & & \vdots & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} & a_h^{(\nu)} \\ a_1^{(k)} & a_2^{(k)} & \cdots & a_\nu^{(k)} & a_h^{(k)} \end{vmatrix}, \tag{12}$$

and therefore vanishes, if $k \leq \nu$, by §. 21, V, because

two rows coincide, while if $k > \nu$, it vanishes by the hypothesis, since (12) is then a $(\nu+1)$-rowed determinant of the matrix (3). Thus from (11), since A is different from zero, we obtain

$$\sum_{\mu=1}^{n} a_\mu^{(k)} x_\mu = 0, \qquad k = 1, 2, \ldots, m, \tag{13}$$

that is, the system of equations (1) is satisfied by (7).

We shall single out from the theorem just proved the special case in which $m = n-1$ and $\nu = n-1$. In this case only one of the unknowns remains arbitrary, and the ratios of the unknowns are completely determined. We may state this result as follows:

Denote the $(n-1)$-rowed determinants of the matrix

$$\begin{matrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(n-1)} & a_2^{(n-1)} & \cdots & a_n^{(n-1)} \end{matrix} \tag{14}$$

taken with alternating sign by $A_1, A_2, \ldots, A_n$, and assume that at least one of these quantities is different from zero. Then the solution of the system

$$\begin{aligned} a_1^{(1)} x_1 + a_2^{(1)} x_2 + \cdots + a_n^{(1)} x_n &= 0, \\ a_1^{(2)} x_1 + a_2^{(2)} x_2 + \cdots + a_n^{(2)} x_n &= 0, \\ &\vdots \\ a_1^{(n-1)} x_1 + a_2^{(n-1)} x_2 + \cdots + a_n^{(n-1)} x_n &= 0 \end{aligned} \tag{15}$$

is given by the ratios

$$x_1 : x_2 : \cdots : x_n = A_1 : A_2 : \cdots : A_n. \tag{16}$$

Thus for $n = 3$ we obtain the solution of the system of equations, often occurring in geometry,

$$ax + by + cz = 0, \qquad a'x + b'y + c'z = 0 \tag{17}$$

in the form

$$x : y : z = bc' - cb' : ca' - ac' : ab' - ba'. \tag{18}$$

§. 25.

Elimination from Linear Equations.

It sometimes happens that, for a given system of linear equations, the question concerns not so much the actual determination of the unknowns as the decision whether their solution is possible, that is, the setting up of the condition equations which must hold among the coefficients if solutions, or solutions of a specified kind, are to exist at all. The setting up of these condition equations is called elimination. The solution of this problem is already contained implicitly in what precedes; but we shall return explicitly to some questions belonging here.

As in the preceding paragraph, we consider a system of m linear equations with n homogeneously occurring unknowns, and ask: when does this system have a solution in which not all the unknowns vanish? We have already seen that this is always the case if $n > m$. But if $n \leq m$, the necessary and sufficient condition for such a solution is that all the n-rowed determinants of the matrix vanish. For if one of them does not vanish, then by §. 24, II the values of the unknowns are necessarily zero; whereas, if they all vanish, a number $\nu < n$ can be found such that all $(\nu+1)$-rowed determinants of the matrix are zero while at least one of the ν-rowed determinants does not vanish, so that by §. 24, III a solution of the required kind exists.

From the matrix one can form, when $n \leq m$,

$$\frac{m(m-1)\cdots(m-n+1)}{1\cdot 2\cdots n}$$

n-rowed determinants, and this, therefore, would be the number of conditions. If $n = m$, this number is $= 1$, and we obtain the case §. 24, II and, as was to be expected, one condition. In general, however, although all these conditions must be fulfilled, their number is greater than necessary, because some of them are necessary consequences of the others.

To obtain a system of necessary, sufficient, and mutually independent conditions, we state the question somewhat more precisely and ask for the conditions:

that from a system of m linear homogeneous equations with n unknowns, ν of the unknowns can be completely determined by $n-\nu$ which remain arbitrary.

This question too is in fact already answered in §. 24. Among the ν-rowed determinants one must be different from zero, while all the $(\nu+1)$-rowed determinants vanish. But it is already sufficient if it is known of a smaller number of the $(\nu+1)$-rowed determinants that they vanish.

Assume that the unknowns $x_{\nu+1}, x_{\nu+2}, \ldots, x_n$ are to remain arbitrary, that $x_1, x_2, \ldots, x_\nu$ are to be determined by them, and take the determinant

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_\nu^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} \end{vmatrix} \tag{1}$$

to be different from zero. We calculate the unknowns $x_1, x_2, \ldots, x_\nu$ according to §. 24, (9), and form the sums (11), whose vanishing says that the given system of equations is actually satisfied. The conditions for this are therefore:

$$\begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} & a_h^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_\nu^{(2)} & a_h^{(2)} \\ \vdots & \vdots & & \vdots & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} & a_h^{(\nu)} \\ a_1^{(k)} & a_2^{(k)} & \cdots & a_\nu^{(k)} & a_h^{(k)} \end{vmatrix} = 0, \tag{2}$$

or, written otherwise,

$$Aa_h^{(k)} - \sum_{i=1}^{\nu}\sum_{\mu=1}^{\nu} a_h^{(i)} a_\mu^{(k)} A_\mu^{(i)} = 0, \tag{3}$$

and these conditions are also sufficient. Equation (2) or (3), however, is identically satisfied when $h = 1, 2, \ldots, \nu$, or $k = 1, 2, \ldots, \nu$, and therefore gives no condition on the coefficients for these values. Such conditions arise only for

$$h = \nu+1, \nu+2, \ldots, n, \qquad k = \nu+1, \nu+2, \ldots, m, \tag{4}$$

and hence

$$(n-\nu)(m-\nu) \tag{5}$$

in number.

These conditions are, however, actually mutually independent, that is, none follows from the others; for the left-hand sides of (3), by suitable assumptions concerning the coefficients a, can receive an entirely arbitrary value for every combination of indices in the series (4), as is seen by setting all $a_h^{(i)}, a_\mu^{(k)}$, with the exception of $a_h^{(k)}$, equal to zero.

§. 26.

Non-homogeneous Linear Equations.

We have simplified the problem of solving linear equations in the preceding considerations, and reduced it to more general laws, to no inconsiderable degree by assuming the equations homogeneous with respect to the unknowns. In applications, however, the unknowns frequently do not occur homogeneously, and although in principle one case is not essentially different from the other, we shall nevertheless consider the case of non-homogeneous equations separately.

We can derive it from the case of homogeneous equations by adding the requirement that one specified unknown shall have the value 1. If this unknown occurs among those which the problem leaves arbitrary, no difficulty arises, because we need only put it = 1. But if it belongs to those determined by the others, certain further conditions must be satisfied, which say that the value 1 is admissible for this unknown.

We shall treat the question here independently and with a somewhat altered notation, but for simplicity restrict ourselves to the most important case, in which the number of unknowns agrees with the number of equations.

Let the following system of equations be solved with respect to the unknowns $x_1, x_2, \ldots, x_n$:

$$\begin{aligned} a_1^{(1)}x_1 + a_2^{(1)}x_2 + \cdots + a_n^{(1)}x_n &= y_1, \\ a_1^{(2)}x_1 + a_2^{(2)}x_2 + \cdots + a_n^{(2)}x_n &= y_2, \\ &\vdots \\ a_1^{(n)}x_1 + a_2^{(n)}x_2 + \cdots + a_n^{(n)}x_n &= y_n. \end{aligned} \tag{1}$$

where the coefficients $a_i^{(k)}$ and the independent terms $y_1, y_2, \ldots, y_n$ are regarded as given. We denote by

$$A = \sum \pm a_1^{(1)} a_2^{(2)} \cdots a_n^{(n)} \tag{2}$$

the determinant of the system, and, as before, denote by $A_i^{(k)}$ the first subdeterminants of A. If k is one of the indices $1, 2, \ldots, n$, and we multiply equations (1) in succession by $A_k^{(1)}, A_k^{(2)}, \ldots, A_k^{(n)}$ and then add them, we obtain

$$x_1 \sum^{i} a_1^{(i)} A_k^{(i)} + x_2 \sum^{i} a_2^{(i)} A_k^{(i)} + \cdots + x_n \sum^{i} a_n^{(i)} A_k^{(i)} = \sum^{i} y_i A_k^{(i)}. \tag{3}$$

By §. 22, (2) and (6), the coefficients of $x_1, x_2, \ldots, x_n$ vanish here with the exception of the single coefficient of x_k, which receives the value A, and consequently:

$$Ax_k = \sum^{i} y_i A_k^{(i)}, \tag{4}$$

and in the case in which A is different from zero, the values of the unknowns can be calculated uniquely from this in such a way that equations (1) are actually satisfied [§. 23, (3), (7)]. If we put $A_k^{(i)} = A\alpha_k^{(i)}$, we can write the solution in the form

$$\begin{aligned} x_1 &= \alpha_1^{(1)} y_1 + \alpha_1^{(2)} y_2 + \cdots + \alpha_1^{(n)} y_n, \\ x_2 &= \alpha_2^{(1)} y_1 + \alpha_2^{(2)} y_2 + \cdots + \alpha_2^{(n)} y_n, \\ &\vdots \\ x_n &= \alpha_n^{(1)} y_1 + \alpha_n^{(2)} y_2 + \cdots + \alpha_n^{(n)} y_n, \end{aligned} \tag{5}$$

whose analogy with system (1) is evident.

But if $A = 0$, equations (4) teach nothing about the values of the x's; they give only a system of conditions which the y's must necessarily satisfy if equations (1) are to be soluble at all. To arrive at a generally valid result, we proceed quite similarly to the case of homogeneous equations. We assume that, besides the determinant A, all subdeterminants down to a certain order are also zero. We denote a non-vanishing ν-rowed subdeterminant by

$$B = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_\nu^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_\nu^{(2)} \\ \vdots & \vdots & & \vdots \\ a_1^{(\nu)} & a_2^{(\nu)} & \cdots & a_\nu^{(\nu)} \end{vmatrix}, \tag{6}$$

denote its subdeterminants by $B_k^{(h)}$, where h, k run only through the values from 1 to ν, and now suppose that all $(\nu+1)$-rowed subdeterminants of A vanish. If we multiply the first ν equations of the system (1) by $B_k^{(1)}, B_k^{(2)}, \ldots, B_k^{(\nu)}$, and add, it follows [§. 22, (2), (6)]:

$$Bx_k = \sum_{i=1}^{\nu} B_k^{(i)} y_i - \sum_{s=\nu+1}^{n} x_s \sum_{i=1}^{\nu} B_k^{(i)} a_s^{(i)}. \tag{7}$$

Hereby ν of the unknowns x are determined by the others, and, to see how far equations (1) are thereby satisfied, we substitute the expressions in (1). For this purpose we multiply (7) by $a_k^{(\lambda)}$, where λ runs through the values $1, 2, \ldots, n$, and sum with respect to k. Thus follows:

$$B \sum_{k=1}^{\nu} a_k^{(\lambda)} x_k = \sum_{i=1}^{\nu} \sum_{k=1}^{\nu} B_k^{(i)} a_k^{(\lambda)} y_i - \sum_{s=\nu+1}^{n} x_s \sum_{i=1}^{\nu} \sum_{k=1}^{\nu} B_k^{(i)} a_s^{(i)} a_k^{(\lambda)}. \tag{8}$$

and if we add to both sides

$$B \sum_{k=\nu+1}^{n} a_k^{(\lambda)} x_k$$

we obtain:

$$B\sum_{k=1}^{n} a_k^{(\lambda)} x_k = \sum_{i=1}^{\nu}\sum_{k=1}^{\nu} B_k^{(i)} a_k^{(\lambda)} y_i + \sum_{s=\nu+1}^{n} x_s \left(B a_s^{(\lambda)} - \sum_{i=1}^{\nu}\sum_{k=1}^{\nu} B_k^{(i)} a_s^{(i)} a_k^{(\lambda)} \right). \tag{9}$$

Now, by §. 23, (12), the coefficient of x_s on the right-hand side is a $(\nu+1)$-rowed subdeterminant of A, and hence is zero by the hypothesis. According to (1),

$$\sum_{k=1}^{n} a_k^{(\lambda)} x_k = y_\lambda$$

is to hold, and consequently the expressions (7) satisfy equations (1) if and only if

$$B y_\lambda = \sum_{i=1}^{\nu}\sum_{k=1}^{\nu} B_k^{(i)} a_k^{(\lambda)} y_i \tag{10}$$

holds. This equation is always satisfied when $\lambda \leq \nu$, by §. 22, (2), (6). But if $\lambda = \nu+1, \nu+2, \ldots, n$, then (10) contains $n-\nu$ conditions on the y's which, since each

condition contains a new one of the quantities y, are independent of one another. If one of these conditions is not fulfilled, the given system of equations has no solution.

If in the system of equations (1) we regard the x's not as unknowns but as variables, then the y's too will be variable quantities. Under this interpretation we call the system (1) a linear substitution, insofar as it mediates the passage from one system of variables to another. Only when the determinant A, which we now call the determinant of the substitution, is different from zero can the y's be regarded as independent variables. If this is the case, the system (5) gives the representation of the variables x in terms of the y's, or the substitution inverse to (1).

§. 27.

Multiplication of Determinants.

The theorem which we still have to prove teaches how one can represent the product of two determinants with the same number of rows by a single determinant with the same number of rows. One is led to it most simply by considering the solution of two systems of linear equations.

Let now the coefficients $a_i^{(k)}, b_i^{(k)}$, when i, k run through the sequence of numbers $1, 2, \ldots, n$, be arbitrary variable quantities; likewise let the quantities x_i, y_i, z_i be bound only by the relations

$$\sum_i a_i^{(k)} x_i = y_k, \tag{1}$$

$$\sum_k b_h^{(k)} y_k = z_h. \tag{2}$$

If the problem is now posed of determining the variables x by the variables z, this problem can be solved in two ways. By § 26, (4) one may solve equations (1) with respect to x, equations (2) with respect to y, and insert the latter expressions into the former. If A, B denote the determinants of the two systems of equations, that is,

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \cdots & \cdots & & \cdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \qquad B = \begin{vmatrix} b_1^{(1)} & b_2^{(1)} & \cdots & b_n^{(1)} \\ b_1^{(2)} & b_2^{(2)} & \cdots & b_n^{(2)} \\ \cdots & \cdots & & \cdots \\ b_1^{(n)} & b_2^{(n)} & \cdots & b_n^{(n)} \end{vmatrix},$$

and if $A_i^{(k)}, B_i^{(k)}$ denote the first subdeterminants, one obtains

$$A x_k = \sum_i A_k^{(i)} y_i, \tag{3}$$

$$B y_i = \sum_h B_h^{(i)} z_h, \tag{4}$$

and, by substitution of (4) in (3),

$$ABx_k = \sum_h z_h \sum_i A_k^{(i)} B_h^{(i)}. \tag{5}$$

But one may also proceed by inserting from (1) the expressions for y into (2), whereby, if

$$c_i^{(h)} = \sum_s a_i^{(s)} b_h^{(s)} \tag{6}$$

is put, one obtains

$$\sum_i c_i^{(h)} x_i = z_h. \tag{7}$$

Now put

$$C = \begin{vmatrix} c_1^{(1)} & c_2^{(1)} & \cdots & c_n^{(1)} \\ c_1^{(2)} & c_2^{(2)} & \cdots & c_n^{(2)} \\ \cdots & \cdots & & \cdots \\ c_1^{(n)} & c_2^{(n)} & \cdots & c_n^{(n)} \end{vmatrix} \tag{8}$$

If $C_i^{(k)}$ denotes the subdeterminants of C, the solution of (7) gives

$$Cx_k = \sum_h z_h C_k^{(h)}, \tag{9}$$

and comparison of (5) with (9) gives

$$AB = C, \tag{10}$$

and for the subdeterminants

$$C_k^{(h)} = \sum_i A_k^{(i)} B_h^{(i)}. \tag{11}$$

In this conclusion, however, there is still a gap, caused by the doubt whether (5) and (9) might not differ by a factor common to both sides. To remove this doubt we shall prove equation (10) directly; the correctness of (11) then follows.

We shall form the principal term of the determinant C from (6). Let $s_1, s_2, \ldots, s_n$ be mutually independent summation letters which run from 1 to n. Then

$$\begin{aligned} c_1^{(1)} c_2^{(2)} \cdots c_n^{(n)} &= \sum_{s_1} a_1^{(s_1)} b_1^{(s_1)} \sum_{s_2} a_2^{(s_2)} b_2^{(s_2)} \cdots \sum_{s_n} a_n^{(s_n)} b_n^{(s_n)} \\ &= \sum_{s_1, s_2, \ldots, s_n} a_1^{(s_1)} a_2^{(s_2)} \cdots a_n^{(s_n)} b_1^{(s_1)} b_2^{(s_2)} \cdots b_n^{(s_n)}. \end{aligned} \tag{12}$$

The permutation of the lower indices of the c corresponds to the permutation of the lower indices of the a, and if, with regard to the rule of signs, one therefore forms the determinant C, one obtains

$$\sum \pm c_1^{(1)} c_2^{(2)} \cdots c_n^{(n)} = \sum_{s_1, s_2, \ldots, s_n} b_1^{(s_1)} b_2^{(s_2)} \cdots b_n^{(s_n)} \sum \pm a_1^{(s_1)} a_2^{(s_2)} \cdots a_n^{(s_n)}. \tag{13}$$

Now

$$\sum \pm a_1^{(s_1)} a_2^{(s_2)} \cdots a_n^{(s_n)}$$

is, by § 21, VI, always zero if the same digit occurs twice among $s_1, s_2, \ldots, s_n$; hence only those terms in (13) keep a non-zero value for which $s_1, s_2, \ldots, s_n$ is an arrangement of the indices $1, 2, \ldots, n$.

This value is $+A$ or $-A$ according as this arrangement belongs to the first or the second kind (§ 21, IV).

Therefore the right-hand side of (13) becomes

$$A \sum_{s_1,s_2,\ldots,s_n} \pm b_1^{(s_1)} b_2^{(s_2)} \cdots b_n^{(s_n)},$$

where the sum is extended over all permutations $s_1, s_2, \ldots, s_n$. But this sum is precisely the determinant B, and hence the formula

$$C = AB$$

is proved.

The law of formation of the elements $c_i^{(h)}$ contained in formula (6) can be expressed in words as follows: To obtain the elements of the determinant C, which is the product of the two determinants A, B, one multiplies the elements of each column of A by the corresponding elements of a column of B and adds the products.

By the theorems on determinants the form of C can be varied in many ways. We record the following.

If two columns in A or in B are interchanged, the elements $c_i^{(h)}$ do not change, but only interchange among themselves. If, however, two rows in A or in B are interchanged, then the $c_i^{(h)}$ change because the factors in the individual products of the sum are grouped differently; but if corresponding interchanges are made simultaneously in the rows of A and of B, the $c_i^{(h)}$ remain unchanged, since only the individual terms of the sum are thereby interchanged.

By turning the rows into columns in A or in B, or in both at once, one obtains three further different ways of forming the product of two determinants in determinant form. This can also be expressed as follows: To form the product of two determinants, one may multiply the elements of the separate rows or columns of one factor by the corresponding elements of the rows or columns of the other factor, add the products, and regard these sums of products as the elements of a new determinant.

The multiplication rule can be applied to a product of two determinants with different numbers of elements by transforming the determinant with the smaller number of rows, by the theorem § 22, IX, into another with more rows.

§. 28.

Determinants of Subdeterminants.

We immediately make an application here of the multiplication law of determinants. Let, as before,

$$A = \begin{vmatrix} a_1^{(1)} & a_2^{(1)} & \cdots & a_n^{(1)} \\ a_1^{(2)} & a_2^{(2)} & \cdots & a_n^{(2)} \\ \cdots & \cdots & & \cdots \\ a_1^{(n)} & a_2^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \tag{1}$$

and let

$$\begin{matrix} A_1^{(1)} & A_2^{(1)} & \cdots & A_n^{(1)} \\ A_1^{(2)} & A_2^{(2)} & \cdots & A_n^{(2)} \\ \cdots & \cdots & & \cdots \\ A_1^{(n)} & A_2^{(n)} & \cdots & A_n^{(n)} \end{matrix} \tag{2}$$

be the system of subdeterminants. If we form from (2) the determinant which we shall denote by Δ, we may apply the multiplication rule to the product $A\Delta$. By § 22, (3) and (7), this gives

$$A\Delta = \begin{vmatrix} A & 0 & \cdots & 0 \\ 0 & A & \cdots & 0 \\ \cdots & \cdots & & \cdots \\ 0 & 0 & \cdots & A \end{vmatrix} = A^n,$$

and, by division by A,

$$\Delta = A^{n-1}. \tag{3}$$

Thus Δ is the $(n-1)$st power of A. In this derivation it has indeed first been assumed that A is different from zero. But since (3) is an identity with respect to the elements $a_i^{(h)}$, that is, since it also holds when these quantities are independent variables, it follows that the formula (3) still holds in the exceptional case; that is, if A vanishes, then Δ also vanishes.

This result is a special case of a more general theorem by which any determinant of the matrix (2) may be formed. Consider the ν-rowed subdeterminant

$$\Delta_\nu = \begin{vmatrix} A_1^{(1)} & A_2^{(1)} & \cdots & A_\nu^{(1)} \\ A_1^{(2)} & A_2^{(2)} & \cdots & A_\nu^{(2)} \\ \cdots & \cdots & & \cdots \\ A_1^{(\nu)} & A_2^{(\nu)} & \cdots & A_\nu^{(\nu)} \end{vmatrix}, \tag{4}$$

from which all other ν-rowed subdeterminants can be derived by permutation of the upper and lower indices. One may apply the multiplication rule by transforming Δ_ν, according to the concluding remark of the last paragraph, into an n-rowed determinant.

Put

$$\Delta_\nu = \begin{vmatrix} A_1^{(1)} & \cdots & A_\nu^{(1)} & A_{\nu+1}^{(1)} & \cdots & A_n^{(1)} \\ \cdots & & \cdots & \cdots & & \cdots \\ A_1^{(\nu)} & \cdots & A_\nu^{(\nu)} & A_{\nu+1}^{(\nu)} & \cdots & A_n^{(\nu)} \\ 0 & \cdots & 0 & 1 & \cdots & 0 \\ \cdots & & \cdots & \cdots & & \cdots \\ 0 & \cdots & 0 & 0 & \cdots & 1 \end{vmatrix}, \tag{5}$$

where $n-\nu$ rows and columns have been adjoined, the former containing only zeros except in the diagonal terms. If one now forms the product $A\Delta_\nu$, it follows that

$$A\Delta_\nu = \begin{vmatrix} A & 0 & \cdots & 0 & a_{\nu+1}^{(1)} & \cdots & a_n^{(1)} \\ 0 & A & \cdots & 0 & a_{\nu+1}^{(2)} & \cdots & a_n^{(2)} \\ \cdots & \cdots & & \cdots & \cdots & & \cdots \\ 0 & 0 & \cdots & A & a_{\nu+1}^{(\nu)} & \cdots & a_n^{(\nu)} \\ 0 & 0 & \cdots & 0 & a_{\nu+1}^{(\nu+1)} & \cdots & a_n^{(\nu+1)} \\ \cdots & \cdots & & \cdots & \cdots & & \cdots \\ 0 & 0 & \cdots & 0 & a_{\nu+1}^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}, \tag{6}$$

and by Theorem IX, § 22, this is

$$= A^\nu \begin{vmatrix} a_{\nu+1}^{(\nu+1)} & \cdots & a_n^{(\nu+1)} \\ \cdots & & \cdots \\ a_{\nu+1}^{(n)} & \cdots & a_n^{(n)} \end{vmatrix}.$$

Dividing here by A and applying the notation of § 23, one obtains

$$\Delta_\nu = A^{\nu-1} A_{1,2,\ldots,\nu}^{1,2,\ldots,\nu}. \tag{7}$$

For $\nu = 2$ this gives the special result

$$A_1^{(1)}A_2^{(2)} - A_1^{(2)}A_2^{(1)} = AA_{1,2}^{1,2}. \tag{8}$$

We shall use formula (8) often later. In the notation by differential quotients it can be represented as

$$A\frac{\partial^2 A}{\partial a_1^{(1)}\partial a_2^{(2)}} = \frac{\partial A}{\partial a_1^{(1)}}\frac{\partial A}{\partial a_2^{(2)}} - \frac{\partial A}{\partial a_1^{(2)}}\frac{\partial A}{\partial a_2^{(1)}}. \tag{9}$$

It is especially important in the case where A is a symmetric determinant, so that $a_i^{(k)} = a_k^{(i)}$; then also

$$\frac{\partial A}{\partial a_1^{(2)}} = \frac{\partial A}{\partial a_2^{(1)}},$$

where in the differentiation no account is taken of the dependence $a_i^{(k)} = a_k^{(i)}$. Then formula (9) becomes

$$A\frac{\partial^2 A}{\partial a_1^{(1)}\partial a_2^{(2)}} = \frac{\partial A}{\partial a_1^{(1)}}\frac{\partial A}{\partial a_2^{(2)}} - \left(\frac{\partial A}{\partial a_1^{(2)}}\right)^2. \tag{10}$$

§. 29.

Interpolation.

In § 9, under a special assumption, we already solved the problem of determining an integral rational function which assumes prescribed values for a sufficient number of prescribed values of the argument. We shall now solve this problem generally by applying determinants. First, however, we shall put forward the following observations.

In § 22, (12), the value of the determinant

$$\begin{vmatrix} \alpha_1^{n-1} & \alpha_1^{n-2} & \cdots & \alpha_1 & 1 \\ \alpha_2^{n-1} & \alpha_2^{n-2} & \cdots & \alpha_2 & 1 \\ \cdots & \cdots & & \cdots & \cdots \\ \alpha_n^{n-1} & \alpha_n^{n-2} & \cdots & \alpha_n & 1 \end{vmatrix}$$

was found to be equal to the product of differences

$$(\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)\cdots(\alpha_1 - \alpha_n)(\alpha_2 - \alpha_3)\cdots(\alpha_2 - \alpha_n)\cdots(\alpha_{n-1} - \alpha_n). \tag{1}$$

For brevity we now denote this product of differences by

$$[\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n], \tag{2}$$

so that the quantity $[\alpha_1, \alpha_2, \alpha_3, \ldots, \alpha_n]$ changes its sign when two of the elements $\alpha_1, \alpha_2, \ldots, \alpha_n$ are interchanged.

We further define an integral rational function $f(x)$ of degree n in the variable x by the equation

$$f(x) = (x - \alpha_1)(x - \alpha_2)\cdots(x - \alpha_n), \tag{3}$$

so that, by § 13,

$$\begin{aligned} f'(\alpha_1) &= (\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)\cdots(\alpha_1 - \alpha_n), \\ f'(\alpha_2) &= (\alpha_2 - \alpha_1)(\alpha_2 - \alpha_3)\cdots(\alpha_2 - \alpha_n), \\ &\vdots \\ f'(\alpha_n) &= (\alpha_n - \alpha_1)(\alpha_n - \alpha_2)\cdots(\alpha_n - \alpha_{n-1}). \end{aligned} \tag{4}$$

In the product of all these expressions each factor of the product (1) occurs twice with opposite sign; since the number of factors of (1) is $\frac{n(n-1)}{2}$, we obtain the equation

$$f'(\alpha_1)f'(\alpha_2)\cdots f'(\alpha_n) = (-1)^{\frac{n(n-1)}{2}}[\alpha_1, \alpha_2, \ldots, \alpha_n]^2. \tag{5}$$

But with the aid of the relations (4) we may also represent our product (1) as follows:

$$\begin{aligned}[\alpha_1, \alpha_2, \ldots, \alpha_n] &= f'(\alpha_1)[\alpha_2, \alpha_3, \ldots, \alpha_n] \\ &= -f'(\alpha_2)[\alpha_1, \alpha_3, \ldots, \alpha_n] \\ &= +f'(\alpha_3)[\alpha_1, \alpha_2, \ldots, \alpha_n] \\ &= \cdots = (-1)^{n-1}f'(\alpha_n)[\alpha_1, \alpha_2, \ldots, \alpha_{n-1}],\end{aligned} \tag{6}$$

where on the right-hand side each bracketed expression contains one element fewer than the bracketed expression on the left.

Now pose the problem of determining an integral rational function of degree $(n-1)$, $\varphi(x)$, which, for the argument values $x = \alpha_1, \alpha_2, \ldots, \alpha_n$, assumes in order the prescribed values $\varphi(\alpha_1), \varphi(\alpha_2), \ldots, \varphi(\alpha_n)$.

If one puts

$$\varphi(x) = a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n, \tag{7}$$

then the n unknown coefficients $a_1, a_2, \ldots, a_n$ are to be determined from the linear equations

$$\begin{aligned}\varphi(\alpha_1) &= a_1\alpha_1^{n-1} + a_2\alpha_1^{n-2} + \cdots + a_{n-1}\alpha_1 + a_n, \\ \varphi(\alpha_2) &= a_1\alpha_2^{n-1} + a_2\alpha_2^{n-2} + \cdots + a_{n-1}\alpha_2 + a_n, \\ &\vdots \\ \varphi(\alpha_n) &= a_1\alpha_n^{n-1} + a_2\alpha_n^{n-2} + \cdots + a_{n-1}\alpha_n + a_n.\end{aligned} \tag{8}$$

The determinant of this system is precisely the quantity (2)

$$[\alpha_1, \alpha_2, \ldots, \alpha_n],$$

and the problem is therefore always soluble by § 26 when this quantity is different from zero, that is, when no two of the n quantities $\alpha_1, \alpha_2, \ldots, \alpha_n$ are equal.

Instead of solving the equations (8) by § 26 and substituting the expressions so found into (7), it is preferable, by § 24, II, to eliminate the homogeneous quantities

$$1, a_1, a_2, \ldots, a_n$$

from the $n+1$ equations (7) and (8). This gives the determinant equation

$$\begin{vmatrix} \varphi(x) & x^{n-1} & x^{n-2} & \cdots & x & 1 \\ \varphi(\alpha_1) & \alpha_1^{n-1} & \alpha_1^{n-2} & \cdots & \alpha_1 & 1 \\ \varphi(\alpha_2) & \alpha_2^{n-1} & \alpha_2^{n-2} & \cdots & \alpha_2 & 1 \\ \cdots & \cdots & \cdots & & \cdots & \cdots \\ \varphi(\alpha_n) & \alpha_n^{n-1} & \alpha_n^{n-1} & \cdots & \alpha_n & 1 \end{vmatrix} = 0. \tag{9}$$

We now expand this determinant according to the elements of the first column. For the subdeterminants which occur there we can use the notation (2), and we obtain

$$\begin{aligned}&\varphi(x)[\alpha_1, \alpha_2, \ldots, \alpha_n] - \varphi(\alpha_1)[x, \alpha_2, \alpha_3, \ldots, \alpha_n] + \varphi(\alpha_2)[x, \alpha_1, \alpha_3, \ldots, \alpha_n] \\ &\quad + \cdots (-1)^n\varphi(\alpha_n)[x, \alpha_1, \alpha_2, \ldots, \alpha_{n-1}] = 0.\end{aligned} \tag{10}$$

Now by (1)

$$[x, \alpha_2, \alpha_3, \ldots, \alpha_n] = (x-\alpha_2)(x-\alpha_3)\cdots(x-\alpha_n)[\alpha_2, \alpha_3, \ldots, \alpha_n],$$

for which we may also write

$$[x, \alpha_2, \alpha_3, \ldots, \alpha_n] = \frac{f(x)}{x-\alpha_1}[\alpha_2, \alpha_3, \ldots, \alpha_n],$$

and likewise

$$[x, \alpha_1, \alpha_3, \ldots, \alpha_n] = \frac{f(x)}{x-\alpha_2}[\alpha_1, \alpha_3, \ldots, \alpha_n],$$
$$\cdots$$
$$[x, \alpha_1, \alpha_2, \ldots, \alpha_{n-1}] = \frac{f(x)}{x-\alpha_n}[\alpha_1, \alpha_2, \ldots, \alpha_{n-1}],$$

Dividing (10) by $[\alpha_1, \alpha_2, \ldots, \alpha_n]$ and taking account of the several expressions (6), one finally obtains

$$\varphi(x) = f(x)\left(\frac{\varphi(\alpha_1)}{(x-\alpha_1)f'(\alpha_1)} + \frac{\varphi(\alpha_2)}{(x-\alpha_2)f'(\alpha_2)} + \cdots + \frac{\varphi(\alpha_n)}{(x-\alpha_n)f'(\alpha_n)}\right). \tag{11}$$

This completely determines the function $\varphi(x)$. This expression for $\varphi(x)$ is called the interpolation formula of Lagrange. That it satisfies the stipulated requirements can be verified afterwards very easily by putting $x_1 = \alpha_1, \alpha_2, \ldots, \alpha_n$. The denominators $x-\alpha_1, x-\alpha_2, \ldots, x-\alpha_n$ are present in it only apparently.

Third Section.

The Roots of Algebraic Equations.

§. 30.

Concept of Roots. Multiple Roots.

After, in the first two sections, the algebraic magnitudes had been considered more from the formal side, where the matter was identical transformations of letter-expressions in which the letters throughout could be conceived as symbols for variable magnitudes, the numerical magnitudes now come more into the foreground.

Here, in accordance with what was fixed in the Introduction, by numbers we understand real or imaginary magnitudes of the form $a+bi$, and for illustration we represent the real magnitudes by points of a straight line and the imaginary magnitudes by points of a plane. By the absolute value of an imaginary magnitude $a+bi$ we understand $\sqrt{a^2+b^2}$ and denote it, following Weierstrass, by

$$|a+bi|.$$

Let now

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n \tag{1}$$

be an integral function of x, in which the coefficients are arbitrary real or imaginary numbers, and the first, a_0, is assumed different from zero. If α is a number which, when put for x, makes the function $f(x)$ equal to zero, that is, which satisfies the condition $f(\alpha) = 0$, then α is called a root of the equation

$$f(x) = 0.$$

We also say briefly that α is a root of $f(x)$. By § 4, $f(x)$ may then be divided by $x-\alpha$ without remainder, so that one may put

$$f(x) = (x-\alpha)f_1(x), \tag{2}$$

where $f_1(x)$ is only of the $(n-1)$st degree and has the same first coefficient a_0 as $f(x)$, thus

$$f_1(x) = a_0 x^{n-1} + a_1' x^{n-2} + \cdots .$$

The coefficients of $f_1(x)$ are given in § 4, (6).

Every root of $f_1(x)$ is therefore at the same time a root of $f(x)$; and conversely every root of $f(x)$ is either $= \alpha$ or is a root of $f_1(x)$. If one root α of $f(x)$ is known, then the task of finding the remaining ones is reduced to the solution of an equation of the $(n-1)$st degree.

The aim of the considerations of this section consists in proving that every function $f(x)$ of the nth degree has at least one root. This theorem is called the fundamental theorem of algebra. First we draw from (2) the important consequence:

I. An equation of the nth degree cannot have more than n roots.

For if $f(x)$ had more than n roots, then $f_1(x)$, which is only of the $(n-1)$st degree, would have more than $n-1$ roots. But an equation of the first degree has not more than one root, whence the correctness of our theorem follows by complete induction.

Sometimes one also gives it the expression:

II. If a function $f(x)$ of the nth degree has more than n roots, then all its coefficients must be zero.

If β is a root of $f_1(x)$, then one may likewise put

$$f_1(x) = (x-\beta) f_2(x), \quad f(x) = (x-\alpha)(x-\beta) f_2(x),$$

where $f_2(x) = a_0 x^{n-2} + \cdots$ is only of the $(n-2)$nd degree. Thus, if one assumes that each of the functions obtained by this division, $f_1(x), f_2(x), \ldots$, has at least one root, one finally obtains

$$f(x) = a_0 (x-\alpha)(x-\beta)\cdots(x-\nu), \tag{3}$$

and we can therefore also express the theorem which we earlier described as the aim of our considerations as follows. It is to be proved:

A function of the nth degree can be decomposed into n linear factors.

The magnitudes $\alpha, \beta, \ldots, \nu$ are then all roots of $f(x)$, and their number is therefore n. It is, however, not excluded that the same number occurs more than once among $\alpha, \beta, \ldots, \nu$. Then $f(x)$ would have fewer than n roots, while the theorem would still remain that $f(x)$ is decomposable into n linear factors. To establish agreement, it has been agreed that, if $x-\alpha$ goes into $f(x)$ several times, α is to be counted several times among the roots, and hence that one speaks of simple, double, triple, etc. roots.

By the concept of the derived functions [§ 13, (2)], if α is an arbitrary magnitude,

$$f(x) = f(\alpha) + (x-\alpha) f'(\alpha) + \frac{(x-\alpha)^2}{1\cdot 2} f''(\alpha) + \frac{(x-\alpha)^3}{1\cdot 2\cdot 3} f'''(\alpha) + \cdots . \tag{4}$$

If it is now again assumed that $f(\alpha)$ vanishes, then there results

$$f_1(x) = f'(\alpha) + \frac{x-\alpha}{1\cdot 2} f''(\alpha) + \frac{(x-\alpha)^2}{1\cdot 2\cdot 3} f'''(\alpha) + \cdots ,$$

and hence

$$f_1(\alpha) = f'(\alpha).$$

Thus α is a double root of $f(x)$ when $f'(\alpha)$ vanishes simultaneously with $f(\alpha)$. Formula (4) also shows that only under this assumption is $f(x)$ divisible by $(x-\alpha)^2$. This conclusion may be carried further and leads to the theorem:

III. The necessary and sufficient condition that α be an m-fold root of $f(x)$ is that $f(\alpha), f'(\alpha), f''(\alpha), \ldots, f^{(m-1)}(\alpha)$ vanish, while $f^{(m)}(\alpha)$ does not vanish.

§. 31.

Continuity of Integral Functions.

If, as before,

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_n$$

is an integral function of x, we first prove the following theorem:

IV. $f(x)$ becomes infinitely large at the same time as x.

This means that, if C is an arbitrarily given positive number, the positive number R can be so chosen that

$$|f(x)| > C$$

as soon as the absolute value of x becomes greater than R; or, expressed geometrically: one can describe in the x-plane a circle around the origin such that outside this circle the absolute value of $f(x)$ no longer falls below C, however large C is assumed. The theorem is evident for the case of a simple power x^n; for if r is the absolute value of x, then r^n is the absolute value of x^n, and r^n, as soon as r has become greater than 1, is greater than r and hence grows with r into the infinite.

To prove the theorem in general, let us denote the absolute values of $x, a_0, a_1, a_2, \ldots, a_n$ by $r, c_0, c_1, c_2, \ldots, c_n$. Then

$$|f(x)| = r^n \left| a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right|, \tag{2}$$

and, since by the theorems proved in the Introduction the absolute value of a sum cannot be smaller than the difference and cannot be greater than the sum of the absolute values of the summands,

$$\left| a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right| \geq c_0 - \left| \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n} \right| \geq c_0 - \frac{c_1}{r} - \frac{c_2}{r^2} - \cdots - \frac{c_n}{r^n}.$$

Since $c_0, c_1, \ldots, c_n$ are fixed constants, one can choose R so large that, as soon as $r > R$,

$$\frac{c_1}{r} + \frac{c_2}{r^2} + \cdots + \frac{c_n}{r^n}$$

becomes arbitrarily small, for example smaller than $\frac{1}{2}c_0$. Then

$$\left| a_0 + \frac{a_1}{x} + \frac{a_2}{x} + \cdots + \frac{a_n}{x^n} \right| > \tfrac{1}{2}c_0,$$

and hence by (2)

$$|f(x)| > \tfrac{1}{2}c_0R^n,$$

so that, if one assumes

$$R^n > \frac{2C}{c_0}$$

$$|f(x)| > C. \tag{3}$$

To this is attached the theorem:

V. $f(x)$ is a continuous function of x.

By this the following is meant. If the absolute value of x is smaller than an arbitrary finite magnitude R, then, however small the positive magnitude ω is assumed, one can find a positive magnitude ε such that

$$|f(x+h) - f(x)| < \omega \tag{4}$$

as long as the absolute value ϱ of h remains smaller than ε, and indeed in such a way that ε depends only on the choice of ω, not on x.

By § 13, (2),

$$\begin{aligned} f(x+h) - f(x) &= hf'(x) + \frac{h^2}{1\cdot 2}f''(x) + \cdots + \frac{h^n}{\Pi(n)}f^{(n)}(x) \\ &= h\left\{f'(x) + \frac{h}{2}f''(x) + \cdots + a_0 h^{n-1}\right\}. \end{aligned} \tag{5}$$

If now we assume the absolute value of h smaller than 1, then the absolute value of the magnitude in parentheses

$$f'(x) + \frac{h}{2}f''(x) + \cdots + a_0 h^{n-1} \tag{6}$$

is smaller than a certain non-zero number k which one obtains if in (6) one replaces h by 1, the coefficients $a_0, a_1, \ldots, a_{n-1}$ by their absolute values and x by R, because thereby every individual term of the sum (6) is replaced by its absolute value or by a larger number. If one chooses ε so small that

$$\varepsilon k < \omega$$

then by (5) the requirement (4) is fulfilled as long as

$$\varrho < \varepsilon$$

remains.

We can give the theorem of the continuity of the function $f(x)$ a somewhat different expression, which is important for what follows. By the theorem that the absolute value of a sum of two terms lies, in magnitude, between the sum and the difference of the absolute values of the terms (cf. Introduction, p. 19), we obtain from (5), if we put

$$\varphi(h) = hf'(x) + \frac{h^2}{1\cdot 2}f''(x) + \cdots + h^n a_0$$

$$|f(x)| - |\varphi(h)| < |f(x+h)| < |f(x)| + |\varphi(h)|,$$

or

$$-|\varphi(h)| < |f(x+h)| - |f(x)| < |\varphi(h)|.$$

Now, if $\varrho < \varepsilon$, then $|\varphi(h)| < \omega$, and we can therefore also say:

VI. The absolute value of the difference

$$|f(x+h)| - |f(x)|$$

is smaller than an arbitrarily given magnitude ω as soon as the absolute value of h is smaller than a sufficiently small ε.

From this we can still conclude, if we put $x = y + iz$ and assume h either real or purely imaginary, that $|f(y+zi)|$ is, for fixed z, a continuous function of y, and for fixed y, a continuous function of z.

VII. If the coefficients $a_0, a_1, \ldots, a_n$ and the variable x are real, then x may be chosen, in absolute value, so large that the sign of $f(x)$ agrees with the sign of the first term $a_0 x^n$; thus, for positive a_0 and even n, it becomes positive, while for odd n it becomes negative for negative x and positive for positive x.

For in

$$f(x) = x^n\left(a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}\right)$$

one can choose x so large in absolute value that the absolute value of

$$\frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}$$

falls below that of a_0, and therefore the expression in parentheses

$$a_0 + \frac{a_1}{x} + \frac{a_2}{x^2} + \cdots + \frac{a_n}{x^n}$$

has the sign of a_0, which it then retains for every absolutely larger x.

§. 32.

Change of Sign of $f(x)$. Roots of Equations of Odd Degree and of Pure Equations.

On the basis of the theorems of the preceding paragraph we can prove the following important theorem.

VIII. If the coefficients of $f(x)$ are real and there exist two real values a, b of x such that $f(a)$ and $f(b)$ have opposite signs, then between a and b there is at least one root ξ of the equation $f(x) = 0$.

We shall assume that $a < b$ and that $f(a)$ is negative, $f(b)$ positive. We divide the numbers between a and b into two parts $\mathfrak{A}$ and $\mathfrak{B}$ as follows: a number α belongs to $\mathfrak{A}$ if, between α and a, including the endpoints, there is no x for which $f(x)$ becomes positive; and a number β belongs to $\mathfrak{B}$ if between β and a there is at least one value of x for which $f(x)$ becomes positive. A third possibility is plainly not possible, and every number between a and b is therefore placed in $\mathfrak{A}$ or in $\mathfrak{B}$. We also count the numbers smaller than a as belonging to $\mathfrak{A}$, and those greater than b as belonging to $\mathfrak{B}$.

If α belongs to $\mathfrak{A}$, then every x smaller than α also belongs to $\mathfrak{A}$; and if β belongs to $\mathfrak{B}$, every larger x belongs to $\mathfrak{B}$. Thus every β is greater than every α, and the two numerical domains $\mathfrak{A}, \mathfrak{B}$ form a cut (Introduction, p. 5), generated by a number ξ, such that every number α smaller than ξ belongs to $\mathfrak{A}$, and every number β greater than ξ belongs to $\mathfrak{B}$.

If now, for some number x_0, the value of $f(x_0)$ is negative or positive, then by V of the preceding paragraph a positive magnitude ε can be so determined that for every x between $x_0 - \varepsilon$ and $x_0 + \varepsilon$ the values of $f(x)$ are always negative or always positive. It follows from this that $f(\xi)$ can be neither negative nor positive and must therefore be zero. For if $f(\xi)$ were negative, then $f(x)$ would also be negative between ξ and $\xi + \varepsilon$; these values would therefore belong to $\mathfrak{A}$, although they are greater than ξ. If $f(\xi)$ were positive, then $f(x)$ would be positive as long as x lay between $\xi - \varepsilon$ and ξ; these values, although smaller than ξ, would therefore belong to $\mathfrak{B}$. Both alternatives are impossible by the definition of

ξ. The same conclusion may be drawn in quite the same way when $f(a)$ is positive and $f(b)$ negative, and so our theorem is proved.

From this we draw two important consequences. If $f(\xi)$ vanishes, then an interval

$$\xi - \varepsilon < x < \xi + \varepsilon$$

can be so determined that $f(x)$ does not vanish in this interval except for $x = \xi$. Then four cases are possible with respect to the signs of $f(x)$ in the two intervals $\xi - \varepsilon < x < \xi$ and $\xi < x < \xi + \varepsilon$, which are put together in the following scheme:

	$\xi - \varepsilon < \xi$	$< \xi + \varepsilon$
1. $f(x)$	$+$	$-$
2. $f(x)$	$-$	$+$
3. $f(x)$	$+$	$+$
4. $f(x)$	$-$	$-$.

If $f^{(\nu)}(x)$ is the first among the derivatives of $f(x)$ which is different from zero for $x = \xi$, then the formula [§ 13, (2)]

$$f(\xi + h) = \frac{h^\nu}{\Pi(\nu)} f^{(\nu)}(\xi) + \cdots ,$$

shows that these four cases are distinguished by the following signs:

1. ν odd $\quad f^{(\nu)}(\xi) < 0$
2. ν odd $\quad f^{(\nu)}(\xi) > 0$
3. ν even $\quad f^{(\nu)}(\xi) > 0$
4. ν even $\quad f^{(\nu)}(\xi) < 0$

We say that in the first case $f(x)$ passes decreasing through zero, and in the second increasing through zero. In case 3, $f(\xi) = 0$ is a minimum, in case 4 a maximum.

In particular, if $f'(\xi)$ is different from zero, so that ξ is a simple root, then the positive or negative sign of $f'(\xi)$ is the characteristic mark for the increase or decrease of the function $f(x)$ as x passes through ξ.

With these aids we still prove two further important theorems.

IX. The equation

$$x^n - a = 0$$

has, when a is a positive number, one and only one positive root.

For if we put

$$f(x) = x^n - a, \tag{1}$$

then by § 31, VII, $f(x)$ is positive for sufficiently large positive x, and negative for $x = 0$; hence there is a positive value of x, denoted by $\sqrt[n]{a}$, for which $f(x)$ vanishes. That there can be only one such value follows from the fact that x^n, as long as x remains positive, increases continually with x.

If n is odd, then also for negative a there exists one and only one negative root of (1), which is proved in the same way.

X. An equation of odd degree with real coefficients always has at least one real root.

For by theorem VII, $f(x)$, if the degree is odd, can become both positive and negative; hence, by VIII, $f(x) = 0$ has a root.

Finally we prove:

XI. A pure equation always has a root.

By a pure equation we understand an equation of the form

$$x^n - a = 0, \tag{2}$$

where a is an arbitrary complex magnitude. That this equation always has a root for real positive a has already been proved above; and for $a = 0$ it is plainly satisfied by $x = 0$.

The general proof can be divided into two parts; it is sufficient to assume first $n = 2$, and then n odd. For if the magnitudes $\sqrt[n]{a}$ and $\sqrt[2]{a}$, or $\sqrt{a}$, exist for every a, then the existence of

$$\sqrt[2n]{a} = \sqrt[2]{\sqrt[n]{a}}, \qquad \sqrt[4n]{a} = \sqrt[2]{\sqrt[2n]{a}} \ldots ,$$

and hence of $\sqrt[n]{a}$ for every arbitrary n, is secured. It would even suffice to prove the existence of $\sqrt[n]{a}$ for the case in which n is a prime number. But from this no particular advantage is at first to be drawn.

We denote the complex magnitude a by $b + ci$ and put, since x will in general also be complex,

$$x = y + iz.$$

Then we first have to show that

$$(y + iz)^2 = b + ic, \tag{3}$$

whatever real values b, c may have, can be satisfied by real y, z. Equation (3), however, is fulfilled if

$$y^2 - z^2 = b, \qquad 2yz = c, \tag{4}$$

and hence

$$(y^2 + z^2)^2 = b^2 + c^2$$

therefore

$$y^2 + z^2 = \sqrt{b^2 + c^2}. \tag{5}$$

By IX there always exists a positive square root $\sqrt{b^2 + c^2}$. From (4) and (5) it now follows that

$$y^2 = \frac{b + \sqrt{b^2 + c^2}}{2}, \qquad z^2 = \frac{-b + \sqrt{b^2 + c^2}}{2},$$

and the two magnitudes $\pm b + \sqrt{b^2 + c^2}$ are plainly positive, since $b^2 + c^2 > b^2$, hence also $\sqrt{b^2 + c^2} > \sqrt{b^2}$. Thus we obtain

$$y = \pm\sqrt{\frac{b + \sqrt{b^2 + c^2}}{2}}, \qquad z = \pm\sqrt{\frac{-b + \sqrt{b^2 + c^2}}{2}},$$

but one sign is determined by the other through the condition

$$2yz = c;$$

one of the two signs remains arbitrary, so that we obtain not merely one, but two opposite solutions of (3).

Further let n be odd; we take the equation to be solved in the form

$$(y + iz)^n = b + ci. \tag{6}$$

If $c = 0$, so that a is real, then we can put $z = 0$ and by IX obtain a real value of x. But if we assume c different from zero, then from (6) (Introduction, p. 19) there follows

$$(y - iz)^n = b - ci. \tag{7}$$

Multiplying the two equations (6), (7) by one another gives

$$(y^2 + z^2)^n = b^2 + c^2,$$

and hence, by IX,

$$y^2 + z^2 = \sqrt[n]{b^2 + c^2}, \tag{8}$$

which determines the absolute value of x. Furthermore, from (6) and (7) one obtains

$$\varphi(y, z) = \frac{(y - iz)^n(b + ic) - (y + iz)^n(b - ic)}{2i} = 0. \tag{9}$$

Now $\varphi(y,z)$ is an integral homogeneous function of the nth degree in the two variables y,z, and indeed with real coefficients, since replacing i by $-i$ changes nothing in $\varphi(y,z)$. If it is ordered by descending powers of y, the first term is cy^n, hence, by assumption, different from zero. Putting

$$y = \lambda z,$$

we obtain from (9)

$$\varphi(\lambda, 1) = 0, \tag{10}$$

therefore an equation of the nth degree for λ, which by X certainly has a root. Once λ is determined, it follows from (8) that

$$z = \sqrt{\frac{\sqrt[n]{b^2+c^2}}{1+\lambda^2}}, \qquad y = \lambda\sqrt{\frac{\sqrt[n]{b^2+c^2}}{1+\lambda^2}}. \tag{11}$$

Through this the equations (8), (9) are in fact satisfied; from (9), however, follows

$$\frac{(y+iz)^n}{b+ic} = \frac{(y-iz)^n}{b-ic},$$

and from (8) that the common value of these two fractions is ± 1. Therefore we have

$$(y+iz)^n = \pm(b+ci), \qquad (y-iz)^n = \pm(b-ci),$$

and thus the sign of the square root in (11) can be so determined that equations (6) and (7) are fulfilled.

§. 33.

Solution of Pure Equations by Trigonometric Functions.

The solvability of a pure equation can be shown far more completely and simply if one presupposes the trigonometric functions and their simplest properties as known. These functions are, it is true, foreign to

algebra proper, and for that reason it is more satisfactory to answer the questions of principle without their aid, as has been done in the preceding and will also be done later. For application and for a more convenient visualization, however, we shall not dispense with this aid.

If, when b, c are real numbers, we put

$$b = r\cos\varphi, \qquad c = r\sin\varphi, \tag{1}$$

then, if r is assumed positive, the angle φ is thereby determined up to a multiple of 2π. It will be completely determined if we fix an interval of length 2π in which it is to lie, for example

$$-\pi < \varphi \leq \pi.$$

In geometrical interpretation r, φ are the polar coordinates of the point whose rectangular coordinates are b, c. The absolute value of the complex magnitude $a = b + ci$ is r, and we shall call φ the phase of this complex magnitude.

The product of two complex magnitudes

$$\begin{aligned} a &= r(\cos\varphi + i\sin\varphi), \\ a' &= r'(\cos\varphi' + i\sin\varphi'), \end{aligned} \tag{2}$$

is obtained in the form

$$aa' = rr'[\cos(\varphi+\varphi') + i\sin(\varphi+\varphi')], \tag{3}$$

and from this, if n is an arbitrary positive whole number, or even a negative one, one obtains the theorem named after Moivre,

$$(\cos\varphi + i\sin\varphi)^n = \cos n\varphi + i\sin n\varphi, \tag{4}$$

which has led to regarding the magnitude $\cos\varphi + i\sin\varphi$ as an exponential function with imaginary exponent, and hence to setting

$$\cos\varphi + i\sin\varphi = e^{i\varphi}$$

Thereby formulas (3) and (4) take the form

$$e^{i\varphi}e^{i\varphi'} = e^{i(\varphi+\varphi')}, \qquad (e^{i\varphi})^n = e^{in\varphi}. \tag{5}$$

If accordingly we put

$$a = r(\cos\varphi + i\sin\varphi) \tag{6}$$

and by $\sqrt[n]{r}$ understand the positive value existing and completely determined by § 32, IX, whose nth power $= r$, then

$$x = \sqrt[n]{r}\left(\cos\frac{\varphi}{n} + i\sin\frac{\varphi}{n}\right) \tag{7}$$

is a magnitude whose nth power is $= a$, hence a root of the equation

$$x^n = a, \tag{8}$$

where n is understood to be an arbitrary positive whole number. But every one of the values

$$x_k = \sqrt[n]{r}\left[\cos\left(\frac{\varphi + 2k\pi}{n}\right) + i\sin\left(\frac{\varphi + 2k\pi}{n}\right)\right], \tag{9}$$

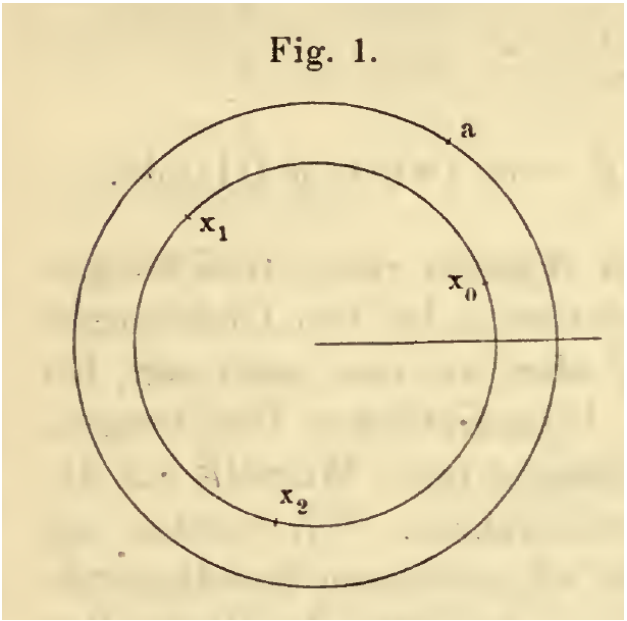
satisfies the same requirement, when k is an arbitrary integer. In (9), however, not more than n different values are contained, namely those obtained by putting

$$k = 0, 1, 2, \ldots, n-1; \tag{10}$$

for if k is increased by a multiple of n, the value of (9) is not changed, while the n values (10) give only different values of x, because they have different phases.

Equation (8) consequently has not merely one, but n different roots. The geometrical images of the values x_k all lie on a circle with radius $\sqrt[n]{r}$, and are separated from one another by the angle $\frac{2\pi}{n}$. The first of these points is obtained by dividing the given angle φ into n equal parts. The radius $\sqrt[n]{r}$ is greater or smaller than r according as r is smaller or greater than 1. The adjacent figure shows these relations for $n = 3$ under the assumption $r > 1$.

Fig. 1.



One can obtain the different values x_k by multiplying the first of them,

$$x_0 = \sqrt[n]{r}\left(\cos\frac{\varphi}{n} + i\sin\frac{\varphi}{n}\right),$$

by the different values of

$$\varepsilon_k = \cos\frac{2k\pi}{n} + i\sin\frac{2k\pi}{n}. \tag{11}$$

The quantities ε_k are roots of the equation

$$x^n = 1, \tag{12}$$

and are called the nth roots of unity. There are n, and only n, different values of them; for odd n only one is real, namely for $k = 0$, and for even n two are real, namely for $k = 0$ and $k = \frac{n}{2}$. The geometrical images of the nth roots of unity are the vertices of a regular n-gon inscribed in the circle of radius 1. Their algebraic determination is the subject of the theory of cyclotomy. If we denote by ε the value of ε_k for $k = 1$, thus put

$$\varepsilon = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n},$$

then, by Moivre's theorem,

$$\varepsilon_k = \varepsilon^k,$$

so that all nth roots of unity are represented as powers of one among them.

§. 34.

Removal of the Second Term of an Equation.

The certainty of the existence of the roots of pure equations puts us in a position to prove the existence of roots for equations of the second, third, and fourth degrees, or, as one also says, for quadratic, cubic, and biquadratic equations, by reducing the determination of their roots to the solution of pure equations. We shall return to this question later from various more general points of view, but here we shall briefly discuss the older methods of solution, which, even if they give little insight into the general connection of this question, are nevertheless very simple in application. They arouse the appearance as if one were dealing with a method capable of generalization to higher equations, which is not the case. First the following general remark.

Let us, for the sake of simplicity, take in the function

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n \tag{1}$$

the coefficient of x^n equal to 1, which in general is obtained by division by the coefficient of x^n. Then by the simple substitution

$$x = y - \frac{a_1}{n}, \tag{2}$$

we can transform $f(x)$ into another function $\varphi(y)$ of the nth degree in which the second term y^{n-1} does not occur; for, by the binomial theorem,

$$x^n = y^n - a_1y^{n-1} + \frac{n-1}{2n}a_1^2y^{n-2} + \cdots,$$

$$x^{n-1} = y^{n-1} - \frac{n-1}{n}a_1y^{n-2} + \cdots, \qquad x^{n-2} = y^{n-2} - \cdots,$$

and consequently

$$f(x) = y^n + \left(a_2 - \frac{n-1}{2n}a_1^2\right) y^{n-2} + \cdots .$$

If we actually carry out the transformation for the cases $n = 2, 3, 4$, we obtain

$$\begin{aligned} &n = 2, \quad \varphi(y) &&= y^2 + a, \\ &n = 3, \quad \varphi(y) &&= y^3 + ay + b, \\ &n = 4, \quad \varphi(y) &&= y^4 + ay^2 + by + c, \end{aligned} \tag{3}$$

provided that we put

$$\begin{aligned} \text{for } n = 2: \qquad a &= -\frac{a_1^2}{4} + a_2, \\ \text{for } n = 3: \qquad a &= -\frac{a_1^2}{3} + a_2, \\ b &= \frac{2a_1^3}{27} - \frac{a_1 a_2}{3} + a_3, \\ \text{for } n = 4: \qquad a &= -\frac{3a_1^2}{8} + a_2, \\ b &= \frac{a_1^3}{8} - \frac{a_1 a_2}{2} + a_3, \\ c &= -\frac{3a_1^4}{256} + \frac{a_1^2 a_2}{16} - \frac{a_1 a_3}{4} + a_4. \end{aligned} \tag{4}$$

Thus the solution of the equation $f(x) = 0$ is reduced by (2) to the solution of $\varphi(y) = 0$. This first gives, for $n = 2$,

$$y = \pm\sqrt{-a},$$

and consequently

$$x = -\frac{a_1}{2} \pm \sqrt{\frac{a_1^2}{4} - a_2} = \frac{-a_1 \pm \sqrt{a_1^2 - 4a_2}}{2}.$$

We therefore have two roots

$$\alpha = \frac{-a_1 + \sqrt{a_1^2 - 4a_2}}{2}, \qquad \beta = \frac{-a_1 - \sqrt{a_1^2 - 4a_2}}{2}. \tag{5}$$

§. 35.

Cubic Equations. Cardano's Formula.

We take the equation of the third degree in the reduced form

$$y^3 + ay + b = 0 \tag{1}$$

and introduce two new unknowns u, v by putting

$$y = u + v. \tag{2}$$

Substitution in (1) gives

$$u^3 + v^3 + (3uv + a)(u + v) + b = 0. \tag{3}$$

We determine one of the two quantities u, v by the other according to the equation

$$3uv = -a, \tag{4}$$

and obtain from (3)

$$u^3 + v^3 = -b. \tag{5}$$

But from (4) and (5), u^3 and v^3 can be determined by a square root. Namely one obtains

$$(u^3 - v^3)^2 = (u^3 + v^3)^2 - 4u^3v^3 = b^2 + \frac{4a^3}{27}. \tag{6}$$

Put, for abbreviation,

$$R = \frac{b^2}{4} + \frac{a^3}{27}; \tag{7}$$

then

$$u^3 - v^3 = 2\sqrt{R}.$$

We give $\sqrt{R}$ one of the two signs and, according to (5), obtain

$$\begin{aligned} u^3 &= -\frac{b}{2} + \sqrt{R}, & v^3 &= -\frac{b}{2} - \sqrt{R}, \\ u &= \sqrt[3]{-\frac{b}{2} + \sqrt{R}}, & v &= \sqrt[3]{-\frac{b}{2} - \sqrt{R}}, \end{aligned} \tag{8}$$

and therefore

$$y = \sqrt[3]{-\frac{b}{2} + \sqrt{R}} + \sqrt[3]{-\frac{b}{2} - \sqrt{R}}. \tag{9}$$

Multiplication of the two expressions (8) gives

$$u^3v^3 = -\frac{a^3}{27}$$

and shows that, if for u a determinate one among the three values of the cube root is taken,

$$v = -\frac{a}{3u} = -\frac{a}{3\sqrt[3]{-\frac{b}{2} + \sqrt{R}}} \tag{10}$$

is in any case one of the three values of $\sqrt[3]{-\frac{b}{2} - \sqrt{R}}$. But it follows from (4) that only when we take this value for v is equation (1) satisfied by $y = u + v$. Thus, corresponding to the three values of u, we obtain three roots of equation (1). From the existence of one root of the cubic equation, however, the existence of three already follows also from the fact that it has already been proved that the quadratic equation has two roots.

To derive the other two roots from the one root (9), put

$$\alpha = u + v$$

and divide $y^3 + ay + b$ by $y - \alpha$. The quotient whose roots are the two other roots β, γ of the cubic equation (1) is

$$y^2 + \alpha y + \alpha^2 + a = 0,$$

whence, by (4) of the preceding paragraph, one finds

$$y = \frac{-\alpha \pm \sqrt{-3\alpha^2 - 4a}}{2}.$$

If here one puts $\alpha = u + v$ and, by (4), $a = -3uv$, then it follows that

$$y = \frac{-(u+v) \pm \sqrt{-3(u+v)^2 + 12uv}}{2} = \frac{-(u+v) \pm \sqrt{-3}(u-v)}{2}.$$

We therefore put

$$\varepsilon = \frac{-1 + \sqrt{-3}}{2}, \tag{11}$$

from which

$$\varepsilon^3 = 1, \qquad \varepsilon^2 + \varepsilon + 1 = 0, \qquad \varepsilon^2 = \frac{-1 - \sqrt{-3}}{2} \tag{12}$$

follows. Thus one obtains the three roots of the cubic equation (6)

$$\begin{aligned} \alpha &= u + v, \\ \beta &= \varepsilon u + \varepsilon^2 v, \\ \gamma &= \varepsilon^2 u + \varepsilon v. \end{aligned} \tag{13}$$

Here ε is a third root of unity different from 1, expressed algebraically without the trigonometric functions. If one substitutes εu or $\varepsilon^2 u$ for u, and hence $\varepsilon^2 v$ or εv for v, the three quantities α, β, γ are interchanged with one another.

The expression (9) for the root of a cubic equation is called Cardano's formula.

§. 36.

Cayley's Expression for Cardano's Formula.

Every cube root has, as we have seen, three different values, which one obtains from one of them by multiplication by $1, \varepsilon, \varepsilon^2$. Thus each of the two quantities u, v, as they are defined by (8), § 35, also has three values, and the sum $u + v$, if one takes account only of the conditions (8), has nine different values. Of these, however, only three give solutions of the cubic equation, and only by adjoining the relation (4), § 35,

$$3uv = -a \tag{1}$$

are the usable values singled out from among the nine.

This is a defect of Cardano's formula, which accordingly by itself does not yet suffice to give the roots of the cubic equation unambiguously. Cayley remedied this defect by the following representation. Define two new quantities ξ, η by the equations

$$u = \xi^2 \eta, \qquad v = \xi \eta^2, \tag{2}$$

from which, by multiplication and use of (1), one obtains

$$\xi\eta = \sqrt[3]{-\frac{a}{3}}. \tag{3}$$

Consequently, if this value is inserted in (2) and the expressions § 35, (8) for u, v are substituted, then

$$\xi = \sqrt[3]{\frac{3b}{2a} + \sqrt{\frac{9b^2}{4a^2} + \frac{a}{3}}}, \qquad \eta = \sqrt[3]{\frac{3b}{2a} - \sqrt{\frac{9b^2}{4a^2} + \frac{a}{3}}}. \tag{4}$$

Thus, if one now puts the root y of the cubic equation in the form

$$y = \xi\eta(\xi + \eta), \tag{5}$$

one obtains an expression which, when ξ is replaced, independently, by $\xi, \varepsilon\xi, \varepsilon^2\xi$ and η by $\eta, \varepsilon\eta, \varepsilon^2\eta$, assumes not nine but only the three values[5]

$$\begin{gathered}\xi^2\eta + \xi\eta^2,\\ \varepsilon\xi^2\eta + \varepsilon^2\xi\eta^2,\\ \varepsilon^2\xi^2\eta + \varepsilon\xi\eta^2\end{gathered}$$

§. 37.

The Biquadratic Equation.

A similar path to that used in solving the cubic equation by Cardano's formula may be taken for the solution of the equation of the fourth degree

$$y^4 + ay^2 + by + c = 0. \tag{1}$$

We put

$$2y = u + v + w$$

and obtain, if we write, for abbreviation,

$$s = u^2 + v^2 + w^2, \qquad t = v^2w^2 + w^2u^2 + u^2v^2, \tag{2}$$

then

$$4y^2 = s + 2(vw + wu + uv),$$
$$16y^4 = s^2 + 4s(vw + wu + uv) + 4t + 8uvw(u + v + w).$$

Substitution in (1) gives

$$s^2 + 4t + 4as + 16c + 8(uvw + b)(u + v + w) + 4(s + 2a)(vw + wu + uv) = 0.$$

This equation is satisfied by the assumptions

$$s + 2a = 0, \qquad uvw + b = 0, \qquad s^2 + 4t + 4as + 16c = 0.$$

With the help of the first of these three equations the third becomes

$$t = a^2 - 4c,$$

and, putting back the values (2) for s, t,

$$\begin{aligned} u^2 + v^2 + w^2 &= -2a,\\ v^2w^2 + w^2u^2 + u^2v^2 &= a^2 - 4c,\\ uvw &= -b. \end{aligned} \tag{3}$$

By § 7 these equations are satisfied if and only if u^2, v^2, w^2 are the roots of the cubic equation

$$z^3 + 2az^2 + (a^2 - 4c)z - b^2 = 0, \tag{4}$$

and if the signs of u, v, w are so chosen that the last of equations (3) is satisfied. This last condition still allows four different determinations of signs, so that the four roots of the biquadratic equation are obtained in the following way:

$$\begin{aligned} 2\alpha &= u + v + w,\\ 2\beta &= u - v - w,\\ 2\gamma &= -u + v - w,\\ 2\delta &= -u - v + w. \end{aligned} \tag{5}$$

Thus the solution of the biquadratic equation has been reduced to that of the cubic equation (4).

This equation is called a cubic resolvent of the biquadratic equation.

[5] Cayley, Phil. Mag. vol. XXI, 1861. Collectet mathematical papers vol. V, No. 310.

§. 38.

Proof of the Fundamental Theorem.

We now proceed to the proof of the fundamental theorem of algebra, namely that every equation $f(x) = 0$ has at least one root. We first put forward some general propositions.

1. If S denotes an arbitrary system of real numbers all greater than a certain positive or negative number C (that is, a system which contains no infinitely negative numbers), then there exists a lower bound for the numbers S.

By a lower bound g one is to understand a number which is not undercut by any number of the system S, but which is of such a nature that, if δ is an arbitrarily given positive quantity, then between g and $g + \delta$ (including the endpoints) there always lies at least one number of the system S, however small δ may be.

The proof follows directly from the possibility of a cut $(\mathfrak{A}, \mathfrak{B})$, constructed as follows: one throws a number α into $\mathfrak{A}$ if it is not undercut by any number of the system S, and a number β into $\mathfrak{B}$ if it is undercut by at least one number in S. The number g determined by this cut is not undercut by any number in S, for otherwise there would also be numbers smaller than g, and therefore belonging to $\mathfrak{A}$, which nevertheless are undercut by numbers in S; while, on the other hand, every number β lying even ever so little above g belongs to $\mathfrak{B}$ and is therefore undercut by a number in S. These are precisely the characteristic marks of the lower bound.

2. If S' is a part of S, then the numbers S' also have a lower bound g', and this is either equal to g or greater than g.

For it cannot be smaller than g, since otherwise numbers would occur in S', and therefore also in S, which lie below g.

Now let $f(x)$ be a real function of x which has only finite values in the interval

$$a \leq x \leq b \tag{1}$$

and which is also continuous in this interval. In agreement with § 31, V and VI, this means that

$$f(x \pm h) - f(x) \tag{2}$$

falls, in absolute value, below an arbitrarily prescribed number η throughout the interval whenever the positive h is smaller than a certain number ε. For $x = a$ we take in (2) only the upper sign, and for $x = b$ only the lower sign, so as not to pass outside the interval with $x \pm h$.

For the values of such a function in the interval there is, by 1., a lower bound g, and we now prove the proposition:

3. The function $f(x)$ assumes the value g for some value ξ of the interval, so that the lower bound becomes a minimum of the function $f(x)$ in the interval.

The proof is as follows. By 2., the lower bound of the function values in a part of the interval (1) is either equal to or greater than g.

If now $f(a) = g$, what is to be proved is true. If, however, $f(a) > g$, then by the continuity of $f(x)$ one can give an interval $a \leq x \leq a + h$ in which $f(x) > g$ remains true, and hence the lower bound of $f(x)$ is greater than g.

We now construct in the interval (1) a cut $(\mathfrak{A}, \mathfrak{B})$ of the following kind: we count a value α of the interval (1) as belonging to $\mathfrak{A}$ if the lower bound of $f(x)$ in the interval $a \leq x \leq \alpha$ is greater than g, and a value β as belonging to $\mathfrak{B}$ if the lower bound in the interval $a \leq x \leq \beta$ is equal to g.

It is possible that $\mathfrak{B}$ consists only of the single value b; but then $f(b) = g$ must hold, and our assertion is fulfilled for $x = b$. For if $f(b) > g$, one could choose a quantity g' between $f(b)$ and g, and, by continuity, an interval $b - h \leq x \leq b$ such that in this interval all function values $f(x)$ are

greater than g', so that their lower bound is equal to or greater than g', and hence certainly greater than g. But since $b-h$ belongs to $\mathfrak{A}$, the lower bound is also greater than g in the interval $a \leq x \leq b-h$, and consequently in the whole interval (1), contrary to the assumption.

If therefore $f(b) > g$, the cut $(\mathfrak{A}, \mathfrak{B})$ defines a number ξ in the interior of the interval (1), and we can now show that

$$f(\xi) = g \tag{3}$$

must hold.

If $f(\xi) > g$ and $f(\xi) > g' > g$, then, by the continuity of $f(x)$, we can determine an interval

$$\xi - h \leq x \leq \xi + h$$

in which all function values $f(x)$ are greater than g', and hence their lower bound is greater than g. But since $\xi - h$ belongs to $\mathfrak{A}$, the lower bound of $f(x)$ in the whole interval

$$a \leq x \leq \xi + h$$

is also greater than g, while $\xi + h$ belongs to $\mathfrak{B}$, and this is the contradiction.

Let $f(x)$ now be an integral rational function with arbitrary complex or real coefficients, and let the variable x also be allowed to take complex values.

By § 31, IV, if C is an arbitrary positive value, the positive quantity R can be so chosen that

$$|f(x)| > C \tag{4}$$

whenever

$$|x| \geq R. \tag{5}$$

For visualization we represent the region (R) bounded by the inequality

$$|x| < R$$

for the variable x by a circular disc of radius R in the plane of the variable $x = y + iz$.

If we take C larger than any value of $|f(x)|$ in the interior of the region (R), for example larger than $|f(0)|$, then $|f(x)|$ will certainly become smaller in the interior of (R) than on its boundary. Since now in the whole interior of (R) the function $|f(x)|$, which for abbreviation we shall now denote by X, is not negative, there is a lower bound g for the values of X, and we have to prove the proposition:

4. There exists a value ξ of x in the interior of the region (R) such that

$$|f(\xi)| = g \tag{6}$$

so that here also the lower bound is a minimum.

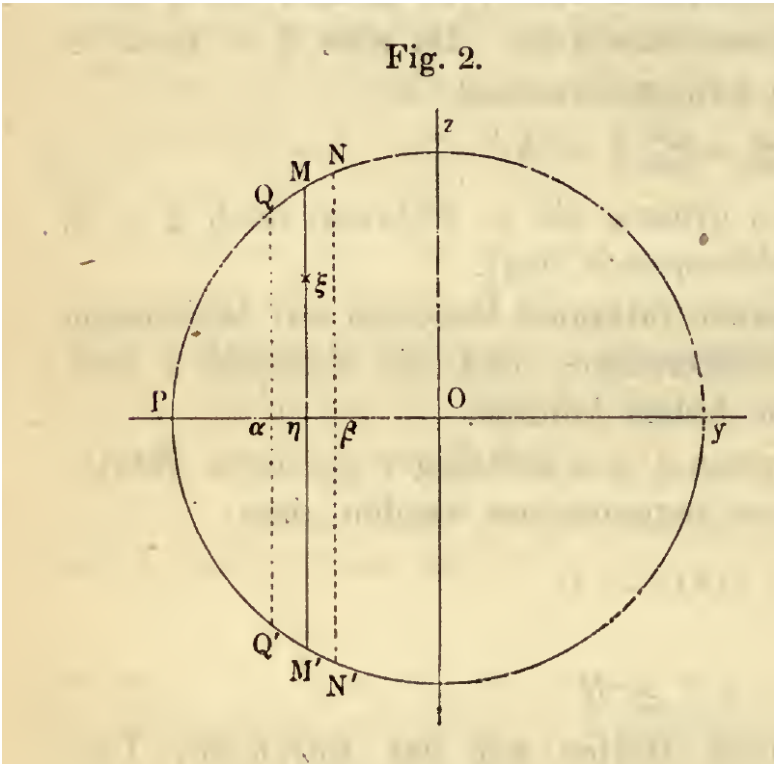
The lower bound of X in any part of the region is either equal to g or greater than g.

A domain of magnitudes determined by the inequalities

$$|x| \leq R, \qquad -R \leq y \leq \alpha \tag{7}$$

is represented in our Figure 2 by the segment $(PQ\alpha Q')$, which we shall call the segment (P, α).

Fig. 2.



We now first determine a value η of y by a cut $(\mathfrak{A}, \mathfrak{B})$ as follows: a number α between $-R$ and $+R$ is admitted into $\mathfrak{A}$ if the lower bound of X in the segment (P, α) is greater than g, and a value β is admitted into $\mathfrak{B}$ if the lower bound of X in the segment (P, β) is equal to g. This cut $(\mathfrak{A}, \mathfrak{B})$ defines a number η with the property that the lower bound of X in the region (P, y) is greater than g if $y < \eta$, and equal to g if $y > \eta$.

Now, by § 31, VI,

$$|f(\eta + iz)|$$

is a continuous function of z in the interval

$$-\sqrt{R^2 - \eta^2} \leq z \leq \sqrt{R^2 - \eta^2}, \tag{8}$$

which is denoted in Figure 2 by M, M'.

Thus by 3. this function receives, for some value ζ of z in this interval, a minimum value γ, so that, if

$$\xi = \eta + i\zeta$$

is put,

$$|f(\xi)| = \gamma. \tag{9}$$

It is now further easy to see that $\gamma = g$ must be the case; for since g is the lower bound of all values of X within (R), γ cannot first of all be smaller than g.

But γ also cannot be greater than g; for all values which X assumes on the segment MM' are $\geq \gamma$. If, however, $\gamma > g$, then one can choose a quantity g' such that $\gamma > g' > g$, and, by the continuity of X proved in § 31, VI, numbers α, β can be so determined that

$$\alpha < \eta < \beta$$

and such that in the whole region $(P, \beta) - (P, \alpha) = (\alpha, \beta)$ ($QNN'Q'$ in Figure 2) X remains greater than g'. Consequently the lower bound of X in (α, β) is greater than g. Since now the lower bound of X in (P, α) is also greater than g, it is also greater than g in (P, β), which is composed of (P, α) and (α, β); but β belongs to $\mathfrak{B}$, and this would give a contradiction with the definition of $\mathfrak{B}$.

Thus there remains only the possibility that $\gamma = g$, and proposition 4 is thereby proved; namely, there is a value ξ in the interior of (R) for which

$$|f(\xi)| = g$$

holds, so that $|f(x)|$ attains a minimum value. We now prove:

5. If α is any value of x for which $f(\alpha)$ is different from zero, then h can be chosen so that

$$|f(\alpha + h)| < |f(\alpha)| \tag{10}$$

results.

It then follows that, if $f(\xi)$ were different from zero, g could not be the minimum of the function $|f(x)|$; but since this has been proved, one must have

$$f(\xi) = 0, \tag{11}$$

and ξ is a root of the equation $f(x) = 0$. The fundamental theorem will therefore be proved.

In the proof of 5. we make use of the theorem already proved (§ 32), that a pure equation always has a root.

Some of the derived functions $f'(\alpha), f''(\alpha), \ldots$ may vanish; the last, $f^{(n)}(\alpha)$, which is equal to $\Pi(n)a_0$, is however

different from zero. Thus suppose that $f'(\alpha), f''(\alpha), \ldots, f^{(m-1)}(\alpha)$ vanish, but $f^{(m)}(\alpha)$ does not vanish. We then have, by § 13,

$$f(\alpha + h) = f(\alpha) + \frac{h^m}{\Pi(m)} f^{(m)}(\alpha) + \frac{h^{m+1}}{\Pi(m+1)} f^{(m+1)}(\alpha) + \cdots . \tag{12}$$

We choose, as is always possible by § 31, V, a positive number ε so that

$$\left| \frac{h}{m+1} \frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)} + \frac{h^2}{(m+1)(m+2)} \frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)} + \cdots \right| < 1 \tag{13}$$

whenever

$$|h| < \varepsilon, \tag{14}$$

and a positive proper fraction δ such that

$$\delta |f(\alpha)| < \left| \frac{\varepsilon^m f^{(m)}(\alpha)}{\Pi(m)} \right|, \tag{15}$$

which, if $f(\alpha)$ does not vanish, is likewise possible. We then determine h (on the basis of § 32) from the equation

$$\frac{h^m f^{(m)}(\alpha)}{\Pi(m)} = -\delta f(\alpha), \tag{16}$$

whence, by (15),

$$|h| < \varepsilon$$

follows. From (12), when the value from (16) is substituted for h^m, we now get

$$f(\alpha + h) = (1 - \delta) f(\alpha) - \delta f(\alpha) \left(\frac{h}{m+1} \frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)} + \frac{h^2}{(m+1)(m+2)} \frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)} + \cdots \right),$$

and hence

$$|f(\alpha+h)| \leq |f(\alpha)|(1-\delta)$$

$$+\delta|f(\alpha)|\left|\frac{h}{m+1}\frac{f^{(m+1)}(\alpha)}{f^{(m)}(\alpha)}+\frac{h^2}{(m+1)(m+2)}\frac{f^{(m+2)}(\alpha)}{f^{(m)}(\alpha)}+\cdots\right|. \tag{17}$$

By (13), therefore,

$$|f(\alpha+h)| < |f(\alpha)|. \tag{18}$$

q.e.d.

Thus the proof of the fundamental theorem is completed.

§. 39.

Algorithm for the Calculation of the Roots.

The proof given in the preceding paragraph for the existence of a root of an algebraic equation, although it leaves nothing to be desired in conclusiveness, nevertheless still has the defect that it does not make clear the steps by which a root can be calculated by a convergent computational procedure. We therefore add the following considerations, intended to remedy this defect, even if only theoretically. They also give a second proof of the fundamental theorem, due in its essentials to Lipschitz (*Lehrbuch der Analysis*, vol. I, § 61 ff.; I owe a simplification to a communication of Dedekind).

If the two integral rational functions $f(x)$ and $f'(x)$ have a common divisor, then this can be found and removed by rational operations according to § 6. We may therefore assume that $f(x)$ and $f'(x)$ have no common divisor, that is, that they do not vanish simultaneously for any value of x.

We now assume that the calculation of the roots has already succeeded for a function of degree $(n-1)$, and accordingly suppose $f'(x)$ resolved into linear factors. Thus, if

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots, \tag{1}$$

then

$$f'(x) = nx^{n-1} + (n-1)a_1x^{n-2} + \cdots = n(x-\beta_1)(x-\beta_2)\cdots(x-\beta_{n-1}), \tag{2}$$

where $\beta_1, \beta_2, \ldots, \beta_{n-1}$ are to be regarded as known numbers, some of which may also be identical. By our assumption, the absolute values

$$|f(\beta_1)|,\ |f(\beta_2)|, \ldots, |f(\beta_{n-1})|, \tag{3}$$

which we denote by

$$b_1, b_2, \ldots, b_{n-1} \tag{4}$$

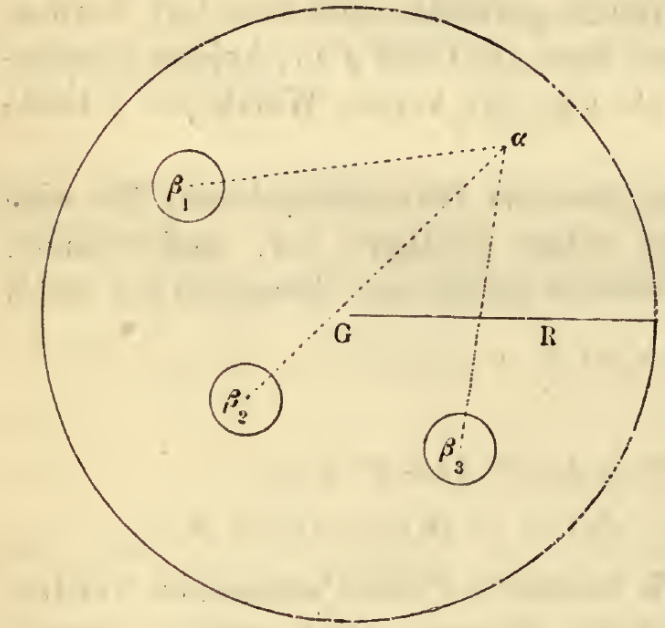
are all different from zero; let the notation be so chosen that b_1 is the smallest among them,

or at least exceeds none of the others. By theorem 5 of § 38 a value α of x can then be determined so that the absolute value a of $f(\alpha)$ is smaller than b_1, hence

$$a < b_1 \leq b_2 \leq b_3 \leq \cdots \leq b_{n-1}. \tag{5}$$

We now bound a domain G for the variable x in such a way that outside this domain the absolute value of $f(x)$ is always greater than a, so that G contains all points x for which $|f(x)| < a$ (but also other points). By § 31, IV this is possible in the following way. About the origin as centre draw a circle (R) of sufficiently large radius R containing the points $\alpha, \beta_1, \beta_2, \ldots, \beta_{n-1}$, and cut out of this circle the points $\beta_1, \beta_2, \ldots, \beta_{n-1}$, at which $|f(x)|$ is certainly greater than a, by circular envelopes of sufficiently small but non-zero radii $\varrho_1, \varrho_2, \ldots, \varrho_{n-1}$, so that within all these circles $(\varrho_1), (\varrho_2), \ldots, (\varrho_{n-1})$ the absolute value $|f(x)|$ remains greater than a (§ 31, V). None of these circles then encloses the point α, for at that point $|f(x)| = a$. The connected piece of surface bounded by the circles $(R), (\varrho_1), \ldots, (\varrho_{n-1})$ is the domain G.

Fig. 3.



We now determine a sequence of numbers

$$\alpha, \alpha', \alpha'', \alpha''', \ldots$$

in such a way that the absolute values of $f(\alpha), f(\alpha'), f(\alpha''), \ldots$, which we denote by

$$a, a', a'', a''', \ldots,$$

always decrease. The points $\alpha, \alpha', \alpha''$ so determined all remain in the interior of the domain G, since for them $|f(x)| < a$.

For this we use a procedure quite similar to that used for the proof of theorem 5 in § 38, but simplified by the fact that $f'(\alpha)$ is already different from zero. If δ denotes a positive proper fraction, still to be determined more precisely, we put in the expansion

$$f(\alpha + h) = f(\alpha) + hf'(\alpha) + \frac{h^2}{1 \cdot 2} f''(\alpha) + \cdots \tag{6}$$

$$h = -\delta \frac{f(\alpha)}{f'(\alpha)}, \tag{7}$$

and obtain

$$f(\alpha + h) = (1 - \delta) f(\alpha) + \delta^2 \left\{ \frac{f(\alpha)^2 f''(\alpha)}{1 \cdot 2\, f'(\alpha)^2} - \delta \frac{f(\alpha)^3 f'''(\alpha)}{1 \cdot 2 \cdot 3\, f'(\alpha)^3} + \cdots \right\}. \tag{8}$$

Now by (2)

$$f'(\alpha) = n(\alpha - \beta_1)(\alpha - \beta_2) \cdots (\alpha - \beta_{n-1});$$

therefore, if the absolute values of $(\alpha - \beta_1), (\alpha - \beta_2), \ldots, (\alpha - \beta_{n-1})$ are denoted by $r_1, r_2, \ldots, r_{n-1}$,

$$|f'(\alpha)| = n r_1 r_2 \cdots r_{n-1}.$$

Since α lies outside the circle ϱ_1 drawn about β_1, we have $r_1 > \varrho_1, r_2 > \varrho_2, \ldots$, and consequently

$$|f'(\alpha)| > n\varrho_1\varrho_2\cdots\varrho_{n-1}.$$

Thus $|f'(\alpha)|$ is greater than a positive number k independent of the position of α in G. It follows that one can choose a sufficiently large positive number Q, likewise independent of the position of α and of the proper fraction δ, so that

$$\left|\frac{f(\alpha)^2 f''(\alpha)}{1\cdot 2\, f'(\alpha)^2} - \delta\frac{f(\alpha)^3 f'''(\alpha)}{1\cdot 2\cdot 3\, f'(\alpha)^3} + \cdots\right| < Q. \tag{9}$$

One can obtain Q by replacing in the denominators on the left of (9) $f'(\alpha)$ by k, replacing δ by 1, replacing all the other terms by their absolute values, replacing finally the absolute value of α by the greater value R, and then making the number so obtained arbitrarily larger. From (8) one then obtains

$$|f(\alpha+h)| < (1-\delta)|f(\alpha)| + \delta^2 Q. \tag{10}$$

Assume now Q so chosen that, for every α in G,

$$2Q > |f(\alpha)| = a$$

which is plainly compatible with the above determination. We may then put

$$\delta = \frac{a}{2Q}, \qquad h = -\frac{af(\alpha)}{2Qf'(\alpha)}$$

and, denoting $\alpha + h$ by α',

$$\alpha' = \alpha - \frac{af(\alpha)}{2Qf'(\alpha)}, \tag{11}$$

and, denoting $|f(\alpha')|$ by a', obtain from (10)

$$a' < a\left(1 - \frac{a}{4Q}\right). \tag{12}$$

By the same procedure we derive from α', a' a second system α'', a'', then from α'', a'' a third system α''', a''', and so on; for the sequence of decreasing positive numbers $a, a', a'', \ldots$ we obtain the inequalities

$$\begin{aligned} a' &< a\left(1 - \frac{a}{4Q}\right), \\ a'' &< a'\left(1 - \frac{a'}{4Q}\right), \\ a''' &< a''\left(1 - \frac{a''}{4Q}\right), \\ &\vdots \end{aligned} \tag{13}$$

We now prove that this sequence approaches the limit zero. If this were not the case, a positive number ω could be given such that all $a^{(\nu)}$ remain above ω; then the differences

$$\left(1 - \frac{a}{4Q}\right), \quad \left(1 - \frac{a'}{4Q}\right), \quad \left(1 - \frac{a''}{4Q}\right)\cdots$$

would all be smaller than the positive proper fraction

$$\Theta = 1 - \frac{\omega}{4Q},$$

and from (13) it would follow that

$$a' < a\Theta, \quad a'' < a'\Theta, \quad a''' < a''\Theta, \quad \ldots, \quad a^{(\nu)} < a^{(\nu-1)}\Theta.$$

Whence, by multiplication,

$$a^{(\nu)} < a\Theta^{\nu}. \tag{14}$$

This is a contradiction, since Θ^{ν} becomes infinitely small as ν increases without bound. The result so obtained is expressed by

$$\lim a^{(\nu)} = 0. \tag{15}$$

If the inequalities (13) are written in the form

$$\begin{aligned} a - a' &> \frac{a^2}{4Q}, \\ a' - a'' &> \frac{a'^2}{4Q}, \\ &\vdots \\ a^{(\nu-1)} - a^{(\nu)} &> \frac{(a^{(\nu-1)})^2}{4Q}, \end{aligned}$$

and added from the $(\mu+1)$-st to the $(\nu+1)$-st, one obtains

$$(a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \cdots + (a^{(\nu)})^2 < 4Q\big(a^{(\mu)} - a^{(\nu+1)}\big),$$

and therefore, when μ and ν both tend to infinity,

$$\lim\big((a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \cdots + (a^{(\nu)})^2\big) = 0. \tag{16}$$

For the sequence of numbers $\alpha, \alpha', \alpha'', \ldots$ the law of formation resulting from (11) is

$$\begin{aligned} \alpha - \alpha' &= \frac{af(\alpha)}{2Qf'(\alpha)}, \\ \alpha' - \alpha'' &= \frac{a'f(\alpha')}{2Qf'(\alpha')}, \\ &\vdots \\ \alpha^{(\nu-1)} - \alpha^{(\nu)} &= \frac{a^{(\nu-1)}f(\alpha^{(\nu-1)})}{2Qf'(\alpha^{(\nu-1)})}. \end{aligned} \tag{17}$$

If an arbitrary number of these equations are added, from the $(\mu+1)$-st to the $(\nu+1)$-st, then

$$\alpha^{(\mu)} - \alpha^{(\nu+1)} = \frac{a^{(\mu)}f(\alpha^{(\mu)})}{2Qf'(\alpha^{(\mu)})} + \cdots + \frac{a^{(\nu)}f(\alpha^{(\nu)})}{2Qf'(\alpha^{(\nu)})}. \tag{18}$$

Since the inequalities

$$|f'(\alpha)| > k$$

are satisfied, it follows for the absolute value of this difference that

$$|\alpha^{(\mu)} - \alpha^{(\nu+1)}| < \frac{(a^{(\mu)})^2 + (a^{(\mu+1)})^2 + \cdots + (a^{(\nu)})^2}{2Qk}, \tag{19}$$

and hence, by (16),

$$\lim|\alpha^{(\mu)} - \alpha^{(\nu+1)}| = 0. \tag{20}$$

This says (according to the concept of a sequence of numbers explained in the Introduction, p. 15) that the numbers $\alpha, \alpha', \alpha'', \ldots$ approach a definite limit, which we shall denote by ξ.

From the continuity of the function $f(x)$ it then follows easily that $f(\xi) = 0$; for if $|f(\xi)| > 0$, then, since the $\alpha^{(\nu)}$ come arbitrarily close to ξ, the absolute values $a^{(\nu)}$ of $f(\alpha^{(\nu)})$ could not sink below every positive value, which is precisely what we have proved for them.

§. 40.

Continuity of the Roots.

We close this section with the proof of the theorem:

The roots of an algebraic equation are continuous functions of the coefficients.

We have first to explain the meaning of this theorem.

Let

$$f(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots \tag{1}$$

be an integral rational function of x of degree n. By what has been proved in the preceding paragraphs, $f(x)$ can be decomposed into n linear factors, some of which may be equal to one another. Combining equal factors and denoting the roots by $\alpha, \beta, \gamma, \ldots$, we put

$$f(x) = (x-\alpha)^a (x-\beta)^b (x-\gamma)^c \cdots , \tag{2}$$

where $a, b, c, \ldots$ are positive whole numbers whose sum is equal to n.

The roots $\alpha, \beta, \gamma, \ldots$ will change with the coefficients $a_1, a_2, a_3, \ldots$; the degree of their multiplicity can also become another one.

We denote the changes of $a_1, a_2, \ldots$ by $\varepsilon_1, \varepsilon_2, \ldots$ and put

$$\varphi(x) = \varepsilon_1 x^{n-1} + \varepsilon_2 x^{n-2} + \cdots , \tag{3}$$

$$f(x) + \varphi(x) = f_1(x). \tag{4}$$

We surround the points $\alpha, \beta, \gamma, \ldots$ by regions of arbitrary smallness, but in such a way that these regions exclude one another; for instance, we enclose the points $\alpha, \beta, \gamma, \ldots$ by circular peripheries with radii $\varrho, \varrho', \varrho'', \ldots$, and denote these regions by $(\varrho), (\varrho'), (\varrho''), \ldots$.

If the absolute values of $\varepsilon_1, \varepsilon_2, \ldots$ lie below sufficiently small values, then we can prove first of the function $f_1(x)$

that it has no roots outside the regions $(\varrho), (\varrho'), (\varrho''), \ldots,$

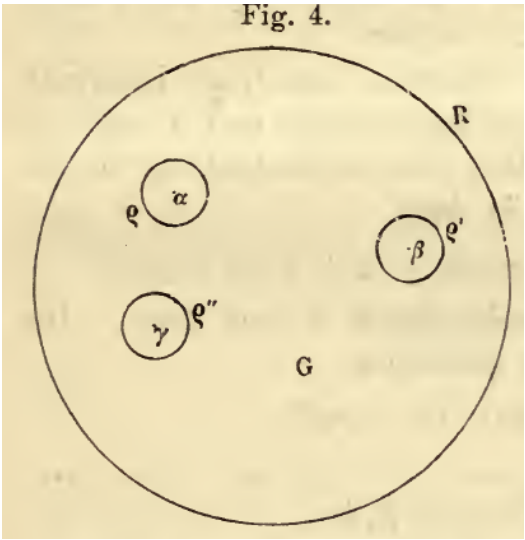
and secondly,

that the number of roots of $f_1(x)$ within (ϱ) is exactly a, within (ϱ') exactly b, within (ϱ'') exactly c, and so on.

In the last part of the theorem, however, it must be observed that, if $f_1(x)$ has multiple roots, these must be counted according to their multiplicity.

We construct, according to §31, IV, in the x-plane a circle with radius R and with the origin as centre, which encloses the regions $(\varrho), (\varrho'), (\varrho''), \ldots$, so that outside this circle no more roots of $f_1(x)$ lie, and we denote by G the region lying within this circle, but outside $(\varrho), (\varrho'), (\varrho''), \ldots$. It must also be remarked that R can be assumed independently of $\varepsilon_1, \varepsilon_2, \ldots$, so long as a definite upper bound is fixed for the absolute values of these quantities.

Fig. 4.



If now x is a point of the region G, then the absolute value of $x - \alpha$ is greater than ϱ, and consequently, by (2),

$$|f(x)| > \varrho^a \varrho'^b \varrho''^c \cdots . \tag{5}$$

If now α_1 is a root of $f_1(x)$, then from (4) it follows that

$$f(\alpha_1) = -\varphi(\alpha_1), \tag{6}$$

and α_1 therefore cannot lie in the region G if one takes care that everywhere within G

$$|\varphi(x)| < \varrho^a \varrho'^b \varrho''^c \cdots , \tag{7}$$

which is always possible by sufficient diminution of the upper bound of $\varepsilon_1, \varepsilon_2, \ldots$. Thus the first part of our assertion is proved, namely that, when the inequality (7) is satisfied, no roots of $f_1(x)$ lie outside the regions $(\varrho), (\varrho'), (\varrho''), \ldots$.

To prove the second part, we put

$$f(x) = \psi(x)(x - \alpha)^a, \tag{8}$$

$$\psi(x) = (x - \beta)^b (x - \gamma)^c \cdots . \tag{9}$$

We now choose a positive number A, which shall however be independent of ϱ, in such a way that, as long as x lies in the interior or on the periphery of the region (ϱ),

$$|\psi(x)| > A \qquad (\text{for } |x - \alpha| \leq \varrho) \tag{10}$$

remains true. Such a number is obtained, for example, if l is taken equal to half the least of the distances $(\alpha, \beta), (\alpha, \gamma), \ldots$ and one puts

$$A = l^{b+c+\cdots};$$

then the requirement expressed in (10) is fulfilled at least as long as ϱ is smaller than l. If only one point α is present, then $\psi(x) = 1$ is to be put, and for A any proper fraction can be chosen.

Let now $\alpha_1, \alpha_2, \ldots$ be the roots of $f_1(x)$ within (ϱ), and let $a_1, a_2, \ldots$ be the degrees of their multiplicity; and let $\beta_1, \beta_2, \ldots$, $b_1, b_2, \ldots$, $\gamma_1, \gamma_2, \ldots$, $c_1, c_2, \ldots$ have the same meaning for the regions $(\varrho'), (\varrho''), \ldots$. Then

$$n = a_1 + a_2 + \cdots + b_1 + b_2 + \cdots + c_1 + c_2 + \cdots , \tag{11}$$

since the total number of all roots must be equal to n. The function $f_1(x)$ can then be represented as

$$f_1(x) = \psi_1(x)(x-\alpha_1)^{a_1}(x-\alpha_2)^{a_2}\cdots, \tag{12}$$

where

$$\psi_1(x) = (x-\beta_1)^{b_1}(x-\beta_2)^{b_2}\cdots(x-\gamma_1)^{c_1}(x-\gamma_2)^{c_2}\cdots. \tag{13}$$

Now we can determine a positive number B independent of $\varrho, \varrho', \varrho''$ such that, as long as x remains within or on the boundary of (ϱ),

$$|\psi_1(x)| < B \qquad (\text{for } |x-\alpha| \le \varrho). \tag{14}$$

For example, one can choose a magnitude L which is greater than twice the distance of the point α from any of the points $\beta, \gamma, \ldots$ and in every case greater than 1, and then assume

$$B > L^n.$$

Now from (4) it follows that

$$|f(x)| < |f_1(x)| + |\varphi(x)|, \tag{15}$$

hence by (8) and (12)

$$|\psi(x)|\,|x-\alpha|^a < |\psi_1(x)|\,|x-\alpha_1|^{a_1}\,|x-\alpha_2|^{a_2}\cdots + |\varphi(x)|. \tag{16}$$

If we now assume x on the periphery of ϱ, and therefore put

$$|x-\alpha| = \varrho,$$

then

$$|x-\alpha_1| < 2\varrho, \qquad |x-\alpha_2| < 2\varrho, \ldots,$$

and consequently, by (10) and (16),

$$\varrho^a A < (2\varrho)^{a_1+a_2+\cdots} B + |\varphi(x)|. \tag{17}$$

We now choose the upper bounds for the coefficients $\varepsilon_1, \varepsilon_2, \ldots$ of φ so small that

$$|\varphi(x)| < \varrho^a A'$$

where A' denotes a number which is smaller than A; then from (17) it follows that[E1]

$$A - A' < 2^a \varrho^{-a+a_1+a_2+\cdots} B. \tag{18}$$

But, for sufficiently small ϱ, this would no longer be possible if a were greater than the sum $a_1 + a_2 + \cdots$. Hence

$$a \le a_1 + a_2 + \cdots. \tag{19}$$

In the same way one proves that

$$b \le b_1 + b_2 + \cdots, \qquad c \le c_1 + c_2 + \cdots, \qquad \ldots. \tag{20}$$

[E1] [Source note:] Set $s = a_1 + a_2 + \cdots$. From (17) one obtains $A - A' < 2^s \varrho^{s-a} B$. The contradiction concerns $s > a$, not $a > s$: for $0 < \varrho < 1$, one has $2^s \varrho^{s-a} B \le 2^n B \varrho \to 0$. Thus the signs in (19), (20) should be $\ge$; summing still gives equality. The source reference (20), (21) means (19), (20). Printed text and formulas are retained.

But since the sums of the left sides as well as of the right sides in the inequalities (20), (21) must be equal to n, only equality signs can hold, and therefore

$$a = a_1 + a_2 + \cdots, \quad b = b_1 + b_2 + \cdots, \quad c = c_1 + c_2 + \cdots, \quad \ldots,$$

whereby the second part of our theorem is also proved.

We shall add to the theorem just proved the following remark concerning the case of multiple roots.

The necessary and sufficient condition that $f(x) = 0$ should have multiple roots at all is that $f(x)$ and $f'(x)$ should have a common factor. Therefore, if the algorithm

of the greatest common divisor is applied to these two functions according to the principles developed in the first section, the condition for common factors is obtained by setting the last constant remainder equal to zero, in the form of an equation between the coefficients,

$$F(a_1, a_2, \ldots, a_n) = 0,$$

where F is an integral rational function of the arguments $a_1, a_2, \ldots, a_n$, which is called the discriminant of f, and whose properties and mode of formation will be treated more fully in the next section. In any case, since equations of degree n without multiple roots do exist at all, F cannot vanish identically.

Consider now

$$F(a_1 + \varepsilon_1, a_2 + \varepsilon_2, \ldots, a_n + \varepsilon_n).$$

This function too, if $a_1, a_2, \ldots, a_n$ are definite and $\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$ are indeterminate numbers, will not vanish identically; and one can choose the ε's so that its value comes out different from zero, even when any arbitrary upper bound has been fixed for the absolute values of the ε's.[1]

One therefore sees from these considerations how an a-fold root α of $f(x)$, under continuous variation of the coefficients, radiates apart into a simple roots, so that from this point of view too one is justified in considering the a-fold root as having arisen from the coincidence of a simple roots.

If we do not assume the coefficient of the highest power of x equal to 1, and therefore examine the equation in the form

$$a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n = 0, \tag{21}$$

then, if $a_n = 0$, one or more of the roots are equal to zero, and equation (21) can be divided by x or by a higher power of x, whereby an equation of correspondingly lower degree arises in which the last term is different from zero. We shall therefore assume from the outset that a_n is different from zero, and now put in (21)

$$x = \frac{1}{y}. \tag{22}$$

By multiplication by y^n and division by a_n, (21) then passes into

$$y^n + \frac{a_{n-1}}{a_n} y^{n-1} + \cdots + \frac{a_1}{a_n} y + \frac{a_0}{a_n} = 0, \tag{23}$$

and we can apply to the roots of this equation the theorem just stated, by assuming $a_0 = 0$, so that one of the roots of (23) vanishes. We thus obtain the theorem:

One can take a_0 in the equation (21) so small that one of its roots becomes large beyond all bounds, while the others differ arbitrarily little from the roots of the equation of degree $(n-1)$

$$a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n = 0.$$

[1] This will only later be rigorously justified. For the time being we presuppose it in these preliminary remarks, on which no further conclusions will be founded.

This is also expressed by saying that, as a_0 vanishes, one of the roots of (21) becomes infinite.

If a_1 also vanishes, or a_1 and a_2, and so on, two or three roots become infinite.

This justifies the expression sometimes used, that an equation of degree n passes, by the becoming-infinite of one root, into an equation of degree $(n-1)$.

Fourth Section.
Symmetric Functions.

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§. 41.
Concept of Symmetric Functions. Symmetric Fundamental Functions.

In this section we consider whole functions, of arbitrary degree, in an arbitrary number n of variables

$$\alpha_1, \alpha_2, \ldots, \alpha_n.$$

Such a function is called symmetric if it remains unchanged when the variables $\alpha_1, \alpha_2, \ldots, \alpha_n$ are subjected to an arbitrary permutation; and it is such symmetric functions whose properties and laws of formation we now have to learn more precisely.

In order that a function $\Phi(\alpha_1, \alpha_2, \ldots, \alpha_n)$ be symmetric, it is sufficient that it not be changed when any two of the arguments $\alpha_1, \alpha_2, \ldots, \alpha_n$ are interchanged, since by § 18 all permutations can be formed by a succession of transpositions.

The function Φ will in general not be homogeneous, but will contain terms of different dimensions. If, however, one collects all terms of the same dimension, every Φ can be represented as a sum of homogeneous functions of different degrees; and, if Φ is to be symmetric, then each homogeneous constituent of a definite degree must by itself be symmetric, since the dimensions of the terms are not changed by the permutations of the variables.

Accordingly we may confine ourselves to the consideration of homogeneous symmetric functions, from which all the others can be put together.

After this it is easy to give the general form of a symmetric function. We obtain it if, in one term of such a function,

$$\alpha_1^{\nu_1}\alpha_2^{\nu_2}\cdots\alpha_n^{\nu_n},$$

as in the formation of determinants, we permute the lower indices in all possible ways and take the sum of all terms so formed. Such a function may be called an elementary constituent of a symmetric function. If we take several such elementary constituents, multiply them by arbitrary factors independent of the α's, and add them, then we obtain the most general symmetric function. The number of terms of one of these elementary constituents is $\Pi(n)$ when the exponents $\nu_1, \nu_2, \ldots, \nu_n$ are all different; but if one and the same exponent ν occurs several times, the permutations which give no different terms are to be omitted. For example, for three variables, if ν_1, ν_2, ν_3 are different, an elementary constituent is

$$\begin{aligned}\alpha_1^{\nu_1}\alpha_2^{\nu_2}\alpha_3^{\nu_3} + \alpha_1^{\nu_1}\alpha_3^{\nu_2}\alpha_2^{\nu_3} + \alpha_2^{\nu_1}\alpha_1^{\nu_2}\alpha_3^{\nu_3} + \alpha_2^{\nu_1}\alpha_3^{\nu_2}\alpha_1^{\nu_3} \\ + \alpha_3^{\nu_1}\alpha_1^{\nu_2}\alpha_2^{\nu_3} + \alpha_3^{\nu_1}\alpha_2^{\nu_2}\alpha_1^{\nu_3},\end{aligned}$$

whereas, if $\nu_3 = \nu_2$,

$$\alpha_1^{\nu_1}\alpha_2^{\nu_2}\alpha_3^{\nu_2} + \alpha_2^{\nu_1}\alpha_1^{\nu_2}\alpha_3^{\nu_2} + \alpha_3^{\nu_1}\alpha_1^{\nu_2}\alpha_2^{\nu_2}.$$

The simplest example of a symmetric function is the sum of the variables

$$\alpha_1 + \alpha_2 + \alpha_3 + \cdots + \alpha_n.$$

Likewise the product $\alpha_1\alpha_2\cdots\alpha_n$ belongs here.

These two are the extreme cases of a series of symmetric functions which we call the symmetric fundamental functions and obtain as follows.

The product

$$f(x) = (x-\alpha_1)(x-\alpha_2)\cdots(x-\alpha_n) \tag{1}$$

is, whatever x may be, provided only that x is independent of the α's, a symmetric function of $\alpha_1, \alpha_2, \ldots, \alpha_n$. If therefore we carry out the multiplication of the several factors and

arrange according to powers of x, then the coefficients of the separate powers of x are likewise symmetric functions. For they change no more under interchange of the α's than does the function $f(x)$ (§ 1). We put

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n. \tag{2}$$

The functions $a_1, a_2, \ldots, a_n$ are the coefficients of the algebraic equation whose roots are $\alpha_1, \alpha_2, \ldots, \alpha_n$. They are called the symmetric fundamental functions. We already formed these symmetric fundamental functions in § 7; as we recall, a_1 is the sum of the α's taken with negative sign, a_2 the sum of the products two at a time, a_3 the sum of the products three at a time taken with negative sign, and so on; finally a_n is the product of all the α's with positive or negative sign according as x is even or odd. We write, for short,

$$\begin{aligned} a_1 &= -\sum \alpha_1, \\ a_2 &= +\sum \alpha_1\alpha_2, \\ a_3 &= -\sum \alpha_1\alpha_2\alpha_3, \\ &\vdots \\ a_n &= \pm\alpha_1\alpha_2\cdots\alpha_n. \end{aligned} \tag{3}$$

The goal of our considerations is the proof of the principal theorem that all symmetric functions of the α's can be expressed rationally by the fundamental functions.

§. 42.

The Power Sums.

We occupy ourselves first with another special kind of symmetric functions, the power sums. Namely, if ν denotes any positive integral exponent, then

$$s_\nu = \alpha_1^\nu + \alpha_2^\nu + \cdots + \alpha_n^\nu \tag{1}$$

obviously belongs to the symmetric functions, and s_ν is called the νth power sum. We shall derive formulae according to which the power sums are expressible by the fundamental functions.

In what follows, whenever $\varphi(x)$ is any function of x, we always denote by

$$S[\varphi(\alpha)]$$

the sum obtained by replacing x in $\varphi(x)$ successively by each of the variables $\alpha_1, \alpha_2, \ldots, \alpha_n$ and adding the functions so formed, thus

$$S[\varphi(\alpha)] = \varphi(\alpha_1) + \varphi(\alpha_2) + \cdots + \varphi(\alpha_n).$$

Accordingly, for example,

$$S(\alpha^\nu) = s_\nu$$

is the νth power sum.

We now divide $f(x)$, by § 4, by an arbitrary linear function $x - \alpha$ and obtain

$$\frac{f(x)}{x-\alpha} = x^{n-1} + f_1(\alpha)x^{n-2} + f_2(\alpha)x^{n-3} + \cdots + f_{n-1}(\alpha) + \frac{f(\alpha)}{x-\alpha}, \tag{2}$$

where, by § 4, (8),

$$\begin{aligned} f_1(\alpha) &= \alpha + a_1, \\ f_2(\alpha) &= \alpha^2 + a_1\alpha + a_2, \\ f_3(\alpha) &= \alpha^3 + a_1\alpha^2 + a_2\alpha + a_3, \\ &\vdots \\ f_{n-1}(\alpha) &= \alpha^{n-1} + a_1\alpha^{n-2} + a_2\alpha^{n-3} + \cdots + a_{n-1}. \end{aligned} \tag{3}$$

We put in (2), for α, each of the quantities $\alpha_1, \alpha_2, \ldots, \alpha_n$, whereby $f(\alpha) = 0$, and form the sum S. The left hand side then gives, by § 14, (11),

$$\begin{aligned} S\left(\frac{f(x)}{x-\alpha}\right) &= f'(x) \\ &= nx^{n-1} + (n-1)a_1x^{n-2} + (n-2)a_2x^{n-3} + \cdots + a_{n-1}, \end{aligned} \tag{4}$$

whereas the right hand side becomes

$$nx^{n-1} + x^{n-2}S[f_1(\alpha)] + x^{n-3}S[f_2(\alpha)] + \cdots + S[f_{n-1}(\alpha)]. \tag{5}$$

The comparison of coefficients of like powers of x in (4) and (5) gives

$$\begin{aligned} S[f_1(\alpha)] &= (n-1)a_1, \\ S[f_2(\alpha)] &= (n-2)a_2, \\ &\vdots \\ S[f_{n-1}(\alpha)] &= a_{n-1}. \end{aligned} \tag{6}$$

But by (1) and (3)

$$\begin{aligned} S[f_1(\alpha)] &= s_1 + na_1, \\ S[f_2(\alpha)] &= s_2 + a_1s_1 + na_2, \\ &\vdots \\ S[f_{n-1}(\alpha)] &= s_{n-1} + a_1s_{n-2} + a_2s_{n-3} + \cdots + na_{n-1}. \end{aligned}$$

Consequently from (6) one obtains the following system of formulae due to Newton[1]:

$$\begin{aligned} 0 &= s_1 + a_1, \\ 0 &= s_2 + a_1s_1 + 2a_2, \\ 0 &= s_3 + a_1s_2 + a_2s_1 + 3a_3, \\ &\vdots \\ 0 &= s_{n-1} + a_1s_{n-2} + a_2s_{n-3} + \cdots + (n-1)a_{n-1}. \end{aligned} \tag{7}$$

[1] Newton, Arithmetica universalis, edit. 's Gravesande, p. 592.

This system of formulae can, however, be continued still further; for, since

$$S[f(\alpha)] = 0, \qquad S[\alpha f(\alpha)] = 0, \qquad S[\alpha^2 f(\alpha)] = 0, \ldots,$$

it follows that

$$\begin{aligned} 0 &= s_n + a_1 s_{n-1} + a_2 s_{n-2} + \cdots + n a_n, \\ 0 &= s_{n+1} + a_1 s_n + a_2 s_{n-1} + \cdots + a_n s_1, \\ 0 &= s_{n+2} + a_1 s_{n+1} + a_2 s_n + \cdots + a_n s_2, \quad \ldots \end{aligned} \tag{8}$$

By the formulae (7), (8) the problem is now solved of representing, one after another, the functions $s_1, s_2, s_3, \ldots$ up to any height as whole rational functions of the symmetric fundamental functions; for example

$$\begin{aligned} s_1 &= -a_1, \\ s_2 &= +a_1^2 - 2a_2, \\ s_3 &= -a_1^3 + 3a_1 a_2 - 3a_3, \\ s_4 &= +a_1^4 - 4a_1^2 a_2 + 4a_1 a_3 + 2a_2^2 - 4a_4, \\ &\vdots \end{aligned} \tag{9}$$

The manner of formation of these expressions shows that the coefficients in these representations are whole numbers.

Conversely, by means of the formulae (7) and (8), one can express the symmetric fundamental functions $a_1, a_2, a_3, \ldots$ rationally by the power sums $s_1, s_2, s_3, \ldots$. This representation is, however, far less simple in so far as the coefficients are not whole but fractional numbers, for example

$$\begin{aligned} a_1 &= -s_1, \\ 2a_2 &= +s_1^2 - s_2, \\ 6a_3 &= -s_1^3 + 3s_1 s_2 - 2s_3, \\ &\vdots \end{aligned} \tag{10}$$

One can also continue the system of formulae (8) in the opposite direction, if one forms the equations

$$S[\alpha^{-1} f(\alpha)] = 0, \qquad S[\alpha^{-2} f(\alpha)] = 0, \ldots.$$

In this way one obtains a means of expressing the power sums s_ν also for negative exponents ν by the fundamental functions. These sums of negative powers likewise belong to the symmetric functions, although no longer to the whole, but to the fractional, ones. They pass into whole functions only after multiplication by powers of the product $\alpha_1 \alpha_2 \cdots \alpha_n$. For completeness we put down here the first two of these formulae:

$$\begin{aligned} 0 &= s_{n-1} + a_1 s_{n-2} + a_2 s_{n-3} + \cdots + n a_{n-1} + a_n s_{-1}, \\ 0 &= s_{n-2} + a_1 s_{n-3} + a_2 s_{n-4} + \cdots + a_{n-1} s_{-1} + a_n s_{-2}. \end{aligned} \tag{11}$$

which, by (7), can also be written as

$$a_{n-1} + a_n s_{-1} = 0, \qquad 2a_{n-2} + a_{n-1} s_{-1} + a_n s_{-2} = 0.$$

§. 43.

Proof of the Principal Theorem for Two Variables.

We now pass to the proof of the fundamental theorem of the theory of symmetric functions, that they are all rationally expressible by the symmetric fundamental functions. Since we use complete induction as the means of proof, we first derive the theorem under the assumption that only two independent variables α, β are given, but in a way which at the same time will bring out the guiding thought for the general proof.

We denote the symmetric fundamental functions by

$$a = -(\alpha + \beta), \qquad b = \alpha\beta \tag{1}$$

and accordingly put

$$f(x) = (x - \alpha)(x - \beta) = x^2 + ax + b. \tag{2}$$

Let $S(\alpha, \beta)$ be any whole rational and symmetric function of α and β. We can substitute for β from (1) the value $-(a + \alpha)$, and obtain, if we arrange according to powers of α,

$$S(\alpha, \beta) = S(\alpha, -a - \alpha) = A_0\alpha^m + A_1\alpha^{m-1} + \cdots + A_{m-1}\alpha + A_m, \tag{3}$$

where the coefficients $A_0, A_1, \ldots, A_m$ depend only on a and on the coefficients still occurring in S. We note expressly, however, that if in $S(\alpha, \beta)$ no fractional numerical coefficients occur, then no fractions occur in the coefficients $A_0, A_1, \ldots, A_m$ either.

We now put

$$\Phi(x) = A_0x^m + A_1x^{m-1} + \cdots + A_{m-1}x + A_m \tag{4}$$

and divide $\Phi(x)$ by $f(x)$, by § 3. We obtain a quotient Q and a remainder which, with respect to x, is at most of the first degree; thus

$$\Phi(x) = Qf(x) + A + Bx; \tag{5}$$

Here A and B are now whole functions of a and b, and they contain no fractional numerical coefficients if none occur in S. If now we put $x = \alpha$, it follows from (5) and (3), since $f(\alpha)$ vanishes, that

$$S(\alpha, \beta) = A + B\alpha. \tag{6}$$

But since $S(\alpha, \beta)$ and likewise A, B are symmetric, it follows by interchanging α and β that

$$S(\alpha, \beta) = A + B\beta. \tag{7}$$

From this one concludes, since α and β are independent variables, that $B = 0$, and consequently

$$S(\alpha, \beta) = A, \tag{8}$$

which proves the fundamental theorem in this case.

§. 44.

General Proof of the Principal Theorem.

We now suppose that the fundamental theorem has been proved for symmetric functions of $n - 1$ variables, and we derive it, by a procedure entirely analogous to that applied in § 43, for n variables.

Let again

$$S = S(\alpha_1, \alpha_2, \ldots, \alpha_n) \tag{1}$$

be a whole symmetric function of the n variables $\alpha_1, \alpha_2, \ldots, \alpha_n$. If we arrange it according to powers of α_1, and accordingly put

$$S = S_0\alpha_1^{\mu} + S_1\alpha_1^{\mu-1} + \cdots + S_{\mu-1}\alpha_1 + S_\mu, \tag{2}$$

then the coefficients $S_0, S_1 \ldots S_\mu$ are whole symmetric functions of the $n-1$ variables $\alpha_2, \ldots \alpha_n$.

If we denote the fundamental symmetric functions of these latter variables by $a_1', a_2' \ldots a_{n-1}'$, then, by our supposition, we can express the coefficients $S_0, S_1 \ldots S_\mu$ rationally in terms of these. But by § 42, (2) and (3),

$$\begin{aligned} a_1' &= f_1(\alpha_1) = \alpha_1 + a_1 \\ a_2' &= f_2(\alpha_1) = \alpha_1^2 + a_1\alpha_1 + a_2 \\ a_3' &= f_3(\alpha_1) = \alpha_1^3 + a_1\alpha_1^2 + a_2\alpha_1 + a_3 \\ &\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots, \end{aligned}$$

that is, the coefficients $S_0, S_1 \ldots S_\mu$ can be expressed wholly and rationally in terms of $\alpha_1, a_1, a_2 \ldots a_n$. If, therefore, after introducing these expressions, we arrange (2) anew according to powers of α_1, we obtain

$$S = A_0\alpha_1^m + A_1\alpha_1^{m-1} + \cdots + A_{m-1}\alpha_1 + A_m, \tag{3}$$

where in general m will be an exponent different from μ, and the coefficients $A_0, A_1 \ldots A_m$ depend wholly and rationally on $a_1, a_2 \ldots a_n$. We put again

$$\Phi(x) = A_0x^m + A_1x^{m-1} + \cdots + A_{m-1}x + A_m \tag{4}$$

and divide $\Phi(x)$ by

$$\begin{aligned} f(x) &= x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n \\ &= (x-\alpha_1)(x-\alpha_2)\ldots(x-\alpha_n). \end{aligned}$$

This gives a quotient and a remainder which, with respect to x, is at most of degree $n-1$. We therefore put

$$\Phi(x) = Qf(x) + \psi(x), \tag{5}$$

$$\psi(x) = C_0x^{n-1} + C_1x^{n-2} + \cdots + C_{n-2}x + C_{n-1}, \tag{6}$$

and here $C_0, C_1 \ldots C_{n-1}$ are whole rational functions of $a_1, a_2 \ldots a_n$ in which, if S in its original form contains no fractional coefficients, no fractions occur either.

But now, since $f(\alpha_1)$ vanishes and $S = \Phi(\alpha_1)$, it follows from (5) that

$$S = \psi(\alpha_1) \tag{7}$$

and here, since S is symmetric, α_1 may be replaced by $\alpha_2, \alpha_3 \ldots \alpha_n$. This gives the following n equations:

$$\begin{aligned} C_0\alpha_1^{n-1} + C_1\alpha_1^{n-2} + \cdots + C_{n-2}\alpha_1 + (C_{n-1} - S) &= 0 \\ C_0\alpha_2^{n-1} + C_1\alpha_2^{n-2} + \cdots + C_{n-2}\alpha_2 + (C_{n-1} - S) &= 0 \\ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\

then its determinant,

$$\begin{vmatrix} \alpha_1^{n-1}, & \alpha_1^{n-2} & \dots & \alpha_1, & 1 \\ \alpha_2^{n-1}, & \alpha_2^{n-2} & \dots & \alpha_2, & 1 \\ \dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots\dots \\ \alpha_n^{n-1}, & \alpha_n^{n-2} & \dots & \alpha_n, & 1 \end{vmatrix},$$

which by § 22, formula (12), equals the product of all differences $\alpha_1 - \alpha_2, \alpha_1 - \alpha_3 \ldots$, is different from zero, and consequently, by Theorem II in § 24,

$$C_0 = 0, \quad C_1 = 0 \ldots \quad C_{n-2} = 0$$

and

$$S = C_{n-1}, \tag{9}$$

which contains the fundamental theorem to be proved.

The same result is also obtained by Theorem II, § 30; for since the function of degree $(n-1)$ in x

$$C_0 x^{n-1} + C_1 x^{n-2} + \cdots + C_{n-2} x + C_{n-1} - S$$

vanishes for $x = \alpha_1, \alpha_2 \ldots \alpha_n$, that is, for more than $n-1$ values, all its coefficients must be zero.

The proof which we have given here for the fundamental theorem, following Cauchy,[1] also provides a means of actually computing, in particular cases, the expression of a symmetric function in terms of the fundamental functions. The other known proofs of the theorem offer the same possibility. We shall give one more proof here, which often leads to a simpler method of computation[2].

§. 45.

Second Proof of the Theorem on Symmetric Functions.

Let S be a whole symmetric function of the variables $\alpha_1, \alpha_2 \ldots \alpha_n$. The individual terms of this function are all of the form

$$M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\ldots\alpha_n^{\nu_n},$$

where M is a coefficient independent of the α. These terms are now to be arranged in a definite manner. After the order of the variables $\alpha_1, \alpha_2 \ldots \alpha_n$ has been fixed, of two terms

$$A = M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\ldots\alpha_n^{\nu_n}, \qquad A' = M'\alpha_1^{\nu_1}\alpha_2^{\nu'_2}\ldots\alpha_n^{\nu'_n}$$

A is to be called the higher when the first of the differences[E1]

$$\nu_1 - \nu'_1, \nu_2 - \nu'_2 \ldots \nu_n - \nu'_n,$$

that is different from zero has a positive value, that is, when either $\nu_1 > \nu'_1$ or $\nu_1 = \nu'_1, \nu_2 > \nu'_2$ or $\nu_1 = \nu'_1, \nu_2 = \nu'_2, \nu_3 > \nu'_3$, and so on.

Since we suppose that all terms in which all the exponents $\nu_1, \nu_2 \ldots \nu_n$ agree have been combined into one term, this decides which of any two terms is higher; and if A is higher than A', and A' higher than A'', then A is also higher than A''.

If now, according to this ordering,

$$A = M\alpha_1^{\nu_1}\alpha_2^{\nu_2}\ldots\alpha_n^{\nu_n}$$

[1] Cauchy, Exercices de mathématiques, 4ème année.

[2] Waring, Meditationes algebraicae. Gauss, Demonstratio nova altera theorematis omnem functionem algebraicam rationalem integram unius variabilis in factores reales primi vel secundi gradus resolvi posse. Werke, vol. III, p. 36.

[E1] [Source note:] The first exponent of the second term is printed as ν_1, without a prime. The following definition requires ν'_1 there; the printed expression is retained unchanged.

is the highest term of our function S, it follows that

$$\nu_1 \geqslant \nu_2$$

must hold; for if $\nu_1 < \nu_2$, the term

$$M\alpha_1^{\nu_2}\alpha_2^{\nu_1}\ldots\alpha_n^{\nu_n},$$

which must also occur in S because of symmetry, would stand higher in the order than A, contrary to the supposition. Likewise, $\nu_2 \geqslant \nu_3$ must hold; for if $\nu_2 < \nu_3$, then

$$M\alpha_1^{\nu_1}\alpha_2^{\nu_3}\alpha_3^{\nu_2}\ldots\alpha_n^{\nu_n}$$

would stand higher than A, and so forth. We therefore conclude that the exponents in A

$$\nu_1, \nu_2, \nu_3 \ldots \nu_n$$

form a decreasing, or at least never increasing, sequence of numbers, or that the differences

$$\nu_1 - \nu_2, \nu_2 - \nu_3, \ldots \nu_{n-1} - \nu_n$$

are all positive or at least not negative.

We conclude secondly that no term of the function S can contain an exponent higher than ν_1; for otherwise a term would also occur in which α_1 had this higher exponent, and this would be of higher order than A, contrary to the supposition. Thus the terms that can occur at all in a symmetric function are, apart from their coefficients, completely determined by the highest term, and the number of possible terms is finite when the highest term is given.

We can use the exponent sequence of its highest term,

$$(\nu_1, \nu_2 \ldots \nu_n)$$

to denote the order of the symmetric function.

Apart from the case in which all exponents are zero, so that the function is a constant, the lowest possible order is

$$(1, 0, 0 \ldots 0),$$

to which the single symmetric function

$$M(\alpha_1 + \alpha_2 + \cdots + \alpha_n) = -Ma_1$$

corresponds.[E2] The next higher order is

$$(1, 1, 0 \ldots 0),$$

to which corresponds the symmetric function

$$Ma_1 + M'a_2$$

and thus in the first n orders we obtain the fundamental functions themselves and their linear combinations, and no others.

The next order is characterized by

$$(2, 0, 0 \ldots 0)$$

and contains the linear combinations of the fundamental functions with the sum of the squares.

In all these cases the representability of the symmetric functions has already been proved, and we shall therefore now give the general proof by deriving the representation

[E2] [Source note:] These examples leave an arbitrary constant summand unstated; adding such a summand does not change the stated order. The formulas are retained as printed.

of the function S of order $(\nu_1,\ \nu_2 \ldots \nu_n)$ by the fundamental functions under the assumption that it is already known for the functions of lower order, that is, by the method of inference of complete induction.

We remark on this that by the multiplication of two symmetric functions S and S', whose highest terms are

$$A = M\,\alpha_1^{\nu_1}\,\alpha_2^{\nu_2} \ldots \alpha_n^{\nu_n}, \qquad A' = M'\,\alpha_1^{\nu'_1}\alpha_2^{\nu'_2} \ldots \alpha_n^{\nu'_n}$$

a symmetric function arises whose highest term is the product

$$A\,A' = M\,M'\,\alpha_1^{\nu_1+\nu'_1}\,\alpha_2^{\nu_2+\nu'_2} \ldots \alpha_n^{\nu_n+\nu'_n}$$

For suppose there were in $S\,S'$ a higher term than $A\,A'$, arising from the multiplication of the two terms

$$B = N\,\alpha_1^{\mu_1}\,\alpha_2^{\mu_2} \ldots \alpha_n^{\mu_n}, \qquad B' = N'\,\alpha_1^{\mu'_1}\alpha_2^{\mu'_2} \ldots \alpha_n^{\mu'_n}$$

so that

$$B\,B' = N\,N'\,\alpha_1^{\mu_1+\mu'_1}\,\alpha_2^{\mu_2+\mu'_2} \ldots \alpha_n^{\mu_n+\mu'_n},$$

and suppose that we do not have simultaneously $B = A$ and $B' = A'$; then the first of the differences

$$\mu_1 - \nu_1 + \mu'_1 - \nu'_1, \quad \mu_2 - \nu_2 + \mu'_2 - \nu'_2, \ldots \mu_n - \nu_n + \mu'_n - \nu'_n,$$

that is different from zero would be positive, which is impossible, since the first non-vanishing one among the differences

$$\mu_1 - \nu_1, \quad \mu_2 - \nu_2 \ldots \mu_n - \nu_n$$

and among the differences

$$\mu'_1 - \nu'_1, \quad \mu'_2 - \nu'_2 \ldots \mu'_n - \nu'_n$$

is by hypothesis negative. By repeating this argument we obtain the more general theorem, that the highest term of a product of several factors is obtained by multiplying the highest terms of the individual factors.

The highest terms of the fundamental symmetric functions

$$a_1,\ a_2,\ a_3 \ldots a_n$$

are now, apart from sign,

$$\alpha_1, \quad \alpha_1\,\alpha_2, \quad \alpha_1\,\alpha_2\,\alpha_3, \ldots, \alpha_1\,\alpha_2\,\alpha_3 \ldots \alpha_n,$$

and if we therefore form the product

$$P = \pm\,M\,a_1^{\nu_1-\nu_2}\,a_2^{\nu_2-\nu_3} \ldots a_{n-1}^{\nu_{n-1}-\nu_n}\,a_n^{\nu_n},$$

we obtain a symmetric function whose highest term is likewise A, as in S.

The difference

$$S - P = S'$$

is therefore again a symmetric function whose highest term is lower than that of S, and which by hypothesis we can represent rationally by the fundamental functions. Since P is represented in the same way, the goal is attained. It is also evident here once again that no fractions are introduced by the representation by means of the fundamental functions, if originally none are contained in S.

This method of computing symmetric functions at the same time gives us information about the degree of the integral rational function of $a_1,\ a_2 \ldots a_n$ by which a given symmetric function of the $\alpha_1,\ \alpha_2 \ldots \alpha_n$ is expressed. Namely, if $(\nu_1,\ \nu_2 \ldots \nu_n)$ is the order of the symmetric function (measured by the sequence of exponents of the first term), then ν_1 is the degree of P with respect to all of $a_1, a_2 \ldots a_n$, and in the developed expression for S no term of higher degree can occur; nor

can the term P be destroyed by any of the following terms, since P is completely determined by the order of S, and the order of S' and of all the following symmetric functions is lower than the order of S.

If, then, in place of $a_1, a_2 \ldots a_n$ we put:

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0} \cdots \frac{a_n}{a_0},$$

then $a_0^{\nu_1} S$ is an integral homogeneous function of degree ν_1 in $a_0,\ a_1 \ldots a_n$, and one cannot turn S into an integral function of the a by multiplication with a lower power of a_0.

§. 46.
Discriminants.

The theorems obtained in the foregoing teach us to express symmetric functions of the roots of an equation rationally by the coefficients of the equation; the results are at first derived under the assumption that the roots, or, what by the third section amounts to the same thing,

the coefficients were independent variables; but they cannot become false when definite numerical values are put in their place, since, at least for the integral symmetric functions, no denominators occur that could vanish.

We first make some applications to particular symmetric functions which are important for what follows. Already in §. 19 it was remarked that the difference product

$$\begin{aligned} P = (\alpha_1 - \alpha_2)\,(\alpha_1 - \alpha_3) \ldots (\alpha_1 - \alpha_n) \\ (\alpha_2 - \alpha_3) \ldots (\alpha_2 - \alpha_n) \\ \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \quad \cdot \\ (\alpha_{n-1} - \alpha_n) \end{aligned} \tag{1}$$

merely changes its sign under an interchange of two elements. The square of this product will therefore remain unchanged under all permutations of two of the variables α, and hence also under all permutations; that is, it will be a symmetric function of the variables. The order of this symmetric function is, as one sees from (1), equal to

$$(2\,n - 2, \quad 2\,n - 4 \ldots 2,\ 0).$$

If, then, we assume the function $f(x)$, whose roots are the $\alpha_1, \alpha_2 \ldots \alpha_n$, in the form

$$f(x) = a_0\,x^n + a_1\,x^{n-1} + a_2\,x^{n-2} + \cdots + a_{n-1}\,x + a_n,$$

so that the highest power of x does not necessarily have the coefficient 1, then P^2 is an integral rational function of

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0} \cdots \frac{a_n}{a_0}, \tag{2}$$

which, on multiplication by a_0^{2n-2}, goes over into a homogeneous function of $a_0, a_1, \ldots a_n$. We call this homogeneous function, which is thus defined by

$$a_0^{2\,n-2}\,P^2 = D \tag{3}$$

the discriminant of the function $f(x)$. If we set $a_0 = 1$, we obtain the inhomogeneous form of the discriminant[6].

The vanishing of the discriminant is the necessary and sufficient condition that two of the values $\alpha_1,\ \alpha_2 \ldots \alpha_n$ become equal to one another.

We can set up yet a second expression for the discriminant.

By §. 29, (4) we have

$$\begin{aligned} f'(\alpha_1) &= a_0\,(\alpha_1-\alpha_2)\,(\alpha_1-\alpha_3)\ldots(\alpha_1-\alpha_n)\\ f'(\alpha_2) &= a_0\,(\alpha_2-\alpha_1)\,(\alpha_2-\alpha_3)\ldots(\alpha_2-\alpha_n)\\ &\cdots\cdots\cdots\cdots\cdots\\ f'(\alpha_n) &= a_0\,(\alpha_n-\alpha_1)\,(\alpha_n-\alpha_2)\ldots(\alpha_n-\alpha_{n-1}), \end{aligned}$$

and consequently, since here every factor $(\alpha_h - \alpha_k)$ occurs twice with opposite signs,

$$D = (-1)^{\frac{n\,(n-1)}{2}}\ a_0^{n-2}\, f'(\alpha_1)\, f'(\alpha_2)\ldots f'(\alpha_n). \tag{5}$$

In order actually to represent the discriminant, we have, first of all a general means. By §. 22 we had

$$(-1)^{\frac{n\,(n-1)}{2}}\ P = \begin{vmatrix} 1,\ \alpha_1,\ \alpha_1^2 \ldots \alpha_1^{n-1}\\ 1,\ \alpha_2,\ \alpha_2^2 \ldots \alpha_2^{n-1}\\ \cdots\cdots\cdots\\ 1,\ \alpha_n,\ \alpha_n^2 \ldots \alpha_n^{n-1} \end{vmatrix}, \tag{6}$$

and if, following §. 27, we represent the square of this determinant, by multiplication along vertical rows, as a new determinant, and set

$$s_m = \alpha_1^m + \alpha_2^m + \cdots \alpha_n^m$$

then

$$D = a_0^{2\,n-2} \begin{vmatrix} s_0, & s_1,\ s_2 & \ldots s_{n-1}\\ s_1, & s_2,\ s_3 & \ldots s_n\\ \cdots & \cdots & \cdots\\ s_{n-1},\ s_n,\ s_{n+1} & \ldots & s_{2\,n-2} \end{vmatrix}. \tag{7}$$

The quantities s_m, however, are the power sums which we have already expressed in §. 42 by the coefficients; only, in place of the $a_1,\ a_2 \ldots a_n$, the quotients (2) are to be put.

Thus we find, e.g., for $n = 2$

$$D = a_0^2\,(s_0\,s_2 - s_1^2) = a_1^2 - 4\,a_0\,a_2 \tag{8}$$

and for $n = 3$

$$D = a_0^4\,(s_0\,s_2\,s_4 - s_0\,s_3^2 + 2\,s_1\,s_2\,s_3 - s_1^2\,s_4 - s_2^3). \tag{9}$$

[6]The discriminant is, apart from the factor $(-1)^{\frac{n\,(n-1)}{2}}\ a_0^{2\,n-2}$, the same as what Gauss calls the determinant of the function (l. c., art. 6).

Herein one is to set, according to §. 42,

$$\begin{aligned} s_0 &= 3, \quad a_0 s_1 = -a_1 \\ a_0^2 s_2 &= a_1^2 - 2a_0 a_2 \\ a_0^3 s_3 &= -a_1^3 + 3a_0 a_1 a_2 - 3a_0^2 a_3 \\ a_0^4 s_4 &= a_1^4 - 4a_0 a_1^2 a_2 + 4a_0^2 a_1 a_3 + 2a_0^2 a_2^2, \end{aligned}$$

whereby one obtains

$$D = a_1^2 a_2^2 + 18 a_0 a_1 a_2 a_3 - 4a_0 a_2^3 - 4a_1^3 a_3 - 27 a_0^2 a_3^2. \tag{10}$$

§. 47.
Discriminants of the Forms of Third and Fourth Order.

For the functions of third and fourth order we have, in the solutions of the equations of these two degrees (§. 35, 37), a simple means of computing the discriminants.

As regards first the equation of third order, the three roots of

$$x^3 + ax + b = 0 \tag{1}$$

were represented in the form:

$$\begin{aligned} \alpha &= u + v \\ \beta &= \varepsilon u + \varepsilon^2 v \\ \gamma &= \varepsilon^2 u + \varepsilon v, \end{aligned} \tag{2}$$

in which

$$\varepsilon = \frac{-1 + \sqrt{-3}}{2}, \quad \varepsilon^2 = \frac{-1 - \sqrt{-3}}{2} \tag{3}$$

are the complex third roots of unity, and u, v are determined by §. 35, (8).

From (3) it follows that

$$\varepsilon - \varepsilon^2 = \sqrt{-3}, \quad 1 - \varepsilon = \varepsilon^2 \sqrt{-3}, \quad 1 - \varepsilon^2 = -\varepsilon \sqrt{-3} \tag{4}$$

and from (2)

$$\begin{aligned} \alpha - \beta &= \sqrt{-3}\,(\varepsilon^2 u - \varepsilon v) \\ \alpha - \gamma &= -\sqrt{-3}\,(\varepsilon u - \varepsilon^2 v) \\ \beta - \gamma &= \sqrt{-3}\,(u - v). \end{aligned}$$

From this, by multiplication,

$$(\alpha - \beta)(\alpha - \gamma)(\beta - \gamma) = 3\sqrt{-3}\,(u^3 - v^3). \tag{5}$$

Squaring, one obtains for the discriminant of the cubic equation (1), by §. 35, (6)

$$D = -(27b^2 + 4a^3). \tag{6}$$

From this one obtains the discriminant of the general function

$$a_0 x^3 + a_1 x^2 + a_2 x + a_3, \tag{7}$$

if for a and b one substitutes the expressions §. 34, (4), in which one replaces a_1, a_2, a_3 by

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \quad \frac{a_3}{a_0}$$

and multiplies [§. 46, (3)] by a_0^4. One thus obtains

$$a = \frac{3a_0a_2 - a_1^2}{3a_0^2},$$
$$b = \frac{2a_1^3 - 9a_0a_1a_2 + 27a_0^2a_3}{27a_0^3},$$

and consequently

$$D = -\frac{1}{27a_0^2}\left\{(2a_1^3 - 9a_0a_1a_2 + 27a_0^2a_3)^2 + 4(3a_0a_2 - a_1^2)^3\right\}. \tag{8}$$

If one works out the square and the cube, one comes back to formula (10) of the preceding paragraph; it is remarkable, however, that formula (8) apparently contains the denominator $27a_0^2$, which cancels out in the computation.

We can proceed in the same way with the biquadratic equations. If α, β, γ, δ are the roots of the biquadratic equation

$$x^4 + ax^2 + bx + c = 0 \tag{9}$$

we obtain, by §. 37, (5)

$$\begin{aligned} \alpha - \beta = v + w, \quad \gamma - \delta = v - w, \\ \alpha - \gamma = w + u, \quad \delta - \beta = w - u, \\ \alpha - \delta = u + v, \quad \beta - \gamma = u - v. \end{aligned} \tag{10}$$

By this the discriminant of equation (9) becomes

$$D = (v^2 - w^2)^2(w^2 - u^2)^2(u^2 - v^2)^2,$$

that is, D is equal to the discriminant of the cubic resolvent §. 37, (4)

$$z^3 + 2az^2 + (a^2 - 4c)z - b^2 = 0, \tag{11}$$

whose roots are u^2, v^2, w^2.

But if one forms this discriminant according to formula (10), §. 46, by

$a_0 = 1, \quad a_1 = 2\,a, \quad a_2 = a^2 - 4\,c, \quad a_3 = -\,b^2$

then it follows that

$$\begin{aligned} D = &-\,4\,a^3b^2 + 144\,acb^2 + 16\,a^4c \\ &- 128\,a^2c^2 + 256\,c^3 - 27\,b^4. \end{aligned} \tag{12}$$

If, on the other hand, one applies formula (8) to form the discriminant of the cubic equation (11), one obtains

$$27\,D = 4\,(a^2 + 12\,c)^3 - (2\,a^3 - 72\,ac + 27\,b^2)^2. \tag{13}$$

From this one derives the discriminant of the general function of the fourth degree

$$f(x) = a_0\,x^4 + a_1\,x^3 + a_2\,x^2 + a_3\,x + a_4 \tag{14}$$

by expressing a, b, c, according to §. 34, (4), by

$$\frac{-3\,a_1^2}{8} + a_2$$

$$\frac{a_1^3}{8} - \frac{a_1 a_2}{2} + a_3$$

$$\frac{-3\,a_1^4}{256} + \frac{a_1^2 a_2}{16} - \frac{a_1 a_3}{4} + a_4$$

then replacing a_1, a_2, a_3, a_4 by

$$\frac{a_1}{a_0}, \quad \frac{a_2}{a_0}, \quad \frac{a_3}{a_0}, \quad \frac{a_4}{a_0}$$

and multiplying by a_0^6.

If, after carrying out this simple calculation, one sets for abbreviation

(15) $$A = a_2^2 - 3\,a_1 a_3 + 12\,a_0 a_4,$$

(16) $$B = 27\,a_1^2 a_4 + 27\,a_0 a_3^2 + 2\,a_2^3 - 72\,a_0 a_2 a_4 - 9\,a_1 a_2 a_3,$$

then one obtains from (13)

(17) $$27\,D = 4\,A^3 - B^2.$$

The quantities A and B, which will occupy us further in the next section, are called the first and second invariant of the biquadratic function f. By (17), then, the discriminant can be expressed rationally through these quantities, though not in such a way that the coefficients become integral; if one carries out the calculation prescribed in (17), in order to represent D through the coefficients a_0, a_1, a_2, a_3 themselves, the factor 27 must still cancel out. We shall not set down here the long expression, whose computation offers no difficulty whatever,

§. 48.
Resultants.

Another application of the theory of symmetric functions, which moreover contains the formation of the discriminant as a special case, is the formation of resultants.

Let

(1) $$f(x) = a_0\,x^n + a_1\,x^{n-1} + a_2\,x^{n-2} + \cdots + a_n,$$

(2) $$\varphi(x) = b_0\,x^m + b_1\,x^{m-1} + b_2\,x^{m-2} + \cdots + b_m$$

be two integral rational functions of degree n and m, and let $\alpha_1, \alpha_2 \ldots \alpha_n$ be the roots of the first, so that the following factorization holds:

(3) $$f(x) = a_0\,(x-\alpha_1)\,(x-\alpha_2)\ldots(x-\alpha_n)$$

Now the product

$$\varphi(\alpha_1)\,\varphi(\alpha_2)\ldots\varphi(\alpha_n)$$

is a symmetric function of the roots of $f(x)$, and can therefore be expressed integrally and rationally in terms of the following quotients:

$$\frac{a_1}{a_0},\quad \frac{a_2}{a_0}\cdots\frac{a_n}{a_0}$$

The order of this symmetric function is $(m, m\ldots m)$ (§. 45), and hence

(4) $$R = a_0^m\,\varphi(\alpha_1)\,\varphi(\alpha_2)\ldots\varphi(\alpha_n)$$

is an integral rational and homogeneous function of order m in $a_0, a_1\ldots a_n$. It is also, as its expression shows, an integral homogeneous function of order n in $b_0, b_1\ldots b_m$. It is called the resultant of the two functions $f(x)$ and $\varphi(x)$. This designation is justified by a second representation.

If we denote the roots of $\varphi(x)$ by $\beta_1, \beta_2\ldots\beta_m$, and thus put

(5) $$\varphi(x) = b_0\,(x-\beta_1)\,(x-\beta_2)\ldots(x-\beta_m),$$

then by (4) we obtain

(6) $$R = a_0^m\,b_0^n \overset{i,k}{\Pi}(\alpha_i-\beta_k),$$

in which the product sign Π extends over all the indices $i = 1, 2\ldots n$, $k = 1, 2\ldots m$,

This expression now shows that R, apart from the factor $(-1)^{mn}$, depends on $\varphi(x)$ in the same way as on $f(x)$, and that we may therefore also set

(7) $$(-1)^{mn}\,R = b_0^n\,f(\beta_1)\,f(\beta_2)\,\ldots\,f(\beta_m).$$

The discriminants are accordingly a special case of the resultants, which one obtains when one takes for $\varphi(x)$ the derivative $f'(x)$.

The vanishing of the resultant of $f(x)$ and $\varphi(x)$ is the necessary and sufficient condition for the two equations

$$f(x) = 0 \quad \text{and} \quad \varphi(x) = 0$$

to have a common root.

The equation that arises by setting the resultant equal to zero is also called the final equation from $f(x) = 0$ and $\varphi(x) = 0$, and the setting up of the final equation the elimination of x.

For the computation of resultants one may apply the algorithm of the greatest common divisor (§. 6). For if $m \leqq n$, one can determine an integral function Q and a like function $\psi(x)$, whose degree is smaller than m, in such a way that

$$f(x) = Q\varphi(x) + \psi(x),$$

and hence, if one puts for x one of the roots β_i,

$$f(\beta_i) = \psi(\beta_i),$$

hence

$$b_0^n \overset{i}{\Pi} f(\beta_i) = b_0^n \overset{i}{\Pi} \psi(\beta_i); \tag{8}$$

if m' is the degree of $\psi(x)$, then

$$b_0^{m'} \overset{i}{\Pi} \psi(\beta_i) = R' \tag{9}$$

is the resultant of $\varphi(x)$ and $\psi(x)$, and therefore

$$R = b_0^{n-m'} R'. \tag{10}$$

The formation of the resultant R is hereby reduced to the formation of R'. One can now divide $\varphi(x)$ again by $\psi(x)$, and thus reduces the formation of R' to the formation of a resultant of functions of ever lower degree, until one finally arrives at a constant, i. e. a function of the coefficients a, b, which must be identical with the resultant sought.

The computation of resultants is usually very laborious. We shall cite only the simplest example, the resultant of two quadratic functions:

$$f(x) = a_0x^2 + a_1x + a_2$$

$$\varphi(x) = b_0x^2 + b_1x + b_2.$$

Here one can compute the product

$$R = a_0^2(b_0\alpha_1^2 + b_1\alpha_1 + b_2)(b_0\alpha_2^2 + b_1\alpha_2 + b_2)$$

directly, and obtains, on setting

$$-a_0(\alpha_1 + \alpha_2) = a_1, \qquad a_0\alpha_1\alpha_2 = a_2$$

the result

$$R = a_0^2b_2^2 + a_2^2b_0^2 - b_0b_1a_1a_2 - a_0a_1b_1b_2 + b_0b_2a_1^2 + a_0a_2b_1^2 - 2a_0b_0a_2b_2, \tag{11}$$

or, what is identical with it,

$$R = (a_0b_2 - a_2b_0)^2 - (a_0b_1 - a_1b_0)(a_1b_2 - a_2b_1). \tag{12}$$

The individual terms of the resultant of the two functions $f(x)$ and $\varphi(x)$ of degree n and m have, apart from a numerical (integer) factor, the form

$$a_0^{\nu_0}a_1^{\nu_1} \dots a_n^{\nu_n} b_0^{\mu_0} b_1^{\mu_1} \dots b_m^{\mu_m}, \tag{13}$$

in which, as follows from the degrees determined above,

$$\begin{aligned} \nu_0 + \nu_1 + \dots + \nu_n &= n \\ \mu_0 + \mu_1 + \dots + \mu_m &= m. \end{aligned} \tag{14}$$

But we can give yet another relation between these exponents.

For since [by (6)] R is a homogeneous function of degree $n\,m$ of the $n+m$ variables α, β, and since further a_0 is of the zeroth, a_1 of the first, a_2 of the second, etc., degree in the α, and in general a_k and b_k are of degree k in these variables, it follows that

$$\begin{aligned}(15)\qquad &\nu_1 + 2\nu_2 + 3\nu_3 + \cdots + n\nu_n \\ &+ \mu_1 + 2\mu_2 + 3\mu_3 + \cdots + m\mu_m = nm,\end{aligned}$$

a relation from which an important conclusion can be drawn, one that we shall discuss somewhat more fully in the following section.

§. 49.
Elimination. Theorem of Bezout.

If in the two functions

$$\begin{aligned}(1)\qquad f(x) &= a_0x^n + a_1x^{n-1} + \cdots + a_n \\ \varphi(x) &= b_0x^m + b_1x^{m-1} + \cdots + b_m\end{aligned}$$

the coefficients a, b are themselves again integral rational functions of a quantity y, then the resultant will also be an integral rational function of y, which we shall denote by $F(y)$. The formation of this equation is the elimination of x from the two equations (1).

The equation

$$(2)\qquad F(y) = 0$$

has as roots all those values of y for which the two equations

$$(3)\qquad f(x) = 0, \qquad \varphi(x) = 0$$

have a common root. If the condition (2) is satisfied, one finds the values of x that satisfy the two equations (3) by seeking the greatest common divisor of $f(x)$ and $\varphi(x)$. This common divisor is of the first degree, and thus determines x rationally, when only one such common root is present; in the other case it will be of correspondingly higher degree, and the determination of x still requires the solution of an equation of higher degree.

If to each root of the resultant there belongs only one common root x, then the number of pairs of values x, y that satisfy the equations (3) is equal to the degree of the resultant with respect to y. Thus if the coefficients a are of degree ν, b of degree μ with respect to y, then by the determination of degree of the preceding paragraph the degree of the resultant is

$$(5)\qquad n\nu + m\mu;$$

so great, then, is the number of common pairs of roots. This number can, however, still be modified by the fact that the coefficients may possibly be so constituted that

the highest power of y drops out of the final equation. One could then save the agreement of the number with (5) only by the arbitrary addition of infinite roots. Multiple roots, too, give cause for misgivings. These drawbacks are met in part by taking as a basis the homogeneous form of the equations, a form in which the elimination problem presents itself especially in geometry.

For if the function $f(x)$ has arisen from a homogeneous function of order n of the three variables x, y, z (a ternary form) by arranging according to powers of x, then the coefficients $a_0, a_1, a_2 \ldots a_n$ are homogeneous forms of the two variables y, z of the degree indicated by the index, so that a_0 is a constant, a_1 a linear, a_2 a quadratic form, and so forth, and the corresponding holds of the coefficients $b_0, b_1 \ldots b_m$ of the function $\varphi(x)$.

If x, y, z denote triangular coordinates in the plane, then $f(x) = 0$, $\varphi(x) = 0$ are the equations of a curve of order n and one of order m, and the common roots of these two equations (the ratios $x : y : z$) denote the points of intersection of the two curves. The final equation, which one obtains by eliminating x, is a homogeneous equation whose degree comes out, by (15), §. 48, equal to nm.

A lowering of the degree cannot take place here; the final equation is always a homogeneous equation of degree nm in the two unknowns y, z. The only exception to which attention must be paid here is that the resultant vanishes identically (for all y, z), so that infinitely many common roots are present. In geometry this case has the meaning that the two curves have a portion of a curve in common, and not merely isolated points.

If the equations $f = 0$, $\varphi = 0$ have only a finite number of common roots, then in the expressions

$$y_1 = y + \beta x, \qquad z_1 = z + \gamma x$$

we can arrange the constants β, γ so that, whenever to one value of the ratio $y : z$ there correspond several distinct values of x satisfying the equations (1), the associated values of $y_1 : z_1$ turn out different from one another.

We can then also regard f and φ as homogeneous functions of degree m and n in the variables x, y_1, z_1, and the resultant then becomes a homogeneous function $R(y_1, z_1)$ of degree mn in y_1 and z_1. If we set in it

$$y_1 = b' + b\lambda, \qquad z_1 = c' + c\lambda,$$

we obtain an equation of degree mn in λ, and, provided $R(y_1, z_1)$ does not vanish identically, we can choose the constants b, b', c, c' so that the coefficient of the highest power of λ, namely $R(b, c)$, is different from zero. Then to each root of the resultant there corresponds only one system of values of the ratio $x : y : z$, and so the number of common roots of f and φ is at most equal to mn. In order to be able to state the theorem that the number of common roots is always equal to mn, one must further adopt a convention according to which certain of these pairs of values are to be counted multiply.

The theorem which, in geometric dress, says that two curves of the mth and nth order intersect in mn points is called Bezout's theorem.

§. 50.
Elimination from several equations.

The principles of elimination, which we have applied in the foregoing to two equations, can be extended without difficulty to several equations with a corresponding number of unknowns.

Let us first take the equations

$$f(x) = 0, \quad \varphi(x) = 0, \quad \psi(x) = 0, \tag{1}$$

of degrees m, n, p, whose coefficients a_i, b_i, c_i are at first to be regarded as independent variables.

We apply to two of them, say to $f(x), \varphi(x)$, the algorithm of the greatest common divisor and obtain a last remainder, which is a rational function of the a and b and which, freed of denominators, agrees with the resultant R of $f(x)$ and $\varphi(x)$, and a last remainder but one, which is a linear function of x. The condition for the existence of a common root of $f = 0$ and $\varphi = 0$ is the vanishing of the resultant, and the last remainder but one

gives, under this assumption, a rational expression for the common root in terms of the coefficients.

If one substitutes this expression into $\psi(x)$, then, on clearing away all denominators, one obtains an integral rational function of the coefficients a, b, c, which we shall denote by S, and whose vanishing, in conjunction with $R = 0$, contains the necessary and sufficient condition for the three equations (1) to have a common root.

Let us now replace the coefficients a, b, c of the functions (1) by integral rational functions of a second unknown y, whose coefficients shall be denoted by α, β, γ and for the time being regarded as independent variables.

R and S then become integral rational functions of α, β, γ and y, and if we therefore regard the equations

$$R = 0, \qquad S = 0 \tag{2}$$

as equations for y, then by the rules of the preceding paragraph we can form their resultant T, which, if it does not vanish identically, will be an integral rational function of the coefficients α, β, γ, and the equation

$$T = 0 \tag{3}$$

is the necessary and sufficient condition for the system of equations (1) to be satisfiable by one or more pairs of values x, y. T is called the resultant of the system (1). One can therefore, from the three equations (1), by elimination of the unknowns x and y, form a final equation for x.

If one now replaces the α, β, γ in turn by integral rational functions of a third unknown z, then T passes over into an equation for z. The functions f, φ, ψ become functions of the three unknowns x, y, z. The degree of equation (3) gives the number of systems of values x, y, z for which the three equations (1) are simultaneously satisfied.

But if the coefficients are so constituted that T vanishes identically (for all values of z), then there are infinitely many systems of values x, y, z which satisfy the equations (1) simultaneously.

If one regards x, y, z as Cartesian coordinates, then the equations (1), each by itself, represent a surface. The degree of the final equation $T = 0$ with respect to z gives the

the number of points of intersection of these three surfaces. But if T is identically zero, then the three surfaces have either a portion of surface or a curve in common.

We can determine the degree of the final equation in the following way, which requires no lengthy considerations at all about the degrees of the functions R, S, T. In order to make ourselves independent of exceptional cases, we introduce homogeneous variables by adding a fourth, u, and thus consider three homogeneous quaternary equations

$$F(x, y, z, u) = 0, \quad \Phi(x, y, z, u) = 0, \quad \Psi(x, y, z, u) = 0 \tag{4}$$

of the orders m, n, p.

If we assume the special case in which each of these three functions splits into purely linear factors, then it is clear that, if the number of common solutions is not infinite. their number must be equal to the product mnp; for we can set one linear factor of each of F, Φ, Ψ equal to zero simultaneously, and so obtain mnp systems of three linear equations each, which, if they are independent of one another, yield just as many systems of values for the ratios $x : y : z : u$.

The conclusion that in the general case the degree of the final equation of (4) is just as great appears questionable at first sight. But following a remark of F. Klein, the doubt can be completely removed by the following simple consideration.

Let us first imagine the final equation formed from the equations (4) under the assumption that the coefficients in the equations (4) are variables independent of one another. Then the resultant which results after elimination of x, y is certainly not identically zero, since it does not vanish identically even for special systems of coefficients, e.g. when F, Φ, Ψ split into linear factors; and we obtain a homogeneous final equation for the two unknowns z, u of a certain degree, which is to be determined.

If one specialises the coefficients in any way whatever, the final equation may indeed possibly be satisfied identically; but if this case does not occur, then its order cannot be changed (since all terms are of the same order). Now since, by the specialisation assumed above, which expresses itself in the splitting of the functions F, Φ, Ψ, the

identical vanishing does not take place, but rather a final equation of order mnp appears, then this must also be the degree of the final equation in the general case.

With this the Bezout theorem is proved for the case of three homogeneous quaternary equations.

That we have restricted ourselves here to three equations was done merely in the interest of simplicity. The extension required for the case of n equations in $n+1$ homogeneous unknowns follows of itself, and may be left to the reader.

With regard to our general problem we wish only to remark further that from these considerations it follows that the problem of algebra, in so far as it relates to the solution of any algebraic equations whatever, is thereby reduced to the treatment of a chain of equations each in a single unknown.

§. 51.
Decomposable and indecomposable functions.

Among the integral rational functions of several variables, decomposable and indecomposable ones must be distinguished.

An integral rational function of any variables whatever is called decomposable if it can be represented as a product of at least two integral rational functions of the same variables; if it cannot be represented as such a product, it is called indecomposable.

If a function is decomposable, then the degree of each of the factors is lower than the degree of the function itself. A linear function is therefore always indecomposable, and every integral rational function can be decomposed into a finite number of indecomposable functions.

We call an integral function W divisible by another w if a third integral function w' exists such that

$$W = w\,w'$$

holds. From this it then follows that, if U is an integral function divisible by w and V is an arbitrary integral function, the product UV

is divisible by w, and that, if U, U', $U''\ldots$ are integral rational functions divisible by w, and V, V', $V''\ldots$ are arbitrary integral rational functions, then $UV + U'V' + U''V''\ldots$ is also divisible by w.

If W is divisible by w, one also says that w goes into W.

Two integral rational functions U, V that are not divisible by one and the same integral rational function are called relatively prime or coprime.

We prove the theorem:

I. If U, V, v are integral rational functions of any variables whatever, if U and v are relatively prime and UV is divisible by v, then V is divisible by v.

This theorem corresponds exactly to a well-known fundamental theorem from the theory of the integers, namely that if a product of two integers is divisible by a third integer which is coprime to one of the factors, then the other factor must be divisible by this number.

We prove it by complete induction. If U, V, v depend on only one variable x, the theorem is true; for by §. 6 (Theorem I) one can in this case, when U and v are relatively prime, determine two other integral functions P and p of x so that

$$PU + pv = 1,$$

from which, by multiplication by V,

$$PUV + pVv = V$$

follows, and from this one sees that if UV is divisible by v, then V must also be divisible by v.

We therefore assume that the theorem we wish to prove has been proved for functions of n and fewer variables, and from this we derive its validity for functions of $n + 1$ variables.

For this it is necessary that we draw from Theorem I, assumed to be true, some consequences, whose conclusion then yields the validity of the theorem for the next higher number of variables.

If v is an indecomposable function, then another function U of the same variables is either relatively prime to v or divisible by v. From this it follows by I. that a product of

two functions UV can be divisible by v only if one of the two factors is divisible by v. The same holds for a product of several functions, and so from I. there follows the theorem:

II. A product of several integral rational functions is divisible by an indecomposable function v only if at least one of the factors of the product is divisible by v.

If an integral rational function U is decomposed in two ways into indecomposable factors,

$$U = v\,v'\,v''\cdots = w\,w'\,w''\ldots,$$

then by II. at least one of the factors $v, v', v''\ldots$ must be divisible by w, say v. But then, since v too is indecomposable, v can differ from w only by a constant factor.

Accordingly, if c is this constant factor,

$$c\,v'\,v''\cdots = w'\,w''\ldots,$$

from which it follows that one of the functions $v', v''\ldots$, say v', is divisible by w', and thus differs from w' only by a constant factor.

We therefore obtain, as a second consequence of Theorem I:

III. An integral rational function can, apart from constant factors, be decomposed into indecomposable factors in only one way.

From this arises the concept of the greatest common divisor of two or more integral rational functions $U, V\ldots$ By this one understands the product of all indecomposable factors that occur in the decompositions of each of the functions $U, V\ldots$, or the function of highest possible degree that divides all the functions $U, V\ldots$ By III. this function is, apart from a constant factor, completely determined for every system of functions $U, V\ldots$

Several functions $U, V, W\ldots$ are called relatively prime if no two of them have a common divisor.

All these definitions and theorems are exactly analogous to very well-known theorems of elementary number theory, which there appear as consequences of the algorithm of the greatest common

divisor occur, only that here the indecomposable functions take over the role of the primes.

We add one more theorem of this kind:

IV. If u, v are relatively prime and Uu, Uv are divisible by w, then U is divisible by w.

For if one decomposes the integral rational functions U, u, v, w into their indecomposable factors, then any factor of w, since it cannot occur in u and v at the same time, must divide U. If one removes it from U and w, one can conclude in the same way for a next factor of w, and so on.

We now consider integral rational functions of one vari able t

$$f(t) = u_0 t^m + u_1 t^{m-1} + \cdots + u_{m-1} t + u_m,$$

whose coefficients are integral rational functions of n vari ables x, of which t is independent.

If the coefficients $u_0, u_1 \ldots u_m$ have no common divisor, then $f(t)$ is called primitive; in the other case the greatest common divisor of the coefficients $u_0, u_1 \ldots u_m$ is called the divisor of the function $f(t)$ (cf. §. 2).

We conclude, always under the assumption of the validity of I:

V. The product of two primitive functions $f(t)$ and $\varphi(t)$ is again a primitive function, and the divisor of a product of two imprimitive functions is equal to the product of the divisors of the two factors.

The proof is exactly as for the corresponding theorem in §. 2.

Let

$$\begin{aligned} f(t) &= u_0 t^m + u_1 t^{m-1} + \cdots + u_m \\ \varphi(t) &= v_0 t^\mu + v_1 t^{\mu-1} + \cdots + v_\mu \end{aligned} \tag{1}$$

be two primitive functions and

$$F(t) = U_0 t^{m+\mu} + U_1 t^{m+\mu-1} + \cdots + U_{m+\mu} \tag{2}$$

their product. Then, if r and s denote any two numbers from the series $0, 1 \ldots m$, and $0, 1, 2 \ldots \mu$ (§. 2), we have

$$\begin{aligned} U_{r+s} = u_r v_s &+ u_{r-1} v_{s+1} + \cdots \\ &+ u_{r+1} v_{s-1} + \cdots \end{aligned} \tag{3}$$

Now if w is any indecomposable function which divides neither all the u_r nor all the v_s, then we choose r and s in (3) so that u_r is the first u not divisible by w, so that $u_{r-1}, u_{r-2} \ldots$ are divisible by w, and likewise so that v_s is the first v not divisible by w. Then U_{r+s} also cannot be divisible by w, because all its terms, with the exception of the first, are divisible by w; that is, $F(t)$ is primitive.

From this it follows immediately, if p and q are any integral rational functions of the x, that pq is the divisor of the product of the two imprimitive functions $pf(t)$, $q\varphi(t)$, and hence the second part of Theorem V. From this it follows further:

VI. If an integral rational function. $F(t)$ of the $n+1$ variables x and t is decomposable into two factors which are integral with respect to t and at least rational with respect to the x, then it is also decomposable into two functions which are integral and rational in x and t.

For by the hypothesis there is an integral rational function w of the x alone and two integral rational functions of the x and t, $f_1(t)$, $\varphi_1(t)$, such that

$$w\, F(t) = f_1(t)\, \varphi_1(t),$$

and by (V) w must divide the product of the divisors of $f_1(t)$ and $\varphi_1(t)$, so that we also obtain

$$F(t) = f(t)\, \varphi(t)$$

where $f(t)$, $\varphi(t)$ are likewise integral rational functions of x and t, and indeed with respect to t of the same degree as $f_1(t)$ and $\varphi_1(t)$, q. e. d. — From this we conclude further that two functions $F(t)$, $f(t)$ which, regarded as integral rational functions of the $n+1$ variables x and t, are relatively

prime in the sense defined above, must also prove to be relatively prime when regarded as functions of t alone and treated by the algorithm of the greatest common divisor (§. 6). For if they had a common divisor which was integral with respect to t but fractional with respect to the x, then an integral function T of x and t and an integral function P of the x alone could be determined so that

$$P\,F(t) = T\,F_1(t), \qquad P\,f(t) = T\,f_1(t)$$

would hold, where F_1, f_1 are integral functions without a common divisor

are. By IV, P would then have to divide the divisor of T, and $F(t)$ and $f(t)$ could not be relatively prime. By §. 6 we can therefore determine two integral rational functions Q, q of x and t and an integral rational function X of the x alone in such a way that the identity

$$QF(t) + qf(t) = X \tag{4}$$

holds.

Now if $\Phi(t)$ is a further integral rational function of x and t, then from (4), by multiplication with $\Phi(t)$, it follows that

$$Q\Phi(t)F(t) + q\Phi(t)f(t) = X\Phi(t),$$

and hence, if $\Phi(t)F(t)$ is divisible by $f(t)$, then by this $X\Phi(t)$ is also divisible by $f(t)$.

Accordingly, if $\varphi(t)$ and $\psi(t)$ again denote two integral functions of x and t, we may set

$$\begin{aligned} X\Phi(t) &= \varphi(t)f(t) \\ F(t)\Phi(t) &= \psi(t)f(t). \end{aligned} \tag{5}$$

If one multiplies the second of these equations by X and substitutes from the first the expression $\varphi(t)f(t)$ for $X\Phi(t)$, then $f(t)$ can be cancelled and it follows that

$$\varphi(t)F(t) = X\psi(t).$$

By IV., X must therefore divide both the divisor of $\varphi(t)f(t)$ and the divisor of $\varphi(t)F(t)$, and since the functions $f(t)$ and $F(t)$, and consequently also their divisors, are relatively prime, X must divide the divisor of $\varphi(t)$. If we accordingly set

$$\varphi(t) = X\varphi_1(t),$$

then from (5) it follows that

$$\Phi(t) = \varphi_1(t)f(t),$$

that is, $\Phi(t)$ is divisible by $f(t)$; hence:

VII. If $F(t)$ and $f(t)$ are relatively prime and $\Phi(t)F(t)$ is divisible by $f(t)$, then $\Phi(t)$ is divisible by $f(t)$.

But this is nothing other than Theorem I for functions of $n+1$ variables, and I. is thus proved in general.

§. 52.
The Tschirnhausen Transformation.

We now make one more important application of the theory of symmetric functions and of the formation of resultants, to the transformation of algebraic equations that goes back to Tschirnhausen[1)].

Let

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n \tag{1}$$

be any function of degree n, and let

$$y = \frac{\chi(x)}{\psi(x)} \tag{2}$$

be any (integral or fractional) rational function of x, subject to the single assumption that $f(x)$ and $\psi(x)$ are to have no common divisor. By the concluding theorem of §. 6 we can then determine the functions $F(x)$ and $\varphi(x)$, the latter of degree at most $n-1$, so that

$$\chi(x) = F(x)f(x) + \varphi(x)\psi(x) \tag{3}$$

holds; consequently, as soon as a root of the equation

$$f(x) = 0 \tag{4}$$

is put for x, we have

$$\frac{\chi(x)}{\psi(x)} = \varphi(x).$$

Since we shall consider the values of y only for those values of x that are roots of (4), we lose nothing in generality if from the outset we set

$$y = \varphi(x),$$

that is, equal to an integral function of degree $(n-1)$. Let us therefore put

$$y = \varphi(x) = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \cdots + \alpha_{n-1}x^{n-1}, \tag{5}$$

in which the coefficients α remain for the present entirely undetermined.

If we denote by

$$x_1, x_2 \ldots x_n \tag{6}$$

1) Acta eruditorum, Leipzig 1683.

the roots of the equation (4), then the corresponding values of y result from (5):

$$y_1 = \varphi(x_1), \quad y_2 = \varphi(x_2) \ldots y_n = \varphi(x_n), \tag{7}$$

and the product

$$\Phi(y) = (y - y_1)(y - y_2)\ldots(y - y_n) \tag{8}$$

is a symmetric function of the quantities (6), which can therefore be expressed rationally in terms of the coefficients of $f(x)$.

Represented in this way, let $\Phi(y)$ have the expression

$$\Phi(y) = y^n + b_1 y^{n-1} + b_2 y^{n-2} + \cdots + b_{n-1}y + b_n. \tag{9}$$

The coefficients b_ν are, by §. 7, determined through the y_i by means of the formulas

$$b_1 = -\Sigma y_1, \quad b_2 = \Sigma y_1 y_2, \quad b_3 = -\Sigma y_1 y_2 y_3 \ldots,$$

from which it follows that b_ν is an integral rational and homogeneous function of degree ν of the variables $\alpha_0, \alpha_1 \ldots \alpha_{n-1}$.

$\Phi(y)$ is nothing other than the resultant of the two equations

$$f(x) = 0, \quad \varphi(x) - y = 0, \tag{10}$$

and by §. 49 one can also, conversely, express the common root of these two equations (10), when y becomes equal to one of the values $y_1, y_2 \dots y_n$, rationally in terms of the coefficients of (10), that is, rationally in terms of a_i, α_i, y, provided that only one such common root exists.

This will occur when the n values $y_1, y_2 \dots y_n$ are different from one another. One then obtains an expression

$$x = \beta_0 + \beta_1 y + \beta_2 y^2 + \cdots + \beta_{n-1} y^{n-1},$$

which is valid when one of the values (6) is put for x and the corresponding value (7) for y. The quantities β_i are composed rationally of the a_i and α_i.

But if among the values $y_1, y_2 \dots y_n$ several equal ones occur, then the determination of the corresponding values of x still requires the solution of an equation of the corresponding degree. This degree, however, is always smaller than n, since the function $\varphi(x)$ can take the same value y for at most $n-1$ values of x. If, then, the equation

$$\Phi(y) = 0 \tag{11}$$

is completely solved, then the equation (1) is thereby also completely solved [if (11) has n distinct roots], or at least reduced to the solution of an equation of lower degree [if (11) has equal roots].

Accordingly we regard (11) as a transformation of the equation (1), and the formation of (11) from (1) is called the Tschirnhausen transformation of the equation (1).

The purpose of such a transformation is to determine the coefficients α_i in such a way that $\Phi(y)$ takes a simpler form than $f(x)$.

One can form the coefficients b_ν if one knows the power sums of the y_i (§. 42).

If we denote the sums of the mth powers of the x_i by s_m, and those of the y_i by σ_m, that is

$$s_m = S x_i^m, \qquad \sigma_m = S y_i^m,$$

then we can form σ_m from the s_m in the following way.

We first expand y^m by the polynomial theorem (§. 12), that is, we set

$$y^m = \sum^{\nu} \frac{\Pi(m)}{\Pi(\nu_0)\,\Pi(\nu_1)\,\dots\,\Pi(\nu_{n-1})}\, \alpha_0^{\nu_0} \alpha_1^{\nu_1} \dots \alpha_{n-1}^{\nu_{n-1}}\, x^{\nu_1 + 2\nu_2 \cdots + (n-1)\nu_{n-1}}, \tag{12}$$

where the sum extends over all non-negative integral values of $\nu_0, \nu_1 \dots \nu_{n-1}$ that satisfy the condition

$$\nu_0 + \nu_1 + \cdots + \nu_{n-1} = m \tag{13}$$

If we then substitute $x_1, x_2 \dots x_n$ for x in (12) and form the sum of the values of $y_1^m, y_2^m \dots y_n^m$ so obtained, it follows that

$$\sigma_m = \sum^{\nu} \frac{\Pi(m)}{\Pi(\nu_0)\,\Pi(\nu_1)\,\dots\,\Pi(\nu_{n-1})}\, s_{\nu_1 + 2\nu_2 + \cdots + (n+1)\nu_{n-1}}\, \alpha_0^{\nu_0} \alpha_1^{\nu_1} \dots \alpha_{n-1}^{\nu_{n-1}}. \tag{14}$$

In this way σ_m is represented as an integral homogeneous function of order m of the $\alpha_0, \alpha_1 \dots \alpha_{n-1}$. A somewhat simplified notation is obtained by the second representation of forms (§. 15). If we let $\mu_1, \mu_2 \dots \mu_m$ run independently of one another through the sequence of numbers $0, 1 \dots n-1$, then, if in a combination the index 0 occurs ν_0 times, the index 1 occurs ν_1 times, the index 2 occurs ν_2 times, … the index $n-1$ occurs ν_{n-1} times, we have

$$\nu_0 + \nu_1 + \nu_2 + \cdots + \nu_{n-1} = m$$

and

$$\mu_1 + \mu_2 + \cdots + \mu_m = \nu_1 + 2\nu_2 + \cdots + (n-1)\nu_{n-1},$$

and accordingly the expression for σ_m can be represented thus:

$$\sigma_m = \sum^{\mu} s_{\mu_1+\mu_2+\ldots+\mu_m}\, \alpha_{\mu_1}\alpha_{\mu_2}\cdots\alpha_{\mu_m}. \tag{15}$$

For example, we obtain

$$\sigma_1 = \alpha_0 s_0 + \alpha_1 s_1 + \cdots + \alpha_{n-1}s_{n-1}$$

$$\begin{aligned} \sigma_2 &= \sum^{i,k} s_{i+k}\, \alpha_i\alpha_k \\ \sigma_3 &= \sum^{h,i,k} s_{h+i+k}\, \alpha_h\alpha_i\alpha_k, \end{aligned} \tag{16}$$

in which h, i, k run through the sequence of numbers $0, 1, 2 \ldots n-1$.

§. 53.
Application to the cubic and biquadratic equations.

The goal that Tschirnhausen already had in view with this transformation consisted in making so many terms of the transformed equation $\Phi(y) = 0$ vanish, by determining the substitution coefficients $\alpha_0, \alpha_1 \ldots \alpha_{n-1}$, that the equation becomes solvable. This succeeds easily for the equations of the third and fourth degree, though it is not simple, and does not seem possible without lengthy calculations, to derive the otherwise well-known formulas in this way.

Let us first take the cubic equation

$$x^3 + ax^2 + bx + c = 0, \tag{1}$$

which we transform by means of the following substitution:

$$y = \alpha + \beta x + \gamma x^2 \tag{2}$$

In order to reduce the equation for y

$$y^3 + Ay^2 + By + C = 0 \tag{3}$$

to a pure cubic equation, we have to determine the ratios $\alpha : \beta : \gamma$ from the two equations

$$A = 0, \quad B = 0 \tag{4}$$

of which the first is linear, the second quadratic.

The equations (4) are equivalent to

$$\sigma_1 = 0, \quad \sigma_2 = 0 \tag{5}$$

and thus, by (16) of the preceding section, they yield

$$\alpha s_0 + \beta s_1 + \gamma s_2 = 0,$$

$$\alpha^2 s_0 + 2\alpha\beta s_1 + 2\alpha\gamma s_2 + \beta^2 s_2 + 2\beta\gamma s_3 + \gamma^2 s_4 = 0,$$

in which $s_0 = 3$. If one multiplies the latter equation by s_0 and subtracts from it the square of the first, it follows that

$$\beta^2(s_0 s_2 - s_1^2) + 2\beta\gamma(s_0 s_3 - s_1 s_2) + \gamma^2(s_0 s_4 - s_2^2) = 0. \tag{6}$$

The discriminant of this quadratic equation

$$(s_0 s_3 - s_1 s_2)^2 - (s_0 s_2 - s_1^2)(s_0 s_4 - s_2^2)$$

gives, when expanded,

$$-s_0(s_0s_2s_4 - s_0s_3^2 - s_1^2s_4 - s_2^3 + 2s_1s_2s_3),$$

and is therefore, if D denotes the discriminant of the cubic equation (1) [§. 46, (9)], equal to $-3D$; and the solution of (6) accordingly gives

$$\frac{\beta}{\gamma} = \frac{-(s_0s_3 - s_1s_2) + \sqrt{-3D}}{s_0s_2 - s_1^2}. \tag{7}$$

With this, equation (3) has been reduced to a pure equation, which for its solution requires only the extraction of a cube root. Which sign we wish to give the square root occurring in (7) is left to our discretion.

To make use of the Tschirnhausen transformation for the biquadratic equation, one may proceed, for instance, by determining the $\alpha_0, \alpha_1, \alpha_2, \alpha_3$ in the transformed equation

$$y^4 + b_1y^3 + b_2y^2 + b_3y + b_4 = 0 \tag{8}$$

so that

$$b_1 = 0, \quad b_3 = 0 \tag{9}$$

whereby (8) passes into a quadratic equation for y^2, from whose roots the square roots must still be extracted, so that, after the equations (9) have been solved, a square root must still be extracted twice more.

If, by means of the first equation (9), one eliminates α_0 from the second, there arises a homogeneous cubic equation among $\alpha_1, \alpha_2, \alpha_3$. One may therefore assume one of the two ratios $\alpha_1 : \alpha_2 : \alpha_3$ arbitrarily, e.g. set $\alpha_3 = 0$, and obtains for the determination of the other a cubic equation. By this, then, the solution of the biquadratic equation is reduced to that of a cubic and to square roots.

§. 54.
The Tschirnhausen Transformation of the Equation of the Fifth Degree.

If one wishes to apply the Tschirnhausen transformation to the equation of the fifth degree

$$x^5 + a_1x^4 + a_2x^3 + a_3x^2 + a_4x + a_5 = 0 \tag{1}$$

one has to set

$$y = \alpha_0 + \alpha_1x + \alpha_2x^2 + \alpha_3x^3 + \alpha_4x^4 \tag{2}$$

and obtains the transformed equation

$$y^5 + b_1y^4 + b_2y^3 + b_3y^2 + b_4y + b_5 = 0. \tag{3}$$

The most obvious thought would now be to determine the ratios of the five unknowns α in such a way that

$$b_1 = 0, \quad b_2 = 0, \quad b_3 = 0, \quad b_4 = 0 \tag{4}$$

would hold, which would reduce (3) to a pure equation; and this was no doubt also the goal that Tschirnhausen had in view. But of the four equations (4) the first is linear, the second quadratic, the third of the third and the fourth of the fourth degree. The final equation that one can derive from them will therefore (by §. 50) be of degree $2.3.4 = 24$, and hence no use can be drawn from it for the solution of the equation of the fifth degree. One therefore seeks at first to satisfy only the two equations

$$b_1 = 0, \quad b_2 = 0, \tag{5}$$

whereby the equation (3) passes into a form which, following F. Klein (Lectures on the Icosahedron), is called a principal equation of the fifth degree.

To satisfy the two equations (5) we have at our disposal the four ratios $\alpha_0 : \alpha_1 : \alpha_2 : \alpha_3 : \alpha_4$, and we can still use these quantities in manifold ways to simplify the equation of the fifth degree.

We shall return to this subject in more detail in a later section, and confine ourselves here to indicating the aims in general terms. For the sake of vividness we make use of a geometrical dress, which, incidentally, is by no means essential to the understanding of the algebraic theory as it is to be given later.

If, with the help of the linear equation $b_1 = 0$, we express one of the five quantities α, say α_0, in terms of the others, then we can regard the four remaining ones $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ as homogeneous coordinates of a point in space. The equation $b_2 = 0$ then represents a surface of the second order, $b_3 = 0$ a surface of the third order, $b_4 = 0$ a surface of the fourth order, $b_5 = 0$ a surface of the fifth order.

If one wished, then, to make b_2, b_3, b_4 vanish simultaneously and thereby reduce the given equation to a pure one, one would have to determine one of the 24 points of intersection of three surfaces of the second, third, and fourth order, which depends on an equation of degree 24.

Instead of proceeding in this way, one seeks to determine a straight line on the surface $b_2 = 0$. Of such straight lines there are on every surface of the second order one or two families (real or imaginary), and one of these lines can be determined by square roots, as will be carried out further below. The determination of a point of intersection of one of these straight lines with the surface $b_3 = 0$ then leads to an equation of the third degree, and if, by determining a common factor of the α, one further makes the coefficient $b_4 = 1$, the equation for y takes the form

$$y^5 + y + b = 0, \tag{6}$$

whereby y is determined as a function of one variable b.

This form of the equation is usually named after the English mathematician Jerrard, who made it known in the year 1834. But it had already been published much earlier (1786) by E. J. Bring in Lund[7]. We therefore call it the Bring-Jerrard form.

Just as well one could make $b_4 = 0$ by an equation of the fourth degree, and would obtain a second normal form of the equation of the fifth degree:

$$y^5 + y^2 + b = 0, \tag{7}$$

which has the same justification as the Bring-Jerrard form.

For the solution of the equation of the fifth degree, to be sure, nothing is gained by these considerations. Their usefulness consists

in this, that the equation of the fifth degree is reduced to a normal form which depends on only one variable coefficient. This can be achieved in infinitely many different ways, and is accomplished most simply, namely without the solution of an equation of the third or fourth degree, if one regards the equation of the fifth degree $b_5 = 0$ itself as a transformation of the given equation of the fifth degree. For if one has solved the equation $b_5 = 0$, then one of the five values of y becomes equal to 0, and the given equation of the fifth degree is thereby solved at the same time.

Fifth Section.
Linear Transformation.

§. 55.
Introduction of the linear transformation.

[7] Cf. F. Klein, Vorlesungen über das Ikosaëder, p. 143.

If, in any functions that depend on the m variables $x_1, x_2 \dots x_m$, we substitute for these variables linear expressions that depend on m new variables $x'_1, x'_2 \dots x'_m$

$$\begin{aligned} x_1 &= \alpha_1^{(1)} x'_1 + \alpha_1^{(2)} x'_2 + \cdots + \alpha_1^{(m)} x'_m, \\ x_2 &= \alpha_2^{(1)} x'_1 + \alpha_2^{(2)} x'_2 + \cdots + \alpha_2^{(m)} x'_m, \\ &\cdots\cdots\cdots\cdots\cdots\cdots \\ x_m &= \alpha_m^{(1)} x'_1 + \alpha_m^{(2)} x'_2 + \cdots + \alpha_m^{(m)} x'_m, \end{aligned} \tag{1}$$

then we call this a linear substitution, and the result a linear transformation.

The quantity

$$r = \begin{vmatrix} \alpha_1^{(1)}, & \alpha_1^{(2)} & \dots & \alpha_1^{(m)} \\ \alpha_2^{(1)}, & \alpha_2^{(2)} & \dots & \alpha_2^{(m)} \\ & \cdots\cdots & & \\ \alpha_m^{(1)}, & \alpha_m^{(2)} & \dots & \alpha_m^{(m)} \end{vmatrix} \tag{2}$$

is called the substitution determinant. It must always be different from zero, since otherwise a dependence among the variables $x_1, x_2 \dots x_m$ would be expressed by (1).

By a linear substitution every homogeneous function of the variables x passes over into a homogeneous function of the same degree in the variables x'.

Thus, for example, every linear function

$$y = \overset{i}{\Sigma} a_i x_i$$

passes into a function of the same kind

$$y = \overset{i}{\Sigma} a'_i x'_i$$

in which

$$a'_k = \overset{i}{\Sigma} a_i \alpha_i^{(k)}. \tag{3}$$

If we consider m such linear functions

$$y_\nu = \overset{i}{\Sigma} a_{i,\nu} x_i \qquad \nu = 1, 2 \dots m,$$

we obtain m transformed functions

$$y_\nu = \overset{i}{\Sigma} a'_{i,\nu} x'_i \qquad \nu = 1, 2 \dots m.$$

The determinants of the systems y_ν, y'_ν

$$\Delta = \Sigma \pm a_{1,1} a_{2,2} \dots a_{m,m}, \qquad \Delta' = \Sigma \pm a'_{1,1} a'_{2,2} \dots a'_{m,m}$$

stand, by the multiplication law of determinants, in the relation

$$\Delta' = r\Delta, \tag{4}$$

so that the one does not vanish without the other.

§. 56.
Quadratic Forms.

Of particular importance is the transformation of the homogeneous functions of the second degree, or the quadratic forms. We denote them, as already in the first section, thus:

$$\varphi(x_1, x_2 \dots x_m) = \overset{i,k}{\Sigma} a_{i,k} x_i x_k, \qquad a_{i,k} = a_{k,i}. \tag{1}$$

By the substitution §. 55, (1), φ passes into

$$\Phi(x'_1, x'_2 \dots x'_m) = \overset{i,k}{\Sigma} a'_{i,k} x'_i x'_k. \tag{2}$$

If we put $x + \xi$ for x and likewise $x' + \xi'$ for x', so that the ξ' stand in the same relation to the ξ as the x' to the x, then, if for abbreviation we set

$$\tfrac{1}{2}\varphi'(x_i) = u_i, \qquad \tfrac{1}{2}\Phi'(x'_i) = u'_i$$

we obtain, by comparison of the terms of the first dimension (§. 16):

$$\Sigma \xi_\nu u_\nu = \Sigma \xi'_\nu u'_\nu, \tag{3}$$

and therefore, as in §. 55, (3)

$$u'_k = \overset{i}{\Sigma} \alpha_i^{(k)} u_i. \tag{4}$$

Now

$$u_k = \overset{i}{\Sigma} a_{k,i} x_i, \qquad u'_k = \overset{i}{\Sigma} a'_{k,i} x'_i, \tag{5}$$

and accordingly (4) gives us the transformation of m linear functions. The coefficients of the given system are $a'_{k,h}$, and those of the transformed system, which one obtains by substituting the first system (5) into (4), are

$$b_{k,h} = \overset{i}{\Sigma} \alpha_i^{(k)} a_{i,h}. \tag{6}$$

The determinant of the former system

$$H' = \Sigma \pm a'_{1,1} a'_{2,2} \dots a'_{m,m}$$

is, by §. 55, (4),

$$= r\Sigma \pm b_{1,1} b_{2,2} \dots b_{m,m},$$

and by (6) the determinant of the $b_{k,h}$ is

$$= r\Sigma \pm a_{1,1} a_{2,2} \dots a_{m,m}.$$

If, then, we set

$$H = \Sigma \pm a_{1,1} a_{2,2} \dots a_{m,m} \tag{7}$$

we have the important theorem

$$H' = r^2 H. \tag{8}$$

The determinant H is called the determinant of the form $\varphi(x)$.

Thus under linear transformation of the function φ it changes only by the factor r^2, and it cannot vanish for the transformed form if it does not do so for the original one.

The vanishing of the determinant H has the meaning that the function $\varphi(x)$ can be regarded as a function of fewer than m variables, which are linear functions of the x.

For if, in the first place, $\varphi(x)$ passes by a linear transformation into the function $\Phi(x')$, which is independent of one of the variables, say of x'_1, then H' vanishes, since the elements in one of the rows (and columns) all vanish, and hence by (8) also $H = 0$.

In the second place, however, H is the determinant of the system of linear equations

$$\varphi'(x_i) = 0 \qquad i = 1, 2, \ldots m, \tag{9}$$

and if this determinant vanishes, then there is (§. 24, III) a system of quantities

$$x_1^{(0)}, x_2^{(0)} \ldots x_m^{(0)}, \tag{10}$$

whose elements do not all vanish, and which, substituted for the x, satisfy the equations (9).

But now, if x_i, x'_i are any two systems of quantities, and λ likewise denotes an arbitrary quantity, one has the identical equations

$$\begin{aligned} \varphi(x + \lambda x') &= \varphi(x) + \lambda \overset{i}{\Sigma} x_i \varphi'(x'_i) + \lambda^2 \varphi(x') \\ \varphi(x') &= \tfrac{1}{2} \Sigma x'_i \varphi'(x'_i), \end{aligned} \tag{11}$$

[§. 16, (12) to (15)], so that, if for x'_i the values $x_i^{(0)}$ are substituted,

$$\varphi(x_i + \lambda x_i^{(0)}) = \varphi(x), \tag{12}$$

becomes entirely independent of λ. If we now assume that $x_1^{(0)}$ is different from zero, and we set

$$\lambda = -\frac{x_1}{x_1^{(0)}},$$

and further:

$$\begin{aligned} y_2 &= x_2 x_1^{(0)} - x_1 x_2^{(0)} \\ y_3 &= x_3 x_1^{(0)} - x_1 x_3^{(0)} \\ &\ldots\ldots\ldots\ldots \\ y_m &= x_m x_1^{(0)} - x_1 x_m^{(0)}, \end{aligned} \tag{13}$$

then it follows that

$$x_1^{(0)\,2} \varphi(x) = \varphi(0, y_2, y_3 \ldots y_m), \tag{14}$$

whereby $\varphi(x)$ is expressed in terms of the $m - 1$ variables $y_2, y_3 \ldots y_m$.

§. 57.
Transformation of Quadratic Forms into a Sum of Squares.

We at once make an important application of the theorem just proved to the discriminant of the function (11) (regarded as a function of the second degree in λ), in which

it is no longer assumed that the determinant of the function φ vanishes. This discriminant is

$$\psi(x) = [\Sigma x_i \varphi'(x_i')]^2 - 4\varphi(x)\varphi(x'). \tag{1}$$

If we put $x_i + \lambda x_i'$ in place of x_i in it, then, substituting for $\varphi(x + \lambda x')$ the expression §. 56, (11) and setting $\Sigma x_i' \varphi'(x_i') = 2\varphi(x')$, we obtain

$$\psi(x + \lambda x') = \psi(x),$$

from which it follows that $\psi(x)$ depends only on $m - 1$ linear combinations of the m variables. If one puts for $x_1', x_2' \ldots x_m'$ arbitrary special values for which $\varphi(x')$ does not vanish, and takes x_1' different from zero, then

$$x_1'^2 \psi(x_1, x_2, \ldots x_n) = \psi(0, x_2 x_1' - x_1 x_2', \ldots x_m x_1' - x_1 x_m'),$$

hence a function of the $m - 1$ variables

$$y_2 = x_2 x_1' - x_1 x_2', \ldots y_m = x_m x_1' - x_1 x_m'.$$

One can also confirm this by observing that

$$\psi'(x_1),\ \psi'(x_2) \ldots \psi'(x_m)$$

all vanish simultaneously for $x_1, x_2 \ldots x_m = x_1', x_2' \ldots x_m'$, so that the determinant of the function ψ must vanish.

Formula (1) now contains a very important result, which can best be surveyed if we give this formula the following form.

We set

$$\begin{aligned} \psi(x) &= -4\varphi(x')\chi(y_2, \ldots y_m) \\ \Sigma x_i \varphi'(x_i') &= 2\sqrt{\varphi(x')}\, Y_1, \end{aligned} \tag{2}$$

and obtain

$$\varphi(x) = Y_1^2 + \chi(y_2, y_3 \ldots y_m), \tag{3}$$

whereby $\varphi(x)$ is represented as the sum of a square of a linear function y_1 and a function χ of $m - 1$ variables.

But from this follows the important theorem

A quadratic form of m variables can always be represented as a sum of squares of m or fewer linear functions.

This theorem is evidently true for a function of one variable, which is itself a square. But from (3) the truth of the theorem follows for a function of m variables, provided its truth is assumed for a function of $m - 1$ variables.

The representation is, as (2) shows, possible in infinitely many different ways, since the entirely arbitrary quantities x' still occur in this formula.

A representation by fewer than m squares is possible only when the determinant of the function φ vanishes.

Since a square root occurs in the expression (2) for Y_1, Y_1 can be real or imaginary, even when the coefficients of φ and the x and x' are assumed to be real.

But if, in the case where Y_1 is imaginary, we put iY_1 in place of Y_1, we can also state our theorem as follows:

A real quadratic form in m variables can be represented as a sum of at most m positive or negative squares of real linear functions of the x[8].

§. 58.
Law of inertia of quadratic forms.

For the decomposition of a quadratic form into squares the following theorem now holds, which received from Sylvester the name of the law of inertia of quadratic forms (Philos. Mag. 1852).

However one may decompose a real quadratic form $\varphi(x)$ into a sum of positive and negative squares of linear functions, the number of the positive and of the negative among these squares, and hence also their total number, is always the same, provided that no linear dependence exists among these linear functions.

The proof of this important theorem is very simple.

Let

$$\begin{aligned}(1)\qquad \varphi(x) &= Y_1^2 + Y_2^2 + \cdots Y_\nu^2 - Y_1'^2 - Y_2'^2 - \cdots - Y_{\nu'}'^2 \\ &= Z_1^2 + Z_2^2 + \cdots Z_\mu^2 - Z_1'^2 - Z_2'^2 - \cdots - Z_{\mu'}'^2\end{aligned}$$

be two decompositions of $\varphi(x)$ into squares. Thus let $Y_1, Y_2 \ldots Y_\nu, Y_1', Y_2' \ldots Y_{\nu'}'$ be homogeneous linear functions of x between which no linear relation with constant coefficients holds, so that $\nu + \nu' \leqq m$; the same assumptions are made about the functions $Z_1, Z_2 \ldots Z_\mu, Z_1', Z_2' \ldots Z_{\mu'}'$, so that $\mu + \mu' \leqq m$ as well.

Suppose that

$$\nu < \mu,$$

then we can subject the variables x to the linear equations

$$\begin{aligned}(2)\qquad Y_1 = 0, \quad Y_2 = 0 \ldots Y_\nu = 0 \\ Z_1' = 0, \quad Z_2' = 0 \ldots Z_{\mu'}' = 0\end{aligned}$$

whose number $\nu + \mu'$ is smaller than m.

Now if all the functions $Z_1, Z_2 \ldots Z_\mu$ were linearly dependent on the functions $Y_1, Y_2 \ldots Y_\nu, Z_1', Z_2' \ldots Z_{\mu'}'$, then from these μ equations one could, by eliminating the ν variables $Y_1, Y_2 \ldots Y_\nu$, derive a linear equation between the $Z_1, Z_2 \ldots Z_\mu, Z_1' \ldots Z_{\mu'}'$, which by hypothesis is not to hold. Hence the vanishing of all the $Z_1, Z_2 \ldots Z_\mu$ is not a necessary consequence of the equations (2), and we can add to the equations (2) one further equation that is not homogeneous, say

$$Z_1 = 1,$$

then for the m unknowns x we have a system of m or fewer equations which are independent of one another and therefore do not contradict one another.

But then the second representation (1) gives a positive value for $\varphi(x)$, while the first gives a negative or vanishing value, and in this lies a contradiction. From this it follows that

$$\nu \geqq \mu,$$

and since one concludes in the same way

$$\mu \geqq \nu,$$

[8]For the general theory of the linear transformation of forms of the second degree, besides the investigations of Jacobi and Hesse (Hesse, "Vorlesungen über analytische Geometrie des Raumes", third edition, edited by Gundelfinger, Leipzig 1876), the works of Weierstrass and Kronecker (Monatsberichte der Berliner Akademie 1868, 1874) are of particular importance. A thorough historical account of the development of the question is given in the "Bericht über den gegenwärtigen Stand der Invariantentheorie" by Franz Meyer (first Jahresbericht der deutschen Mathematiker-Vereinigung, Berlin 1892).

only $\nu = \mu$ remains. In the same way one can prove that $\nu' = \mu'$, whereby our theorem is completely proved.

§. 59.
Transformation of forms of degree n.

When a form of degree n in m variables, $F(x)$, is transformed by a linear substitution [§. 55, (1)] into $\Phi(x')$, then at the same time the linear function of the variables $\xi_1, \xi_2 \ldots \xi_m$

$$\overset{i}{\Sigma}\, \xi_i F'(x_i) \tag{1}$$

is transformed, by the same substitution applied to the ξ_i, into

$$\overset{i}{\Sigma}\, \xi_i' \Phi(x_i') \tag{2}$$

because both expressions represent the coefficient of the first power of t in the expansion of

$$F(x_i + t\xi_i) = \Phi(x_i' + t\xi_i')$$

in powers of t (§. 16). Likewise the quadratic function

$$\Sigma\, \xi_i \xi_k F''(x_i x_k) \tag{3}$$

is transformed into

$$\Sigma\, \xi_i' \xi_k' \Phi''(x_i', x_k') \tag{4}$$

If we apply the first result to a system of m functions $F_1(x), F_2(x) \ldots F_m(x)$ of arbitrary degrees, and denote by Δ the determinant

$$\begin{vmatrix} F_1'(x_1), & F_1'(x_2) \ldots F_1'(x_m) \\ F_2'(x_1), & F_2'(x_2) \ldots F_2'(x_m) \\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot & \\ F_m'(x_1), & F_m'(x_2) \ldots F_m'(x_m) \end{vmatrix},$$

while we denote by Δ' the corresponding determinant formed from the functions $\Phi_1, \Phi_2 \ldots \Phi_m$, then by §. 55, (4), if we again denote by r the substitution determinant, we obtain

$$\Delta' = r\Delta. \tag{5}$$

The function Δ is called the functional determinant of the m functions $F_1, F_2 \ldots F_m$.

Likewise we obtain from the transformation (3), (4), if we denote by H the determinant

$$\begin{vmatrix} F''(x_1, x_1), & F''(x_1, x_2) \ldots F''(x_1, x_m) \\ F''(x_2, x_1), & F''(x_2, x_2) \ldots F''(x_2, x_m) \\ \ldots\ldots\ldots\ldots\ldots & \\ F''(x_m, x_1), & F'''(x_m, x_2) \ldots F''(x_m, x_m) \end{vmatrix}$$

and by H' the corresponding determinant for the transformed function Φ, then by §. 56, (8)

(6) $$H' = r^2 H.$$

H is called the Hessian determinant[9] of the function F. In both formulas (5) and (6), r denotes the substitution determinant.

§. 60.
Invariants and Covariants.

In the linear transformation of homogeneous functions there always occur certain functions of the coefficients which, when one replaces the coefficients of the given form by the coefficients of the transformed form, change in no other way than that they take on a factor which is a power of the substitution determinant. Such functions also occur in the simultaneous transformation of a system of several forms, and they depend on the coefficients of all the forms of the system. Such functions are, for example, in a system of n linear forms the determinant of the system, and in a quadratic form the determinant of this form.

These functions are called invariants of the form, or, when they depend on several forms, simultaneous invariants of the system.

If, then, $I(a)$ is a function of the coefficients a of a form $F(x)$, and if a' are the coefficients of the transformed form $\Phi(x')$, then $I(a)$ is called an invariant if the identical relation

(1) $$I(a') \doteq r^\lambda I(a)$$

holds. The exponent λ we shall call the weight of the invariant. If this exponent is zero, then I is called an absolute invariant.

One mostly considers only integral rational invariants, that is, functions I which depend rationally on the coefficients a and are moreover integral homogeneous functions of them. The simultaneous invariants are required to be homogeneous with respect to the coefficients of each of the forms on which they depend.

Besides the invariants there are also functions which, in addition to the coefficients, still contain the variables themselves, but in other respects have the same properties as the invariants (1), "the invariant property"; thus, if $C(x, a)$ is such a function,

(2) $$C(x', a') \doteq r^\lambda C(x, a).$$

Such functions are called covariants of the form $F(x)$ The functional determinant is a simultaneous covariant of a system of m functions, and the Hesse determinant of a function of higher than the second degree is a covariant of a single form. Among the covariants too one considers preferably integral rational and homogeneous functions of x.

If m is the number of the variables x, n the degree of the function $F(x)$, and ν and μ the degrees of $C(x, a)$ with respect to the x and a, then among the numbers λ, μ, ν, m, n there holds the relation

(3) $$n\mu = m\lambda + \nu,$$

which one proves by comparing the degree of the right and left sides of (2) with respect to the substitution coefficients. The a' are namely of degree n in the substitution coefficients, r of degree m, x of the first degree, from which (3) follows. The corresponding formula for the invariants is

[9] Named after Hesse, who made frequent use of this determinant in geometrical investigations, in particular in the theory of curves of the third order and in elimination problems, and who recognised its significance.

obtained by setting $\nu = 0$; indeed, the invariants may altogether be regarded as a special case of the covariants.

A general law for the formation of invariants and covariants, which we have already used in the particular cases of §. 59, we shall now briefly discuss.

If, following §. 16, (4), we arrange the function

$$F(x_1 + t\xi_1, x_2 + t\xi_2 \dots x_m + t\xi_m)$$

in powers of t, we obtain for the coefficient of t^ν

$$F_\nu(x,\xi) = \frac{1}{\Pi(\nu)} \sum^{\alpha} \frac{\Pi(\nu)}{\Pi(\alpha_1)\dots\Pi(\alpha_m)} \xi_1^{\alpha_1}\xi_2^{\alpha_2}\dots\xi_m^{\alpha_m} D_{\alpha_1,\alpha_2\dots\alpha_m}(F), \tag{4}$$

$$\alpha_1 + \alpha_2 + \dots + \alpha_m = \nu.$$

If we apply the linear transformation (1), §. 55 to the variables ξ and x simultaneously, then $F_\nu(x,\xi)$ passes over into $\Phi_\nu(x',\xi')$, which is obtained from F_ν by simultaneously replacing all the letters x, ξ, a by x', ξ', a' and F by Φ.

From this it follows:

I. If one regards $F_\nu(x,\xi)$ as a form of degree ν in ξ and forms an invariant of this form, whose coefficients therefore still depend on the x, one obtains a covariant of the form F.

And since the variables x and ξ are subject to the same transformation, it likewise follows:

II. If one forms a covariant of the function $F_\nu(x,\xi)$ regarded as a function of ξ, and in it replaces the ξ by the x, one obtains a covariant of the original function $F(x)$.

The same theorems hold also for simultaneous invariants and covariants.

For the functions $F_\nu(x,\xi)$ we have the theorem

$$F_\nu(x,\xi) = F_{n-\nu}(\xi,x), \tag{5}$$

which follows by comparing corresponding powers of t on the two sides of the following identity:

$$F(x_1 + t\xi_1 \dots x_m + t\xi_m) = t^n F(t^{-1}x_1 + \xi_1 \dots t^{-1}x_m + \xi_m)$$

The functions

$$P_\nu(x,\xi) = \frac{\Pi(\nu)\,\Pi(n-\nu)}{\Pi(n)} F_\nu(x,\xi)$$

are called the polars of the function $F(x)$, and specifically $P_\nu(x,\xi)$ is called the νth polar. For them too the following theorem holds:

$$P_\nu(x,\xi) = P_{n-\nu}(\xi,x).$$

The constant factor $\Pi(\nu)\Pi(n-\nu) : \Pi(n)$ is added so that, when the form $F(x)$ is written with the polynomial coefficients, the functions P_ν receive the simplest possible coefficients.

For the binary form $f(x,y)$, to which we shall chiefly apply these definitions in what follows, the following expressions result for the polars:

$$P_1(x,\xi) = \frac{1}{n}\left[\xi f'(x) + \eta f'(y)\right]$$

$$P_2(x,\xi) = \frac{1}{n(n-1)}\left\{\xi^2 f''(x,x) + 2\,\xi\,\eta\, f''(x,y) + \eta^2 f''(y,y)\right\},$$

and in general

$$n(n-1)\dots(n-\nu+1)\,P_\nu(x,\xi) =$$
$$\xi^\nu \frac{\partial^\nu f}{\partial x^\nu} + \nu\,\xi^{\nu-1}\eta\,\frac{\partial^\nu f}{\partial x^{\nu-1}\partial y} + \dots + \eta^\nu \frac{\partial^\nu f}{\partial y^\nu}.$$

§. 61.
Linear Transformation of Binary Forms.

An integral rational function of degree n in x

$$f(x) = a_0x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_{n-1}x + a_n \tag{1}$$

is turned into a binary form by replacing x with $x : y$ and multiplying by y^n:

$$f(x,y) = a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \cdots + a_ny^n. \tag{2}$$

If one knows the roots of $f(x)$, one can decompose $f(x)$ into n linear factors:

$$f(x) = a_0(x-\alpha_1)(x-\alpha_2)\ldots(x-\alpha_n), \tag{3}$$

and from this it follows that

$$f(x,y) = a_0(x-\alpha_1y)(x-\alpha_2y)\ldots(x-\alpha_ny). \tag{4}$$

But if one also replaces $\alpha_1, \alpha_2 \ldots \alpha_n$ by ratios $\alpha_1 : \beta_1, \alpha_2 : \beta_2, \ldots \alpha_n : \beta_n$, then one obtains

$$f(x,y) = A(x\beta_1 - y\alpha_1)(x\beta_2 - y\alpha_2)\ldots(x\beta_n - y\alpha_n), \tag{5}$$

if

$$a_0 = A\beta_1\beta_2\ldots\beta_n$$

is set.

For the factors of $f(x,y)$, which we shall now encounter more often, we introduce as an abbreviation the sign

$$x\beta_1 - y\alpha_1 = (x\beta_1). \tag{6}$$

Likewise we shall set

$$x_1y_2 - x_2y_1 = (x_1y_2), \tag{7}$$

in which x_1, y_1 and x_2, y_2 are two arbitrary pairs of variables. When we employ the inhomogeneous representation, we set still more briefly

$$\begin{gathered} x-\alpha_1 = (0,1),\ x-\alpha_2 = (0,2)\ldots x-\alpha_n = (0,n) \\ \alpha_1-\alpha_2 = (1,2)\ldots\alpha_i-\alpha_k = (i,k). \end{gathered} \tag{8}$$

If we make in the form (2) a linear substitution

$$\begin{aligned} x &= \alpha x' + \beta y' \\ y &= \gamma x' + \delta y' \end{aligned} \tag{9}$$

with the determinant, different from zero,

$$r = \alpha\delta - \beta\gamma, \tag{10}$$

then the form $f(x, y)$ passes over into a form $F(x', y')$ of degree n, and we have

$$F(x', y') = a'_0 x'^n + a'_1 x'^{n-1} y' + a'_2 x'^{n-2} y'^2 + \cdots a'_n y'^n. \tag{11}$$

The coefficients $a'_0, a'_1 \ldots a'_n$ of the transformed form are obtained as linear and homogeneous expressions in the coefficients $a_0, a_1 \ldots a_n$, and as homogeneous expressions of degree n in the substitution coefficients $\alpha, \beta, \gamma, \delta$. One thus obtains, on setting $x' = 1, y' = 0$ or $x' = 0, y' = 1$,

$$a'_0 = f(\alpha, \gamma), \qquad a'_n = f(\beta, \delta). \tag{12}$$

The remaining coefficients a' can be expressed by means of the derivatives of the function $f(x, y)$, e.g.

$$(n-1)a'_1 = \beta f'(\alpha) + \delta f'(\gamma),$$

which we shall not carry out further.

If we apply the transformation (9) to the two pairs of variables x, y and α_k, β_k simultaneously, we obtain the linear factors of $f(x, y)$ transformed, namely

$$(x'\beta'_k) = r(x\beta_k), \tag{13}$$

as one finds from the multiplication rule for determinants, or also by direct computation. These factors $(x\beta_k)$ are therefore also covariants of $f(x, y)$, though to be sure irrational ones, since they do not depend rationally on the coefficients of $f(x)$. In the same way one obtains

$$(\alpha'_h \beta'_k) = r(\alpha_h \beta_k), \tag{14}$$

according to which the determinants $(\alpha_h \beta_k)$ are to be regarded as irrational invariants. We now wish to set out how rational invariants and covariants can be formed from them.

For this purpose it is best to consider the linear transformation in the inhomogeneous form

$$x = \frac{\alpha x' + \beta}{\gamma x' + \delta} \tag{15}$$

and to apply it to the function $f(x)$ in (1). It then follows, if

$$F(x') = a'_0 x'^n + a'_1 x'^{n-1} + a'_2 x'^{n-2} + \cdots + a'_n$$

is set, that

$$F(x') = (\gamma x' + \delta)^n f(x). \tag{16}$$

We also write this equation thus

$$F(x') = (\alpha x' + \beta)^n \left(a_0 + a_1 \frac{1}{x} + a_2 \frac{1}{x^2} \cdots \right),$$

from which, if one sets $x' = -\delta : \gamma$, and hence $x = \infty$,

$$a_0 r^n = (-\gamma)^n F\left(-\frac{\delta}{\gamma}\right),$$

or, if $\alpha'_1, \alpha'_2, \ldots \alpha'_n$ depend on $\alpha_1, \alpha_2, \ldots \alpha_n$ in the same way as x' does on x, so that

$$F(x') = a'_0(x' - \alpha'_1)(x' - \alpha'_2)\ldots(x' - \alpha'_n)$$

is set,

$$r^n a_0 = a'_0(\gamma\alpha'_1 + \delta)(\gamma\alpha'_2 + \delta)\ldots(\gamma\alpha'_n + \delta) \tag{17}$$

follows.

Now by the transformation (15) we have

$$\begin{aligned} x - \alpha_i &= \frac{r(x' - \alpha'_i)}{(\gamma x' + \delta)(\gamma\alpha'_i + \delta)} \\ \alpha_i - \alpha_k &= \frac{r(\alpha'_i - \alpha'_k)}{(\gamma\alpha'_i + \delta)(\gamma\alpha'_k + \delta)}. \end{aligned} \tag{18}$$

We now form a product P out of any of the factors

$$\begin{aligned} &(0,1), (0,2)\ldots(0,n) \\ &(1,2), (1,3)\ldots(n-1,n), \end{aligned}$$

but in such a way that each of the digits $1, 2 \ldots n$ occurs in it equally often in all, say μ times. The index 0, that is, the variable x, may occur ν times, and the total number of the factors shall be q. Then, since each factor contains two indices,

$$\nu + n\mu = 2q. \tag{19}$$

Then from (17) and (18), if P' arises from P by the permutation of x, α_i with x', α'_i, it follows that

$$a_0'^{\mu} P' = a_0^{\mu} r^{n\mu - q}(\gamma x' + \delta)^{\nu} P. \tag{20}$$

We now form

$$C(x, y, a) = a_0^{\mu} y^{\nu} \Sigma P\left(\frac{x}{y}\right), \tag{21}$$

the sum being taken over all the values that one obtains from P when the indices $1, 2, 3 \ldots n$ are permuted with one another in all possible ways. This sum is then a symmetric function of the roots of (1), and can therefore be expressed as an integral rational function of the ratios $a_1 : a_0, a_2 : a_0 \ldots a_n : a_0$, in such a way that no term exceeds the order μ. The function $C(x, y, a)$ is therefore an integral homogeneous function of $a_0, a_1 \ldots a_n$ of order μ, and an integral homogeneous function of x, y of order ν. Since by (19), (20) it satisfies the condition

$$C(x', y'a') = r^{\lambda} C(x, y, a), \tag{22}$$

$$\lambda = n\mu - q, \quad 2\lambda = n\mu - \nu \tag{23}$$

it is a covariant, and in the particular case where $\nu = 0$, an invariant.

The weight λ is, as the second expression shows, always positive or zero, since $n\mu$ is at least equal to ν. The extreme value $n\mu = \nu$, or $\lambda = 0$, occurs only when in P each of the indices $1, 2, 3 \ldots n$ occurs joined only with 0, so that P is a power of $f(x)$ itself.

§. 62.
Binary cubic forms.

A binary quadratic form gives no invariant formations other than the discriminant. With these we therefore do not concern ourselves, and pass at once to the consideration of the cubic forms

$$f(x,y) = a_0x^3 + a_1x^2y + a_2xy^2 + a_3y^3 \tag{1}$$

over. Here we form, as the first quadratic covariant, the Hessian determinant. We shall here always follow the rule of writing the forms with integer numerical coefficients having no common factor. Then, in the determinant formed from the second derivatives of (1), we must discard the factor 4, and we obtain as the first covariant

$$H = \begin{vmatrix} 3\,a_0\,x + a_1\,y, & a_1\,x + a_2\,y \\ a_1\,x + a_2\,y, & a_2\,x + 3\,a_3\,y \end{vmatrix}.$$

or

$$H = A_0\,x^2 + A_1\,x\,y + A_2\,y^2, \tag{2}$$

if for abbreviation

$$\begin{aligned} A_0 &= 3\,a_0\,a_2 - a_1^2 \\ A_1 &= 9\,a_0\,a_3 - a_1\,a_2 \\ A_2 &= 3\,a_1\,a_3 - a_2^2 \end{aligned} \tag{3}$$

is set.

The discriminant of this quadratic form is an invariant of the given cubic form. It contains, however, the factor 3 in all its numerical coefficients, and accordingly we set

$$3\,D = 4\,A_0\,A_2 - A_1^2, \tag{4}$$

and then D agrees with the discriminant of the cubic form [§. 46, (10)]:

$$D = a_1^2\,a_2^2 + 18\,a_0\,a_1\,a_2\,a_3 - 4\,a_0\,a_2^3 - 4\,a_1^3\,a_3 - 27\,a_0^2\,a_3^2. \tag{5}$$

We obtain a further cubic covariant if we form the functional determinant of $f(x,y)$ and $H(x,y)$:

$$Q(x,y) = f'(x)\,H'(y) - f'(y)\,H'(x) \tag{6}$$

or

$$Q = \begin{vmatrix} 3\,a_0\,x^2 + 2\,a_1\,x\,y + a_2\,y^2, & 2\,A_0\,x + A_1\,y \\ a_1\,x^2 + 2\,a_2\,x\,y + 3\,a_3\,y^2, & A_1\,x + 2\,A_2\,y \end{vmatrix}, \tag{7}$$

in which no numerical factor can be cancelled. We set down here, in order, the computed coefficients of x^3, x^2y, xy^2, y^3, whose formation presents no difficulty:

$$\begin{aligned} &27\,a_0^2\,a_3 - 9\,a_0\,a_1\,a_2 + 2\,a_1^3, \\ &27\,a_0\,a_1\,a_3 - 18\,a_0\,a_2^2 + 3\,a_1^2\,a_2, \\ -&27\,a_0\,a_2\,a_3 + 18\,a_1^2\,a_3 - 3\,a_1\,a_2^2, \\ -&27\,a_0\,a_3^2 + 9\,a_1\,a_2\,a_3 - 2\,a_2^3. \end{aligned}$$

The behaviour of H, D, Q under linear transformation follows from §. 60, (2), (3); if H', D', Q' are the corresponding constructions for a transformed form, one has

$$H' = r^2\,H, \quad D' = r^6\,D, \quad Q' = r^3\,Q. \tag{8}$$

We can use the covariants to transform the cubic form to a normal form which at the same time gives the solution of the cubic equation.

We choose the normal form

$$f(x,y) = F(\xi,\eta) = \xi^3 + \eta^3, \tag{9}$$

in which ξ, η are linear functions of x, y; a form which, as will appear presently, can always be produced when D does not vanish, that is, when the equation $f = 0$ does not have two equal roots. The solutions of $f = 0$ are then obtained from the linear equations

$$\xi + \eta = 0, \quad \xi + \varepsilon\,\eta = 0, \quad \xi + \varepsilon^2\,\eta = 0,$$

in which ε is a third root of unity.

To form the expressions H', D', Q' for the normal form (9), we have to set $a'_0 = a'_3 = 1$, $a'_1 = a'_2 = 0$, and obtain

$$H' = 9\,\xi\,\eta, \quad D' = -27, \quad Q' = 27\,(\xi^3 - \eta^3).$$

If the normal form (9) is derived from the general form $f(x,y)$ by linear transformation, then by (8) we obtain

$$\begin{aligned} 9\,\xi\,\eta &= r^2\,H \\ -27 &= r^6\,D \\ 27\,(\xi^3 - \eta^3) &= r^3\,Q \\ \xi^3 + \eta^3 &= f. \end{aligned} \tag{10}$$

From this it follows that

$$r^6 = \frac{-27}{D}, \quad 2\,\xi^3 = \frac{r^3\,Q}{27} + f, \quad 2\,\eta^3 = -\frac{r^3\,Q}{27} + f,$$

hence, if for r^3 the value $-9 : \sqrt{-3\,D}$ from the first equation is substituted,

$$\begin{aligned} 6\,\xi^3\,\sqrt{-3\,D} &= +Q + 3\,\sqrt{-3\,D}\,f \\ 6\,\eta^3\,\sqrt{-3\,D} &= -Q + 3\,\sqrt{-3\,D}\,f, \end{aligned} \tag{11}$$

from which ξ and η are determined by two cube roots, of which, by the first equation (10), the one is determined by the other.

In this also lies the proof that, if D is different from zero, the normal form can be produced by a linear transformation. For under this assumption H decomposes into two linear factors distinct from one another, and if one chooses these, apart from constant factors, as the new variables ξ, η of a linear transformation, then $A'_0 = 0$, $A'_2 = 0$; but it follows from the identical relations

$$3\,a_3\,A_0 + a_1\,A_2 = a_2\,A_1, \quad a_2\,A_0 + 3\,a_0\,A_2 = a_1\,A_1,$$

unless at the same time $A'_1 = 0$, which is excluded by the non-vanishing D, that $a'_1 = 0$, $a'_2 = 0$, i. e. the transformed form of f contains only the cubes of ξ and η.

If one raises the first equation (10) to the cube, and eliminates ξ^3, η^3, r^6 by means of (11) and the second equation (10), one obtains among the covariants the following identical relation

$$4\,H^3 + Q^2 + 27\,D\,f^2 = 0. \tag{12}$$

To carry out the transformation into the normal form, i. e. actually to find the functions ξ, η, one decomposes the function H into its linear factors

$$4\,A_0\,H = (2\,A_0\,x + A_1\,y + \sqrt{-3\,D}\,y)\,(2\,A_0\,x + A_2\,y - \sqrt{-3\,D}\,y).$$

Then ξ, η differ from these two factors of H only by a constant factor each. We therefore set, if we denote two constant factors by h, k,

$$\begin{aligned} 2\,\xi &= h\,(2\,A_0\,x + A_1\,y - \sqrt{-3\,D}\,y) \\ 2\,\eta &= k\,(2\,A_0\,x + A_1\,y + \sqrt{-3\,D}\,y), \end{aligned} \tag{13}$$

from which, by multiplication, with regard to the first two equations (10), there results

$$3\,h\,k\,A_0\,\sqrt[3]{-D} = 1, \tag{14}$$

and the equations (11) give, by comparison of the coefficients of x^3,

$$\begin{aligned} 6\,h^3\,A_0^3\,\sqrt{-3\,D} &= +\,q_0 + 3\,\sqrt{-3\,D}\,a_0 \\ 6\,k^3\,A_0^3\,\sqrt{-3\,D} &= -\,q_0 + 3\,\sqrt{-3\,D}\,a_0, \end{aligned} \tag{15}$$

where q_0 is the coefficient of x^3 in Q, that is

$$q_0 = 27\,a_0^2\,a_3 - 9\,a_0\,a_1\,a_2 + 2\,a_1^3 \tag{16}$$

.

That the signs of $\sqrt{-3\,D}$ are to be taken, with regard to the signs in (11), as has been done in the formulas (13), follows most simply if one carries out the computation for the special case $a_0 = 0$, $a_3 = 0$.

§. 63.

The full system of forms of the binary cubic form.

We wish to show further that the invariants and covariants of the cubic form are exhausted by the functions D, f, H, Q, i. e. that all invariants and covariants of a cubic form can be expressed rationally in terms of these four forms.

So let

$$C = C(x, y, a)$$

be a covariant of degree ν in the variables and of degree μ in the coefficients, of which we shall assume that it has only integer numerical coefficients.

Then, for any transformed form,

$$C' = C(\xi, \eta, a') = r^\lambda\,C(x, y, a), \tag{1}$$

$$\lambda = \frac{3\,\mu - \nu}{2}. \tag{2}$$

We now understand by ξ, η the variables of the normal form considered above. If ε denotes a third root of unity, then

$$\xi = \varepsilon\,\xi', \quad \eta = \varepsilon^2\,\eta'$$

is a linear substitution with determinant 1, by which the normal form §. 62, (9) passes into itself. It follows, therefore, that C' cannot change when ξ, η are replaced by $\varepsilon\,\xi$, $\varepsilon^2\,\eta$; hence C' can depend rationally only on $\xi\,\eta$, ξ^3, η^3.

The permutation of ξ with η corresponds to a substitution with determinant -1; under it, therefore, C' remains unchanged or changes its sign, according as λ is even or odd. Hence C' consists of a sum of terms of the form

$$M\,(\xi\,\eta)^h\,(\xi^{3k} \pm \eta^{3k}),$$

in which M is an integer factor (since we assume the numerical coefficients in C, and hence also in C', to be

have set) and the upper or lower sign holds according as λ is even or odd. In the latter case this expression is divisible by $\xi^3 - \eta^3$, and the quotient can be expressed rationally and in integers through $\xi^3 + \eta^3$ and $\xi\eta$ (because a symmetric function of two quantities can be represented in this way through the sum and the product). If we therefore set $\tau = 0$ or $= 1$ according as λ is even or odd, there results for C' an expression of the form

$$C'(\xi, \eta) = (\xi^3 - \eta^3)^\tau \, \Sigma M(\xi\eta)^\alpha (\xi^3 + \eta^3)^\beta,$$

in which the M are integer coefficients and α, β run through a series of positive integers which satisfy the condition

$$3\tau + 3\beta + 2\alpha = \nu \tag{3}$$

From (1) there results, then, if we multiply by a sufficiently high power of 3 and express $\xi\eta$, $\xi^3 + \eta^3$, $\xi^3 - \eta^3$ by H, f, Q according to §. 62, (10), and r by $D^{-\frac{1}{6}}$,

$$3^\sigma C = Q^\tau \, \Sigma M D^\gamma H^\alpha f^\beta, \tag{4}$$

in which

$$6\gamma = \lambda - 3\tau - 2\alpha, \tag{5}$$

and in which the M are likewise integer factors.

To each value of γ there belongs, by (5), a definite value of α, and by (3) a definite value of β; likewise the two remaining numbers are in each case determined by a value of α or of β, so that in (4) every exponent of D, H or f occurs only once.

That γ is an integer follows easily from (2), (3), (5); for since by the definition $\lambda - 3\tau$ is an even number, 3γ is first of all an integer, and since further, by (3),

$$\lambda - 2\alpha = \frac{3}{2}(\mu - \nu + 2\beta + 2\tau),$$

6γ, and hence 3γ, is divisible by 3. From this it follows that σ too is an integer.

That γ must, however, also be positive, we see as follows:

Suppose that on the right-hand side of (4) negative powers of D also occur; then we multiply this equation on both sides by so high a power of D that on the right no negative powers of D appear any longer, but that the right-hand side also does not acquire the factor D. If in the equation so obtained we set

$$a_0 = 0, \quad a_1 = 1, \quad a_2 = 0, \quad a_3 = 0, \tag{6}$$

then

$$D = 0, \quad H = -x^2, \quad f = x^2 y, \quad Q = 2x^3. \tag{7}$$

Thus the left side would vanish while the right side does not vanish, and in this lies a contradiction.

But finally we can also prove that on the left side of (4) no power of 3 can occur which is not contained in all the factors M, and which therefore cancels out.

In §. 2 we proved the theorem that the product of two primitive integral rational functions of arbitrary variables is again a primitive function. Here a primitive function is understood to be one whose coefficients are integers without a common divisor. The point now is to show that a sum of the form

$$Q^\tau \, \Sigma M D^\gamma H^\alpha f^\beta,$$

in which the M are integers without a common divisor, cannot become imprimitive, and in particular cannot acquire the factor 3, when Q, D, H, f are replaced by their expressions in the a, x, y. By the theorem just mentioned it suffices, since Q is a primitive function, to show this for the form

$$\Sigma M D^{\gamma} H^{\alpha} f^{\beta}. \tag{8}$$

In this we may moreover omit all terms whose M already has the factor 3, and finally, again by the theorem mentioned, we may assume that the expression is not divisible by D, and hence that it contains a term with $\gamma = 0$. Suppose then that under these assumptions the expanded expression (8) has the divisor 3, and now substitute the particular values (6), (7), i. e. set $D = 0$, $H = -x^2$, $f = x^2 y$; then the expression reduces to the single term in which $\gamma = 0$, $2\alpha = \lambda$,

$$(-1)^{\alpha} M x^{2(\alpha+\beta)} y^{\beta}$$

and it would therefore follow that this M must have the divisor 3, which is contrary to the assumption. Thus we have proved:

Every integer covariant of the cubic form can be expressed integrally and rationally and with integer coeffi-

cients M be represented by one of the two expressions

$$\Sigma M D^{\gamma} H^{\alpha} f^{\beta}, \quad Q \Sigma M D^{\gamma} H^{\alpha} f^{\beta}.$$

The invariants are contained among the covariants as a special case, and are all powers of D.

§. 64.
Biquadratic Forms.

We now pass on to the consideration of the following biquadratic form:

$$f(x, y) = a_0 x^4 + a_1 x^3 y + a_2 x^2 y^2 + a_3 x y^3 + a_4 y^4 \tag{1}$$

We have first the Hessian determinant as a covariant of the fourth order, which, in order to free it of a numerical factor, we divide by 3:

$$H = \frac{1}{3} \begin{vmatrix} 12a_0 x^2 + 6a_1 xy + 2a_2 y^2, & 3a_1 x^2 + 4a_2 xy + 3a_3 y^2 \\ 3a_1 x^2 + 4a_2 xy + 3a_3 y^2, & 2a_2 x^2 + 6a_3 xy + 12a_4 y^2 \end{vmatrix}, \tag{2}$$

or

$$H = A_0 x^4 + A_1 x^3 y + A_2 x^2 y^2 + A_3 x y^3 + A_4 y^4,$$

in which

$$\begin{aligned} A_0 &= 8a_0 a_2 - 3a_1^2, & A_4 &= 8a_2 a_4 - 3a_3^2, \\ A_1 &= 24a_0 a_3 - 4a_1 a_2, & A_3 &= 24a_1 a_4 - 4a_2 a_3, \\ A_2 &= 48a_0 a_4 + 6a_1 a_3 - 4a_2^2, \end{aligned}$$

and a covariant of the sixth degree

$$T = \frac{1}{12} \begin{vmatrix} f'(x), & f'(y) \\ H'(x), & H'(y) \end{vmatrix}. \tag{3}$$

The invariants of the biquadratic form are most simply formed, following §. 61, from the differences of the roots.

We thus obtain an invariant of the second order and one of the third order in the coefficients:

$$A = \tfrac{1}{2} a_0^2 [(12)^2 (34)^2 + (13)^2 (24)^2 + (14)^2 (23)^2], \tag{4}$$

$$B = a_0^3\,\Sigma\,(12)^2(34)^2(13)(42), \tag{5}$$

expressions which can be written more perspicuously if one sets

$$U = (12)(34), \quad V = (13)(42), \quad W = (14)(23) \tag{6}$$

namely

$$\begin{aligned} A &= \tfrac{1}{2}a_0^2(U^2 + V^2 + W^2) \\ B &= a_0^3[U^2(V - W) + V^2(W - U) + W^2(U - V)] \\ &= a_0^3(W - V)(U - W)(V - U). \end{aligned} \tag{7}$$

For the calculation of these quantities the necessary formulas have already been developed in §. 47. By the formulas (10) given there we obtain

$$U = v^2 - w^2, \quad V = w^2 - u^2, \quad W = u^2 - v^2,$$

if u^2, v^2, w^2 are the roots of the cubic resolvent

$$z^3 + 2az^2 + (a^2 - 4c)z - b^2 = 0$$

and from this it follows that

$$\begin{aligned} A &= a_0^2[(u^2 + v^2 + w^2)^2 - 3(v^2w^2 + w^2u^2 + u^2v^2)] \\ &= a_0^2(a^2 + 12c) \\ B &= a_0^3(2u^2 - v^2 - w^2)(2v^2 - w^2 - u^2)(2w^2 - u^2 - v^2) \\ &= a_0^3(3u^2 + 2a)(3v^2 + 2a)(3w^2 + 2a) \\ &= a_0^3(2a^3 - 72ac + 27b^2), \end{aligned}$$

so that A, B have the same meaning as in §. 47, and, expressed in terms of the coefficients a_0, a_1, a_2, a_3, a_4, are represented thus:

$$A = a_2^2 - 3a_1a_3 + 12a_0a_4 \tag{8}$$

$$B = 27a_1^2a_4 + 27a_0a_3^2 + 2a_2^3 - 72a_0a_2a_4 - 9a_1a_2a_3. \tag{9}$$

The discriminant of the biquadratic equation as well, as a third invariant, though one dependent on A and B,

$$D = a_0^6\,U^2V^2W^2 \tag{10}$$

we have already formed at the place mentioned, and found

$$27D = 4A^3 - B^2. \tag{11}$$

By the formula of §. 67, $2\lambda = n\mu - \nu$, we obtain for the invariant constructions found so far, which we denote for a transformed form with accents, the following relations

$$\begin{aligned} H' &= r^2H, \quad T' = r^3T, \\ A' &= r^4A, \quad B' = r^6B, \quad D' = r^{12}D. \end{aligned} \tag{12}$$

§. 65.
Solution of the biquadratic equation.

We wish to produce a normal form by a linear substitution with determinant 1; for its derivation we assume the given biquadratic form decomposed into linear factors :

$$f(x,\,y) = a_0\,(x-\alpha y)\,(x-\beta y)\,(x-\gamma y)\,(x-\delta y). \tag{1}$$

We set

$$\begin{aligned} x\,\sqrt{\alpha-\beta} &= \beta\,\xi - \alpha\,\eta, \quad r=1,\\ y\,\sqrt{\alpha-\beta} &= \xi-\eta, \end{aligned} \tag{2}$$

hence

$$\begin{aligned} x-\alpha y &= -\sqrt{\alpha-\beta}\,\xi\\ x-\beta y &= -\sqrt{\alpha-\beta}\,\eta\\ (x-\gamma y)\,\sqrt{\alpha-\beta} &= (\beta-\gamma)\,\xi - (\alpha-\gamma)\,\eta\\ (x-\delta y)\,\sqrt{\alpha-\beta} &= (\beta-\delta)\,\xi - (\alpha-\delta)\,\eta, \end{aligned} \tag{3}$$

and thereby $f(x,\,y)$ passes over into

$$F(\xi,\,\eta) = a_1'\,\xi^3\eta + a_2'\,\xi^2\eta^2 + a_3'\,\xi\eta^3. \tag{4}$$

The coefficients a_1', a_2', a_3' are easily expressed by (3) in terms of the $\alpha, \beta, \gamma, \delta$:

$$\begin{aligned} a_1' &= \quad a_0\,(\beta-\gamma)\,(\beta-\delta)\\ a_3' &= \quad a_0\,(\alpha-\gamma)\,(\alpha-\delta)\\ a_2' &= -a_0\,[(\alpha-\gamma)\,(\beta-\delta) + (\alpha-\delta)\,(\beta-\gamma)] \end{aligned} \tag{5}$$

By permutation of the roots we can produce the normal form (4) in six different ways, which however yield only three different values of a_2'. These three values are, if as above

$$\begin{aligned} U &= (\alpha-\beta)\,(\gamma-\delta),\\ V &= (\alpha-\gamma)\,(\delta-\beta),\\ W &= (\alpha-\delta)\,(\beta-\gamma),\\ U+V+W &= 0 \end{aligned} \tag{6}$$

is set,

$$\begin{aligned} a_2' &= a_0\,(V-W),\\ a_2'' &= a_0\,(W-U),\\ a_2''' &= a_0\,(U-V). \end{aligned} \tag{7}$$

If the a_2', a_2'', a_2''' are known, then the U, V, W can also be determined by

$$\begin{aligned} 3\,a_0\,U &= a_2''' - a_2''\\ 3\,a_0\,V &= a_2' - a_2'''\\ 3\,a_0\,W &= a_2'' - a_2', \end{aligned} \tag{8}$$

from which it appears that if two of the three quantities a_2', a_2'', a_2''' are equal to each other, then one of the quantities U, V, W vanishes , and hence two of the roots of the biquadratic equation are equal to each other, and consequently its discriminant vanishes.

If we further set

$$\begin{aligned} \alpha\beta + \gamma\delta &= u,\\ \alpha\gamma + \delta\beta &= v,\\ \alpha\delta + \beta\gamma &= w, \end{aligned} \tag{9}$$

then

$$a_0\,(u+v+w) = a_2,$$

[the sum of the products of the roots of $f(x,\,y)$ taken two at a time],

$$\begin{aligned} U &= v - w, \\ V &= w - u, \\ W &= u - v, \end{aligned}$$

and hence

$$\begin{aligned} 3\,a_0\,u &= a_2 - a_2', \\ 3\,a_0\,v &= a_2 - a_2'', \\ 3\,a_0\,w &= a_2 - a_2'''. \end{aligned} \tag{10}$$

Thus with the quantities a_2', a_2'', a_2''' the u, v, w are known at the same time. But if one knows the quantities u, v, w, then the solution of the biquadratic equation is reduced to square roots. For we have

$$\alpha\beta + \gamma\delta = u, \quad a_0\,\alpha\beta\,.\,\gamma\delta = a_4$$

and from this

$$2\,\alpha\beta = v_1 + \sqrt{\frac{a_0\,u^2 - 4\,a_4}{a_0}}, \quad 2\,\gamma\delta = v_1 - \sqrt{\frac{a_0\,u^2 - 4\,a_4}{a_0}}.$$

Since one can now in the same way determine the products

$$\alpha\gamma, \quad \alpha\delta, \quad \beta\gamma, \quad \beta\delta$$

can be determined, then α^2, for example, can be computed from the following quotient:

$$\frac{\alpha\beta\,.\,\alpha\gamma}{\beta\gamma}$$

It now remains, therefore, to form the cubic equation whose roots are a_2', a_2'', a_2'''. But this is obtained at once from the invariants.

If one forms the invariants A', B' from the transformed form (4), then by §. 64, (8), (9), (12), since $r = 1$, it follows that

$$\begin{aligned} A &= a_2'^2 - 3\,a_1'\,a_3', \\ B &= 2\,a_2'^3 - 9\,a_1'\,a_2'\,a_3', \\ D &= a_1'^2\,a_2'^2\,a_3'^2 - 4\,a_1'^3\,a_3'^3, \end{aligned} \tag{11}$$

and from these, by elimination of $a_1'\,a_3'$,

$$a_2'^3 - 3\,a_2'\,A + B = 0.$$

Hence a_2', a_2'', a_2''' are the roots of the equation, cubic in z,

$$z^3 - 3\,A\,z + B = 0. \tag{12}$$

This is surely the simplest form that can be given to the cubic resolvent of the biquadratic equation.

§. 66.
The Covariants.

If we form the covariant H, according to §. 64, (2), for the transformed form

$$f(x,\,y) = F(\xi,\,\eta) = a_1'\,\xi^3\eta + a_2'\,\xi^2\eta^2 + a_3'\,\xi\eta^3 \tag{1}$$

we obtain

$$\begin{aligned} -H = {} & 3\,a_1'^2\,\xi^4 + 4\,a_1'\,a_2'\,\xi^3\eta - (6\,a_1'\,a_3' - 4\,a_2'^2)\,\xi^2\eta^2 \\ & + 4\,a_2'\,a_3'\,\xi\eta^3 + 3\,a_3'^2\,\eta^4, \end{aligned} \tag{2}$$

and from this it follows

$$H + 4\,a_2'\,f = -3\,(a_1'\,\xi^2 - a_3'\,\eta^2)^2. \tag{3}$$

This combination is thus, apart from the factor -3, the square of a quadratic form, and the same holds if we replace a_2' by a_2'', a_2'''. We therefore set, as an abbreviation,

$$\begin{aligned} H + 4\,a_2'\,f &= -3\,\psi_1^2 \\ H + 4\,a_2''\,f &= -3\,\psi_2^2 \\ H + 4\,a_2'''\,f &= -3\,\psi_3^2. \end{aligned} \tag{4}$$

Of the three functions ψ_1, ψ_2, ψ_3 no two can have a common divisor, provided the discriminant D is assumed different from zero. For then the three quantities a_2', a_2'', a_2''' are also different from one another. If then ψ_1 and ψ_2 vanish, H and f must vanish, i. e. a common divisor of ψ_1 and ψ_2 would have to be a common divisor of H and f. But now ξ was an arbitrary linear divisor of f. By (2) this can be a divisor of H only if $a_3' = 0$, that is, again, only if the discriminant D vanishes [§. 65, (11)].

Now the functional determinant T of f and H is identical with the functional determinant of f and $H + \lambda f$, whatever λ may be, and from this it follows, when $\lambda = 4\,a_2'$ is set,

$$T = -\tfrac{1}{2}\,\psi_1\,[F'(\xi)\,\psi_1'(\eta) - F'(\eta)\,\psi_1'(\xi)].$$

Thus T is divisible by ψ_1, and for the same reasons by ψ_2 and ψ_3, and consequently T is, up to a factor independent of ξ, η, identical with the product $\psi_1\,\psi_2\,\psi_3$. If we accordingly form the product of the three equations (4), it follows, having regard to the cubic equation §. 65, (12), which the quantities a_2', a_2'', a_2''' satisfy:

$$H^3 - 48\,A\,H\,f^2 - 64\,B\,f^3 = c\,T^2,$$

where c is a constant still to be determined. By §. 64, (12) c remains unchanged under a linear transformation, and one therefore finds its value by setting the terms with the highest power of ξ on the two sides of the normal form (1), (2) equal to each other, $c = -27$.

We thus have between the invariants and covariants the following identical relation

$$H^3 - 48\,A\,H\,f^2 - 64\,B\,f^3 = -27\,T^2. \tag{5}$$

The functions ψ_1, ψ_2, ψ_3 are likewise covariants, though with coefficients which are not rational in the coefficients of f. We can easily express them in terms of the roots $\alpha, \beta, \gamma, \delta$ of $f = 0$. If for simplification one sets $y = 1$, then by §. 65, (3) and (5) it follows that

$$\begin{aligned} \psi_1 &= a_1'\xi^2 - a_3'\eta^2 \\ &= a_0\,\frac{(\beta-\gamma)(\beta-\delta)(x-\alpha)^2 - (\alpha-\gamma)(\alpha-\delta)(x-\beta)^2}{\alpha-\beta}. \end{aligned}$$

In this expression, however, $\alpha - \beta$ can be cancelled from numerator and denominator, and ψ_1 can be represented in two forms; to obtain ψ_2 and ψ_3, one then need only permute β, γ, δ cyclically.

One thus finds

$$\begin{aligned}\frac{\psi_1}{a_0} &= (\gamma-\beta)(x-\alpha)(x-\delta) - (\alpha-\delta)(x-\beta)(x-\gamma)\\ &= (\delta-\beta)(x-\alpha)(x-\gamma) - (\alpha-\gamma)(x-\beta)(x-\delta)\end{aligned}$$

$$\begin{aligned}\frac{\psi_2}{a_0} &= (\delta-\gamma)(x-\alpha)(x-\beta) - (\alpha-\beta)(x-\gamma)(x-\delta)\\ &= (\beta-\gamma)(x-\alpha)(x-\delta) - (\alpha-\delta)(x-\gamma)(x-\beta)\\ \frac{\psi_3}{a_0} &= (\beta-\delta)(x-\alpha)(x-\gamma) - (\alpha-\gamma)(x-\delta)(x-\beta)\\ &= (\gamma-\delta)(x-\alpha)(x-\beta) - (\alpha-\beta)(x-\delta)(x-\gamma).\end{aligned} \tag{6}$$

The quantities ψ_1^2, ψ_2^2, ψ_3^2 may be regarded as the roots of a cubic equation whose coefficients are covariants. One obtains this equation easily from (4) in the form

$$\left(z + \frac{H + 4a_2' f}{3}\right)\left(z + \frac{H + 4a_2'' f}{3}\right)\left(z + \frac{H + 4a_2''' f}{3}\right) = 0.$$

By §. 65, (12), and making use of the expression (5) for T^2, this equation yields

$$z^3 + Hz^2 + \tfrac{1}{3}(H^2 - 16Af^2)z - T^2 = 0. \tag{7}$$

It is also of interest to form the discriminant of this equation. One obtains it most simply from the expression

$$\Delta = (\psi_1^2 - \psi_2^2)^2(\psi_1^2 - \psi_3^2)^2(\psi_2^2 - \psi_3^2)^2,$$

if one substitutes the expressions (4) and then, for $(a_2' - a_2'')^2(a_2' - a_2''')^2(a_2'' - a_2''')^2$, puts the discriminant of the equation §. 65, (12)

$$27(4A^3 - B^2) = 3^6 D$$

where then D is the discriminant of f. One thus obtains

$$\Delta = 2^{12} D f^6. \tag{8}$$

§. 67.
The full system of invariants of the binary biquadratic form.

We wish further to prove that with the constructs A, B the system of independent invariants of the binary biquadratic form is exhausted. But whereas one usually contents oneself with the proof that every invariant is an integral rational function of A, B, we wish, as we have already done in §. 63 for the covariants of the cubic form, also to touch on the question of the numerical coefficients, and here a new phenomenon shows itself.

The invariant D, for example, is indeed by §. 64, (11) expressed rationally in terms of A, B. But the numerical coefficients are not integers; rather they have the denominator 27, although the coefficients in the expanded function D are all integers.

We therefore now consider as integral invariants integral rational homogeneous functions of degree μ of the five variables a_0, a_1, a_2, a_3, a_4

$$I(a_0, a_1, a_2, a_3, a_4) = I(a), \tag{1}$$

whose coefficients are integers, and to which the invariant property

$$I' = r^{2\mu} I \tag{2}$$

belongs, when I' or $I(a')$ is the same function of the coefficients of a transformed function.

Such invariants are A, B, D, between which there holds the relation

$$27D = 4A^3 - B^2, \tag{3}$$

and our goal is to prove that all integral invariants are integral rational functions, with integer coefficients, of these three.

If we form the function I' in the normal form (4) of §. 65, then we first obtain from (2), since $r = 1$,

$$I(a) = I'(0, a_1', a_2', a_3', 0),$$

and this function I can depend only on a_2' and the product $a_1' a_3'$; for the substitution

$$\xi = \lambda \xi', \quad \eta = \frac{1}{\lambda} \eta',$$

whose determinant is $= 1$, leaves the normal form §. 65, (4), and in it a_2', unchanged, while a_1', a_3' pass into $\lambda^2 a_1'$, $a_3' : \lambda^2$, and here λ is an arbitrary quantity.

If, then, we set

$$a_2' = z$$

we have

$$I = \varphi(a_1' a_3', z), \tag{4}$$

where φ is an integral, rational function with integer coefficients of the two arguments $a_1' a_3'$ and z.

But now, by §. 65, (11),

$$3 a_1' a_3' = z^2 - A,$$

and if we therefore multiply (4) by a suitable power of 3, say 3^ν, we obtain

$$3^\nu I = \psi(A, z), \tag{5}$$

where ψ is again a function of A and z with integer coefficients. But by §. 65, (12),

$$z^3 = 3Az - B,$$

and by means of this we can express all higher powers of z through the first and second, and thus obtain

$$3^\nu I = \chi(A, B, z), \tag{6}$$

where χ is at most of the second degree with respect to z.

But the left side of (6) remains unchanged when z is replaced by a_2', a_2'', a_2''', that is, by three different values, and consequently the function χ cannot contain the variable z at all any longer (§. 30, II).

Therefore

$$3^\nu I = \chi(A, B), \tag{7}$$

where χ is an integral, rational function with integer coefficients.

If in (7), by (3), we set

$$B^2 = 4A^3 - 27D$$

it follows that (7) has one of the two following forms

$$3^\nu I = \Phi(A, D), \quad B\Phi(A, D), \tag{8}$$

according as the degree of I is even or odd.

If I is a primitive function of the a, that is, one whose numerical coefficients have no common divisor, then the coefficients of the functions Φ can in any case have no common divisor other than a power of 3.

What we have to prove is that they are all divisible by 3^ν, or, what is the same thing, that if we assume $\Phi(A, D)$ to be a primitive function, then $\nu = 0$ must hold.

The question is therefore this: Can functions with the divisor 3 arise from the primitive function $\Phi(A, D)$ or $B\Phi(A, D)$ when A, B, D are replaced by their expressions in the a? Since B is primitive, by §. 2 the function $B\Phi(A, D)$, expressed in the a, can have the divisor 3 only if $\Phi(A, D)$ has it.

The question is thus reduced to this: Can the primitive function $\Phi(A, D)$ acquire the divisor 3 when A, D are replaced by their expressions?

That this question must be answered in the negative we can easily see as follows. We first imagine all those terms removed from $\Phi(A, D)$ whose coefficients are divisible by 3; for evidently the remaining terms must still retain the same property of acquiring the divisor 3 upon substitution of the expressions for A, D.

Secondly, we may assume that $\Phi(A, D)$ does not have the factor D, for by omitting this factor the property in question would, by the theorem already mentioned (§. 2), not be destroyed.

But then a number divisible by 3 would have to arise from $\Phi(A, D)$ when all the a, with the exception of a_2, are set equal to zero and $a_2 = 1$. This gives $D = 0$, $A = 1$, and so $\Phi(1, 0)$ would have to be a number divisible by 3; but this is the coefficient of the highest power of A in $\Phi(A, D)$, which by hypothesis is not divisible by 3. With this we have therefore proved:

Every integral invariant of the biquadratic form can be represented rationally and with integer coefficients by A, B, D, and indeed in one of the two forms

$$\Phi(A, D), \quad B\Phi(A, D).$$

That conversely every such expression, if it is homogeneous in the coefficients a, represents an invariant with integer coefficients, is clear of itself[10].

Sixth Section.
Tschirnhausen Transformation.

§. 68.
The Hermite Form of the Tschirnhausen Transformation.

We have already become acquainted with the basic idea of the Tschirnhausen transformation in the fourth section.

The task was this: to transform an algebraic equation of degree n

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0 \tag{1}$$

by a substitution of degree $(n - 1)$

$$y = \alpha_0 + \alpha_1 x + \alpha_2 x^2 + \cdots + \alpha_{n-1}x^{n-1} \tag{2}$$

[10]The theory of invariants is presented in detail in the works: Clebsch, "Theorie der binären algebraischen Formen". Leipzig 1872. Faà di Bruno, "Einleitung in die Theorie der binären Formen". German by Walter. Leipzig 1881. P. Gordans, "Vorlesungen über Invariantentheorie", edited by Kerschensteiner. Leipzig 1885. Cf. also Franz Meyer, "Bericht über den gegenwärtigen Stand der Invariantentheorie" in the Jahresbericht der deutschen Mathematiker- Vereinigung 1890/91 (Berlin 1892).

so as to gain, in the arbitrary coefficients α, means of simplifying the equation.

Hermite, by giving the substitution (2) a particular form, greatly simplified this task and brought it into connection with the theory of invariants[11].

In §. 4 we became acquainted with a series of functions, $f_0, f_1, f_2 \dots f_{n-1}$, by which the powers of x up to the $(n-1)$st can be expressed rationally, and it is these functions that Hermite uses for the representation of the substitution (2).

Here we denote by x a root of the equation (1), while t is to be understood as an indeterminate variable. Then by §. 4

$$\frac{f(t)}{t-x} = t^{n-1}f_0(x) + t^{n-2}f_1(x) + \cdots + tf_{n-2}(x) + f_{n-1}(x) \tag{3}$$

and

$$\begin{aligned} f_0(x) &= a_0 \\ f_1(x) &= a_0x + a_1 \\ f_2(x) &= a_0x^2 + a_1x + a_2 \\ &\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\ f_{n-1}(x) &= a_0x^{n-1} + a_1x^{n-2} + a_2x^{n-3} + \cdots + a_{n-1}. \end{aligned} \tag{4}$$

We now assume the substitution (2) in the form

$$y = t_{n-1}f_0(x) + t_{n-2}f_1(x) + \cdots + t_1f_{n-2}(x) + t_0f_{n-1}(x), \tag{5}$$

in which the $t_{n-1}, t_{n-2} \dots t_1, t_0$ are the indeterminate quantities that have taken the place of the α, and are not to be confused with the powers of t. But y arises from the expression (3) when t^k is replaced by t_k.

If, as in §. 42, we denote by the sign S placed before a function that the sum is to be taken over all the roots x of equation (1), then we have

$$\begin{aligned} S[f_0(x)] &= na_0 \\ S[f_1(x)] &= (n-1)a_1 \\ S[f_2(x)] &= (n-2)a_2 \\ &\cdots\cdots\cdots\cdots\cdots \\ S[f_{n-1}(x)] &= a_{n-1}, \end{aligned} \tag{6}$$

hence

$$S(y) = na_0t_{n-1} + (n-1)a_1t_{n-2} + \cdots + 2a_{n-2}t_1 + a_{n-1}t_0, \tag{7}$$

an expression which results from $f'(t)$ when t^k is replaced by t_k.

If with the help of (7) we eliminate the variable t_{n-1} from (5), it follows that

$$y - \frac{1}{n}S(y) = t_{n-2}F_0(x) + t_{n-3}F_1(x) + \cdots + t_0F_{n-2}(x), \tag{8}$$

if

$$\begin{aligned} F_0 &= f_1 - \frac{n-1}{n}a_1 = a_0x + \frac{a_1}{n}, \\ F_1 &= f_2 - \frac{n-2}{n}a_2 = a_0x^2 + a_1x + \frac{2a_2}{n}, \\ &\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\ F_{n-2} &= f_{n-1} - \frac{1}{n}a_{n-1} = a_0x^{n-1} + a_1x^{n-2} + \cdots + \frac{n-1}{n}a_{n-1}, \end{aligned} \tag{9}$$

[11] Hermite, Sur quelques théorèmes d'algèbre et la résolution de l'équation du quatrième degré, published separately from the Comptes rendus of the Paris Academy. Paris 1859.

so that

$$S[F_0(x)] = 0, \quad S[F_1(x)] = 0, \ \ldots \ S[F_{n-2}(x)] = 0. \tag{10}$$

If, in accordance with (3), we set

$$\begin{aligned} F(t,x) &= \frac{f(t)}{t-x} - \frac{1}{n}f'(t) \\ &= t^{n-2}F_0(x) + t^{n-3}F_1(x) + \cdots + tF_{n-3}(x) + F_{n-2}(x), \end{aligned} \tag{11}$$

then the right-hand side of (8) arises from $F(t,x)$ by the replacement of t^k by t_k.

If we assume y from the outset in the form

$$y = t_{n-2}F_0(x) + t_{n-3}F_1(x) + \cdots + t_0F_{n-2}(x) \tag{12}$$

then the equation

$$S(y) = 0 \tag{13}$$

is identically satisfied. What advantage this form of the substitution affords, the next considerations will show.

§. 69.
Invariant Property of the Tschirnhausen Transformation.

We have now to investigate the influence which a linear transformation, to which we subject the function $f(x)$, has on the Tschirnhausen transformation.

We make in $f(x)$ the linear substitution

$$x = \frac{\alpha\xi + \beta}{\gamma\xi + \delta}, \quad \alpha\delta - \beta\gamma = r, \tag{1}$$

whereby we obtain

$$\varphi(\xi) = (\gamma\xi + \delta)^n f\left(\frac{\alpha\xi + \beta}{\gamma\xi + \delta}\right), \tag{2}$$

so that $\varphi(\xi)$ is an integral rational function of degree n, and $\varphi(\xi) = 0$ is the equation $f(x) = 0$ transformed by (1).

We now derive a function $\Phi(\tau, \xi)$ from $\varphi(\xi)$ in exactly the same way as we derived $F(t,x)$ from $f(x)$ in the preceding paragraph, namely

$$\begin{aligned} \Phi(\tau,\xi) &= \frac{\varphi(\tau)}{\tau - \xi} - \frac{1}{n}\varphi'(\tau) \\ &= \tau^{n-2}\Phi_0(\xi) + \tau^{n-3}\Phi_1(\xi) + \cdots + \tau\Phi_{n-3}(\xi) + \Phi_{n-2}(\xi), \end{aligned} \tag{3}$$

in which τ, just like t, is a variable, and the functions $\Phi_0(\xi), \Phi_1(\xi) \ldots \Phi_{n-2}(\xi)$, likewise from ξ and the coefficients

of $\varphi(\xi)$, just as the corresponding functions $F_0(x), F_1(x) \cdots F_{n-2}(x)$ are formed from x and the coefficients of $f(x)$.

If now, on the other hand, we carry out in $F(t,x)$, simultaneously with the substitution (1), the substitution

$$t = \frac{\alpha\tau + \beta}{\gamma\tau + \delta} \tag{4}$$

then we obtain

$$(\gamma\tau+\delta)^n f(t)=\varphi(\tau), \tag{5}$$

$$(\gamma\xi+\delta)(\gamma\tau+\delta)(t-x)=r(\tau-\xi) \tag{6}$$

and, by forming from (5) the derivative, most simply by differentiation by means of the formula

$$\frac{dt}{d\tau}=\frac{r}{(\gamma\tau+\delta)^2}$$

we obtain

$$r(\gamma\tau+\delta)^{n-1}\frac{1}{n}f'(t)=(\gamma\tau+\delta)\frac{1}{n}\varphi'(\tau)-\gamma\varphi(\tau). \tag{7}$$

From (5), (6) and (7) it follows that

$$r(\gamma\tau+\delta)^{n-2}\frac{f(t)}{t-x}=\frac{\varphi(\tau)}{\tau-\xi}\frac{\gamma\xi+\delta}{\gamma\tau+\delta}$$

$$r(\gamma\tau+\delta)^{n-2}\frac{1}{n}f'(t)=-\frac{\gamma(\tau-\xi)\varphi(\tau)}{(\gamma\tau+\delta)(\tau-\xi)}+\frac{1}{n}\varphi'(\tau),$$

and by subtraction of the two formulas

$$r(\gamma\tau+\delta)^{n-2}\left(\frac{f(t)}{t-x}-\frac{1}{n}f'(t)\right)=\frac{\varphi(\tau)}{\tau-\xi}-\frac{1}{n}\varphi'(\tau),$$

hence

$$r(\gamma\tau+\delta)^{n-2}F(t,x)=\Phi(\tau,\xi) \tag{8}$$

or, written more explicitly,

$$\tau^{n-2}\Phi_0+\tau^{n-3}\Phi_1+\cdots+\Phi_{n-2}=r[(\alpha\tau+\beta)^{n-2}F_0+(\alpha\tau+\beta)^{n-3}(\gamma\tau+\delta)F_1+\cdots+(\gamma\tau+\delta)^{n-2}F_{n-2}]. \tag{9}$$

We now set, replacing in $F(t,x)$ the powers t^k by arbitrary variables t_k, and likewise in $\Phi(\tau,k)$ τ^k by τ_k,

$$\begin{aligned} Y(t,x)&=t_{n-2}F_0+t_{n-3}F_1+\cdots+t_0F_{n-2},\\ H(\tau,\xi)&=\tau_{n-2}\Phi_0+\tau_{n-3}\Phi_1+\cdots+\tau_0\Phi_{n-2}, \end{aligned} \tag{10}$$

so that the substitutions of the Tschirnhausen transformation

$$y=Y(t,x),\qquad \eta=H(\tau,\xi) \tag{11}$$

read [§. 68, (12)].

Equation (9), which is an identity with respect to τ, remains valid, however, when after carrying out the indicated raising to powers τ^k is replaced on the right by τ_k, and it therefore teaches that between the functions $Y(t,x)$, $H(\tau,\xi)$ the relation holds

(12) $$H(\tau,\xi) = r\,Y(t,x),$$

if we let the t depend on the τ in the following way: Set

(13) $$\begin{aligned} t_{n-2} &= (\alpha\tau+\beta)^{n-2} \\ t_{n-3} &= (\alpha\tau+\beta)^{n-3}(\gamma\tau+\delta) \\ &\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\ t_0 &= (\gamma\tau+\delta)^{n-2}, \end{aligned}$$

and after carrying out the powers replace τ^k by τ_k.

Now let z be an arbitrary variable, and before we carry out the raising to powers in (13), let us multiply these equations in order by

$$1, \quad -(n-2)z, \quad \frac{(n-2)(n-3)}{1\,.\,2}z^2, \cdots \pm z^{n-2};$$

then on the left we obtain an integral rational function of z of degree $(n-2)$

(14) $$\begin{aligned} T(z) &= t_{n-2} - (n-2)t_{n-3}z \\ &\quad + \frac{(n-2)(n-3)}{1\,.\,2}t_{n-4}z^2 - \cdots \pm t_0 z^{n-2}, \end{aligned}$$

and the right side yields, by the binomial theorem,

$$[(\alpha\tau+\beta) - z(\gamma\tau+\delta)]^{n-2} = [\tau(\alpha-\gamma z) + (\beta-\delta z)]^{n-2},$$

which, if we set

(15) $$z = \frac{\alpha\xi+\beta}{\gamma\xi+\delta}, \qquad \xi = \frac{\delta z-\beta}{-\gamma z+\alpha}$$

goes over into

$$(\alpha-\gamma z)^{n-2}(\tau-\xi)^{n-2} = \frac{r^{n-2}(\tau-\xi)^{n-2}}{(\gamma\xi+\delta)^{n-2}}$$

If we carry out the power $(\tau-\xi)^{n-2}$ here by the binomial theorem, and then replace τ^k by τ_k, we obtain a transformation of the function $T(z)$. We therefore set

$$\begin{aligned} \Theta(\xi) &= \tau_{n-2} - (n-2)\tau_{n-3}\xi \\ &\quad + \frac{(n-2)(n-3)}{1\,.\,2}\tau_{n-4}\xi^2 + \cdots \pm \tau_0\xi^{n-2} \end{aligned}$$

and obtain the identical transformation

(16) $$(\gamma\xi+\delta)^{n-2}\,T\left(\frac{\alpha\xi+\beta}{\gamma\xi+\delta}\right) = r^{n-2}\Theta(\xi),$$

in which the connection between the coefficients t and τ is expressed by the symbolic equations (13), and is therefore the same as in the functions Y and H.

Accordingly, $r^{n-2}\Theta(\xi)$ is the transformation of the function $(\gamma\xi+\delta)^{n-2}T(z)$ by the same linear substitution by which $(\gamma\xi+\delta)^n f(z)$ passes over into $\varphi(\xi)$.

If we denote the coefficients of this transformed function, that is, the quantities $r^{n-2}\tau_k$, by t', then equation (12), which is linear with respect to τ, yields

$$H(t',\xi) = r^{n-1}Y(t,x). \tag{17}$$

Now from $Y(t,x)$ there arise n functions when one puts for x the n roots of $f(x)$. Every homogeneous, rational, symmetric function of these n functions is rationally expressible in terms of the coefficients of $f(x)$; and if we therefore denote such a function by $K(t,a)$, then equation (17) yields, if we denote the coefficients of $\varphi(\xi)$ by a',

$$K(t',a') = r^{\nu(n-1)}K(t,a), \tag{18}$$

where ν denotes the degree of the symmetric function. With this the beautiful theorem of Hermite is proved:

The coefficients in the equation $f(x)=0$ transformed by the Tschirnhausen transformation

$$y = Y(t,x)$$

and all symmetric functions of the n values y are simultaneous invariants of the two functions

$$f(z), \quad T(z).$$

§. 70.
Developments concerning the Hermite theorem.

From the considerations of the last paragraph it has emerged that if we set

$$y = t_{n-2}F_0(x) + t_{n-3}F_1(x)\cdots + t_0F_{n-2}(x) \tag{1}$$

then the symmetric functions of the n values $y_1, y_2\cdots y_n$, which one obtains from y when one puts for x the n roots of

$f(x)=0$, are simultaneous invariants of two forms of degrees n and $(n-2)$, $f(z)$, $T(z)$.

Such an integral rational and homogeneous symmetric function of order ν

$$P(y_1, y_2\cdots y_n)$$

is, by (1), evidently a homogeneous function of order ν of the variables t. It is likewise a homogeneous function of order ν in the a, since the expression (1), so far as it contains the a explicitly, is linear in them, and since the symmetric functions of the x depend only on the ratios $a_1 : a_0, a_2 : a_0, \ldots a_n : a_0$. In the expression of P by the a, however, a power of a_0 might possibly remain in the denominator, and we still have to show that this does not occur.

If we set

$$a_0^\lambda P(y_1, \ldots y_n) = K(t,a)$$

and determine the power a_0^λ so that $K(t,a)$ is an integral function of the a, but no longer vanishes for $a_0 = 0$, then it will follow that $\lambda = 0$ must hold, if we show that for $a_0 = 0$ all the y retain finite values.

If we let $a_0 = 0$, while the remaining a stay unchanged, then one of the roots x will, as we have seen in §. 40, grow infinite. But that the corresponding y nevertheless remains finite follows from the equation §. 68, (11), which is an identity with respect to t,

$$\frac{f(t)}{t-x} - \frac{1}{n}f'(t) = t^{n-2}F_0(x) + t^{n-3}F_1(x) + \cdots + F_{n-2}(x), \tag{2}$$

from which it can be seen that, when $a_0 = 0$ and x becomes infinite, the functions

$$F_0(x), \quad F_1(x) \cdots F_{n-2}(x)$$

become equal to the coefficients of $-\frac{1}{n}f'(t)$, that is, equal to

$$0, \quad -\frac{n-1}{n}a_1 \ldots, \quad -\frac{1}{n}a_{n-1}$$

Thus the y also remain finite, and $P(y_1, y_2 \cdots y_n)$ is equal to an integral rational invariant of degree ν both in the t and in the a.

If we therefore assume the equation for y in the form

$$y^n + P_2 y^{n-2} + P_3 y^{n-3} + \cdots + P_n = 0, \tag{3}$$

then P_ν is such an invariant of degree ν.

An invariant of degree $n(n-1)$ is also the discriminant Δ of equation (3), or the square of the difference product

$$\begin{aligned} \Pi = (y_1 - y_2)(y_1 - y_3) \cdots \\ (y_2 - y_3) \cdots \end{aligned} \tag{4}$$

To obtain closer insight into the formation of this quantity, we recall that we obtain y when, in

$$\frac{f(t)}{t-x} - \frac{1}{n}f'(t)$$

we replace t^k by t_k. We therefore obtain $y_1 - y_2$ from the formula

$$y_1 - y_2 = (x_1 - x_2)\frac{f(t)}{(t-x_1)(t-x_2)}, \tag{5}$$

when we make the same replacement.

The quotient

$$\frac{f(t)}{(t-x_i)(t-x_k)}$$

is an integral rational function of t of degree $n-2$. If we replace t^k in it by t_k, let it pass over into $t_{i,k}$; we then have

$$y_i - y_k = (x_i - x_k)t_{i,k}. \tag{6}$$

If we therefore set

$$a_0^{n-1}\Theta = t_{1,2}t_{1,3}\dots t_{1,n}\,t_{2,3}\dots t_{2,n}\dots t_{n-1,n} \tag{7}$$

then Θ is a homogeneous integral function of degree $\frac{1}{2}n(n-1)$ with respect to the t, and moreover, as a symmetric function of the x, it can be expressed rationally in terms of the coefficients a.

But from (6) it follows, if D, as in §. 46, (3), denotes the discriminant of $f(x)$, that

$$\Delta = D\Theta^2, \tag{8}$$

from which it is to be concluded that Θ is an invariant which no longer contains the denominator a_0. Δ is of degree $n(n-1)$ in the coefficients a, while D by §. 46 is of degree $2n-2$; from this it follows that Θ is of degree $\frac{1}{2}(n-1)(n-2)$ in the coefficients a.

Θ is a so-called decomposable form of degree $\frac{1}{2}n(n-1)$ in the variables t; for by (7) it can be decomposed entirely into linear factors, which, to be sure, are not rational in the a.

We have already pointed out in §. 52 that in the Tschirnhausen transformation x is also expressible rationally in terms of y; and the same therefore holds for every rational function of x as well.

If we consider any such function $\varphi(x)$, which may also contain the coefficients a and t, but is always assumed to be integral and homogeneous in both kinds of variables, then we may set

$$\varphi(x) = C_0 + C_1 y + C_2 y^2 + \dots + C_{n-1}y^{n-1}, \tag{9}$$

in which the $C_0, C_1, \dots C_{n-1}$ are expressible rationally in a and in t.

If we write down equation (9) for $x = x_1, x_2 \dots x_n$, $y = y_1, y_2, \dots y_n$, we obtain for the determination of the C a system of linear equations whose determinant

$$\begin{vmatrix} 1, & y_1, & y_1^2 & \dots & y_1^{n-1} \\ 1, & y_2, & y_2^2 & \dots & y_2^{n-1} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 1, & y_n, & y_n^2 & \dots & y_n^{n-1} \end{vmatrix}, \tag{10}$$

is therefore equal to $\Theta\sqrt{D}$. The numerator of the expression for C_ν arises from the determinant (10) by replacing the elements of the $(\nu+1)$th column by $\varphi(x_1), \varphi(x_2) \dots \varphi(x_n)$, and consequently the numerator, when everything is expressed in terms of the x_i, has the factor $\sqrt{D}$ (because it vanishes when two of the x_i become equal to each other).

We may therefore set

$$\Theta C_\nu = Q_\nu,$$

in which Q_ν is an integral rational function of the t and the a, which can at most still contain a power of a_0 in the denominator. We then obtain

$$\Theta\varphi(x) = Q_0 + Q_1 y + Q_2 y^2 + \dots + Q_{n-1}y^{n-1}. \tag{11}$$

The coefficients Q_ν, however, are not invariants, and their computation is in most cases difficult.

§. 71.
Transformation of the cubic equation.

The theorem of Hermite (§. 69) suffices, for the cubic equation, to carry out the transformation without further computation.

For this purpose, if we employ homogeneous variables z_1, z_2, we have to form simultaneous invariants of the cubic form

$$f(z_1, z_2) = a_0 z_1^3 + a_1 z_1^2 z_2 + a_2 z_1 z_2^2 + a_3 z_2^3 \tag{1}$$

and of the linear form

$$T = -t_0 z_1 + t_1 z_2 \tag{2}$$

But if we apply the linear substitution

$$\begin{aligned} z_1 &= \alpha z_1' + \beta z_2' \\ z_2 &= \gamma z_1' + \delta z_2' \end{aligned} \tag{3}$$

to T, there results

$$-t_0 z_1 + t_1 z_2 = -t_0' z_1' + t_1' z_2',$$

provided that

$$\begin{aligned} t_0' &= +t_0\alpha - t_1\gamma, \\ t_1' &= -t_0\beta + t_1\delta, \end{aligned}$$

or

$$\begin{aligned} r t_1 &= \alpha t_1' + \beta t_0' \\ r t_0 &= \gamma t_1' + \delta t_0'. \end{aligned} \tag{4}$$

But this substitution arises from (3) when one replaces z_1, z_2, z_1', z_2' by rt_1, rt_0, t_1', t_0'. Now if $I(a, t_1, t_0)$ is a simultaneous invariant of f and T, homogeneous and of degree ν in t_0, t_1 and of weight λ, so that

$$I(a', t_1', t_0') = r^\lambda I(a, t_1, t_0),$$

then by this permutation it follows that

$$I(a', z_1', z_2') = r^{\lambda-\nu} I(a, z_1, z_2),$$

i. e. $I(a, z_1, z_2)$ is a covariant of f.

One therefore obtains all simultaneous invariants of f and T from the covariants of f, by replacing z_1, z_2 in them by t_1, t_0.

But the covariants of f we have come to know completely in §. 62, 63.

After this it is easy to form the cubic equation which, from the substitution §. 68, (12),

$$y = t_1\Big(a_0 x + \frac{a_1}{3}\Big) + t_0\Big(a_0 x^2 + a_1 x + \frac{2a_2}{3}\Big) \tag{5}$$

results for y. If we write the equation in the form

$$y^3 + P_2 y + P_3 = 0,$$

then P_2, P_3 are covariants of f, and indeed P_2 of the second, P_3 of the third order, both in t and in a.

Now since P_2 and P_3 are covariants of the cubic form, by §. 62 and 63 they can differ from the two functions defined there,

$$H(t_1, t_0), \qquad Q(t_1, t_0)$$

only by constant, that is, numerical factors. These constant factors can be determined by some special assumption.

We may, for example, assume

$$t_1 = 1, \quad t_0 = 0, \quad a_0 = 1, \quad a_1 = 0,$$

and then $y = x$, hence

$$P_2 = a_2, \qquad P_3 = a_3.$$

On the other hand, by the formulas (2), (3), (7) given in §. 62, this particular assumption yields

$$H(1, 0) = 3a_2, \qquad Q(1, 0) = 27a_3,$$

from which one concludes in general

$$P_2 = \frac{1}{3} H(t_1, t_0), \qquad P_3 = \frac{1}{27} Q(t_1, t_0),$$

so that we obtain for y the cubic equation

$$y^3 + \frac{1}{3} H(t_1, t_0) y + \frac{1}{27} Q(t_1, t_0) = 0. \tag{6}$$

The discriminant Δ of this cubic equation is

$$\Delta = -\frac{1}{27}(4H^3 + Q^2),$$

hence, applying the relation §. 62, (12),

$$4H^3 + Q^2 + 27Df^2 = 0 \tag{7}$$

$$\Delta = Df(t_1, t_0)^2, \tag{8}$$

if D is the discriminant of the given cubic equation, in agreement with the general results of the preceding section.

If we wish to base the solution of the cubic equation on this, we must first, following the prescriptions of the preceding section, express x rationally in terms of y. We set

$$a_0 f(t_1, t_0) x = Q_0 + Q_1 y + Q_2 y^2. \tag{9}$$

The computation of the coefficients Q is easy to carry out if one sets up this equation for x_1, x_2, x_3 and correspondingly y_1, y_2, y_3.

We shall give a few indications of the course of the computation, which requires only the representation of symmetric functions of the roots of a cubic equation by the coefficients, according to the prescriptions of the fourth section (§. 42, 45). If one first takes the sum of the three equations (9), one obtains by (6)

$$3Q_0 = \frac{2}{3}H(t_1, t_0)Q_2 - a_1 f(t_1, t_0),$$

so that only Q_1 and Q_2 still need to be computed; these one finds by multiplying the equation (9) formed for $x = x_1$ by

$$y_2 - y_3 = a_0(x_2 - x_3)(t_1 - t_0 x_1)$$

and by

$$y_2^2 - y_3^2 = -a_0(x_2 - x_3)y_1(t_1 - t_0 x_1)$$

and adding the three analogous equations formed by cyclic permutation of the indices 1, 2, 3.

One then has only to make use of the two formulae

$$a_0^2 \Sigma x_1^2(x_2 - x_3) = -\sqrt{D}$$

$$a_0^3 \Sigma x_1^3(x_2 - x_3) = a_1\sqrt{D},$$

and thus finds

$$Q_0 = \frac{-a_1}{3}f(t_1, t_0) - \frac{2}{3}t_0 H(t_1, t_0)$$

$$Q_1 = a_0 t_1^2 + \frac{2}{3}a_1 t_0 t_1 + \frac{1}{3}a_2 t_0^2$$

$$Q_2 = -t_0.$$

If for simplification we set $t_1 = t$, $t_0 = 1$, thus passing to the inhomogeneous expressions, we obtain

$$(3a_0 x + a_1)f(t) = -\frac{2}{3}H(t) + yf'(t) - 3y^2, \tag{10}$$

while the equation for y reads

$$y^3 + \frac{1}{3}H(t)y + \frac{1}{27}Q(t) = 0, \tag{11}$$

if we write $H(t)$, $Q(t)$ for $H(t, 1)$, $Q(t, 1)$.

Now, in order to solve the cubic equation, we determine t from the quadratic equation

$$H(t) = 0.$$

For the moment, as an abbreviation, we shall set

$$\varrho = \sqrt{-3D} \tag{12}$$

then from (7) it follows that

$$Q = 3\varrho f(t), \tag{13}$$

and from (11)

$$y = -\frac{1}{3}\sqrt[3]{3\varrho f(t)}, \tag{14}$$

and the expression (10) yields

$$(3a_0x + a_1)f(t) = yf'(t) - 3y^2. \tag{15}$$

In order to derive the agreement of this result with the Cardano formula, we go back to §. 62 and set $x = t$, $y = 1$ in the formulas there.

On account of (13), by §. 62, (11), (13), we get

$$\eta = 0, \quad \xi = -h\varrho,$$

and by §. 62, (10)

$$f(t) = \xi^3, \quad f'(t) = 3\xi^2\xi',$$

in which $\xi' = hA_0$ is, by §. 62, (13), the derivative of ξ with respect to t,

$$f(t) = -h^3\varrho^3, \quad f'(t) = 3h^3A_0\varrho^2.$$

Accordingly it follows from (14) that

$$y = \frac{h\varrho\sqrt[3]{3\varrho}}{3}$$

and from (15)

$$3a_0x + a_1 = -hA_0\sqrt[3]{3\varrho} - \frac{1}{h\sqrt[3]{3\varrho}},$$

or, by §. 62, (14), (15),

$$3a_0x + a_1 = A_0(k - h)\sqrt[3]{3\varrho}, \tag{16}$$

in which, by §. 62, (15),

$$hA_0\sqrt[3]{3\varrho} = \sqrt[3]{\frac{-q_0 + 3\varrho a_0}{2}}$$

$$kA_0\sqrt[3]{3\varrho} = \sqrt[3]{\frac{+q_0 + 3\varrho a_0}{2}}.$$

If we assume $a_0 = 1$, $a_1 = 0$, then from (16) we obtain the Cardano formula (§. 35)

$$x = \sqrt[3]{\frac{-a_3}{2} + \sqrt{\frac{a_2^3}{27} + \frac{a_3^2}{4}}} + \sqrt[3]{\frac{-a_3}{2} - \sqrt{\frac{a_2^3}{27} + \frac{a_3^2}{4}}}. \tag{17}$$

§. 72.

General execution of the transformation.

In the general carrying-out of the Tschirnhausen transformation in the Hermite form we can go a significant step further.

We first consider the general substitution §. 68, (5)

$$y = t_{n-1}f_0 + t_{n-2}f_1 + t_{n-3}f_2 + \cdots + t_0f_{n-1}, \tag{1}$$

from which we obtain the more special form §. 68 (12) if we bind the variables t to the one linear condition

$$S(y) = na_0t_{n-1} + (n-1)a_1t_{n-2} + \cdots + a_{n-1}t_0 = 0 \tag{2}$$

— which, however, is not yet to be done for the present.

By means of the functions $f_0, f_1, f_2 \dots f_{n-1}$, according to §. 68, (4), all powers of x, and hence also all rational functions of x, can be represented linearly and homogeneously.

If we can find this representation for the various powers of y, then by elimination of the f_i the equation of degree n for y can be formed.

But we attain the same end still more simply if we represent the n products yf_s linearly in terms of the f. For if we have the equations

$$yf_s = E_{0,s}f_0 + E_{1,s}f_1 + E_{2,s}f_2 + \cdots + E_{n-1,s}f_{n-1} \tag{3}$$
$$s = 0, 1, 2 \dots n-1,$$

then by elimination of $f_0, f_1 \dots f_{n-1}$ from the system (3) we obtain

$$E = \begin{vmatrix} E_{0,0} - y, & E_{1,0}, & E_{2,0} & \cdots & E_{n-1,0} \\ E_{0,1}, & E_{1,1} - y, & E_{2,1} & \cdots & E_{n-1,1} \\ \cdots & \cdots & \cdots & \cdots & \cdots \\ E_{0,n-1}, & E_{1,n-1}, & E_{2,n-1} & \cdots & E_{n-1,n-1} - y \end{vmatrix} = 0, \tag{4}$$

that is, an equation of degree n for y, which is the transformed equation sought. Everything is therefore reduced to the determination of the coefficients $E_{r,s}$, which, as appears from (1), are in any case linear functions of the variables t. In particular,

$$E_{0,0} = a_0 t_{n-1}, \quad E_{1,0} = a_0 t_{n-2}, \quad \dots \quad E_{n-1,0} = a_0 t_0. \tag{5}$$

To compute the remaining E, we observe that by the definition §. 68, (4) the following relations hold between the functions f:

$$\begin{aligned} xf_0 &= f_1 - a_1 \\ xf_1 &= f_2 - a_2 \\ &\cdots\cdots\cdots\cdots \\ xf_{n-2} &= f_{n-1} - a_{n-1} \\ xf_{n-1} &= -a_n; \end{aligned} \tag{6}$$

of which the last is a consequence of the equation $f(x) = 0$.

If we now continue the series of variables $t_0, t_1 \dots t_{n-1}$ by defining the new variables $t_n, t_{n+1}, t_{n+2} \dots$ through the following equations

$$\begin{aligned} a_0 t_n + a_1 t_{n-1} + a_2 t_{n-2} + \cdots + a_n t_0 &= 0, \\ a_0 t_{n+1} + a_1 t_n + a_2 t_{n-1} + \cdots + a_n t_1 &= 0, \\ a_0 t_{n+2} + a_1 t_{n+1} + a_2 t_n + \cdots + a_n t_2 &= 0, \\ &\cdots\cdots\cdots\cdots\cdots\cdots \end{aligned} \tag{7}$$

then, multiplying the equations (6) in order by $t_{n-1}, t_{n-2} \dots t_0$ and adding, we obtain

$$xy = t_n f_0 + t_{n-1} f_1 + \cdots + t_1 f_{n-1},$$

and if we now multiply (6) in the same way by $t_n, t_{n-1} \dots t_1$,

$$x^2 y = t_{n+1} f_0 + t_n f_1 + \cdots + t_2 f_{n-1}.$$

Thus the following system results

$$\begin{aligned} y &= t_{n-1} f_0 + t_{n-2} f_1 + \cdots + t_0 f_{n-1} \\ xy &= t_n f_0 + t_{n-1} f_1 + \cdots + t_1 f_{n-1} \\ x^2 y &= t_{n+1} f_0 + t_n f_1 + \cdots + t_2 f_{n-1} \\ &\cdots\cdots\cdots\cdots\cdots\cdots \\ x^s y &= t_{n+s-1} f_0 + t_{n+s-2} f_1 + \cdots + t_s f_{n-1}, \end{aligned} \tag{8}$$

and if we multiply these equations in turn by $a_s, a_{s-1} \dots a_0$ and add, there follows the expression sought

$$yf_s = E_{0,s}f_0 + E_{1,s}f_1 + \cdots + E_{n-1,s}f_{n-1},$$

in which, by (7),

$$\begin{aligned}
E_{0,s} &= a_0t_{n+s-1} + a_1t_{n+s-2} + \cdots + a_st_{n-1} \\
&= -a_{s+1}t_{n-2} \cdots - a_nt_{s-1} \\
E_{1,s} &= a_0t_{n+s-2} + a_1t_{n+s-3} + \cdots + a_st_{n-2} \\
&= -a_{s+1}t_{n-3} \cdots - a_nt_{s-2} \\
&\cdots\cdots\cdots\cdots\cdots\cdots \\
E_{s-1,s} &= a_0t_n + a_1t_{n-1} + \cdots + a_st_{n-s} \\
&= -a_{s+1}t_{n-s-1} \cdots - a_nt_0 \\
E_{s,s} &= a_0t_{n-1} + a_1t_{n-2} + \cdots + a_st_{n-s-1} \\
E_{s+1,s} &= a_0t_{n-2} + a_1t_{n-3} + \cdots + a_st_{n-s-2} \\
&\cdots\cdots\cdots\cdots\cdots\cdots \\
E_{n-1,s} &= a_0t_s + a_1t_{s-1} + \cdots + a_st_0.
\end{aligned} \tag{9}$$

To this it should further be remarked that the superfluously introduced variables $t_n, t_{n+1} \dots$ have already been eliminated again in the second form of $E_{0,s}, E_{1,s} \dots E_{s-1,s}$, and that the variable t_{n-1} occurs only in $E_{s,s}$.

§. 73.
The Bezoutiant.

Equation (4) of the preceding section, when expanded, will have the form

$$y^n + P_1y^{n-1} + P_2y^{n-2} + \cdots + P_n = 0,$$

or, if we determine t_{n-1} so that $S(y) = 0$,

$$y^n + P_2y^{n-2} + \cdots + P_n = 0; \tag{1}$$

in it, then,

$$-2P_2 = S(y^2)$$

is a function of the second degree with respect to the t and with respect to the a. This function is especially important for many applications and has received from Sylvester the name Bezoutiant of the function $f(x)$, in honour of the French mathematician Bezout, who already in the last century gave the first correct treatments of elimination.

By the formulas of the last section this function can be computed comparatively simply. We first introduce, alongside the variables $t_{n-1}, t_{n-2} \dots t_0$, a second system of variables independent of them, $\tau_{n-1}, \tau_{n-2} \dots \tau_0$, and set

$$\begin{aligned}
y &= t_{n-1}f_0 + t_{n-2}f_1 + \cdots + t_0f_{n-1} \\
z &= \tau_{n-1}f_0 + \tau_{n-2}f_1 + \cdots + \tau_0f_{n-1}.
\end{aligned} \tag{2}$$

Now instead of forming $S(y^2)$, we first compute $S(yz)$, from which $S(y^2)$ then results when one sets the τ equal to the t.

If one multiplies formula (3) of the preceding section

$$yf_s = E_{0,s}f_0 + E_{1,s}f_1 + \cdots + E_{n-1,s}f_{n-1} \tag{3}$$

by τ_{n-s-1} and then sums with respect to s from 0 to $n-1$, one finds

$$
\begin{aligned}
yz = f_0 &\sum_{0,\,n-1}^{s} E_{0,s}\tau_{n-s-1} \\
&+ f_1 \sum_{0,\,n-1}^{s} E_{1,s}\tau_{n-s-1} + \cdots + f_{n-1} \sum_{0,\,n-1}^{s} E_{n-1,s}\tau_{n-s-1}.
\end{aligned} \tag{4}
$$

Now by §. 68, (6),

$$Sf_0 = na_0, \quad Sf_1 = (n-1)a_1, \ldots Sf_{n-1} = a_{n-1},$$

so that one obtains

$$
\begin{aligned}
S(yz) = na_0 &\sum_{0,\,n-1}^{s} E_{0,s}\tau_{n-s-1} \\
&+ (n-1)a_1 \sum_{0,\,n-1}^{s} E_{1,s}\tau_{n-s-1} \\
&\cdots\cdots\cdots\cdots \\
&+ a_{n-1} \sum_{0,\,n-1}^{s} E_{n-1,s}\tau_{n-s-1}.
\end{aligned}
$$

In this, now, the coefficient of τ_{n-1} is

$$na_0E_{0,0} + (n-1)a_1E_{1,0} + \cdots + a_{n-1}E_{n-1,0},$$

hence by (3)

$$= a_0S(y).$$

Now since $S(z)$ begins with the term $na_0\tau_{n-1}$, in the difference

$$S(yz) - \frac{1}{n}S(y)S(z) \tag{5}$$

no term will occur that is multiplied by τ_{n-1}. But since this expression is moreover symmetric with respect to t and τ, it also does not contain t_{n-1}, and in forming it we may simply assume t_{n-1} and $\tau_{n-1} = 0$.

We now set

$$S(y) = na_0t_{n-1} + \varphi(t),$$

in which

$$\varphi(t) = (n-1)a_1t_{n-2} + (n-2)a_2t_{n-3} + \cdots + a_{n-1}t_0. \tag{6}$$

Then we have

$$
\begin{aligned}
S(yz) - \frac{1}{n}S(y)S(z) = na_0 &\sum_{1,\,n-1}^{s} E_{0,s}\tau_{n-s-1} \\
&+ (n-1)a_1 \sum_{1,\,n-1}^{s} E_{1,s}\tau_{n-s-1} \\
&\cdots\cdots\cdots\cdots \\
&+ a_{n-1} \sum_{1,\,n-1}^{s} E_{n-1,s}\tau_{n-s-1} - \frac{1}{n}\varphi(t)\varphi(\tau),
\end{aligned} \tag{7}
$$

where, however, it is to be noted that $t_{n-1} = 0$ is to be assumed, and hence

$$E_{s,s} = a_1t_{n-2} + \cdots + a_st_{n-s-1}$$

is to be set, while the remaining E are determined by the formulas (9) of the preceding section.

The expression on the right-hand side of (7) is a bilinear function of the t and τ; when developed, let it have the form

$$\sum_{0,\,n-2}^{h}\sum_{0,\,n-2}^{k} B_{h,k}t_h\tau_k,$$

with the condition $B_{i,k} = B_{k,i}$. The Bezoutiant is then obtained by setting $\tau = t$, thus

$$B = \sum_{0,\,n-2}^{h,k} B_{h,k}t_h t_k, \tag{8}$$

in which the coefficients $B_{i,k}$ are quadratic functions of the a; and from (7) it follows

$$S(yz) - \frac{1}{n}S(y)S(z) = \sum_{0,\,n-2}^{h,k} B_{h,k}t_h\tau_k, \tag{9}$$

$$S(y^2) = \frac{1}{n}[S(y)]^2 + B. \tag{10}$$

By formula (8), §. 68, we have

$$y - \frac{1}{n}S(y) = t_{n-2}F_0 + t_{n-3}F_1 + \cdots + t_0F_{n-2}$$
$$z - \frac{1}{n}S(z) = \tau_{n-2}F_0 + \tau_{n-3}F_1 + \cdots + \tau_0F_{n-2},$$

and if we multiply the two together,

$$yz - \frac{1}{n}[yS(z) + zS(y)] + \frac{1}{n^2}S(y)S(z) = \sum_{0,\,n-2}^{h,k} \tau_h t_k F_{n-h-2}F_{n-k-2}.$$

If we sum this formula over all the roots x, that is, take the sum denoted by S, it follows that

$$S(yz) - \frac{1}{n}S(y)S(z) = \sum^{h,k} \tau_h t_k S(F_{n-h-2}F_{n-k-2}),$$

and comparison with (9) shows

$$B_{h,k} = S(F_{n-h-2}F_{n-k-2}) \tag{11}$$

We denote by Δ the determinant of the function B, and thus set

$$\Delta = \sum \pm B_{0,0}B_{1,1}\ldots B_{n-2,\,n-2}.$$

By the multiplication theorem for determinants, and having regard to the formulas

$$SF_0 = 0, \quad SF_1 = 0, \ldots SF_{n-2} = 0$$

is, however, if we set

$$\Phi = \begin{vmatrix} 1, & F_0(x_1), & F_1(x_1) & \ldots & F_{n-2}(x_1) \\ 1, & F_0(x_2), & F_1(x_2) & \ldots & F_{n-2}(x_2) \\ & \cdots\cdots\cdots\cdots\cdots\cdots & & & \\ 1, & F_0(x_n), & F_1(x_n) & \ldots & F_{n-2}(x_n) \end{vmatrix}$$

by (11)

$$\Phi^2 = n\Delta. \tag{12}$$

If one now takes account of the expressions (9) in §. 68 for the functions $F_0, F_1 \dots F_{n-2}$, then by the theorem that in a determinant a column multiplied by an arbitrary factor may be added to another column (§. 22, VII), one easily obtains for Φ the expression

$$a_0^{n-1} \begin{vmatrix} 1, & x_1, & x_1^2 & \dots & x_1^{n-1} \\ 1, & x_2, & x_2^2 & \dots & x_2^{n-1} \\ & \dots\dots\dots\dots\dots & & & \\ 1, & x_n, & x_n^2 & \dots & x_n^{n-1} \end{vmatrix},$$

whose square, by §. 46, is the discriminant of $f(x)$. Accordingly, by (12), we thus have the important theorem:

The determinant of the Bezoutiant of $f(x)$ is the nth part of the discriminant of $f(x)$.

The computation of the Bezoutiant presents, by our formulas, no further difficulty at all, and turns out to be fairly simple. For $n = 3$ one obtains the result, already to be inferred from the formulas of §. 71,

$$B = -\frac{2}{3} H(t_1, t_0). \tag{13}$$

For the computation of the Bezoutiant of the biquadratic form we shall, to illustrate the foregoing, set out the system of formulas in full, leaving the carrying out of the slightly longer computation for the form of the fifth degree to the reader.

For the form of the fourth degree we have, by (7) and §. 72, (9)

$$f(x) = a_0x^4 + a_1x^3 + a_2x^2 + a_3x + a_4$$

$E_{0,1}$	=	$-a_2t_2 - a_3t_1 - a_4t_0$	$4a_0\tau_2$
$E_{1,1}$	=	a_1t_2	$3a_1\tau_2$
$E_{2,1}$	=	$a_0t_2 + a_1t_1$	$2a_2\tau_2$
$E_{3,1}$	=	$a_0t_1 + a_1t_0$	$a_3\tau_2$
$E_{0,2}$	=	$-a_3\,t_2 - a_4\,t_1$	$4\,a_0\,\tau_1$
$E_{1,2}$	=	$-a_3\,t_1 - a_4\,t_0$	$3\,a_1\,\tau_1$
$E_{2,2}$	=	$a_1\,t_2 + a_2\,t_1$	$2\,a_2\,\tau_1$
$E_{3,2}$	=	$a_0\,t_2 + a_1\,t_1 + a_2\,t_0$	$a_3\,\tau_1$
$E_{0,3}$	=	$-a_4\,t_2$	$4\,a_0\,\tau_0$
$E_{1,3}$	=	$-a_4\,t_1$	$3\,a_1\,\tau_0$
$E_{2,3}$	=	$-a_4\,t_0$	$2\,a_2\,\tau_0$
$E_{3,3}$	=	$a_1\,t_2 + a_2\,t_1 + a_3\,t_0$	$a_3\,\tau_0$.

One has to multiply these expressions by the factors placed to the right of them and to add, then further append

$$-\frac{1}{4}\,\varphi(t)\,\varphi(\tau)$$
$$= -\frac{1}{4}\,.(3\,a_1\,t_2 + 2\,a_2\,t_1 + a_3\,t_0)\,(3\,a_1\,\tau_2 + 2\,a_2\,\tau_1 + a_3\,\tau_0)$$

and to seek out the coefficients of $\tau_h t_k$. In this way one obtains

$$\begin{aligned} B_{2,2} &= -\frac{1}{4}(8\,a_0\,a_2 - 3\,a_1^2), & B_{0,1} &= -\frac{1}{2}\,(6\,a_1\,a_4 - a_2\,a_3) \\ B_{1,1} &= -4\,a_0\,a_4 - 2\,a_1\,a_3 + a_2^2, & B_{1,2} &= -\frac{1}{2}\,(6\,a_0\,a_3 - a_1\,a_2) \\ B_{0,0} &= -\frac{1}{4}(8\,a_2\,a_4 - 3\,a_3^2), & B_{0,2} &= -\frac{1}{4}(16\,a_0\,a_4 - a_1\,a_3). \end{aligned} \tag{14}$$

In the same way one can proceed with a form of the fifth degree, and one finds for the coefficients of the Bezoutiant the following expressions

$$\begin{aligned}
5\,B_{0,0} &= 4\,a_4^2 - 10\,a_3\,a_5\\
5\,B_{1,1} &= 6\,a_3^2 - 10\,a_2\,a_4 - 20\,a_1\,a_5\\
5\,B_{2,2} &= 6\,a_2^2 - 10\,a_1\,a_3 - 20\,a_0\,a_4\\
5\,B_{3,3} &= 4\,a_1^2 - 10\,a_0\,a_2
\end{aligned}$$

$$\begin{aligned}
5\,B_{0,1} &= 3\,a_3\,a_4 - 15\,a_2\,a_5\\
5\,B_{0,2} &= 2\,a_2\,a_4 - 20\,a_1\,a_5\\
5\,B_{0,3} &= a_1\,a_4 - 25\,a_0\,a_5\\
5\,B_{1,2} &= 4\,a_2\,a_3 - 15\,a_1\,a_4 - 25\,a_0\,a_5\\
5\,B_{1,3} &= 2\,a_1\,a_3 - 20\,a_0\,a_4\\
5\,B_{2,3} &= 3\,a_1\,a_2 - 15\,a_0\,a_3.
\end{aligned}$$

§. 74.
Transformation of the equation of the fifth degree.

These developments shall now be applied in order to carry out the transformation of the equation of the fifth degree, as we sketched it in §. 54.

A system of values t_0, t_1, t_2, t_3 can be determined in infinitely many ways so that the Bezoutiant B vanishes; in general the solution of a quadratic equation is required for this. We can, for example, set

$$t_2 = 0, \quad t_3 = 0$$

and determine the ratio $t_0 : t_1$ from the quadratic equation

$$B_{0,0}\,t_0^2 + 2\,B_{0,1}\,t_0\,t_1 + B_{1,1}\,t_1^2 = 0\,.$$

These values of t then contain the square root

$$\sqrt{B_{0,1}^2 - B_{0,0}\,B_{1,1}}\,.$$

Instead of this, however, one can also make other choices, and one obtains ever different square roots. In the selection of this square root there lies something arbitrary and undetermined.

Among these various modes of determination we pick out one and with it transform the given equation of the fifth degree into a principal equation, that is, into one in which the third and fourth powers of the unknown do not occur; or we assume, following F. Klein's precedent, that the equation to be transformed is from the outset a principal equation

$$(1) \qquad f(x) = a_0\,x^5 + a_3\,x^2 + a_4\,x + a_5 = 0\,.$$

But we wish to emphasise expressly that, by §. 70, (8), through this preliminary Tschirnhausen transformation the discriminant of the given equation is changed only by a factor Θ^2, in which Θ depends rationally on the t employed, and hence also on the square root used for their determination.

Under the assumption (1) the coefficients of the Bezoutiant, by the formulas of the preceding paragraph, now become the following:

$$(2) \qquad \begin{aligned}
5\,B_{0,0} &= 4\,a_4^2 - 10\,a_3\,a_5, & 5\,B_{0,1} &= 3\,a_3\,a_4\\
5\,B_{1,1} &= 6\,a_3^2, & 5\,B_{0,2} &= 0\\
5\,B_{2,2} &= -20\,a_0\,a_4, & 5\,B_{0,3} &= -25\,a_0\,a_5\\
5\,B_{3,3} &= 0 & 5\,B_{1,2} &= -25\,a_0\,a_5\\
5\,B_{1,3} &= -20\,a_0\,a_4, & 5\,B_{2,3} &= -15\,a_0\,a_3.
\end{aligned}$$

If one computes the determinant from these, one obtains without difficulty, by §. 73, (12), for the discriminant of the principal equation of the fifth degree

$$(3) \qquad D = a_0^2(5^5\, a_0^2\, a_5^4 + 2^8\, a_0\, a_4^5 + 2250\, a_0\, a_3^2\, a_4\, a_5^2 - 1600\, a_0\, a_3\, a_4^3\, a_5 + 108\, a_3^5\, a_5 - 27\, a_3^4\, a_4^2)\,.$$

Now since $B_{3,3} = 0$ here, we have a system of values of the t for which the Bezoutiant vanishes, namely

$$t_0 = 0, \quad t_1 = 0, \quad t_2 = 0\,.$$

We now take our Tschirnhausen substitution in the following form [§. 68, (12)]

$$(4) \qquad y = t_3\, F_0 + t_2\,(\alpha_2\, F_1 + \alpha_1\, F_2 + \alpha_0\, F_3)$$

and we wish to dispose of the $\alpha_0, \alpha_1, \alpha_2$ in such a way that the equation of the fifth degree for y becomes a principal equation independently of t_3, t_2 — that is, that $S(y^2)$ vanishes identically for all t_2, t_3.

Since $S(F_0^2) = B_{3,3}$ vanishes here, this requirement is satisfied, by §. 73, (11), if we determine $\alpha_0, \alpha_1, \alpha_2$ so that

$$(5) \qquad \alpha_0^2\, B_{0,0} + \alpha_1^2\, B_{1,1} + \alpha_2^2\, B_{2,2} + 2\,\alpha_0\,\alpha_1\, B_{0,1} + 2\,\alpha_0\,\alpha_2\, B_{0,2} + 2\,\alpha_1\,\alpha_2\, B_{1,2} = 0\,,$$

$$(6) \qquad \alpha_0\, B_{0,3} + \alpha_1\, B_{1,3} + \alpha_2\, B_{2,3} = 0\,.$$

These equations lead, by elimination of one of the three quantities α, to a quadratic equation for the ratio of the other two. In itself it is not necessary to exclude any particular case, not even the one in which $B_{0,3}, B_{1,3}, B_{2,3}$ all three vanish; one would then merely have to take for the α some solution of equation (5).

But in order not to be long-winded, we shall assume that $B_{0,3}$ is different from zero, because the cases thereby excluded, in which a root of the given equation becomes zero or infinite, really offer no interest here, since they come back to an equation of lower degree. The treatment, moreover, remains entirely the same if

we assume that another of the three quantities $B_{0,3}, B_{1,3}, B_{2,3}$ is different from zero.

In order now to treat the equations (5), (6) in a symmetric way, and in particular to recognise which square root is required for their solution, we proceed as follows. For brevity we set

$$(7) \qquad \begin{aligned} B_0 &= \alpha_0\, B_{0,0} + \alpha_1\, B_{0,1} + \alpha_2\, B_{0,2} \\ B_1 &= \alpha_0\, B_{1,0} + \alpha_1\, B_{1,1} + \alpha_2\, B_{1,2} \\ B_2 &= \alpha_0\, B_{2,0} + \alpha_1\, B_{2,1} + \alpha_2\, B_{2,2}\,. \end{aligned}$$

Then we can represent the equations (5) and (6) as follows:

$$(8) \qquad \begin{array}{rcrcrcl} \alpha_0\, B_0 & + & \alpha_1\, B_1 & + & \alpha_2\, B_2 & = & 0 \\ \alpha_0\, B_{0,3} & + & \alpha_1\, B_{1,3} & + & \alpha_2\, B_{2,3} & = & 0\,. \end{array}$$

We multiply the second by an indeterminate factor α_3 and add it to the first, whereby we obtain

$$(9) \qquad \alpha_0\,(B_0 + \alpha_3\, B_{0,3}) + \alpha_1\,(B_1 + \alpha_3\, B_{1,3}) + \alpha_2\,(B_2 + \alpha_3\, B_{2,3}) = 0\,,$$

an equation which is satisfied if we set

$$(10) \qquad \begin{aligned} B_0 + \alpha_3\, B_{0,3} &= 0 \\ B_1 + \alpha_3\, B_{1,3} &= \sigma\,\alpha_2 \\ B_2 + \alpha_3\, B_{2,3} &= -\sigma\,\alpha_1\,. \end{aligned}$$

From these three equations, in combination with the equation (6), we have to determine the unknowns $\alpha_0, \alpha_1, \alpha_2, \alpha_3, \sigma$. We first write the equations out here in full

$$(11) \qquad \begin{array}{rcrcrcrcl} \alpha_0\, B_{0,0} & + & \alpha_1\, B_{0,1} & + & \alpha_2\, B_{0,2} & + & \alpha_3\, B_{0,3} & = & 0 \\ \alpha_0\, B_{1,0} & + & \alpha_1\, B_{1,1} & + & \alpha_2\,(B_{1,2} - \sigma) & + & \alpha_3\, B_{1,3} & = & 0 \\ \alpha_0\, B_{2,0} & + & \alpha_1\,(B_{2,1} + \sigma) & + & \alpha_2\, B_{2,2} & + & \alpha_3\, B_{2,3} & = & 0 \\ \alpha_0\, B_{3,0} & + & \alpha_1\, B_{3,1} & + & \alpha_2\, B_{3,2} & & & = & 0\,, \end{array}$$

and if we eliminate $\alpha_0, \alpha_1, \alpha_2, \alpha_3$ from these, there results a quadratic equation for σ, after whose solution the ratios $\alpha_0 : \alpha_1 : \alpha_2 : \alpha_3$ can be determined from linear equations.

The quadratic equation for σ, however, reads in determinant form

$$\begin{vmatrix} B_{0,0}, & B_{0,1}, & B_{0,2}, & B_{0,3} \\ B_{1,0}, & B_{1,1}, & B_{1,2} - \sigma, & B_{1,3} \\ B_{2,0}, & B_{2,1} + \sigma, & B_{2,2}, & B_{2,3} \\ B_{3,0}, & B_{3,1}, & B_{3,2}, & 0 \end{vmatrix} = 0 .$$

Since the left side is not changed by the permutation of σ with $-\sigma$, it is a pure quadratic equation, and, having regard to §. 73, (12), it gives

$$B_{0,3}^2 \, \sigma^2 - 5 \, D = 0,$$

where D is the discriminant of the given equation. By (2) we therefore obtain

$$5 \, a_0 \, a_5 \, \sigma = \sqrt{5 \, D}, \tag{12}$$

in which the sign is arbitrary.

§. 75.
Normal form of the equation of the fifth degree.

As already remarked earlier, the principal aim of these considerations is to establish a normal form of the equation of the fifth degree which depends on only one undetermined coefficient, one parameter. Such a normal form is the Bring-Jerrard form. To obtain it, by (4) of the preceding paragraph we have to set

$$y = t_3 \, F_0 + t_2 \, (\alpha_2 \, F_1 + \alpha_1 \, F_2 + \alpha_0 \, F_3) \tag{1}$$

so that identically

$$S(y) = 0, \quad S(y^2) = 0$$

holds, and then the ratio $t_3 : t_2$ is to be formed from the cubic equation

$$S(y^3) = 0$$

This cubic equation can actually be formed, even though its expression becomes long. The formulae needed for this have been computed by Cayley[1].

It is the result of a remarkable investigation of Gordan that one can obtain another normal form, the Brioschi form, without any new irrationality, and to conclude these considerations on the Tschirnhausen transformation we shall now derive this result[2].

We hold fast to the assumptions of the preceding paragraph and, for the moment, set as an abbreviation

$$u = F_0, \quad v = \alpha_2 \, F_1 + \alpha_1 \, F_2 + \alpha_0 \, F_3, \tag{2}$$

in which $\alpha_0, \alpha_1, \alpha_2$ are to have the values determined in the preceding paragraph.

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[1] Cayley, on Tschirnhausen's Transformation, Phil. Trans. 1861, Mathematical Papers, Tom. IV, Nr. 275.

[2] Gordan, Mathematische Annalen, Bd. 28, 1886.

By the formula §. 73, (4) the three functions

$$u^2, \quad u\,v, \quad v^2$$

can be represented linearly and homogeneously in terms of f_0, f_1, f_2, f_3, f_4, and since

$$S(u^2) = 0, \quad S(u\,v) = 0, \quad S(v^2) = 0 \tag{3}$$

holds, these expressions also become linear and homogeneous in F_0, F_1, F_2, F_3.

The calculation is easy to carry out by the formulas of §§. 72, 73, but it will not be pursued further here, since we are concerned only with setting out the fundamental idea. We wish only to remark that the coefficients in the expressions for u^2, $u\,v$, v^2 are linear in the coefficients of the original equation of the fifth degree and quadratic in the $\alpha_0, \alpha_1, \alpha_2$. If now, with the aid of (2), we eliminate the functions F_0, F_1, F_2, F_3 from these expressions, we obtain a relation of the form

$$p\,u^2 + 2\,q\,u\,v + r\,v^2 = a\,u + b\,v, \tag{4}$$

in which $p, q, r, a : b$ depend rationally on the coefficients a_i and α_i, and in any case do not all vanish at once.

If for the moment we set

$$p\,u^2 + 2\,q\,u\,v + r\,v^2 = \varphi(u, v),$$

then by §. 57

$$4\,\varphi(u', v')\,\varphi(u, v) - [u'\,\varphi'(u) + v'\,\varphi'(v)]^2$$

is the square of a linear function of u, v, so that $\varphi(u, v)$ is thereby decomposed into the sum of two squares. If we choose $u' = b$, $v' = -a$, it follows that

$$\varphi(b, -a)\,\varphi(u, v) = [b\,(p\,u + q\,v) - a\,(q\,u + r\,v)]^2 + (p\,r - q^2)\,(a\,u + b\,v)^2,$$

or, if for abbreviation one sets

$$\begin{aligned} p\,b^2 - 2\,q\,a\,b + r\,a^2 &= m, \\ b\,p - a\,q &= a', \\ b\,q - a\,r &= b', \\ q^2 - p\,r &= c \end{aligned} \tag{5}$$

then

$$m\,(p\,u^2 + 2\,q\,u\,v + r\,v^2) = (a'\,u + b'\,v)^2 - c\,(a\,u + b\,v)^2, \tag{6}$$

and by (4) this equation can also be represented thus:

$$m\,(a\,u + b\,v) = (a'\,u + b'\,v)^2 - c\,(a\,u + b\,v)^2. \tag{7}$$

Now set

$$y = \frac{a'\,u + b'\,v}{a\,u + b\,v}, \tag{8}$$

and form the equation of the fifth degree whose roots are the five values of y:

$$y^5 + c_1\,y^4 + c_2\,y^3 + c_3\,y^2 + c_4\,y + c_5 = 0; \tag{9}$$

in it the coefficients c can be determined more closely in the following way.

From (8) we derive

$$y + \sqrt{c} = \frac{a'\,u + b'\,v + \sqrt{c}\,(a\,u + b\,v)}{a\,u + b\,v}$$

and from this, by (7),

$$y+\sqrt{c}=\frac{m}{a'\,u+b'\,v-\sqrt{c}\,(a\,u+b\,v)},$$
$$\frac{m}{y+\sqrt{c}}=a'\,u+b'\,v-\sqrt{c}\,(a\,u+b\,v).$$

But from this it follows, by (3), that

$$S\left[\frac{1}{y+\sqrt{c}}\right]=0,\quad S\left[\frac{1}{(y+\sqrt{c})^2}\right]=0.$$

If, then, we derive from (9) the equation whose roots are the values

$$\xi=\frac{1}{y+\sqrt{c}}$$

that is, if we set

$$y=\frac{1-\xi\,\sqrt{c}}{\xi}$$

an equation for ξ must arise in which the coefficients of ξ^4 and ξ^3 vanish, and indeed whichever sign we give to the square root $\sqrt{c}$. In this way one obtains four equations between the coefficients c_ν of equation (9).

The equation for ξ becomes, namely,

$$(1-\xi\,\sqrt{c})^5+c_1\,\xi\,(1-\xi\,\sqrt{c})^4+c_2\,\xi^2\,(1-\xi\,\sqrt{c})^3+c_3\,\xi^3\,(1-\xi\,\sqrt{c})^2+c_4\,\xi^4\,(1-\xi\,\sqrt{c})+c_5\,\xi^5=0.$$

If one here sets the coefficients of ξ^4 and ξ^3 equal to 0, it follows that

$$5\,\sqrt{c}^{\,4}-4\,c_1\,\sqrt{c}^{\,3}+3\,c_2\,\sqrt{c}^{\,2}-2\,c_3\,\sqrt{c}+c_4=0,$$
$$-10\,\sqrt{c}^{\,3}+6\,c_1\,\sqrt{c}^{\,2}-3\,c_2\,\sqrt{c}+c_3=0,$$

and on account of the double sign of $\sqrt{c}$ these equations split into the four

$$5\,c^2+3\,c_2\,c+c_4=0,$$
$$4\,c_1\,c+2\,c_3=0,$$
$$10\,c+3\,c_2=0,$$
$$6\,c_1\,c+c_3=0.$$

From the second and fourth of these equations it follows that

$$c_1=0,\quad c_3=0,$$

and then from the third and first

$$c_2=-\frac{10}{3}\,c,\quad c_4=-5\,c^2,$$

so that the equation for y thus takes the form

$$y^5-\frac{10}{3}\,c\,y^3+5\,c^2\,y+c_5=0. \tag{10}$$

This equation still depends on the two parameters c_5 and c; one can reduce it to an equation with one parameter by the substitution

$$y=\sqrt{\frac{c}{3}}\,z,\quad \gamma=c_5\,\sqrt{\frac{3^5}{c^5}}, \tag{11}$$

whereby it takes the simple and elegant form

$$z^5 - 10\,z^3 + 45\,z + \gamma = 0. \tag{12}$$

This is the Brioschi normal form. The substitution (11), however, suffers from the drawback that it still contains a square root; whereas in the formulas (8) and (10) only the two square roots discussed in the previous section occur. But, following a remark of Klein, one can also pass from (10) by rational means to a normal form that contains only one parameter.

If, for example, one sets

$$y = \frac{3\,c_5}{c^2}\,\frac{1}{z}, \quad \gamma = \frac{9\,c_5^2}{c^5}, \tag{13}$$

then from (10) one obtains the equation

$$z^5 + 15\,z^4 - 10\,\gamma\,z^2 + 3\,\gamma^2 = 0. \tag{14}$$

In this transformation the assumption has been tacitly made that the quantities denoted by m and c in (5) are not zero.

If $m = 0$, then in formula (4) $p\,u^2 + 2\,q\,u\,v + r\,v^2$ is divisible by $a\,u + b\,v$; hence, at least for some of the roots, the equation

$$a\,u + b\,v = 0,$$

that is, an equation of degree no higher than the fourth, must hold.

If $c = 0$, so that $p\,u^2 + 2\,q\,u\,v + r\,v^2$ is the square of a linear function, then it follows from (7) that, if we set

$$y = a'\,u + b'\,v$$

then

$$S(y) = 0, \quad S(y^2) = 0, \quad S(y^3) = 0$$

and hence that the Bring-Jerrard form can be produced in a rational way.

Finally, we may summarise the results of these considerations as follows:

The principal equation of the fifth degree can be transformed, by a transformation which contains as its only irrationality the square root of the discriminant, into a normal form with one parameter.

BOOK TWO.

THE ROOTS.

Seventh Section.
Reality of the Roots.

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§. 76.
Generalities on the reality of roots of equations and on the discriminant.

In this section we shall occupy ourselves with the question of how many roots of an algebraic equation are real. The coefficients of the equation are thereby assumed to be real numbers. We shall call such equations, for short, real equations, and begin with some general considerations.

The real equation of degree n

$$f(x) = 0$$

has, as we have seen in the third section, n roots, which are either real or imaginary, i. e. of the form $\xi + i\eta$, with ξ, η real. But this count of n roots is obtained in general only if, in certain circumstances, we count a root multiply, namely $(m+1)$-fold when, together with $f(x)$, the first m derivatives of $f(x)$ vanish. Now since a complex expression of the form $X + iY$ is equal to zero only when the two real constituents X and Y vanish separately, that is, when together with $X + iY$ also $X - iY$ vanishes, and since further, with real coefficients, if $f(\xi + i\eta) = X + iY$, one obtains $f(\xi - i\eta) = X - iY$, it follows from $f(\xi + i\eta) = 0$ that at the same time $f(\xi - i\eta) = 0$ must hold, and hence that to every imaginary root $\xi + i\eta$ there belongs a second imaginary root distinct from it, $\xi - i\eta$, i. e. the conjugate imaginary root

belongs. Since, together with the derivatives $f'(\xi + i\eta), f''(\xi + i\eta) \ldots$, the conjugate quantities $f'(\xi - i\eta), f''(\xi - i\eta) \ldots$ vanish at the same time, it follows that conjugate imaginary roots have the same degree of multiplicity.

From this it follows that the imaginary roots always occur in even number, and that an equation of odd degree must always have at least one real root.

Our next aim will be to determine the number of real roots, without solving the equation, directly from the values of certain rational functions of the coefficients of the equation.

A very important part is played here by the discriminant, whose significance for our question we must first consider.

In §. 46 we defined the discriminant as the product of the squares of all the root differences

$$(x_1 - x_2)^2,$$

further multiplied by a_0^{2n-2}, where a_0 is the coefficient of the highest power of the unknown in the given equation, and we showed there how the discriminant can be computed as a rational function of the coefficients. The factor a_0^{2n-2} is always positive, and could here also be assumed equal to 1 without essential restriction of generality. As before, we shall use the sign D for the discriminant.

The discriminant vanishes when, and only when, two equal roots occur among the roots.

If we assume that $f(x)$ has been freed of multiple factors by separating off the greatest common divisor of $f(x)$ and $f'(x)$, which is done by rational computation, then D will not vanish.

If x_1 and x_2 are real, then $(x_1 - x_2)^2$ is positive. If x_1 is real and x_2 imaginary, then there is a root x_2' conjugate to x_2, and the product

$$(x_1 - x_2)(x_1 - x_2')$$

is positive, being the product of two conjugate imaginary quantities, and hence so is its square.

If x_1 and x_2 are both imaginary, but not conjugate, then the product

$$(x_1 - x_2)(x_1' - x_2')$$

and its square are positive.

But if, finally, x_1 and x_2 are conjugate imaginary, then their difference is purely imaginary and its square negative.

Thus in the product by which we have defined the discriminant there occur exactly as many negative factors as there are pairs of conjugate imaginary roots, and from this we conclude the important fundamental theorem:

If $f(x) = 0$ is a real equation without multiple roots, then the discriminant is positive or negative according as the number of pairs of conjugate imaginary roots is even or odd.

The case in which only real roots are present belongs to those in which the discriminant is positive.

For the cases of the quadratic and cubic equations, in which only one pair of imaginary roots can occur, this principal theorem already completes the determination among the various cases possible with regard to the reality of the roots.

§. 77.
Discussion of the quadratic and cubic equation.

For the quadratic equation

$$a_0x^2 + a_1x + a_2 = 0$$

we have (§. 46)

$$D = a_1^2 - 4a_0a_2 \begin{cases} > 0 & \text{real roots.} \\ < 0 & \text{imaginary roots.} \end{cases}$$

For the cubic equation

$$a_0x^3 + a_1x^2 + a_2x + a_3 = 0$$

we have

$$D = a_1^2a_2^2 + 18a_0a_1a_2a_3 - 4a_0a_2^3 - 4a_1^3a_3 - 27a_0^2a_3^2,$$

$D > 0$ three real roots,
$D < 0$ one real, two imaginary roots.

The case $D = 0$ likewise gives no occasion here for further distinctions; for in this case the two roots of the quadratic equation are equal to one another and real, and of the three roots of the cubic equation two are equal to one another and all three are real; for an imaginary root cannot occur doubly here, since otherwise its conjugate would also occur doubly, and just as little can the single root be imaginary, since otherwise a second one would have to be present.

For the cubic equation it only remains to seek the condition under which all three roots are equal to one another.

In this case the left-hand side of the cubic equation must be a complete cube, so that

$$a_0x^3 + a_1x^2 + a_2x + a_3 = a_0(x - \alpha)^3.$$

Comparison of the two sides of this identical equation yields

$$a_1 = -3a_0\alpha, \quad a_2 = 3a_0\alpha^2, \quad a_3 = -a_0\alpha^3,$$

from which, by elimination of α, one obtains

$$\begin{aligned} a_1^2 - 3a_0a_2 &= 0, \\ a_1a_2 - 9a_0a_3 &= 0. \end{aligned} \tag{1}$$

From this there results as a consequence (by multiplying the first of these equations by a_2, the second by a_1, and subtracting)

$$a_2^2 - 3a_1a_3 = 0;$$

and when these conditions are satisfied, there follows from them conversely

$$f(x) = a_0\left(x + \frac{a_1}{3a_0}\right)^3,$$

i. e. the equality of all three roots.

For the cubic equation

$$x^3 + a_1x^2 + a_2x + a_3 = 0$$

we can decide completely not only about the reality but also about the signs of the roots.

Namely, if we denote by α, β, γ the three roots, then

$$a_1 = -(\alpha + \beta + \gamma), \quad a_2 = \alpha\beta + \alpha\gamma + \beta\gamma, \quad a_3 = -\alpha\beta\gamma. \tag{2}$$

Now if $D < 0$, so that only one root, say α, is real, then $\beta\gamma$ is positive and α will be negative or positive according as a_3 is positive or negative. But if D is positive, so that all three roots are real, then, when α, β, γ are positive, by (2)

$$a_1 < 0, \quad a_2 > 0, \quad a_3 < 0; \tag{3}$$

these conditions are necessary in order that all three roots be positive. But they are also sufficient; for if one or three roots are negative, then $a_3 > 0$. If, however, two roots, say β, γ, are negative, then either

$$a_1 \geqq 0 \quad \text{or} \quad \alpha > -(\beta + \gamma).$$

But in the latter case it follows that

$$a_2 < \beta\gamma - (\beta + \gamma)^2 = -\beta^2 - \beta\gamma - \gamma^2,$$

so a_2 is negative.

Finally, if one root is equal to 0, then a_3 must necessarily vanish.

By replacing x with $-x$, one concludes that for three negative roots the necessary and sufficient condition is

$$a_1 > 0, \quad a_2 > 0, \quad a_3 > 0. \tag{4}$$

We therefore obtain, under the assumption of a positive discriminant, the following table:

a_1	a_2	a_3	
$-$	$+$	$-$	3
$-$	$-$	$-$	
			}1
$+$	$+$	$-$	
$+$	$-$	$-$	
$+$	$+$	$+$	0
$-$	$-$	$+$	
			}2
$-$	$+$	$+$	
$+$	$-$	$+$	

where in the last column the number of positive roots is given. We can express the result of this consideration as follows:

For a positive discriminant, the number of positive roots of the cubic equation is equal to the number of sign changes in the sequence

$$1,\ a_1,\ a_2,\ a_3,$$

where by a sign change we understand the succession of a positive and a negative or of a negative and a positive quantity.

If one of the two quantities a_1, a_2 vanishes, then, since in that case not all the roots can be of the same sign, we have one or two positive roots.

If $a_3 = 0$, the discriminant reduces to

$$a_2^2(a_1^2 - 4a_2),$$

and if this is positive, the cubic equation has one vanishing root and two further real roots. These are positive if

$$a_1 < 0, \quad a_2 > 0,$$

negative if

$$a_1 > 0, \quad a_2 > 0,$$

and one of them is positive and one negative if

$$a_1 \gtreqless 0, \quad a_2 < 0$$

holds.

§. 78.
Discussion of the biquadratic equation.

For equations of the fourth and fifth degree there exist either no imaginary roots, or two, or four. If the discriminant is negative, there are two imaginary roots. With a positive discriminant, either four imaginary roots or none may be present. To distinguish these two cases will be our task in what follows. But first we turn to an elementary consideration of the biquadratic equation, which for the sake of simplicity we shall assume in the abbreviated form

$$x^4 + ax^2 + bx + c = 0 \tag{1}$$

from which we can easily return to the general form.

If we denote the roots by α, β, γ, δ and set, as in §. 47,

$$\begin{aligned} \alpha - \beta &= v + w, \quad \gamma - \delta = v - w, \\ \alpha - \gamma &= w + u, \quad \delta - \beta = w - u, \\ \alpha - \delta &= u + v, \quad \beta - \gamma = u - v, \end{aligned} \tag{2}$$

then u^2, v^2, w^2 are the roots of the cubic resolvent

$$y^3 + 2ay^2 + (a^2 - 4c)y - b^2 = 0, \tag{3}$$

and the discriminant D of this cubic equation, which by

$$27D = 4(a^2 + 12c)^3 - (2a^3 - 72ac + 27b^2)^2 \tag{4}$$

is determined, is at the same time the discriminant of the biquadratic equation (1), and the signs of the quantities u, v, w are restricted by the condition

$$uvw = -b \tag{5}$$

[§. 37, (3)].

If the discriminant is negative, then the equation (3), just like (1), has two conjugate imaginary roots.

If D is positive and all four roots α, β, γ, δ are real, then u, v, w also become real, and hence their squares, i. e. the roots of (3), are positive.

If all four roots are imaginary, say α conjugate to β and γ to δ, then it follows from (2) that v and w are purely imaginary and u is real; hence in this case the equation (3) has one positive and two negative roots.

In both cases, however, if b vanishes, one of the quantities u, v, w may also be equal to zero.

Having regard to the results of the last section on the cubic equation, we thus arrive at the following result:

The necessary and sufficient condition for the existence of four distinct real roots is

$$D > 0, \quad a < 0, \quad a^2 - 4c > 0. \tag{6}$$

In all other cases in which D is positive, the equation has four imaginary roots.

We can also carry the discussion through completely for $D = 0$, that is, in the case of the equality of two roots.

If we assume $\gamma = \delta$, then, since we have made the assumption $\alpha + \beta + \gamma + \delta = 0$, it follows from (2) that

$$2v = 2w = \alpha - \beta$$
$$2u = \alpha + \beta - 2\gamma = 2(\alpha + \beta)$$

and from this, by (3),

$$\begin{aligned} -4a &= 2(\alpha+\beta)^2 + (\alpha-\beta)^2 \\ 16(a^2-4c) &= (\alpha-\beta)^2[8(\alpha+\beta)^2 + (\alpha-\beta)^2]. \end{aligned} \tag{7}$$

To dispose first of the case in which also $\alpha = \beta$, we obtain from (7) for this the condition $a^2 - 4c = 0$, and the first equation (7) shows that a is negative if α and γ

real, but positive when α and γ are conjugate (and then, because $\alpha + \gamma = 0$, purely imaginary). From this it follows:

The biquadratic equation has two pairs of equal real roots when

$$D = 0, \quad a < 0, \quad a^2 - 4c = 0 \tag{8}$$

and two pairs of equal imaginary roots when

$$D = 0, \quad a > 0, \quad a^2 - 4c = 0. \tag{9}$$

But if α is different from β, then γ must be real, and (7) shows that when α and β are real, a is negative and $a^2 - 4c$ is positive; that, on the other hand, when α and β are conjugate imaginary, either a must be positive or $a^2 - 4c$ negative, since in this case $(\alpha - \beta)^2$ is negative, and if $2(\alpha+\beta)^2 + (\alpha-\beta)^2$ is positive, then certainly $8(\alpha+\beta)^2 + (\alpha-\beta)^2$ must also be positive.

We can therefore supplement (6) by saying:

The necessary and sufficient condition for the existence of four real roots, of which two (but not more) may also be equal to one another, is

$$D \geqq 0, \quad a < 0, \quad a^2 - 4c > 0. \tag{10}$$

The necessary and sufficient condition for three equal roots, which are necessarily all real, is by (2)

$$u^2 = v^2 = w^2,$$

hence the cubic resolvent (3) must have three equal roots, and for this, by §. 77 (1), the necessary and sufficient conditions are

$$\begin{aligned} a^2 + 12c &= 0 \\ 2a^3 - 8ac + 9b^2 &= 0. \end{aligned} \tag{11}$$

By §. 47, (15), (16) the two invariants A and B of the biquadratic equation are

$$A = a^2 + 12c, \quad B = 2a^3 - 72ac + 27b^2.$$

Hence we can also write the equations (11) thus

$$A = 0, \quad B + 4aA = 0,$$

so that (11) is equivalent to

$$A = 0, \quad B = 0.$$

By elimination of c one can also derive from (11) the equation:

$$8a^3 + 27b^2 = 0, \tag{12}$$

Finally, there remains the condition for the equality of all four roots

$$a = 0, \quad b = 0, \quad c = 0. \tag{13}$$

These relations can be made very vivid by a geometric interpretation, and although geometric considerations do not properly lie within our plan, we shall nevertheless not neglect to point them out to the reader from time to time.

If one interprets a, b, c as rectangular coordinates in space, then every point of space is to be regarded as the carrier of a certain biquadratic equation of the form (1), $f(x) = 0$; all equations that have a given number x as a root are represented by the points of a plane $[f(x) = 0]$. The equation $D = 0$ is the equation of a curved surface (of the fifth degree), the discriminant surface, which is generated by the lines of intersection of the planes $f(x) = 0$, $f'(x) = 0$, and is therefore a developable surface. It is the envelope of all the planes $f(x) = 0$.

The surface has an edge of regression consisting of two branches, which is represented by the equations (11), (12), and a double line, which is determined by the equations $b = 0$, $a^2 - 4c = 0$ and therefore has the shape of a parabola. On the side of negative a the surface crosses itself in this parabola. On the side of positive a the parabola continues as an isolated line.

The discriminant surface divides the whole of space into three compartments, of which two are connected only along the double parabola, and these compartments contain the points to which no, two, and four real roots correspond. If for the moment we call these three compartments [0], [2], [4], then [0] borders [4] only along the double parabola, while [0] as well as [4] border [2] along the parts of the surface.

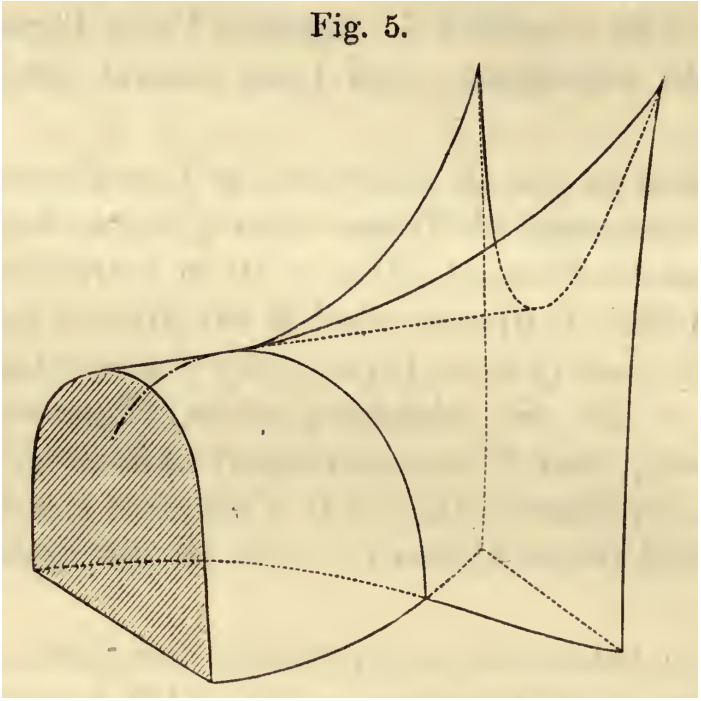
We shall denote by $[0, 2]$ and $[4, 2]$ the boundary surfaces of [0], [2] and of [0], [4]. The isolated part of the parabola continues into the interior of [0]. The edges of regression lie on the part of the surface that separates [4] from [2].

At the origin of coordinates the three parts of space and their boundary surfaces and boundary lines meet. The points of the surface parts $[0, 2]$ represent equations with two equal and

two imaginary roots, and the points of $[2, 4]$ equations with two equal roots and two real roots distinct from them.

On the double parabola two equal roots occur twice, and indeed on the part where [0] borders on [4] they are real, on the isolated part imaginary.

Fig. 5.



The points of the cuspidal edges represent equations with three equal roots, and the origin of coordinates the equation with four equal roots.

It was Kronecker who first drew attention to the discriminant surface and its significance for the discussion of the biquadratic equation. A model of the surface has been constructed by Kerschensteiner.[12]

The analytic expression for the conditions of reality of the roots can be brought into a form in which only the covariants of the biquadratic form occur.[13]

The starting point for this is furnished by the expressions for the functions ψ_1, ψ_2, ψ_3 given in §. 66, (6). If we, in

those expressions we interchange α with β, then ψ_1, ψ_2, ψ_3 pass into ψ_1, $-\psi_3$, $-\psi_2$, and if we simultaneously interchange α with β and γ with δ, then ψ_1, ψ_2, ψ_3 pass into ψ_1, $-\psi_2$, $-\psi_3$. From this the following results:

If one sets an arbitrary real value for the variable x, and if the roots α, β, γ, δ of the biquadratic form are all real, then ψ_1, ψ_2, ψ_3 are also real.

If α, β are conjugate imaginary and γ, δ real, then the interchange of α and β is equivalent to the interchange of i with $-i$. In this case, therefore, ψ_1 is real, and ψ_2, ψ_3 are conjugate imaginary.

If, finally, α, β and γ, δ are two pairs of conjugate imaginary roots, then α is interchanged with β and γ with δ when i passes into $-i$; hence in this case ψ_1 is real, and ψ_2, ψ_3 are purely imaginary.

If four roots are real, then ψ_1^2, ψ_2^2, ψ_3^2 become real and positive.

If two roots are real, then ψ_1^2 is real and positive, and ψ_2^2, ψ_3^2 are conjugate imaginary.

If four roots are imaginary, then ψ_1^2 is positive, and ψ_2^2, ψ_3^2 are real and negative.

Now in §. 66, (7) we have set up the cubic equation whose roots are ψ_1^2, ψ_2^2, ψ_3^2, and have also formed its discriminant. In the previous section we became acquainted with a criterion for the number of positive roots of a cubic equation, from which the following result is obtained:

If $D < 0$, the biquadratic equation has two real and two imaginary roots.

[12] Kronecker, Monatsbericht der Berliner Akademie of 14 February 1878. A description of the model is to be found in the catalogue of mathematical models edited by Dyck, Munich 1892, from which Fig. 5 above is taken.

[13] Clebsch, "Theorie der binären Formen" §. 47. Faà di Bruno, "Theorie der binären Formen", German translation by Walter (Leipzig 1881), §. 20, where an essential addition by Nöther is to be found.

If $D > 0$, then, if four real roots are to exist, in the sequence

$$1, \quad H, \quad H^2 - 16Af^2, \quad -T^2$$

three changes of sign must occur, i.e. we must have

$$H < 0, \quad H^2 - 16Af^2 > 0$$

hold. In the three other still possible combinations of signs all four roots are imaginary. Here, however, an arbitrary real value may be set for x.

§. 79.
The Bezoutiant and its significance for the reality of the roots.

For the investigation of the reality of the roots of an arbitrary real equation in more general cases, the function which we designated in §. 73 as the Bezoutiant of the equation can be applied with advantage.

If

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n = 0 \tag{1}$$

is any real equation of degree n, then the Bezoutiant was defined as follows:

One sets

$$y = t_{n-1}f_0(x) + t_{n-2}f_1(x) + \cdots + t_0f_{n-1}(x), \tag{2}$$

in which the $t_0, t_1 \ldots t_{n-1}$ denote indeterminate variables and

$$f_0(x) = 1, \quad f_1(x) = x + a_1, \quad f_2(x) = x^2 + a_1x + a_2 \ldots$$

Then [§. 73, (10)]

$$S(y^2) = \frac{1}{n}[S(y)]^2 + B, \tag{3}$$

where B is a quadratic form of the $n-1$ variables $t_0, t_1 \ldots t_{n-2}$ with coefficients that are composed rationally from the coefficients a_i. This function B we have defined as the Bezoutiant, and we have actually constructed it for the cases $n = 4$ and $n = 5$.

Now in the sum on the left-hand side of (3)

$$y_1^2 + y_2^2 + y_3^2 + \cdots + y_n^2, \tag{4}$$

the $y_1, y_2 \ldots y_n$ are linear functions of the variables $t_0, t_1 \ldots t_{n-1}$, and thus in the formula (3) we have a transformation of the quadratic form

$$\frac{1}{n}[S(y)]^2 + B = \Phi(t_0, t_1 \ldots t_{n-1}) \tag{5}$$

into a sum of n squares.

For the present we make no restrictive assumptions whatever about equality or distinctness of the roots of $f(x)$, and must now first investigate to what extent the linear functions $y_1, y_2 \ldots y_n$ are independent of one another. In this regard the following theorem holds:

If $x_1, x_2 \ldots x_\nu$ are roots of equation (1) that are distinct from one another, then $y_1, y_2 \ldots y_\nu$ are linearly independent.

For suppose there existed between $y_1, y_2 \ldots y_\nu$ a linear relation, identical with respect to t,

$$\alpha_1y_1 + \alpha_2y_2 + \cdots + \alpha_\nu y_\nu = 0,$$

whose coefficients α are independent of the t and not all equal to zero; then this would break up into the following equations

$$\begin{aligned}
\alpha_1 f_0(x_1) + \alpha_2 f_0(x_2) + \cdots + \alpha_\nu f_0(x_\nu) &= 0,\\
\alpha_1 f_1(x_1) + \alpha_2 f_1(x_2) + \cdots + \alpha_\nu f_1(x_\nu) &= 0,\\
\cdots\cdots\cdots

If the determinant of B, and hence also the discriminant of $f(x)$ (§. 73), is different from zero, then $\pi + \nu = n - 1$ and ν is less than or at most equal to $\frac{1}{2}n$.

The Bezoutiant is thus a quadratic form of a special nature, which expresses itself precisely in the fact that among the squares into which it can be decomposed, at most $\frac{1}{2}n$ negative ones can occur.

§. 80.
The Inertia of Forms of the Second Degree.

By the theorem of the preceding section we are directed to the investigation of the homogeneous functions of the second degree with real coefficients. Concerning these functions we became acquainted in §. 58, under the name of the law of inertia, with a theorem which forms the foundation for what follows.

The theorem consisted in this: however one may transform a quadratic form of m variables into a sum of positive and negative squares of linearly independent linear functions, which is possible in infinitely many different ways, the number of the positive squares, and likewise that of the negative ones, is always the same.

If we denote by π the number of the positive squares and by ν the number of the negative ones, then $\pi + \nu$ is at most equal to m.

If we denote the difference $m - \pi - \nu$ by ϱ, then, when we represent the general quadratic form of m variables as a sum of m squares, ϱ can be designated as the number of squares with vanishing coefficients.

The three numbers π, ν, ϱ, of which none can be negative and whose sum is equal to the number of the variables,

$$\pi + \nu + \varrho = m, \tag{1}$$

are therefore unalterable for a given quadratic form.

We have to seek means of determining the numbers π, ν, ϱ from the coefficients of the quadratic form. If we apply these means to the Bezoutiant, we obtain criteria for inferring, from the coefficients of an equation, the number of its real roots.

Now, then, let, as in §. 56

$$\varphi(x_1, x_2 \dots x_m) = \sum_{1,m}^{i,k} a_{i,k} x_i x_k, \qquad a_{i,k} = a_{k,i} \tag{2}$$

be a quadratic form of m variables with real coefficients $a_{i,k}$.

The determinant

$$R = \Sigma \pm a_{1,1},\, a_{2,2} \dots a_{m,m} \tag{3}$$

shows by its vanishing that φ can be transformed by a linear transformation into a function of fewer than m variables, and hence that ϱ is greater than zero.

The determinant R is a symmetric determinant. Among its subdeterminants of various orders there occur certain ones which are again symmetric determinants, namely those which one obtains from R by striking out rows and columns which intersect in diagonal terms; those, therefore, which, if $\alpha, \beta, \gamma \dots$ denote any of the indices $1, 2 \dots m$, are to be denoted by

$$\Sigma \pm a_{\alpha,\alpha}\, a_{\beta,\beta}\, a_{\gamma,\gamma} \dots \tag{4}$$

These we shall call the principal subdeterminants of R.

We shall now determine the number of the principal subdeterminants.

The number of μ-rowed principal subdeterminants is equal to the number of ways in which groups of μ different indices can be selected from the sequence of indices $1, 2, 3 \ldots m$, hence equal to the number of combinations of m elements taken μ at a time, and this number is equal to the binomial coefficient B_μ^m. The number of all the principal subdeterminants is therefore, if we count the determinant R itself and in addition unity as a zero-rowed determinant,

$$1 + m + \frac{m(m-1)}{1.2} + \cdots + m + 1 = 2^m.$$

But usually only a small part of these actually needs to be taken into account.

The indices $1, 2, 3 \ldots m$ can be arranged in $\Pi(m) = 1.2.3 \ldots m$ different ways; if we form the determinant with any one of these arrangements

$$\begin{vmatrix} a_{1,1}, & a_{1,2}, & a_{1,3} & \cdots & a_{1,m} \\ a_{2,1}, & a_{2,2}, & a_{2,3} & \cdots & a_{2,m} \\ a_{3,1}, & a_{3,2}, & a_{3,3} & \cdots & a_{3,m} \\ \cdot & \cdot & \cdot & \cdots & \cdot \\ a_{m,1}, & a_{m,2}, & a_{m,3} & \cdots & a_{m,m} \end{vmatrix},$$

then from it a series of principal subdeterminants can be

derive, as is indicated by the strokes, that is, in such a way that one obtains each preceding one from the one that follows it by omitting the last row and column; thus

$$R_0 = 1, \quad R_1 = a_{1,1}, \quad R_2 = a_{1,1}a_{2,2} - a_{1,2}^2,$$

$$R_3 = \begin{vmatrix} a_{1,1}, & a_{1,2}, & a_{1,3} \\ a_{2,1}, & a_{2,2}, & a_{2,3} \\ a_{3,1}, & a_{3,2}, & a_{3,3} \end{vmatrix}, \ldots R_m = R.$$

Such a system, when it is ordered in the descending sequence

$$R_m,\ R_{m-1} \ldots R_1,\ R_0$$

shall be called a chain of principal subdeterminants. Of such chains, $\Pi(m)$ different ones can be formed, in all of which the first term is R and the last term is 1.

§. 81.
Quadratic Forms with Vanishing Determinant.

If the determinant R of the quadratic form $\varphi(x_1, x_2 \ldots x_m)$, defined in (3), §. 80, vanishes, then, as we have already seen in §. 57, φ can be expressed by fewer than m mutually independent linear functions of x. If we denote by $m - \varrho$ the smallest number of variables by which φ can be expressed, then ϱ has the same meaning as in the preceding section.

Now if among the principal subdeterminants of R there is found a k-rowed one which is different from zero, say

$$R_k = \Sigma \pm a_{1,1}a_{2,2} \ldots a_{k,k},$$

then, if we set

$$x_{k+1} = 0, \quad x_{k+2} = 0, \ldots x_m = 0$$

the expression

$$\varphi(x_1, x_2 \ldots x_k, 0, 0, 0)$$

is a quadratic form of k variables $x_1, x_2 \ldots x_k$ whose determinant R_k is different from zero, and which accordingly cannot be expressed by fewer than k variables. Still less, then, can the function φ, when $x_1, x_2 \ldots x_m$ vary without restriction, depend on fewer than k variables, and it follows that

(1) $$\varrho \leqq m - k.$$

We now assume that the form $\varphi(x_1, x_2 \dots x_m)$ can be expressed by means of k mutually independent linear forms in x, which we shall denote as follows:

$$
\begin{aligned}
z_1 &= \beta_1^{(1)} x_1 + \beta_2^{(1)} x_2 + \cdots + \beta_k^{(1)} x_k + \beta_{k+1}^{(1)} x_{k+1} + \cdots + \beta_m^{(1)} x_m \\
&\quad \cdots\cdots\cdots\cdots\cdots\cdots \\
z_k &= \beta_1^{(k)} x_1 + \beta_2^{(k)} x_2 + \cdots + \beta_k^{(k)} x_k + \beta_{k+1}^{(k)} x_{k+1} + \cdots + \beta_m^{(k)} x_m
\end{aligned} \tag{2}
$$

Since $z_1, z_2 \dots z_k$ are to be linearly independent of one another, at least one of the k-rowed determinants that can be formed from the matrix

$$
\begin{matrix}
\beta_1^{(1)}, & \beta_2^{(1)} & \dots & \beta_m^{(1)} \\
\cdot & \cdot & \cdot & \cdot \\
\beta_1^{(k)}, & \beta_2^{(k)} & \dots & \beta_m^{(k)}
\end{matrix}
$$

must be different from zero. For if they were all equal to zero, then the unknowns $h_1, h_2 \dots h_k$, which are not all to vanish, could be determined from the equations

$$
h_1 \beta_s^{(1)} + h_2 \beta_s^{(2)} + \cdots + h_k \beta_s^{(k)} = 0, \quad s = 1, 2, 3 \dots m
$$

(§. 24), and $z_1, z_2 \dots z_k$ would satisfy an equation

$$
h_1 z_1 + h_2 z_2 + \cdots + h_k z_k = 0
$$

and so would not be independent.

We may, however, without restricting the generality, assume that the determinant

$$
\Sigma \pm \beta_1^{(1)} \beta_2^{(2)} \dots \beta_k^{(k)}
$$

is different from zero, and then we can solve the equations (2) for the variables $x_1, x_2 \dots x_k$.

To these solutions we give the form

$$
\begin{aligned}
x_1 &= y_1 + \alpha_{k+1}^{(1)} x_{k+1} + \cdots + \alpha_m^{(1)} x_m \\
x_2 &= y_2 + \alpha_{k+1}^{(2)} x_{k+1} + \cdots + \alpha_m^{(2)} x_m \\
&\quad \cdots\cdots\cdots\cdots \\
x_k &= y_k + \alpha_{k+1}^{(k)} x_{k+1} + \cdots + \alpha_m^{(k)} x_m,
\end{aligned} \tag{3}
$$

in which the y_s are linear homogeneous combinations of the z_s, which are determined from the equations

$$
\begin{aligned}
\beta_1^{(1)} y_1 + \beta_2^{(1)} y_2 + \cdots + \beta_k^{(1)} y_k &= z_1, \\
\beta_1^{(2)} y_1 + \beta_2^{(2)} y_2 + \cdots + \beta_k^{(2)} y_k &= z_2, \\
&\quad \cdots\cdots\cdots\cdots \\
\beta_1^{(k)} y_1 + \beta_2^{(k)} y_2 + \cdots + \beta_k^{(k)} y_k &= z_k
\end{aligned}
$$

be determined, while the α are new coefficients, which are derived from the β in an easily surveyed manner, and with whose formation we are not further concerned here.

Now by hypothesis the function $\varphi(x_1, x_2 \dots x_m)$ can be expressed by $z_1, z_2 \dots z_k$ alone, and hence also by $y_1, y_2 \dots y_k$ alone. If we denote this expression by $\Phi(y_1, y_2 \dots y_k)$, then [by the mediation of (3)] we have the identity

$$
\varphi(x_1, x_2 \dots x_k, x_{k+1} \dots x_m) = \Phi(y_1, y_2 \dots y_k), \tag{4}
$$

and if we set $x_{k+1}, x_{k+2}, \dots x_m = 0$ in it,

$$
\Phi(x_1, x_2 \dots x_k) = \varphi(x_1, x_2 \dots x_k, 0, 0 \dots 0), \tag{5}
$$

whereby, since the designation of the variables does not matter, Φ is completely determined; we may therefore set

$$\Phi(y_1, y_2 \dots y_k) = \varphi(y_1, y_2 \dots y_k, 0, 0 \dots 0) \tag{6}$$

$$= \varphi(x_1, x_2 \dots x_m); \tag{1}$$

Φ thus arises from φ by setting the $m-k$ last variables (in the arrangement chosen here) equal to zero:

$$\Phi(x_1, \dots x_k) = \sum_{1,k}^{r,s} a_{r,s} x_r x_s. \tag{7}$$

The determinant of Φ is one of the principal subdeterminants of R, and indeed one obtains it by striking out the $m-k$ last rows and columns in R.

If we assume that φ cannot be expressed by still fewer than k variables, so that $k = m - \varrho$, then the determinant of this function Φ cannot vanish.

From this and from what was proved above there now results the theorem:

II. If all principal subdeterminants of R of more than k rows vanish, while among the principal subdeterminants of k rows at least one is different from zero, then

$$\varrho = m - k.$$

For it follows from (7) that, if $\varrho = m-k$, at least one k-rowed principal subdeterminant must be different from zero, and from (1) that, if in addition a principal subdeterminant of more than k rows is different from zero, then

$$\varrho < m - k$$

holds.

If π, ν, ϱ are the number of positive, negative, vanishing squares into which φ can be decomposed, then π, ν are the numbers of positive and negative squares into which the form Φ determined by (7) can be decomposed, and the determination of the numbers π, ν therefore need only be carried out for the latter function, whose determinant is different from zero.

§. 82.
Quadratic forms with non-vanishing determinant.

In the investigation of the forms $\varphi(x_1, x_2 \dots x_m)$ with non-vanishing determinant we make use of a theorem on determinants which we proved at the end of §. 28, and which we wish to recall here.

If A is a determinant of m rows, and if $A_i^{(k)}$ are its first, $A_{i,i'}^{k,k'}$ its second subdeterminants, then

$$A_1^{(1)} A_i^{(i)} - A_1^{(i)} A_i^{(1)} = A A_{1,i}^{1,i}. \tag{1}$$

If A is a symmetric determinant different from zero, then $A_1^{(i)} = A_i^{(1)}$ and we can write formula (1) thus:

$$A_1^{(1)} A_i^{(i)} - A_1^{(i)2} = A A_{1,i}^{1,i}. \tag{2}$$

In this, i may denote any of the indices $2, 3 \dots m$. If $A_1^{(1)} = 0$ and $A_{1,i}^{1,i} = 0$, it follows from this that $A_1^{(i)} = 0$ must also hold, and from this we conclude that

$$A_1^{(1)},\ A_{1,2}^{1,2},\ A_{1,3}^{1,3} \dots A_{1,m}^{1,m}$$

cannot vanish simultaneously, since otherwise

$$A_1^{(2)},\ A_1^{(3)} \dots A_1^{(m)},$$

and hence also, contrary to the hypothesis, A would vanish (§. 22).

If we apply this to the determinant $R = R_m$ of our function φ, now assumed to be different from zero, it follows that though the first principal subdeterminant R_{m-1} may vanish, not all the second principal subdeterminants R_{m-2} can vanish. The same mode of reasoning can be applied when we replace R by a non-vanishing $R_{m-1}, R_{m-2} \dots$, and we thus arrive at the following important theorem:

III. If R is different from zero, one can arrange the indices $1, 2 \dots m$ so that in the chain of principal subdeterminants

$$R_m,\ R_{m-1} \dots R_1,\ R_0 \tag{3}$$

no two consecutive terms vanish.

The determinant relation (2) yields, if one sets

$$A = R_{k+1}, \quad (k = 1, 2 \dots m-1)$$

an equation between three consecutive terms of the chain (3), which we can write thus:

$$R_k S_k - T_k^2 = R_{k-1} R_{k+1}, \tag{4}$$

in which S_k and T_k are certain subdeterminants of R, and hence integral rational functions of the coefficients $a_{i,k}$.

More precisely described, R_k, S_k, T_k are first subdeterminants of R_{k+1}, and indeed

$$R_k = \frac{\partial R_{k+1}}{\partial a_{k+1,k+1}}, \quad S_k = \frac{\partial R_{k+1}}{\partial a_{k,k}}, \quad T_k = \frac{\partial R_{k+1}}{\partial a_{k,k+1}}.$$

An arrangement of the indices corresponding to theorem (3) we shall suppose chosen for what follows. Then, if R_k vanishes, R_{k-1} and R_{k+1} are different from zero, and (4) shows that they have opposite signs; hence:

IV. If an inner term of a chain of principal subdeterminants vanishes, then the two adjoining terms have opposite signs.

§. 83.

Number of the positive and negative squares.

In order to determine the number of positive and negative squares of a form with non-vanishing determinant, we assume first that in the chain of principal subdeterminants

$$R_m,\ R_{m-1},\ R_{m-2} \dots R_1,\ R_0 = 1$$

no term vanishes. We can then determine the coefficient λ so that the determinant of the form

$$\psi = \varphi(x_1, x_2, x_3 \dots x_m) - \lambda x_m^2 \tag{1}$$

vanishes. For the determinant is

$$\begin{vmatrix} a_{1,1},\ a_{1,2} \dots a_{1,m} \\ a_{2,1},\ a_{2,2} \dots a_{2,m} \\ \dots\dots\dots\dots \\ a_{m,1},\ a_{m,2} \dots a_{m,m} - \lambda \end{vmatrix} = R_m - \lambda R_{m-1},$$

and it vanishes when we set

$$\lambda = \frac{R_m}{R_{m-1}}$$

The function ψ can then be expressed in terms of $m-1$ variables y, and by formula (6), §. 81, one obtains

$$\psi(x_1, x_2 \dots x_m) = \psi(y_1, y_2 \dots y_{m-1}, 0),$$

or, by (1),

$$\varphi(x_1, x_2 \dots x_m) = \frac{R_m}{R_{m-1}} x_m^2 + \varphi(y_1, y_2 \dots y_{m-1}, 0), \tag{2}$$

and the investigation of the function φ of m variables is thereby reduced to the investigation of $\varphi_1(y) = \varphi(y_1, y_2 \dots y_{m-1}, 0)$ of $m-1$ variables, whose determinant is equal to R_{m-1} and hence different from zero. According as $R_m : R_{m-1}$ is positive or negative, the function $\varphi(x)$ will have one more positive or one more negative square than $\varphi_1(y)$.

By applying the same procedure to $\varphi_1(y)$ and the following functions, we obtain the theorem:

The number of positive and negative terms of the sequence

$$\frac{R_m}{R_{m-1}}, \quad \frac{R_{m-1}}{R_{m-2}}, \dots \frac{R_1}{R_0} \tag{3}$$

agrees with the number of positive and negative squares into which the function $\varphi(x)$ can be decomposed.

We wish to give this theorem a somewhat different expression, but first premise the following explanation.

If a sequence of real numbers different from zero is given in a definite arrangement, the signs of these quantities can change in manifold ways; if two quantities of the same sign follow one another, there is a succession of signs (permanence), but if a quantity is followed by another of opposite sign, we have a change of sign (variation).

If we now consider the sequence of quantities from this point of view,

$$R_m,\ R_{m-1},\ R_{m-2} \dots R_1,\ R_0,$$

then in passing from R_k to R_{k-1} there occurs a permanence of sign or a change of sign, according as the quotient $R_k : R_{k-1}$ is positive or negative.

We can therefore also state the following theorem:

V. If π is the number of positive, ν that of the negative squares of φ, then π is equal to the number of permanences of sign, ν equal to the number of changes of sign in the chain

$$R_m,\ R_{m-1},\ R_{m-2} \dots R_1,\ R_0. \tag{4}$$

This formulation of the theorem has the advantage that it can be carried over to the case in which individual interior terms of the sequence (4) vanish, provided only that no two consecutive terms are zero (which by §. 82, III. may always be assumed). For if $R_k = 0$, while R_{k-1}, R_{k+1} are different from zero, then by §. 82, Theorem IV, R_{k-1} and R_{k+1} have different signs. In the sequence

$$R_{k+1},\ R_k,\ R_{k-1}$$

there thus occurs one change of sign and one permanence of sign, no matter whether we replace the vanishing R_k by a positive or by a negative quantity.

If now we replace the chain (4)

$$R_m,\ R_{m-1} \dots R_1,\ R_0,$$

in which individual terms vanish, by another,

$$R'_m,\ R'_{m-1}\dots R'_1,\ R'_0, \tag{5}$$

in which no term vanishes, and in which to the non-vanishing terms of the sequence (4) there correspond terms of the same sign, then the sequences (4) and (5) have equally many changes of sign, whatever signs the R' corresponding to the vanishing R may have.

Now let $\psi(x)$ be a second arbitrary quadratic form of the variables x, with which we form the form

$$\varphi' = \varphi + \varepsilon\psi \tag{6}$$

in which ε is a still undetermined coefficient. We shall now show at once that we can choose ε in such a way that φ' and φ have the same number of positive and negative squares

have; that, however, in the chain of the principal subdeterminants R'_k of the form φ' no vanishing terms occur; and that, finally, to a non-vanishing R_k there corresponds an R'_k of the same sign. Then the numbers π, ν for φ and for φ' can be determined both from the series (4) and from the series (5), and the number of changes of sign, which is the same in both, gives the number ν of the negative squares.

Now, in order to prove that the coefficient ε can be determined in the manner indicated, we assume that φ has somehow been transformed into a sum of squares

$$\varphi = \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_m y_m^2,$$

and that ψ, expressed in the variables $y_1, y_2 \dots y_m$, has the expression

$$\psi = \Sigma \beta_{i,k} y_i y_k.$$

The numbers π, ν for the function φ' are then determined from the chain of the principal subdeterminants of

$$\begin{vmatrix} \lambda_1 + \varepsilon\beta_{1,1}, & \varepsilon\beta_{1,2}\dots & \varepsilon\beta_{1,m} \\ \varepsilon\beta_{2,1}, & \lambda_2 + \varepsilon\beta_{2,2}\dots & \varepsilon\beta_{2,m} \\ \cdot & \cdot & \cdot \\ \varepsilon\beta_{m,1}, & \varepsilon\beta_{m,2}\dots & \lambda_m + \varepsilon\beta_{m,m} \end{vmatrix}$$

according to Theorem V.

Now, however, one may take ε so small that these principal subdeterminants agree in sign with

$$\lambda_1\lambda_2\lambda_3\dots\lambda_m, \quad \lambda_1\lambda_2\lambda_3\dots\lambda_{m-1}, \quad \dots, \quad \lambda_1\lambda_2, \quad \lambda_1, \quad 1,$$

and then the number of positive and negative squares of φ' is equal to the number of positive and negative ones among the coefficients λ, of which none vanishes, i. e. the numbers π and ν are the same for φ and φ'.

If now the coefficients of ψ, expressed in the original variables, are $b_{i,k}$, so that

$$\psi = \Sigma b_{i,k} x_i x_k,$$

then one of the principal subdeterminants of φ' is

$$R'_k = \begin{vmatrix} a_{1,1} + \varepsilon b_{1,1} & \dots & a_{1,k} + \varepsilon b_{1,k} \\ \cdot & \cdot & \cdot \\ a_{k,1} + \varepsilon b_{k,1} & \dots & a_{k,k} + \varepsilon b_{k,k} \end{vmatrix},$$

and is thus an integral rational function of degree k in ε,

$$R'_k = R_k + \varepsilon M_1 + \varepsilon^2 M_2 + \dots \varepsilon^k M_k,$$

in which the $M_1, M_2 \dots M_k$ depend rationally on the $a_{i,k}, b_{i,k}$. In particular,

$$M_k = \Sigma \pm b_{1,1} b_{2,2} \dots b_{k,k},$$

and one can always choose the $b_{i,k}$ so that M_k is different from zero. By §. 31 one can therefore choose ε so that, if R_k is different from zero, R'_k has the same sign as R_k, and if R_k vanishes, R'_k does not vanish.

We can combine what has thus been proved with the results of §. 81 into a general rule for determining the number of negative squares, valid also in the case of vanishing determinant.

VI. To determine the numbers π, ν, ϱ of the function $\varphi(x)$, arrange the variables $x_1, x_2 \dots x_m$ so that in the chain of principal subdeterminants

$$R_m, R_{m-1} \dots R_1, R_0 \tag{7}$$

as small a number of initial terms as possible vanishes, and so that of the following terms no two standing next to one another vanish; ϱ is then the number of vanishing initial terms, ν the number of changes of sign, and $\pi = m - \nu - \varrho$.

Finally let it be remarked that the number of changes of sign in the series (7) can also be counted from right to left, that is, one finds the same number of changes of sign if the series is written in the reverse order.

§. 84.
Application to the Bezoutiant.

The exact discussion of the inertia of quadratic forms had for us only the one purpose of determining the number of real and imaginary roots of an algebraic equation by counting the positive and negative squares of the Bezoutiant. If n is the degree of the equation, its Bezoutiant is a quadratic form in $n-1$ variables, and in §. 73 we formed it completely for $n = 3, 4, 5$

and have learned the ways by which one can arrive at their computation in other cases as well.

We have already remarked earlier that the Bezoutiant is not a general quadratic form, but that it has the special peculiarity that the number ν of its negative squares can never become greater than $\frac{1}{2}n$. This peculiarity must find its expression in certain inequality conditions between the coefficients, which, however, are not yet known at present. Only in the case $n = 3$ can we state the algebraic character of this restriction completely.

For the cubic equation

$$f(x) = a_0 x^3 + a_1 x^2 + a_2 x + a_3 = 0$$

the Bezoutiant is, by §. 73, (13), nothing other than the Hessian covariant

$$-\tfrac{2}{3} H(t_1, t_0) = -\tfrac{2}{3}(A_0 t_1^2 + A_1 t_1 t_0 + A_2 t_0^2).$$

But if D is the discriminant, so that

$$3D = 4A_0 A_2 - A_1^2$$

there holds the relation [§. 62, (4), (12)]

$$4H^3 + Q^2 + 27 D f^2 = 0,$$

and hence, if we set $t_0 = 0$ in it,

$$-4A_0^3 + q_0^2 + 27 D a_0^2 = 0, \tag{1}$$

where the coefficient appearing above is given by

$$q_0 = 27a_0^2a_3 - 9a_0a_1a_2 + 2a_1^3.$$

This relation shows that A_0 and D cannot both be positive at the same time; and hence that the form $-H$ is not decomposable into two negative squares.

We wish to apply the general theorem of the preceding section to the biquadratic equation as well, which for the sake of simplicity we assume in the form

$$x^4 + ax^2 + bx + c = 0 \tag{2}$$

To form the coefficients of the Bezoutiant, we have, in the formulas §. 73, (14), to replace

$$a_0,\ a_1,\ a_2,\ a_3,\ a_4$$

by

$$1,\ 0,\ a,\ b,\ c$$

We thus obtain

$$B_{2,2} = -2a, \quad B_{1,1} = a^2 - 4c, \quad B_{0,0} = -\tfrac{1}{4}(8ac - 3b^2),$$
$$B_{0,1} = \tfrac{1}{2}ab, \quad B_{2,0} = -4c, \quad B_{1,2} = -3b.$$

We arrange the determinant thus:

$$\begin{vmatrix} -2a, & -3b, & -4c \\ -3b, & a^2 - 4c, & \tfrac{1}{2}ab \\ -4c, & \tfrac{1}{2}ab, & -\tfrac{1}{4}(8ac - 3b^2) \end{vmatrix}$$

and obtain the chain of principal subdeterminants

$$D,\ -2a^3 + 8ac - 9b^2,\ -2a,\ 1, \tag{3}$$

where D denotes the discriminant, which by §. 64, (11) is determined by

$$27D = 4(a^2 + 12c)^3 - (2a^3 - 72ac + 27b^2)^2. \tag{4}$$

The criterion for all four roots to be real is accordingly

$$D > 0, \quad -2a^3 + 8ac - 9b^2 > 0, \quad -2a > 0. \tag{5}$$

This criterion is apparently different from, and in any case less simple than, the one set up in §. 78

$$D > 0, \quad a < 0, \quad a^2 - 4c > 0. \tag{6}$$

But we now wish to show further that the two say exactly the same thing.

First of all it is immediately seen that (6) must be satisfied when (5) is satisfied, for

$$-2a(a^2 - 4c) - 9b^2$$

cannot be positive when a and $a^2 - 4c$ are negative; but to see conversely that the conditions (5) follow from (6), we must take the value of D into consideration.

For brevity we set

$$\alpha = -2a, \quad \beta = -6a^3 + 24ac - 27b^2, \quad \gamma = 3a^2 - 12c, \tag{7}$$

so that the conditions (5) read

$$\alpha > 0, \quad \beta > 0, \quad D > 0,$$

while the conditions (6) read

$$\alpha > 0, \quad \gamma > 0, \quad D > 0.$$

It is therefore to be shown that if α, γ and D are positive, β must also be positive. To this end we express D in (4), by means of (7), in terms of α, β, γ and obtain

$$27D = 4(\alpha^2 - \gamma)^3 - [2\alpha(\alpha^2 - \gamma) - \beta]^2.$$

If D is positive, then by this $\alpha^2 - \gamma$ must certainly be positive. If we now set

$$\alpha^2 - \gamma = \delta^2,$$

then $\alpha^2 > \delta^2$, and hence, if δ is taken positive,

$$\alpha > \delta$$

and

$$27D = 4\delta^6 - (2\alpha\delta^2 - \beta)^2.$$

By this, D cannot be positive when β is negative; for if β is negative, then

$$2\alpha\delta^2 - \beta > 2\alpha\delta^2 > 2\delta^3,$$

and so $4\delta^6 - (2\alpha\delta^2 - \beta)^2$ is negative.

It is hereby directly proved that the criteria (5) and (6) say exactly the same thing.

According to the geometric interpretation given in §. 78, $\beta = 0$ means a surface of the third order which passes through the parabola $b = 0$, $\gamma = 0$, and which, so far as negative values of a come into consideration, runs entirely within the region of space in which D is negative.

In order to have before our eyes an example for the equation of the fifth degree, we consider the equation

$$x^5 + x^2 + a = 0.$$

We therefore have to set, in our general expressions of §. 73 or also of §. 74, (2), (3),

$$a_0, a_1, a_2, a_3, a_4, a_5 =$$

$$1, 0, 0, 1, 0, a,$$

and we obtain for the determinant of the Bezoutiant

$$\begin{vmatrix} -2a, & 0, & 0, & -5a \\ 0, & \frac{6}{5}, & -5a, & 0 \\ 0, & -5a, & 0, & -3 \\ -5a, & 0, & -3, & 0 \end{vmatrix}$$

and for the discriminant

$$D = 108a + 3125a^4.$$

A chain of principal subdeterminants is

$$D,\ 50a^3,\ \frac{-12a}{5},\ -2a,\ 1. \tag{8}$$

The discriminant is negative when

$$0 < -a < \sqrt[3]{\frac{108}{3125}}, \tag{9}$$

otherwise positive, and so in particular positive for all positive a. One sees that when the discriminant is positive there are always two

sign changes, whereas for negative discriminant, where a is also negative, one sign change takes place.

So long as a lies in the interval (9), our equation therefore has two imaginary roots; if it lies outside this interval, four imaginary roots, never four real roots.

A purely numerical example is offered by the equation, drawn from the complex multiplication of the elliptic functions,

$$x^5 - x^3 - 2x^2 - 2x - 1 = 0.$$

For the determinant of the Bezoutiant we obtain from §. 73

$$\begin{vmatrix} \frac{-4}{5}, & \frac{-3}{5}, & \frac{4}{5}, & 5 \\ \frac{-3}{5}, & \frac{4}{5}, & \frac{33}{5}, & 8 \\ \frac{4}{5}, & \frac{33}{5}, & \frac{46}{5}, & 6 \\ 5, & 8, & 6, & 2 \end{vmatrix}$$

the chain of principal subdeterminants

$$\frac{47^2}{5},\ \frac{94}{5},\ -1,\ \frac{-4}{5},\ 1,$$

so that the discriminant of our equation is 47^2. The equation therefore has two pairs of imaginary roots and one real root. The second term of this chain, $94 : 5$, need not be computed if one knows the discriminant; indeed, the mere knowledge of the negative sign in the penultimate term already suffices to reach the foregoing decision on the reality of the roots.

Eighth Section.
Sturm's Theorem.

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§. 85.
Sturm's Problem.

The question treated in the previous section, of the number of real roots of an equation, is a special case of a more general problem, which is to form the subject of this section, and which is the foundation of all methods for the approximate numerical computation of the roots of equations.

The question is this: how many real roots of a real numerical equation lie between two given real values a and b?

Once this is decided, the further task is to narrow the bounds a, b until only one root of the given equation lies between them, and finally to bring them so close to one another that each of them can be regarded as an approximate value of this root.

If we take $a = -\infty$, $b = +\infty$, or at any rate a negative and b positive and so large that no roots can lie beyond these bounds, then this problem coincides with the one treated in the previous section, of determining the number of all the real roots.

We shall first show how the general problem can be reduced to the special one, and thus how the question we have now raised is, in principle at least, answered by the considerations of the previous section. Let us first consider

the problem of determining the number of positive roots of an equation $f(x) = 0$. If we set $x = y^2$, then to every real value of y there corresponds a positive value of x, and conversely to every positive value of x there correspond two real, opposite values of y. The number of positive roots of $f(x) = 0$ is therefore half as great as the number of real roots of $f(y^2) = 0$. If we further set

$$y = \frac{x - a}{b - x},$$

then, while x goes from a to b, y will run through positive values from 0 to infinity; and if by this substitution we transform $f(x)$ into $F(y)$, then $f(x) = 0$ will have just as many roots between a and b as $F(y) = 0$ has positive roots, and hence half as many as $F(z^2) = 0$ has real roots. Our task is thus in fact reduced to determining the number of real roots of a certain other equation, whose degree is twice as great as the degree of the given equation. But this reduction of the problem does not give its solution in the simplest form, and we must look for a simpler answer to the question.

§. 86.
The Sturm chains.

The following consideration leads us to a solution of the problem that we posed in the preceding section.

Let α and β be any two real numbers with $\alpha < \beta$. Further, let $f(x)$ be a given integral rational function of x, for which it is to be determined how many of its roots lie in the interval Δ

$$\alpha < x < \beta.$$

We assume that α, β do not themselves belong to the roots of $f(x) = 0$, so that $f(\alpha)$ and $f(\beta)$ are different from zero. Moreover, we shall assume for the present that $f(x)$ has no multiple root, at least in the interval Δ, so that for no value of the interval are $f(x)$ and $f'(x)$ to vanish simultaneously.

We assume that we can in some way construct a series of $m + 1$ continuous functions

$$f(x), f_1(x), f_2(x) \dots f_m(x), \quad (\mathfrak{R}) \tag{1}$$

possessing the following properties:

1. Of the functions f_ν, no two consecutive ones shall vanish simultaneously in the interval Δ.

2. The last of them, $f_m(x)$, shall not vanish at all in the interval Δ, and hence shall keep an unchanging sign.

3. If a middle term, say $f_\nu(x)$, vanishes for some x in the interval Δ, then for this x the two adjacent functions $f_{\nu-1}(x)$ and $f_{\nu+1}(x)$ shall have opposite signs.

4. If $f(x)$ vanishes in the interval Δ, then $f_1(x)$ shall have the same sign for this value of x as $f'(x)$.

Such a series of functions $(\mathfrak{R})$ we shall call a Sturm chain. How one can form it we shall see later; first, consequences are to be drawn from the definition.

For every value of x for which none of the functions of $(\mathfrak{R})$ vanishes, each of these functions has a definite sign. We count a sign change whenever, in running through the chain from left to right, a positive term is followed by a negative one, or a negative term by a positive one. The number of sign changes (variations) counted in this way for a given value of x we shall denote by $V(x)$. If a middle term $f_\nu(x)$ vanishes, then by 3. we are to count one sign change in passing from $f_{\nu-1}(x)$ to $f_{\nu+1}(x)$.

If α, β are two values for which none of the functions f_ν vanishes, and if we now let x increase continuously from α to β, then a change in the number of sign changes can occur only when one of the functions of $(\mathfrak{R})$ changes its sign, and hence passes through the value zero (because of the assumed continuity).

But if this is a middle function, the number of sign changes does not alter (by 3). For in

$$f_{\nu-1},\ f_\nu,\ f_{\nu+1}$$

a change of sign always takes place before and after the passage of f_ν through zero, because $f_{\nu+1}$ and $f_{\nu-1}$ have opposite signs.

But when $f(x)$ passes through zero, then by 4. a change of sign is lost in the passage through x. For if $f(x)$ goes from negative to positive values, then (cf. §. 32) $f'(x)$, and hence $f_1(x)$, is positive, and the signs change thus:

$$\begin{array}{cc} f(x), & f_1(x) \\ - & + \\ + & +, \end{array}$$

and when $f(x)$ passes from positive to negative values, then $f'(x)$ and $f_1(x)$ are negative, so the signs are thus:

$$\begin{array}{cc} f(x), & f_1(x) \\ + & - \\ - & -, \end{array}$$

hence in both cases a change of sign has been lost.

Now since $f_m(x)$ by 2. does not change its sign, it follows that $V(\alpha) - V(\beta)$ is equal to the number of roots of $f(x) = 0$ lying between α and β, or:

The number of roots of $f(x) = 0$ between α and β is equal to the excess of the number of changes of sign of the chain for $x = \alpha$ over the number of changes of sign for $x = \beta$.

If for $x = \alpha$ or $x = \beta$ one or several of the intermediate functions of $(\mathfrak{R})$ should vanish, then by 3. it makes no difference whether we replace this vanishing value by a positive or by a negative one.

§. 87.
First example: spherical functions.

Before we now pass on to the general methods by which Sturm chains are to be formed, we shall consider a few examples in which such chains present themselves through special circumstances.

We consider first the so-called spherical functions, which are used in many ways in mathematical physics and mechanics.

By spherical functions one understands a system of integral rational functions of increasing degree, which is defined as follows:

$$\begin{aligned} P_0(x) &= 1 \\ P_1(x) &= x \\ P_2(x) &= \tfrac{3}{2}(x^2 - \tfrac{1}{3}) \\ P_3(x) &= \tfrac{5}{2}(x^3 - \tfrac{3}{5}x) \\ &\quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \quad . \end{aligned}$$

$$P_n(x) = \frac{1.3 \ldots (2n-1)}{1.2 \ldots n} \left(x^n - \frac{n(n-1)}{2(2n-1)} x^{n-2} + \frac{n(n-1)(n-2)(n-3)}{2.4(2n-1)(2n-3)} x^{n-4} - \cdots \right),$$

so that $P_n(x)$ is an integral rational function of degree n, and, according as n is even or odd, contains only the even or only the odd powers. By use of the sign $\Pi(n)$ one can also represent the general expression for the spherical functions thus:

$$P_n(x) = \sum^{\mu} \frac{(-1)^\mu}{2^n} \frac{\Pi(2n-2\mu) x^{n-2\mu}}{\Pi(\mu)\,\Pi(n-2\mu)\,\Pi(n-\mu)}, \tag{1}$$

in which the sum with respect to μ is to be extended from $\mu = 0$ so far as $n - 2\mu$ does not become negative.

Between these functions there hold the following relations, which, after substituting the expression (1), can be verified by simple comparison of the coefficients of equal powers of x,

$$\begin{aligned} nP_n(x) - (2n-1)xP_{n-1}(x) + (n-1)P_{n-2}(x) &= 0 \\ P_1(x) - xP_0(x) &= 0 \end{aligned} \tag{2}$$

(3) $$(1-x^2)P_n'(x) + nxP_n(x) - nP_{n-1}(x) = 0.$$

From the last equation it follows for $x = \pm 1$

$$\pm P_n(\pm 1) = P_{n-1}(\pm 1),$$

hence

(4) $$P_n(1) = 1, \quad P_n(-1) = (-1)^n.$$

From (2) and (3) it now follows that the functions P_n, ordered downwards from an arbitrary m,

(5) $$P_m,\ P_{m-1},\ P_{m-2} \ldots P_0$$

have, in the interval from -1 to $+1$, the properties of a Sturm chain. For if P_n and P_{n-1} were to vanish simultaneously for any x, then by (2) P_{n-2}, and consequently P_{n-3} etc. down to P_0, would also have to vanish, which is impossible, since $P_0 = 1$. Accordingly the condition §. 86, 2. is also satisfied for every arbitrary interval.

If $P_{n-1} = 0$, then by (2) P_n and P_{n-2} are of opposite sign, as the condition §. 86, 3. requires.

From (3) it follows that $P_n(x)$ and $P_n'(x)$ never vanish simultaneously, since otherwise $P_{n-1}(x)$ would also have to vanish; and finally it follows from (3) that, if $P_n(x)$ vanishes and $-1 < x < 1$, then $P_n'(x)$ and $P_{n-1}(x)$ have the same sign.

If we now substitute the values $-1, +1$ for x in (5), it follows from (4) that for $x = -1$ there are nothing but sign changes in (5), and for $x = +1$ no sign change at all; and from this it follows:

The equation $P_m(x) = 0$ has m real roots between $x = -1$ and $x = +1$, and since P_m is of degree m, these are all the roots.

But we can draw a conclusion that goes somewhat further still. If we take an interval $\Delta = (\alpha, \beta)$ in which two roots ξ, η of $P_m = 0$, and no more, are contained, then in the passage from α to β two sign changes are lost in the chain (5). We take α so close to ξ, and β so close to η, that $P_{m-1}(\alpha)$ has the same sign as $P_{m-1}(\xi)$, and $P_{m-1}(\beta)$ the same sign as $P_{m-1}(\eta)$. If P_m passes from negative to positive as it goes through ξ, then it passes from positive to negative as it goes through η; hence $P_m'(\xi)$, and consequently $P_{m-1}(\xi)$ and $P_{m-1}(\alpha)$, are positive, while $P_m'(\eta)$, and consequently $P_{m-1}(\eta)$ and $P_{m-1}(\beta)$, are negative; and the reverse takes place if P_m passes from positive to negative as it goes through ξ. One sees from this, then, that in the passage from α to β, in

$$P_m, \quad P_{m-1}$$

one sign change is lost; hence in the same passage, and consequently also in the passage from ξ to η, there must, in the chain

$$P_{m-1}, \quad P_{m-2} \ldots P_0$$

a change of sign is likewise lost; from this we conclude the theorem:

Between two successive roots of $P_m = 0$ there lies one and only one root of $P_{m-1} = 0$.

§. 88.
Second example.

As a second example we consider an equation which, for brevity, we shall call the secular equation, because the investigation of the secular perturbations of the planets first led to it[14]). It

[14] Laplace, Histoire de l'Académie des Sciences 1772. Among more recent works one may consult on this Dziobek, Theorie der Planetenbewegung, Leipzig 1888.

is indeed otherwise well known under this name, but it also occurs in many other investigations, e.g. in the determination of the principal axes of a surface of the second degree, and in the theory of small oscillations; its general analytic significance lies in the fact that it furnishes a particular transformation, the so-called orthogonal transformation, of a quadratic form into a sum of squares. We shall here take the equation as it stands, without reference to any application, and shall discuss it according to our general theorems.

Let $a_{i,k}$, where i and k run through the series of indices $1, 2, 3 \ldots n$, be any system of real quantities, and let

$$a_{i,k} = a_{k,i} \tag{1}$$

be assumed. We consider the symmetric determinant

$$L_n(x) = \begin{vmatrix} a_{1,1} - x, & a_{1,2} & \ldots & a_{1,n} \\ a_{2,1}, & a_{2,2} - x & \ldots & a_{2,n} \\ \cdot & \cdot & \cdot & \cdot \\ a_{n,1}, & a_{n,2} & \ldots & a_{n,n} - x \end{vmatrix} \tag{2}$$

which is an integral rational function of degree n in x, in which x^n has the coefficient $(-1)^n$. The equation $L_n(x) = 0$ shall be the object of our consideration.

We form the chain of principal subdeterminants taken with alternating signs

$$L_n(x),\ -L_{n-1}(x),\ L_{n-2}(x) \ldots,\ (-1)^{n-1} L_1(x),\ (-1)^n, \tag{3}$$

and we shall now show that if we add the assumption that of the quantities (3) no two consecutive ones vanish at the same time, we have before us a Sturm chain.

This is a simple consequence of the formula which we already used in §. 82 for a similar purpose, and which we can represent thus:

$$L_k S_k - T_k^2 = L_{k-1} L_{k+1}, \tag{4}$$

in which S_k and T_k are integral rational functions of x.

We have only to show that the requirements §. 86, 1. to 4. are satisfied. Of these, however, 1. is included in the assumption, and 2. is fulfilled, since the last element is equal to ± 1. From the formula (4) it follows that when $L_k = 0$, L_{k-1} and L_{k+1} have opposite signs, so that condition 3. is fulfilled.

Only condition 4. now remains. But if we apply the formula (4) to $k = n - 1$, it reads

$$\frac{\partial L_n}{\partial a_{n,n}} \cdot \frac{\partial L_n}{\partial a_{n-1,n-1}} - \left(\frac{\partial L_n}{\partial a_{n,n-1}} \right)^2 = L_n L_{n-2},$$

and from this it follows that, when $L_n = 0$,

$$\frac{\partial L_n}{\partial a_{n,n}}, \quad \frac{\partial L_n}{\partial a_{n-1,n-1}}$$

have equal signs; and in the same way one can conclude that all the principal subdeterminants of L_n

$$\frac{\partial L_n}{\partial a_{i,i}}$$

have the same sign.

Now, however, the derivative of L_n is (cf. §. 22)

$$L'_n(x) = -\sum_{1,n}^{i} \frac{\partial L_n}{\partial a_{i,i}},$$

and hence, for a value x for which $L_n(x)$ vanishes, it has the opposite sign to L_{n-1}, which is precisely what requirement 4. demands. From this follows the theorem:

I. If α, β are two real values, $\alpha < \beta$, then the number of roots of $L_n(x)$ lying between α and β is equal to the excess of the number of sign changes of the chain (3) for $x = \alpha$ over the number of sign changes for $x = \beta$.

The highest power of x that occurs in $L_k(x)$ is, as already remarked above,

$$(-1)^k x^k,$$

and when the absolute value of x is sufficiently large, the sign of this term will decide the sign of $L_k(x)$. If, then, we take for α a sufficiently large negative value, and for β a sufficiently large positive value, then in (3) there occur only changes of sign for $x = \alpha$, and only permanences of sign for $x = \beta$. Hence n roots will lie between α and β, and from this follows the theorem:

II. The equation $L_n(x) = 0$ has only real roots.

But these two theorems are of only limited applicability so long as we cannot free ourselves from the assumption that in the chain of the L_k no two consecutive terms are to vanish at the same time. From this restriction, however, we can free the theorem by the following simple consideration.

Suppose that for some value of x, L_k vanishes but L_{k-1} does not; we can then determine quantities $a_{k+1,i}$, which do not occur in L_k, in such a way that L_{k+1} does not vanish for this value of x; for L_{k+1} cannot vanish identically for all $a_{k+1,i}$, because it contains the term

$$-a_{k+1,k}^2 L_{k-1}$$

[§. 23, (12)].

Accordingly, if in the chain

$$L_n,\ -L_{n-1},\ L_{n-2} \cdots \pm L_1,\ \mp 1 \tag{5}$$

some consecutive terms vanish for some value of x, we can, by altering the $a_{i,k}$, derive another sequence

$$L'_n,\ -L'_{n-1},\ L'_{n-2} \cdots \pm L'_1,\ \mp 1 \tag{6}$$

in which no two consecutive terms vanish for any value x (of the interval $\alpha \ldots \beta$).

At the same time the $a'_{i,k}$, on which the L' depend, can be so chosen that they differ from the $a_{i,k}$ by less than an arbitrarily given quantity ω.

If now the numbers α, β are so chosen that in the sequence (5) no term vanishes for $x = \alpha$ or $x = \beta$, then we can take ω so small that corresponding terms of (5) and (6) have the same sign. But by our theorem the number of roots of $L'_n = 0$ between α and β is determined by the signs of the sequence (6).

Now on the other hand ω can again be taken so small that the roots of $L'_n = 0$ differ as little as we please from those of $L_n = 0$ (§. 40).

A double root of $L_n = 0$ may indeed pass over into two simple roots of $L'_n = 0$; but since $L'_n = 0$ has no imaginary roots, these are real; and the same holds when $L_n = 0$ has multiple roots.

Accordingly the theorems I, II retain their validity when the assumption is given up that in the chain of the L_k no consecutive terms vanish; only the multiple roots must then be counted according to their multiplicity.

§. 89.
The Sturm functions.

After these particular examples we turn to the consideration of the procedure by which one obtains a Sturm chain in all cases[15]. The manner of forming these functions is in principle extraordinarily simple, even though in practical execution it is usually not feasible.

We restrict ourselves here to the consideration of equations without multiple roots, or we assume that before the application $f(x)$ has been freed of every common factor with its derivative $f'(x)$.

If we then set

$$f_1(x) = f'(x) \tag{1}$$

the condition §. 86, 4. is certainly satisfied. We now proceed as if it were a matter of finding the greatest common divisor of $f(x)$ and $f_1(x)$, reversing each time the sign of the remainder; we thus form by division the equations

$$\begin{aligned} f &= q_1 f_1 - f_2 \\ f_1 &= q_2 f_2 - f_3 \\ &\cdots\cdots \\ f_{m-2} &= q_{m-1} f_{m-1} - f_m, \end{aligned} \tag{2}$$

in which the $q_1, q_2 \ldots q_{m-1}$, and likewise the $f_1, f_2 \ldots f_m$, are integral rational functions of x; the degrees of the functions $f, f_1, f_2, \ldots f_m$ decrease, and one can therefore continue the operation until f_m is constant, or at least no longer vanishes in the interval under consideration. That f_m cannot become zero is a consequence of the assumption that f and f_1 are without common divisor.

That in the sequence

$$f,\ f_1,\ f_2 \ldots f_m \tag{3}$$

one then really has a Sturm chain follows immediately on going through the criteria of §. 86, 1. to 4. For if two consecutive ones among the functions (3) vanish simultaneously, then by (2) all the following ones vanish as well; but this is impossible, because f_m is different from zero.

If however $f_\nu(x) = 0$, then from (2) it follows that

$$f_{\nu-1}(x) = -f_{\nu+1}(x),$$

with which all the requirements of §. 86 are satisfied.

It is, as goes without saying, permitted to multiply the functions of the sequence (3) by positive, e.g. constant, factors, without their ceasing to form a Sturm chain.

For $n = 2$ we can accordingly take as the Sturm functions the following:

$$\begin{aligned} f(x) &= x^2 + ax + b \\ f_1(x) &= 2x + a \\ f_2(x) &= a^2 - 4b. \end{aligned}$$

[15] Sturm, Mém. sur la résolution des équations numériques. Mém. de l'académie de Paris. Sav. étrang. VI, 1835. Extract in Bull. de Ferussac XI, 1829.

§. 90.
Hermite's Solution of Sturm's Problem.

Another way of solving Sturm's problem, which leads to simpler results, at least as regards the carrying out of the calculation in detail, has been taken by Hermite, who connects it with the law of inertia of quadratic forms and the Tschirnhausen transformation[16].

Let

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0 \tag{1}$$

be the given equation and

$$x_1,\ x_2 \ldots x_n \tag{2}$$

its roots, which we assume to be distinct from one another.

We use the functions defined in §. 68

$$f_0(x),\ f_1(x),\ f_2(x) \ldots f_{n-1}(x) \tag{3}$$

and, understanding by $t_0, t_1 \ldots t_{n-1}$ independent variables, we set, as there,

$$y = t_{n-1}f_0(x) + t_{n-2}f_1(x) + \cdots + t_1f_{n-2}(x) + t_0f_{n-1}(x). \tag{4}$$

Let $y_1, y_2 \ldots y_n$ be the values that y takes for $x = x_1, x_2 \ldots x_n$.

Now if α is an arbitrary real value, we set

$$H_\alpha = (x_1 - \alpha)y_1^2 + (x_2 - \alpha)y_2^2 + \cdots + (x_n - \alpha)y_n^2. \tag{5}$$

This is a quadratic form in the n variables t, and we wish first to determine the number of its negative and positive terms. If x_1 is a real root, so that y_1 is also real, then the square $(x_1 - \alpha)y_1^2$ is positive or negative according as x_1 is greater or smaller than α. But if x_1, x_2 form an imaginary pair, then y_1, y_2 are also conjugate imaginary, and

$$(x_1 - \alpha)y_1^2 + (x_2 - \alpha)y_2^2$$

decomposes into a positive and a negative square. This follows as in §. 79, if one sets $y_1\sqrt{x_1 - \alpha} = u + iv$, $y_2\sqrt{x_2 - \alpha} = u - iv$.

From this it follows that the number N_α of negative squares in H_α is equal to the number of imaginary pairs, increased by the number of real roots that are smaller than α.

If we now take a second real number $\beta > \alpha$, form the function H_β, and denote by N_β the number of its negative squares, then the difference

$$N_\beta - N_\alpha$$

is equal to the number of real roots between α and β.

One thus obtains a means of determining this number, that is, of solving Sturm's problem, if one expresses the coefficients of the function H_α as functions of the coefficients of $f(x)$ and of α, and then investigates the number N_α,

§. 91.
Determination of the Hermite form H.

[16] Hermite, Remarques sur le théorème de M. Sturm. Comptes rendus of the Paris Academy, T. 36 (1853).

To determine the Hermite form H we can apply the methods that we became acquainted with in Section VI. In §. 72 we derived the formula:

$$yf_s = E_{0,s}f_0 + E_{1,s}f_1 + \cdots + E_{n-1,s}f_{n-1}. \tag{1}$$

Making use of the relations

$$xf_0 = f_1 - a_1, \quad xf_1 = f_2 - a_2, \ldots xf_{n-1} = -a_n,$$

it follows from this that

$$\begin{aligned} xyf^s = {} & E_{0,s}f_1 + E_{1,s}f_2 + \cdots + E_{n-2,s}f_{n-1} \\ & - a_1E_{0,s} - a_2E_{1,s} - \cdots - a_nE_{n-1,s}. \end{aligned} \tag{2}$$

We therefore determine a further series of functions $E_{-1,s}$ by the equation

$$a_0E_{-1,s} + a_1E_{0,s} + a_2E_{1,s} + \cdots + a_nE_{n-1,s} = 0 \tag{3}$$

and then obtain from (1) and (2)

$$\begin{aligned} (x-\alpha)yf_s = {} & (E_{-1,s} - \alpha E_{0,s})f_0 + (E_{0,s} - \alpha E_{1,s})f_1 + \cdots \\ & + (E_{n-2,s} - \alpha E_{n-1,s})f_{n-1}. \end{aligned} \tag{4}$$

The function $E_{-1,s}$ can, however, by means of the relation [§. 72. (7)]

$$a_0t_{n+s} + a_1t_{n+s-1} + \cdots + a_nt_s = 0$$

be expressed in entirely the same way as the remaining E, so that we obtain the following system of formulas [§. 72, (9)]:

$$\begin{aligned} E_{-1,s} &= a_0t_{n+s} + a_1t_{n+s-1} + \cdots + a_st_n \\ &= -a_{s+1}t_{n-1} \cdots - a_nt_s \\ E_{0,s} &= a_0t_{n+s-1} + a_1t_{n+s-2} + \cdots + a_st_{n-1} \\ &= -a_{s+1}t_{n-2} \cdots - a_nt_{s-1} \\ E_{1,s} &= a_0t_{n+s-2} + a_1t_{n+s-3} + \cdots + a_st_{n-2} \\ &= -a_{s+1}t_{n-3} \cdots - a_nt_{s-2} \\ &\vdots \\ E_{s-1,s} &= a_0t_n + a_1t_{n-1} + \cdots + a_st_{n-s} \\ &= -a_{s+1}t_{n-s-1} \cdots - a_nt_0 \\ E_{s,s} &= a_0t_{n-1} + a_1t_{n-2} + \cdots + a_st_{n-s-1} \\ E_{s+1,s} &= a_0t_{n-2} + a_1t_{n-3} + \cdots + a_st_{n-s-2} \\ &\vdots \\ E_{n-1,s} &= a_0t_s + a_1t_{s-1} \cdot + \cdots + a_st_0. \end{aligned} \tag{5}$$

Let us introduce a second system of variables τ and set

$$z = \tau_{n-1}f_0 + \tau_{n-2}f_1 + \cdots + \tau_0f_{n-1},$$

then from (4), making use of the formulae [§. 68, (6)]

$$S(f_s) = (n-s)\,a_s$$

we obtain

$$
\begin{aligned}
S[(x-\alpha)\,y\,z] = n\,a_0 \sum_{0,\,n-1}^{s} (E_{-1,s} - \alpha\,E_{0,s})\,\tau_{n-s-1} \\
+ (n-1)\,a_1 \sum_{0,\,n-1}^{s} (E_{0,s} - \alpha\,E_{1,s})\,\tau_{n-s-1} \\
\cdots\cdots\cdots\cdots\cdots\cdots \\
+ a_{n-1} \sum_{0,\,n-1}^{s} (E_{n-2,s} - \alpha\,E_{n-1,s})\,\tau_{n-s-1},
\end{aligned} \tag{6}
$$

and this passes directly into the function H when we set $\tau = t$.

If we take as an example the case $n = 3$, we obtain

$$
\begin{aligned}
E_{-1,0} &= -a_1t_2 - a_2t_1 - a_3t_0, & E_{0,0} &= a_0t_2, \\
E_{-1,1} &= -a_2t_2 - a_3t_1, & E_{0,1} &= -a_2t_1 - a_3t_0, \\
E_{-1,2} &= -a_3t_2, & E_{0,2} &= -a_3t_1, \\
E_{1,0} &= a_0t_1, & E_{2,0} &= a_0t_0, \\
E_{1,1} &= a_0t_2 + a_1t_1, & E_{2,1} &= a_0t_1 + a_1t_0, \\
E_{1,2} &= -a_3t_0, & E_{2,2} &= a_0t_2 + a_1t_1 + a_2t_0,
\end{aligned}
$$

and from these the coefficients of the form H can easily be computed according to (6):

$$
\begin{aligned}
H_{2,2} &= -a_0(a_1 + 3a_0\alpha) \\
H_{1,1} &= -3a_0a_3 - a_1a_2 - 2(a_1^2 - a_0a_2)\alpha \\
H_{0,0} &= -a_2a_3 + (2a_1a_3 - a_2^2)\alpha \\
H_{1,0} &= -2a_1a_3 + (3a_0a_3 - a_1a_2)\alpha \\
H_{2,1} &= -2a_0(a_2 + a_1\alpha) \\
H_{0,2} &= -a_0(3a_3 + a_2\alpha).
\end{aligned}
$$

§. 92.
The Determinant of the Hermite Form.

For the question of the number of negative squares of the form H, a knowledge of its determinant is of importance. This determinant can be computed in general in the following way.

If we set

$$H = \sum_{0,\,n-1}^{i} \sum^{k} H_{i,k}\,t_i\,t_k,$$

then the coefficients $H_{i,k}$, whose determinant is to be formed, are given by the formula §. 91, (6). Here $H_{i,k}$ is the coefficient of $t_i\tau_k$ in that formula, hence

$$H_{i,k} = S(x-\alpha)\,f_{n-i-1}\,f_{n-k-1}.$$

By the multiplication theorem, however, the determinant of these n^2 quantities can be put into the form:

$$
\Delta = \begin{vmatrix}
(x_1-\alpha)f_0(x_1), & (x_1-\alpha)f_1(x_1), & \dots & (x_1-\alpha)f_{n-1}(x_1) \\
\cdots & \cdots & \cdots & \cdots \\
(x_n-\alpha)f_0(x_n), & (x_n-\alpha)f_1(x_n), & \dots & (x_n-\alpha)f_{n-1}(x_n)
\end{vmatrix} \times \begin{vmatrix}
f_0(x_1), & f_1(x_1), & \dots & f_{n-1}(x_1) \\
\cdots & \cdots & \cdots & \cdots \\
f_0(x_n), & f_1(x_n), & \dots & f_{n-1}(x_n)
\end{vmatrix},
$$

or also, since

$$a_0(x_1-\alpha)(x_2-\alpha)\cdots(x_n-\alpha) = (-1)^n f(\alpha)$$

into the form

$$\frac{(-1)^n f(\alpha)}{a_0}\begin{vmatrix} f_0(x_1), & f_1(x_1) & \dots & f_{n-1}(x_1) \\ \dots & \dots & \dots & \dots \\ f_0(x_n), & f_1(x_n) & \dots & f_{n-1}(x_n) \end{vmatrix}^2.$$

The determinant of the $f_k(x_i)$ still occurring here is the product of the two following:

$$\begin{vmatrix} a_0, & 0 & \dots & 0 \\ a_1, & a_0 & \dots & 0 \\ \dots & \dots & \dots & \dots \\ a_{n-1}, & a_{n-2} & \dots & a_0 \end{vmatrix}\begin{vmatrix} 1, & x_1, & x_1^2 & \dots & x_1^{n-1} \\ 1, & x_2, & x_2^2 & \dots & x_2^{n-1} \\ \dots & \dots & \dots & \dots & \dots \\ 1, & x_n, & x_n^2 & \dots & x_n^{n-1} \end{vmatrix}.$$

But the square of this product, if D denotes the discriminant of the function $f(x)$ (§. 46), is equal to $a_0^2 D$, so that for Δ we obtain

$$\Delta = (-1)^n a_0 f(\alpha)\, D. \tag{1}$$

Since by hypothesis D is not to vanish, Δ will therefore vanish only when one of the roots of $f(x) = 0$ is put for α, and this too is to be excluded.

If we denote a chain of principal subdeterminants of Δ, beginning with the lowest, by

$$\Delta_1(\alpha),\ \Delta_2(\alpha), \dots, \Delta_n(\alpha) = \Delta,$$

then by §. 83, V. the number of sign changes in

$$1,\ \Delta_1(\alpha), \Delta_2(\alpha) \dots \Delta_n(\alpha) \tag{2}$$

is equal to the number N_α of negative squares in H, and if we therefore count the corresponding number N_β in

$$1,\ \Delta_1(\beta), \Delta_2(\beta) \dots \Delta_n(\beta)$$

then $N_\beta - N_\alpha$ is the number of roots of $f(x)$ lying between α and β.

For the cubic equation the expressions follow from the end of the preceding section. We form the chain

$$a_0, \quad \frac{1}{a_0} H_{2,2}, \quad \frac{1}{a_0}\left(H_{2,2}H_{1,1} - H_{1,2}^2\right), \quad \frac{1}{a_0}\Delta,$$

which evidently has the same number of sign changes as (2), and thus obtain the four functions

$$\begin{aligned} &a_0, \qquad -3a_0\alpha - a_1, \\ &2\alpha^2 a_0(a_1^2 - 3a_0a_2) + \alpha(2a_1^3 + 9a_0^2a_3 - 7a_0a_1a_2) \\ &\quad + 3a_0a_1a_3 + a_1^2a_2 - 4a_0a_2^2, \qquad -f(\alpha)\, D. \end{aligned}$$

If as a test we take $\alpha = -\infty$, $\beta = +\infty$, then, when moreover $a_0 = 1$ is set, we obtain the determination of signs

$$\begin{aligned} &1, \quad +1, \quad (a_1^2 - 3a_2), \quad D, \\ &1, \quad -1, \quad (a_1^2 - 3a_2), \quad -D. \end{aligned}$$

In the enumeration one must observe that when D is positive, $a_1^2 - 3a_2$ cannot be negative [§. 47, (8)].

§. 93.
Outlines of the theory of characteristics[17]).

[17] Monatsberichte der Berliner Akademie, March and August 1869, February 1873, February 1878.

The theorems of Sturm are generalised in an extraordinary manner by the theory of characteristics of Kronecker, which pursues the same aim as Sturm's theorem for systems of equations in arbitrarily many variables. We confine ourselves here to the consideration of the simplest case, which we intend to apply to the enclosure of the complex roots of an equation; moreover, we here avail ourselves un-

restricted use of geometric intuition and of the notation of the differential calculus.

We first consider two real functions $\varphi(x, y)$, $\psi(x, y)$, and interpret the real variables x, y as rectangular coordinates in a plane. The equations

$$\varphi(x, y) = 0, \qquad \psi(x, y) = 0 \tag{1}$$

then represent two curves, which we shall call briefly the curve φ and the curve ψ. We assume first that these curves are closed and do not extend to infinity. We assume moreover that φ is negative in the interior of the curve φ and positive in the exterior; likewise that ψ is negative in the interior of ψ and positive in the exterior.

If, as Fig. 6 shows, we denote by ϑ the angle which the normal n at any point of φ, drawn in the direction in which φ increases, that is, outward, makes with the direction of the positive x-axis, then, if $\varphi'(x)$ and $\varphi'(y)$ are the partial derivatives of φ, and the square root is taken positive,

$$\cos\vartheta = \frac{\varphi'(x)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}, \qquad \sin\vartheta = \frac{\varphi'(y)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}.$$

If we draw the tangent t so that it lies to the normal n just described as the positive y-axis lies to the positive x-axis, and denote by ξ the angle which it makes with the positive x-axis, then

$$\cos\xi = -\frac{\varphi'(y)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}, \qquad \sin\xi = \frac{\varphi'(x)}{\sqrt{\varphi'(x)^2 + \varphi'(y)^2}}.$$

If we designate the advance along φ in the direction t as the positive one, and if dx, dy are the projections of this advance, then dx and dy are proportional to, and agree in sign with, $-\varphi'(y)$, $\varphi'(x)$; and if Φ is any other arbitrary function, then

$$d\Phi = \Phi'(x)\,dx + \Phi'(y)\,dy$$

agrees in sign with the functional determinant

$$[\varphi, \Phi] = \varphi'(x)\Phi'(y) - \varphi'(y)\Phi'(x). \tag{2}$$

Under the usual assumption about the direction of the coordinates, the positive direction of advance is the one for which the interior of the surface lies to the left.

Now if, at one of the points of intersection of the curves φ, ψ, the positive direction of advance on φ leads into the interior of ψ, then $d\psi$, and hence also the functional determinant $[\varphi, \psi]$, is negative, and in the opposite case positive (Fig. 7). We call a point of intersection of φ and ψ a point of exit $A(\varphi, \psi)$ or a point of entry $E(\varphi, \psi)$, according as

Fig. 6.

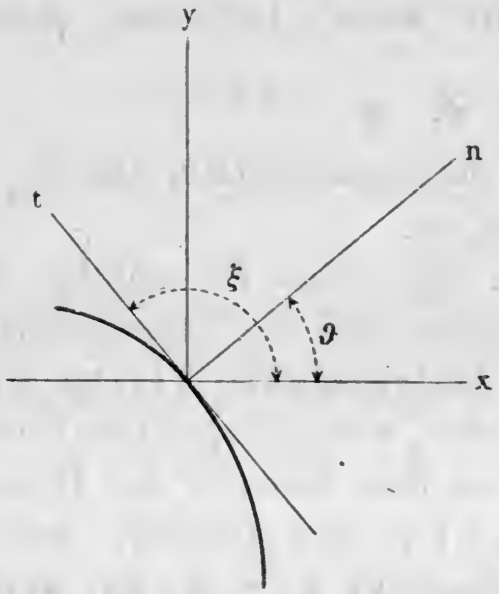
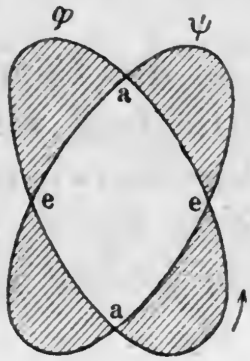


Fig. 7.



the positive direction of advance of φ leads from the interior of ψ to the exterior or from the exterior to the interior, and thus we have the theorem:

(4)
$$\begin{array}{llll} \text{at an } A(\varphi,\psi) & \text{is} & & [\varphi,\psi] > 0 \\ \text{„ \quad „} & & E(\varphi,\psi) & \text{„ } [\varphi,\psi] < 0. \end{array}$$

The number of the $A(\varphi,\psi)$ is just as great as that of the $E(\varphi,\psi)$; and if we interchange φ with ψ, the $A(\varphi,\psi)$, $E(\varphi,\psi)$ pass into $E(\psi,\varphi)$ and $A(\psi,\varphi)$.

The two functions φ, ψ determine in a unique manner a region of the plane, which is characterised by the fact that in it the product $\varphi\psi$ is negative. We shall call it the inner region (φ,ψ) or (ψ,φ). In our Fig. 7 it is the shaded area.

§. 94.
Characteristic of a system of three functions.

We now adjoin to the two functions φ, ψ a third, $f(x,y)$. The curve represented by the equation $f=0$, or the curve f, shall likewise be closed in the finite part of the plane

and we shall further assume that it passes through none of the points of intersection of φ and ψ.

We give these three functions a definite cyclic order

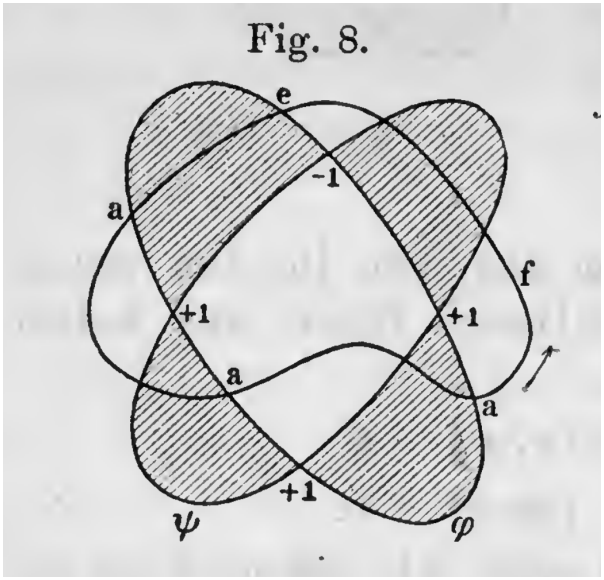
$$f,\ \varphi,\ \psi,$$

so that f is again to follow ψ, and we associate with this order a definite direction of advance on each of the three curves, say the positive one. We shall stipulate that with the reversed order

$$f,\ \psi,\ \varphi$$

the opposite, that is the negative, direction of advance shall be associated on each of the three curves.

We now traverse, in the first order, the curve f in the positive sense, and attend to the points of intersection of f and φ. Such a point shall be called an exit point $A(f;\varphi,\psi)$



if the positive direction of f leads from the interior of the inner region (φ, ψ) into the exterior, and an entry point $E(f; \varphi, \psi)$ if it leads from the exterior into the interior.

In Fig. 8 the points a are the exit points, the point e an entry point.

The total number of entry and exit points agrees with the number of points of intersection of f and φ, and is therefore, since both curves are closed, an even number. If a is the number of points $A(f; \varphi, \psi)$, e the number of points $E(f; \varphi, \psi)$, then $e - a$ is likewise an even number, and

$$\tfrac{1}{2}(e - a) = k$$

is an integer, which after Kronecker is called the characteristic of the system of functions f, φ, ψ. (In the case of Fig. 8 it is equal to -1).

If we permute the functions f, φ, ψ cyclically, we obtain three determinations of the characteristic, and if we reverse the order and thereby, in accordance with the convention adopted, also replace the positive directions by the negative ones, three more. These determinations are, if we now use the symbols $A(f; \varphi, \psi)$, $E(f; \varphi, \psi)$ at the same time as a designation for the numbers of the points concerned,

1. $\frac{1}{2}\big[E(f; \varphi, \psi) - A(f; \varphi, \psi)\big]$,
2. $\frac{1}{2}\big[E(\varphi; \psi, f) - A(\varphi; \psi, f)\big]$,
3. $\frac{1}{2}\big[E(\psi; f, \varphi) - A(\psi; f, \varphi)\big]$,
4. $\frac{1}{2}\big[E(f; \psi, \varphi) - A(f; \psi, \varphi)\big]$,
5. $\frac{1}{2}\big[E(\psi; \varphi, f) - A(\psi; \varphi, f)\big]$,
6. $\frac{1}{2}\big[E(\varphi; f, \psi) - A(\varphi; f, \psi)\big]$,

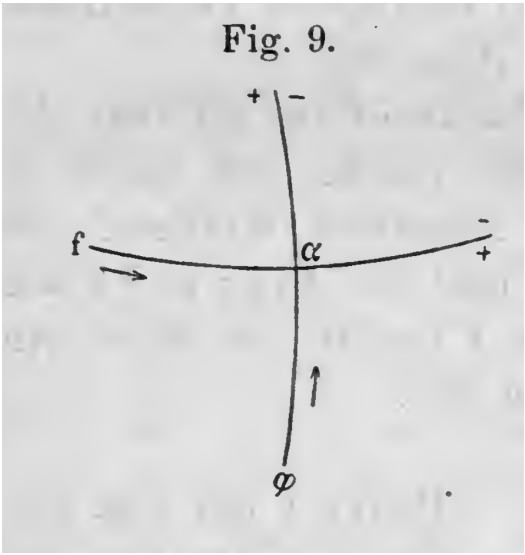
and there now holds the fundamental theorem that these six determinations yield the same number.

To prove it, we traverse the curve f in the positive sense, and attend to all the intersection points with φ and ψ. The path along the curve f must in doing so enter the inner region (φ, ψ) just as often as it steps out of it, because it must return again to its starting point. But the curve f enters at the points $E(f; \varphi, \psi)$ and $A(f; \psi, \varphi)$ and steps out at the points $A(f; \varphi, \psi)$ and $E(f; \psi, \varphi)$. Hence

$$E(f; \varphi, \psi) + A(f; \psi, \varphi) = A(f; \varphi, \psi) + E(f; \psi, \varphi),$$

by which the agreement of 1. and 4. is established.

Fig. 9.



If now, secondly, we traverse the curve φ in the negative sense, and attend to its intersection points with f, it turns out that every point $E(f;\varphi,\psi)$ is at the same time a point $E(\varphi;f,\psi)$, and every point $A(f;\varphi,\psi)$ a point $A(\varphi;f,\psi)$. For in Fig. 9, in which the point α represents any one of the intersection points of f and φ, this point is an $E(f;\varphi,\psi)$ and an $E(\varphi;f,\psi)$ when ψ is positive at α, and an $A(f;\varphi,\psi)$ and an $A(\varphi;f,\psi)$ when ψ is negative at α.

From this follows the agreement of 1. with 6. There follows further, by a renewed application of the first argument, the agreement of 6. with 2., and then by application of the second argument that of 2. with 5., and finally by the first that of 5. with 3., hence the agreement of all six expressions.

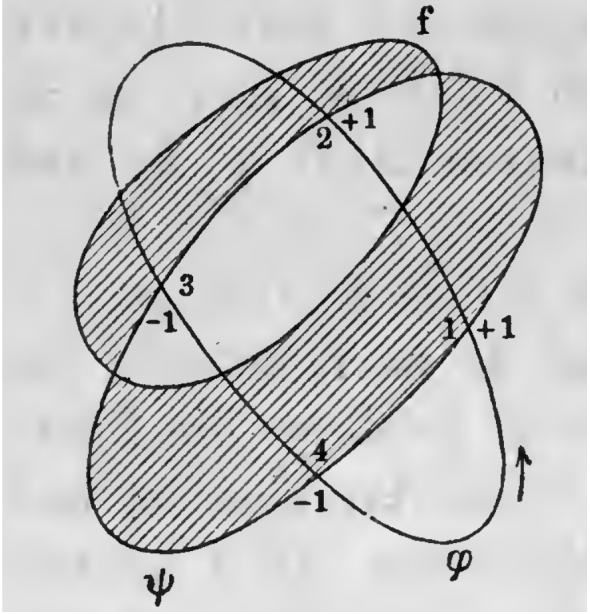
One is therefore justified in designating the number so determined simply as the characteristic of the system of functions (f,φ,ψ).

§. 95.

Relation of the characteristic to the points of intersection.

One must now observe that the designation of the points of intersection of two curves as points of entry or of exit in §. 93 is essentially different from that in §. 94. There only the relation of the points to the two intersecting curves came into consideration, while in §. 94 the relation to a third curve was also in question; this is completely expressed in the chosen designations $E(\varphi,\psi)$, $A(\varphi,\psi)$ and

Fig. 10.



$E(f;\varphi,\psi)$, $A(f;\varphi,\psi)$. But now we must investigate more precisely how the two modes of designation are related to one another.

For the sake of clarity we choose a somewhat simpler figure than above, which at the same time illustrates all possible circumstances (Fig. 10).

At a point $E(\varphi,\psi)$ the φ-curve enters the region of the curve ψ, i. e. ψ passes from positive to negative values. If at the same time f is positive, then this point is an $E(\varphi;\psi,f)$, like the point 1 in our figure; but if f is negative, then it is an $A(\varphi;\psi,f)$, like the point 3. Thus we have:

A point $E(\varphi,\psi)$ is an
$E(\varphi;\psi,f)$, if $f>0$, (point 1 of Fig. 10)
$A(\varphi;\psi,f)$, if $f<0$, (point 3 of Fig. 10).
A point $A(\varphi,\psi)$ is an
$E(\varphi;\psi,f)$, if $f<0$, (point 2 of Fig. 10)
$A(\varphi;\psi,f)$, if $f>0$, (point 4 of Fig. 10).

Now at the points $E(\varphi,\psi)$ the functional determinant $[\varphi,\psi]$ is negative, and at $A(\varphi,\psi)$ positive (§. 93), and we can therefore summarise what has been proved as follows:

$E(\varphi;\psi,f)$ is the number of points of intersection of φ, ψ at which $[\varphi,\psi]\,f<0$,

$A(\varphi;\psi,f)$ the number of points of intersection of φ, ψ at which $[\varphi,\psi]\,f>0$;

thus the characteristic is equal to half the excess of the first number over the second.

To this theorem we can give the following expression, in which it appears entirely independent of relations of magnitude and depends only on the relations of position of the curves and of their points of intersection.

If we distinguish the points of intersection of φ, ψ, according as they are exit points $A(\varphi,\psi)$ or entry points $E(\varphi,\psi)$, by the signs α and ε, and likewise, according as they are exterior or interior points with respect to the curve f, by α and ε, and then give the points α,ε and ε,α the character $+1$, and the points α,α and ε,ε the character -1[1)], then the characteristic of the system of functions (f,φ,ψ) is equal to half the sum of the characters of all the points of intersection of φ,ψ.

The theorems and definitions that we have given here remain valid unchanged even when the curves considered consist of several closed circuits; double points may also be present on the curves; it is only necessary that, for every direction of advance on one of the curves, it be completely

determined whether it is to be regarded as positive or as negative, i. e. every piece of a curve f or φ or ψ must separate portions of surface from one another in which the functions f or φ or ψ have opposite signs.

Only those cases are to be excluded in which one of the curves passes through a double point of the other, or in which the curves touch one another, or in which the three curves f, φ, ψ pass through one point; for in these cases the determination of a point as an entry point or exit point would become doubtful.

If the curves extend to infinity, then, in order to make our theorems applicable, one must close them off by arbitrarily added pieces of curve, as we shall see in the next section.

---

[1] Kronecker understands by the character of a point something different.

§. 96.
Application of the characteristics to the bounding of the complex roots of an equation.

Now let

$$z = x + yi$$

be a complex variable and

$$F(z) = \varphi(x, y) + i\,\psi(x, y) \tag{1}$$

an integral rational function of z with real or complex coefficients, and in it let $\varphi(x, y)$, $\psi(x, y)$ be real functions of the real variables x, y. We assume that $F(z)$ and $F'(z)$ do not vanish simultaneously. $F(z)$ vanishes only at the intersection points of the two curves φ and ψ. Differentiation of (1) yields

$$F'(z) = \varphi'(x) + i\,\psi'(x) = -\,i\,\varphi'(y) + \psi'(y),$$

hence

$$\varphi'(x) = \psi'(y), \quad \varphi'(y) = -\,\psi'(x), \tag{2}$$

and consequently

$$\begin{aligned}[\varphi, \psi] &= \varphi'(x)\,\psi'(y) - \varphi'(y)\,\psi'(x) \\ &= \varphi'(x)^2 + \varphi'(y)^2 = \psi'(x)^2 + \psi'(y)^2.\end{aligned} \tag{3}$$

From this it follows that the functional determinant $[\varphi, \psi]$ never becomes negative, and vanishes only where $\varphi'(x)$ and $\varphi'(y)$, and hence also $\psi'(x)$ and $\psi'(y)$, vanish simultaneously.

But this never occurs at an intersection point of φ and ψ, since otherwise $F(z)$ and $F'(z)$ would vanish there simultaneously. The positive direction of advance is completely determined, in every part of the curve φ or ψ, by the sign of the function in the adjoining portions of the surface; even any double points, at which φ, $\varphi'(x)$ and $\varphi'(y)$ vanish simultaneously, form no exception to this.

Since the functional determinant $[\varphi, \psi]$ is positive at all intersection points of the curves φ, ψ, all these points are exit points $A(\varphi, \psi)$, and from this it follows that the curves φ and ψ cannot be closed, but must extend to infinity, since otherwise an exit point would necessarily have to be followed by an entry point. For the rest, here too the curves φ, ψ determine an inner region in which the product $\varphi\psi$ is negative.

We now add to the two functions φ, ψ a third function f, which, set equal to zero, represents a closed curve, and we determine the characteristic of the system of functions entirely in the earlier manner, by proceeding along the curve f

$$k = \tfrac{1}{2}\,[E(f; \varphi, \psi) - A(f; \varphi, \psi)] = \tfrac{1}{2}\,[E(f; \psi, \varphi) - A(f; \psi, \varphi)]. \tag{1}$$

Thus, for example, in Fig. 11 the characteristic is $k = 2$.

Fig. 11.

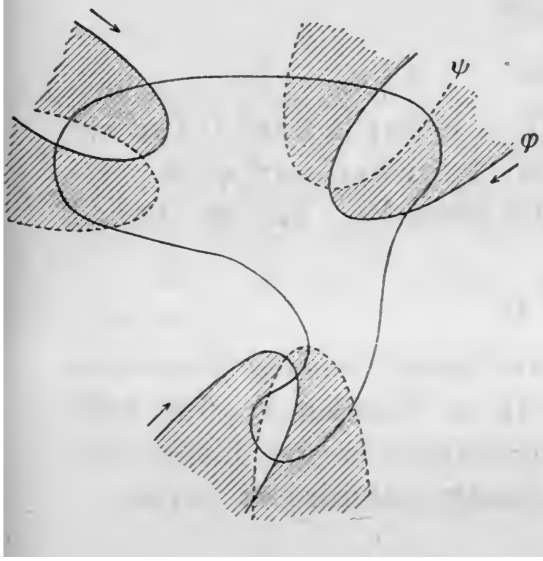
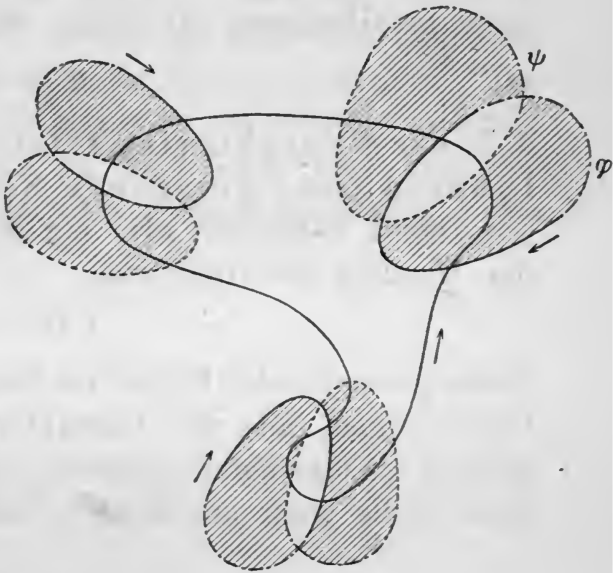


Fig. 12.



In order to be able to apply our theorems, we must make the curves φ, ψ closed in some way by arbitrarily constructed connections, all of which, however, are to run outside f. The characteristic of this system is then likewise determined by (1). As an example we complete Fig. 11 by Fig. 12, in which these closing connections are drawn. The points of intersection added in this way may be partly exit points $A(\varphi, \psi)$, partly entry points $E(\varphi, \psi)$. We shall denote their numbers by

$$A'(\varphi, \psi), \quad E'(\varphi, \psi)$$

while $A(\varphi, \psi)$ denotes the number of the points of intersection originally present. We must now always have

$$A(\varphi, \psi) + A'(\varphi, \psi) = E'(\varphi, \psi) \tag{2}$$

We divide the points $A(\varphi, \psi)$ into two groups $A_\alpha(\varphi, \psi)$, $A_\varepsilon(\varphi, \psi)$, of which the first are to lie outside, the others inside f. Then we have the following characters of the points of intersection

$$\begin{aligned} A_\alpha(\varphi, \psi) &\quad -1, \\ A_\varepsilon(\varphi, \psi) &\quad +1, \\ A'(\varphi, \psi) &\quad -1, \\ E'(\varphi, \psi) &\quad +1. \end{aligned}$$

Twice the value of the characteristic is then, if for the sake of simplicity we omit the designation (φ, ψ) (§. 95),

$$-A_\alpha + A_\varepsilon - A' + E' = 2k; \tag{3}$$

adding to this the equation (2) yields

$$k = A_\varepsilon, \tag{4}$$

i. e. the characteristic is equal to the number of the intersection points of φ, ψ lying in the interior of f.

With this we have thus found a means of determining, from the characteristic, the number of roots of the equation

$$F(z) = 0,$$

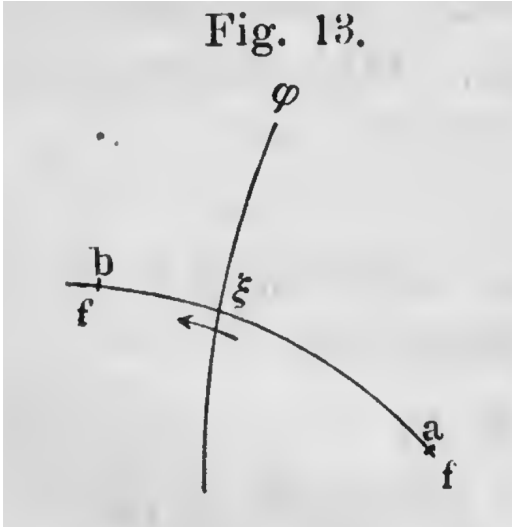
whose geometric images lie in the interior of an arbitrarily given curve. The outer connections added at will play no further part here at all, and can be suppressed entirely.

§. 97.

Determination of the Characteristic.

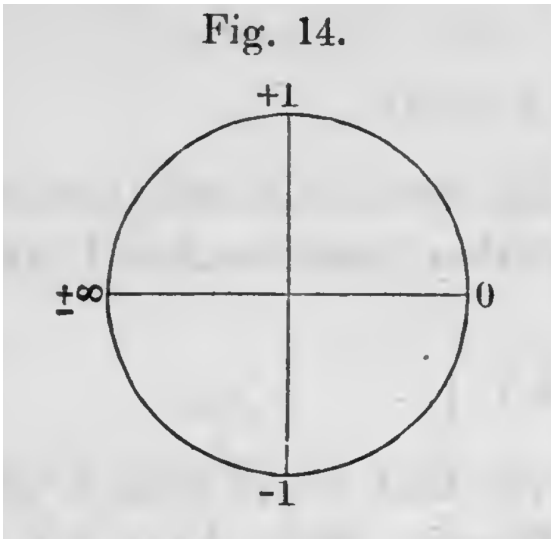
To make the theorem proved in the foregoing applicable, we must still show how we can actually determine the characteristics. For this purpose, let us imagine the curve f divided into segments in such a way that in any one of them there lies only one intersection point with the curve φ or one intersection point with the curve ψ; and let us denote by a (Fig. 13) the beginning, by b the end of such a segment, and by ξ the intersection point of f with φ lying in it. Then the point ξ will be a point $E(f;\varphi,\psi)$ if the product $\varphi\psi$ is positive at a, and a point $A(f;\varphi,\psi)$ if $\varphi\psi$ is negative at a; and with this the

Fig. 13.



the possibility of determining the characteristic completely from the signs of the functions φ,ψ at the points of division of the segments. But how we are to divide the curve f into such subsegments is taught us by the Sturm theorem. For this purpose we imagine x and y represented as functions of a variable t, which runs along the curve f through all real values from $-\infty$ to $+\infty$ (or also only through the values of a finite interval). Thereby φ and ψ also pass into functions of the variable t, and one has only to investigate the position of their roots according to the Sturm theorem.

Fig. 14.



If, for example, as in Fig. 14, the curve f is a circle with the equation

$$(x-\alpha)^2+(y-\beta)^2=\varrho^2,$$

then we can set:

$$x - \alpha = \varrho \frac{1 - t^2}{1 + t^2},$$
$$y - \beta = \varrho \frac{2t}{1 + t^2},$$

and t simply runs through all values from $-\infty$ to $+\infty$, while the point x, y moves over the whole circle in the positive sense. φ and ψ, multiplied by a suitable power of $1 + t^2$, pass into integral rational functions of t, which permit the application of the Sturm theorem.

§. 98.
Gauss' first proof of the fundamental theorem of algebra.

Gauss has given three different proofs of the principal theorem of algebra, that an equation of degree n has n roots. The first of these, which is found in his doctoral dissertation, and which he later carried out still further and presented differently, rests, though in a different garb,

on the idea that underlies the concept of the characteristic [18]. We therefore fittingly close this section with an exposition of this proof.

Let, then, $z = x + yi$ and

$$F(z) = \varphi(x, y) + i\,\psi(x, y) \tag{1}$$

be an integral rational function of z of degree n with real or complex coefficients, in which z^n has the coefficient 1, thus

$$F(z) = z^n + a_1 z^{n-1} + a_2 z^{n-2} + \cdots + a_n. \tag{2}$$

We assume again that $F(z)$ and $F'(z)$ have no common divisor, and the proof that $F(z)$ vanishes for n values of z now rests on the fact that, if we choose for f a circle of sufficiently large radius, the characteristic of the system of functions f, φ, ψ can be determined directly and is found to be equal to n. It then follows by §. 96 that in the circle so chosen there lie n points z at which $F(z)$ vanishes.

We make use of polar coordinates, and set, since $a_1, a_2, a_3 \ldots a_n$ may also be complex,

$$a_1 = p_1 e^{iq_1}, \quad a_2 = p_2 e^{iq_2}, \ldots a_n = p_n e^{iq_n}$$
$$z = Re^{i\vartheta},$$

in which $p_1, p_2, \ldots p_n, R$ are positive and the angles $q_1, q_2, \ldots q_n, \vartheta$ can be chosen in any interval of extent 2π. Hereby we obtain

$$\begin{aligned} \varphi(x, y) &= R^n \cos n\vartheta + p_1 R^{n-1} \cos[(n-1)\vartheta + q_1] \\ &\quad + \cdots + p_n \cos q_n \\ \psi(x, y) &= R^n \sin n\vartheta + p_1 R^{n-1} \sin[(n-1)\vartheta + q_1] \\ &\quad + \cdots + p_n \sin q_n, \end{aligned} \tag{3}$$

and we also form the derivatives taken with respect to ϑ

$$\begin{aligned} \frac{d\varphi}{d\vartheta} &= -nR^n \sin n\vartheta - (n-1)p_1 R^{n-1} \sin[(n-1)\vartheta + q_1] \\ &\quad - \cdots - p_{n-1} \sin(\vartheta + q_{n-1}) \\ \frac{d\psi}{d\vartheta} &= nR^n \cos n\vartheta + (n-1)p_1 R^{n-1} \cos[(n-1)\vartheta + q_1)] \\ &\quad + \cdots + p_{n-1} \cos(\vartheta + q_{n-1}). \end{aligned} \tag{4}$$

[18] Gauss, Demonstratio nova theorematis omnem functionem algebraicam rationalem integram unius variabilis in factores reales primi vel secundi gradus resolvi posse (1799). Beiträge zur Theorie der algebraischen Gleichungen (1849). Kronecker, Monatsberichte der Berliner Akademie, 21. Februar 1878.

We now divide the periphery of the circle of radius R into $4n$ parts of the angular magnitude

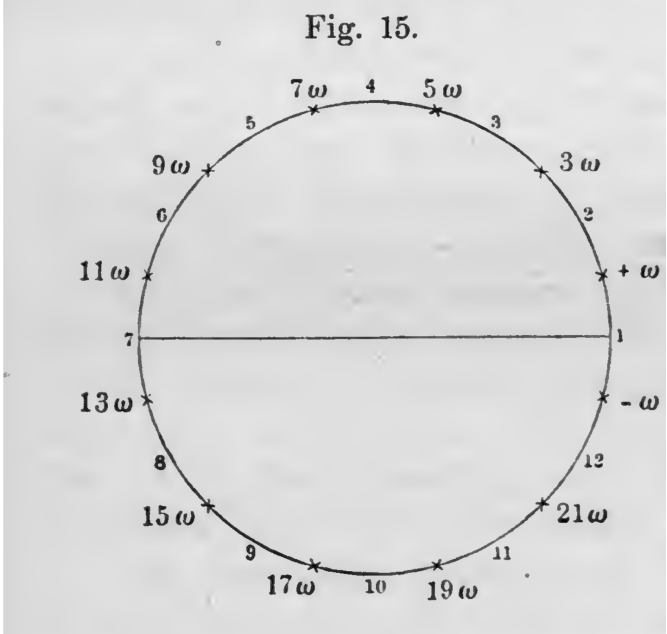
$$2\omega = 2\,\frac{\pi}{4n},$$

beginning at $\vartheta = -\omega$. The points of division are therefore

$$-\omega,\ \omega,\ 3\omega,\ 5\omega,\ \ldots (8n-3)\omega,$$

and let the intervals be denoted in order by $1, 2, 3 \ldots 4n$. Fig. 15 shows this division for $n = 3$.

Fig. 15.



In the intervals

$$1, 3, 5 \ldots (4n-1)$$

$\cos n\vartheta$, taken absolutely, is greater than $1 : \sqrt{2}$, and is alternately of positive and negative sign. In the intervals

$$2, 4, 6 \ldots 4n$$

$\sin n\vartheta$ is absolutely greater than $1 : \sqrt{2}$ and likewise alternately of positive and negative sign.

But R may be taken so large that in the intervals $1, 3, 5 \ldots (4n-1)$ the sign of φ and of $\dfrac{d\psi}{d\vartheta}$ agrees with that of $\cos n\vartheta$, and in the intervals $2, 4, 6 \ldots 4n$ the sign of ψ and of $\dfrac{d\varphi}{d\vartheta}$ agrees with that of $\sin n\vartheta$.

Since, then, φ changes its sign in the intervals $2, 4, 6 \ldots 4n$, it must pass through zero in each of these intervals; and since the derivative $\dfrac{d\varphi}{d\vartheta}$ does not change its sign, it can pass through zero only once in each.

In the same way it is to be concluded that ψ passes through zero once and only once in each of the intervals $1, 3, 5 \ldots 4n-1$. We thus have a division of the circumference of the circle such as was required in the preceding section for the determination of the characteristic, and since at the points of division $\omega, 5\omega, 9\omega \ldots (4n-3)\omega$ the product $\varphi\psi$ is positive, only points $E(f; \varphi, \psi)$ occur on the circumference of the circle, and their number is $2n$; the number of the roots of $F(z)$ is therefore equal to n, which was to be proved.

Ninth Section.
Estimation of the Roots.

§. 99.
The Budan-Fourier Theorem.

The functions of the Sturm chains do indeed solve completely and without exception the problem of determining the number of roots of an equation between given limits; but their actual computation is usually difficult and often impracticable. A number of theorems are known for the estimation of the number of roots between given limits, which furnish much simpler rules, but which, to be sure, also do not answer the question completely, giving only a maximal number beyond which the number of roots cannot go. These rules are often very useful and sufficient for applications, and they must therefore not be omitted here.

We consider first a procedure which originates with Budan, but which was then used and extended by Fourier [1]).

Let us consider, in place of a Sturm chain, the sequence of derivatives of a real function $f(x)$ of degree n

$$f(x),\ f'(x),\ f''(x)\ \ldots\ f^{(n)}(x), \tag{1}$$

in which, then, $f^{(n)}(x)$ is a constant, which we assume to be different from zero and positive. It is, apart from a positive numerical factor, the coefficient of x^n in $f(x)$.

[1]) Budan, Nouvelle methode pour la résolution des équations numériques, (presented to the Paris Academy in 1803). Fourier, Analyse des équations déterminées. Paris 1831. On the history of this question compare Lagrange, Traité de la résolution des équations, Note VIII (Works, Vol. 8).

Let α and $\beta > \alpha$ again be two real numbers, which determine the interval (α, β) in which the number of real roots of $f(x)$ is to be counted.

We shall first assume that in the interval (α, β) no two terms of the sequence (1) vanish simultaneously, and that for $x = \alpha$, $x = \beta$ no term of the sequence vanishes.

If we now let x increase continuously from α to β, a change can occur in the succession of signs of the sequence (1) only when one of the functions passes through zero. When $f^{(\nu)}(x)$ passes through zero for $x = \xi$, then $f^{(\nu)}(x)$ passes from negative to positive or from positive to negative values, according as $f^{(\nu+1)}(x)$ is positive or negative, and thus a change of sign between $f^{(\nu)}$ and $f^{(\nu+1)}$ is lost in the passage through ξ. This holds also when $f^{(\nu)}$ is the function $f(x)$ itself. But when $f^{(\nu)}$ is one of the derivatives, it is preceded by a function $f^{(\nu-1)}$, and since $f^{(\nu-1)}$ is by hypothesis not zero for $x = \xi$, and $f^{(\nu)}$ changes its sign in the passage through ξ, either a loss or a gain of one change of sign takes place between $f^{(\nu-1)}$ and $f^{(\nu)}$ in the passage through ξ. Thus when an inner term of the sequence (1) passes through zero, the number of changes of sign remains unaltered, or two changes of sign are lost.

But if $f(x)$ itself passes through zero, one change of sign is lost.

From this follows the theorem:

I. The number of roots of $f(x)$ lying between α and β is at most as great as the number of changes of sign lost between α and β, and if it is smaller, the difference is an even number.

Making use of a formula, we can also say:

If $V(x)$ is the number of changes of sign (variations) that the sequence (1) presents for any value x, then the number of roots of $f(x)$ lying between α and β is

$$V(\alpha) - V(\beta) - 2h, \tag{2}$$

where h is a non-negative integer.

The proof of theorem I. still requires a supplement for the case in which several successive terms of the sequence (1) vanish simultaneously.

Suppose then that for a value ξ of x between α and β, in the sequence

$$f^{(\nu)}(x),\ f^{(\nu+1)}(x),\ \ldots\ f^{(\nu+\mu-1)}(x),\ f^{(\nu+\mu)}(x) \tag{3}$$

all the terms, with the exception of the last, vanish, and suppose the last, $f^{(\nu+\mu)}(x)$, has, say, a positive value.

We mark off about ξ two intervals δ_1, δ_2, such that all values of x in the interval δ_1 are smaller than ξ, and in the interval δ_2 greater, and we take these intervals so small that the functions (3) do not vanish in them except at ξ, so that in particular $f^{(\nu+\mu)}(x)$ remains positive.

Now since, when $f^{(\nu+\mu)}(x)$ is positive, $f^{(\nu+\mu-1)}(x)$ increases together with x, we have

$f^{(\nu+\mu-1)}(x)$ in δ_1 negative,
in δ_2 positive.

From this it follows that $f^{(\nu+\mu-2)}(x)$ decreases in δ_1 and increases in δ_2, hence is positive in both intervals; and continuing in this way we infer the succession of signs in the sequence (3):

$$\begin{array}{llllllll}
(4) & f^{(\nu)}(x), & f^{(\nu+1)}(x), & f^{(\nu+2)}(x), & \ldots & f^{(\nu+\mu-1)}(x), & f^{\nu+\mu}(x) \\
\delta_1, & (-1)^\mu & -(-1)^\mu & (-1)^\mu & \ldots & - & + \\
\delta_2, & + & + & + & \ldots & + & +
\end{array}$$

that is, in the sequence (3), on passing through ξ, nothing but changes of sign become successions of signs, and μ changes of sign are lost in the sequence (3).

If $f^{(\nu+\mu)}(\xi)$ is negative, then all the signs in (4) are the opposite ones, and the conclusion otherwise remains the same.

Now if $\nu = 0$, that is, if $f^{(\nu)}(x)$ is the original function $f(x)$ itself, which then has a $(\mu+1)$-fold root at ξ, then on passing through ξ a loss of μ changes of sign also takes place in the sequence (1).

But if $\nu > 0$ and the function $f^{(\nu-1)}(x)$ preceding $f^{(\nu)}(x)$ is different from zero, then we have the following signs:

1. μ even:

$$\begin{array}{llll}
\text{a)}\ f^{(\nu-1)}(x), & f^{(\nu)}(x), & \text{b)}\ f^{(\nu-1)}(x), & f^{(\nu)}(x) \\
\delta_1, \quad + & + & - & + \\
\delta_2. \quad + & + & - & +
\end{array}$$

2. μ odd:

$$\begin{array}{llll}
\text{a)}\ f^{(\nu-1)}(x), & f^{(\nu)}(x), & \text{b)}\ f^{(\nu-1)}(x), & f^{(\nu)}(x) \\
\delta_1, \quad + & - & - & - \\
\delta_2, \quad + & + & - & +
\end{array}$$

Thus for even μ no change of sign is lost between $f^{(\nu-1)}$ and $f^{(\nu)}$; for odd μ a change of sign is lost or one is gained. Hence the loss of changes of sign on passing through ξ in the sequence

$$f^{(\nu-1)},\ f^{(\nu)},\ f^{(\nu+1)}\ \ldots\ f^{(\nu+\mu)} \tag{5}$$

is equal to μ for even μ, and to $\mu \pm 1$ for odd μ; it is therefore always an even number and never negative.

We can give these results a clear analytic expression, which lets their meaning be recognised better.

As before, we denote by $V(x)$ the number of changes of sign in the sequence (1) for a definite value of x, but with the further stipulation that, if for a value of x some of the terms of the sequences vanish, these are simply passed over in counting the changes of sign. We can then regard $V(x)$ as a function of x, which however can change only by integers, and is therefore discontinuous for those values of x for which some terms of the sequence (1) vanish. We shall then denote by $V(x-0)$ and $V(x+0)$ the values of the function $V(x)$ immediately before and immediately after such a place. Then the consideration of (4) shows that in all cases

$$V(\xi+0) = V(\xi) \tag{6}$$

and that, if ξ is a μ-fold root of $f(x)$,

$$V(\xi-0) = V(\xi) + \mu + 2h, \tag{7}$$

where h is a non-negative integer.

From this it follows, then, if for the present we hold to the assumption that for $x=\alpha$ and $x=\beta$ none of the functions (1) vanishes, that $V(\alpha) - V(\beta)$ differs from the sum of all the numbers μ, that is, from the number of the roots lying in the interval (α,β), each counted according to the degree of its multiplicity, by an even, non-negative number; and with this Theorem I is thus freed from its restriction.

But we can also give up the assumption that none of the functions (1) vanishes for α and β, provided that α and β do not occur among the roots of $f(x)$.

For then our theorem is applicable to two values, of which the first is a little greater than α, the other a little smaller than β, and which enclose the same roots as α and β; that is, if we denote by $\mathfrak{z}$ the number of these roots, we have

$$V(\alpha+0) - V(\beta-0) = \mathfrak{z} + 2h,$$

and since

$$V(\alpha+0) = V(\alpha), \quad V(\beta-0) = V(\beta) + 2h',$$

it follows that

$$V(\alpha) - V(\beta) = \mathfrak{z} + 2(h+h'),$$

which is again the statement of our theorem.

But if α or β itself belongs to the roots of $f(x)$, then our theorem is correct only when these limiting values are not counted as belonging to the interval.

The Budan-Fourier theorem does not, it is true, give a completely certain decision about the number of roots in an interval, as the Sturm theorem does; but it can nevertheless in many cases replace the Sturm theorem; for if one has restricted the interval (α,β) so far that no change of sign, or only one, is lost from α to β, then it follows with certainty that in the first case no root, and in the second case one and only one root, lies in the interval.

The separation of the roots can therefore be completely given by the Budan sequence only when, in the passage through a value ξ, several changes of sign are not lost simultaneously. This behaviour cannot be recognised with certainty, however small the interval (α,β) may be chosen, and so, if after a suitable narrowing of the interval the separation has not succeeded, one will after all have to resort to another procedure, in the last instance to the Sturm theorem.

§. 100.
Newton's Rule.

A supplement to the preceding rule, which permits a further approximation to the exact determination of the limits of the roots, is given by the prescription now to be discussed, which originates with Newton but was first proved by Sylvester[19].

Again let

$$f(x) = a_0x^n + a_1x^{n-1} + \cdots + a_{n-1}x + a_n = 0 \tag{1}$$

be the equation to be investigated. Denoting by $\Pi(m)$, as always, the product $1.2.3\ldots m$, and by $f^{(\nu)}(x)$ the νth derivative, we set

$$f_\nu(x) = \frac{\Pi(n-\nu)}{\Pi(n)} f^{(\nu)}(x), \tag{2}$$

$$F_\nu(x) = f_\nu^2(x) - f_{\nu+1}(x)f_{\nu-1}(x), \tag{3}$$

with the addition that

$$F_0 = F = 1, \quad F_n = f_n^2, \tag{4}$$

and hence, which is the essential point, are to be positive.

For the derivatives of the functions f_ν, F_ν we obtain from (2) and (3)

$$f'_\nu(x) = (n-\nu)f_{\nu+1}, \tag{5}$$

$$f_\nu F'_\nu(x) = (n-\nu-1)(F_\nu f_{\nu+1} + F_{\nu+1}f_{\nu-1}). \tag{6}$$

Since the positive numerical factors do not matter, we can, in place of the Budan sequence, take the sequence of functions

$$f, f_1, f_2 \ldots f_n$$

instead.

Newton's rule now makes use of the double sequence:

$$\begin{array}{l} f,\ f_1,\ f_2 \ldots f_n \\ F,\ F_1,\ F_2 \ldots F_n. \end{array} \tag{7}$$

If we consider two successive terms of this double sequence, such as

$$\begin{array}{ll} f_\nu, & f_{\nu+1} \\ F_\nu, & F_{\nu+1} \end{array}$$

for a value of x at which none of the four functions vanishes, then with regard to the succession of signs the following cases are possible:

$$\text{a)} \quad \begin{matrix} + & + \\ + & + \end{matrix}, \quad \begin{matrix} + & + \\ - & - \end{matrix}, \quad \begin{matrix} - & - \\ + & + \end{matrix}, \quad \begin{matrix} - & - \\ - & - \end{matrix}, \quad PP,$$

$$\text{b)} \quad \begin{matrix} + & - \\ + & - \end{matrix}, \quad \begin{matrix} + & - \\ - & + \end{matrix}, \quad \begin{matrix} - & + \\ + & - \end{matrix}, \quad \begin{matrix} - & + \\ - & + \end{matrix}, \quad VV,$$

$$\text{c)} \quad \begin{matrix} + & + \\ + & - \end{matrix}, \quad \begin{matrix} + & + \\ - & + \end{matrix}, \quad \begin{matrix} - & - \\ + & - \end{matrix}, \quad \begin{matrix} - & - \\ - & + \end{matrix}, \quad PV,$$

[19] Newton, Arithmetica universalis. Sylvester, Transactions of the R. Irish Academy, t. 24 (1871). Phil. Mag., 4. ser., t. 31. Cf. also Petersen, Theorie der algebr. Gleichungen. Kopenhagen 1878.

$$\text{d)} \quad \begin{matrix} + - \\ + + \end{matrix}, \quad \begin{matrix} + - \\ - - \end{matrix}, \quad \begin{matrix} - + \\ + + \end{matrix}, \quad \begin{matrix} - + \\ - - \end{matrix}, \quad VP.$$

We denote this behaviour by the following expressions and symbols:

a) succession-succession (permanence-permanence) PP,

b) change-change (variation-variation) VV,

c) succession-change (permanence-variation) PV,

d) change-succession (variation-permanence) VP.

The two theorems that we wish to prove read:

II. The number of roots of $f(x)$ lying between α and β is either exactly as great as, or smaller by an even number than, the number of change-successions lost in the double sequence (7) in the passage from α to β.

III. The number of roots of $f(x)$ lying between α and β is either exactly as great as, or smaller by an even number than, the number of succession-successions gained in the double sequence (7) in the passage from α to β.

Of these two theorems, which do not always give the same upper bound for the number of roots, one will apply the one that gives the lowest bound.

In the proof of this theorem we make the following assumptions, from some of which we shall later free ourselves again.

1. $f(x)$ has no double or multiple roots.

2. Of the functions $f_\nu(x)$, no two adjacent ones vanish for the same value of x.

3. Likewise, of the functions $F_\nu(x)$, no two adjacent ones vanish for the same value of x.

4. A consequence of 2. is [by formula (3)] that $f_\nu(x)$ and $F_\nu(x)$ do not vanish for the same value of x.

5. For $x = \alpha$ and $x = \beta$ themselves, none of the functions f_ν, F_ν is to vanish.

The proof of Theorems II and III is now as follows:

If we let x go, continuously increasing, from α to β, then a change in the succession of signs can occur only when one of the functions f_ν or F_ν passes through zero, and it is only a matter of investigating the effect that such a passage through zero produces in the various cases. Let us first assume that one of the middle functions f, say f_ν, passes through zero, where therefore $0 < \nu < n$. According as $f_{\nu+1}$ is positive or negative, f_ν will increase or decrease. Thus in the first case $f_\nu(x-0)$ will be negative and $f_\nu(x+0)$ positive, and in the second case the reverse; moreover it follows from (3) that, when $f_\nu = 0$, $F_{\nu-1}$ and $F_{\nu+1}$ are positive, and that F_ν has the sign opposite to that of $f_{\nu-1}f_{\nu+1}$. We collect the possibilities that remain after this in tabular form, but omit from the enumeration those cases which arise from the ones listed by a simultaneous change of sign of the three functions $f_{\nu-1}$, f_ν, $f_{\nu+1}$, since these evidently yield no different results. There remain to us after this only two cases:

	$\nu-1$,	ν,	$\nu+1$
$f(x-0)$	$+$	$-$	$+$
$f(x+0)$	$+$	$+$	$+$
F	$+$	$-$	$+$

$\nu-1$,	ν,	$\nu+1$
$-$	$-$	$+$
$-$	$+$	$+$
$+$	$+$	$+$

.

In these cases we have accordingly

$$\left|\begin{matrix} x-0 & VV, & VV \\ x+0 & PV, & PV \end{matrix}\right| \qquad \left|\begin{matrix} PP, & VP \\ VP, & PP \end{matrix}\right|.$$

Thus neither a VP nor a PP is lost or gained.

But if $f(x)$ itself passes through zero, then, assuming $f_1(x)$ positive, which suffices in view of

the remark made above, we have

	0	1
$f(x-0)$	$-$	$+$
$f(x+0)$	$+$	$+$
F	$+$	$+$

.

Here, then, a VP passes into a PP; that is, a VP is lost and a PP gained.

If, finally, F_ν passes through zero, where of course we must have $0 < \nu < n$, the behaviour is as follows:

If $F_\nu = 0$, then $f_{\nu-1}$ and $f_{\nu+1}$ must necessarily have the same sign, since otherwise F_ν would consist of two positive terms and could not vanish. Further, by (6), $F'_\nu(x)$ has the same sign as the product $f_{\nu-1} f_\nu F_{\nu+1}$, and hence $F_\nu(x-0)$ has the opposite sign to the product, and $F_\nu(x+0)$ the same sign as it.

Here too we may omit from the enumeration those cases which arise from the ones listed by a simultaneous change of sign in all three functions $F_{\nu-1}, F_\nu, F_{\nu+1}$; there remain, then, the following cases:

f	$\nu-1$,	ν,	$\nu+1$	$\nu-1$,	ν,	$\nu+1'$
	$+$	$+$	$+$	$+$	$+$	$+$
$F(x-0)$	$+$	$-$	$+$	$-$	$-$	$+$
$F(x+0)$	$+$	$+$	$+$	$-$	$+$	$+$

f	$\nu-1$,	ν,	$\nu+1$	$\nu-1$,	ν,	$\nu+1$
	$+$	$-$	$+$	$+$	$-$	$+$
$F(x-0)$	$+$	$+$	$+$	$-$	$+$	$+$
$F(x+0)$	$+$	$-$	$+$	$-$	$-$	$+$

.

The changes of sign in these four cases are:

$x-0$	PV, PV	PP, PV	VP, VP	VV, VP
$x+0$	PP, PP	PV, PP	VV, VV	VP, VV

.

In the first case two PP are gained, in the second and fourth cases no change occurs, in the third case two VP are lost.

But with this the theorems II, III are proved.

The assumptions 2., 3., that no two adjacent ones of the functions f_ν and F_ν shall vanish for the same x, we may now give up. For if we hold fast to the first assumption, that no multiple roots are present, we can impart to the coefficients a_ν changes so small that, first, the assumptions 2., 3. are satisfied, and secondly, the roots lying between α and β change so little that none of them leaves the interval. Since the roots occur only singly, real roots cannot in this process pass into imaginary ones either.

Whether the theorem, with correct counting of the multiple roots, remains valid also in the case of multiple roots may be left undecided.

§. 101.
The Cartesian Theorem.

The Cartesian theorem, which is also named after Harriot[20], gives a rule for estimating the number of positive roots. We can regard it simply as a special case of Budan's theorem, by taking $\alpha = 0$ and $\beta = \infty$, or at least so large that the functions

(1) $$f(x), f'(x), f''(x) \dots f^{(n)}(x)$$

[20]Harriot, Artis analyticae praxis ad aequationes algebraicas resolvendas 1631. Gauss' Werke, Bd. III, S. 67. Lagrange, l. c.

all have the same sign for $x > \beta$. This sign agrees with the sign of the coefficient of the highest power of x in $f(x)$, and may be assumed positive.

If

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n,$$

so the functions (1), apart from positive numerical factors, take for $x = 0$ the values

$$a_n, a_{n-1}, a_{n-2} \ldots a_0, \tag{2}$$

and from this, accordingly, we obtain the theorem:

IV. The number of positive roots of the equation $f(x) = 0$ is equal to, or less by an even number than, the number of changes of sign in the sequence of the coefficients of $f(x)$, where any vanishing coefficients are simply to be passed over, and the root $x = 0$, if it is present, is not counted.

To estimate the number of negative roots as well, one can either apply the Budan-Fourier theorem to the interval $(-\infty, 0)$, or one replaces x by $-x$, that is, one changes the signs of $a_1, a_3, a_5 \ldots$ and then applies Theorem IV.

In a similar way we shall specialise the Newton criterion of §. 100.

We remark for this purpose that

$$f_\nu(0) = \frac{\Pi(n-\nu)\,\Pi(\nu)}{\Pi(n)}\, a_{n-\nu},$$

and consequently

$$F_\nu(0) = \frac{\Pi(\nu)\,\Pi(\nu-1)\,\Pi(n-\nu)\,\Pi(n-\nu-1)}{\Pi(n)\,\Pi(n)} \left[\nu(n-\nu)\, a_{n-\nu}^2 - (\nu+1)(n-\nu+1)\, a_{n-\nu-1}\, a_{n-\nu+1}\right]$$

and further that the sequence of the $f_\nu(x)$ presents, for $x = +\infty$, only successions of signs, and for $x = -\infty$ only changes of sign.

If, then, we set

$$A_\nu = \nu(n-\nu)\, a_\nu^2 - (\nu+1)(n-\nu+1)\, a_{\nu-1}\, a_{\nu+1}, \tag{3}$$

$$A_0 = 1, \quad A_n = 1,$$

so that A_ν, apart from a positive factor, agrees with $F_{n-\nu}(0)$, it follows that for $x = 0$ the double sequence (7), §. 100, written in reverse order, agrees in sign with

$$\begin{matrix} a_0,\ a_1,\ a_2 \ldots a_n \\ A_0,\ A_1,\ A_2 \ldots A_n. \end{matrix} \tag{4}$$

On passing to $x = +\infty$, in the sequence $f(x), f_1(x) \ldots f_n(x)$, all the changes of sign are, and on passing to

$x = -\infty$ all successions of signs have been lost; hence in the double sequence of the f_ν, F_ν, for $x = +\infty$ all VP, and for $x = -\infty$ all PP.

Accordingly, by §. 100, II, III, we can state the following two theorems:

V. The number of positive roots of $f(x)$ is as great as, or less by an even number than, the number of change-successions in the double sequence (4).

VI. The number of negative roots of $f(x)$ is as great as, or less by an even number than, the number of succession-successions in the double sequence (4).

But the sum of the numbers of the VP and of the PP is equal to the number of successions of signs in the simple sequence

$$A_0, A_1, A_2 \ldots A_n, \tag{5}$$

and we can therefore combine V. and VI. into the single theorem:

VII. The number of all real roots of $f(x)$ is as great as, or less by an even number than, the number of successions of signs in the sequence (5).

The total number of changes of sign and successions of signs in the sequence (5) is as great as the number of all the real and imaginary roots taken together, namely n. For the case in which some terms of the sequence (5) vanish is here excluded. Since, moreover, the first and last terms of the sequence (5) have the same sign, the number of changes of sign in this sequence is an even number. We can therefore, without saying anything new, add to VII. the further theorem:

VIII. The number of imaginary roots of $f(x)$ is at least equal to the number of changes of sign in the sequence (5).

In the formulation of the last two theorems it is immaterial whether some of the coefficients $a_1, a_2 \ldots a_n$ vanish, provided only that none of the A_ν is zero.

For the cubic equation

$$a_0x^3 + a_1x^2 + a_2x + a_3 = 0$$

we have

$$A_1 = 2(a_1^2 - 3a_0a_2)$$
$$A_2 = 2(a_2^2 - 3a_1a_3),$$

and in the sequence

$$1,\ A_1,\ A_2,\ 1$$

there thus occur two changes of sign whenever, of the two quantities A_1, A_2, one or both are negative, and in these cases the equation accordingly has two imaginary roots. But if A_1 and A_2 are positive, the decision cannot be made from this theorem alone; it is then necessary to take into consideration also the quantity

$$B = 9\,a_0\,a_3 - a_1\,a_2$$

and to examine the sign of

$$3\,D = A_1\,A_2 - B^2$$

(§. 62).

In the case of the biquadratic equations one has to consider the three expressions:

$$A_1 = 3\,a_1^2 - 8\,a_0\,a_2,$$
$$A_2 = 4\,a_2^2 - 9\,a_1\,a_3,$$
$$A_3 = 3\,a_3^2 - 8\,a_2\,a_4.$$

If the signs are, in order, $- + -$, one has four imaginary roots. If all three signs are positive, then four or two or no imaginary roots may be present; in all other cases one has two or four imaginary roots.

§. 102.
The Jacobi Criterion.

In order to estimate the number of roots that lie between two limits α, β, Jacobi derived a criterion from the Cartesian theorem. If, namely,

$$f(x) = 0 \tag{1}$$

is the proposed equation of degree n, of which it is to be decided how many roots it has between the two limits α, β, then Jacobi[21] sets

$$y = \frac{x-\alpha}{\beta-x}, \qquad x = \frac{\alpha+\beta y}{1+y} \tag{2}$$

[21] Observatiunculae ad theoriam aequationum algebraicarum pertinentes. Crelle's Journal, Bd. 13. Gesammelte Werke, Bd. III.

and now forms the equation

$$(3) \qquad (1+y)^n f\left(\frac{\alpha+\beta y}{1+y}\right) = \varphi(y) = b_0\, y^n + b_1\, y^{n-1} + \cdots + b_n,$$

in which b_0, $b_1 \ldots b_n$ are integral homogeneous functions of degree n in α and β, and linear functions of the coefficients of $f(x)$. Now when x increases from α to β, y, likewise increasing, goes from 0 to ∞. Hence $f(x)$ has exactly as many roots between α and β as $\varphi(y)$ has positive roots.

From the Cartesian theorem it therefore follows:

IX. The number of roots of $f(x)$ between α and β is as great as, or less by an even number than, the number of changes of sign in the sequence

$$b_0,\ b_1,\ b_2 \ldots b_n.$$

While, then, in the Budan theorem the changes of sign have to be counted and compared in two sequences of values, the Jacobi criterion requires only the counting of the changes of sign in one sequence of values.

In the same way the Newton rule can also be transformed, if one takes into consideration the quantities

$$B_\nu = \nu(n-\nu)\, b_\nu^2 - (\nu+1)(n-\nu+1)\, b_{\nu-1}\, b_{\nu+1}$$

and one thus obtains the theorem:

X. The number of roots of $f(x)$ between α and β is as great as, or less by an even number than, the number of changes of succession in the double sequence

$$b_0,\ b_1,\ b_2 \ldots b_n,$$
$$B_0,\ B_1,\ B_2 \ldots B_n.$$

§. 103.

Klein's geometric comparison of the various criteria.

In order to compare the scope of the various criteria with one another, Klein has employed a geometric interpretation, which we shall now discuss. If one does not wish to operate with spaces of more dimensions, and thereby to lose geometric intuition, one must here

confine ourselves to the first cases, and so we begin with the quadratic equations

$$(1) \qquad x^2 + a\,x + b = 0.$$

If we regard a and b as rectangular coordinates in a plane, then every point of this plane is the image of one and only one quadratic equation, and conversely every quadratic equation with real coefficients has a point of the plane as its image.

The equation

$$(2) \qquad a^2 - 4\,b = 0$$

represents a parabola, on which lie the images of the equations with equal roots. The inner points, that is, those lying in the hollow of the parabola, represent the equations with imaginary roots, the outer points the equations with real roots.

We now form, according to §. 89, the Sturm chain

$$(3) \qquad x^2 + a\,x + b, \quad 2\,x + a, \quad a^2 - 4\,b$$

and examine the changes of sign. If x is constant and a, b variable, then

$$x^2 + a\,x + b = 0$$

is the equation of the tangent to the parabola at the point

$$a = -2\,x, \quad b = x^2,$$

and indeed $x^2 + a\,x + b$ is positive in the part of the plane that also contains the parabola.

By this tangent and by the parabola the plane is divided into four regions, in which, as is indicated in Fig. 16 (on the following page), 0, 1, 2 changes of sign occur in the sequence (3).

If we now set $x = \alpha$ and $x = \beta$, we obtain two such tangents and a division of the plane into four regions, in which lie the representative points of those equations that have no root, one root, or two roots between α and β. In Fig. 17 these four regions are marked by the corresponding numerals.

This is the difference between those points from which no tangent, one tangent, or two tangents can be drawn to the parabola having their point of contact between α and β. Fig. 17 thus gives the exact and complete distinction of the various cases.

If we now compare with this the theorem of Budan, we have the three functions $f(x)$, $f'(x)$, $f''(x)$, and thus have

Fig. 16.

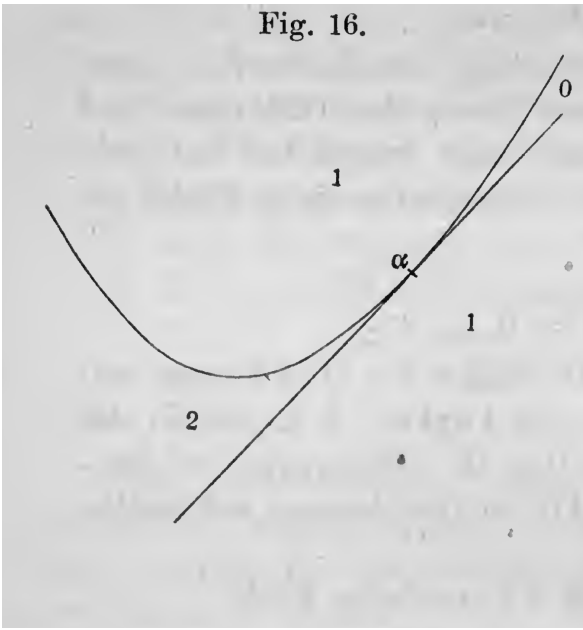
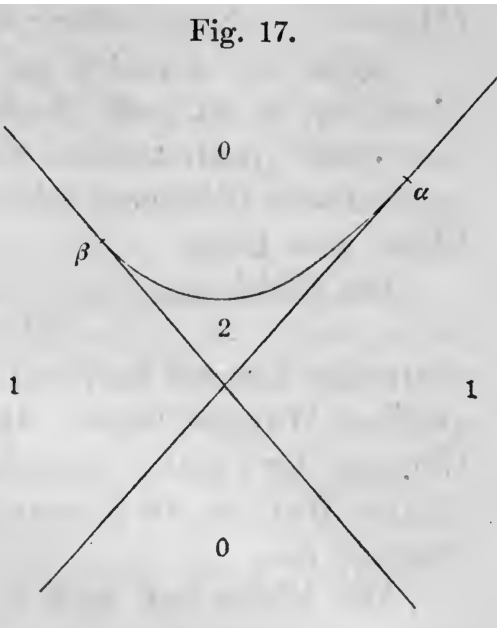


Fig. 17.



(4) $$x^2 + a\,x + b, \quad 2\,x + a, \quad 1$$

to examine with regard to the changes of sign.

The two straight lines

$$x^2 + a\,x + b = 0, \quad 2\,x + a = 0$$

divide the plane into three regions, in which the sequence (4) presents the changes of sign indicated in Fig. 18.

Fig. 18.

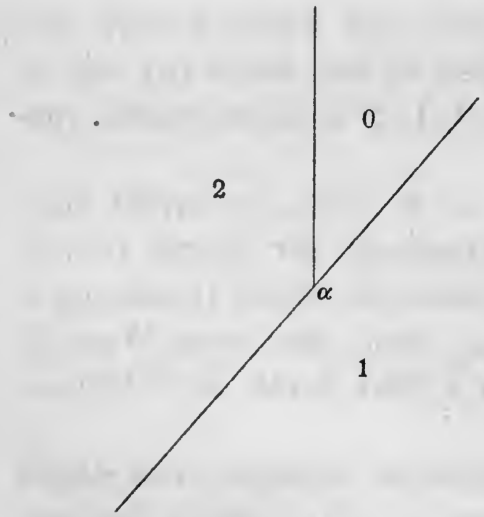
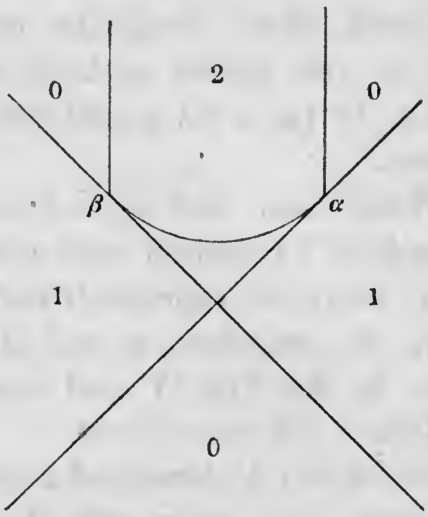


Fig. 19.



If we again take two values $x = \alpha$ and $x = \beta$, six regions result, in which the loss of changes of sign is given in Fig. 19. In the regions marked 0 and 1

regions, then, the loss of changes of sign gives the correct number of roots. In the region 2, on the other hand, the decision between two roots and none is not made.

Let us now take the Jacobi criterion (§. 102). According to it, in order to estimate the number of roots lying between α and β, we must first transform the equation (1) into

$$(\alpha + \beta y)^2 + a(\alpha + \beta y)(1 + y) + b(1 + y)^2 = 0$$

or

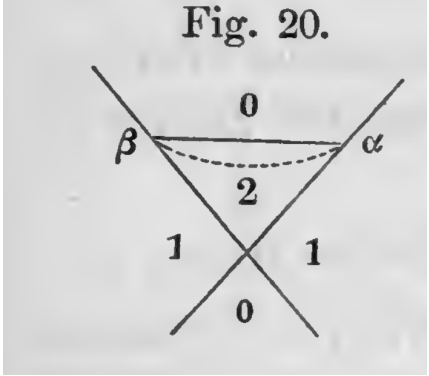
$$\begin{aligned}(\beta^2 + a\beta + b)y^2 + (2\alpha\beta + a(\alpha + \beta) + 2b)y \\ + (\alpha^2 + a\alpha + b) = 0,\end{aligned}$$

and now count the number of changes of sign in the three functions

$$\begin{aligned}&\beta^2 + a\beta + b \\ &2\alpha\beta + a(\alpha + \beta) + 2b \\ &\alpha^2 + a\alpha + b\end{aligned}$$

The first and the third of these expressions, set equal to zero, represent the two parabola tangents considered above. The middle one, $2\alpha\beta + a(\alpha + \beta) + 2b$, vanishes for $a = -2\alpha$, $b = \alpha^2$ and for $a = -2\beta$, $b = \beta^2$ and thus represents the line joining the two points α, β, and indeed in such a way that it becomes negative on the side that contains the point of intersection of the two tangents [at this point of intersection itself it is equal to $-(\beta - \alpha)^2$].

Fig. 20.



We obtain here five regions with no, one or two changes of sign, as Fig. 20 indicates.

One sees from this that this figure confines the indecision to a much smaller region than the previous one. From this point of view, then, it does not appear justified that Jacobi so decidedly gives the Budan-Fourier criterion preference over his own[22].

§. 104.
Determination of an upper limit for the roots.

We have already, in the third section, made use of the theorem that one can always find a positive number which is greater than the absolute values of all the roots of a given equation. But now the point is to give a determination of such an upper limit that is as simple and at the same time as accurate as possible.

We consider at first only the real roots, and thus seek a positive number which is greater than the greatest positive root of a given equation $f(x) = 0$.

Such a determination is given to us by the theorem of Budan.

If we take the coefficient of the highest power of x in $f(x)$ equal to 1 (or at least positive), then the positive number α can always be taken so large that

$$f(\alpha),\ f'(\alpha),\ f''(\alpha) \dots f^{(n)}(\alpha) \tag{1}$$

all become positive. Then in the sequence of functions

$$f(x),\ f'(x),\ f''(x) \dots f^{(n)}(x)$$

no change of sign is lost any more between $x = \alpha$ and $x = \infty$, and hence, by Theorem I, no root of $f(x)$ can lie between α and ∞.

If one wishes to obtain from the same theorem a lower limit for the negative roots, one takes the positive number β so that

$$(-1)^n f(-\beta),\ (-1)^{n-1} f'(-\beta),\ \dots f^{(n)}(-\beta) \tag{2}$$

all become positive; then the equation $f(x) = 0$ certainly has no negative root below $-\beta$.

This determination of the limits goes back already to Newton.

Another determination of an upper limit, which is simpler for computation, has been given by Laguerre[23].

Instead of the derivatives $f(x), f'(x), f''(x) \dots$ he uses the functions $f_0(x), f_1(x), f_2(x), \dots f_{n-1}(x)$, which have already

many ways, and in particular have rendered good service in the Tschirnhausen transformation.

We defined these functions in §. 4 by

$$\begin{aligned} f_0(x) &= 1 \\ f_1(x) &= x + a_1 \\ f_2(x) &= x^2 + a_1 x + a_2 \\ &\dots\dots\dots\dots\dots\dots\dots \\ f_{n-1}(x) &= x^{n-1} + a_1 x^{n-2} + a_2 x^{n-3} + \dots + a_{n-1} \\ f(x) = f_n(x) &= x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n. \end{aligned}$$

and for them set up the recursion formula

$$f_v(x) - x f_{v-1}(x) = a_v.$$

[22] Cf. F. Klein, Geometrisches zur Abzählung der reellen Wurzeln einer algebraischen Gleichung in Dyck's Catalog mathematischer Modelle (München 1892).

[23] Laguerre, Nouvelles Annales d. math., 2. ser., t. XIX, 1880. Journal de math., 3. ser., t. IX, 1883.

One now has, as has already been shown in the place cited,

$$f(x) = (x-\alpha)\left[x^{n-1}f_0(\alpha) + x^{n-2}f_1(\alpha) + \cdots + f_{n-1}(\alpha)\right] + f(\alpha). \tag{3}$$

A glance at this formula shows that if α is a number which makes the functions

$$f_0(\alpha),\ f_1(\alpha),\ f_2(\alpha)\dots f_{n-1}(\alpha),\ f(\alpha) \tag{4}$$

positive, then no positive x greater than α can exist which makes $f(x)$ zero, since in that case nothing but positive terms stand on the right-hand side of (3).

If, then, a positive α is so determined that the functions (4) are all positive, this is an upper limit for the positive roots of $f(x)$.

In the same way one obtains a lower limit $-\beta$ for the negative roots, if one determines the positive quantity β so that

$$f_0(-\beta),\ -f_1(-\beta),\ f_2(-\beta),\ \dots (-1)^n f(-\beta) \tag{5}$$

all become positive.

This determination of the limits is indeed simpler to handle in calculation, but under some circumstances it gives less accurate limits than the Newton determination.

For if one differentiates (3) repeatedly with respect to x and then sets $x = \alpha$, one obtains, under the assumption that the quantities (4) are all positive, for the derivatives $f'(x)f''(x)\dots$ expressions which for $x = \alpha$ consist of nothing but positive terms. If, then, the expressions (4) are positive, the derivatives (1) are also necessarily all positive, while the converse is not necessarily the case.

In order to find an upper limit for the absolute value of the imaginary roots which remains valid also in the case of complex coefficients, we consider the function

$$f(x) = x^n + a_1x^{n-1} + a_2x^{n-2} + \cdots + a_n,$$

in which the coefficients $a_1, a_2 \dots a_n$ are real or imaginary. We denote the absolute value of a complex quantity a, as before, by $|a|$, and then, if we assume that the absolute value of x is greater than 1, we have

$$\left|a_1 + \frac{a_2}{x} + \cdots + \frac{a_n}{x^{n-1}}\right| < |a_1| + |a_2| + \cdots + |a_n| = \alpha.$$

If we therefore take x so that its absolute value is greater than each of the two numbers 1 and α, the expression

$$x^{1-n}f(x) = x + a_1 + \frac{a_2}{x} + \cdots + \frac{a_n}{x^{n-1}}$$

cannot vanish, since the absolute value of x is greater than the absolute value of the sum of all the remaining terms of the right-hand side.

If, then, according as α is greater or less than 1, we draw a circle with the radius α or 1 about the origin as centre, certainly no roots of $f(x)$ lie outside this circle any longer.

§. 105.
Estimation of the imaginary roots.

We have seen in §. 96 how, from the characteristic of a certain system of curves, we can determine the exact number of the complex roots of an equation that lie in the interior of a closed curve. This criterion too can be simplified if one no longer wishes to find the exact number of the roots, but only an upper limit for this number, which in many cases also suffices for the exact determination.

Let, then, as in §. 96

$$F(x) = \varphi(x,y) + i\psi(x,y) \tag{1}$$

be a real or complex function of the complex argument

$z = x + yi$; therein let φ and ψ be real functions of x and y, and now let a closed curve f be drawn. We then have the theorem:

XI. If the curve f is divided by the curve φ into $2m$ segments, in which φ is alternately positive and negative, then the number of roots of $F(z)$ lying within f is at most equal to m; and the same also holds when f is divided by ψ into $2m$ such segments.

The theorem is an immediate consequence of the theorem on characteristics II, §. 96; for by that theorem

$$k = \tfrac{1}{2}[E(f;\varphi,\psi) - A(f;\varphi,\psi)];$$

and if now the curve f is cut by the φ-curve in $2m$ points, then

$$m = \tfrac{1}{2}[E(f;\varphi,\psi) + A(f;\varphi,\psi)],$$

hence

$$k = m - A(f;\varphi,\psi).$$

But here, by the theorem just mentioned, k is equal to the number of roots of F lying in the interior of f, and $A(f;\varphi,\psi)$ is in any case a non-negative number. In exactly the same way one can proceed with the formula

$$k = \tfrac{1}{2}[E(f;\psi,\varphi) - A(f;\psi,\varphi)]$$

§. 106.
The Theorem of Rolle.

If the equation $f(x) = 0$ has two real roots α and β, of which the first is an a-fold root and the second a b-fold root, then we can set

$$f(x) = (x-\alpha)^a (x-\beta)^b f_1(x) \tag{1}$$

Now if $\alpha < \beta$, and no root of $f(x) = 0$ lies between α and β, then $f(x)$ and $f_1(x)$ will have an unchanged sign throughout this whole interval. If we form the derivative of (1), we obtain

$$\frac{(x-\alpha)(x-\beta)f'(x)}{f(x)} = a(x-\beta) + b(x-\alpha) + (x-\alpha)(x-\beta)\frac{f_1'(x)}{f_1(x)}. \tag{2}$$

The right-hand side of this expression becomes negative for $x = \alpha$, namely $a(\alpha - \beta)$, and positive for $x = \beta$, namely $b(\beta - \alpha)$, and must therefore, while x goes from α to β, pass through zero once, or more generally an odd number of times. On the left-hand side of (2), $(x-\alpha)(x-\beta)$ and $f(x)$ have an unchanged sign, and hence $f'(x)$ must pass through zero an odd number of times. From this follows Rolle's theorem:

XII. If α and β are two successive roots of $f(x) = 0$, then between α and β (not counting the limits) there lie roots of $f'(x) = 0$ in odd number, and hence at least one.

Important consequences follow from this for the case in which the equation $f(x) = 0$ has only real roots. If these, ordered by magnitude, are $\alpha, \beta, \gamma, \ldots$, and if α is an a-fold, β a b-fold, γ a c-fold root, etc., then $f'(x) = 0$ has at α an $(a-1)$-fold, at β a $(b-1)$-fold, at γ a $(c-1)$-fold root, and between α and β, between β and γ, and so on, at least one real root each. But the total number of these roots is

$$a + b + c + \cdots - 1 = n - 1,$$

and since $f'(x)$ is of degree $(n-1)$, this minimal number is also not exceeded. From this follows the theorem:

XIII. If $f(x) = 0$ has only real roots, then $f'(x) = 0$ also has only real roots, and of these, those that do not coincide with multiple roots of $f(x)$ are simple roots, which are separated by the roots of $f(x)$.

If we apply this theorem repeatedly, we can generalise it as follows:

XIV. If $f(x)$ has only real roots, then $f'(x), f''(x), f'''(x) \ldots$ also have only real roots. If one of these functions $f^{(\nu)}(x)$ has an a-fold root α, and if $a > 1$, then α is an $(a+\nu)$-fold root of $f(x)$.

These theorems are substantially generalised if one applies a linear transformation to them[24].

This is more easily carried out if one uses the homogeneous notation, that is, understands by $f(x, y)$ an integral homogeneous function of degree n. If, for an arbitrary value $\xi : \eta$, one forms the polars of this function (§. 60):

$$P_1(x,\xi) = \frac{1}{n}\{\xi f'(x) + \eta f'(y)\}$$

$$P_2(x,\xi) = \frac{1}{n(n-1)}\{\xi^2 f''(x,x) + 2\xi\eta f''(x,y) + \eta^2 f''(y,y)\} \tag{3}$$

$$P_3(x,\xi) = \frac{1}{n(n-1)(n-2)}\{\xi^3 f'''(x,x,x) + 3\xi^2\eta f'''(x,x,y) + 3\xi\eta^2 f'''(x,y,y) + \eta^3 f'''(y,y,y)\}$$

$$\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots,$$

then these functions remain unchanged if one transforms ξ, η and x, y simultaneously by a linear transformation and puts for f and its derivatives the corresponding transformed functions. With respect to the reality of the roots, nothing is changed by a real linear transformation. But now, if ξ, η are real, the real linear substitution can be so determined that the transformed values become $\xi' = 1$, $\eta' = 0$, and thereby, if one also sets $y = 1$, the polars pass into the derivatives of the function $f(x)$ with respect to x (apart from numerical factors).

From this it follows that in the theorems XIII, XIV the derivatives $f'(x), f''(x) \ldots$ can be replaced by the polars $P_1(x,\xi), P_2(x,\xi) \ldots$ for arbitrary real ξ, η.

We make an application of this to the equation which one obtains by setting the $(n-2)$th polar equal to zero:

$$x^2 f''(\xi,\xi) + 2xy f''(\xi,\eta) + y^2 f''(\eta,\eta) = 0. \tag{4}$$

If $f(x, y) = 0$ has, as we assume, only real roots, then the quadratic equation (4) must also have real roots, i. e. the Hesse determinant

$$f''(\xi,\xi) f''(\eta,\eta) - f''(\xi,\eta)^2 \tag{5}$$

must be negative for all real ξ, η. It can only vanish if (4) has two equal roots, and by XIV this is possible only when $f(x, y)$ is the nth power of a linear function. In this case the determinant (5) vanishes identically: if we therefore set this case aside, it follows that for an equation with only real roots, the equation obtained by setting the Hesse determinant equal to zero has only imaginary roots.

If we wish to return to the inhomogeneous representation, we can, making use of the Euler theorem

$$\begin{aligned} xf'(x) + yf'(y) &= nf(x) \\ xf''(x,x) + yf''(x,y) &= (n-1)f'(x) \\ xf''(y,x) + yf''(y,y) &= (n-1)f'(y) \end{aligned} \tag{6}$$

[24] Laguerre, Nouvelles Annales de Mathématiques, 2. Sér., Vol. 19, 1880.

eliminate $f'(y), f''(x,y), f''(y,y)$ by means of it. One obtains in this way, on setting $y = 1$,

$$\begin{aligned} f'(y) &= nf(x) - xf'(x) \\ f''(x,y) &= (n-1)f'(x) - xf''(x) \\ f''(y,y) &= n(n-1)f(x) - 2(n-1)xf'(x) + x^2 f''(x), \end{aligned} \tag{7}$$

and hence, if

$$(n-1)H(x,y) = f''(x,x)f''(y,y) - f''(x,y)^2 \tag{8}$$

is set:

$$H(x,y) = H(x) = nf(x)f''(x) - (n-1)f'(x)^2. \tag{9}$$

$H(x)$ is, with respect to x, at most of degree $2n-4$.

§. 107.
The theorems of Laguerre for equations with only real roots.

Of the theorems last derived, Laguerre has made an application to equations with only real roots, which we shall now become acquainted with.

Let $f(x) = 0$ be an equation of degree n with n real roots $\alpha, \alpha_1, \alpha_2 \ldots$, which we assume to be distinct from one another.

If x is a variable quantity, we have the formula, already applied several times [§. 14, (11)]:

$$S\left(\frac{1}{x-\alpha}\right) = \frac{f'(x)}{f(x)}, \tag{1}$$

in which the summation sign S refers to the different roots $\alpha, \alpha_1, \alpha_2 \ldots$ If we take the derivative of this once more with respect to x, it follows that

$$S\frac{1}{(x-\alpha)^2} = \frac{f'(x)^2 - f(x)f''(x)}{f(x)^2}, \tag{2}$$

and if on the right-hand side we introduce the function H [§. 106, (9)],

$$S\frac{1}{(x-\alpha)^2} = \frac{f'(x)^2 - H(x)}{nf(x)^2}. \tag{3}$$

If one applies to this formula a linear transformation under which the value $x = \infty$ corresponds to an arbitrary other real value of the new variable, one obtains a more general formula, which we shall first derive. We introduce homogeneous variables by replacing x by $x : y$ and α by $\alpha : \beta$. Thereby (3) becomes

$$S\frac{\beta^2}{(x\beta - y\alpha)^2} = \frac{f'(x)^2 - y^2 H(x,y)}{nf(x,y)^2}; \tag{4}$$

now we apply an arbitrary linear transformation:

$$\begin{aligned} x &= ax' + by', \quad \alpha = a\alpha' + b\beta', \\ y &= cx' + dy', \quad \beta = c\alpha' + d\beta', \end{aligned} \tag{5}$$

or, solved:

$$\begin{aligned} rx' &= dx - by, \\ ry' &= -cx + ay, \quad r = ad - bc. \end{aligned} \tag{6}$$

If by this transformation

$$f(x, y) = \varphi(x', y')$$

then

$$rf'(x) = d\varphi'(x') - c\varphi'(y'),$$

and, on account of the covariant property of the function H (§. 59),

$$r^2 H(x, y) = H'(x', y'),$$

where H' is derived from φ in the same way as H is from f. From this it follows from (4):

$$S\frac{(c\alpha' + d\beta')^2}{(x'\beta' - y'\alpha')^2} = \frac{[d\varphi'(x') - c\varphi'(y')]^2 - (cx' + dy')^2 H'(x', y')}{n\varphi(x', y')^2}. \tag{7}$$

In this formula, since this is only a matter of notation, the accents may be omitted at x', y', α', β', H', and f may be

be put in place of φ. The two arbitrary quantities c and d remain in it.

If, for the sake of a more elegant notation, we further set $c = -\eta$, $d = \xi$, we thus obtain:

$$S\left(\frac{\xi\beta - \eta\alpha}{x\beta - y\alpha}\right)^2 = \frac{[\xi f'(x) + \eta f'(y)]^2 - (x\eta - \xi y)^2 H(x, y)}{nf(x, y)^2}, \tag{8}$$

and in this form both the right-hand and the left-hand side remain completely unchanged if we apply an arbitrary linear transformation simultaneously to the pairs of variables x, y; ξ, η; α, β.

The formula (8) is an identity; it holds for every function $f(x, y)$ and for all values of the variables x, y; ξ, η. But now we make the assumption that $f(x, y)$ has only real roots, that is, that the α, β are real, and we also suppose ξ, η and x, y to be real. Both sides of the equation (8) are then essentially positive. We denote their common value by P, and now determine a quantity $X : Y$ by the quadratic equation

$$P = \left(\frac{X\eta - Y\xi}{Xy - Yx}\right)^2,$$

or

$$\Phi = (Xy - Yx)^2 P - (X\eta - Y\xi)^2 = 0. \tag{9}$$

This equation has two real roots, which are obtained from

$$X\eta - Y\xi = \pm\sqrt{P}\,(Xy - Yx)$$

About the position of the roots of this equation we can, however, assert the following:

If we set $X = x$, $Y = y$, then Φ becomes negative.

But if we set $X = \alpha$, $Y = \beta$, where $\alpha : \beta$ is any one of the roots of $f = 0$, then

$$\left(\frac{X\eta - Y\xi}{Xy - Yx}\right)^2 = \left(\frac{\xi\beta - \eta\alpha}{x\beta - y\alpha}\right)^2,$$

that is, equal to one term of the sum

$$P = S\left(\frac{\xi\beta - \eta\alpha}{x\beta - y\alpha}\right)^2$$

and therefore smaller than P. From this it follows that Φ is positive for $X : Y = \alpha : \beta$.

If we now arrange the values of all the $\alpha : \beta$ and $x : y$, ordered cyclically by magnitude, say on a circle,

and represent the value $X : Y$ by a variable point of this circle, and denote the points in question by α, x, X, then Φ passes through zero when X goes from x forwards or backwards to the nearest points, which we shall denote by α_1 and α_2. If then X_1, X_2 are the roots of equation (9), we have the following arrangement on the circle

(10) $$\alpha_1,\ X_1,\ x,\ X_2,\ \alpha_2,$$

and (X_1, X_2) represents an interval in which the arbitrarily chosen point x lies, but no root point of the equation $f = 0$. X_1 and X_2 are therefore nearer to the two nearest roots α_1, α_2 than the initial value x.

This holds whatever position the second arbitrary point ξ may have. The points X_1, X_2 will indeed depend on it; but they will always lie in the interval (α_1, x) and (x, α_2).

There now arises the question: how is the point ξ to be chosen so that X_1 may lie as near as possible to α_1, or X_2 as near as possible to α_2; or rather, since the knowledge of ξ itself does not matter, which is the point X_1 lying as near as possible to α_1, and the point X_2 lying as near as possible to α_2?

If we think of the point X as given, then (9) is a quadratic equation for the point ξ; to every position of X there thus belong two positions of ξ. But only such values of X_1 or X_2 are admissible in (10) for which this quadratic equation yields real roots. Hence only such values of X_1 and X_2 can occur for which the discriminant of (9) with respect to ξ is positive, and we obtain the limiting positions of X_1 and X_2 when we set the discriminant of (9) equal to zero. But the coefficients of ξ^2, $2\xi\eta$, η^2 in (9) are

$$\frac{f'(x)^2 - y^2H}{nf^2}(Xy - Yx)^2 - Y^2$$

$$\frac{f'(x)f'(y) + xyH}{nf^2}(Xy - Yx)^2 + XY$$

$$\frac{f'(y)^2 - x^2H}{nf^2}(Xy - Yx)^2 - X^2,$$

and the discriminant is obtained when from the square

the product of the two outer ones from that of the middle one, namely, using the relation $xf'(x) + yf'(y) = nf$:

$$\frac{(Xy - Yx)^2}{nf^2}\left\{(n-1)H(Xy - Yx)^2 + [Xf'(x) + Yf'(y)]^2\right\},$$

and one therefore obtains the limiting values of $X : Y$ from the equation

(11) $$[Xf'(x) + Yf'(y)]^2 + (n-1)(Xy - Yx)^2H = 0,$$

which is quadratic in $X : Y$.

Since H is negative, this equation has two real roots, which one finds from

$$Xf'(x) + Yf'(y) = \pm(Xy - Yx)\sqrt{(1-n)H} \tag{12}$$

.

This formula comprises several particular results, which one obtains on abandoning the homogeneous form.

Let us first take $y = 0$, so that $x : y$ is infinite, and set $Y = 1$; then there result two values X_1, X_2, between which the totality of all the roots of $f(x)$ is contained. If we set

$$f(x, y) = a_0x^n + a_1x^{n-1}y + a_2x^{n-2}y^2 + \cdots,$$

then

$$H = [2na_0a_2 - (n-1)a_1^2]x^{2n-4}, \tag{13}$$

and one obtains these two values from

$$na_0X = -a_1 \pm \sqrt{(n-1)^2a_1^2 - 2n(n-1)a_0a_2}. \tag{14}$$

If one leaves $x : y$ undetermined, but returns to the inhomogeneous form by setting $y = 1$, $Y = 1$ and applying formulas (7) and (9) of §. 106, one obtains from (12)

$$x - X = \frac{nf(x)}{f'(x) \pm \sqrt{(n-1)^2f'(x)^2 - n(n-1)f(x)f''(x)}}, \tag{15}$$

and these values have the following meaning:

If x lies between two roots α_1, α_2 of $f(x)$, then of the two values of X represented by (15), one lies nearer to α_1, the other nearer to α_2, but both still within the interval (α_1, α_2).

But if x is greater than the greatest, or less than the least root of $f(x)$, then all the roots lie between the two values of X, while x lies outside.

If x lies near a root, then one of the two values of X gives a still more accurate value of this root.

Just as an interval is determined by X_1, X_2 in which no root of the equation lies at all, so one can also determine an interval in which roots certainly lie. Here too one best starts from the homogeneous form and makes use of the linear transformation.

Let $f(x, y)$ then be, as before, a form of degree n with none but real roots; and let further $x : y$, $\xi : \eta$, $\xi' : \eta'$ be three real values, of which two are arbitrary, and of which the third, through the relation

$$\begin{aligned}[\xi f'(x) + \eta f'(y)]\,[\xi' f'(x) + \eta' f'(y)] \\ - H(\xi y - \eta x)\,(\xi' y - \eta' x) = 0\end{aligned} \tag{16}$$

depends on the two others. The equation (16) remains unchanged if we subject the three pairs of quantities x, y; ξ, η; ξ', η' simultaneously to the same linear transformation. Hence all conclusions that we draw from a special form of this equation can be generalised. The linear substitution can be so determined that for the new variables $y = 0$, $\xi = 0$; then (16) yields for ξ', η' the equation

$$a_1(n\,a_0\,\xi' + a_1\,\eta') = [2\,n\,a_0\,a_2 - (n-1)\,a_1^2]\eta',$$

or, if one assumes $a_0 = 1$, $\eta' = 1$,

$$\xi' = -\frac{a_1^2 - 2\,a_2}{a_1}.$$

But if, as above, we denote by α the roots of f, we have

$$a_1^2 - 2\,a_2 = S(\alpha^2), \quad a_1 = -S(\alpha);$$

and thus

$$\xi' = \frac{S(\alpha^2)}{S(\alpha)}.$$

For this we may also write:

$$S[\alpha(\alpha - \xi')] = 0,$$

and from this it follows, since all the α are to be real, that the products $\alpha(\alpha - \xi')$ can be neither all positive nor all negative, or in other words, one part of the roots must

between 0 and ξ', and another part outside this interval.

From this it now follows generally, by applying the linear transformation, that on the circle carrying the points α, x, ξ, ξ', if ξ, ξ' are connected by (16), at least one of the roots α lies in each of the two parts into which the circle is divided by ξ and ξ'. The point x and one of the points ξ, ξ' are arbitrary; the third point is determined by (16).

We shall in particular make the assumption that the point ξ' coincides with one of the two points X determined by the equation (12), and thus set, say,

$$Xf'(x) + Yf'(y) + (Xy - Yx)\sqrt{(1-n)H} = 0 \tag{17}$$

If we introduce this into (16), by setting $\xi', \eta' = X, Y$ and then removing the factor $Xy - Yx$, we obtain

$$(n-1)\,[\xi f'(x) + \eta f'(y)] + (\xi y - \eta x)\sqrt{(1-n)H} = 0, \tag{18}$$

by which the point ξ is uniquely determined.

If we pass to the inhomogeneous form, we obtain [§. 106, (7) and (9)] from (17) and (18)

$$x - X = \frac{nf(x)}{f'(x) + (n-1)\sqrt{f'(x)^2 - \dfrac{n}{n-1}f(x)f''(x)}}, \tag{19}$$

$$x - \xi = \frac{nf(x)}{f'(x) + \sqrt{f'(x)^2 - \dfrac{n}{n-1}f(x)f''(x)}}, \tag{20}$$

in which the square root can have either sign.

The position of the points can then be characterised as follows. There is at least one root α such that

$$\xi,\ \alpha,\ X,\ x, \tag{21}$$

follow one another in order of magnitude, and such that no root lies between x and X. We can also choose α so that no further root lies between α and X. Between α and ξ, however, other roots may possibly lie.

But if we now let x approach the root α, then X will approach the same root, and (20) shows that ξ also tends to the same limit. It follows, therefore, that if x lies sufficiently near a root, this nearest root lies in the interval (ξ, X), in which no second root is contained, and that this interval closes more and more about α.

As an example we take (with Laguerre) the spherical functions, of which we have already shown earlier (§. 87) that they have only real roots, which all lie between -1 and $+1$. The function

$$P_n(x) = \frac{1\,.\,3\,\ldots\,(2\,n-1)}{1\,.\,2\,\ldots\,n}\left(x^n - \frac{n(n-1)}{2(2\,n-1)}\,x^{n-2} + \cdots\right)$$

has the property that

$$P_n(1) = 1, \qquad P_n'(1) = \frac{n(n+1)}{2},$$
$$P_n''(1) = \frac{n(n-1)(n+1)(n+2)}{8}$$

holds. One proves this easily by complete induction, by verifying the formulas directly in the first cases $n = 2, 3\ldots$, and then, from the recursion formula

$$(1-x^2)\,P_n'(x) + n\,x\,P_n(x) - n\,P_{n-1}(x) = 0$$

by differentiating twice, establishing their correctness for the index n, provided it is assumed for the index $n-1$.

If we then set $x = 1$ in (19) and (20), we obtain two limits between which the root coming nearest to 1 lies, namely

$$X = 1 - \frac{2}{n+1+(n-1)\sqrt{\dfrac{n(n+1)}{2}}},$$
$$\xi = 1 - \frac{2}{n+1+\sqrt{\dfrac{n(n+1)}{2}}}.$$

For $n = 7$, for example, this gives

$$X = 0{,}94967\ldots$$
$$\xi = 0{,}84952\ldots$$

The more exact values of the two roots lying nearest to 1, which we cite for the sake of comparison, are, according to the computation of Gauss (Gauss Werke, Bd. III, S. 195):

$$0,94911\ldots; \quad 0,74153\ldots$$

Thus only one root lies here between X and ξ, and it is only in the fourth decimal place that X shows the excess over the value of this root.

Tenth Section.
Approximate Calculation of the Roots.

§. 108.
Interpolation. Regula falsi.

With the delimitation of the intervals in each of which only one root of an algebraic equation lies, the possibility is given of calculating the real roots numerically with arbitrary accuracy, by narrowing the interval more and more, e.g. by repeated halving. If one divides an interval of size 1 repeatedly into ten parts, then each new step yields a further decimal place.

Once a root has first been separated from all the others, then in the further narrowing of the interval one has only to take into account the sign of the function $f(x)$ itself. The calculations needed for this can be greatly facilitated by the application of the interpolation formula which we came to know in the first section.

Namely, in §. 10, (7) we derived a formula by which one can determine a function of degree n, which we shall now denote by $\varphi(\xi)$, when $n+1$ successive values $\varphi(0), \varphi(1) \ldots \varphi(n)$ are given. This formula is, if

$$B_\nu^{(\xi)} = \frac{\xi(\xi-1)\ldots(\xi-\nu+1)}{1.2.3\ldots\nu} \tag{1}$$
$$\Delta_\xi = \varphi(\xi+1) - \varphi(\xi),$$

$$\begin{aligned} \Delta'_\xi &= \Delta_{\xi+1} - \Delta_\xi. \\ &\cdots\cdots\cdots \\ \Delta_\xi^{(n-1)} &= \Delta_{\xi+1}^{(n-2)} - \Delta_\xi^{(n-2)} \end{aligned} \tag{2}$$

is set,

$$\varphi(\xi) = \varphi(0) + \Delta_0 B_1^{(\xi)} + \Delta'_0 B_2^{(\xi)} + \cdots + \Delta_0^{(n-1)} B_n^{(\xi)}. \tag{3}$$

The value of the formula lies in this, that the sequence of differences $\Delta_0, \Delta'_0, \Delta''_0 \ldots$ in many cases presents numerical values decreasing so rapidly that one can content oneself with a few of the first terms of the series (3).

Now if $f(x)$ is the function to be investigated and (α, β) the interval in which one of the sought roots lies, then, understanding by m any suitable integer, we set

$$\delta = \frac{\beta-\alpha}{m}; \qquad x = \alpha + \delta\xi,$$

and if we accordingly set

$$\begin{aligned} \Delta_x &= f(x+\delta) - f(x) \\ \Delta'_x &= \Delta_{x+\delta} - \Delta_x \\ &\cdots\cdots\cdots \end{aligned}$$

then

$$f(x) = f(\alpha) + \frac{\Delta_\alpha}{\delta}(x-\alpha) + \frac{\Delta'_\alpha}{\delta^2}\frac{(x-\alpha)(x-\alpha-\delta)}{2} + \cdots \tag{4}$$

One could just as well start from the other end point β of the interval, and would then have to set in (3)

$$x = \beta - \delta\xi$$

One would then obtain

$$\begin{aligned} \Delta_x &= f(x-\delta) - f(x) \\ \Delta'_x &= \Delta_{x-\delta} - \Delta_x \\ &\cdots\cdots\cdots \end{aligned}$$

$$f(x) = f(\beta) + \frac{\Delta_\beta}{\delta}(x-\beta) + \frac{\Delta'_\beta}{\delta^2}\frac{(x-\beta)(x-\beta+\delta)}{2} + \cdots \tag{5}$$

If one interprets the equation

$$y = f(x)$$

as the equation of a curve in a rectangular coordinate system x, y, then the relations can be illustrated geometrically.

If, for example, one applies equation (4) to the interval from α to $(\alpha+\delta)$ and contents oneself with taking account of the first difference, that means, in the language of geometry,

that the arc of the curve running between the two curve points $\alpha, f(\alpha)$ and $\alpha+\delta, f(\alpha+\delta)$ is replaced by the chord. If the second difference is also taken into account, then the arc of the curve passing through the three successive points with the abscissae $\alpha, \alpha+\delta, \alpha+2\delta$ is replaced by the arc of a parabola which passes through the same points and whose axis is parallel to the axis of ordinates; and this parabola will fit the curve still more closely than the chord. The smaller the interval δ is, the less influence the higher differences will have.

How far one has to go in the approximation, then, depends not only on the accuracy with which one wishes to know $f(x)$, but essentially also on the density of the values $\alpha, \alpha+\delta, \alpha+2\delta \ldots$ for which the function is known. On these theorems rests the arrangement of our works of tables, especially of the tables of logarithms. There, to be sure, we are not dealing with integral rational functions; but for continuous functions in general the same laws hold here. In the tables one therefore also finds, beside the values $f(\alpha), f(\alpha+\delta), f(\alpha+2\delta)\ldots$, the values of the first difference, or of the first two differences, given. For the customary seven-place tables the first difference suffices. In the ten-place table "Thesaurus logarithmorum" of Vega the second differences are also given, and must be taken into account in quite precise computations.

Our interpolation formulae can be applied with advantage to compute the roots of equations, or, more exactly expressed, to improve the approximate values found by other means.

We can formulate the problem thus: for a given value of $f(x)$ lying between $f(\alpha)$ and $f(\alpha+\delta)$, the corresponding value of x is to be found.

We regard $x = \alpha$ as a first approximate value. If we now set

$$f(x) - f(\alpha) = \Delta,$$

then Δ is a given value of the same sign as $\Delta_\alpha = f(\alpha+\delta) - f(\alpha)$ and absolutely smaller than Δ_α. If we further set $x - \alpha = u$, formula (4) gives

$$\Delta = \frac{\Delta_\alpha u}{\delta} + \frac{\Delta'_\alpha}{\delta^2}\frac{u(u-\delta)}{2} + \cdots \tag{6}$$

If we stop, to begin with, at the first difference, we obtain as first correction

$$u = \delta\frac{\Delta}{\Delta_\alpha}, \tag{7}$$

and if we now set

$$u = \delta\frac{\Delta}{\Delta_\alpha} + u'$$

then it follows from (6), if one omits u' in the second term, that

$$0 = \frac{\Delta_\alpha}{\delta}u' + \frac{\Delta'_\alpha\Delta(\Delta-\Delta_\alpha)}{2\Delta_\alpha^2},$$

and hence as second correction:

$$u' = \delta\frac{\Delta'_\alpha\Delta(\Delta_\alpha-\Delta)}{2\Delta_\alpha^3}. \tag{8}$$

The first correction is obtained by replacing the arc of the curve running between α and $\alpha+\delta$ by the chord, as above; the second by replacing the arc between $\alpha, \alpha+\delta, \alpha+2\delta$ by a parabola which passes through the same points, but which now has its axis parallel to the x-axis.

Let us take as an example the equation

$$f(x) = x^3 - 2x - 2 = 0,$$

which has a real root between 1,7 and 1,8.

One computes

$$\begin{array}{llll} x = 1,7, & f(x) = -0,487, & \Delta_\alpha = 0,719, & \Delta'_\alpha = 0,108, \\ 1,8, & = +0,232, & =0,827, & \\ 1,9, & & & =1,059. \end{array}$$

Since $f(x) = 0$ is to hold, we must set

$$\Delta = 0,487$$

and the first correction to $\alpha = 1,7$ is, by (7),

$$u = 0,06773,$$

the second correction gives, by (8),

$$u' = 0,0164,$$

hence

$$\alpha + u + u' = 1,76937.$$

The more accurate value, computed in another way, is 1,76929... We thus have a result accurate in the first three decimals. But we have here no other means of judging the accuracy in advance than the decrease of the differences $\Delta, \Delta', \Delta'' \ldots$ If Δ' is so small that it falls outside the limits of the intended accuracy, then taking the first difference into account gives an exact result. The rule for the computation of roots set forth here is called the Regula falsi.

The simplest form of this rule, suited to the first approximation, is this:

If a root x of the equation $f(x) = 0$ lies between α and β, and if $f(\alpha) = -a, f(\beta) = b$, then

$$x = \frac{b\alpha + a\beta}{a+b} \tag{9}$$

is an approximate value of x. This is only another way of writing formula (7).

§. 109.
The Newton method of approximation.

A method which, for the approximate computation of the roots of an equation, is usually better than interpolation is due to Newton and was developed and more closely investigated by Fourier.

The method consists simply in the following. Let

$$f(x) = 0 \tag{1}$$

be the equation to be solved, and let a value $x = \alpha$ have been found which may be regarded as a certain approximation to a root. We set in (1)

$$x = \alpha + h,$$

and obtain

$$f(\alpha+h) = f(\alpha) + hf'(\alpha) + \frac{h^2}{1.2}f''(\alpha) + \cdots \tag{2}$$

If one now determines h from the equation

$$f(\alpha) + hf'(\alpha) = 0, \tag{3}$$

which presupposes that $f'(\alpha)$ is different from zero, then

$$f(\alpha + h) = \frac{h^2}{1.2} f''(\alpha) + \cdots,$$

and thus becomes, if h is a small number, since only the square

and higher powers of h occur, has a small value. Under suitable hypotheses, then,

$$\alpha' = \alpha - \frac{f(\alpha)}{f'(\alpha)} \tag{4}$$

is to be regarded as a better approximation to the true value of the root.

If one then replaces α by α', one will obtain in

$$\alpha'' = \alpha' - \frac{f(\alpha')}{f'(\alpha')}$$

a still better approximation, and so on.

But there still remain here the following two questions to answer:

1. Under what hypotheses is α' really a better value than α?
2. How can one estimate the degree of accuracy that one attains in this way?

These questions were answered by Fourier; but in doing so he makes the following assumption:

A root of $f(x)$ is enclosed in an interval (α, β) that contains no second root.

In the interval (α, β), $f'(x)$ is different from zero and $f''(x)$ is also different from zero.

As regards the latter assumption, that $f''(x)$ is different from zero in the interval, it is made only in order to obtain conditions for the application of the method that can be expressed more simply. In itself the usefulness of the method, given sufficient narrowing of the interval, does not depend on it. But if $f(x)$ and $f''(x)$ have no common divisor, and so do not vanish simultaneously, then by narrowing the interval one can always satisfy the Fourier assumption; and if $f(x)$ and $f''(x)$ do have a common divisor, one can first separate it off and then apply the approximation methods to the individual factors of $f(x)$.

One surveys the situation most simply in geometry, by interpreting $y = f(x)$ as the equation of a plane curve in a rectangular coordinate system.

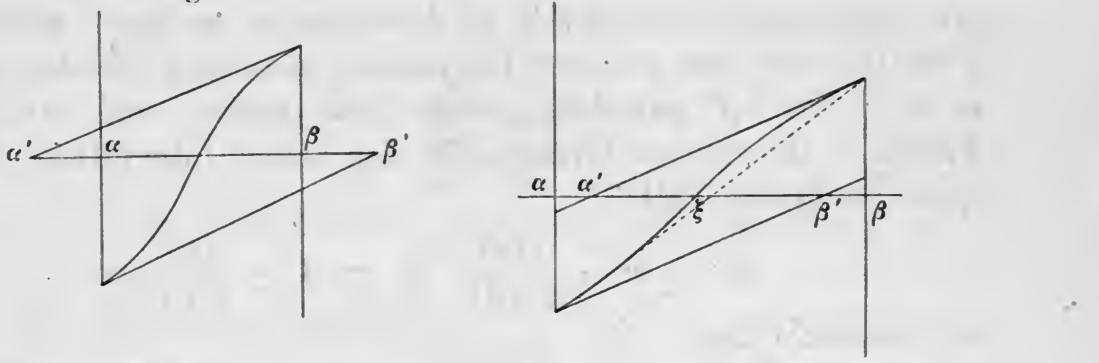
The equation

$$y = f(\alpha) + (x - \alpha) f'(\alpha) \tag{3}$$

is the equation of the tangent to the curve at the point $\alpha, f(\alpha)$

and the Newton approximation method therefore comes down to this, that in a first approximation one replaces the curve in the interval (α, β) by the tangent at one of the endpoints, instead of by its chord, as in the interpolation method.

Fig. 21.



If the second differential quotient vanishes in the interval (α, β), so that the curve has a point of inflection, then the case can occur that both end tangents lead out of the interval; in that case the Newton method is accordingly not ap-

Fig. 23.

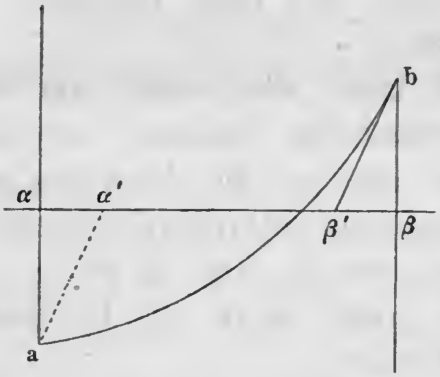
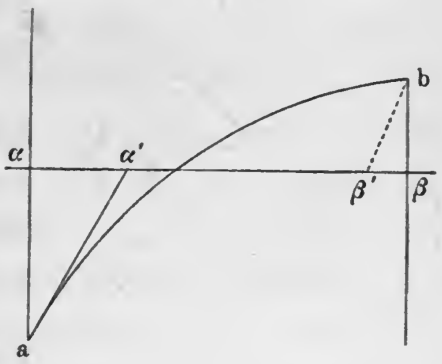


Fig. 24.



plicable (Fig. 21). But it can also happen, when the interval (α, β) has already been sufficiently narrowed, that the other case occurs, in which both tangents lead to interior points of the interval (Fig. 22). In these cases, however, the regula falsi will always give a better approximation.

But we shall now assume that $f'(x)$ and $f''(x)$ do not vanish in the interval (α, β). Then the curve does not change the sense of its curvature, and then certainly one of the two end tangents always has its point of intersection with the x-axis in the interior of the interval. This certainly holds when the tangent is taken at that endpoint of the interval,

in which $f(x)$ and $f''(x)$ have the same sign. The two Figs. 23 and 24 illustrate the situation for positive $f'(x)$ and positive and negative $f''(x)$.

Thus in the first case β', and in the second α', is a better approximate value than β and α.

If one wishes at the same time to shift the other limit, so as to obtain a new narrower interval, one may, following Fourier, draw from the other end point — that is, in the case of Fig. 23, at a — the straight line parallel to $b\beta'$, and one obtains the point α' as the lower limit point of the new interval. Then in the first case

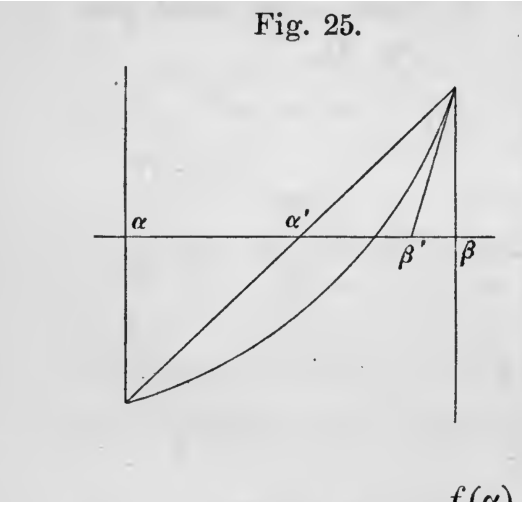
$$\alpha' = \alpha - \frac{f(\alpha)}{f'(\beta)}, \quad \beta' = \beta - \frac{f(\beta)}{f'(\beta)},$$

in the second case

$$\alpha' = \alpha - \frac{f(\alpha)}{f'(\alpha)}, \quad \beta' = \beta - \frac{f(\beta)}{f'(\alpha)},$$

and (α', β') is a narrower interval in which the sought root lies.

Fig. 25.



It is hardly necessary to consider separately the two other cases, in which $f'(x)$ is negative in the interval.

But one may also, in order to obtain two new limits, combine the Newton method with the interpolation method, with still better success, as Fig. 25 shows.

One then obtains as the two new limits:

$$\alpha' = \alpha - \frac{f(\alpha)(\beta-\alpha)}{f(\beta)-f(\alpha)} = \frac{\alpha f(\beta) - \beta f(\alpha)}{f(\beta)-f(\alpha)},$$
$$\beta' = \beta - \frac{f(\beta)}{f'(\beta)}.$$

We summarise the results in the following theorem:

If an interval (α, β) has been marked off, in which $f(x)$ changes its sign once, and $f'(x)$ and $f''(x)$ not at all, and if β is the end point of the interval in which $f(\beta)$ and $f''(\beta)$ have the same sign, no matter whether β is smaller or greater than

α, then one obtains a narrower interval (α', β') with the same properties, if one

(4) $$\alpha' = \alpha - \frac{f(\alpha)(\beta-\alpha)}{f(\beta)-f(\alpha)}, \quad \beta' = \beta - \frac{f(\beta)}{f'(\beta)}$$

sets.

To prove these results of geometric intuition analytically, we assume that in the interval $f'(x)$ and $f''(x)$ are positive, hence $\alpha < \beta$, and

$$f(\alpha) < 0, \quad f(\beta) > 0.$$

We restrict ourselves to the consideration of this one case, since the three others are to be treated in exactly the same way.

We form the function

$$\varphi(x) = \frac{f(\beta) - f(x)}{\beta - x}$$

and its first derivative

$$\varphi'(x) = \frac{-(\beta-x)f'(x) + f(\beta) - f(x)}{(\beta-x)^2}.$$

The numerator of this expression

$$\psi(x) = -(\beta-x)f'(x) + f(\beta) - f(x)$$

vanishes for $x = \beta$, and its derivative is

$$\psi'(x) = -(\beta - x)f''(x),$$

hence negative in the interval. From this it follows that $\psi(x)$ decreases in the interval as x increases, and therefore, because it vanishes for the greatest value $x = \beta$, remains positive. Consequently $\varphi'(x)$ is also positive, and $\varphi(x)$ increases steadily in the interval as x increases. We accordingly have

$$\frac{f(\beta) - f(\alpha)}{\beta - \alpha} < \frac{f(\beta) - f(x)}{\beta - x} < f'(\beta) \tag{5}$$
$$\alpha < x < \beta.$$

If we now set

$$\alpha' = \alpha - \frac{f(\alpha)(\beta - \alpha)}{f(\beta) - f(\alpha)}, \quad \beta' = \beta - \frac{f(\beta)}{f'(\beta)},$$

then

$$\beta' - \alpha' = f(\beta)\left\{\frac{\beta - \alpha}{f(\beta) - f(\alpha)} - \frac{1}{f'(\beta)}\right\}, \tag{6}$$

and from (5) it follows that this difference is positive, hence

$$\alpha < \alpha' < \beta' < \beta.$$

If one then sets in (5)

$$x = \alpha' \quad \text{and} \quad x = \beta'$$

and observes that

$$\beta - \alpha' = \frac{(\beta - \alpha)f(\beta)}{f(\beta) - f(\alpha)}, \quad \beta - \beta' = \frac{f(\beta)}{f'(\beta)},$$

it follows that

$$f(\alpha') < 0, \quad f(\beta') > 0.$$

The root sought is therefore contained between α' and β', and the interval (α', β') is smaller than (α, β).

If we put α', β' in place of α, β in these expressions, we obtain a new, still narrower interval, and so on.

As for the convergence of this procedure, we can assure ourselves of it in the following way. Following (6), we set:

$$\frac{\beta' - \alpha'}{\beta - \alpha} = \Phi(\alpha, \beta) = \frac{f(\beta)\{f(\alpha) - f(\beta) + (\beta - \alpha)f'(\beta)\}}{(\beta - \alpha)f'(\beta)\{f(\beta) - f(\alpha)\}}. \tag{7}$$

Numerator and denominator are here divisible by $(\beta - \alpha)^2$, and if we cancel the common factor $(\beta - \alpha)^2$ and put ξ, η for α, β, we obtain an expression of the form

$$\Phi(\xi, \eta) = \frac{\psi_1(\xi, \eta)}{\psi_2(\xi, \eta)},$$

in which ψ_1 and ψ_2 are integral rational functions of the two variables ξ, η.

If x is the root lying in the interval (α, β), then by our assumptions about $f(x)$ the two functions ψ_1 and ψ_2 do not become zero when

$$\alpha \leqq \xi < x, \quad x < \eta \leqq \beta. \tag{8}$$

The function $\Phi(\xi, \eta)$ is less than 1 for $\xi = \alpha, \eta = \beta$, but also for every other pair of values in the interval (α, β) for which $f(\xi)$ is negative and $f(\eta)$ positive, since every such interval (ξ, η) could be taken in place of (α, β).

Now since $\Phi(\xi,\eta)$ is a continuous function of the two variables ξ,η so long as these remain in the region (8), the function $\Phi(\xi,\eta)$ must have a maximum value in this region, and this must be less than 1, because $\Phi(\xi,\eta)$ remains less than 1 even in the limiting cases $\xi = x, \eta = x$.

A positive proper fraction Θ can therefore be assigned such that

$$\Phi(\xi,\eta) < \Theta$$

holds, so long as ξ,η belong to the domain (8). Then from (7) it follows that

$$\beta' - \alpha' < (\beta - \alpha)\Theta.$$

In the same way it follows, when we proceed in the same manner from the interval (α',β') to a narrower interval (α'',β''), that

$$\beta'' - \alpha'' < (\beta' - \alpha')\Theta',$$

in which Θ' has the same meaning for α',β' as Θ has for α,β. But since α',β' belong to the domain (8), Θ' is not greater than Θ, and Θ may also be put in place of Θ'. It follows then:

$$\beta'' - \alpha'' < (\beta - \alpha)\Theta^2,$$

and so we conclude further

$$\beta^{(\nu)} - \alpha^{(\nu)} < (\beta - \alpha)\Theta^{\nu}.$$

The intervals therefore decrease at least as strongly as the terms of a decreasing geometric progression.

As an example may serve the equation:

$$x^3 - 2x^2 - 2 = 0,$$

which has a root between

$$\alpha = 2{,}35 \quad \text{and} \quad \beta = 2{,}36$$

From the formulae (4) one obtains for

$$\beta' = 2{,}35931\ldots, \quad \alpha' = 2{,}359298\ldots,$$

so that a value of the root accurate in the fourth decimal is

$$2{,}3593$$

For this value itself, as a more exact calculation shows, $f(x)$ is still negative, so that it can be taken for α'. The next step of the approximation gives.

$$2{,}359304.$$

§. 110.

The approximation method of Daniel Bernoulli and related methods.

The method for the approximate solution of an equation which is due to Daniel Bernoulli rests on the fact that, if one has a sequence of real quantities, the powers of the greatest among them will outweigh the equally high powers of the rest the more, the higher the powers are.

If $\alpha,\beta,\gamma\ldots$ are arbitrary real or complex quantities, subject however to the condition that the absolute value of α is greater than the absolute value of all the others, so that the absolute values of the fractions $\beta:\alpha, \gamma:\alpha\ldots$ are proper fractions, then

$$\frac{\alpha^m + \beta^m + \gamma^m + \cdots}{\alpha^{m-1} + \beta^{m-1} + \gamma^{m-1} + \cdots} = \alpha\,\frac{1 + \left(\frac{\beta}{\alpha}\right)^m + \left(\frac{\gamma}{\alpha}\right)^m + \cdots}{1 + \left(\frac{\beta}{\alpha}\right)^{m-1} + \left(\frac{\gamma}{\alpha}\right)^{m-1} + \cdots}, \tag{1}$$

and the greater m becomes, the more this expression will approach the limit α, as the second representation shows.

If $\alpha, \beta, \gamma \ldots$ are the roots of an algebraic equation, then the left side of (1) is the quotient of the mth and the $(m-1)$th power sum, and we thus obtain the theorem:

The quotient of the mth and the $(m-1)$th power sum approaches, as m increases, the absolutely greatest among the roots.

If one takes m negative, and supposes α absolutely smaller than $\beta, \gamma \ldots$, it follows in the same way:

The quotient of the $-m$th and the $-(m+1)$th power sum approaches, as m increases, the absolutely smallest among the roots.

Since the power sums can be computed as symmetric functions from the coefficients, one need only take a sufficiently large m in order to obtain an approximate value of the absolutely greatest and the absolutely smallest root.

But if among the roots of the equation there occur some which have the same absolute value, and this is the greatest or the smallest, then this method is not applicable — thus, for example, not to real equations in which a pair of imaginary roots occurs that surpasses the real ones in absolute value.

But if the greatest root, though it occurs singly, does not much surpass the next following one, then the procedure will yield some accuracy only slowly.

Jacobi has supplemented the Bernoulli method in one direction[25].

Let us assume that among the roots $x_1, x_2 \ldots x_n$ there are

$$x_1, x_2 \ldots x_k$$

whose absolute values are greater than the absolute values of the following $x_{k+1}, x_{k+2} \ldots x_n$, and let us set

$$\begin{array}{rcl} p_m & = & x_1^m + x_2^m + \cdots + x_k^m \\ p_{m+1} & = & x_1^{m+1} + x_2^{m+1} + \cdots + x_k^{m+1} \\ \cdots\cdots & \cdots\cdots & \cdots\cdots \\ p_{m+k} & = & x_1^{m+k} + x_2^{m+k} + \cdots + x_k^{m+k}, \end{array} \tag{2}$$

then, if we set

$$(x - x_1)(x - x_2) \ldots (x - x_k) = x^k + A_1 x^{k-1} + \cdots + A_k \tag{3}$$

the coefficients $A_1, A_2 \ldots A_k$ will satisfy the equations

$$\begin{array}{rcl} p_{m+k} + p_{m+k-1}A_1 + p_{m+k-2}A_2 + \cdots + p_m A_k & = & 0 \\ p_{m+k+1} + p_{m+k}A_1 + p_{m+k-1}A_2 + \cdots + p_{m+1}A_k & = & 0 \\ \cdots\cdots & \cdots\cdots & \cdots\cdots \\ p_{m+2k-1} + p_{m+2k-2}A_1 + p_{m+2k-3}A_2 + \cdots + p_{m+k-1}A_k & = & 0. \end{array} \tag{4}$$

If one adjoins to the equations (4) the further equation

$$x^k + A_1 x^{k-1} + A_2 x^{k-2} + \cdots + A_k = 0$$

[25] Observatiunculae etc. Werke, Bd. 3, S. 280.

and eliminates the $A_1, A_2 \dots A_k$, there results the equation of degree k

$$\begin{vmatrix} x^k, & x^{k-1}, & \cdots & 1 \\ p_{m+k}, & p_{m+k-1}, & \cdots & p_m \\ p_{m+k+1}, & p_{m+k}, & \cdots & p_{m+1} \\ \cdots & \cdots & \cdots & \cdots \\ p_{m+2k-1}, & p_{m+2k-2}, & \cdots & p_{m+k-1} \end{vmatrix} = 0,$$

whose roots are $x_1, x_2 \dots x_k$. The quantities $p_m, p_{m+1} \dots$ can, however, be replaced by the corresponding power sums of all the roots, and with all the greater accuracy the greater m is.

Thus one obtains, for example, for $k = 2$:

$$x^2 + \frac{p_m p_{m+3} - p_{m+1} p_{m+2}}{p_{m+1}^2 - p_m p_{m+2}}\, x + \frac{p_{m+2}^2 - p_{m+1} p_{m+3}}{p_{m+1}^2 - p_m p_{m+2}} = 0,$$

an equation which one can apply when, in a real equation, a pair of conjugate roots have the absolutely greatest value.

If one applies the Bernoulli method to the sum of the negative powers, one obtains, as has already been

marked, an approximation to the absolutely smallest root. But by shifting the origin one can make any root the absolutely smallest, though for this, admittedly, the knowledge of a value approximated to a certain degree is necessary.

A formula that suffices for all cases has been given by Fr. Meyer[26]. Let $\alpha_1, \alpha_2, \alpha_3 \dots \alpha_n$ be the roots of the equation $f(x) = 0$, which may be real or imaginary. One seeks a point p in the x-plane such that a circle can be described about p as centre which contains only one of the points $\alpha_1, \alpha_2 \dots \alpha_n$, say the point α_1, so that the absolute values of

$$\frac{p-\alpha_1}{p-\alpha_2}, \quad \frac{p-\alpha_1}{p-\alpha_3}, \quad \cdots \quad \frac{p-\alpha_1}{p-\alpha_n} \tag{5}$$

are proper fractions. Such points always exist; if need be, they must be found by the Sturm method. If, in addition, one chooses an arbitrary function $\varphi(x)$ that has no common divisor with $f(x)$, e.g. $\varphi(x) = 1$, and forms the symmetric functions

$$y_m = \frac{\dfrac{\alpha_1\varphi(\alpha_1)}{(p-\alpha_1)^m} + \dfrac{\alpha_2\varphi(\alpha_2)}{(p-\alpha_2)^m} + \cdots + \dfrac{\alpha_n\varphi(\alpha_n)}{(p-\alpha_n)^m}}{\dfrac{\varphi(\alpha_1)}{(p-\alpha_1)^m} + \dfrac{\varphi(\alpha_2)}{(p-\alpha_2)^m} + \cdots + \dfrac{\varphi(\alpha_n)}{(p-\alpha_n)^m}}, \tag{6}$$

then with increasing m these approach the limit α_1, and the more rapidly, the smaller the absolute values of the fractions (5) already are.

The correctness of this shows itself at once if one writes the formula (6) in the following way:

$$y_m = \frac{\alpha_1\varphi(\alpha_1) + \alpha_2\varphi(\alpha_2)\left(\dfrac{p-\alpha_1}{p-\alpha_2}\right)^m + \cdots + \alpha_m\varphi(\alpha_m)\left(\dfrac{p-\alpha_1}{p-\alpha_n}\right)^m}{\varphi(\alpha_1) + \varphi(\alpha_2)\left(\dfrac{p-\alpha_1}{p-\alpha_2}\right)^m + \cdots + \varphi(\alpha_m)\left(\dfrac{p-\alpha_1}{p-\alpha_n}\right)^m}.$$

§. 111.
The approximation method of Gräffe.

On an idea similar to that of the Bernoulli approximation there also rests a procedure that Gräffe

[26] Mathematische Annalen, Bd. 33.

given, and further developed by Encke[27], and which lends itself particularly well to a practical execution of the numerical computations.

If, as above, $\alpha, \beta, \gamma \ldots$ are the roots of an equation, and α is the absolutely greatest among them, so that none of the others equals the root α in absolute value, then

$$\sqrt[m]{\alpha^m + \beta^m + \gamma^m + \cdots} \tag{1}$$

has, as m grows without bound, the value α as its limit; indeed the convergence toward α is even better than for the quotient considered in the preceding section,

$$\frac{\alpha^m + \beta^m + \gamma^m + \cdots}{\alpha^{m-1} + \beta^{m-1} + \gamma^{m-1} + \ldots},$$

as one recognises when, according to the generalised binomial theorem, one sets:

$$\begin{aligned}\sqrt[m]{\alpha^m + \beta^m + \gamma^m + \cdots} &= \alpha \sqrt[m]{1 + \left(\frac{\beta}{\alpha}\right)^m + \left(\frac{\gamma}{\alpha}\right)^m + \cdots} \\ &= \alpha \left[1 + \frac{1}{m}\left(\frac{\beta}{\alpha}\right)^m + \frac{1}{m}\left(\frac{\gamma}{\alpha}\right)^m + \cdots\right]\end{aligned}$$

The absolutely smallest among the roots is obtained by the same principle as the limit of

$$\frac{1}{\sqrt[m]{\frac{1}{\alpha^m} + \frac{1}{\beta^m} + \frac{1}{\gamma^m} + \cdots}}, \tag{2}$$

and if p is an arbitrary value, one obtains the root lying nearest to the value p as the limit of

$$p + \frac{1}{\sqrt[m]{\frac{1}{(\alpha - p)^m} + \frac{1}{(\beta - p)^m} + \cdots}}. \tag{3}$$

If one transforms the equation $f(x)$ by the substitution

$$y = \frac{1}{x}, \qquad y = \frac{1}{x - p}$$

then the expressions (2), (3) arise from the expression (1), to which we shall therefore now confine ourselves.

If one applies the formula (1) to a real equation, then $\alpha^m + \beta^m + \gamma^m + \cdots$ is a real number; and if the absolutely greatest root α is real, then the mth root is also to be taken real. But if m is an even number, then, in order to decide about the sign, one must further know whether α is positive or negative. If necessary, this is decided by substituting the approximate value found into the given equation.

Likewise, if one applies the formula (1) to an imaginary equation, one will have to decide which of the various mth roots gives the correct approximation.

The expressions

$$s_m = \alpha^m + \beta^m + \gamma^m + \cdots$$

can be computed especially simply when one takes for m the successive powers of 2, that is $m = 2, 4, 8, 16 \ldots$, and they then yield very good results.

Now s_2 is the negative coefficient of the $(n-1)$th power of the unknown in the equation whose roots are the squares of the roots of $f(x)$. This equation, however, is easily formed in the following way.

[27] Crelle's Journal, Bd. 22 (1841).

In $f(x)$ collect the terms with even and with odd powers of x, and accordingly set

$$f(x) = \varphi(x^2) + x\,\psi(x^2).$$

Then

$$f_1(x) = \varphi(x)^2 - x\,\psi(x)^2 = 0$$

is the equation whose roots are the squares of the roots of $f(x)$. If one treats $f_1(x)$ in the same way, one obtains an equation whose roots are the fourth powers of the roots of $f(x)$, and so on.

The carrying out of this computation is very simple, and for the most part leads after a few steps to a good approximation.

We consider some examples.

First we take the equation

$$x^3 - 2x - 2 = 0, \tag{4}$$

which has one real positive and two imaginary roots. The real root is greater than 1,7, and since the product of all three roots is equal to 2 and $(1,7)^3 > 2$, the absolute value of the two imaginary roots is smaller than the real root. Hence the real root is the absolutely greatest, and the Gräffe-

method of approximation must lead to it. One now obtains the equations whose roots are the second, fourth, eighth ... power of equation (4):

$$\begin{aligned} x^3 - 4x^2 + 4x - 4 &= 0 \\ x^3 - 8x^2 - 16x - 16 &= 0 \\ x^3 - 96x^2 - 16^2 &= 0 \end{aligned}$$

and

$$x = \sqrt[8]{96} = 1{,}7692\ldots$$

is already a value accurate in the first four decimals. The next step would yield no different result (because the first power of the unknown is missing in the last equation); but the step after that yields the value, accurate in the sixth decimal,

$$x = \sqrt[32]{85032960} = 1{,}769293\ldots$$

We shall consider yet a second example of an equation of the fifth degree, one which offers the opportunity to apply several of the theorems of the preceding section. The example is taken, like the earlier ones, from the theory of the complex multiplication of the elliptic functions. It is the equation of the fifth degree

$$x^5 - x^3 - 2x^2 - 2x - 1 = 0. \tag{5}$$

From this we derive the equation whose roots are the squares of the roots of (5), by substituting, in

$$(x^5 - x^3 - 2x)^2 - (2x^2 - 1)^2 = 0$$

x for x^2. This gives

$$x^5 - 2x^4 - 3x^3 - 1 = 0, \tag{6}$$

and if we now apply the same procedure twice more, we obtain the equations whose roots are the fourth and eighth powers of the roots of (5):

$$x^5 - 10x^4 + 9x^3 - 4x^2 - 1 = 0, \tag{7}$$

$$x^5 - 82x^4 + x^3 - 36x^2 - 8x - 1 = 0. \tag{8}$$

The last of these equations lends itself to discussion. By the Cartesian theorem (§. 101) it has at least one positive root.

If, for equation (8), we form the functions occurring in the Newton criterion (§. 101)

$$1, \quad 4a_1^2 - 10a_0a_2, \quad 6a_2^2 - 12a_1a_3$$
$$6a_3^2 - 12a_2a_4, \quad 4a_4^2 - 10a_3a_5, \quad 1$$

we obtain the signs

$$+ \quad + \quad - \quad + \quad - \quad +$$

and (8) therefore has, by §. 101, Theorem VIII, four imaginary roots. The positive root is in any case greater than 1, since the left-hand side of (8) is negative for $x = 0$ and $x = 1$.

In order now to decide which of the roots has the greatest absolute value, we shall set $x = 1 : z$ in (8) and then investigate, according to §. 96, 97, how many roots lie in the interior of the unit circle in the z-plane. If, then, in

$$z^5 + 8z^4 + 36z^3 - z^2 + 82z - 1$$

we set $z = \cos\vartheta + i\sin\vartheta$, we obtain the real and imaginary parts:

$$\varphi = \cos 5\vartheta + 8\cos 4\vartheta + 36\cos 3\vartheta - \cos 2\vartheta + 82\cos\vartheta - 1$$

$$\psi = \sin 5\vartheta + 8\sin 4\vartheta + 36\sin 3\vartheta - \sin 2\vartheta + 82\sin\vartheta.$$

It can now be shown that ψ is positive when $0 < \vartheta < \pi$, and consequently negative when $0 > \vartheta > -\pi$, so that the unit circle is divided into two segments in which ψ has opposite signs. From this it follows, by §. 96, that only one root can lie in the interior of the circle, and this is the real root, while the imaginary roots lie outside. It then follows from this, for the equation (8), that the real root has the greatest absolute value, and that it is therefore found by the Gräffe method.

In order now to establish this property of the function ψ, we set $2\cos\vartheta = \xi$, and apply the goniometric formulae:

$$\sin 5\vartheta = \sin\vartheta(\xi^4 - 3\xi^2 + 1)$$
$$\sin 4\vartheta = \sin\vartheta(\xi^3 - 2\xi)$$
$$\sin 3\vartheta = \sin\vartheta(\xi^2 - 1)$$
$$\sin 2\vartheta = \sin\vartheta\,\xi.$$

By this means we obtain

$$\psi = \sin\vartheta[\xi^4 + \xi^2(8\xi + 33) + (47 - 17\xi)],$$

from which one sees at once that the factor of $\sin\vartheta$ remains positive so long as ξ lies between -2 and $+2$.

Equation (8) itself now already gives a fairly good approximate value for the real root

$$x = \sqrt[8]{82} = 1{,}73471.$$

A more accurate value, which one obtains by going two steps further, is 1,73469, which is still correct in the fifth decimal place.

This method does not give an enclosure of the sought root between two limits. The criterion for whether a sufficient degree of accuracy has been reached consists in this: that if s_μ, $s_{\mu'}$ are two successive power sums of the roots used in the computation, these are related, with sufficient accuracy, as the μth to the μ'th power of a quantity, or that approximately

$$\mu \log s_{\mu'} - \mu' \log s_\mu = 0$$

holds. One must, however, carry out this test on several successive terms, since in special cases this relation may appear to be satisfied and then cease to hold again in the later computation.

§. 112.
Trigonometric Solution of Cubic Equations.

We now discuss a few further methods of numerical solution applicable to equations of special forms. The aim of these methods is to make the widely available tables of the trigonometric functions and of logarithms useful for the solution of equations. We turn first to the consideration of the cubic equations, which we always assume in the reduced form

$$x^3 + ax + b = 0 \tag{1}$$

in which a and b are real numbers. Since the interchange of x with $-x$ is equivalent to the interchange of b with $-b$, we may restrict ourselves to the assumption that b is negative, and there then remain three different cases to consider. We set $b = -g$ and, according as a is positive or negative, $a = \pm e$. Then we have

$$\begin{aligned} &\text{a)} \quad x^3 + ex - g = 0 \\ &\text{b)} \quad x^3 - ex - g = 0, \quad \frac{e^3}{27} < \frac{g^2}{4} \\ &\text{c)} \quad x^3 - ex - g = 0, \quad \frac{e^3}{27} > \frac{g^2}{4}. \end{aligned}$$

We exclude the limiting cases in which one of the quantities e, g, $4e^3 - 27g^2$ vanishes.

To apply the Cardan formula (§. 35), we set

$$R = \frac{g^2}{4} \pm \frac{e^3}{27}, \tag{2}$$

where the upper sign holds in case a), the lower in cases b) and c). In cases a) and b) only one root is real; in case c) (the Casus irreducibilis) all three roots are real. The Cardan formula gives

$$x = \sqrt[3]{\frac{g}{2} + \sqrt{R}} + \sqrt[3]{\frac{g}{2} - \sqrt{R}}. \tag{3}$$

We now proceed in the three cases as follows:

a) We introduce an angle ϑ, which we can choose between 0^0 and 90^0, by the equation:

$$\frac{g}{2} = \sqrt{\frac{e^3}{27}} \operatorname{cotg} \vartheta, \tag{4}$$

hence

$$\sqrt{R} = \sqrt{\frac{e^3}{27}} \frac{1}{\sin \vartheta},$$

$$\begin{aligned} x &= \sqrt{\frac{e}{3}} \left(\sqrt[3]{\operatorname{cotg} \vartheta + \frac{1}{\sin \vartheta}} + \sqrt[3]{\operatorname{cotg} \vartheta - \frac{1}{\sin \vartheta}} \right) \\ &= \sqrt{\frac{e}{3}} \left(\sqrt[3]{\operatorname{cotg} \frac{\vartheta}{2}} - \sqrt[3]{\operatorname{tg} \frac{\vartheta}{2}} \right), \end{aligned}$$

and if we further determine an angle φ from

$$\operatorname{tg}\frac{\vartheta}{2} = \operatorname{tg}\varphi^3 \tag{5}$$

then we obtain

$$x = 2\sqrt{\frac{e}{3}}\operatorname{cotg} 2\varphi. \tag{6}$$

The angles ϑ, φ and finally x are found from the logarithmic-trigonometric tables by the formulas (4), (5), (6).

To obtain the imaginary roots, one replaces $\operatorname{tg}\varphi$ by $\varrho\operatorname{tg}\varphi$, where ϱ denotes an imaginary third root of unity.

b) In the second case we determine the angle ϑ, likewise in the first quadrant, from

$$\frac{g}{2} = \sqrt{\frac{e^3}{27}}\frac{1}{\sin\vartheta} \tag{7}$$

$$\sqrt{R} = \sqrt{\frac{e^3}{27}}\operatorname{cotg}\vartheta$$

$$x = \sqrt{\frac{e}{3}}\left(\sqrt[3]{\operatorname{cotg}\frac{\vartheta}{2}} + \sqrt[3]{\operatorname{tg}\frac{\vartheta}{2}}\right),$$

$$\operatorname{tg}\frac{\vartheta}{2} = \operatorname{tg}\varphi^3, \tag{8}$$

$$x = 2\sqrt{\frac{e}{3}}\frac{1}{\sin 2\varphi}. \tag{9}$$

c) In the last case, finally, where three real roots are present, we set

$$\frac{g}{2} = \sqrt{\frac{e^3}{27}}\cos\vartheta \tag{10}$$

$$\sqrt{R} = i\sqrt{\frac{e^3}{27}}\sin\vartheta$$

$$x = \sqrt{\frac{e}{3}}\left(\sqrt[3]{\cos\vartheta + i\sin\vartheta} + \sqrt[3]{\cos\vartheta - i\sin\vartheta}\right),$$

or, by the theorem of Moivre:

$$x = 2\sqrt{\frac{e}{3}}\cos\frac{1}{3}\vartheta. \tag{11}$$

If one takes ϑ again in the first quadrant, one obtains for the two other, likewise real, roots

$$-2\sqrt{\frac{e}{3}}\cos\frac{\pi+\vartheta}{3}, \qquad -2\sqrt{\frac{e}{3}}\cos\frac{\pi-\vartheta}{3}.$$

All these formulae are arranged for logarithmic computation.

§. 113.
Gauss's method for the solution of trinomial equations.

Gauss has given a method for solving numerically, in a simple way, the roots of an equation that contains only three terms. Such equations occur frequently and include as special cases all quadratic equations and the reduced cubic equations. Of such an equation, whose general form is

$$x^{m+n} + ax^m + b = 0 \tag{1}$$

only the positive root, if it exists, is sought at first. Any negative root is obtained by replacing x by $-x$.

According to the signs of a, b one has three cases to distinguish, since, when both signs are positive, no positive root is present. Denoting by e and g positive numbers, we therefore consider the three cases:

$$\begin{aligned} &\text{a)} \quad x^{m+n} + ex^m - g = 0, \\ &\text{b)} \quad x^{m+n} - ex^m - g = 0, \\ &\text{c)} \quad x^{m+n} - ex^m + g = 0. \end{aligned}$$

In the first two cases we have, by the Cartesian theorem, one positive root each; in the third, two positive roots or none may be present.

Following Gauss, we now set

$$\lambda = \frac{g^n}{e^{m+n}},$$

and seek to reduce the three equations a), b), c) by suitable substitutions to the form:

$$\sin^2 \Theta + \cos^2 \Theta = 1$$

We set:

$$\begin{aligned} &\text{a)} \quad \frac{x^{m+n}}{g} = \sin \Theta^2, \quad \frac{ex^m}{g} = \cos \Theta^2, \quad \lambda = \frac{\sin \Theta^{2m}}{\cos \Theta^{2m+2n}}, \\ &\text{b)} \quad gx^{-m-n} = \sin \Theta^2, \quad ex^{-n} = \cos \Theta^2, \quad \lambda = \frac{\sin \Theta^{2n}}{\cos \Theta^{2m+2n}}, \\ &\text{c)} \quad \frac{x^n}{e} = \sin \Theta^2, \quad \frac{gx^{-m}}{e} = \cos \Theta^2, \quad \lambda = \sin \Theta^{2m} \cos \Theta^{2n}. \end{aligned} \tag{2}$$

The last equation yields the distinction between the two cases in which the equation c) has two positive roots or none.

For by the rules of the differential calculus one obtains for the maximum of $\sin \Theta^{2m} \cos \Theta^{2n}$

$$\frac{m^m n^n}{(m+n)^{m+n}},$$

which is attained for $\operatorname{tg} \Theta^2 = m : n$. If therefore λ lies below this bound, we have two real roots in c), otherwise none. In the formulae (2), λ is a given quantity, and one now has to seek from the tables the angle Θ that satisfies these equations. If one as yet has no knowledge at all of the approximate position of this angle, it is expedient, for the first approximation, to use a table proceeding say from degree to degree and abridged to two decimal places, in order then to improve the value so found with the help of more exact tables.

Gauss uses not the trigonometric tables, but the addition and subtraction logarithms first introduced by him. We shall show this briefly in one of the cases. The details for the practical application of the method are to be sought in Gauss's memoir [1]).

The tables of addition and subtraction logarithms, as they were first introduced and computed by Gauss, and as they are now also to be found in the tables in common use, give, for three numbers which are greater than 1:

$$a, \quad b = 1 + \frac{1}{a}, \quad c = 1 + a,$$

the Briggian logarithms:

$$A, \quad B, \quad C.$$

If Θ is an angle in the first octant, one may set:

$$(3) \quad a = \operatorname{cotg} \Theta^2, \quad b = \frac{1}{\cos \Theta^2}, \quad c = \frac{1}{\sin \Theta^2}, \quad 0 < \Theta < 45^0,$$

and if Θ lies in the second octant:

$$(4) \quad a = \operatorname{tg} \Theta^2, \quad b = \frac{1}{\sin \Theta^2}, \quad c = \frac{1}{\cos \Theta^2}, \quad 45^0 < \Theta < 90^0.$$

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[1]) Beiträge zur Theorie der algebraischen Gleichungen, second part (1849). Gauss Werke, Vol. III, p. 85.

We wish to apply this to the case b); here we must distinguish whether λ is smaller or greater than 2^m, because on this depends whether Θ lies in the first or in the second octant. So there are again two cases to distinguish:

$$\alpha) \quad \lambda < 2^m,$$

$$\lambda = \frac{b^m}{a^n} = \frac{c^m}{a^{m+n}} = \frac{b^{m+n}}{c^n},$$

$$(5) \quad x^{m+n} = gc, \quad x^n = eb, \quad x^m = \frac{ga}{e}.$$

$$\beta) \quad \lambda > 2^m,$$

$$\lambda = a^{m+n} b^m = a^n c^m = \frac{c^{m+n}}{b^n},$$

$$x^{m+n} = gb, \quad x^n = ec, \quad x^m = \frac{g}{ea}.$$

In the case α) one would therefore set

$$(6) \quad \log \lambda = mB - nA$$

and then look up in the table the corresponding values of A and B. Once these have been found, we obtain

$$(7) \quad m \log x = A + \log g - \log e.$$

To illustrate the use of these formulas by an example, we shall consider the equation

$$x^3 - 2x - 2 = 0,$$

and thus set $e = g = 2$, $m = 1$, $n = 2$, $\lambda = \frac{1}{2}$. The formula (6) therefore gives

$$(8) \quad 2A - B = 0{,}3010300$$

and (7)

$$(9) \quad \log x = A.$$

To find the first approximate value of A, one makes use of a small extract from the table:

$$\left|\begin{array}{ll} A=0, & B=0{,}301 \\ A=0{,}1, & B=0{,}254 \\ A=0{,}2, & B=0{,}212 \\ A=0{,}3, & B=0{,}176 \end{array}\right|.$$

From this we obtain:

A	$2A-B$	Error
0,2	0,188	−0,113
0,3	0,424	+0,123

One can now apply the method of interpolation to this in order to obtain a more accurate value of A, by distributing the total error of 0,236 over the two values of A in proportion to the partial errors, that is, by putting

$$A = 0{,}2 + \frac{0{,}1}{0{,}236}\,0{,}113 = 0{,}247\ldots$$

With the help of a seven-place table of Zach one now obtains:

A	B	$2A-B$	Error
0,247	0,1948581	0,2991419	−0,0018881
0,248	0,1944969	0,3015031	+0,0004731

and by a further application of the interpolation

$$A = 0{,}2477996$$

or

$$x = 1{,}769292.$$

§. 114.

Computation of the imaginary roots of a trinomial equation.

Gauss has also given a procedure for the computation of the imaginary roots of a trinomial equation, which we shall briefly discuss here. We make the assumption of real coefficients, although the method is applicable to the general case as well. We shall put the equation in the form:

$$(1)\quad x^{m+n} - ex^m - g = 0,$$

but here we need not restrict ourselves to positive values of e and g.

We set

$$x = re^{i\vartheta}$$

and, by setting the imaginary part in (1) equal to zero by itself, we obtain

$$(2)\quad r^n = \frac{e\sin m\vartheta}{\sin(n+m)\vartheta};$$

of such equations, however, we obtain two more by multiplying (1) by x^{-m} and by x^{-m-n}. This gives

$$(3)\quad r^{n+m} = \frac{-g\sin m\vartheta}{\sin n\vartheta},$$

$$(4)\quad r^m = \frac{-g\sin(n+m)\vartheta}{e\sin n\vartheta},$$

and of these three equations each follows from the other two. If one eliminates r from two of them, there results, if, as above,

$$\lambda = \frac{g^n}{e^{n+m}}$$

is set,

$$(5) \quad \lambda = (-1)^n \frac{\sin n\vartheta^n \sin m\vartheta^m}{\sin(n+m)\vartheta^{n+m}}.$$

Since one need compute only one of the two conjugate roots, one may restrict ϑ to the first quadrant, but must then in some circumstances also take r negative.

How one computes from formula (5), by means of the trigonometric tables, the angle ϑ, and then from one of the three formulas (2), (3), (4) the corresponding value of r, we shall now demonstrate on the example considered above

$$x^3 - 2x - 2 = 0$$

as well.

Here equation (5) becomes

$$(6) \quad \frac{1}{2} = \frac{\sin 2\vartheta^2 \sin \vartheta}{\sin 3\vartheta^3},$$

or

$$(7) \quad 3\log\sin 3\vartheta - 2\log\sin 2\vartheta - \log\sin\vartheta = 0{,}3010300,$$

from which it is first of all evident that the angle ϑ lies in the interval from 0 to 60^0. After a few trials one finds first that ϑ lies between 33^0 and 34^0, and if one then computes for these two values the difference

$$(8) \quad 3\log\sin 3\vartheta - 2\log\sin 2\vartheta - \log\sin\vartheta - 0{,}3010300$$

one obtains first, to three decimals,

$$0{,}026, \quad -0{,}013,$$

and from this, by interpolation, there results the more exact value

$$\vartheta = 33^0\, 40'.$$

One next computes the difference (8) with somewhat greater accuracy, say to five places, for a few angles in the neighbourhood of the values $30^0\, 40'$, proceeding from minute to minute, and finds from the change of sign that ϑ lies between

$$33^0\, 41' \text{ and } 33^0\, 42'$$

If one now carries out the computation to seven places for these two values, there results again, by interpolation, the more exact value

$$\vartheta = 33^0 \quad 41' \quad 20{,}6''.$$

From the formulae (3) or (4) one sees that r is negative here, and one obtains r very simply from one of these equations. One finds the Briggian logarithms:

$$\log(-r) = 0{,}0266148$$
$$\log\cos\vartheta = 9{,}9201547 - 10$$
$$\log\sin\vartheta = 9{,}7440458 - 10.$$

The two imaginary roots are therefore:

$$-0{,}884646 \pm 0{,}589740\, i.$$

Eleventh Section.

Continued Fractions.

§. 115.

Transformation of rational fractions into continued fractions.

If one divides a natural number m by another n according to the ordinary rules, one obtains a quotient and a remainder, which is positive and smaller than the divisor n. If a denotes the quotient and r the remainder, then

$$m = an + r. \tag{1}$$

All numbers $m, m', m'' \ldots$ which give the same remainder are called of equal residue or congruent with respect to the modulus n, and following Gauss one expresses this by the sign:

$$m \equiv m' \pmod{n}. \tag{2}$$

Such a formula is called a congruence.

If, then, we consider the system of numbers

$$r = 0, 1, 2 \ldots n-1$$

every arbitrary (positive or negative) integer m is congruent to one and only one of these numbers with respect to the modulus n. But the same property is also possessed by every system of numbers

$$s = m_0, m_1, m_2 \ldots m_{n-1},$$

chosen in such a way that among their remainders all the numbers r occur (and each only once). Then every arbitrary number m is, with respect to the modulus n, congruent to one and only one of the numbers s

congruent. Such a system of numbers we shall call a complete system of residues for the modulus n. Such a complete system of residues is formed, for example, when n is odd, by the numbers

$$\varrho = \frac{-n+1}{2}, \ \frac{-n+3}{2} \cdots -1, 0, 1, 2 \cdots \frac{n-1}{2}.$$

The numbers r are called the least (or least positive) residues, the numbers ϱ the absolutely least residues.

Now, in order to continue the division indicated in (1), we shall denote m, n, r by m, m_1, m_2, and then in turn the quotient and the remainder of the division of m_1 by m_2 by a_1, m_3, and so forth. If, moreover, we also write a_0 for a, there results a series of equations:

$$\begin{aligned} m &= a_0\, m_1 + m_2 \\ m_1 &= a_1\, m_2 + m_3 \\ m_2 &= a_2\, m_3 + m_4 \\ &\cdots\cdots\cdots\cdots \\ m_{\nu-1} &= a_{\nu-1}\, m_\nu + m_{\nu+1}, \end{aligned} \tag{3}$$

which can be continued so long as none of the remainders vanishes, that is, so long as none of the divisions is exact. But because

$$m_1 > m_2 > m_3 > \cdots$$

it follows, since there is only a finite number of positive numbers smaller than m_1, that after a finite number of divisions the division must be exact. If m_ν is divisible by $m_{\nu+1}$, then $m_{\nu+1}$ is the greatest common divisor of m and m_1.

For $m_{\nu+1}$ is then, by the equations (3), a divisor of all the preceding m_μ, and conversely every common divisor of two adjacent m_μ is also a divisor of all the following ones, and hence also of $m_{\nu+1}$.

If therefore m and m_1 have no common divisor other than 1 (that is, if they are coprime or relatively prime), then we must have $m_{\nu+1} = 1$, and we can represent the equations (3) as follows:

$$\frac{m}{m_1} = a_0 + \frac{m_2}{m_1}, \quad \frac{m_1}{m_2} = a_1 + \frac{m_3}{m_2}, \cdots \frac{m_{\nu-1}}{m_\nu} = a_{\nu-1} + \frac{1}{m_\nu}. \tag{4}$$

This gives the development of the fraction $m : m_1$ or $m : n$ into a continued fraction. We obtain successively

$$\frac{m}{n} = a_0 + \frac{1}{\frac{m_1}{m_2}}, \quad \frac{m}{n} = a_0 + \frac{1}{a_1 + \frac{1}{\frac{m_2}{m_3}}} \cdots , \tag{5}$$

and finally

$$\frac{m}{n} = a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cdots + \cfrac{1}{a_{\nu-1} + \cfrac{1}{m_\nu}}}}, \tag{6}$$

for which, for the sake of more convenient printing, we also

$$\begin{gathered} \frac{m}{n} = \left(a_0, \frac{m_1}{m_2}\right), \quad \frac{m}{n} = \left(a_0, a_1, \frac{m_2}{m_3}\right) \\ \frac{m}{n} = (a_0, a_1, a_2 \ldots a_{\nu-1}, m_\nu) \end{gathered} \tag{7}$$

write.

To this the following remark must still be added: every positive rational fraction can be put into the form $m : n$ in such a way that m and n are relatively prime, and hence we can develop every such fraction into a continued fraction of the form (6) or (7). This also holds for negative rational fractions, if we admit negative values for the number a_0 as well. The $a_1, a_2 \ldots a_{\nu-1}, m_\nu$ are, however, positive integers even then. These numbers are completely and unambiguously determined. Only in one respect is an arbitrariness still open to us.

Namely, according to the determination made up to this point, the last of the partial denominators m_ν is greater than 1, since we have assumed that $m_{\nu+1}$ was to be the first among the residues that is equal to 1. Accordingly, if a_ν is a positive integer, we can set either

$$m_\nu = a_\nu,$$

or

$$m_\nu = a_\nu + \frac{1}{1}$$

and therefore obtain the two representations by continued fractions:

$$\frac{m}{n} = (a_0, a_1, a_2 \ldots a_\nu) = (a_0, a_1, a_2 \ldots a_\nu, 1). \tag{8}$$

I. Thus every rational fraction can be developed into a continued fraction which has, at pleasure, an even or an odd number of partial denominators.

The numbers occurring in the formulae (5),

$$\frac{m_1}{m_2},\ \frac{m_2}{m_3},\ \frac{m_3}{m_4}\cdots,$$

we call the closing numbers of the continued fraction; the numbers $a_0, a_1, a_2 \ldots$ shall be called the partial denominators (although a_0 does not properly occur as a denominator).

If we replace m by a number m' congruent to it modulo n, the two fractions $m : n$ and $m' : n$ differ only by an integer; the continued fraction developments therefore differ only in their first terms a_0.

§. 116.
Continued fraction development of irrational numbers.

If x is a real irrational number, there will always be one and only one integer a_0 such that x lies between a_0 and $a_0 + 1$. a_0 is positive when x is greater than 1, zero when x is a positive proper fraction, and negative when x is negative. If we therefore set

$$x = a_0 + \frac{1}{x_1},$$

then x_1 is a positive improper fraction; hence, if a_1 is the integer lying next below x_1, a_1 is positive. We set

$$x_1 = a_1 + \frac{1}{x_2},$$
$$x_2 = a_2 + \frac{1}{x_3},$$
$$\cdots\cdots\cdots$$
$$x_{n-1} = a_{n-1} + \frac{1}{x_n},$$

so that $x_1, x_2 \ldots x_n$ are positive improper fractions, and $a_1, a_2 \ldots a_{n-1}$ are positive integers.

By the notation of the preceding section, x is thus equal to the continued fraction

$$(a_0, a_1, a_2 \ldots a_{n-1}, x_n).$$

But if x is irrational, this continued-fraction development can be continued without limit; that is, we may take n as large as we please. Each such development arises from the preceding one by setting

$$x_n = a_n + \frac{1}{x_{n+1}}$$

Here too we call the integers $a_0, a_1, a_2 \ldots a_{n-1}$ the partial denominators, and x_n the closing number.

The continued-fraction development of the rational numbers is contained in this as a special case. The continued fraction can (by Theorem I.) be continued by at most one term when the closing number is an integer.

§. 117.
The Convergents.

If we convert a continued fraction

(1) $$x = (a_0, a_1, a_2 \ldots a_{n-1}, x_n),$$

in which we may regard x_n as a variable quantity, into an ordinary fraction, we obtain in the numerator and in the denominator a linear expression in x_n. Thus we have

$$x = \frac{P_n x_n + P_{n-1}}{Q_n x_n + Q_{n-1}}, \tag{2}$$

in which P_n, Q_n, P_{n-1}, Q_{n-1} are integral rational functions of $a_0, a_1, a_2 \ldots a_{n-1}$. That this is correct follows for the first values $n = 1, 2 \ldots$ by direct calculation:

$$\begin{aligned} x &= a_0 + \frac{1}{x_1} = \frac{a_0 x_1 + 1}{x_1} \\ &= a_0 + \frac{1}{a_1 + \dfrac{1}{x_2}} = \frac{(a_0 a_1 + 1)x_2 + a_0}{a_1 x_2 + 1}. \end{aligned}$$

If we assume the formula (2) as already proved for n, then, setting

$$x_n = a_n + \frac{1}{x_{n+1}}$$

we obtain:

$$x = \frac{(P_n a_n + P_{n-1})x_{n+1} + P_n}{(Q_n a_n + Q_{n-1})x_{n+1} + Q_n}.$$

But this is the formula that results from (2) by permutation of n with $n+1$, if we adopt the following law of formation for the P_n, Q_n:

$$\begin{aligned} P_{n+1} &= a_n P_n + P_{n-1} \\ Q_{n+1} &= a_n Q_n + Q_{n-1}. \end{aligned} \tag{3}$$

These formulas completely determine the P_n, Q_n for $n = 2, 3, 4 \ldots$, if we further add the determination:

$$\begin{aligned} P_0 &= 1, \quad P_1 = a_0, \\ Q_0 &= 0, \quad Q_1 = 1. \end{aligned} \tag{4}$$

The fractions

$$\frac{P_0}{Q_0}, \ \frac{P_1}{Q_1}, \ \frac{P_2}{Q_2} \ldots,$$

of which the first, with the denominator zero, is introduced only formally, for the sake of uniformity, are called the convergents of the continued fraction (1), P_n the numerator and Q_n the denominator of the nth convergent.

The law of formation (3) shows that, if the a are integers, then P_n and Q_n are also integers. If the partial denominators from a_1 on are positive, then the denominators $Q_1, Q_2, Q_3 \ldots$ are all positive and increase with n, and indeed, since they are integers, to infinity. Only in the special case where $a_1 = 1$ is $Q_1 = Q_2 = 1$, and the increase begins only from Q_3 on, and we can therefore formulate the theorem:

II. If Q_{n-1}, Q_n are the denominators of two successive convergents, then

$$0 \leqq Q_{n-1} \leqq Q_n, \quad Q_n = \infty \text{ for } n = \infty,$$

where equality with the lower limit 0 can occur only for $n = 1$, and with the upper limit Q_n only for $n = 2$.

The $P_1, P_2, P_3 \ldots$ are, if a_0 is positive, likewise all positive; if a_0 is negative, then they are all negative, with the possible exception of $P_2 = a_0 a_1 + 1$, which can be equal to zero. The Q_n are entirely independent of a_0. In what manner the P_n depend on a_0 we can recognise in the following way.

Let us take the sequence of numbers

$$a_1,\ a_2,\ a_3,\ a_4 \ldots$$

as given and consider the recursion formula

$$T_{n+1} = a_n T_n + T_{n-1}, \tag{5}$$

then T_n is thereby completely determined for all n, once T_0 and T_1 are given.

If we take two special cases Q_n and R_n, which are determined by the conditions

$$\begin{aligned} R_0 &= 1, \quad R_1 = 0 \\ Q_0 &= 0, \quad Q_1 = 1 \end{aligned} \tag{6}$$

then one obtains the general solution of equation (5) in the form

$$T_n = T_0 R_n + T_1 Q_n \tag{7}$$

and in this is therefore also contained

$$P_n = R_n + a_0 Q_n \tag{8}$$

If we consider two solutions of (5), T_n and S_n, thus:

$$\begin{aligned} T_{n+1} &= a_n T_n + T_{n-1} \\ S_{n+1} &= a_n S_n + S_{n-1}, \end{aligned}$$

then by elimination of a_n it follows that

$$T_{n+1} S_n - T_n S_{n+1} = -(T_n S_{n-1} - T_{n-1} S_n),$$

or, if we set

$$T_n S_{n+1} - S_n T_{n+1} = \Delta_n$$

,.

$$\Delta_n = -\Delta_{n-1} = \Delta_{n-2} = \cdots = (-1)^n \Delta_0. \tag{9}$$

But for $T_n = R_n$ and $S_n = Q_n$ we have $\Delta_0 = 1$, [by (6)], hence by (9)

$$R_n Q_{n+1} - Q_n R_{n+1} = (-1)^n,$$

or, by changing n into $n-1$ and reversing the sign:

$$R_n Q_{n-1} - Q_n R_{n-1} = (-1)^n. \tag{10}$$

If one applies this to two functions T_n, S_n, which are defined by

$$T_n = T_0 R_n + T_1 Q_n, \quad S_n = S_0 R_n + S_1 Q_n$$

then it follows that

$$T_n S_{n-1} - S_n T_{n-1} = (-1)^n (T_0 S_1 - T_1 S_0),$$

and so in particular

$$P_n Q_{n-1} - Q_n P_{n-1} = (-1)^n. \tag{11}$$

This formula, of which we shall yet make manifold application, shows that the numbers P_n, Q_n have no common divisor, and hence that the convergents $P_n : Q_n$ cannot be reduced by cancellation.

If the continued fraction is a finite one, that is, if the closing number $x_{n-1} = a_{n-1}$ is an integer, then we carry the formation of the convergents no further than to

$$\frac{P_n}{Q_n} = \frac{P_{n-1}a_{n-1} + P_{n-2}}{Q_{n-1}a_{n-1} + Q_{n-2}},$$

and formula (2) shows that then the last convergent agrees with the value of the continued fraction.

§. 118.
Solution of indeterminate equations with two unknowns.

The formulas last found lead to the solution of a problem that occurs in many applications:

Let α, β be two given integers with no common divisor; two other integers are to be found which satisfy the condition:

$$\alpha y - \beta x = 1. \tag{1}$$

This problem, if it is solvable at all, has infinitely many solutions, all of which can be derived from any one of them.

For if x_0, y_0 is a solution, so that

$$\alpha y_0 - \beta x_0 = 1, \tag{2}$$

then by subtraction of (1) and (2) it follows that

$$\alpha(y - y_0) = \beta(x - x_0),$$

from which it is to be concluded, since α has with β no common divisor

that $x - x_0$ is divisible by α; if accordingly we set, denoting by h an arbitrary integer,

$$x = x_0 + h\alpha,$$

it follows that:

$$y = y_0 + h\beta,$$

and all solutions of (1) are contained in this form.

We therefore need only to seek one solution of (1), which we can always obtain in the following way. We assume β to be positive, which does not essentially restrict the generality, since if need be we can change x into $-x$, and, following §. 115, we develop the rational number $\alpha : \beta$ into a continued fraction:

$$\frac{\alpha}{\beta} = (a_0, a_1, a_2 \ldots a_{n-1}).$$

From this we form the convergents

$$\frac{P_0}{Q_0}, \quad \frac{P_1}{Q_1}, \quad \frac{P_2}{Q_2} \cdots \frac{P_{n-1}}{Q_{n-1}}, \quad \frac{P_n}{Q_n},$$

so that

$$\frac{\alpha}{\beta} = \frac{P_n}{Q_n}.$$

But since α, β as well as P_n, Q_n are without common divisor, and since β and Q_n are positive, it follows that

$$\alpha = P_n, \quad \beta = Q_n.$$

Since further, by §. 117,

$$P_n Q_{n-1} - Q_n P_{n-1} = (-1)^n$$

it follows that

$$x = (-1)^n P_{n-1}, \quad y = (-1)^n Q_{n-1} \tag{3}$$

are integers which satisfy the equation (1), and with this the problem is completely solved.

For moderately large numbers the calculation is quite simple.

If, for example, we set $\alpha = 24335$, $\beta = 3588$, we first obtain by §. 115 the continued fraction:

$$\frac{24335}{3588} = (6,\ 1,\ 3,\ 1,\ 1,\ 2,\ 6,\ 2,\ 1,\ 1,\ 4),$$

and the convergents

$$\frac{1}{0}, \quad \frac{6}{1}, \quad \frac{7}{1}, \quad \frac{27}{4}, \quad \frac{34}{5}, \quad \frac{61}{9}, \quad \frac{156}{23}, \quad \frac{997}{147}, \quad \frac{2150}{317}, \quad \frac{3147}{464},$$
$$\frac{5297}{781}, \quad \frac{24335}{3588},$$

and since here $n = 11$, and thus odd, one must set:

$$x = -5297, \quad y = -781.$$

These are the absolutely smallest values of x, y that satisfy equation (1).

The smallest positive solution is obtained from this by adding α and β to them, thus

$$x = 19038, \quad y = 2807.$$

To this is also reduced the solution of the more general equation

$$\alpha y - \beta x = \gamma \tag{4}$$

in integers x, y.

First of all it is clear that, if (4) is to be solvable at all, every common divisor of α and β must also be a divisor of γ. Such a common divisor is then removed by division. We therefore assume here too that α and β have no common divisor. Then it follows, just as above, that all solutions of (4) are obtained from one of them, x_0, y_0, in the form

$$x = x_0 + h\alpha, \quad y = y_0 + h\beta,$$

in which h is an undetermined integer. If one then sets in (4)

$$x = \gamma\xi, \quad y = \gamma\eta,$$

and divides by γ, then (4) passes into

$$\alpha\eta - \beta\xi = 1,$$

that is, into an equation of the form (1).

The solution becomes a completely determinate one when a further condition is given from which h can be determined, e.g. the condition that y shall lie between 0 and β, with the inclusion of one of the two limits. We shall summarise this in the following theorem:

III. The indeterminate equation

$$\alpha y - \beta x = \gamma$$

has, if α, β, γ are integers and α and β are without common divisor, always an integral solution x, y, and only one if the condition

$$0 \leqq y < \beta,$$

or

$$0 < y \leqq \beta$$

is added.

By what was remarked above, however, there are also always solutions of the equation (4) when γ is divisible by the greatest common divisor of α and β. In particular we can state the theorem: The greatest common divisor δ of two numbers α, β can be represented in the form

$$\alpha x - \beta y = \delta$$

in which x, y are integers.

Theorem III can be generalised in the following way:

If α_1, α_2, $\alpha_3 \ldots$ are given integers in arbitrary number, without a divisor common to them all, and m is an arbitrarily given number, then the integers x_1, x_2, $x_3 \ldots$ can always be determined so that

$$m = \alpha_1 x_1 + \alpha_2 x_2 + \alpha_3 x_3 + \cdots \tag{5}$$

holds.

If the right-hand side of (5) consists of only two terms, this theorem coincides with Theorem III. If, then, we assume the possibility of satisfying a formula (5) as proved whenever the right-hand side consists of fewer terms, we can, if δ is the greatest common divisor of α_2, $\alpha_3 \ldots$, since δ must be relatively prime to α_1, satisfy the two equations:

$$\delta = \alpha_2 y_2 + \alpha_3 y_3 + \cdots$$

$$m = \alpha_1 x_1 + \delta x$$

by integral x, x_1, y_2, $y_3 \ldots$ But then the equation (5) is satisfied by

$$x_1, \quad x_2 = x y_2, \quad x_3 = x y_3 \ldots$$

The problem solved by Theorem III is also expressed in number theory as follows: If α, β, γ are given numbers, a number y is to be found which satisfies the congruence

$$\alpha y \equiv \gamma \pmod{\beta}$$

This problem therefore always has a solution when γ is divisible by the greatest common divisor of α and β.

§. 119.
Convergence of the Convergents.

We now assume that the sequence of numbers $a_1, a_2, a_3 \ldots$ is an unlimited one, and form the difference of two successive convergents:

$$\frac{P_n}{Q_n} - \frac{P_{n-1}}{Q_{n-1}} = \frac{(-1)^n}{Q_n Q_{n-1}}. \tag{1}$$

This difference therefore has the positive sign for even n and the negative sign for odd n, and decreases without limit in absolute value as n increases to infinity. If we further form from (1):

$$\frac{P_n}{Q_n} - \frac{P_{n-2}}{Q_{n-2}} = \frac{(-1)^n}{Q_n Q_{n-1}} + \frac{(-1)^{n+1}}{Q_{n-1}Q_{n-2}} = \frac{(-1)^n}{Q_{n-1}}\left(\frac{1}{Q_n} - \frac{1}{Q_{n-2}}\right), \tag{2}$$

then, since $Q_{n-2} < Q_n$, it follows that this difference is negative for even n and positive for odd n.

From this it follows that the sequence of convergents with even index

$$\frac{P_2}{Q_2}, \quad \frac{P_4}{Q_4}, \quad \frac{P_6}{Q_6} \cdots \tag{3}$$

is a decreasing one, and the sequence of convergents with odd index

$$\frac{P_1}{Q_1}, \quad \frac{P_3}{Q_3}, \quad \frac{P_5}{Q_5} \cdots \tag{4}$$

an increasing one.

Now by §. 117

$$x = \frac{P_n x_n + P_{n-1}}{Q_n x_n + Q_{n-1}},$$

hence

$$\frac{P_n}{Q_n} - x = \frac{(-1)^n}{Q_n(Q_n x_n + Q_{n-1})}, \tag{5}$$

and since Q_n, Q_{n-1}, x_n are positive, all the numbers of the sequence (3) are greater than x, all the numbers of the sequence (4) less than x.

The difference (5) sinks below every limit as n increases without bound, and since $x_n > a_n$, and hence $Q_n x_n + Q_{n-1} > Q_{n+1}$, it is in absolute value smaller than

$$\frac{1}{Q_n Q_{n+1}} \tag{6}$$

and all the more smaller than $1 : Q_n^2$.

The numbers of the sequence (3) thus approach the limit x decreasingly, those of the sequence (4) increasingly. The expression (6) gives a measure of the error one commits when x is replaced by the convergent $P_n : Q_n$.

The convergents are thus approximate expressions of irrational numbers by rational fractions.

That these convergents, for an equal degree of approximation to the irrational number x, are the simplest possible is expressed by the following theorem:

IV. Between the two rational fractions

$$\frac{P_n}{Q_n}, \quad \frac{P_{n-1}}{Q_{n-1}}$$

no other rational fraction can be inserted whose denominator is smaller than Q_n or even only equal to Q_n.

Suppose that the rational fraction $M : N$ lies between the convergents $P_n : Q_n$ and $P_{n-1} : Q_{n-1}$. Then the difference

$$\frac{P_n}{Q_n} - \frac{P_{n-1}}{Q_{n-1}}$$

is absolutely greater than, and of the same sign as,

$$\frac{M}{N} - \frac{P_{n-1}}{Q_{n-1}},$$

hence

$$(-1)^n \left(\frac{P_n}{Q_n} - \frac{P_{n-1}}{Q_{n-1}} \right) > (-1)^n \left(\frac{M}{N} - \frac{P_{n-1}}{Q_{n-1}} \right).$$

If one multiplies both sides by NQ_{n-1}, then by (1) it follows that

$$\frac{N}{Q_n} > (-1)^n (MQ_{n-1} - NP_{n-1}),$$

and since on the right stands a positive integer, which is therefore at least equal to 1, it follows that

$$N > Q_n.$$

§. 120.
Equivalent Numbers.

The continued fractions lead us to the consideration of a special kind of linear substitutions, which are of great importance in algebra and number theory generally, and which we shall consider somewhat more closely.

If x and y are two numbers, or also variable quantities, which stand in the mutual dependence

$$y = \frac{\alpha x + \beta}{\gamma x + \delta}, \tag{1}$$

in which $\alpha, \beta, \gamma, \delta$ are integers satisfying the condition

$$\alpha\delta - \beta\gamma = \varepsilon = \pm 1 \tag{2}$$

then we call x and y equivalent to one another[1]. We call them properly or improperly equivalent, according as $\varepsilon = +1$ or $\varepsilon = -1$. This relation is a reciprocal one, for from (1) it follows that

$$x = \frac{\delta y - \beta}{-\gamma y + \alpha}. \tag{3}$$

If two quantities are equivalent to a third, then they are also equivalent to one another.

For if

$$x = \frac{\alpha' z + \beta'}{\gamma' z + \delta'}, \tag{4}$$

then from (1) it follows that

$$y = \frac{\alpha'' z + \beta''}{\gamma'' z + \delta''}, \tag{5}$$

where

$$\begin{aligned} \alpha'' &= \alpha\alpha' + \beta\gamma', \quad \beta'' = \alpha\beta' + \beta\delta', \\ \gamma'' &= \gamma\alpha' + \delta\gamma', \quad \delta'' = \gamma\beta' + \delta\delta', \end{aligned} \tag{6}$$

(7) $$\alpha''\delta'' - \beta''\gamma'' = (\alpha\delta - \beta\gamma)(\alpha'\delta' - \beta'\gamma'),$$

or

(8) $$\varepsilon'' = \varepsilon\varepsilon'.$$

The substitution (5) is said to be composed of (1) and (4).

The need for an abbreviated notation for these substitutions makes itself felt. Since it is often not the variables that matter, but only the substitution numbers $\alpha, \beta, \gamma, \delta$, one denotes the whole substitution (1) by the substitution numbers, or also merely by a single letter

(9) $$S = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix},$$

and then, when it is necessary, writes the equation (1) thus:

$$y = S(x) = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}(x).$$

The composition of two substitutions is denoted by setting the symbols side by side, in which, however, attention must be paid to the order; thus

(10) $$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}\begin{pmatrix} \alpha', & \beta' \\ \gamma', & \delta' \end{pmatrix} = \begin{pmatrix} \alpha'', & \beta'' \\ \gamma'', & \delta'' \end{pmatrix},$$

or

(11) $$SS' = S''.$$

The formulas (6) contain the rule according to which a composite substitution is to be formed. One can derive it from the multiplication rule for determinants, but must observe that in the first factor the summation is to be taken by rows, in the second by columns, just as the formulas (6) show.

Just as one composes two substitutions, one can also form the composition of several:

$$SS_1S_2S_3\ldots$$

In forming the composite substitution the components may not, in general at least, be interchanged; but one may at will combine two adjacent ones into one, then again two, and so on; that is, the commutative law does not hold, but the associative law does.

This follows immediately from the fact that, if x is expressed in terms of x_1, x_1 in terms of x_2, and x_2 in terms of x_3, then the expression of x in terms of x_3 can be found either in this way, that one first expresses x_1 in x by means of x_2 and then x_2 by means of x_3, or in this way, that

one first expresses x_1 in terms of x_3 and substitutes this into the expression of x in terms of x_1.

By (8), a composite equivalence is a proper or an improper one according as an even or an odd number of improper ones is found among the components.

If one composes the two substitutions

(12) $$\begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}, \quad \begin{pmatrix} \varepsilon\delta, & -\varepsilon\beta \\ -\varepsilon\gamma, & \varepsilon\alpha \end{pmatrix}$$

[1] Cf. Dedekind, letter to Herr Borchardt on the theory of elliptic modular functions. Crelle's Journal, Vol. 83 (1877).

then one obtains, no matter which of the two one puts in the first place,

$$\begin{pmatrix} 1, & 0 \\ 0, & 1 \end{pmatrix}. \tag{13}$$

This substitution has, by (1), the same meaning as $y = x$; it changes nothing, and is called the identical substitution, and indeed is also denoted briefly by (1). Accordingly one calls the two substitutions (12) reciprocal to one another, and denotes them by

$$S,\ S^{-1},$$

or one sets

$$\begin{pmatrix} \varepsilon\delta, & -\varepsilon\beta \\ -\varepsilon\gamma, & \varepsilon\alpha \end{pmatrix} = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}^{-1}. \tag{14}$$

Under composition with the identical substitution (13) every other substitution remains unchanged.

From this we very simply derive the proof of the theorem:

V. All rational numbers are equivalent to one another, and indeed both properly and improperly.

For if $m : n$ and $m' : n'$ are two rational fractions, and m and n, as well as m' and n', are without common divisor, then, if $\varepsilon, \varepsilon'$ are ± 1 at pleasure, we can determine the numbers μ, ν and μ', ν' according to §. 118 so that

$$m\nu - n\mu = \varepsilon, \qquad m'\nu' - n'\mu' = \varepsilon'$$

holds, and so that therefore

$$M = \begin{pmatrix} m, & \mu \\ n, & \nu \end{pmatrix}, \quad M' = \begin{pmatrix} m', & \mu' \\ n', & \nu' \end{pmatrix}$$

are two linear substitutions. It is then also

$$MM'^{-1} = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix}$$

a linear substitution, and

$$\begin{pmatrix} m, & \mu \\ n, & \nu \end{pmatrix} = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \begin{pmatrix} m', & \mu' \\ n', & \nu' \end{pmatrix},$$

from which it follows that

$$m = \alpha m' + \beta n', \qquad n = \gamma m' + \delta n',$$

hence

$$\frac{m}{n} = \frac{\alpha\dfrac{m'}{n'} + \beta}{\gamma\dfrac{m'}{n'} + \delta},$$

which was to be proved. The equivalence is a proper or an improper one, according as $\varepsilon = \varepsilon'$ or $\varepsilon = -\varepsilon'$.

Every number is improperly equivalent to its opposite and to its reciprocal, for we have

$$-x = \begin{pmatrix} -1, & 0 \\ 0, & 1 \end{pmatrix}(x), \qquad \frac{1}{x} = \begin{pmatrix} 0, & 1 \\ 1, & 0 \end{pmatrix}(x).$$

§. 121.
Development of equivalent numbers in continued fractions.

If of two equivalent numbers one is irrational, then so is the other; and if one is real, then so is the other. We make both of these assumptions, and thus in

$$y = \frac{\alpha x + \beta}{\gamma x + \delta}, \qquad \alpha\delta - \beta\gamma = \varepsilon \tag{1}$$

we suppose x to be irrational and real, and we now wish to investigate how the continued-fraction development of y can be derived from the continued-fraction development of x.

Let x, then, be developed in a continued fraction with the closing number x_n:

$$x = (a_0, a_1, a_2 \ldots a_{n-1}, x_n), \tag{2}$$

in which we leave n undetermined for the time being.

Expressed by means of the convergents, this becomes

$$x = \frac{P_n x_n + P_{n-1}}{Q_n x_n + Q_{n-1}}, \qquad P_n Q_{n-1} - Q_n P_{n-1} = (-1)^n. \tag{3}$$

If we substitute this in (1), it follows that

$$y = \begin{pmatrix} \alpha, & \beta \\ \gamma, & \delta \end{pmatrix} \begin{pmatrix} P_n, & P_{n-1} \\ Q_n, & Q_{n-1} \end{pmatrix} (x_n) = \begin{pmatrix} R_n, & R_{n-1} \\ S_n, & S_{n-1} \end{pmatrix} (x_n), \tag{4}$$

in which, by §. 120, (10) and (6), (7),

$$\begin{aligned} R_n &= \alpha P_n + \beta Q_n, \quad R_{n-1} = \alpha P_{n-1} + \beta Q_{n-1}, \\ S_n &= \gamma P_n + \delta Q_n, \quad S_{n-1} = \gamma P_{n-1} + \delta Q_{n-1}, \end{aligned} \tag{5}$$

$$R_n S_{n-1} - S_n R_{n-1} = (-1)^n \varepsilon. \tag{6}$$

If one puts S_n into the form

$$S_n = Q_n \left(\gamma \frac{P_n}{Q_n} + \delta \right),$$

then, since $P_n : Q_n$ approaches the value x to any arbitrary degree and Q_n is positive, it follows that, if n is chosen large enough, S_n agrees in sign with

$$(\gamma x + \delta)$$

Since x is irrational, this quantity is different from zero, and we shall assume it to be positive. Were it negative, we should only have to change the signs of the four numbers $\alpha, \beta, \gamma, \delta$ simultaneously, whereby (1) remains unaltered, in order to make it positive.

Hence S_n will be positive for sufficiently large n. But now from (5), having regard to §. 117, it follows that

$$S_{n+1} = a_n S_n + S_{n-1},$$

from which one concludes that, if n is so large that S_{n-2} and the following S are positive, S_n increases together with n, and hence:

$$S_n > S_{n-1} > 0. \tag{7}$$

We now develop the rational fraction $R_n : S_n$ into a finite continued fraction

$$\frac{R_n}{S_n} = (b_0, b_1, b_2 \dots b_{m-1}), \tag{8}$$

and denote the next-to-last convergent by $R' : S'$, so that we have the relation:

$$R_n S' - S_n R' = (-1)^m. \tag{9}$$

By Theorem I in §. 115, however, we may assume m to be even or odd at pleasure, and we shall dispose of it in such a way that

$$(-1)^m = (-1)^n \varepsilon$$

holds. Moreover, as we know from §. 117, II,

$$S_n \geqq S' \geqq 0, \tag{10}$$

in which, however, the equality sign can occur in the lower limit only for $m = 1$, and in the upper limit only for $m = 2$ and $b_1 = 1$, hence in general only when $S_n = 1$; and this can be avoided, by (7), if n is taken large enough.

But since, by §. 118, III, the numbers S', R' are completely determined by the conditions (9), (10), and S_{n-1}, R_{n-1} satisfy the same conditions by (6), (7), it follows that

$$S' = S_{n-1}, \qquad R' = R_{n-1}.$$

From this it now results further that the continued fraction with the closing number x_n

$$(b_0, b_1, b_2 \dots b_{m-1}, x_n) \tag{11}$$

has the value

$$\frac{R_n x_n + R_{n-1}}{S_n x_n + S_{n-1}},$$

and therefore agrees with y.

If one now develops x_n further into a continued fraction

$$x_n = (a_n, a_{n+1}, a_{n+2} \dots),$$

one obtains from (2) and (11):

$$x = (a_0, a_1, a_2 \dots a_{n-1}, a_n, a_{n+1}, a_{n+2} \dots)$$

$$y = (b_0, b_1, b_2 \dots b_{m-1}, a_n, a_{n+1}, a_{n+2} \dots),$$

or, expressed in words, the theorem:

VI. The continued fraction developments of two equivalent numbers agree with one another from a certain partial denominator on.

We may add that, if the equivalence is a proper one ($\varepsilon = +1$), the number of partial denominators preceding the agreeing ones is even in both, or odd in both; and that, if the equivalence is improper, this number is even in the one and odd in the other.

That the theorem also holds conversely is easy to see. For if two continued fractions agree from a certain partial denominator on, they can be written in such a way that they have the same closing number. But the value of a continued fraction is always equivalent to each of its closing numbers [§. 117, (2)]; hence two continued fractions with the same closing number are also equivalent to one another.

§. 122.
Quadratic Irrational Numbers.

Simple and beautiful laws emerge when one applies the continued fraction expansion to the determination of the roots of a quadratic equation. The root of a quadratic equation with integral coefficients has the form

$$\omega = x + y\sqrt{d}, \tag{1}$$

in which x, y, d are integral or fractional rational numbers. We may, however, without restricting the generality, assume d to be an integer and without square divisor; for if d has a denominator, we can multiply through by this denominator, and may reckon the root of the square denominator, and of any square divisor of the numerator, to y; we further assume that d is not equal to 1, since otherwise ω would be rational; d may be positive or negative, and on this it depends whether ω is real or imaginary.

If we change the sign of the root, we obtain

$$\omega' = x - y\sqrt{d}, \tag{2}$$

which is called the number conjugate to ω. The product of the two numbers ω, ω'

$$\omega\omega' = x^2 - y^2 d$$

is called the norm of ω (or also of ω'). The norm of a product of two quadratic irrational numbers is equal to the product of the norms.

The numbers ω and ω' are the roots of an equation with rational coefficients:

$$\omega^2 - 2x\omega + (x^2 - y^2 d) = 0. \tag{3}$$

To transform it into an equation with integral coefficients, we set

$$x^2 - y^2 d = -\frac{a}{c}, \quad 2x = \frac{b}{c}, \tag{4}$$

in which a, b, c are integers without common divisor, and obtain from (3)

$$c\omega^2 = a + b\omega, \tag{5}$$

that is, a primitive equation with integral coefficients

The numbers a, b, c are determined by (4) only up to a common sign.

The discriminant of (3) is

$$D = b^2 + 4ac = 4c^2 y^2 d, \tag{6}$$

and shall also be called the discriminant of the irrational number ω. D is thus an integer which agrees in sign with d and which is divisible by d; for if p is a prime divisor of d, which therefore by hypothesis divides d only once, then p cannot divide the denominator of $2cy$, and hence by (6) must divide D. Therefore $2cy$ is also an integer, and the quotient $D : d$ is a square number.

By (4) and (6), ω and ω' can be represented thus:

$$\omega = \frac{\sqrt{D} + b}{2c}, \quad \omega' = \frac{-\sqrt{D} + b}{2c}, \tag{7}$$

or, what is the same thing:

$$\omega = \frac{2a}{\sqrt{D} - b}, \quad \omega' = \frac{-2a}{\sqrt{D} + b}, \tag{8}$$

if the sign of the root is determined from

$$\sqrt{D} = 2cy\sqrt{d} \tag{9}$$

Since an even square number is divisible by 4, while an odd one, divided by 4, leaves the residue 1, it follows from (6) that

$$D \equiv 0 \quad \text{or} \quad \equiv 1 \pmod 4 \tag{10}$$

must hold.

We wish further to investigate how the numbers a, b, c, D change when we pass from ω to an equivalent number ω_1. Let, then,

$$\omega_1 = \frac{\alpha\omega + \beta}{\gamma\omega + \delta}, \quad \omega = \frac{\delta\omega_1 - \beta}{-\gamma\omega_1 + \alpha_1} \quad \cdot \quad \alpha\delta - \beta\gamma = \varepsilon. \tag{11}$$

From (5) one obtains for ω_1 the quadratic equation

$$c_1\omega_1^2 = a_1 + b_1\omega_1, \tag{12}$$

if we put

$$\begin{aligned} -a_1 &= -a\alpha^2 + b\alpha\beta + c\beta^2 \\ b_1 &= -2a\alpha\gamma + b(\alpha\delta + \beta\gamma) + 2c\beta\delta, \\ c_1 &= -a\gamma^2 + b\gamma\delta + c\delta^2 \end{aligned} \tag{13}$$

From (13) one obtains, by solving for a, b, c:

$$\begin{aligned} a &= a_1\delta^2 + b_1\beta\delta - c_1\beta^2 \\ b &= 2a_1\gamma\delta + b_1(\alpha\delta + \beta\gamma) - 2c_1\alpha\beta, \\ -c &= a_1\gamma^2 + b_1\alpha\gamma - c_1\alpha^2, \end{aligned} \tag{14}$$

and further

$$b^2 + 4ac = b_1^2 + 4a_1c_1 = D. \tag{15}$$

Since a, b, c are without common divisor, a_1, b_1, c_1 also have no common divisor, as one sees from (14), and a_1, b_1, c_1 have the same meaning for ω_1 as a, b, c have for ω.

VII. One sees further from (15) that for equivalent numbers not only the irrationality $\sqrt{d}$, but also the discriminant is the same.

If one sets

$$\omega_1 = x_1 + y_1\sqrt{d} = \frac{\alpha(x + y\sqrt{d}) + \beta}{\gamma(x + y\sqrt{d}) + \delta}, \tag{16}$$

and, in order to determine x_1, y_1, multiplies the numerator and denominator of the last fraction by $\gamma(x - y\sqrt{d}) + \delta$, then by (4) and (13) there results

$$2x_1 = \frac{b_1}{c_1}, \qquad y_1 = \frac{\varepsilon yc}{c_1}, \tag{17}$$

hence, if the $\sqrt{D}$ is understood in the same way as in (7) and (8), by (9):

$$\omega_1 = \frac{b_1 + \varepsilon\sqrt{D}}{2c_1}. \tag{18}$$

§. 123.
Reduced Numbers with Negative Discriminant.

The considerations up to this point are applicable uniformly to the two cases of real and imaginary quadratic irrational numbers, that is, to positive and negative discriminants. Now, however, the two cases must be separated from one another, and we begin with the negative discriminants. We are concerned, then, with imaginary numbers:

$$\omega = x + y\sqrt{d} = \xi + i\eta, \tag{1}$$

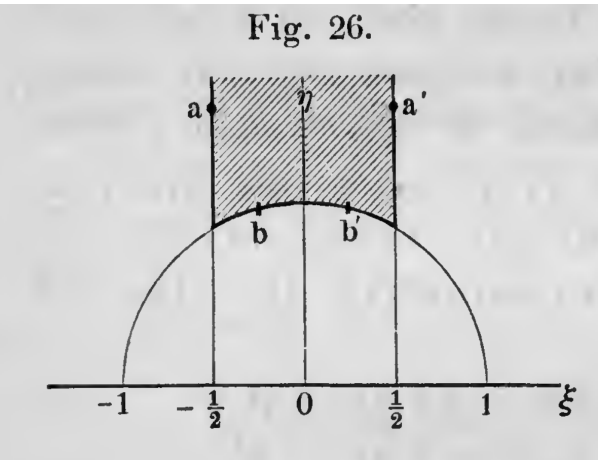
in which ξ and η are real; it suffices to consider one of the two conjugate numbers $\xi + i\eta$, $\xi - i\eta$. We therefore assume η to be positive and lay down the following definition:

A complex number $\omega = \xi + i\eta$ with positive η is called reduced if

$$-\tfrac{1}{2} \leqq \xi \leqq \tfrac{1}{2}, \quad \xi^2 + \eta^2 \geqq 1. \tag{2}$$

If one regards ξ, η as rectangular coordinates in a plane, then every number ω is represented by a point of this plane, and the position of the points that correspond to the reduced numbers is represented by the region shaded in Fig. 26 (with inclusion of the boundaries), which we call the fundamental region.

Fig. 26.



From the conditions (2) it then further follows, as is also easily confirmed in the figure, that

$$\eta^2 \geqq \tfrac{3}{4}. \tag{3}$$

If one now passes from a number ω to an equivalent number ω_1, formula (16) of the preceding section gives

$$\omega_1 = \frac{(\alpha\xi + \beta) + i\alpha\eta}{(\gamma\xi + \delta) + i\gamma\eta} = \frac{[(\alpha\xi + \beta)(\gamma\xi + \delta) + \alpha\gamma\eta^2] + i\varepsilon\eta}{(\gamma\xi + \delta)^2 + \gamma^2\eta^2}, \tag{4}$$

and this formula shows that η_1 has the same sign as η or the opposite one, according as $\varepsilon = +1$ or $= -1$, that is, according as the equivalence is a proper or an improper one.

If we restrict ourselves, then, to numbers with positive imaginary part, only proper equivalence comes into consideration, and we now prove the fundamental theorem:

1. Every imaginary quadratic irrational number $x + y\sqrt{d}$ with positive imaginary part is equivalent to a reduced number.

If we understand by α the integer nearest to ξ, then $\omega - \alpha$ will satisfy the first of the conditions (2), namely that $-\frac{1}{2} \leqq \xi - \alpha \leqq \frac{1}{2}$.

If now

$$r^2 = (\xi - \alpha)^2 + \eta^2 = (\omega - \alpha)(\omega' - \alpha) \tag{5}$$

is greater than or equal to 1, then $\omega - \alpha$, which is equivalent to ω, is already reduced; otherwise we set, applying a continued-fraction expansion,

$$\omega = \alpha - \frac{1}{\omega_1}, \tag{6}$$

so that $\omega_1 = \xi_1 + i\eta_1$ is also equivalent to ω. We now treat the number ω_1 again in the same way as ω, by setting, if $\omega_1 - \alpha_1$ is not yet reduced,

$$\omega_1 = \alpha_1 - \frac{1}{\omega_2} \tag{7}$$

and so on. If we consider the sequence of the quantities formed on the analogy of (5),

$$r_1^2 = (\xi_1 - \alpha_1)^2 + \eta_1^2, \quad r_2^2 = (\xi_2 - \alpha_2)^2 + \eta_2^2 \ldots,$$

then all that now matters is to show that after a finite number of steps of this kind we come to an r_ν^2 that is equal to or greater than 1.

But by (6) we have

$$r^2 = (\omega - \alpha)(\omega' - \alpha) = \frac{1}{\xi_1^2 + \eta_1^2},$$

and from this the comparison of the imaginary parts on the two sides of (6) gives:

$$\eta = r^2 \eta_1.$$

In the same way there follows the sequence of equations

$$\eta = r^2\eta_1, \quad \eta_1 = r_1^2\eta_2, \quad \eta_2 = r_2^2\eta_3 \ldots \tag{8}$$

Now, however, $i\eta = y\sqrt{d}$, and hence by the formula (6), §. 122

$$i\eta = \frac{\sqrt{D}}{2c},$$

in which c is an integer. But because equivalent numbers have the same discriminant, it follows in the same way:

$$i\eta_1 = \frac{\sqrt{D}}{2c_1}, \quad i\eta_2 = \frac{\sqrt{D}}{2c_2}, \ldots,$$

in which $c, c_1, c_2 \ldots$ is a sequence of positive integers. From this one obtains from (8):

$$c_1 = r^2 c, \quad c_2 = r_1^2 c_1, \quad c_3 = r_2^2 c_2 \ldots \tag{9}$$

But so long as the numbers $r^2, r_1^2 \ldots r_\nu^2$ are less than 1, it follows from this

$$c > c_1 > c_2 \ldots > c_{\nu+1}, \tag{10}$$

and because there can be only a finite number of positive numbers that are less than c, this sequence must terminate, and our Theorem 1. is thereby proved.

The reduced numbers whose images lie on the boundary of the fundamental region are equivalent in pairs, namely

a) $\omega = -\frac{1}{2} + i\eta$ and $\omega_1 = \omega + 1 = \frac{1}{2} + i\eta$

(a and a' in the figure),

b) $\omega = -\xi + i\eta$ and $\omega_1 = -\dfrac{1}{\omega} = \xi + i\eta$

when $\xi^2 + \eta^2 = 1$ (b and b' in the figure). But there holds further the theorem:

2. Apart from the cases a), b), no two reduced numbers are equivalent.

For let us assume two reduced numbers that are not identical,

$$\omega = \xi + \eta i, \quad \omega_1 = \xi_1 + \eta_1 i, \tag{11}$$

and suppose, which does not impair the generality,

$$\eta_1 \geqq \eta, \tag{12}$$

then it follows from formula (4):

$$\eta_1 = \frac{\eta}{(\gamma\xi + \delta)^2 + \gamma^2\eta^2}, \tag{13}$$

hence, on account of (12):

$$(\gamma\xi + \delta)^2 + \gamma^2\eta^2 \leqq 1, \tag{14}$$

hence also, by (3):

$$1 \geqq \gamma^2\eta^2 \geqq \tfrac{3}{4}\gamma^2; \tag{15}$$

thus there are two possibilities:

$$\alpha)\ \gamma = 0, \qquad \beta)\ \gamma = \pm 1.$$

In the case α) we have $\alpha\delta = 1$, hence $\alpha = \delta = \pm 1$; from (12) and (13) it results that $\eta_1 = \eta$, and further from (4):

$$\xi_1 = \xi \pm \beta.$$

But this is compatible with the conditions (2) only when either $\xi_1 = \xi$, $\beta = 0$ and hence ω_1 is identical with ω, or when $\beta = \pm 1$ and $\xi_1 = -\xi = \pm\frac{1}{2}$, that is, in the case a).

In the case β) we have by (14) $(\xi \pm \delta)^2 \leqq 1$, and hence either: $\delta = 0$, $\beta\gamma = -1$ and $\xi^2 + \eta^2 = 1$, $\eta_1 = \eta$, further by (4) $\xi_1 = \pm\alpha - \xi$; hence, since $\xi_1 = \xi$ is excluded, $\xi_1 = -\xi$, $\alpha = 0$, which is the exceptional case b),

or: $\delta = \pm 1$, hence $(\xi \pm 1)^2 + \eta^2 \leqq 1$, and consequently by (3) $(\xi \pm 1)^2 \leqq \frac{1}{4}$. But since $(\xi \pm 1)$ must by (2) be at least equal to $\frac{1}{2}$ in absolute value, it follows from this that $\xi = \mp\frac{1}{2}$, $\eta^2 = \frac{3}{4}$; hence by (13) $\eta_1^2 = \eta^2 = \frac{3}{4}$, and by

(2) $\xi_1^2 = \frac{1}{4}$, $\xi_1 = \mp\frac{1}{2}$, which is contained both under case a) and under case b);

and with this, then, Theorem 2. is proved. To this we now add, as a third theorem:

3. For a given negative discriminant D there is only a finite number of reduced irrational numbers.

From the representation of the irrational numbers §. 122, (7):

$$\omega = \frac{\sqrt{D}+b}{2c}, \qquad \xi = \frac{b}{2c}, \qquad \eta = \frac{\sqrt{-D}}{2c}$$

and from the inequality (3) $\eta^2 \geqq \frac{3}{4}$ it follows that

$$\frac{-D}{4c^2} \geqq \frac{3}{4},$$

or

$$c \leqq \sqrt{\frac{-D}{3}};$$

thus, when D is given, there is only a finite number of values of the positive integer c. Further, from $-\frac{1}{2} \leqq \xi \leqq \frac{1}{2}$ it follows that:

$$-c \leqq b \leqq c.$$

Thus for each value of c there is only a finite number of values for b, q. e. d.

If one unites all numbers equivalent to one another into a number class, then by 1. we have accordingly proved:

4. The number of classes of imaginary quadratic irrational numbers of a given discriminant is finite.

§. 124.
Reduced numbers with positive discriminant.

We now have to carry out a similar investigation for a positive d, that is, for real quadratic irrational numbers. Such a number $\omega = x + y\sqrt{d}$ shall be called reduced if it satisfies the following conditions:

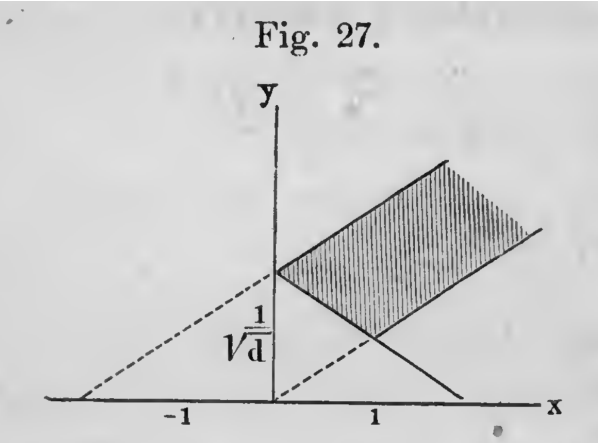
ω is positive and greater than 1, and the number ω' conjugate to ω is negative and smaller than 1 in absolute value,

or in symbols:

(1) $$0 < y\sqrt{d} - x < 1 < y\sqrt{d} + x.$$

If one interprets x, y in a plane as rectangular coordinates, so that the points of the plane become the images of the numbers ω, then the images of the reduced numbers lie in that part of the plane which is shaded in Fig. 27, which corresponds to the assumption $d = 2$.

Fig. 27.



We here take proper and improper equivalence together and first prove the theorems:

1. Every number ω is equivalent to a reduced number.

One sees this easily if one develops ω into a continued fraction and goes as far as the closing number ω_n:

$$\omega = (\alpha_0, \alpha_1, \alpha_2 \ldots \alpha_{n-1}, \omega_n). \tag{2}$$

Then ω is equivalent to ω_n, and ω_n, as soon as n is greater than 0, is positive and greater than 1. Since ω_n is equivalent to ω, it only remains to prove that, when n is sufficiently large, the number ω_n' conjugate to ω_n is negative and absolutely smaller than 1. But this one sees from the following formulae: From (2) it follows, if $P_n : Q_n$ are the convergents,

$$\omega = \frac{P_n\omega_n + P_{n-1}}{Q_n\omega_n + Q_{n-1}},$$

or by solving

$$\omega_n = -\frac{Q_{n-1}\omega - P_{n-1}}{Q_n\omega - P_n} = -\frac{Q_{n-1}}{Q_n} - \frac{(-1)^n}{Q_n(Q_n\omega - P_n)}. \tag{3}$$

If in this equation we change $\sqrt{d}$ into $-\sqrt{d}$, then at the same time ω passes into ω', ω_n into ω_n', and from (3) we obtain

$$\omega_n' = -\frac{Q_{n-1}\omega' - P_{n-1}}{Q_n\omega' - P_n} = -\frac{Q_{n-1}}{Q_n}\frac{\omega' - \dfrac{P_{n-1}}{Q_{n-1}}}{\omega' - \dfrac{P_n}{Q_n}}, \tag{4}$$

$$\omega_n' + 1 = \frac{1}{Q_n}\left\{Q_n - Q_{n-1} - \frac{(-1)^n}{Q_n\left(\omega' - \dfrac{P_n}{Q_n}\right)}\right\}. \tag{5}$$

If now n is taken sufficiently large, then

$$\omega' - \frac{P_n}{Q_n}, \quad \omega' - \frac{P_{n-1}}{Q_{n-1}}$$

differ arbitrarily little from the non-vanishing difference $\omega' - \omega = 2y\sqrt{d}$, whence (4) shows that ω_n' is negative. But from (5) it follows that $\omega_n' + 1$ is positive, since $Q_n - Q_{n-1}$, being a positive integer, is at least equal to 1, and

$$\frac{1}{Q_n\left(\omega' - \dfrac{P_n}{Q_n}\right)}$$

becomes arbitrarily small for a sufficiently large n; and with this the proof of Theorem 1. is complete.

Since the conjugate number of the conjugate ω' is the original number ω, it follows from the definition of the reduced numbers that ω and $-1 : \omega'$ are reduced simultaneously.

2. If ω is reduced and, for an integer α,

$$\omega = \alpha + \frac{1}{\omega_1}, \tag{6}$$

then ω_1 is reduced if and only if α is the greatest integer contained in ω, that is, if

$$\alpha < \omega < \alpha + 1. \tag{7}$$

If the condition (7) is fulfilled, then ω_1 is positive and greater than 1, and

$$\omega_1' = \frac{1}{\omega' - \alpha} \tag{8}$$

is, since ω' is negative, a negative proper fraction; hence ω_1 is reduced. On the other hand, ω_1 cannot be reduced if $\alpha > \omega$, since then ω_1 would be negative, or if $\alpha + 1 < \omega$, since otherwise ω_1 would be less than 1.

If one puts (8) in the form

$$-\frac{1}{\omega_1'} = \alpha + \frac{1}{-\frac{1}{\omega'}}, \tag{9}$$

then, since $-1 : \omega'$ and $-1 : \omega_1'$ are also reduced, α is the greatest integer contained in $-1 : \omega_1'$, and ω' is completely determined by ω_1' according to (9), and hence also ω by ω_1.

From this it follows:

3. If a reduced number ω is developed in a continued fraction, then all the closing numbers ω_n are again reduced numbers, and by any one of these closing numbers ω_n both the following ω_{n+1} and the preceding ω_{n-1} are completely determined.

§. 125.
Development of real quadratic irrational numbers in continued fractions.

4. For a given discriminant there is only a finite number of reduced numbers.

The correctness of this theorem is easily seen if, according to §. 122, one puts the irrational number ω in the form:

$$\omega = \frac{b + \sqrt{D}}{2c} = \frac{-2a}{b - \sqrt{D}}, \quad D = b^2 + 4ac. \tag{1}$$

The conjugate number is

$$\omega' = \frac{b - \sqrt{D}}{2c} = \frac{-2a}{b + \sqrt{D}}. \tag{2}$$

If ω is to be reduced, then

$$0 < \frac{\sqrt{D} - b}{2c} < 1 < \frac{\sqrt{D} + b}{2c}. \tag{3}$$

Hence $\sqrt{D}$ must have the same sign as c; if we take it to be positive, then

(4) $$0 < \sqrt{D} - b < 2c < \sqrt{D} + b.$$

A second form of these conditions we obtain from the second representation

(5) $$0 < \sqrt{D} - b < 2a < \sqrt{D} + b.$$

Hence a, b, c are positive and b is less than $\sqrt{D}$.

If λ denotes the greatest integer contained in $\sqrt{D}$, so that

$$\lambda < \sqrt{D} < \lambda + 1,$$

so b has one of the values 1, 2, 3 $\ldots \lambda$, but with the restriction that for even D only the even, and for odd D only the odd among these numbers are to be taken for b. Then a and c are to be determined so that

(6) $$\frac{D - b^2}{4} = ac,$$

and so that

(7) $$\frac{\lambda - b + 1}{2} \leqq c \leqq \frac{\lambda + b}{2},$$

and a must lie between the same limits. Only one of these two limiting values is an integer; the other may be replaced, for the lower limit, by the next greater integer, and for the upper limit, by the next smaller. Thus for b only a finite number of values is admissible; then $(D - b^2) : 4$ is decomposable into two factors in only a finite number of ways, and of these decompositions only those are to be retained in which both factors satisfy the condition (7). Moreover, such combinations are also to be discarded in which a, b, c acquire a common divisor. Herein lies the means of actually determining all reduced numbers for a given D.

For what follows we shall, for the numbers ω so determined, use the symbol

(8) $$\omega = \frac{\sqrt{D} + b}{2c} = \frac{2a}{\sqrt{D} - b} = \{a, b, c\}$$

to denote them.

From 3. §. 124 and 4. §. 125 there now results the following important theorem:

5. The continued-fraction expansion of a reduced number ω is periodic, that is, if ω is expanded in a continued fraction

(9) $$\omega = (\alpha_0, \alpha_1, \alpha_2 \ldots)$$

then after a certain number of partial denominators the same partial denominators always recur in the same order; thus, if the period consists of ν terms, then

$$\begin{array}{ccccc} \alpha_0 & = & \alpha_\nu & = & \alpha_{2\nu} \ldots \\ \alpha_1 & = & \alpha_{\nu+1} & = & \alpha_{2\nu+1} \ldots \\ \multicolumn{5}{c}{\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots} \\ \alpha_{\nu-1} & = & \alpha_{2\nu-1} & = & \alpha_{3\nu-1} \ldots, \end{array}$$

and it therefore suffices to denote the whole continued fraction by the period, say thus:

$$\omega = [\alpha_0, \alpha_1 \dots \alpha_{\nu-1}]. \tag{10}$$

This theorem is evidently equivalent to saying that the closing number ω_ν of the continued fraction (9) is identical with ω itself, and in this also lies the proof of the assertion. For the closing numbers ω_ν, being equivalent numbers, all belong to the same discriminant, and hence by 4. the number of all possible closing numbers is only a finite one. In the sequence of closing numbers $\omega, \omega_1, \omega_2 \dots$ one that has already occurred must therefore at some point appear for the second time. If ω_ν is the first that appears for the second time, then $\omega_\nu = \omega$ must hold, since otherwise by 3. §. 124 the number $\omega_{\nu-1}$ would also appear for the second time.

Thus all the reduced numbers belonging to one discriminant can be arranged in periods:

$$\omega, \omega_1, \omega_2 \dots \omega_{\nu-1}, \tag{11}$$

so that, if

$$\alpha_0, \alpha_1, \alpha_2 \dots \alpha_{\nu-1}$$

are the greatest integers contained in them, we have

$$\begin{aligned} \omega &= [\alpha_0, \alpha_1 \dots \alpha_{\nu-1}] \\ \omega_1 &= [\alpha_1, \alpha_2 \dots \alpha_0] \\ &\dots\dots\dots\dots\dots\dots\dots \end{aligned}$$

by which the continued fraction expansion is given for each of these numbers.

If the whole system of reduced numbers is not yet exhausted by one period, one forms a second period, and so on. Since by a number ω both the number preceding it in the period and the one following it are completely determined, two different periods never contain a common number.

Of these periods there now holds the theorem:

6. Reduced numbers from the same period are equivalent; from different periods they are not equivalent.

The first part of the assertion is clear from the outset, since reduced numbers of the same period are closing numbers of one another; the second part is proved by the theorem §. 121, VI, that equivalent numbers, when the continued fraction expansion is carried sufficiently

fraction development finally obtain the same closing numbers. If, then, both continued fractions are periodic, they must belong to the same period.

If for a given discriminant D one has developed the complete system of reduced numbers according to the rules given at the beginning of the section, it is easy to arrange these numbers in periods and to find the partial denominators α of the continued fraction. Namely, let

$$\omega = \frac{\sqrt{D} + b}{2c} = \frac{2a}{\sqrt{D} - b} \tag{12}$$

be one of these numbers and

$$\omega_1 = \frac{\sqrt{D} + b_1}{2c_1} = \frac{2a_1}{\sqrt{D} - b_1} \tag{13}$$

the number immediately following it in the period, so that

$$\omega = \alpha + \frac{1}{\omega_1}, \quad \omega_1 = \frac{1}{\omega - \alpha} \tag{14}$$

where α denotes the greatest integer contained in ω. By (12) and (14) we then have

$$\omega_1 = \frac{2c}{\sqrt{D} + b - 2c\alpha},$$

and comparison with (13) gives

$$a_1 = c, \quad b_1 = 2c\alpha - b. \tag{15}$$

If, conversely, two of the reduced numbers $\{a, b, c\}$, $\{a_1, b_1, c_1\}$ belonging to D stand in the relation represented by (15), in which α is any integer, then the second immediately follows the first in the period. For from (15) follows (14), and hence by 2. α must be the greatest number contained in ω.

The number ω_1 is therefore completely and uniquely determined from ω by the conditions (15).

7. One therefore arranges the numbers $\{a, b, c\}$ from left to right in such a way that the last number c of the preceding one becomes at the same time the first number a_1 of the following, and that the sum of the two middle numbers $b + b_1$ is divisible by $2c$. This arrangement is possible in only one way, and the quotient $b + b_1 : 2c$ is the number α that appears as partial denominator in the continued fraction.

Let us now consider any primitive quadratic equation with integer coefficients

$$A + B\Omega + C\Omega^2 = 0, \tag{16}$$

in which A, B, C are integers without common divisor, satisfying the condition

$$B^2 - 4AC = D$$

The two roots Ω, Ω' of this equation are quadratic irrational numbers belonging to the discriminant D, and if we develop them in continued fractions, then by §. 124 we shall at last come upon closing numbers ω_1, ω_2 which belong to the reduced numbers, and which must therefore be contained in the periods discussed above.

It remains to determine whether these closing numbers ω_1, ω_2 are contained in the same period or in different periods.

By 6. they are contained in the same period if and only if Ω and Ω' are equivalent to each other. If we call such quadratic irrational numbers as are equivalent to their conjugates two-sided numbers[28], then we can say that the continued-fraction development of the roots of equation (16) leads to one or to two different periods, according as the roots are two-sided or not. Which of the two cases occurs can be recognised from the period itself.

If, namely, ω is a reduced number, and ω' conjugate to ω, then $-1 : \omega'$ is also reduced; and if Ω is equivalent to ω, then Ω' is equivalent to ω', and hence also to $-1 : \omega'$ (§. 120).

If

$$\omega = \{a, b, c\},$$

then by (12):

$$-\frac{1}{\omega'} = \{c, b, a\}.$$

By §. 124, (9), $-1 : \omega'$ therefore follows $-1 : \omega_1'$ in the same way as ω_1 follows ω. If therefore the period of ω is

$$\omega, \omega_1, \omega_2 \ldots \omega_{\nu-1},$$

then the period of $-1 : \omega'$ is:

$$\frac{-1}{\omega_{\nu-1}'}, \frac{-1}{\omega_{\nu-2}'} \cdots \frac{-1}{\omega_1'}, \frac{-1}{\omega'}.$$

[28] "Two-sided" we use, following Dedekind's proposal, in place of the Gaussian anceps or the otherwise usual ambig: Dirichlet-Dedekind, Vorlesungen über Zahlentheorie. 4th edition, 1894, p. 139.

If the period of the continued fraction development of ω is

$$[\alpha_0, \alpha_1 \ldots \alpha_{\nu-1}], \tag{17}$$

then for $-1 : \omega'$ it is

$$[\alpha_{\nu-1}, \alpha_{\nu-2} \ldots \alpha_0], \tag{18}$$

that is, the reverse.

Now if ω is equivalent to ω', and hence also to $-1 : \omega'$, then by 6. these two periods must agree with one another when one begins at a suitable term, or the period of ω must be reversible.

This means: in the continued fraction development for ω two elements must admit of being so chosen that the development reads the same whether one reads it proceeding from the one of these numbers to the left or from the other to the right. In symbols: for some suitably determined k and for $i = 0, 1, 2 \ldots \nu - 1$ we must have

$$\alpha_i = \alpha_{\nu+k-i}$$

If the period is reversible, then conversely (by 6.) ω is equivalent to $-1 : \omega'$, and hence also to ω'. From this, then, the result:

8. Two-sided numbers have in their continued fraction development a reversible period, and conversely numbers which have a reversible period in their continued fraction development are two-sided.

Finally we wish to remark further that by the property of admitting development into a periodic continued fraction, the real quadratic irrational numbers are distinguished from all other numbers. For if ω is a number developed into a continued fraction, and ω_n, ω_m are two closing numbers of this continued fraction, then

$$\omega = \frac{P_n\omega_n + P_{n-1}}{Q_n\omega_n + Q_{n-1}} = \frac{P_m\omega_m + P_{m-1}}{Q_m\omega_m + Q_{m-1}}.$$

Now if the continued fraction is periodic, no matter whether the periodicity begins right from the start or only in the course of the development, then for two different values of n and m we have $\omega_n = \omega_m$, and one thus obtains, by elimination of

ω_n a quadratic equation for ω, which can be put in the form:

$$\begin{vmatrix} P_n - \omega Q_n, & P_{n-1} - \omega Q_{n-1} \\ P_m - \omega Q_m, & P_{m-1} - \omega Q_{m-1} \end{vmatrix} = 0.$$

§. 126.
Examples.

We shall here carry out some examples of the determination of the reduced numbers and periods, in order to show the application of the method; the examples can of course be multiplied quite at pleasure.

1. $D = 29$.

We first determine, according to §. 125, 4., all the reduced numbers belonging to D. The greatest integer λ contained in $\sqrt{D}$ is here $= 5$; hence b can have only the values:

$$b = 1, 3, 5,$$

whence

$$ac = \frac{D - b^2}{4} = 7, 5, 1.$$

The limits within which a and c must lie are, by §. 125, (7), for the three values of b:

$$3, 3; \quad 2, 4; \quad 1, 5.$$

One therefore obtains only the single reduced number:

$$\{1, 5, 1\}.$$

The period consists of a single term, and the continued fraction for ω is:

$$\omega = \frac{5 + \sqrt{29}}{2} = (5, 5, 5 \ldots)$$

2. $D = 116 = 4.29.$

Here $\lambda = 10$ and b has one of the values

$$b = 2, 4, 6, 8, 10,$$

hence

$$ac = 28, 25, 20, 13, 4;$$

the limits for a and c are

$$5, 6; \quad 4, 7; \quad 3, 8; \quad 2, 9; \quad 1, 10,$$

hence, since the cases in which a, b, c have the common factor

2, are still to be omitted, the reduced numbers are found to be:

$$\{5, 4, 5\}, \quad \{4, 6, 5\}, \quad \{5, 6, 4\}, \quad \{1, 10, 4\}, \quad \{4, 10, 1\},$$

which arrange themselves into a period as follows, the first term being appended once more at the end:

$$\{4, 10, 1\}, \quad \{1, 10, 4\}, \quad \{4, 6, 5\}, \quad \{5, 4, 5\}, \quad \{5, 6, 4\}, \quad \{4, 10, 1\},$$

and for ω the continued-fraction period is found:

$$5 + \sqrt{29} = [10, 2, 1, 1, 2].$$

If one begins with the fourth term, that is, considers the continued fraction $[1, 2, 10, 2, 1]$, which is the expansion of $\dfrac{2 + \sqrt{29}}{5}$, one sees that the period is reversible.

3. $D = 76 = 4.19, \quad \lambda = 8$

$$b = \quad 2, \quad 4, \quad 6, \quad 8$$

$$a\,c = 18, \quad 15, \quad 10, \quad 3.$$

Limits for a and c:

$$4, 5; \quad 3, 6; \quad 2, 7; \quad 1, 8;$$

hence the reduced numbers are:

$$\{3, 4, 5\}, \quad \{5, 4, 3\}, \quad \{2, 6, 5\}, \quad \{5, 6, 2\}, \quad \{1, 8, 3\}, \quad \{3, 8, 1\}.$$

We obtain a period:

$$\{3, 8, 1\}, \quad \{1, 8, 3\}, \quad \{3, 4, 5\}, \quad \{5, 6, 2\}, \quad \{2, 6, 5\}, \quad \{5, 4, 3\}, \quad \{3, 8, 1\}$$

and the period of the continued fraction for $4 + \sqrt{19}$ becomes

$$[8, 2, 1, 3, 1, 2].$$

This continued fraction too has a reversible period, since one obtains the same succession of numbers whether one reads from the first 2 to the right or from the second 2 to the left.

4. $D = 37, \qquad \lambda = 6,$

$$b = 1, \quad 3, \quad 5$$

$$a\,c = 9, \quad 7, \quad 3;$$

Limits for a, c:

$$3,3; \quad 2,4; \quad 1,5;$$

reduced numbers:

$$\{3,1,3\}, \quad \{1,5,3\}, \quad \{3,5,1\};$$

Period:

$$\{3,5,1\}, \quad \{1,5,3\}, \quad \{3,1,3\}, \quad \{3,5,1\};$$

Continued fraction with reversible period $[1,5,1]$:

$$\frac{5+\sqrt{37}}{2} = [5,1,1].$$

5. $D = 148 = 4.37 \quad \lambda = 12$

$$b = \quad 2, \quad 4, \quad 6, \quad 8, \quad 10, \quad 12$$

$$a\,c = 36, \quad 33, \quad 28, \quad 21, \quad 12, \quad 1;$$

limits for a, c:

$$6,7; \quad 5,8; \quad 4,9; \quad 3,10; \quad 2,11; \quad 1,12;$$

reduced numbers:

$$\{4,6,7\}, \quad \{7,6,4\}, \quad \{3,8,7\}, \quad \{7,8,3\}$$

$$\{3,10,4\}, \quad \{4,10,3\}, \quad \{1,12,1\}.$$

Here we obtain three periods of 1, 3, and 3 terms:

$$\{1,12,1\}, \quad \{1,12,1\},$$

$$\{3,10,4\}, \quad \{4,6,7\}, \quad \{7,8,3\}, \quad \{3,10,4\},$$

$$\{4,10,3\}, \quad \{3,8,7\}, \quad \{7,6,4\}, \quad \{4,10,3\}$$

and the continued-fraction periods

$$[12], \quad [2,1,3], \quad [3,1,2];$$

of these the first is reversible, and the other two pass into one another under reversal.

6. $D = 136 = 4.34, \quad \lambda = 11,$

$$b = \quad 2, \quad 4, \quad 6, \quad 8, \quad 10,$$

$$a\,c = 33, \quad 30, \quad 25, \quad 18, \quad 9;$$

limits for a, c:

$$5,6; \quad 4,7; \quad 3,8; \quad 2,9; \quad 1,10;$$

reduced numbers:

$$\{5,4,6\}, \quad \{6,4,5\}, \quad \{5,6,5\}, \quad \{2,8,9\}, \quad \{9,8,2\}$$

$$\{3,8,6\}, \quad \{6,8,3\}, \quad \{1,10,9\}, \quad \{9,10,1\}, \quad \{3,10,3\};$$

periods:

$$\{5,4,6\}, \quad \{6,8,3\}, \quad \{3,10,3\}, \quad \{3,8,6\}, \quad \{6,4,5\}, \quad \{5,6,5\}, \quad \{5,4,6\},$$

$$\{9,10,1\}, \quad \{1,10,9\}, \quad \{9,8,2\}, \quad \{2,8,9\}, \quad \{9,10,1\};$$

continued-fraction periods:

$$[1,3,3,1,1,1], \quad [10,1,4,1].$$

Here, then, we have two different periods, both of which are reversible.

§. 127.
The Pell Equation.

If ω is a reduced number belonging to the discriminant D, then by the continued fraction development one obtains

$$\omega = \frac{P_n\omega_n + P_{n-1}}{Q_n\omega_n + Q_{n-1}}, \tag{1}$$

and $\omega_n = \omega$ holds always when n is a multiple of the number of terms ν of the period. But then it follows from (1):

$$Q_n\omega^2 = (P_n - Q_{n-1})\omega + P_{n-1}. \tag{2}$$

Now let ω satisfy the quadratic equation §. 122, (5):

$$c\omega^2 = a + b\omega, \quad D = b^2 + 4ac, \tag{3}$$

and from this, since a, b, c are without common divisor, equation (2) must follow by multiplication with an integer. If we denote this integer by u, we thus have

$$Q_n = uc, \quad P_{n-1} = ua, \quad P_n - Q_{n-1} = ub. \tag{4}$$

If we further set

$$P_n + Q_{n-1} = t, \tag{5}$$

so that t too is a positive integer, then from (4) and (5) it follows that

$$P_n = \frac{t + ub}{2}, \quad Q_n = uc,$$

$$P_{n-1} = ua, \quad Q_{n-1} = \frac{t - ub}{2}, \tag{6}$$

and hence, having regard to the equation

$$P_nQ_{n-1} - Q_nP_{n-1} = (-1)^n:$$

$$t^2 - Du^2 = (-1)^n 4. \tag{7}$$

Equation (7) is called the Pell equation. The problem is this: for a given D, to find all its integral solutions t, u. Since, if t, u satisfy the equation, $\pm t$, $\pm u$ also satisfy it, we may restrict ourselves to the determination of its positive solutions.

The formulae (4), (5) give us an infinite number of such solutions; we have only to choose for n an arbitrary multiple of the period number ν, and for u the greatest common divisor of Q_n, P_{n-1}, $P_n - Q_{n-1}$, or also simply the number

$Q_n : c$, and to determine t from (5), or also, after u has been determined, directly from (7).

If ν is odd and n is an odd multiple of ν, then

$$t^2 - Du^2 = -4, \tag{8}$$

but if ν is even or n is an even multiple of ν, then

$$t^2 - Du^2 = 4. \tag{9}$$

Equation (9) has the self-evident solution $t = 2$, $u = 0$; in all other solutions of (8) or (9), t and u are different from zero, and we consider here only the positive solutions.

We shall now show that from an arbitrary equation of the form (3), by the formulae (4), (5), one obtains all solutions of (8) and (9).

So let us assume that t, u is any positive solution of equation (8) or (9), and that $\omega = \{a, b, c\}$ is a reduced irrational number belonging to the discriminant D. By the equations (6) we determine P_n, Q_n, P_{n-1}, Q_{n-1}, which evidently become integers, since t and ub are either both even or both odd. From equation (3), which ω satisfies, (2) then results, from which in turn

$$\omega = \frac{P_n\omega + P_{n-1}}{Q_n\omega + Q_{n-1}} \tag{10}$$

is to be inferred; and from (8) or (9) it results that

$$P_nQ_{n-1} - Q_nP_{n-1} = \mp 1, \tag{11}$$

in which, as also in the following formulae, the upper or the lower sign holds according as t, u satisfies equation (8) or (9). But now from (6) it follows that

$$Q_n - Q_{n-1} = \frac{(2c + b)u - t}{2},$$

and because $\{a, b, c\}$ is a reduced number, we have $2c + b > \sqrt{D}$ [§. 125, (4)], hence

$$Q_n - Q_{n-1} > \frac{u\sqrt{D} - t}{2} = \frac{u^2D - t^2}{2(u\sqrt{D} + t)} = \frac{\pm 2}{u\sqrt{D} + t};$$

but since u and t are positive integers and $D > 1$, we have $u\sqrt{D} + t > 2$, hence $Q_n - Q_{n-1} > 0$ for the upper and > -1 for the lower sign.

Now because $Q_n - Q_{n-1}$ must be an integer, it is accordingly equal to or greater than zero, and we have

$$Q_{n-1} \leqq Q_n,$$

where the equality sign is possible only in the case of the lower sign, that is, in the case of equation (9).

Further (since $b < \sqrt{D}$) we have

$$Q_{n-1} = \frac{t - ub}{2} > \frac{t - u\sqrt{D}}{2} = \frac{t^2 - u^2D}{2(t + u\sqrt{D})} = \frac{\mp 2}{t + u\sqrt{D}},$$

so Q_{n-1} is zero or positive, and zero can occur only in the case of the upper sign, that is, in the case of equation (8). From this we conclude

$$0 \leqq Q_{n-1} \leqq Q_n, \tag{12}$$

where the equality sign can occur at the lower limit only in the case of equation (8), that is, of the upper signs, and at the upper limit only in the case of equation (9), that is, of the lower signs.

We now develop the rational fraction $P_n : Q_n$ into a finite continued fraction

$$\frac{P_n}{Q_n} = (\alpha_0, \alpha_1, \alpha_2 \ldots \alpha_{n-1}),$$

by taking n, according to §. 115, I, so that

$$(-1)^n = \mp 1$$

If then $P' : Q'$ denotes the next-to-last convergent, we have

$$P_n Q' - Q_n P' = \mp 1$$

and, by §. 117, II:

$$0 \leqq Q' \leqq Q_n,$$

where the equality sign can occur at the lower limit only for $n = 1$, that is, in the case of the upper signs, and at the upper limit only for $n = 2$, that is, of the lower signs, just as in (12). From this it follows by §. 118, III:

$$Q' = Q_{n-1}, \quad P' = P_{n-1},$$

and hence, by (10),

$$(\alpha_0, \alpha_1, \alpha_2 \ldots \alpha_{n-1}, \omega) = \omega,$$

that is, $[\alpha_0, \alpha_1, \alpha_2 \ldots \alpha_{n-1}]$ is a period of the continued fraction for ω, from which, by the formulae (4), (5), the solution t, u of (7) is derived. It is thereby proved that in this way we find all solutions of equation (7).

From this some further important consequences can be drawn.

From (7) it follows that when u increases, t must increase as well, and hence that when u has the smallest possible positive value, t is also as small as possible. One may therefore speak of a smallest positive solution, which we shall denote by T, U. It is obtained by taking n in the formulae (4), (5) as small as possible, that is, equal to the number of terms ν of the period. Hence if ν is even, only the equation (9), and not the equation (8), is solvable; but if ν is odd, both (8) and (9) are solvable. Since this can depend only on the discriminant D, and not on the particular number ω, it follows

9. that for a given discriminant the various periods contain either all an even or all an odd number of terms.

If D is even, and hence divisible by 4, then t must also be even, and the equation (7) can be divided term by term by 4. If D is odd, then t, u are likewise either both even or both odd; if they are both odd, then their squares, like all odd square numbers, are congruent to 1 modulo 8, and hence we must have $D \equiv 5 \pmod 8$. But it is not always possible, when $D \equiv 5 \pmod 8$, to solve the equation (7) in odd numbers[29]. If $D \equiv 1 \pmod 8$, then t, u must be even.

The examples of §. 126, treated in this way, yield the following smallest positive solutions of the Pell equation, where the factor 4 has been cancelled wherever possible.

$$\begin{aligned} 5^2 - 29.1^2 &= -4 \\ 70^2 - 29.13^2 &= -1 \\ 170^2 - 19.39^2 &= 1 \\ 6^2 - 37.1^2 &= -1 \\ 35^2 - 34.6^2 &= 1. \end{aligned}$$

§. 128.

Derivation of all solutions of the Pell equation from the least positive one.

If t, u is any solution of the Pell equation, then the expression

$$\frac{t^2 - Du^2}{4},$$

[29]Cf. Cayley, Note sur l'équation $x^2 - Dy^2 = \pm 4$, $D \equiv 5 \pmod 8$ Crelle's Journal, Bd. 53. Mathematical papers, Vol. IV. Tables for the solutions of the Pell equation in the form $y^2 = ax^2 + 1$ have been computed by Degen (Canon Pellianus, Havniae 1817). Such a table is also to be found in Legendre's number theory (German translation by Maser).

which has the value ± 1, is the norm of the two conjugate numbers

$$\frac{t+u\sqrt{D}}{2}, \quad \frac{t-u\sqrt{D}}{2}.$$

We shall call them the units belonging to $\sqrt{D}$. If, then, (t_1, u_1), (t_2, u_2) are any two (positive or negative) solutions of the Pell equation

$$t^2 - Du^2 = \pm 4,$$

then the two numbers

$$\Theta_1 = \frac{t_1 + u_1\sqrt{D}}{2}, \quad \Theta_2 = \frac{t_2 + u_2\sqrt{D}}{2}$$

have the norm ± 1, and the same therefore holds also of their product

$$\Theta_3 = \Theta_1\Theta_2 = \frac{t_3 + u_3\sqrt{D}}{2}.$$

Here

$$t_3 = \frac{t_1t_2 + u_1u_2D}{2}, \quad u_3 = \frac{t_1u_2 + t_2u_1}{2},$$

and t_3 and u_3 are, as one sees at once, integers.

For if D is even, then t_1, t_2 are even, and if D is odd, then t_1, u_1 are either both even or both odd, and likewise t_2, u_2. Thus Θ_3 is likewise a unit belonging to $\sqrt{D}$, and it follows that the product of two (and consequently also of several) units is again a unit.

If Θ is an arbitrary unit, then $\pm\Theta^{-1}$ is the unit conjugate to Θ. Among the units different from ± 1 there always belong four together,

$$\pm\Theta, \quad \pm\Theta^{-1},$$

among which one is positive and greater than 1. This one corresponds

the positive values of t, u; a second is positive and smaller than 1, and the two others are negative. If T, U is the smallest positive solution of the Pell equation, then

$$\Theta = \frac{T + U\sqrt{D}}{2}$$

is the smallest improperly fractional positive unit belonging to $\sqrt{D}$, and we can show that all units belonging to $\sqrt{D}$ are contained in the form $\pm\Theta^n$, in which n denotes an integer. For this it suffices to show that the expression Θ^n, for a positive n, yields all the units that are greater than 1. But this is very simple; for if Θ_1 is any one of these units, then it will lie between two successive powers of Θ, since these powers grow to infinity with the exponent; thus

$$\Theta^n \leqq \Theta_1 < \Theta^{n+1},$$

hence

$$1 \leqq \Theta_1\Theta^{-n} < \Theta.$$

But since $\Theta_1\Theta^{-n}$ is likewise a unit, it follows, unless the equality sign holds, that $\Theta_1\Theta^{-n}$ is greater than 1 and smaller than Θ, which is contrary to the assumption that Θ is the smallest improperly fractional unit; hence we must have

$$\Theta_1 = \Theta^n$$

which was to be proved.

Remark concerning the Gauss theory of quadratic forms[30].

[30] Cf. Disq. ar. art. 183 ff., Dirichlet-Dedekind, Vorlesungen über Zahlentheorie. Fourth edition. §. 72 ff.

In order to establish the connection of our considerations with the otherwise well-known Gauss theory of quadratic forms, we add a few remarks, which in our context are not strictly required, and which may therefore also be passed over.

Two quadratic forms,

$$(A, B, C), \quad (-A, B, -C)$$

in the notation of Gauss, are equivalent to one another in the sense of Gauss then and only then, when the period

of the continued fraction development consists of an odd number of terms. Now a reduced number $\omega = \{a, b, c\}$ is the first root of

$$(a, \tfrac{1}{2}b, -c) \quad \text{or} \quad (2a, b, -2c)$$

(according as D is even or odd) and the second root of

$$(-a, \tfrac{1}{2}b, c) \quad \text{or} \quad (-2a, b, 2c).$$

If these two forms are equivalent, that is, if the periods of ω consist of an odd number of terms, then to each two-sided class of forms there corresponds one, and to each pair of opposite classes two systems of equivalent numbers ω; hence the class number in the Gaussian sense is just as great as the number of periods of the reduced numbers ω.

But when the number of terms of the period is even, the forms $(A, B, C), (-A, B, -C)$ are not equivalent, and the Gaussian class number is twice as great as the number of periods of the ω.

§. 129.
Approximate calculation of the real roots of a numerical equation by continued fractions.

On the theory of continued fractions is founded a procedure due to Lagrange for calculating the real roots of a numerical equation with any desired degree of approximation, a procedure which can now be set forth in a few words.

Let $f(x) = 0$ be a real equation which has at least one real root. By one of the various methods of Sections VIII and IX we can then always find an integer a_0 such that one or more of the roots of $f(x)$ lie between a_0 and $a_0 + 1$. If, then, we transform $f(x)$ by the substitution

$$x = a_0 + \frac{1}{x_1}$$

into $f_1(x_1)$, then $f_1(x_1)$ will have one or more roots which are greater than 1. We therefore determine a positive integer a_1 such that one or more of the roots of $f_1(x_1)$ lie between a_1 and $a_1 + 1$, and transform $f_1(x_1)$ by the substitution

$$x_1 = a_1 + \frac{1}{x_2}$$

in $f_2(x_2)$. The equation $f_2 = 0$ then again has at least one root that is greater than 1, and which therefore lies between the two positive integers a_2 and $a_2 + 1$, and so on. Finally one will in this way come to isolate a single root, and one obtains in the sequence of convergents

$$(a_0), \quad (a_0, a_1), \quad (a_0, a_1, a_2), \quad (a_0, a_1, a_2, a_3) \ldots$$

a sequence of rational numbers which lie alternately above and below one of the roots x and approach this root without limit. If one has advanced in this way to the convergent

$$(a_0, a_1, a_2 \ldots a_{\nu-1}) = \frac{P_\nu}{Q_\nu}$$

then this differs from the true value of x by less than $1 : Q_\nu^2$. The size of the denominator thus gives directly a measure of the accuracy already attained.

In the ordinary cases one will obtain the numbers $a_0, a_1, a_2 \ldots$ simply by attending to the changes of sign of the functions $f, f_1, f_2 \ldots$

To test this procedure on an example, we take the cubic equation which we have already considered earlier:

$$x^3 - 2x - 2 = 0.$$

It has a root between 1 and 2, so we have to set $a_0 = 1$.

If one actually carries out the transformation indicated above, one obtains the following chain of equations:

$$\begin{aligned}
x^3 - 2x - 2 = 0, &\quad a_0 = 1\\
3x^3 - x^2 - 3x - 1 = 0, &\quad a_1 = 1\\
2x^3 - 4x^2 - 8x - 3 = 0, &\quad a_2 = 3\\
9x^3 - 22x^2 - 14x - 2 = 0, &\quad a_4 = 1\\
46x^3 - 6x^2 - 32x - 9 = 0, &\quad a_3 = 2\\
x^3 - 94x^2 - 132x - 46 = 0, &\quad a_5 = 95\\
3561x^3 - 9183x^2 - 191x - 1 = 0, &\quad a_6 = 2,
\end{aligned}$$

hence

$$x = (1, 1, 3, 2, 1, 95, 2 \ldots)$$

and the convergents:

$$\frac{1}{1}, \frac{2}{1}, \frac{7}{4}, \frac{16}{9}, \frac{23}{13}, \frac{2201}{1244}, \frac{4425}{2501} \cdots$$

If we convert the two last fractions into decimal fractions, we obtain for x the two limits:

$$1{,}7692926; \quad 1{,}76929228,$$

and the error in the latter, a number slightly too small, is less than

$$0{,}00000016.$$

That one here obtains a good result after few steps rests on the circumstance that a larger partial denominator 95 occurs fairly early. In most cases one obtains small numbers for the partial denominators; the denominators of the convergents then grow slowly, and one only laboriously obtains a result of greater accuracy.

The finding of larger partial denominators, e.g. 95, is made easier by the remark that, when a root of an equation has a large value, the negative coefficient of the second-highest power of the unknown gives an approximation to this root, even if only a rough one, as e.g. in the next to last of the above equations 94 to 95.

§. 130.

Rational Roots of Equations with Integral Coefficients. Reducible Equations.

We close this section with a consideration of equations with integral coefficients which, even though it does not stand in immediate relation to the theory of continued fractions, yet most fittingly finds a place here.

When one has to do with equations whose coefficients are rational numbers, one will first look for possible rational roots. If the equation

$$f(x) = a_0 x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n = 0$$

has rational coefficients, we can always assume that these coefficients are integers; to reduce the case of fractional coefficients to this one, one need only multiply the whole equation by a common multiple of all the denominators.

We can, however, further assume that $a_0 = 1$; for if in the product $a_0^{n-1} f(x)$ we set

$$a_0 x = x_1,$$

the equation becomes

$$x_1^n + a_1 x_1^{n-1} + a_0 a_2 x_1^{n-2} + \cdots + a_0^{n-1} a_n = 0.$$

We shall therefore now assume that in the equation

$$x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n = 0 \tag{1}$$

the coefficients $a_1, a_2 \ldots a_n$ are integers. A rational root of (1) cannot be a fractional number; for if we assume that (1) is satisfied by the fraction

$$x = \frac{p}{q},$$

where p and q are integers without a common divisor, and q is positive and greater than 1, then multiplication by q^{n-1} gives

$$\frac{p^n}{q} + a_1 p^{n-1} + a_2 q p^{n-2} + \cdots + a_n q^{n-1} = 0.$$

Thus $p^n : q$ would have to be an integer, which is impossible.

Now if the equation (1) can be satisfied by putting an integer for x, then this integer, as the equation (1) shows, must necessarily be a divisor of a_n, and one has therefore only to substitute the various divisors of a_n, taken with positive and negative sign, into the equation by way of trial.

To the question of rational roots of an equation with integral coefficients there also comes back the general question whether an integral rational function

$$f(x) = x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n,$$

whose coefficients $a_1, a_2 \ldots a_n$ are integers, can be divisible by a rational function of lower degree ν:

$$\varphi(x) = x^\nu + \alpha_1 x^{\nu-1} + \alpha_2 x^{\nu-2} + \cdots + \alpha_{\nu-1} x + \alpha_\nu$$

with rational coefficients $\alpha_1, \alpha_2 \ldots \alpha_\nu$. If $f(x)$ is divisible by $\varphi(x)$, then the quotient $\psi(x)$ is again an integral rational function with rational coefficients of degree $n - \nu$:

$$\psi(x) = x^{n-\nu} + \beta_1 x^{n-\nu-1} + \cdots + \beta_{n-\nu-1} x + \beta_{n-\nu},$$

and since $f(x)$ is also divisible by $\psi(x)$, we can, in

the decision of our question assume $\nu \leqq \frac{1}{2} n$. Moreover it follows from §. 2 that the coefficients α and β of φ and ψ must be integers.

Now in order to decide on the possibility of such a division, one takes the function $\varphi(x)$ with indeterminate coefficients α, and divides $f(x)$ by $\varphi(x)$ (according to §. 3). There then results a remainder of degree $\nu - 1$, and if one sets its coefficients, which are all rational functions of the α, equal to zero, one obtains ν equations which these coefficients must satisfy. By elimination (§. 50) one can produce an equation which contains only one of these unknowns, and which must then have an integral root. It will often be more expedient, however, not actually to carry out this elimination,. but to attempt directly to satisfy the system of the ν equations by integral values of the α. The selection of the numbers to be tried can be greatly restricted by many an artifice resting on theorems of number theory.

As an example we shall take an equation of the tenth degree from the theory of elliptic functions, where such problems very frequently occur in such a way that the decomposability into factors of a definite degree is theoretically established, and where it is then a matter of finding these factors [1]). Such an equation is

$$x^{10} - 36\,x^8 + 528\,x^6 - 3897\,x^4 + 14354\,x^2 - 21025 = 0.$$

We ask whether the left side is decomposable into two factors of the fifth degree. If one of the two factors is

$$x^5 + \alpha x^4 + \beta x^3 + \gamma x^2 + \delta x + \varepsilon$$

then, since no odd powers occur in the given equation, the other is

$$x^5 - \alpha x^4 + \beta x^3 - \gamma x^2 + \delta x - \varepsilon,$$

and hence we must have:

$$\begin{aligned} x^{10} - 36\,x^8 + 528\,x^6 - 3897\,x^4 + 14354\,x^2 - 21025 \\ = (x^5 + \beta x^3 + \delta x)^2 - (\alpha x^4 + \gamma x^2 + \varepsilon)^2. \end{aligned}$$

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[1]) Cf. the author's work: "Elliptische Functionen und algebraische Zahlen", Braunschweig 1891. The example treated here, from the theory of the transformation of degree 47, is taken from the paper: "Ein Beitrag zur Transformationstheorie der elliptischen Functionen etc." by H. Weber (1893). Mathematische Annalen, Bd. 43.

If one sets corresponding coefficients equal to one another, it follows that

1. $\alpha^2 - 2\beta = 36,$
2. $\beta^2 + 2\delta - 2\alpha\gamma = 528,$
3. $\gamma^2 + 2\alpha\varepsilon - 2\beta\delta = 3897,$
4. $\delta^2 - 2\gamma\varepsilon = 14354,$
5. $\varepsilon^2 = 21025,$

and it is now a matter of seeking integral values of $\alpha, \beta, \gamma, \delta, \varepsilon$ that satisfy these five equations. First of all, from 5. one obtains

$$\varepsilon = \sqrt{21025} = 145 = 5\,.\,29;$$

and one may take ε positive, because the term independent of x is in any case positive in one of the two factors. Equation 4. yields, if one takes on the right the residue with respect to the modulus 145,

$$\delta^2 \equiv -1 \pmod{145}, \tag{4}$$

and this congruence has the four solutions

$$\delta \equiv \pm 12,\ \pm 17 \pmod{145}.$$

Since δ must moreover be even, by 4., δ could have the following values:

$$\delta = \pm 12,\ \pm 128,\ \pm 162.$$

For the time being one will go no further, since large numerical values for δ are improbable from the outset. Corresponding to these values of δ one obtains from 4. for γ:

$$\gamma = -49,\ 7,\ 41.$$

If we apply in addition the modulus 5, it follows that:

$$\gamma^2 \equiv 1,\ -1,\ 1 \pmod 5$$

and from 3.:

$$\beta \equiv \pm 1,\ \pm 2,\ \pm 1 \pmod 5$$

and from 1.:

$$\alpha^2 \equiv 1 \pm 2,\ 1 \mp 1,\ 1 \pm 2 \pmod 5.$$

From this it follows, since only 0, 1 and 4, not 2 and 3, can be congruent to a square with respect to the modulus 5, that the signs in δ must be taken thus:

$$\delta = -12,\ +128,\ -162$$
$$\alpha \equiv \pm 2,\ 0,\ \pm 2 \pmod 5.$$

Equation 2. shows further that the middle case is to be excluded, since the right-hand side is not divisible by 5.

That $\delta = -12$ must also be excluded follows from equation 2., which for $\delta = -12$, $\gamma = -49$ has as a consequence the congruence

$$\beta^2 \equiv 6 \pmod 7$$

which, however, is not solvable.

There thus remains only:

$$\delta = -162, \quad \gamma = 41.$$

From 3. there follows for β the congruence:

$$17\,\beta \equiv 93 \pmod{145}$$

or

$$\beta \equiv -1 \pmod 5, \quad 2\,\beta \equiv -1 \pmod{29},$$

hence

$$\beta \equiv 14 \pmod{145}.$$

If one takes $\beta = 14$, then $\alpha = -8$ follows, and all the equations 1. to 5. are satisfied. With this, then, the decomposition has been obtained, which is easily verified afterwards.

$$x^{10} - 36\,x^8 + 528\,x^6 - 3897\,x^4 + 14354\,x^2 - 21025 =$$
$$(x^5 - 8\,x^4 + 14\,x^3 + 41\,x^2 - 162\,x + 145)$$
$$(x^5 + 8\,x^4 + 14\,x^3 - 41\,x^2 - 162\,x - 145).$$

Twelfth Section.

Theory of the Roots of Unity.

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§. 131.

The Roots of Unity.

By a root of unity one understands in general a real or imaginary number of such a nature that some power of it with an integral exponent is equal to 1. We have already become acquainted with these roots of unity in §. 33 and represented them by trigonometric and exponential functions. We

have, moreover, made use of special cases, e. g. in §. 35 the third roots of unity, which were there expressed without trigonometric functions.

In all deeper investigations of algebraic equations a more precise knowledge of the roots of unity and of their properties is now indispensable. In this section we shall therefore occupy ourselves more thoroughly with the elementary part of the algebraic theory of the roots of unity, without making use of their representation by trigonometric functions. On account of the geometric application to the construction of the regular polygons, to which we have already pointed in §. 33, the theory of the roots of unity is also called the theory of the division of the circle. It is in essence a creation of Gauss [1]).

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[1]) Gauss, Disq. ar. Sectio VII.

When the nth power of a number r is equal to 1, that is, when the equation

$$r^n = 1 \tag{1}$$

is satisfied for a positive integer n, then r is called an nth root of unity, or a root of unity of degree n. Thus all the nth roots of unity, and only these, are roots of the equation of degree n

$$f(x) = x^n - 1 = 0. \tag{2}$$

We have

$$f'(x) = n\,x^{n-1}, \tag{3}$$

and consequently $f(x)$ has no divisor in common with $f'(x)$; hence $f(x)$ has no multiple roots, and there are n, and no more, distinct nth roots of unity.

If r is an nth root of unity, then so is every integral power of r; for from $r^n = 1$ it follows that $r^{kn} = 1$, if k is any positive or negative integer ($k = 0$ not being excluded); hence r^k is also an nth root of unity. But since there are only n roots of unity of degree n, the powers r^k are not all distinct from one another. On this point the following now holds:

If two numbers k, k' differ by a multiple of n, that is, if

$$k' \equiv k \pmod{n} \tag{4}$$

then also

$$r^k = r^{k'}. \tag{5}$$

For if $k' = k + h\,n$, then

$$r^{k'} = r^k r^{hn},$$

from which, since $r^n = 1$, equation (5) follows.

Thus in the sequence of numbers

$$1, r, r^2, r^3 \ldots r^{n-1} \tag{6}$$

all the r^k that are distinct from one another are certainly contained; but conversely the quantities (6) need not all be distinct from one another.

Let us assume that k and $k' = k + \mu$ are two numbers of the sequence

$$0, 1, 2 \ldots n-1$$

and

$$r^k = r^{k+\mu},$$

it follows that $r^\mu = 1$ must hold. Thus no term that has already occurred can recur in the sequence (6) before the first term 1 occurs for the second time, and if μ is the least positive number for which $r^\mu = 1$, then the numbers:

$$1, r, r^2 \ldots r^{\mu-1} \tag{7}$$

are all different from one another.

Then μ must be a divisor of n. For by division the integers h, μ' can be determined so that

$$n = h\mu + \mu'; \quad 0 \leqq \mu' < \mu.$$

But then, as follows from

$$r^n = r^{h\mu} r^{\mu'}$$

we also have $r^{\mu'} = 1$, that is, since μ is to be the least positive number for which $r^\mu = 1$, $\mu' = 0$, or n is divisible by μ.

Thus r is at the same time a μth root of unity, but not a root of unity of any still lower degree.

The number r is called a primitive nth root of unity if it is not at the same time a root of unity of a lower degree.

From this definition it follows that the numbers of the sequence (6), if r is a primitive nth root of unity, are all different from one another, and that all the nth roots of unity are contained among them, but that they make up only a part of the nth roots of unity if r is an imprimitive nth root of unity.

Every root of unity whose degree is a divisor of n different from n is at the same time an imprimitive nth root of unity.

If r is at the same time an nth and an mth root of unity, then it is also a μth root of unity, if μ is the greatest common divisor of n and m.

For by §. 118 the integers x, y can be determined so that

$$mx + ny = \mu$$

and consequently

$$r^\mu = r^{mx} r^{ny} = 1.$$

To this corresponds the other theorem:

If $r_1, r_2 \ldots$ are roots of unity of the degrees $n_1, n_2 \ldots$, then they are all at the same time roots of unity of the degree m, if m denotes any common multiple of $n_1, n_2 \ldots$.

§. 132.
Primitive Roots of Unity.

That primitive roots of unity exist for every degree n we have not yet proved in the preceding section. We must first make good this omission, and in doing so we shall also determine the exact number of the primitive roots of unity.

Let the degree n be decomposed into two factors a, b which are relatively prime to one another, so that

$$n = ab,$$

and let α be an ath and β a bth root of unity. Then the product

$$r = \alpha\beta \tag{1}$$

is an nth root of unity. If α', β' are two other ath and bth roots of unity, then $r' = \alpha'\beta'$ is also an nth root of unity, and it is to be shown that r' is different from r unless at the same time $\alpha = \alpha'$ and $\beta = \beta'$.

For since a, b are relatively prime, one can by §. 118 determine the integers x, y so that

$$ax + by = 1 \tag{2}$$

holds; and then it follows from (1) that

$$\alpha = r^{by}, \quad \beta = r^{ax}.$$

Accordingly α and β are completely determined by r, and if $\alpha'\beta'$ is also to be equal to r, then we must have $\alpha = \alpha'$, $\beta = \beta'$.

If, therefore, we let α in (1) run through all the ath, and β through all the bth roots of unity, then r takes exactly $ab = n$ different values, and it follows:

I. that in the form $\alpha\beta$ all the nth roots of unity can be represented,

and further:

II. that r is a primitive nth root of unity if and only if α is a primitive ath and β a primitive bth root of unity.

For, first, let μ be the smallest positive exponent for which $r^\mu = 1$; then also $\alpha^\mu\beta^\mu = 1$, and from this it follows by (2), if we raise both sides to the powers by and ax,

$$\alpha^\mu = 1, \quad \beta^\mu = 1.$$

Now if $\mu < n$, it cannot be divisible at once by a and by b, and hence a and b cannot both be the least positive exponents of the powers of α and β that become equal to 1; that is to say, if r is not a primitive nth root of unity, then α and β are not at the same time primitive ath and bth roots of unity; or, if α and β are primitive ath and bth roots of unity, then their product r is a primitive nth root of unity. On the other hand it is clear that if α or β is at the same time a root of unity of lower degree than a or b, then r too will be a root of unity of degree lower than the nth.

If n decomposes into several factors $a, b, c \ldots$, of which every two are relatively prime to one another, and if $\alpha, \beta, \gamma \ldots$ are roots of unity of the degrees $a, b, c \ldots$, and one sets

$$r = \alpha\beta\gamma \ldots, \tag{3}$$

then one concludes, by repeated application of the preceding theorems, that in (3) all the nth roots of unity, and each only once, are contained, and further that r is a primitive nth root of unity then and only then, when $\alpha, \beta, \gamma \ldots$ are primitive roots of unity of the degrees $a, b, c \ldots$.

If we now denote the number of primitive nth roots of unity by $\varphi(n)$, it follows from what has here been proved that

$$\varphi(n) = \varphi(a)\varphi(b)\varphi(c)\ldots, \tag{4}$$

if

$$n = abc\ldots$$

and $a, b, c\ldots$ are numbers which, two by two, are relatively prime to one another.

Now every number n can be decomposed in one and only one way into a product of prime powers

$$n = p^{\pi} p_1^{\pi_1} p_2^{\pi_2} \ldots,$$

in which $p, p_1, p_2\ldots$ are distinct primes and $\pi, \pi_1, \pi_2\ldots$ are positive exponents, so that from the formula (4) it follows:

$$\varphi(n) = \varphi(p^{\pi})\,\varphi(p_1^{\pi_1})\,\varphi(p_2^{\pi_2})\ldots, \tag{5}$$

and that it therefore only remains to decide whether and how many primitive roots of unity of the degree p^{π} exist.

But this question is very simple to decide. For if ϱ is a root of unity of the degree p^{π}, then the lowest degree to which ϱ belongs as a root of unity is a divisor of p^{π}, hence a power of p, and if it is therefore not equal to

p^{π}, a divisor of $p^{\pi-1}$; that is, every non-primitive root of unity of degree p^{π} is at the same time a root of unity of degree $p^{\pi-1}$. But since there are p^{π} roots of unity of degree p^{π} and only $p^{\pi-1}$ roots of unity of degree $p^{\pi-1}$, there must be

$$p^{\pi} - p^{\pi-1} = p^{\pi}\left(1 - \frac{1}{p}\right)$$

primitive roots of unity of degree p^{π}. From this, by (5), one obtains the number of all primitive nth roots of unity

$$\varphi(n) = n\,\Pi\left(1 - \frac{1}{p}\right), \tag{6}$$

in which the product sign Π extends over all the mutually distinct primes dividing n. Only in the case $n = 1$ does the formula (6) no longer apply; in this case $\varphi(1) = 1$ is to be set. The number $\varphi(n)$ is therefore never equal to zero. Since, by this, for every degree n there is at least one primitive root of unity r, all nth roots of unity can be represented by the powers of r:

$$1, r, r^2, r^3 \ldots r^{n-1}. \tag{7}$$

If r^k is any power of r, then $r^{km} = 1$ will hold if and only if km is divisible by n. If, then, $n = n'n''$ and n' is the greatest common divisor of k and n, then m must be divisible by n'', and n'' is the exponent of the lowest power of r^k which becomes equal to 1; that is, r^k is a primitive n''th root of unity. From this follows the theorem:

III. If r is a primitive nth root of unity, then r^k is a primitive nth root of unity if and only if k is relatively prime to n.

If we take k from the sequence of numbers $1, 2, 3\ldots n$, there results the theorem of number theory that $\varphi(n)$ is equal to the number of the numbers which are not

greater than n and relatively prime to n. This is the original definition of the symbol $\varphi(n)$ in general use in number theory.

If k is relatively prime to n, one can determine an integer x so that $kx \equiv 1 \pmod{n}$. If then r, r_1 are two nth roots of unity, $r^k = r_1^k$ can hold only when $r = r_1$, as follows by raising to the power x. From this it follows, by III.:

IV. If k is relatively prime to n and r runs through the sequence of the primitive nth roots of unity, then r^k runs through the same sequence of numbers, though in a different order.

Finally we cite the further theorem: –

V. If n is a prime, then every nth root of unity, with the exception of 1, is a primitive nth root of unity.

§. 133.

Equations for the primitive roots of unity of degree n.

All nth roots of unity are, as we have seen, roots of an equation $f_n(x) = 0$, where

$$f_n(x) = x^n - 1 \tag{1}$$

is understood. If now we free the function $f_n(x)$ from all factors which it has in common with other functions of the same form $f_{n_1}(x)$, which is done by rational operations (§. 6), we obtain an equation $X_n = 0$, which is satisfied by the primitive nth roots of unity and by these only, and X_n has the form

$$X_n = x^\nu + a_1 x^{\nu-1} + \cdots + a_\nu. \tag{2}$$

The degree ν is equal to $\varphi(n)$, and the coefficients $a_1, a_2 \ldots a_\nu$ are rational numbers.

In seeking the common factors of f_n and f_{n_1} we may, for n_1, restrict ourselves to the divisors of n. How the function X_n may be formed in a simple way we shall presently see more closely. But first we prove a general theorem concerning these functions.

The nth roots of unity comprise all primitive μth roots of unity, where μ is any divisor of n, n itself and 1 included, since it is precisely the unity 1 itself that is to be regarded as the primitive first root of unity. If, then, we let μ run through all divisors of n, and observe that two different X_μ never have a common divisor, and that neither $f_n(x)$ nor X_μ contains multiple factors, it follows that

$$f_n(x) = \Pi X_\mu, \tag{3}$$

in which the product sign Π extends over all divisors μ of n.

From this we obtain, incidentally, a proof of the number-theoretic theorem

$$n = \sum^{\mu} \varphi(\mu), \tag{4}$$

if we set the degrees of the functions on the right and the left side of (3) equal to one another.

On the other hand we conclude, by Gauss's theorem (§. 2), that the coefficients of all the X_μ are integers. For if fractional numbers occurred among them, the product could not consist solely of integral coefficients, as by (3) and (1) it nevertheless must.

All the functions $f_n(x)$ have the divisor $x - 1$, and if we carry out the division, we obtain

$$\frac{x^n - 1}{x - 1} = x^{n-1} + x^{n-2} + \cdots + x + 1. \tag{5}$$

From this we conclude that, if r is any primitive or non-primitive nth root of unity, with the sole exception of $r = 1$, always

$$1 + r + r^2 + \cdots + r^{n-1} = 0, \tag{6}$$

while for $r = 1$ the sum on the left side of (6) obviously has the value n.

If n is a prime, then apart from 1 there are no imprimitive nth roots of unity, and therefore, when n is a prime, which we shall indicate by writing p for it, we have

$$X_p = x^{p-1} + x^{p-2} + \cdots + x + 1. \tag{7}$$

Just as simply, X_n can be formed when n is a prime power, that is,

$$n = p^{\pi},$$

or, if for brevity we set $p^{\pi-1} = p'$:

$$n = p'p.$$

The nth roots of unity consist in this case of the primitive nth roots of unity and of the p'th roots of unity, and thus:

$$X_n = \frac{x^{pp'} - 1}{x^{p'} - 1} = x^{p'(p-1)} + x^{p'(p-2)} + \cdots + x^{p'} + 1 \tag{8}$$

If $p' > 1$, and hence $\pi > 1$, then the term with the $(\nu - 1)$th power of the unknown, whose coefficient is equal to the negative sum of the roots, is missing from this equation. We therefore have the theorem:

The sum of all primitive nth roots of unity, when n is a higher power of a prime, is always equal to zero.

For the general formation of X_n we shall apply a recurrent procedure; we assume X_n as already formed, denote by p a prime which does not divide n, by p', as above, the $(\pi - 1)$th power of p, and now form $X_{np'p}$, in which of course p' may also be equal to 1.

If we denote by r the primitive nth roots of unity, by α every root of unity of degree pp', and by α' every root of unity of degree p', then one obtains all the primitive roots of unity of degree npp' by taking away from all the $r\alpha$ the $r\alpha'$ (§. 132, II). But the $r\alpha$ are the roots of the equation:

$$X_n(x^{pp'}) = 0,$$

because the quantities

$$(r\alpha)^{pp'} = r^{pp'},$$

agree, apart from their order, with the r themselves (§. 132, IV). Likewise the $r\alpha'$ are the roots of the equation

$$X_n(x^{p'}) = 0;$$

and from this it follows that

$$X_{npp'}(x) = \frac{X_n(x^{pp'})}{X_n(x^{p'})}. \tag{9}$$

The value $n = 1$ forms no exception to this, if we understand by X_1 the function $x - 1$. According to this we can form X_n comparatively simply in all cases. We consider some special cases.

Let us suppose that n contains only two distinct primes p, q, so that

$$n = pp'qq',$$

then the formulae (8) and (9) yield:

$$X_n = \frac{(x^n - 1)\left(x^{\frac{n}{pq}} - 1\right)}{\left(x^{\frac{n}{p}} - 1\right)\left(x^{\frac{n}{q}} - 1\right)},$$

a formula which can easily be generalised by complete induction as follows.

If one denotes by μ_1 all the numbers that arise from n when one divides n by an even number of distinct prime divisors of n (n itself included), and by μ_2 the numbers that arise from n when one divides n by an odd number of such prime divisors, then

$$X_n = \frac{\prod_{\mu_1}(x^{\mu_1} - 1)}{\prod_{\mu_2}(x^{\mu_2} - 1)}. \tag{10}$$

The proof results from (9) for npp', if one assumes that the correctness has already been proved for n.

If r denotes each of the primitive nth roots of unity, then for odd n the $-r$ are the primitive $2n$th roots of unity (§. 132, I, II). Accordingly, for odd n,

$$X_{2n}(x) = X_n(-x). \tag{11}$$

If n is a power of 2, then by (8):

$$X_n = \frac{x^n - 1}{x^{\frac{n}{2}} - 1} = x^{\frac{n}{2}} + 1. \tag{12}$$

If we set $x = 1$ in formula (8), then X_n takes the value p if n is a power of p. On the other hand, formula (9) yields, when $n > 1$, the value 1 for $X_{npp'}$ (for $n = 1$ the numerator and denominator on the right-hand side would be $= 0$).

We thus obtain the theorem:

V. The function X_n takes for $x = 1$ the value p if n is a power of the prime p, and the value 1 if n has more than one prime as divisor.

§. 134.
Irreducibility.

The functions X_n, which are defined in the preceding section, have the property of irreducibility, which is very important for their deeper study and all applications, i.e.

It is not possible to decompose a function X_n of x into factors of lower degree which themselves

integral rational functions of x with rational numerical coefficients.

This important theorem was set up and first proved by Gauss for the case that n is a prime. Later many other proofs have been given, some of them more general and simpler. We follow here a proof by Kronecker.

In the function X_n, as we have seen above, the highest power of x has the coefficient 1, while the remaining coefficients are integers. If we now assume that X_n can be decomposed into two factors with rational coefficients

$$X_n = \varphi(x)\,\psi(x) \tag{1}$$

then the product of the coefficients of the highest powers of x in φ and ψ must be equal to 1, and we may therefore assume that these highest coefficients in φ and ψ themselves have the value 1 (since otherwise we would divide by the product of the two coefficients). But then it follows from

the theorem of Gauss (§. 2) that the remaining coefficients of $\varphi(x)$ and $\psi(x)$ must be integers, and hence that $\varphi(x)$ and $\psi(x)$, when an integer is put for x, must themselves pass into integers.

If we therefore set $x = 1$, it follows that:

$$\varphi(1)\,\psi(1) = X_n(1),$$

and so by Theorem V. this is equal to 1 or to a prime p. Accordingly, of the two factors $\varphi(1)$, $\psi(1)$, one must certainly have the value ± 1, while the other is equal to ± 1 or $\pm p$. We therefore assume that

$$\varphi(1) = \pm 1. \tag{2}$$

The roots r of the equation $X_n = 0$ now distribute themselves between the two equations $\varphi = 0$, $\psi = 0$, and since neither of the two factors φ, ψ is independent of x, there must certainly be one among the primitive nth roots of unity r, which we shall denote by ϱ, for which

$$\varphi(\varrho) = 0$$

holds.

Now if r denotes any primitive nth root of unity, then among the powers

$$r,\ r^2,\ r^3\ \ldots\ r^{n-1}$$

all nth roots of unity with the exception of 1, and therefore also ϱ, are contained; and it follows therefore that the product

$$\Phi(x) = \varphi(x)\,\varphi(x^2)\,\varphi(x^3)\ldots\varphi(x^{n-1}) \tag{3}$$

vanishes for $x = r$. But since r may be any root of X_n, and these roots are all different from one another, $\Phi(x)$ must be divisible by X_n, hence

$$\Phi(x) = X_n\,\Psi(x). \tag{4}$$

The function $\Phi(x)$, as appears from its mode of formation, has the coefficient 1 for the highest power of x and otherwise integer coefficients; likewise X_n. Hence $\Psi(x)$ must also have the coefficient 1 for the highest power of x and otherwise integer coefficients. But it follows from (2) and (3) that $\Phi(x)$ takes the value ± 1 for $x = 1$, hence

$$\pm 1 = X_n(1)\,\Psi(1), \tag{5}$$

wherein $\Psi(1)$ must be an integer.

Now if n is a prime power p^π, then $X_n(1) = p$, and this contradicts equation (5). Thus the irreducibility of X_n is proved for the case in which n is a prime or a power of a prime.

The general proof therefore still requires the following: If a and b are two integers without a common divisor, and if it is further assumed that X_a and X_b are irreducible, it follows that X_{ab} is also irreducible.

To prove this, we set $n = ab$ and assume that $\varphi(x)$ is a rational factor of X_n. The roots of $\varphi(x) = 0$ are then contained among the primitive nth roots of unity, and are therefore of the form

$$\varrho = \alpha\beta, \tag{6}$$

wherein α and β denote certain primitive ath and bth roots of unity (§. 132).

Now (by a special case of the Tschirnhausen transformation) a function $\Phi(x)$ with rational coefficients can be derived from $\varphi(x)$, whose roots are the ath powers of the roots of $\varphi(x)$. The roots of $\Phi(x)$ are therefore

$$\varrho^a = \beta^a,$$

that is, since a is relatively prime to b, all primitive bth roots of unity; and if therefore any ϱ exists at all, $\Phi(x)$ must have a non-constant factor in common with X_b. But now we have assumed, as already proved, that X_b has no rational factor other than itself; hence $\Phi(x)$ must be divisible by

X_b, and the β^a, and consequently also the β themselves, which occur in (6), together comprise all the primitive bth roots of unity. In the same way it is now proved that α in (6) runs through all the primitive ath roots of unity, and that therefore ϱ runs through all the primitive nth roots of unity.

The roots of $\varphi(x)$ thus agree completely with the totality of the roots of X_n, and consequently, if for the determination of a numerical factor we further stipulate, say, that the highest power of x in $\varphi(x)$ shall have the coefficient 1, $\varphi(x)$ must be identical with X_n.

Under the assumptions made, therefore, X_n is not decomposable into factors of lower degree with rational coefficients.

By the same means we can prove the following theorem of Kronecker, which expresses the irreducibility of the functions X_n in a wider sense[31].

The function X_n cannot be decomposed into factors which are integral and rational with respect to x, and which contain in their coefficients only rational numbers and roots of unity whose degree is relatively prime to n.

Suppose that a is relatively prime to n, and that r is a primitive nth, α a primitive ath root of unity, and let $\varphi(x, \alpha)$ be a divisor of X_n, so that at least for one r

$$\varphi(r, \alpha) = 0. \tag{7}$$

Now let ϱ be a primitive nath root of unity, hence a root of $X_{na} = 0$; then both r and α are powers of ϱ (§. 131), say

$$r = \varrho^\xi, \quad \alpha = \varrho^\eta,$$

where ξ is relatively prime to n but divisible by a, and η relatively prime to a but divisible by n. Hence

$$\varphi(\varrho^\xi, \varrho^\eta) = 0. \tag{8}$$

Now, as we have just proved, X_{na} is irreducible, and consequently the function $\varphi(x^\xi, x^\eta)$, which has a divisor in common with X_{na}, must be divisible by this function; that is, for every integer h that is relatively prime to na:

$$\varphi(\varrho^{h\xi}, \varrho^{h\eta}) = 0. \tag{9}$$

Now if s is an arbitrary number coprime to n, we can determine h so that

$$h \equiv s \pmod{n}, \quad h \equiv 1 \pmod{a}$$

and accordingly it follows from (9) that

$$\varphi(r^s, \alpha) = 0,$$

[31] Mém. sur les facteurs irréductibles de l'expression $(x^n - 1)$. Liouville's Journal, Vol. XIV.

that is, equation (7) is satisfied for all roots of X_n. Hence $\varphi(x,\alpha)$ is divisible by X_n, and since on the other hand X_n was assumed to be divisible by $\varphi(x,\alpha)$, it must be identical with X_n up to a numerical factor.

§. 135.
The Discriminant of the Cyclotomic Equation.

The roots of unity of degree n can be represented in transcendental form by

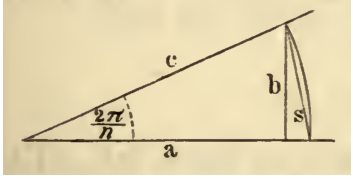
$$r = e^{\frac{2\pi ik}{n}} = \cos\frac{2\pi k}{n} + i\sin\frac{2\pi k}{n}. \tag{1}$$

This root of unity is primitive or not primitive according as k is coprime to n or not. The simplest among the primitive ones is obtained for $k = 1$, namely

$$r_0 = e^{\frac{2\pi i}{n}} = \cos\frac{2\pi}{n} + i\sin\frac{2\pi}{n} = a + bi. \tag{2}$$

The angle $2\pi : n$ is the nth part of the whole circumference of the circle. The various points of division, which one obtains by laying off this angle on the circumference from an arbitrary initial point, determine the regular n-gon inscribed in the circle.

Fig. 28.



The points of division can be constructed if one knows r, and with it a and b (or even only one of these two quantities). In particular, for the side s of the regular n-gon, if the radius c of the circumscribed circle is taken equal to 1, we obtain

$$s = 2\sin\frac{\pi}{n} = \sqrt{2\left(1 - \cos\frac{2\pi}{n}\right)}.$$

The remaining quantities r stand in the same relation to the other points of division. On account of this geometric

meaning we call the equation $X_n = 0$, whose roots are the quantities (1), the cyclotomic equation.

We now determine the discriminant of the cyclotomic equation $X_n = 0$, but restrict ourselves to the case in which n is an odd prime.

We then have

$$X_n = x^{n-1} + x^{n-2} + \cdots + x + 1, \tag{3}$$

and since only the one imprimitive nth root of unity 1 exists, we have

$$f_n(x) = x^n - 1 = (x-1)X_n, \tag{4}$$

hence

$$f_n'(x) = nx^{n-1} = X_n + (x-1)X_n'. \tag{5}$$

Now if D is the discriminant of X_n, then by §. 46

$$D = (-1)^{\frac{n-1}{2}} \Pi X_n'(r), \tag{6}$$

where the product sign Π extends over all roots r of the equation $X_n = 0$, and where, since n is odd, the equality

$$(-1)^{\frac{n(n-1)}{2}} = (-1)^{\frac{n-1}{2}}$$

could be used. But by (5) we have

$$nr^{n-1} = (r-1)X_n'(r), \tag{7}$$

and since the product of all the r is equal to 1 (because it is equal to the term of X_n independent of x), and since

$$\Pi(x-r) = \Pi(r-x) = X_n,$$

hence

$$\Pi(r-1) = X_n(1) = n, \qquad (\S.\,133)$$

it follows from (6) and (7) that

$$D = (-1)^{\frac{n-1}{2}} n^{n-2}. \tag{8}$$

The roots of X_n, if r is one of them, are all contained in the form

$$r, r^2, r^3 \ldots r^{n-1},$$

and if we therefore form the difference product

$$\begin{gathered} P = (r-r^2)(r-r^3)\cdots(r-r^{n-1}) \\ (r^2-r^3)\cdots(r^2-r^{n-1}) \\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot \\ (r^{n-2}-r^{n-1}) \end{gathered} \tag{9}$$

introduce, then (§. 46)

$$D = P^2.$$

By (8) it is therefore also

$$P = n^{\frac{n-3}{2}} \sqrt{(-1)^{\frac{n-1}{2}} n}, \tag{10}$$

whereby P is determined, apart from the sign. P is real when $n \equiv 1$, and imaginary when $n \equiv -1 \pmod 4$, hence real for $n = 5, 13, 17, 29, 37 \ldots$, imaginary for $n = 3, 7, 11, 19, 23, 31 \ldots$ The sign that we have to give the root in (10) depends on which r we have chosen in the expression (9).

If we choose a definite r, e.g. $r = r_0$, then we can still determine the sign in (10). To carry this out, we divide the binomial factors of P,

$$r^\nu - r^\mu, \quad \nu < \mu,$$

whose number amounts to $\frac{n(n-1)}{2}$, into two classes. The differences of the one class form the product

$$Q = (r-r^{n-1})(r^2-r^{n-2})\cdots\left(r^{\frac{n-1}{2}} - r^{\frac{n+1}{2}}\right), \tag{11}$$

which thus contains all those factors in which $\mu + \nu = n$.

The remaining factors can be gathered into pairs of the following form

$$R = (r^\nu - r^\mu)(r^{n-\mu} - r^{n-\nu}). \tag{12}$$

The number of factors in Q is $\frac{n-1}{2}$, and hence the number of pairs R is

$$\frac{1}{2}\left(\frac{n(n-1)}{2} - \frac{n-1}{2}\right) = \left(\frac{n-1}{2}\right)^2.$$

Now if

$$r = e^{\frac{2\pi i k}{n}}$$

then

$$R = -2 + r^{\mu-\nu} + r^{\nu-\mu} = -2\left(1 - \cos\frac{2\pi(\mu-\nu)}{n}\,k\right),$$

that is, R is always negative. Consequently the product of all the R has the sign

$$(-1)^{\frac{n-1}{2}}.$$

Further,

$$Q = (2i)^{\frac{n-1}{2}} \sin\frac{2\pi k}{n} \sin\frac{4\pi k}{n} \cdots \sin\frac{(n-1)\pi k}{n}, \tag{13}$$

and the sign of this depends on k. But if we take $k = 1$, and hence $r = r_0$, then all the angles

$$\frac{2\pi}{n}, \frac{4\pi}{n} \cdots \frac{(n-1)\pi}{n}$$

lie between zero and π, and the sines are all positive. We conclude from this that in this case

$$P = (-i)^{\frac{n-1}{2}} n^{\frac{n-2}{2}} \sqrt{n} \tag{14}$$

and that $\sqrt{n}$ is to be taken positive.

The sign of the product of the R does not depend on the choice of r, but only on the nature of n. It is therefore of interest to determine by itself the product Q, whose sign does depend on the choice of r. We obtain it in the following way:

If we multiply the expression (11) by

$$r \cdot r^2 \cdot r^3 \cdots r^{\frac{n-1}{2}} = r^{\frac{n^2-1}{8}}$$

and then by

$$r^{-1} \cdot r^{-2} \cdot r^{-3} \cdots r^{-\frac{n-1}{2}} = r^{-\frac{n^2-1}{8}},$$

we obtain

$$r^{\frac{n^2-1}{8}} Q = (r^2-1)(r^4-1)\cdots(r^{n-1}-1)$$

$$r^{-\frac{n^2-1}{8}} Q = (1-r^{n-2})(1-r^{n-4})\cdots(1-r).$$

Now since the exponents $2, 4 \ldots n-1, n-2, n-4 \ldots 1$ together comprise all the numbers $1, 2 \ldots n-1$, multiplication of these two expressions yields:

$$Q^2 = (-1)^{\frac{n-1}{2}} \Pi(r-1) = (-1)^{\frac{n-1}{2}} n,$$

hence

$$Q = \pm i^{\frac{n-1}{2}} \sqrt{n}. \tag{15}$$

Comparison with (13) yields

$$2^{\frac{n-1}{2}} \sin\frac{2\pi k}{n} \sin\frac{4\pi k}{n} \cdots \sin\frac{(n-1)\pi k}{n} = \pm\sqrt{n}. \tag{16}$$

The sign in this formula, as in (15), still depends on k; but it is the positive one when $k = 1$. For the rest, we shall derive one more important theorem concerning this sign, which will be investigated more closely later on.

By the definition (11), Q is an integral rational function of r which, when one orders it according to powers of r, has integral rational numerical coefficients. If we therefore set

$$r = r_0^k,$$

then

$$Q = F(r_0^k),$$

where F is the symbol for a rational function whose coefficients do not depend on k. If we now set for k in turn the values

$$k = 1,\ 2 \ldots n-1,$$

then r_0^k represents all the roots of the cyclotomic equation $X_n = 0$, and the sum

$$\overset{k}{\Sigma} Q = F(r_0) + F(r_0^2) + \cdots + F(r_0^{n-1})$$

is a symmetric function of the roots of this equation; it can therefore be expressed rationally in terms of the coefficients of the equation, that is, in terms of rational numbers, and is consequently itself a rational number. On the other hand, by the formula (15),

$$\overset{k}{\Sigma} Q = h\, i^{\frac{n-1}{2}} \sqrt{n},$$

where h denotes an integer, namely the number of cases in which the positive sign is to be taken in (15), diminished by the number of cases in which the negative sign holds. But the two are compatible with one another only when $h = 0$; and with this the following theorem is proved:

1. If k runs through the sequence of numbers 1, 2, 3, ... $n-1$, then in the formula (16) the positive sign holds just as often as the negative.

It hardly needs special mention that one may also take for k another complete system of residues of n, excluding the number divisible by n.

§. 136.
Primitive congruence roots.

Parallel to the theory of the roots of unity runs a theory of the so-called binomial congruences, without which, in the theory of the cyclotomic equations, further steps cannot

can be made, the fundamentals of which may fittingly be inserted here, where the analogy with the theory of the roots of unity shows itself clearly.

So let n now be a prime and

$$f(x) = a_0 x^m + a_1 x^{m-1} + \cdots + a_{m-1} x + a_m \tag{1}$$

an integral function of x, whose coefficients a are integers, and a_0 not divisible by n. If we put for x an integer α such that $f(\alpha)$ becomes divisible by n, then we say (by analogy with equations) that α is a root of the congruence of degree m:

$$f(x) \equiv 0 \pmod{n}. \tag{2}$$

If α is increased by a multiple of n, it remains a root of the congruence (2). Such roots, congruent with respect to the modulus n, are not counted as distinct. Under this assumption we can state the theorem:

1. A congruence of degree m for a prime modulus cannot have more than m distinct roots.

The theorem is true for $m = 1$, for the congruence $a_0 x + a \equiv 0 \pmod{n}$ has only one root, namely, if a_0' is determined so that $a_0 a_0' \equiv 1 \pmod{m}$ holds, $\alpha \equiv -a a_0' \pmod{n}$. We therefore now assume our theorem to be proved for the degree $m-1$ and deduce its validity for the degree m from this. By §. 4 we can, if at first α is arbitrary, set

$$\frac{f(x) - f(\alpha)}{x - \alpha} = f_1(x), \tag{3}$$

in which $f_1(x)$ is an integral function of degree $(m-1)$, which, if α is an integer, has integral coefficients. If therefore $f(\alpha) \equiv 0$, it follows from (3) that

$$f(x) \equiv (x-\alpha)\, f_1(x) \pmod{n}. \tag{4}$$

Every root β of the congruence (2) must therefore satisfy the condition $(\beta-\alpha)\, f_1(\beta) \equiv 0 \pmod{n}$. Hence either $\beta-\alpha$ or $f_1(\beta)$ is divisible by n.

Now by hypothesis there are at most $m-1$ values β for which $f_1(\beta)$ becomes divisible by n; if therefore there is still an mth root of (4), it must be equal to α.

If therefore a congruence of the form $f(x) \equiv 0 \pmod{n}$ has more roots than its degree amounts to, then we conclude that

the congruence is identical, i.e. that all coefficients of $f(x)$ must be divisible by n.

This theorem is, as one sees, entirely analogous to the algebraic theorem that an equation cannot have more roots than its degree indicates. But the other theorem, that every equation also actually has that many roots, cannot be carried over. A congruence of degree m can have fewer roots, indeed none at all. All the more remarkable is a particular congruence in which the number of roots always equals the degree, on the basis of a theorem which is called Fermat's theorem, and which we formulate here as follows.

2. The congruences

$$x^n - x \equiv 0, \quad x^{n-1} - 1 \equiv 0 \pmod{n} \tag{5}$$

have, if n is a prime, as many roots as their degree amounts to, namely n and $n-1$.

The two assertions are not essentially different, for the first of the congruences (5) has all the roots of the second and in addition the root 0, which does not satisfy the second. Likewise the second has all the roots of the first, with the exception of the root 0.

Now since for the modulus n there are altogether only n different numbers, it is thus to be proved that for every integer α the congruence holds

$$\alpha^n \equiv \alpha \pmod{n}. \tag{6}$$

This congruence is correct for $\alpha = 0$ and $\alpha = 1$. We therefore prove it again by complete induction, deriving the congruence

$$(\alpha+1)^n \equiv \alpha + 1 \pmod{n} \tag{7}$$

from congruence (6), assumed to be correct. This follows from the binomial theorem, if one observes that all binomial coefficients, with the exception of the first and the last, which are equal to 1, namely

$$n, \quad \frac{n(n-1)}{1\,.\,2}, \quad \frac{n(n-1)(n-2)}{1\,.\,2\,.\,3}, \ldots$$

are integers divisible by n, because the numerator, but not the denominator, is divisible by n. Accordingly

$$(\alpha+1)^n \equiv \alpha^n + 1 \pmod{n},$$

hence the formula (7) follows from the formula (6).

The difference

$$x(x-1)(x-2)\ldots(x-n+1) - x^n + x$$

is a function of degree at most $n-1$. But it is congruent to zero for n values of x, namely for $x = 0, 1, 2, \ldots n-1$, and hence we have identically

$$x(x-1)(x-2)\ldots(x-n+1) \equiv x^n - x \pmod{n}.$$

From this there results Wilson's theorem:

$$1 \cdot 2 \cdot 3 \ldots (n-1) \equiv -1 \pmod{n},$$

if one compares the coefficients of the first power of x on both sides.

We now restrict ourselves to the second of the congruences (5)

$$x^{n-1} \equiv 1 \pmod{n} \tag{8}$$

and first prove the theorem:

3. If a is a divisor of $n-1$, then the congruence

$$x^a \equiv 1 \pmod{n} \tag{9}$$

has exactly a different roots.

For, if $n-1 = ab$, we have

$$x^{n-1} - 1 = (x^a - 1)(x^{a(b-1)} + x^{a(b-2)} + \cdots + x^a + 1),$$

and since every root of the left-hand side must be a root of one or the other factor on the right-hand side, it follows that if $x^a - 1 \equiv 0$ had fewer than a roots, the second factor would have to have more roots than its degree indicates, contrary to Theorem 1.

If α is a number not divisible by n, and a is the least positive number for which $\alpha^a \equiv 1 \pmod{n}$, and if further h is any positive integer for which $\alpha^h \equiv 1 \pmod{n}$, then a is necessarily a divisor of h, and so in particular a is a divisor of $n-1$.

For if we set $h = qa + a'$, where q is an integer and $a' < a$, then also $\alpha^{a'} \equiv 1$; hence we must have $a' = 0$, because a was to be the least positive number for which this congruence is satisfied.

A root α of the congruence (9) with the property that no power lower than the ath becomes congruent to unity is called a primitive root of this congruence, or a primitive ath congruence root of the prime n, and a primitive root g of the congruence (8) we also call, for short, a primitive root of the prime n.

We wish to prove that primitive roots exist for every prime n, and to determine their number; and in doing so we follow the same path as with the roots of unity.

Let a and b be two divisors of $n-1$ which are relatively prime to each other, and let α be an ath, β a bth primitive congruence root of n. If we set $ab = c$, then $\gamma = \alpha\beta$ is a primitive cth congruence root, and conversely every primitive cth congruence root can also be represented in the form $\alpha\beta$.

For, first: $(\alpha\beta)^c = \alpha^c\beta^c \equiv 1 \pmod{n}$; and secondly: if $(\alpha\beta)^h = \alpha^h\beta^h \equiv 1 \pmod{n}$, then also $\alpha^{hb}\beta^{hb} \equiv 1$, hence $\alpha^{hb} \equiv 1$, hence hb is divisible by a, hence h too is divisible by a; and in the same way one concludes that h must be divisible by b, and hence also by $ab = c$; that is, $\alpha\beta$ is a primitive cth congruence root.

If conversely γ is a primitive cth congruence root, then we determine the positive integers x, y so that

$$ay + bx \equiv 1 \pmod{c} \tag{10}$$

holds, by first determining x from $bx \equiv 1 \pmod{a}$ and then y from $y \equiv \dfrac{1-bx}{a} \pmod{b}$. Then y is relatively prime to b and x relatively prime to a. Accordingly

$$\gamma \equiv \gamma^{bx}\gamma^{ay} = \alpha\beta \pmod{n}$$

and $\gamma^{bx} = \alpha$ is a primitive ath congruence root, $\gamma^{ay} = \beta$ a primitive bth congruence root (because $\alpha^h = \gamma^{bxh}$ can be $\equiv 1$ only when h is divisible by a).

It remains to be shown that the products $\alpha\beta$ are all distinct from one another, that is, that the congruence

$$\alpha\beta \equiv \alpha'\beta' \pmod{n}, \tag{11}$$

when α, α'; β, β' are primitive ath and bth congruence roots, can be satisfied only when

$$\alpha \equiv \alpha', \quad \beta \equiv \beta' \pmod{n}$$

holds. But this follows by (10), if one raises (11) to the power

$$bx \equiv 1 \pmod{a}$$

from which $\alpha \equiv \alpha'$ results, and then also $\beta \equiv \beta'$.

If, then, we denote the number of primitive ath congruence roots by $\varphi(a)$, it follows from what has been proved that

$$\varphi(c) = \varphi(a)\varphi(b). \tag{12}$$

It still remains, when p is a prime and pp_1 is a power of p dividing $n-1$, to determine $\varphi(pp_1)$.

If α is a non-primitive congruence root of degree p^π and $p^{\pi-1} = p_1$, then, if α^h is the lowest power of α congruent to 1 with respect to the modulus n, h is a divisor of pp_1, i. e. a power of p; but since h is to be smaller than pp_1, it is also a divisor of p_1, and consequently $\alpha^{p_1} \equiv 1 \pmod{n}$; thus: a non-primitive congruence root of degree pp_1 is at the same time a congruence root of degree p_1. Conversely, every congruence root of degree p_1 is at the same time an imprimitive congruence root of degree pp_1. Now since the two congruences $x^{pp_1} \equiv 1$, $x^{p_1} \equiv 1$ have as many roots as their degree amounts to, there remain $pp_1 - p_1$ primitive congruence roots of the first, and thus we have

$$\varphi(pp_1) = p_1(p-1). \tag{13}$$

The function $\varphi(a)$ therefore has the same meaning as in §. 132, and with this the number of primitive roots of the prime n is at the same time found to be equal to $\varphi(n-1)$. We therefore state the further theorem:

4. A prime n has $\varphi(n-1)$ primitive roots.

With this the existence of primitive roots for every prime is proved, and at the same time their number is determined. For finding the primitive roots we have, to be sure, no other general method than trial, which can be somewhat shortened by a few artifices.

If g is a primitive root of the prime n, then the residues of the powers

$$1, g, g^2, g^3 \ldots g^{n-2} \tag{14}$$

are all different from one another, since, if $g^m \equiv g^{m+\mu}$ held, $g^\mu \equiv 1$ would have to hold, which is not possible so long as μ is smaller than $n-1$. Hence among the residues of the sequence (14), among which the residue 0 does not occur, each of the numbers

$$1, 2, 3 \ldots n-1 \tag{15}$$

must be contained, and each only once, or in other words:

5. If a is a number not divisible by n, there is one and only one number α from the sequence of the numbers $0, 1, 2 \ldots n-2$, by which the congruence

$$g^\alpha \equiv a \pmod{n} \tag{16}$$

is satisfied. But the same congruence is also satisfied if a number congruent to α modulo $n-1$ is put as exponent, and in every complete system of residues one finds only one such number α.

The exponent α is called the index of a with respect to the base g, and one writes

$$\alpha \equiv \operatorname{ind} a \pmod{n-1}. \tag{17}$$

One thus has, for every number not divisible by n,

$$g^{\operatorname{ind} a} \equiv a \pmod{n}. \tag{18}$$

From the two formulae

$$g^{\operatorname{ind} a + \operatorname{ind} b} \equiv ab, \quad g^{m \operatorname{ind} a} \equiv a^m \pmod{n} \tag{19}$$

there follows the theorem:

6. The index of a product is equal to the sum of the indices of the factors; the index of the mth power of a is equal to m times the index of a.

These theorems are quite analogous to the corresponding theorems that relate to computation with logarithms, and the analogy can be pursued still further. Thus, for example, the transition from one base g to another base g' is expressed by the relation

$$\overset{g}{\operatorname{ind}} a \equiv \overset{g}{\operatorname{ind}} g' \overset{g'}{\operatorname{ind}} a \pmod{n-1} \tag{20}$$

between the two systems of indices. We shall further adduce the easily proved theorem that among the numbers a not divisible by n, those and only those are primitive roots whose indices are relatively prime to $n-1$.

For practical computations one conveniently makes use of so-called index tables, which correspond to the tables of logarithms.

In the congruence (17) one calls α the index and a the numerus, and it is best to set up two tables, of which the one gives for each index the numerus, the other for each numerus the index, where the second is obtained from the first by rearrangement. Here it is to be observed that for the indices the modulus $n-1$, and for the numeri the modulus n, comes into consideration, and congruent numbers count as equivalent. Thus one obtains, e.g., for $n = 13$ and the base 2

I	0	1	2	3	4	5	6	7	8	9	10	11
N	1	2	4	8	3	6	12	11	9	5	10	7

N	1	2	3	4	5	6	7	8	9	10	11	12
I	0	1	4	2	9	5	11	3	8	10	7	6

.

In the Canon Arithmeticus of Jacobi (1839) index tables are compiled for all primes in the first thousand. On a smaller scale, after computations of Ostrogradsky, in the "Theory of Congruences" of Tschebyscheff (German by Schapira, Berlin 1889).

It is still to be remarked that the index of 1 is always equal to 0, and that the index of -1 (or of $n-1$) is always equal to $\frac{n-1}{2}$. For since $(-1)^2$ has the index 0 or $n-1$, -1 must have the index $\frac{n-1}{2}$, since it does not have the index 0, and twice its index is equal to 0 or $n-1$; in formulae:

$$\operatorname{ind} 1 \equiv 0, \quad \operatorname{ind}(-1) \equiv \frac{n-1}{2} \pmod{n-1}. \tag{21}$$

§. 137.

Multiplication and division of the trigonometric functions.

We have already pointed out the connection that exists between the theory of the roots of unity and the geometric problem of the division of the circumference of the circle. The generalisation of this problem leads to the division of an arbitrary angle. This problem too leads to algebraic equations, of which we have already used the one for trisection earlier, for the general solution of the cubic equations (§. 112).

We now begin with the setting up of the general formulae.

By the theorem of Moivre (§. 33), if n is an arbitrary positive integer and φ an arbitrary angle, we have

$$\cos n\varphi + i \sin n\varphi = (\cos\varphi + i\sin\varphi)^n, \tag{1}$$

and if we apply the binomial theorem on the right-hand side, then by separation of the real from the imaginary it follows that

$$\begin{aligned}\cos n\varphi = \cos^n\varphi - B_2^{(n)}\cos^{n-2}\varphi\sin^2\varphi \\ + B_4^{(n)}\cos^{n-4}\varphi\sin^4\varphi - \cdots\end{aligned}$$

$$\begin{aligned}\frac{\sin n\varphi}{\sin\varphi} = n\cos^{n-1}\varphi - B_3^{(n)}\cos^{n-3}\varphi\sin^2\varphi \\ + B_5^{(n)}\cos^{n-5}\varphi\sin^4\varphi - \cdots\end{aligned} \tag{2}$$

wherein $B_\nu^{(n)}$ are, as before, the binomial coefficients, and the sums on the right-hand side are to be continued as far as no negative exponents of $\cos\varphi$ occur. Since now $\sin^2\varphi = 1 - \cos^2\varphi$, these two expressions can be represented rationally in terms of $\cos\varphi$; we set

$$2\cos\varphi = x, \tag{3}$$

and introduce the notation

$$2\cos n\varphi = A_n(x), \qquad \frac{\sin n\varphi}{\sin\varphi} = B_n(x), \tag{4}$$

wherein, then, $A_n(x)$ and $B_n(x)$ are integral rational functions of x, $A_n(x)$ of degree n, $B_n(x)$ of degree $n-1$. We wish to determine the general law of these functions. We set them up for the first values $n = 1, 2, 3\ldots$, and then derive a recursion formula for the computation of the higher functions. One finds

$$\begin{aligned} A_1(x) &= x & B_1(x) &= 1 \\ A_2(x) &= x^2 - 2, & B_2(x) &= x \\ A_3(x) &= x^3 - 3x, & B_3(x) &= x^2 - 1. \end{aligned} \tag{5}$$

But now, by well-known formulae,

$$\begin{aligned}\cos(n+1)\varphi + \cos(n-1)\varphi &= 2\cos\varphi\cos n\varphi \\ \sin(n+1)\varphi + \sin(n-1)\varphi &= 2\cos\varphi\sin n\varphi,\end{aligned}$$

from which the recursion formulae for A_n and B_n follow:

$$\begin{aligned} A_{n+1}(x) &= xA_n(x) - A_{n-1}(x), \\ B_{n+1}(x) &= xB_n(x) - B_{n-1}(x). \end{aligned} \tag{6}$$

From this one can compute A_n, B_n for any arbitrary n, if these functions are known for $n = 1$ and $n = 2$. Thus one finds

$$A_4(x) = x^4 - 4x^2 + 2$$
$$A_5(x) = x^5 - 5x^3 + 5x$$
$$A_6(x) = x^6 - 6x^4 + 9x^2 - 2$$

$$B_4(x) = x^3 - 2x$$
$$B_5(x) = x^4 - 3x^2 + 1$$
$$B_6(x) = x^5 - 4x^3 + 3x,$$

and so one may continue. By induction one arrives at the following general law:

$$A_n(x) = \overset{\nu}{\Sigma}(-1)^\nu \frac{n}{n-\nu} B_\nu^{(n-\nu)} x^{n-2\nu}, \quad 0 \leqq \nu \leqq \frac{n}{2}, \tag{7}$$

$$B_n(x) = \overset{\nu}{\Sigma}(-1)^\nu B_\nu^{(n-\nu-1)} x^{n-2\nu-1}, \quad 0 \leqq \nu \leqq \frac{n-1}{2}, \tag{8}$$

where the sums begin with $\nu = 0$, and are to be continued so long as the exponents of x do not become negative.

Since these formulae prove to be correct in the first cases, in order to prove them generally it is only necessary to show that they verify the formulae (6). If for this purpose we form

$$A_{n-1}(x) = \overset{\nu}{\Sigma}(-1)^\nu \frac{n-1}{n-\nu-1} B_\nu^{(n-\nu-1)} x^{n-2\nu-1},$$

and put $\nu - 1$ in place of ν, it follows that

$$A_{n-1}(x) = -\overset{\nu}{\Sigma}(-1)^\nu \frac{n-1}{n-\nu} B_{\nu-1}^{(n-\nu)} x^{n-2\nu+1}, \quad 1 \leqq \nu \leqq \frac{n+1}{2}. \tag{9}$$

If we assume $B_{-1}^{(n)} = 0$, we can let ν increase from 0 here as well.

The values that ν takes in formula (7) then differ from those in formula (9) only when n is odd, because then the value $\nu = (n+1) : 2$ occurs in (9), but not in (7). We can, however, add this value in (7), provided that

$$B_{\frac{n+1}{2}}^{\left(\frac{n-1}{2}\right)} = 0$$

is assumed. Then there results

$$xA_n(x) - A_{n-1}(x) = \overset{\nu}{\Sigma}(-1)^\nu x^{n-2\nu+1} \frac{nB_\nu^{(n-\nu)} + (n-1)B_{\nu-1}^{(n-\nu)}}{n-\nu}, \quad 0 \leqq \nu \leqq \frac{n+1}{2}. \tag{10}$$

Now, as follows easily from the expressions for the binomial coefficients (§. 8),

$$\frac{nB_\nu^{(n-\nu)} + (n-1)B_{\nu-1}^{(n-\nu)}}{n-\nu} = \frac{n+1}{n-\nu+1} B_\nu^{(n-\nu+1)},$$

also in the two limiting cases $\nu = 0$, $\nu = (n+1) : 2$, and thus (10) agrees with what results from the right-hand side of (7) by interchanging n with $n+1$.

In just the same way, formula (8) yields

$$xB_n(x) - B_{n-1}(x) = \overset{\nu}{\Sigma}(-1)^\nu x^{n-2\nu} \left(B_\nu^{(n-\nu-1)} + B_{\nu-1}^{(n-\nu-1)} \right),$$

and since

$$B_\nu^{(n-\nu-1)} + B_{\nu-1}^{(n-\nu-1)} = B_\nu^{(n-\nu)}$$

holds, (8) is thereby also proved in general.

For the sake of a clearer view we set out the formulae somewhat more fully:

$$A_n(x) = x^n - nx^{n-2} + \frac{n(n-3)}{1\,.\,2}x^{n-4} - \frac{n(n-4)(n-5)}{1\,.\,2\,.\,3}x^{n-6} + \cdots \tag{11}$$

$$B_n(x) = x^{n-1} - (n-2)x^{n-3} + \frac{(n-3)(n-4)}{1\,.\,2}x^{n-5} - \frac{(n-4)(n-5)(n-6)}{1\,.\,2\,.\,3}x^{n-7} + \cdots \tag{12}$$

If we take $\cos n\varphi$ and $\sin n\varphi$ as given, then the equation of degree n

$$2\cos n\varphi = A_n(x) \tag{13}$$

serves to determine the unknown $x = 2\cos\varphi$. The meaning of the n roots of this equation follows from the fact that the cosine does not change when the angle increases by a multiple of 2π. Accordingly, equation (13) is satisfied by every value

$$x = 2\cos\left(\varphi + \frac{2\nu\pi}{n}\right), \tag{14}$$

where ν denotes an arbitrary positive or negative integer. But one obtains all the mutually distinct values of (14) by letting ν run through a complete system of residues with respect to n, thus setting, for example, $\nu = 0, 1, 2 \ldots n-1$, and one

one also sees easily that these n values are all different from one another, if one leaves aside the special case that φ is a multiple of π. For two cosines are equal to one another only when the sum or the difference of the angles is a multiple of 2π.

The equation

$$\sin n\varphi = \sin\varphi\, B_n(x) \tag{15}$$

determines $\sin\varphi$ uniquely in terms of x, when $B_n(x)$ is not equal to zero.

If we seek in equation (13) the term independent of x, then with the use of (6) there results for an even n

$$2(-1)^{\frac{n}{2}} - 2\cos n\varphi,$$

and for an odd n

$$-2\cos n\varphi,$$

and if one sets this equal to the positive or negative product of the roots, one obtains for an even n

$$2^{n-1}\overset{\nu}{\Pi}\cos\left(\varphi + \frac{2\nu\pi}{n}\right) = (-1)^{\frac{n}{2}} - \cos n\varphi, \tag{16}$$

and for an odd n

$$2^{n-1}\overset{\nu}{\Pi}\cos\left(\varphi + \frac{2\nu\pi}{n}\right) = \cos n\varphi, \tag{17}$$

and these formulae hold, since on the right and on the left stand continuous functions of φ, also for the case at first excluded, where φ is a multiple of π.

Herein we may put νm for ν under the product sign, if m is any number coprime to n, because then νm runs, together with ν, through a complete system of residues to the modulus n. If we separate off the value $\nu = 0$, and then let ν run through both the positive and the negative numbers that are absolutely smaller than $n : 2$, we obtain from (17) the formula, valid for every odd n,

$$2^{n-1}\underset{1,\,\frac{n-1}{2}}{\overset{\nu}{\Pi}}\cos\left(\varphi + \frac{2\nu m\pi}{n}\right)\cos\left(\varphi - \frac{2\nu m\pi}{n}\right) = \frac{\cos n\varphi}{\cos\varphi}. \tag{18}$$

If we set $\varphi = 0$ herein and take the square root, there results

$$2^{\frac{n-1}{2}} \prod_{1,\,\frac{n-1}{2}}^{\nu} \cos\frac{2\nu m\pi}{n} = \pm 1, \tag{19}$$

in which for the time being the sign still remains undetermined.

But if in (18) we let φ pass into $\pi : 2$, the right-hand side, which presents itself under the indeterminate form $0 : 0$, acquires the limiting value $(-1)^{\frac{n-1}{2}} n$, and we obtain again, generally for every odd n, the formula which we derived in the preceding section for a prime n

$$2^{\frac{n-1}{2}} \prod_{1,\,\frac{n-1}{2}}^{\nu} \sin\frac{2\nu m\pi}{n} = \pm\sqrt{n}, \tag{20}$$

and for $m = 1$ the positive sign is to be taken.

The function B_n, which with respect to x is of degree $(n-1)$, vanishes when $n\varphi$, but not φ, is a multiple of π, that is, for

$$x = 2\cos\frac{h\pi}{n}, \tag{21}$$

in which h is an integer not divisible by n. If we first assume n even, we obtain all the distinct values contained in (21) by taking for m an arbitrary number coprime to n and setting

$$h = m,\ 2m,\ 3m \ldots (n-1)m \tag{22}$$

for

$$\cos\frac{h\pi}{n} = \cos\frac{h'\pi}{n},$$

holds only when either $h + h'$ or $h - h'$ is a multiple of $2n$, which cannot occur when h and h' are both taken from the sequence (22).

Among the values (21) there is one, namely the one corresponding to the value $h = nm : 2$, which is equal to zero, and the rest are opposite in pairs. Accordingly, for an even n we obtain

$$B_n(x) = x \prod_{1,\,\frac{n}{2}-1}^{\nu} \left(x^2 - 4\cos^2\frac{\nu m\pi}{n}\right). \tag{23}$$

If n is odd, then $B_n(x)$ depends only on x^2; hence the values (21) are equal and opposite in pairs (for h and $h + n$). If, again supposing m relatively prime to n, one

$$h = 2m,\ 4m,\ 6m,\ \ldots (n-1)m$$

assumes, one obtains in (21) neither two equal nor two opposite values, and all the mutually distinct values of x^2 result.

Thus for an odd n we have

$$B_n(x) = \prod_{1,\,\frac{n-1}{2}}^{\nu} \left(x^2 - 4\cos^2\frac{2\nu m\pi}{n}\right). \tag{24}$$

If in this formula we set

$$x = 2\cos\varphi, \quad x^2 = 4(1 - \sin^2\varphi), \quad \cos^2\frac{2\nu m\pi}{n} = 1 - \sin^2\frac{2\nu m\pi}{n},$$

we can represent it thus:

$$\frac{\sin n\varphi}{\sin\varphi} = (-1)^{\frac{n-1}{2}} 2^{n-1} \prod_{1,\,\frac{n-1}{2}}^{\nu} \left(\sin^2\varphi - \sin^2\frac{2\nu m\pi}{n}\right), \tag{25}$$

from which again, for $\varphi = 0$, formula (20) follows.

Since $\cos n\varphi$ vanishes for odd n when

$$\cos\varphi = \pm\sin\frac{2\nu m\pi}{n}$$

holds, we have thereby found the decomposition into factors of the function $A_n(x)$ for this case also. For odd n we have

$$A_n(x) = x \prod_{1,\,\frac{n-1}{2}}^{\nu} \left(x^2 - 4\sin^2\frac{2\nu m\pi}{n}\right). \tag{26}$$

The quantities

$$\alpha = \sin^2\frac{2\nu m\pi}{n} \tag{27}$$

are thus the roots of an algebraic equation of degree $(n-1):2$

$$\Phi_n(x) = 0,$$

and we obtain the function $\Phi_n(x)$ if, in place of the variable x^2 in the function $A_n(x) : x$, we put a new variable $4x$ and divide by 2^{n-1}. It therefore becomes [formula (11)]

$$\Phi_n(x) = \overset{\alpha}{\Pi}(x-\alpha) = x^{\frac{n-1}{2}} - \frac{n}{4}x^{\frac{n-3}{2}} + \frac{n(n-3)}{1.2}\frac{1}{16}x^{\frac{n-5}{2}} - \cdots, \tag{28}$$

and for (25) we can write

$$\frac{\sin n\varphi}{\sin\varphi} = (-1)^{\frac{n-1}{2}} 2^{n-1} \overset{\alpha}{\Pi}(\sin^2\varphi - \alpha). \tag{29}$$

§. 138.
Determination of signs. Quadratic residues.

In the preceding section we derived two formulae in which signs still remained to be determined, namely

$$2^{\frac{n-1}{2}} \overset{\nu}{\Pi} \cos\frac{2\nu m\pi}{n} = \pm 1, \tag{1}$$

$$2^{\frac{n-1}{2}} \overset{\nu}{\Pi} \sin\frac{2\nu m\pi}{n} = \pm\sqrt{n}, \tag{2}$$

where m is relatively prime to the odd number n and ν runs through the following sequence.

$$1,\ 2,\ \cdots\ \frac{n-1}{2} \tag{ν}$$

We shall here retain the symbol ν for the numbers of this sequence.

Every integer whatsoever gives, upon division by n, as residue one of the numbers ν or 0 or $n-\nu$. Instead of $n-\nu$ we may, if we also admit negative residues, choose $-\nu$; and if we accordingly

set aside the numbers divisible by n, there remains one of the numbers $\pm\nu$ as residue. We call this the absolutely least residue (in distinction from the least residue in the ordinary sense, which is taken from the numbers $1, 2, \ldots n-1$).

Now let m be a number coprime to n; we consider the sequence of numbers

$$(m\nu) \qquad m,\ 2m,\ 3m \cdots \frac{n-1}{2}\, m$$

and form for each the absolutely least residue ϱ:

$$(\varrho) \qquad \pm\nu_1,\ \pm\nu_2,\ \pm\nu_3 \cdots \pm\nu_{\frac{1}{2}(n-1)};$$

among these numbers ϱ no two equal ones occur, and also no two opposite ones; for if the sum or the difference of two numbers (ϱ), that is $\varrho + \varrho'$ or $\varrho - \varrho'$, were equal to zero, then for two distinct numbers ν, ν' from (ν)

$$m(\nu \pm \nu')$$

would have to be divisible by n, and hence $\nu \pm \nu'$ would also have to be divisible by n, which is impossible, since each of the numbers ν, ν' is less than $n : 2$,

The totality of the numbers (ϱ) thus agrees, apart from sign and from order, with the totality of the numbers (ν).

In the formulae (1) and (2) we can, however, on account of the periodicity of sine and cosine, replace νm by ϱ, and since $\cos(-\varphi) = \cos\varphi$, we can in formula (1) also replace $m\nu$ by ν, i.e. the sign does not depend on m. To determine it, we take into account that the cosine of an angle in the first quadrant is positive, in the second quadrant negative, and thus that the sign in (1) is positive or negative according as an even or an odd number of the angles

$$\frac{2\nu\pi}{n}$$

lies in the second quadrant, or according as an even or an odd number of the numbers ν lies between $\frac{n}{4}$ and $\frac{n}{2}$.

If x is any number that is not an integer, we denote by $E(x)$ the greatest integer that is less than x, so that x lies between $E(x)$ and $E(x)+1$. The number of integers lying between two numbers x and y is then equal to $E(y) - E(x)$, and we therefore have to investigate whether

$$E\left(\frac{n}{2}\right) - E\left(\frac{n}{4}\right)$$

is an even or an odd number. We must distinguish four cases, as appears from the following compilation, in which k denotes a non-negative integer:

$$\begin{aligned}
n &= 8k+1, \quad E\left(\frac{n}{2}\right) = 4k, && E\left(\frac{n}{4}\right) = 2k,\\
n &= 8k+3, \quad E\left(\frac{n}{2}\right) = 4k+1, && E\left(\frac{n}{4}\right) = 2k,\\
n &= 8k+5, \quad E\left(\frac{n}{2}\right) = 4k+2, && E\left(\frac{n}{4}\right) = 2k+1,\\
n &= 8k+7, \quad E\left(\frac{n}{2}\right) = 4k+3, && E\left(\frac{n}{4}\right) = 2k+1.
\end{aligned}$$

Thus in (1) the positive sign is to be taken when n is of the form $8k+1$ or $8k+7$, the negative when n is of the form $8k+3$ or $8k+5$. In the first cases

$\frac{n^2-1}{8}$ is an even number, in the last two an odd one, and accordingly we obtain the exact formula (1):

(3) $$2^{\frac{n-1}{2}} \overset{\nu}{\Pi} \cos \frac{2\nu m\pi}{n} = (-1)^{\frac{n^2-1}{8}}.$$

In formula (2) the sign depends on m; it is positive or negative according as, among the numbers (ϱ), an even or an odd number of negative ones occurs.

Now, in order to express in the notation the dependence of the sign on m and n, we write for (2)

(4) $$2^{\frac{n-1}{2}} \overset{\nu}{\underset{1,\,\frac{n-1}{2}}{\Pi}} \sin \frac{2\nu m\pi}{n} = \left(\frac{m}{n}\right)\sqrt{n},$$

in which

$$\left(\frac{m}{n}\right) = \pm 1$$

holds.

The symbol $\left(\frac{m}{n}\right)$ has been introduced into number theory from another side, and in a more special form, by Legendre, and has been generalised by Jacobi. We shall call it the Legendre symbol. Its significance for number theory will appear presently; first we have the theorem:

1. We have $\left(\frac{m}{n}\right)$ equal to $+1$ or equal to -1, according as the number of negative ones among the absolutely least residues of $m\nu$ is even or odd.

For $m = 1$, as we have already seen, the positive sign holds in formula (2); hence we have, as formula (4) also shows directly,

$$2. \quad \left(\frac{1}{n}\right) = +1.$$

If we change m into $-m$, the sign changes in all factors of the product on the left-hand side of (4); since the number of factors is $\frac{n-1}{2}$, it follows that

$$3. \quad \left(\frac{-m}{n}\right) = (-1)^{\frac{n-1}{2}} \left(\frac{m}{n}\right)$$

and for $m = 1$

$$4. \quad \left(\frac{-1}{n}\right) = (-1)^{\frac{n-1}{2}}.$$

If we put $2m$ for m and use the formula

$$\sin \frac{4\nu m\pi}{n} = 2 \sin \frac{2\nu m\pi}{n} \cos \frac{2\nu m\pi}{n},$$

then from (3) it follows that

$$5. \quad \left(\frac{2m}{n}\right) = (-1)^{\frac{n^2-1}{8}} \left(\frac{m}{n}\right)$$

and for $m = 1$

$$6. \quad \left(\frac{2}{n}\right) = (-1)^{\frac{n^2-1}{8}}.$$

If m is altered by a multiple of n, all the factors of the product (4) remain unchanged, and from this it follows:

7. We have

$$\left(\frac{m}{n}\right) = \left(\frac{m'}{n}\right),$$

if $m \equiv m' \pmod{n}$.

We now assume not only n but also m to be odd and positive, and apply the formula (29) of §. 137, in which we replace n by m:

$$\frac{\sin m\varphi}{\sin \varphi} = (-1)^{\frac{m-1}{2}} 2^{m-1} \overset{\beta}{\Pi}(\sin^2 \varphi - \beta), \tag{5}$$

in which β runs through all the roots of the equation $\Phi_m(x) = 0$.

$$\beta = \sin^2 \frac{2\mu\pi}{m} \tag{6}$$

Now if m' is a number coprime to n, then

$$\alpha = \sin^2 \frac{2\nu m'\pi}{n} \tag{7}$$

runs through all the roots of the equation $\Phi_n(x) = 0$, and if we set

$$\varphi = \frac{2\nu m'\pi}{n} \tag{8}$$

in (5) and take the product with respect to all ν, it follows that:

$$\frac{\overset{\nu}{\Pi} \sin \frac{2\nu m m'\pi}{n}}{\overset{\nu}{\Pi} \sin \frac{2\nu m'\pi}{n}} = (-1)^{\frac{(m-1)(n-1)}{4}} 2^{\frac{(m-1)(n-1)}{2}} \overset{\alpha}{\Pi} \overset{\beta}{\Pi}(\alpha - \beta). \tag{9}$$

But the right-hand side of this formula is entirely independent of m'. The value of the left-hand side is therefore the same as if $m' = 1$, and if one substitutes for the products the values from (4), it follows that

$$\left(\frac{mm'}{n}\right) : \left(\frac{m'}{n}\right) = \left(\frac{m}{n}\right) : \left(\frac{1}{n}\right),$$

or

$$8. \quad \left(\frac{mm'}{n}\right) = \left(\frac{m}{n}\right)\left(\frac{m'}{n}\right).$$

Here it was at first assumed that m and m' are positive and odd. But by 3. and 5., 8. still remains valid when m or m' is negative or even, provided only that m and m' are relatively prime to n.

If we again take m odd and positive, it now follows from (4) and (9)

$$\left(\frac{m}{n}\right) = (-1)^{\frac{(m-1)(n-1)}{4}} 2^{\frac{(m-1)(n-1)}{2}} \overset{\alpha}{\Pi} \overset{\beta}{\Pi}(\alpha - \beta). \tag{10}$$

If we interchange m with n here, then in the double product on the right-hand side every factor changes its sign; everything else remains unchanged. But the number of these factors is

$$\frac{n-1}{2} \cdot \frac{m-1}{2}$$

and from this it follows

$$9. \quad \left(\frac{m}{n}\right) = (-1)^{\frac{(m-1)(n-1)}{4}} \left(\frac{n}{m}\right),$$

or $\left(\frac{m}{n}\right)$ is equal to $\left(\frac{n}{m}\right)$, if of the two odd numbers m, n at least one is of the form $4k+1$, and $\left(\frac{m}{n}\right)$ is opposite to $\left(\frac{n}{m}\right)$, if both numbers are of the form $4k+3$.

This famous theorem is known under the name of the law of reciprocity. The proof given here is due to Eisenstein. With its help, and by 5. and 6., one can determine the value of the symbol $\left(\frac{m}{n}\right)$ very quickly, by

one proceeds as if it were a matter of determining the greatest common divisor of m and n.

From 8. and 9. there follows yet one last theorem, which holds when n and n' are two positive odd numbers coprime to m:

$$10. \quad \left(\frac{m}{n}\right)\left(\frac{m}{n'}\right) = \left(\frac{m}{nn'}\right).$$

Its correctness follows, when m is odd and positive, on the ground of the congruence

$$(7) \qquad (n-1)(n'-1) = (nn'-1) - (n-1) - (n'-1) \equiv 0 \pmod 4,$$

if one applies formula 9. and then 8. to both sides of 10.; and, when m is even or negative, from 4. and 6.

By these theorems one can reduce $\left(\frac{m}{n}\right)$ to a product of values $\left(\frac{q}{p}\right)$, in which p, q are primes. The computation of $\left(\frac{m}{n}\right)$, however, is carried out more easily by 9. without this decomposition.

We now wish to learn another meaning of the symbol $\left(\frac{m}{p}\right)$ in the case where p is an odd prime, for which Legendre originally introduced the symbol.

If we raise to the square all the integers x not divisible by p and seek out the residues of the division by p, we obtain in all only $\frac{p-1}{2}$ different residues; for, first, the squares of congruent numbers give the same residues, and secondly the squares of opposite numbers also give the same residues. We therefore certainly obtain all the residues if we take the $\frac{p-1}{2}$ numbers ν

$$(\nu) \qquad 1,\ 2,\ 3 \cdots \frac{p-1}{2}$$

and raise them to the square. On the other hand, the squares of these numbers ν also yield residues that are all different; for, if ν and ν' are different ones among them, the quantity

$$\nu^2 - \nu'^2 = (\nu - \nu')(\nu + \nu')$$

can never

be divisible by p, because both $\nu - \nu'$ and $\nu + \nu'$ are smaller than p. Among the numbers

(s) $$1, 2, 3 \ldots p-1$$

half therefore occur among the residues of x^2, and half do not. The former numbers, and all those congruent to them with respect to p, are called the quadratic residues of p, the others the quadratic non-residues of p.

Now if m is a quadratic residue, the congruence

$$x^2 \equiv m \pmod{p}$$

is possible, and from 7. and 8. it follows that

$$\left(\frac{m}{p}\right) = \left(\frac{x^2}{p}\right) = \left(\frac{x}{p}\right)^2 = +1.$$

Thus $\left(\frac{m}{p}\right) = +1$ when m is a quadratic residue of p.

But in §. 135, 1. the theorem was stated that, if for m the sequence of numbers 1, 2, ... p - 1 is put, $\left(\frac{m}{p}\right)$ has the positive sign just as often as the negative one; and from this one sees that the negative sign holds when m is a quadratic non-residue. It therefore follows:

11. If p is an odd prime and m is not divisible by p, then $\left(\frac{m}{p}\right) = +1$ or $= -1$, according as m is a quadratic residue or a quadratic non-residue of p. In this all the principal theorems on the quadratic residues are contained, e. g. the theorem:

The product of two quadratic residues or of two non-residues is a residue; the product of a residue and a non-residue is a non-residue.

THIRD BOOK.

ALGEBRAIC QUANTITIES.

Thirteenth Section.
The Galois Theory.

§. 139.
The Concept of a Field.

A system of numbers is called a number field when it is so complete and closed in itself that the four fundamental operations of reckoning (the four species) — addition, subtraction, multiplication and division — carried out with any numbers of the system, excepting division by zero, always lead to numbers that belong to the same system. This concept, which gives a classification of the kinds of numbers according to a natural point of view, was introduced by Dedekind (Dirichlet-Dedekind, Vorlesungen über Zahlentheorie, 2nd ed. 1871, §. 159). It is

of the greatest importance for algebra, and it is not a matter of indifference to have a distinctive and expressive name for it. The word Zahlkörper (number field) was formed by Dedekind on the model of numerous analogies, in which the word Körper (corpus, corps) in a similar way signifies a union of things belonging together, to which a certain completeness attaches.

The concept of the number field can be extended and carried over to all magnitudes with which one can reckon according to the rules of the four species; in particular, therefore, to rational functions of any variables.

We accordingly call a function field a system of functions of one or more variables of such a nature that in this system the reckonings with the four species can be carried out without limit and always lead to a definite function of the same system (always with the exception of division by zero). The variables

may be entirely independent of one another; but the case is not excluded that certain relations are fixed between them, which are to be taken into account in calculation.

A function of a function field counts as zero only when it vanishes identically, that is, for all values of the variables that come into consideration.

Since for the present we do not wish to restrict our considerations, we shall now speak of fields simply, and shall designate the objects with which the calculations are to be carried out, which may be numbers as well as functions, as quantities or also as the elements of the field [1]).

A field is then a system of quantities of such completeness that in it the quantities can be added, subtracted, multiplied and divided.

When all the quantities of one field are contained in a second field, the first field is called a divisor of the second.

If Ω is a divisor of Ω', we shall also say that Ω' is a field over Ω.

Strictly speaking, the single number "zero" forms a field by itself. But for the sake of brevity we shall exclude this once and for all.

Then the most obvious example of a field is the totality of all rational numbers.

This field is a divisor of every other field; for every field contains at least one quantity ω different from zero, hence also the quotient $\omega : \omega = 1$, that is, the number 1; hence also, since all integers arise by addition and subtraction of units, all integers; and from the integers one can again derive all fractions by division.

Other examples of number fields are: the totality of all complex numbers of the form $x + yi$, in which $i = \sqrt{-1}$, and x and y denote all rational numbers; further the totality of all (rational and irrational) real numbers, or the totality of all complex numbers $x + yi$ that exist at all.

As an example of a function field may serve the totality of all integral and fractional rational functions of a

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[1]) Cf. the author's paper: "Die allgemeinen Grundlagen der Galois'schen Gleichungstheorie." Mathematische Annalen, Bd. 43.

variables (with the inclusion of the constants), in which one may restrict the constant coefficients to an arbitrary number field, say to that of the rational numbers, or may also take them entirely unrestricted.

We do not wish to multiply examples further here, since the exact investigation of fields of special kinds will be one of the principal tasks of our later expositions.

§. 140.
Adjunction.

If to a field of any magnitudes whatever, which we shall denote by Ω, there is added any magnitude α that is not contained in it, there arises a new system of magnitudes

$$\Omega,\ \alpha,$$

which, however, will not be a field; in order to derive a field from it, one must add all magnitudes that can be derived by combining α with the numbers of Ω by means of the fundamental operations of arithmetic. One thus obtains an extended field Ω', which contains both α and Ω, and which is completely determined by α and Ω. We shall call it the field Ω, α.

This adding of a new magnitude α to a field Ω is called adjoining, and one says that the field Ω' arises from Ω by adjunction of α.

To Ω' one can again adjoin a magnitude α', and obtains a third field Ω'', which then contains Ω, α and α', and so on. But one can also adjoin two or more numbers to Ω simultaneously, and thus obtains the same field Ω'' if one adjoins α and α' to Ω simultaneously. One can also adjoin to one field a second field, and thus obtains a field that contains the numbers of both fields at once. In the above example one would, by adjunction of the field Ω, α' to the field $\Omega' = \Omega, \alpha$, again obtain the same field Ω''.

If one adjoins to a field one of its parts, no new field arises.

Thus, for example, by adjoining to the field of rational numbers, which may be denoted by $\mathfrak{R}$, the number $i = \sqrt{-1}$

one obtains the field of complex numbers $x + yi$, with rational x, y, which we shall call $\mathfrak{J}$. By the adjunction of a variable u to the field $\mathfrak{R}$ one obtains the field which consists of all integral and fractional rational functions of u with rational coefficients; by the adjunction of u and i to $\mathfrak{R}$, the field of the rational functions of u whose coefficients are numbers in $\mathfrak{J}$, and so on.

Let it finally be remarked that a field of quantities is often also called, following Kronecker's example, a domain of rationality. It is sometimes useful to employ the expression domain of rationality for a field when it is to be expressed thereby that, in a particular problem, the quantities of this field are to be regarded as known or rational.

§. 141.
Functions in a Field.

We often have occasion to consider integral rational functions of a variable x

$$f(x) = a_0 x^m + a_1 x^{m-1} + \cdots + a_{m-1} x + a_m \tag{1}$$

whose coefficients belong to a particular field Ω, and which we shall call functions contained in Ω, or briefly functions in Ω. They are not to be confused with functions which, say, belong to the field Ω, when Ω is a function field.

In the third section we saw that, when the coefficients $a_0, a_1 \ldots a_m$ have definite numerical values, there exist m roots $\alpha_1, \alpha_2 \ldots \alpha_m$ of $f(x)$, which likewise have definite numerical values, among which, however, several equal to one another occur when the discriminant $\Delta(a)$ of the function (1) vanishes. If the coefficients a of $f(x)$ are independent variables, then for every definite system of values of the a there is a definite system of values of the α, and the α vary continuously with the a (§. 40). So long, therefore, as we restrict the variation of the a to a sufficiently narrow region, in which no such values occur for which the discriminant $\Delta(a)$ vanishes, the m quantities α, as functions of the a, are completely determined and distinguished from one another as soon as the value of each of them is given for some particular system of values of the a.

If the a are not independent variables, but are themselves in turn functions of other variables, all this remains valid, so long only as the changes of the a take place continuously.

Accordingly, in all these cases we may speak of the m roots of an equation of degree m, which are to be understood as numbers or as algebraic functions of their coefficients.

Now if Ω is an arbitrary field and $f(x)$ is a function in Ω, two cases are possible.

The function $f(x)$ is either decomposable or not decomposable into factors of lower degree whose coefficients belong to the field Ω. In the first case the function $f(x)$ is called reducible, in the second irreducible in Ω. When the field Ω may be regarded as known and fixed, for

instance when it is the field of rational numbers, one sometimes also speaks simply of reducible or irreducible functions, without expressly adding the designation "in the field Ω". That in itself, however, the precise specification of the field to which the reducibility refers, or which is regarded as the domain of rationality, is very essential, follows from the following remarks.

A linear function is of course irreducible in every field. If one multiplies several functions of the form (1) with one another, there arises a function of the same form, which, however, is then naturally reducible, since it can again be decomposed into the factors from which it arose.

A function $f(x)$ which is irreducible in Ω can become reducible in an extended field Ω' which arises from Ω by any adjunction; thus every function $f(x)$ becomes reducible in the field Ω' which arises from Ω by the adjunction of a root α of $f(x) = 0$; for a linear factor $x - \alpha$ can be separated off from $f(x)$. In the field; which consists of all numbers, every function with numerical coefficients is reducible.

I. An irreducible function $f(x)$ can have no common divisor with another function $F(x)$ whose coefficients belong to the same field, unless $F(x)$ is divisible by $f(x)$.

This theorem, which will render us very important services in what follows, is almost self-evident; for the greatest common divisor of two functions $F(x)$ and $f(x)$ can be found by rational computation, and is therefore also contained in Ω. But since $f(x)$ has no divisor contained in Ω except itself or a constant (i. e. a quantity contained in Ω), this greatest common divisor must therefore be either a constant or $f(x)$ itself.

It follows from this theorem that an irreducible function can never have multiple factors; for otherwise it would have to have a common divisor with its derivative $f'(x)$; thus $f'(x)$ would have to be divisible by $f(x)$, which is impossible, since the degree of $f'(x)$ is lower than that of $f(x)$.

We can also give Theorem I the following expression:

II. If $f(x)$ is irreducible, and $F(x)$ vanishes for one root of $f(x) = 0$, then it also vanishes for all other roots of $f(x) = 0$.

In particular we can conclude from this:

III. If $F(x)$ is of lower degree than the irreducible function $f(x)$, and $F(x)$ vanishes for a root of $f(x)$, then $F(x)$ must vanish identically, i. e. all coefficients of $F(x)$ must be zero.

Integral rational functions of several variables, $f(x, y, z \ldots)$, whose coefficients belong to the field Ω, are likewise called functions in Ω. These too are designated as reducible and irreducible functions, according as they are decomposable or not decomposable into several factors which are themselves functions in Ω, and actually contain at least some of the variables $x, y, z \ldots$.

Here, however, attention must be called to a distinction.

Among the functions of more than one independent variable there are some that are not decomposable at all into factors of lower degree which depend rationally on the variables; and again others that are indeed decomposable, but only into factors whose coefficients are not contained in Ω; and finally, thirdly, functions that are decomposable into several functions in Ω; only the last kind shall we call reducible in Ω, while those of the second kind are described as decomposable, those of the first kind being designated as indecomposable functions. If, for example, Ω is the field of rational numbers, then $x^2 - y^2 = (x - y)(x + y)$ is reducible in Ω. The function $x^2 - 2y^2$ is indeed decomposable into the factors $x + y\sqrt{2}, x - y\sqrt{2}$, but these factors are not functions in Ω. The function $x^2 + y^2 + 1$, finally, is not decomposable at all. A function irreducible in a field Ω, but decomposable, becomes reducible in an extended field.

The fundamental theorems on irreducible functions we came to know in §. 51. In those developments, it is true, the field of all numbers was at first assumed as the domain of rationality, in accordance with the standpoint there taken, in which field the distinction between indecomposable and irreducible functions disappears. But only the reducibility or irreducibility of a function is ever made use of, and the particular nature of the field Ω does not come into consideration. Thus there

is contained in it the proof of the theorem:

IV. A reducible function in Ω can be decomposed into irreducible factors in only one way, if rational functions which differ only by a constant factor are regarded as not distinct.

§. 142.
Algebraic Fields.

If Ω is an arbitrary field, and $F(x)$ a function in Ω, then the equation

$$F(x) = 0 \tag{1}$$

is called an equation in Ω. This equation is called reducible or irreducible according as the function $F(x)$ is reducible or irreducible. By the adjunction of a root α of such an equation to Ω there arises (if α does not itself already belong to Ω) a new field Ω', which we shall call an algebraic field "over" Ω, or also, when doubt as to the meaning is excluded, briefly, an algebraic field. For such a field we use the sign

$$\Omega(\alpha).$$

If $\beta, \gamma \ldots$ are roots of other equations in Ω, or also other roots of the same equation, then by simultaneous adjunction of $\alpha, \beta, \gamma \ldots$ we likewise obtain algebraic fields over Ω, which we denote by

$$\Omega(\alpha, \beta, \gamma \ldots)$$

We shall see presently that this extension of the concept of algebraic fields is only an apparent one, and so for the time being we stop at the adjunction of one quantity α, that is, at the consideration of $\Omega(\alpha)$.

The equation (1), whose root is α, may be reducible. But among the irreducible factors of $F(x)$ there is at least one that vanishes for $x = \alpha$, which we shall denote by $f(x)$; this function $f(x)$ is, if we take the coefficient of the highest power of x equal to 1, completely determined, because by the preceding section $x = \alpha$ cannot be the root of two different irreducible equations.

The equation $f(x) = 0$ thus has the form

$$x^n + a_1 x^{n-1} + a_2 x^{n-2} + \cdots + a_{n-1} x + a_n = 0, \tag{2}$$

in which $a_1, a_2 \ldots a_n$ are quantities in Ω. If n is the degree of this equation, then we also call n the degree of the field $\Omega(\alpha)$.

All quantities Θ of the field $\Omega(\alpha)$ can be derived by addition, subtraction, multiplication and division from α and from numbers in Ω; they can therefore be represented as rational functions of α with coefficients in Ω, or in the form

$$\Theta = \frac{\chi(\alpha)}{\psi(\alpha)}, \tag{3}$$

in which $\chi(x)$ and $\psi(x)$ are functions in Ω. But since $\psi(\alpha)$ must be different from zero, $\psi(x)$ is not divisible by $f(x)$, and since $f(x)$ is assumed irreducible, $\psi(x)$ is relatively prime to $f(x)$.

Accordingly, by §. 6, Theorem II, we can determine the functions $Q(x)$, $\varphi(x)$ in Ω, and indeed $\varphi(x)$ at most of degree $n - 1$, so that

$$Q(x) f(x) + \varphi(x)\psi(x) = \chi(x) \tag{4}$$

holds, and if we then set $x = \alpha$ herein, so that $f(\alpha) = 0$, it follows that

$$\frac{\chi(\alpha)}{\psi(\alpha)} = \varphi(\alpha). \tag{5}$$

If we denote the coefficients of $\varphi(x)$, which, as we have seen, are quantities in Ω, by $c_0, c_1 \ldots c_{n-1}$, then every quantity Θ in $\Omega(\alpha)$ can accordingly be represented in the form:

$$\Theta = c_0 + c_1\alpha + c_2\alpha^2 + \cdots + c_{n-1}\alpha^{n-1}.$$

This representation is possible in only one way; for if at the same time

$$\Theta = c_0' + c_1'\alpha + c_2'\alpha^2 + \cdots + c_{n-1}'\alpha^{n-1}, \tag{6}$$

then the function of degree $(n-1)$

$$c_0 - c_0' + (c_1 - c_1')x + (c_2 - c_2')x^2 + \cdots + (c_{n-1} - c_{n-1}')x^{n-1}$$

vanishes for $x = \alpha$. But by Theorem III of the preceding section this is possible only if

$$c_0 = c_0', \quad c_1 = c_1', \ldots c_{n-1} = c_{n-1}'$$

holds.

If we regard the quantities of the field Ω as the given ones, its quantities as rational, and the field $\Omega(\alpha)$ as the one derived from them, then we also call a rational function in Ω briefly a rational function, and can accordingly also say:

The field $\Omega(\alpha)$ consists of all rational functions of α.

§. 143.
Simultaneous adjunction of several algebraic quantities.

To a more exact investigation of the field $\Omega(\alpha)$ we prefix a general theorem, very simple but yet rich in consequences.

1. If $\Phi_1(x, y, z \ldots)$, $\Phi_2(x, y, z \ldots)$, $\Phi_3(x, y, z \ldots) \ldots$ are integral rational functions of the variables $x, y, z \ldots$ with arbitrary coefficients, which do not all vanish at once in any of the functions, then one can substitute for the variables, in infinitely many ways, such rational numerical values that none of the functions $\Phi_1, \Phi_2, \Phi_3 \ldots$ vanishes.

The theorem is first of all evident when the functions $\Phi_1, \Phi_2, \Phi_3 \ldots$ depend on only one variable; for then there are altogether only a finite number of numerical values for this variable for which one of these functions becomes zero.

But then we can easily see the correctness of the theorem for functions of $n+1$ variables, provided we regard it as proved for functions of n variables. For if we order the functions according to the $(n+1)$th variable t, then by hypothesis we can put for the remaining n variables such rational values that in none of the functions do all the coefficients of the powers of t vanish; we then have functions of the single variable t, and for it we can put such a rational value that none of the functions vanishes. The theorem is thus proved in general.

It shall now first be shown that the simultaneous adjunction of several algebraic quantities can be replaced by the adjunction of a single one; in other words, that every field $\Omega(\alpha, \beta, \gamma \ldots)$ can be regarded as a field $\Omega(\alpha)$.

Let then $\alpha, \beta, \gamma \ldots$ be algebraic quantities in arbitrary number, and let

$$A(x) = 0, \quad B(x) = 0, \quad C(x) = 0 \ldots \tag{1}$$

be equations in Ω whose roots are $x = \alpha$, $x = \beta$, $x = \gamma \ldots$. None of these equations is to have a multiple root, an assumption by which the generality is not restricted.

We have seen in the preceding section that every quantity of a field $\Omega(\alpha)$ can be represented as an integral rational function in Ω of α. If in this theorem we put in place of the field Ω the field $\Omega(\beta, \gamma \ldots)$, it follows that every number in $\Omega(\alpha, \beta, \gamma \ldots)$ can be represented as an integral rational function of α with coefficients from $\Omega(\beta, \gamma \ldots)$; and if we repeat the same mode of inference with respect to the coefficients, we obtain:

2. Every quantity of the field $\Omega(\alpha, \beta, \gamma \ldots)$ can be represented as an integral rational function in Ω of $\alpha, \beta, \gamma \ldots$.

We form a linear function of the $\alpha, \beta, \gamma \ldots$

$$\xi = x\alpha + y\beta + z\gamma + \cdots,$$

and observe that each of the equations (1) has not merely one root, but several, whose number is equal to the degree of the equation. If then $\alpha', \beta', \gamma' \ldots$ is any combination, different from $\alpha, \beta, \gamma \ldots$, of one root from each of the equations (1), we set:

$$\xi' = x\alpha' + y\beta' + z\gamma' + \ldots$$

and form in the same way $\xi'', \xi''' \ldots$. The number of the quantities ξ so formed is, if a is the degree of $A(x)$, b the degree of $B(x)$, c the degree of $C(x)$, and so on,

$$m = abc \ldots$$

The differences $\xi - \xi'$, $\xi - \xi''$, $\xi' - \xi'' \ldots$ are linear functions of $x, y, z \ldots$, and none of them is identically zero, since we have assumed that none of the equations (1) has equal roots. By Theorem 1 we can therefore put for $x, y, z \ldots$ such rational numbers that none of these differences vanishes, so that the m values $\xi, \xi', \xi'' \ldots$ are all different from one another.

Now every rational function that is symmetric with respect to the roots of each of the equations (1) is rationally expressible in terms of the coefficients of these equations, i.e. it is a number in Ω. To these belong also the coefficients of the function of degree m in t:

$$F(t) = (t - \xi)(t - \xi')(t - \xi'') \ldots, \tag{2}$$

and $F(t) = 0$ is therefore an equation in Ω that has no equal roots, and one of whose roots is ξ.

If, further, Θ is a quantity in $\Omega(\alpha, \beta, \gamma \ldots)$, and hence an integral rational function of $\alpha, \beta, \gamma \ldots$, and if we denote by $\Theta', \Theta'' \ldots$ the quantities that arise from this rational function when the variables are replaced by $\alpha', \beta', \gamma' \ldots$ or $\alpha'', \beta'', \gamma'' \ldots$ and so on, then

$$F(t)\left(\frac{\Theta}{t - \xi} + \frac{\Theta'}{t - \xi'} + \frac{\Theta''}{t - \xi''} + \cdots\right) = \psi(t)$$

is, as a symmetric function of the α and of the $\beta \ldots$, an integral rational function of degree $(m - 1)$ in t in Ω, and if one sets $t = \xi$, it follows

$$\Theta = \frac{\psi(\xi)}{F'(\xi)}, \tag{3}$$

and Θ is therefore contained in $\Omega(\xi)$. Since conversely every quantity in $\Omega(\xi)$ is at the same time contained in $\Omega(\alpha, \beta, \gamma \ldots)$, the two fields are identical, i. e. we have

$$\Omega(\xi) = \Omega(\alpha, \beta, \gamma \ldots).$$

But with this our assertion is proved.

§. 144.
Primitive and imprimitive fields.

By the theorem last proved, we now restrict ourselves to the consideration of algebraic fields $\Omega(\alpha)$, where α is the root of an equation in Ω. Let this equation, freed from multiple factors, be denoted by

$$F(x) = 0 \tag{1}$$

and be of degree m. Its roots are as follows.

$$\alpha, \alpha_1, \alpha_2 \ldots \alpha_{m-1}. \tag{2}$$

It may be that $F(x)$ is reducible in Ω; then there will be one and only one irreducible factor $f(x)$ of $F(x)$, so that α is a root of the following equation.

$$f(x) = 0 \tag{3}$$

Let the degree of $f(x)$ be n. The roots of (3), which are all contained among the roots (2), are as follows.

$$\alpha, \alpha_1, \alpha_2 \ldots \alpha_{n-1}. \tag{4}$$

The degree of the field $\Omega(\alpha)$ is then, by the stipulation made in §. 142, likewise n.

Each of the quantities (4) gives rise to an algebraic field, and thus arise the fields

$$\Omega(\alpha), \Omega(\alpha_1) \ldots \Omega(\alpha_{n-1}), \tag{5}$$

which we call the fields conjugate to $\Omega(\alpha)$. It may be that these fields are all or in part identical, but they may also all be different from one another, of which more later.

By §. 142 we obtain every quantity Θ in $\Omega(\alpha)$ if, in an integral rational function $\chi(t)$ in Ω, of degree at most $(n-1)$, we put the number α for the variable t.

If we substitute for t in $\chi(t)$ the various quantities (4), we obtain n quantities, one from each of the conjugate fields

$$\Theta = \chi(\alpha),\ \Theta_1 = \chi(\alpha_1) \ldots \Theta_{n-1} = \chi(\alpha_{n-1}), \tag{6}$$

and these are called the quantities conjugate to Θ.

Every symmetric function of these quantities is at the same time a symmetric function of the roots of the equation (3), and hence contained in Ω.

Among these symmetric functions we wish to single out two, which occur frequently, by special names and notations; there is the sum

$$S(\Theta) = \Theta + \Theta_1 + \Theta_2 + \cdots + \Theta_{n-1}, \tag{7}$$

which we call the trace of Θ, and the product

$$N(\Theta) = \Theta\Theta_1\Theta_2 \ldots \Theta_{n-1}, \tag{8}$$

which is called the norm of Θ. Conjugate numbers accordingly have the same traces and norms.

The product

$$(t-\Theta)(t-\Theta_1) \ldots (t-\Theta_{n-1}) = \Phi(t) \tag{9}$$

is an integral rational function of degree n of t in Ω, and its roots are Θ and the quantities conjugate to Θ. From this results the theorem:

1. Every quantity in $\Omega(\alpha)$ is a root of an equation of degree n in Ω, whose remaining roots are the quantities conjugate to Θ.

The computation of the function $\Phi(t)$ from $f(x)$ is nothing other than the Tschirnhausen transformation considered from another point of view in the fourth and sixth sections.

We now have to examine the function $\Phi(t)$ as to its irreducibility.

If the function $\Phi(t)$ is reducible, then it has an irreducible factor $\varphi(t)$, in which we may assume the coefficient of the highest power of t equal to 1, and every such factor vanishes for at least one of the values $\Theta, \Theta_1 \dots \Theta_{n-1}$.

Let us then have

$$\varphi(\Theta) = \varphi[\chi(\alpha)] = 0,$$

that is, the equations $\varphi[\chi(x)] = 0$ and $f(x) = 0$ have a common root. But since $f(x)$ is assumed irreducible, $\varphi[\chi(x)]$ must, by the Theorem II, §. 141, for all roots

$\alpha, \alpha_1, \alpha_2 \dots \alpha_{n-1}$ of $f(x) = 0$ vanish, i. e. we have

$$\varphi(\Theta) = 0, \quad \varphi(\Theta_1) = 0, \quad \varphi(\Theta_2) = 0 \dots \varphi(\Theta_{n-1}) = 0.$$

If, then, the values conjugate to Θ are all different from one another, $\varphi(t)$ is identical with $\Phi(t)$, i. e. $\Phi(t)$ is irreducible. But if among the values conjugate to Θ only n_1 different ones are present, say $\Theta, \Theta_1, \Theta_2 \dots \Theta_{n_1-1}$, then

$$\varphi(t) = (t-\Theta)(t-\Theta_1)\dots(t-\Theta_{n_1-1}), \tag{10}$$

and every other irreducible factor of $\Phi(t)$, since it must vanish for at least one of the conjugate values Θ and consequently for all of them, is identical with $\varphi(t)$. Hence $\Phi(t)$ is a power of $\varphi(t)$, say

$$\Phi(t) = \varphi(t)^{n_2} \tag{11}$$

and

$$n = n_1 n_2. \tag{12}$$

From this, then, follows the theorem:

2. The function $\Phi(t)$ is either irreducible or it is a power of an irreducible function. The n numbers conjugate to a number in $\Omega(\alpha)$ are either all different from one another, or they fall into n_1 systems of n_2 numbers each, equal among themselves. In the first case $\Phi(t)$ is irreducible, in the second it is the n_2th power of an irreducible function of degree n_1 in Ω.

A quantity Θ in $\Omega(\alpha)$ that is different from all its conjugate numbers, and that therefore satisfies an irreducible equation of degree n, is called a primitive quantity of the field. By the theorem §. 143, 1. infinitely many such primitive quantities can be determined, indeed in such a way that the coefficients of $\Theta = \chi(\alpha)$ are rational numbers. One need only dispose of the coefficients of χ in such a way that among the conjugate quantities $\chi(\alpha)$ no equal ones occur.

3. Every quantity ω of the field $\Omega(\alpha)$ can be expressed rationally in terms of an arbitrary primitive quantity Θ of the field.

For if $\omega, \omega_1, \omega_2 \dots \omega_{n-1}$ are the numbers conjugate to ω, and likewise $\Theta, \Theta_1, \Theta_2 \dots \Theta_{n-1}$ those conjugate to Θ, and

$$\Phi(t) = (t-\Theta)(t-\Theta_1)\dots(t-\Theta_{n-1}),$$

then

$$\Phi(t)\left(\frac{\omega}{t-\Theta} + \frac{\omega_1}{t-\Theta_1} + \dots + \frac{\omega_{n-1}}{t-\Theta_{n-1}}\right) = \Psi(t)$$

is an integral rational function of degree $n-1$ of t, whose coefficients are numbers in Ω, and from this it follows, on setting $t = \Theta$,

$$\omega = \frac{\Psi(\Theta)}{\Phi'(\Theta)},$$

in which $\Phi'(\Theta)$ is different from zero.

Accordingly the field $\Omega(\Theta)$ is identical with the field $\Omega(\alpha)$.

4. If Θ is not primitive, then not every quantity in $\Omega(\alpha)$ can be expressed rationally in terms of Θ. The field $\Omega(\Theta)$ is a divisor of the field $\Omega(\alpha)$, and the degree of $\Omega(\Theta)$ is a divisor of the degree of $\Omega(\alpha)$.

For every number of the field $\Omega(\Theta)$ satisfies an equation in Ω of degree n_1, where n_1 is a divisor of n and less than n. Hence a primitive quantity of the field $\Omega(\alpha)$, which satisfies an irreducible equation of degree n, cannot be contained in $\Omega(\Theta)$.

The field $\Omega(\alpha)$ is called primitive if, apart from the quantities in Ω, it contains no imprimitive quantities; imprimitive, if it contains still other imprimitive quantities.

From this definition it follows first of all that a field whose degree is a prime is necessarily primitive; for an imprimitive quantity Θ in $\Omega(\alpha)$ satisfies an equation whose degree is a divisor of the degree n of $\Omega(\alpha)$ different from n. If this divisor is equal to 1, then Θ is contained in Ω.

We shall now become acquainted with some of the most important properties of the imprimitive fields.

If $\Omega(\alpha)$ has a divisor Ω' different from Ω, which in its turn contains Ω as a divisor (so that Ω' is a field over Ω), and if β is a quantity that belongs to the field Ω' but not to Ω, then the field $\Omega(\beta)$ is an algebraic field over Ω and at the same time a divisor of Ω' and of $\Omega(\alpha)$, and according to what we have just

proved, the degree n_1 of the field $\Omega(\beta)$ is a divisor of n. If now Ω' is not exhausted by $\Omega(\beta)$, we take a quantity γ in Ω', but not in $\Omega(\beta)$; then the field $\Omega(\beta,\gamma)$ is a divisor of Ω' and of $\Omega(\alpha)$ and has, for its part, $\Omega(\beta)$ as a divisor. The degree of $\Omega(\beta,\gamma)$ is therefore greater than that of $\Omega(\beta)$ and smaller than that of $\Omega(\alpha)$. If Ω' is not yet exhausted by this, we continue in the same way, but must finally come to an end, since the degrees of the fields $\Omega(\beta), \Omega(\beta,\gamma)\ldots$ always increase and yet remain smaller than n. From this it follows:

5. Every divisor of $\Omega(\alpha)$ which contains the field Ω is an algebraic field $\Omega(\beta)$ over Ω. The degree of $\Omega(\beta)$ is a divisor of the degree of $\Omega(\alpha)$, and if both fields have the same degree, they are identical.

We can therefore also state our definition thus:

6. An algebraic field over Ω is primitive if it has no field over Ω except Ω and itself as a divisor.

§. 145.
Normal fields. Galois resolvent.

If $\Omega(\alpha)$ is an algebraic field, then the conjugate fields

$$\Omega(\alpha), \Omega(\alpha_1)\ldots\Omega(\alpha_{m-1})$$

are all of the same degree, and hence if one of them is contained in another, the two are identical.

A field which is identical with all its conjugate fields is called a normal field. In the normal fields much simpler laws prevail, and the great advance which algebra owes to Galois rests essentially on the fact that arbitrary fields are reduced to normal fields. The normal fields are therefore also called Galois fields.

If a field $\Omega(\varrho)$ of degree μ has the property that the numbers $\varrho_1, \varrho_2\ldots\varrho_{\mu-1}$ conjugate to ϱ are all contained in $\Omega(\varrho)$, then it is a normal field; for then the fields $\Omega(\varrho_1), \Omega(\varrho_2)\ldots\Omega(\varrho_{\mu-1})$ are also all contained in $\Omega(\varrho)$ and hence all identical with $\Omega(\varrho)$.

We shall call an equation a normal equation if it is irreducible and has the property that all its roots can be expressed rationally (in Ω) by one of them. Then a primitive element of a normal field of degree μ is a root of a normal equation of degree μ. But the converse also holds, namely, that if ϱ is a root of a normal equation, then $\Omega(\varrho)$ is a normal field of the same degree. For if ϱ_0 is the root of the normal equation by which all the others are rationally expressed, then $\Omega(\varrho_0)$ is certainly a normal field. But $\Omega(\varrho)$ is conjugate to this field, and hence identical with it. From this it follows further that in a normal equation every root is rationally expressible not only by one determinate root among them, but by any one whatever.

One can now reduce arbitrary algebraic fields to normal fields in the following way.

Let

$$F(x) = 0 \tag{1}$$

be an arbitrary reducible or irreducible equation in Ω of degree m, of which we shall only assume that it has no multiple roots. Its roots are as follows.

$$\alpha,\ \alpha_1 \dots \alpha_{m-1}. \tag{2}$$

From these one obtains m algebraic fields

$$\Omega(\alpha),\ \Omega(\alpha_1) \dots \Omega(\alpha_{m-1}), \tag{3}$$

and one can evidently choose the function $F(x)$ so that among the fields (3) any given algebraic fields over Ω occur. We call the field over Ω derived from all the quantities (2), i. e. from all the roots of the equation (1),

$$N = \Omega(\alpha, \alpha_1 \dots \alpha_{m-1}) \tag{4}$$

the Galois field of the equation $F(x) = 0$.

If $F(x)$ is irreducible, the fields (3) are the fields conjugate to $\Omega(\alpha)$. In this case the field N shall be called the norm of each of the fields $\Omega(\alpha), \Omega(\alpha_1) \dots \Omega(\alpha_{m-1})$.

If $\Omega(\alpha)$ is a normal field, it is identical with its norm. In the general case it is to be shown that N is a normal field. We choose a primitive element ϱ of the field N (according to §. 143), and can then also denote the field N

denote by $\Omega(\varrho)$. If μ is the degree of N, then ϱ satisfies an irreducible equation of degree μ

$$g(t) = 0, \tag{5}$$

of which it is to be shown that it is a normal equation. For this purpose we first remark that the one root ϱ of this equation is a rational function of $\alpha, \alpha_1 \dots \alpha_{m-1}$, because it was contained in N. To indicate this, let us set

$$\varrho = \varrho(\alpha, \alpha_1 \dots \alpha_{m-1}),$$

and now form all the different arrangements of the digits

$$0,\ 1,\ 2 \dots m-1,$$

whose number amounts to $\Pi(m)$:

$$(0, 1, 2 \dots m-1),\ (0', 1', 2' \dots (m-1)'),\ (0'', 1'', 2'' \dots (m-1)'') \dots, \tag{6}$$

in which the digits with one, two, etc. accents are the same as those without accent, only in a different order, and from these form the functions

$$\varrho = \varrho(\alpha_0, \alpha_1 \dots \alpha_{m-1}),\ \varrho' = \varrho(\alpha_{0'}, \alpha_{1'} \dots \alpha_{(m-1)'}),\ \varrho'' = \varrho(\alpha_{0''}, \alpha_{1''} \dots \alpha_{(m-1)''}) \dots, \tag{7}$$

without concern as to whether some among them may be equal to one another or not.

If in all the arrangements (6) we carry out one and the same permutation, e.g. 0 with 1, the totality of these arrangements does not change; only their order becomes a different one. For, first, such a permutation can produce nothing other than arrangements of the digits, and secondly, two different arrangements cannot pass into the same arrangement by one and the same permutation.

Hence by every such permutation the functions (7) will likewise only be permuted among one another. If, then, we form the product

$$G(t) = (t - \varrho)\,(t - \varrho')\,(t - \varrho'')\,\ldots$$

for a variable t, its coefficients, which are certainly functions of $\alpha, \alpha_1 \ldots \alpha_{m-1}$, remain unchanged when these quantities are permuted in any way; that is, they are symmetric functions of the roots of the function $F(x)$, and hence $G(t)$ is a function of t in Ω. All the roots of $G(t)$ are quantities in N, since they are rationally expressed by the α.

Now $G(t)$ and $g(t)$ have a root in common. But since $g(t)$ is irreducible, $G(t)$ must be divisible by $g(t)$; hence all the roots of $g(t)$ are also contained in N, i. e. N is a normal field, q. e. d.

Every equation $g(t) = 0$ is called a Galois resolvent of the equation $F(x) = 0$, and a Galois resolvent is accordingly defined by the following specification:

An equation $g(t) = 0$ is a Galois resolvent of a given equation $F(x) = 0$ in Ω if 1) $g(t)$ is irreducible, if 2) all the roots of $F(x)$ are rationally expressible through a root ϱ of $g(t)$, and 3) one root of $g(t)$ is rationally expressible through the roots of $F(x)$.

For by 2), N is contained in $\Omega(\varrho)$, and by 3) one of the fields conjugate to $\Omega(\varrho)$, namely $\Omega(\varrho_1)$, is contained in N. The degrees of N and $\Omega(\varrho)$ therefore cannot be different, and consequently $N = \Omega(\varrho)$.

Every Galois resolvent is a normal equation, and a normal equation is its own Galois resolvent.

§. 146.
The Substitutions of a Normal Field.

Now let $\Omega(\varrho)$ be a normal field of degree μ and ϱ one of its primitive numbers, and further let

$$g(t) = 0 \tag{1}$$

be the irreducible equation of degree μ of which one root is $t = \varrho$. The elements conjugate to ϱ are as follows.

$$\varrho,\ \varrho_1,\ \varrho_2 \ldots \varrho_{\mu-1}. \tag{2}$$

Since by the definition of the normal field the quantities (2) are all contained in $\Omega(\varrho)$, we can determine $\Theta_1(t), \Theta_2(t) \ldots \Theta_{\mu-1}(t)$ as integral rational functions in Ω, of degree at most $\mu - 1$, such that

$$\varrho_1 = \Theta_1(\varrho),\ \varrho_2 = \Theta_2(\varrho) \ldots \varrho_{\mu-1} = \Theta_{\mu-1}(\varrho) \tag{3}$$

becomes. If ω is an arbitrary quantity in $\Omega(\varrho)$, then the quantities conjugate to ω can be represented thus:

$$\omega = \varphi(\varrho),\ \omega_1 = \varphi(\varrho_1) \ldots \omega_{\mu-1} = \varphi(\varrho_{\mu-1}), \tag{4}$$

in which $\varphi(t)$ is a rational function in Ω.

Now since $g(t)$ is irreducible, the theorem of §. 141, II holds, which we now state as follows:

1. If a rational function $\Phi(t)$ in Ω has a root in common with $g(t)$, then there vanish all the conjugate quantities

$$\Phi(\varrho),\ \Phi(\varrho_1)\dots\Phi(\varrho_{\mu-1}).$$

If in one of the functions $\Theta_k(\varrho)$, by which according to (3) the roots of (1) are expressed, ϱ is replaced by another root ϱ_h, then there arises from it again one of the roots; for if

$$g[\Theta_k(\varrho)] = 0,$$

then by 1. also

$$g[\Theta_k(\varrho_h)] = 0,$$

and the sequence of quantities

$$\varrho_h,\ \Theta_1(\varrho_h),\ \Theta_2(\varrho_h)\dots\Theta_{\mu-1}(\varrho_h) \tag{5}$$

agrees, apart from the arrangement, with the following sequence.

$$\varrho,\ \Theta_1(\varrho),\ \Theta_2(\varrho)\dots\Theta_{\mu-1}(\varrho) \tag{6}$$

This will be proved when we show that no two equal values occur in (5). If, for the sake of uniformity, we denote by $\Theta_0(\varrho)$ or $\Theta(\varrho)$ the root ϱ itself, then from the equality of two of the quantities (5)

$$\Theta_i(\varrho_h) = \Theta_k(\varrho_h) \tag{7}$$

Theorem 1 gives

$$\Theta_i(\varrho) = \Theta_k(\varrho), \tag{8}$$

which, however, if i is different from k, is not possible. Hence, together with the quantities (6), the quantities (5) are also different from one another.

We can sum this up as a theorem:

2. If one interchanges ϱ with ϱ_h, then at the same time every quantity conjugate to ϱ passes over into a definite other one, and never two different ones into the same.

If in all functions of ϱ we put another root ϱ_a in place of ϱ, we carry out a substitution. We denote this substitution by

$$\sigma_a = (\varrho, \varrho_a), \qquad a = 0, 1, 2 \dots \mu - 1,$$

where by σ_0 or σ is understood the so-called identical substitution (ϱ, ϱ), which consists in ϱ being replaced by itself, so that all numbers in $\Omega(\varrho)$ remain unchanged.

If in any one of the roots (3)

$$\varrho_h = \Theta_h(\varrho)$$

we carry out the substitution σ_a, then ϱ_h passes into another root ϱ_k, which is determined by

$$\varrho_k = \Theta_h(\varrho_a) = \Theta_h\Theta_a(\varrho) = \Theta_k(\varrho).$$

Thus ϱ_k is the same function of ϱ_a as ϱ_h is of ϱ.

An arbitrary quantity $\omega = \varphi(\varrho)$ of the field $\Omega(\varrho)$ passes, by the substitution σ_a, into $\omega_a = \varphi(\varrho_a)$. If we express ω in terms of ϱ_h and ω_a in terms of ϱ_k:

$$\omega = \psi(\varrho_h), \qquad \omega_a = \psi(\varrho_k),$$

then one sees that ω passes into ω_a by the substitution

$$\sigma_a = (\varrho_h, \varrho_k) = (\varrho_h, \Theta_h \Theta_a(\varrho)), \tag{9}$$

and this is only another expression for the substitution (ϱ, ϱ_a).

Here ϱ_h is an arbitrary root of $g(t)$, and the associated ϱ_k is determined by σ_a. But for a given σ_a one may also take ϱ_k arbitrarily and determine the associated ϱ_h, by letting the index h in $\Theta_h(\varrho_a)$ run through the values $0, 1, 2 \ldots \mu - 1$, whereby each root ϱ_k makes its appearance once. Hence:

3. Every substitution σ_a can be represented in the form (ϱ_h, ϱ_k), in which either ϱ_h or ϱ_k is an arbitrarily given one of the μ roots ϱ, while the other is completely determined by σ_a.

The number of all mutually distinct substitutions σ_a is therefore, the identical substitution included, equal to the degree of the field $\Omega(\varrho)$, which we have already denoted above by μ. Each of these substitutions carries the totality of the quantities of the field $\Omega(\varrho)$ into itself, in such a way that each passes into a definite other one, and never two distinct quantities into the same one [formulae (7), (8)].

We therefore call the σ_a the substitutions of the field $\Omega(\varrho)$.

If $\omega = \varphi(\varrho)$ remains unchanged in the field $\Omega(\varrho)$ when ϱ is replaced by ϱ_a, that is, if

$$\varphi(\varrho) = \varphi(\varrho_a)$$

then we say that ω permits or admits the substitution (ϱ, ϱ_a) or σ_a.

The quantities ω which, apart from the identical substitution, admit no substitution σ_a, are the primitive elements of the field $\Omega(\varrho)$.

4. A quantity which admits all the substitutions σ_a is necessarily an element of Ω.

For such a quantity is identical with all its conjugate quantities, and hence by §. 144 satisfies an equation of the first degree in Ω.

§. 147.

Composition of the Substitutions.

If in any function of ϱ we replace the root ϱ first by ϱ_a and then ϱ_a by ϱ_b, the result is the same as if we had from the outset interchanged ϱ with ϱ_b. If therefore we set

$$\sigma = (\varrho, \varrho_a), \quad \sigma' = (\varrho_a, \varrho_b), \quad \sigma'' = (\varrho, \varrho_b), \tag{1}$$

then it is indifferent for the result whether in all the numbers of the field $\Omega(\varrho)$ we first carry out the substitution σ and then the substitution σ', or whether we make the substitution σ'' in a single step.

1. We therefore call σ'' compounded or composed of σ and σ', and denote this relation by the symbolic equation

$$\sigma\sigma' = \sigma''. \tag{2}$$

Since by Theorem 3. of the preceding section we may, in the designation of a substitution (ϱ_h, ϱ_k), choose the first or the second element arbitrarily, ϱ_a, ϱ_b can be so selected that σ, σ' are any two arbitrarily given ones among the μ substitutions of the field $\Omega(\varrho)$; but σ'' is thereby completely determined. But likewise, if σ and σ'' are given, σ' is uniquely determined. For if we choose ϱ arbitrarily, then ϱ_a is completely determined by σ, and ϱ_b by σ'', whereby $\sigma' = (\varrho_a, \varrho_b)$ is also given. Finally, if σ' and σ'' are given, then σ is uniquely determined, since first ϱ_b is determined by σ'' and then ϱ_a by σ'. We therefore have:

2. Of the three substitutions of the field $\Omega(\varrho)$ connected by the symbolic equation $\sigma\sigma' = \sigma''$, any two may be given arbitrarily, while the third is thereby uniquely determined.

We have expressed the composition of the substitutions symbolically by the sign of multiplication. Here, however, one must pay careful attention to the order of the factors or components, because $\sigma\sigma'$ can be different from $\sigma'\sigma$; that is, for the symbolic multiplication expressing composition, the commutative law of ordinary multiplication does not hold; but the so-called associative law does hold, which is expressed in the following theorem:

3. If $\sigma, \sigma', \sigma''$ are any three of the μ substitutions of the field $\Omega(\varrho)$, then

$$(\sigma\sigma')\sigma'' = \sigma(\sigma'\sigma''). \tag{3}$$

The proof is very simple; for by Theorem 3., §. 146, we can determine $\varrho_a, \varrho_b, \varrho_c$ so that

$$\sigma = (\varrho, \varrho_a), \quad \sigma' = (\varrho_a, \varrho_b), \quad \sigma'' = (\varrho_b, \varrho_c), \tag{4}$$

and then

$$\sigma\sigma' = (\varrho, \varrho_b), \qquad (\sigma\sigma')\sigma'' = (\varrho, \varrho_b)(\varrho_b, \varrho_c) = (\varrho, \varrho_c),$$
$$\sigma'\sigma'' = (\varrho_a, \varrho_c), \qquad \sigma(\sigma'\sigma'') = (\varrho, \varrho_a)(\varrho_a, \varrho_c) = (\varrho, \varrho_c).$$

We therefore denote briefly the substitution composed of the three substitutions $\sigma, \sigma', \sigma''$ by

$$\sigma\sigma'\sigma'',$$

and can in general compose from arbitrarily many components a definite substitution

$$\sigma\sigma'\sigma''\sigma'''\ldots$$

by uniting, one after another, two adjacent components at a time into a single substitution, until the whole symbolic product has contracted to a single substitution. For by §. 146, 3. we can set

$$\sigma = (\varrho, \varrho_a), \quad \sigma' = (\varrho_a, \varrho_b), \quad \sigma'' = (\varrho_b, \varrho_c), \quad \sigma''' = (\varrho_c, \varrho_e),$$

and always obtain

$$(\varrho, \varrho_a)(\varrho_a, \varrho_b)(\varrho_b, \varrho_c)(\varrho_c, \varrho_e) = (\varrho, \varrho_e),$$

which evidently can be extended to an arbitrary number of components.

On the basis of the properties 1., 2., 3. we call the totality of the μ substitutions of the normal field $\Omega(\varrho)$ a group, and indeed a finite group of degree μ. It is also called the group of the substitutions of the field $\Omega(\varrho)$.

With the general theory of groups we shall occupy ourselves in detail in later sections.

§. 148.

Permutation groups.

In §. 145 we have seen how, for an arbitrary equation in Ω of degree m

$$F(x) = 0, \tag{1}$$

of which it is only assumed that it has no equal roots, one can determine a normal field $\Omega(\varrho)$ to which all the roots of (1) belong. If, then, the roots of (1) are denoted by

$$\alpha, \alpha_1, \alpha_2 \ldots \alpha_{m-1} \tag{2}$$

then

$$\alpha = \chi(\varrho), \quad \alpha_1 = \chi_1(\varrho), \dots \alpha_{m-1} = \chi_{m-1}(\varrho), \tag{3}$$

and the χ are rational functions in Ω.

But if we set in place of ϱ a quantity ϱ_1 conjugate to ϱ, that is, if we carry out a substitution $\sigma = (\varrho, \varrho_1)$, then every root α passes over into a definite other one, and never do two different ones pass into the same; for if

$$F(\alpha) = F[\chi(\varrho)] = 0$$

then by §. 146, 1:

$$F[\chi(\varrho_1)] = 0,$$

that is, $\chi(\varrho_1)$ is also a root of the equation (1). But that two different quantities of the field $\Omega(\varrho)$ cannot pass over into the same quantity through a substitution σ we have already shown in general in §. 146.

1. A substitution σ therefore calls forth a certain permutation among the roots α, in that each of the indices $0, 1, 2 \dots m-1$ is replaced by a definite other one.

This leads us, then, to study more closely the permutations of m figures

$$0, 1, 2 \dots m-1$$

and their properties in general.

If the figure 0 passes into a_0, 1 into a_1, 2 into a_2, and so on, $m-1$ into a_{m-1}, then

$$a_0, a_1, a_2 \dots a_{m-1} \tag{4}$$

must be some other arrangement of the figures $0, 1, 2 \dots m-1$. The transition from the one arrangement to the other we denote by a symbol, such as

$$\pi_a = \begin{pmatrix} 0, & 1, & 2 & \dots & m-1 \\ a_0, & a_1, & a_2 & \dots & a_{m-1} \end{pmatrix} \tag{5}$$

and we call this transition a permutation.

The number of all possible permutations of m elements is

$$\Pi(m) = 1 \,.\, 2 \,.\, 3 \dots m,$$

among which the identical permutation

$$\begin{pmatrix} 0, & 1, & 2 & \dots & m-1 \\ 0, & 1, & 2 & \dots & m-1 \end{pmatrix},$$

which leaves every element in its place, is also counted.

The symbol (5) does not change its meaning if one arranges the individual pairs

$$\begin{matrix} 0, & 1, & 2, & \dots & m-1 \\ a_0, & a_1, & a_2, & \dots & a_{m-1} \end{matrix}$$

differently. If then

$$b_0, b_1, b_2 \dots b_{m-1}$$

is any arrangement of the figures $0, 1, 2 \dots m-1$, we may also put, in place of (5):

(6) $$\pi_a = \begin{pmatrix} b_0, & b_1, & b_2 & \dots & b_{m-1} \\ a_{b_0}, & a_{b_1}, & a_{b_2} & \dots & a_{b_{m-1}} \end{pmatrix},$$

which is entirely equivalent to (5).

From this the very important concept of the composition of permutations can be derived. If we replace an index a by an index b, and then b by c, the effect is the same as if a were replaced at once by c. If, then, we carry out two permutations one after the other, the result is a definite permutation, which we can represent in the following way. If

$$\pi_a = \begin{pmatrix} 0, & 1 & \dots & m-1 \\ a_0, & a_1 & \dots & a_{m-1} \end{pmatrix}, \quad \pi_b = \begin{pmatrix} 0, & 1 & \dots & m-1 \\ b_0, & b_1 & \dots & b_{m-1} \end{pmatrix}$$

are any two permutations, then by (6) we can also write the second thus:

$$\pi_b = \begin{pmatrix} a_0, & a_1 & \dots & a_{m-1} \\ b_{a_0}, & b_{a_1} & \dots & b_{a_{m-1}} \end{pmatrix},$$

and if we therefore carry out both one after the other, first π_a, then π_b, there arises a third permutation, which we call composed (componirt) from π_a and π_b, and which we represent as the symbolic product of π_a and π_b:

(7) $$\pi_a\pi_b = \begin{pmatrix} 0, & 1 & \dots & m-1 \\ b_{a_0}, & b_{a_1} & \dots & b_{a_{m-1}} \end{pmatrix}.$$

In this composition the factors usually cannot be interchanged, since $\pi_a\pi_b$ does not in general agree with $\pi_b\pi_a$. The following symbolic equation expresses the associative principle.

(8) $$(\pi_a\pi_b)\pi_c = \pi_a(\pi_b\pi_c) = \pi_a\pi_b\pi_c$$

One convinces oneself of this by the following simple consideration.

If an index p is changed by π_a into q, the resulting index is changed by π_b into r, and r by π_c into s, then p passes by $\pi_a\pi_b$ into r, and q by $\pi_b\pi_c$ into s; hence p is changed into s both by $(\pi_a\pi_b)\pi_c$ and by $\pi_a(\pi_b\pi_c)$. But formula (8) says nothing else.

To every permutation π_a there is a definite opposite (inverse) permutation, which consists in this, that the change produced by π_a is undone again, and which therefore, composed with π_a, produces the identical permutation. We denote this permutation by π_a^{-1}, and we have, if π_a has the expression (5),

(9) $$\pi_a^{-1} = \begin{pmatrix} a_0, & a_1 & \dots & a_{m-1} \\ 0, & 1 & \dots & m-1 \end{pmatrix}.$$

For by the rule of composition in (7) we have

$$\pi_a\pi_a^{-1} = \begin{pmatrix} 0, & 1 & \dots & m-1 \\ 0, & 1 & \dots & m-1 \end{pmatrix}.$$

But in the same way we also have

$$\pi_a^{-1}\pi_a = \begin{pmatrix} a_0, & a_1 & \dots & a_{m-1} \\ a_0, & a_1 & \dots & a_{m-1} \end{pmatrix},$$

which is again the identical permutation. Thus the permutation opposite to π_a^{-1} is again the original permutation π_a.

If $\pi_a\pi_b = \pi_c$, then

$$\pi_c^{-1} = \pi_b^{-1}\pi_a^{-1},$$

for $\pi_a\pi_b\pi_b^{-1}\pi_a^{-1}$ is the identical permutation.

If we denote the identical permutation by π_0, then

$$\pi_a\pi_a^{-1} = \pi_a^{-1}\pi_a = \pi_0, \tag{10}$$

and π_0 is opposite to itself (which does not exclude the possibility that there are still other permutations opposite to themselves). By composition with the identical permutation no other permutation is changed; thus

$$\pi_0\pi_a = \pi_a\pi_0 = \pi_a. \tag{11}$$

2. If π_c is composed of π_a and π_b, that is,

$$\pi_a\pi_b = \pi_c, \tag{12}$$

then not only is π_c completely and uniquely determined by π_a and π_b, but also π_a by π_b and π_c, and π_b by π_a and π_c.

For it follows from (12) that

$$\pi_a = \pi_c\pi_b^{-1}, \quad \pi_b = \pi_a^{-1}\pi_c. \tag{13}$$

3. A system

$$Q = \pi, \pi', \pi'', \pi''' \ldots,$$

selected from the totality of all $\Pi(m)$ permutations of m elements, which is so closed and complete in itself that any two, and consequently also arbitrarily many, permutations of Q always yield by composition a permutation of the same system, is called a permutation group. By the degree of a permutation group we understand the number of permutations that it contains.

The totality of all $\Pi(m)$ permutations forms such a group. Likewise the single identical permutation is a group by itself. What may still lie between these two extreme cases of groups is what will occupy us in the following sections.

As a simple example we mention here only the group of cyclic permutations, briefly the cyclic group, which one obtains when, in any arrangement of digits $(1, 2, 3 \ldots n)$, one replaces each digit by the following one and the last by the first, and when one repeats this permutation until one comes back to the original arrangement. The degree of the cyclic group is equal to the number of digits, e. g. with three digits:

$$\begin{pmatrix}1, & 2, & 3\\ 2, & 3, & 1\end{pmatrix}, \quad \begin{pmatrix}1, & 2, & 3\\ 3, & 1, & 2\end{pmatrix}, \quad \begin{pmatrix}1, & 2, & 3\\ 1, & 2, & 3\end{pmatrix}.$$

There are special groups, which will occupy us still further later on, that consist only of such permutations in whose composition, besides the laws mentioned above, the commutative law also holds, for which therefore always $\pi_1\pi_2 = \pi_2\pi_1$, and in consequence of which the analogy with multiplication is a much more complete one. Such groups, to which the cyclic groups belong, are called commutative or Abelian groups. When all the elements of a permutation group Q_1 are contained in a group Q, then Q_1 is called a divisor of Q, and Q divisible by Q_1.

§. 149.
Galois Group.

If, in the roots $\alpha, \alpha_1, \alpha_2 \ldots \alpha_{m-1}$ of an equation $F(x) = 0$, as they are expressed by §. 148, (3) as quantities of the normal field $\Omega(\varrho)$ of degree μ, we carry out one of the μ substitutions σ of the normal field, then a certain permutation takes place among these roots, as has already been shown above. To every substitution σ there corresponds therefore a certain permutation π of the m digits $0, 1, 2 \ldots m-1$. Two different substitutions σ_h, σ_k lead also to two different permutations π_h, π_k.

For by §. 145, ϱ is a rational function of $\alpha, \alpha_1 \ldots \alpha_{m-1}$; we use the following notation.

$$\varrho = \Phi(\alpha, \alpha_1 \ldots \alpha_{m-1})$$

If we here set

$$\alpha_s = \chi_s(\varrho)$$

it follows that

$$\varrho = \Phi[\chi(\varrho), \chi_1(\varrho) \ldots \chi_{m-1}(\varrho)], \tag{1}$$

and by §. 146, 1. we thus have, for every index k,

$$\varrho_k = \Phi[\chi(\varrho_k), \chi_1(\varrho_k) \ldots \chi_{m-1}(\varrho_k)]. \tag{2}$$

Now if the substitutions σ_h, σ_k produce the same permutation among the α, then for every index $s = 0, 1 \ldots m-1$

$$\chi_s(\varrho_h) = \chi_s(\varrho_k),$$

and from this it follows by (2) that we must also have $\sigma_h = \sigma_k$.

If the substitution σ_h produces the permutation π_h among the roots of $F(x) = 0$, then we say that the permutation π_h corresponds to the substitution σ_h.

To every substitution there corresponds a definite permutation, and the number of these permutations — counting in the identical permutation, which corresponds to the identical substitution — is therefore equal to μ.

If the permutations π_h and π_k correspond to the substitutions σ_h and σ_k, then to the composed substitution $\sigma_h\sigma_k$ there corresponds the composed permutation $\pi_h\pi_k$. For the permutations are the result of the corresponding substitutions on certain quantities of the field $\Omega(\varrho)$, and here it makes no difference whether we carry out two substitutions one after the other or carry out the substitution composed of both at one stroke (by §. 147). We therefore have the theorem:

I. To the μ substitutions of the field $\Omega(\varrho)$

$$\sigma, \sigma_1, \sigma_2 \ldots \sigma_{\mu-1}, \tag{3}$$

there correspond μ permutations of m figures

$$\pi, \pi_1, \pi_2 \ldots \pi_{\mu-1} \tag{4}$$

in such a way that to the substitution composed of two substitutions there corresponds the permutation composed of the corresponding permutations.

But from this it follows that the permutations (4) form a group. We denote this group by P and call it the Galois group of the equation $F(x) = 0$, or also the Galois group of each of the fields $\Omega(\alpha), \Omega(\alpha_1) \ldots \Omega(\alpha_{m-1})$.

This group stands to the group of substitutions of the field $\Omega(\varrho)$ in the relation expressed by Theorem I., for which reason the two groups are called isomorphic.

To find the characteristic properties of the Galois group, we have only to express the theorems §. 146, 1. to 4. somewhat differently. Of a function of the m roots which remains unchanged under any permutation we say that it admits this permutation, and thus obtain:

a) Every rational equation in Ω that holds between the m roots of $F(x)$ remains valid when the roots are subjected to any permutation of the Galois group.

b) Every rational function in Ω of the m roots of $F(x)$ that admits all the permutations of the Galois group is a number in Ω.

For if we express the roots of $F(x)$ as rational functions of ϱ, then a function of the roots α passes into a function $\varphi(\varrho)$. If a permutation of the Galois group has first been carried out, one obtains one of the conjugate numbers $\varphi(\varrho_a)$, which arises from $\varphi(\varrho)$ by a substitution $\sigma_a = (\varrho, \varrho_a)$. Now if $\varphi(\varrho) = 0$, then by the theorem §. 146, 1. all the conjugates $\varphi(\varrho_a) = 0$ as well, whereby a) is proved; and if the conjugate quantities $\varphi(\varrho_a)$ are all equal to one another, then their common value is, by §. 146, 4., contained in Ω, whereby b) is proved.

To a) and b) a third statement is added:

c) If any permutation π of the roots of $F(x)$ is applicable to all rational equations in Ω that hold between the roots, then π belongs to the Galois group, and the Galois group can therefore also be defined as the totality of all permutations that are applicable to all the rational equations between the roots.

For by §. 143 one can take ϱ to be a rational (e. g. linear) function of the m quantities $\alpha, \alpha_1, \alpha_2 \ldots \alpha_{m-1}$ in such a way that all the $\Pi(m)$ values which one obtains from it by the $\Pi(m)$ permutations of the α are different from one another. If then $g(t) = 0$ is the Galois resolvent of $F(x) = 0$ whose root is this ϱ, then

$$g(\varrho) = 0.$$

Hereupon, if ϱ is expressed in terms of the roots α, we can by hypothesis apply the permutation π, and thus, if ϱ thereby passes into ϱ_a, we obtain $g(\varrho_a) = 0$; that is, ϱ_a is also a root of the resolvent, and the permutation π therefore corresponds to a substitution (ϱ, ϱ_a) of the field $\Omega(\varrho)$, i. e. π belongs to the Galois group.

From this we conclude further the following important theorem:

d) If P is a group of permutations of the m roots α which possesses the properties a) and b), then P is the Galois group of the equation $F(x) = 0$.

For, in the first place, by c) every permutation of P belongs to the Galois group, and P is therefore certainly a divisor of the latter.

But if P comprises only ν permutations, we use the following notation.

$$\pi_1, \pi_2 \ldots \pi_\nu \tag{5}$$

If we apply these permutations to ϱ, let the result be:

$$\varrho_1, \varrho_2 \ldots \varrho_\nu. \tag{6}$$

If we apply one of the permutations (5), say π_k, to one of the quantities (6), say to ϱ_i, the effect is the same as if $\pi_i \pi_k$ had been applied to ϱ; let the result of this permutation be ϱ_i'. Now it lies in the hypothesis that P is a group that the composed permutation $\pi_i \pi_k$ also belongs to P, and hence that ϱ_i' is contained among the quantities (6). At the same time, by §. 146, 2., ϱ_i' is different from ϱ_h' as soon as ϱ_i is different from ϱ_h. From this it follows that the quantities

$$\varrho_1', \varrho_2' \ldots \varrho_\nu' \tag{7}$$

agree with the quantities (6), apart from the order.

The product

$$g'(t) = (t - \varrho_1)(t - \varrho_2) \ldots (t - \varrho_\nu)$$

therefore remains unchanged by the permutations of the group P, and is consequently, since we presuppose the property b) of P, a rational function of t in Ω. At the same time $g'(t)$ is a divisor of $g(t)$, and must therefore, since $g(t)$ is irreducible, agree with $g(t)$; hence ν is not smaller than the degree of the Galois group, and P is identical with the Galois group.

If, as above, we choose the quantity ϱ as a rational function of the α in such a way that the $\Pi(m)$ quantities arising through the permutations of the α,

$$\varrho, \varrho', \varrho'' \dots$$

are all different from one another, then

$$G(t) = (t - \varrho)(t - \varrho')(t - \varrho'') \dots$$

is a function in Ω. Now each of the quantities $\varrho, \varrho', \varrho'' \dots$ is a primitive quantity of the normal field $N = \Omega(\alpha, \alpha_1 \dots \alpha_{m-1})$, and each of them is therefore the root of a Galois resolvent of degree μ. These quantities, μ at a time, are the roots of one such equation. Hence $G(t)$, which has no equal roots, must split into nothing but irreducible factors of degree μ, and from this it follows further that μ is a divisor of $\Pi(m)$. At the same time μ is the degree of the Galois group of $F(x)$.

If the degree of the Galois resolvent of an equation of degree m has the greatest value $\Pi(m)$, we say, with an expression originating with Kronecker, that the equation has no affect. It has an affect that is the higher, the lower the degree μ of the Galois resolvent is. The quotient $\Pi(m) : \mu$, which must always be an integer, at most equal to $\Pi(m)$ and at least equal to 1, we shall call the degree of the affect; for an equation without affect it accordingly has the value 1. If the affect has the highest possible degree $\Pi(m)$, then the roots of the equation are themselves contained in the domain of rationality Ω, and the equation is thus solved.

If by the adjunction of an algebraic quantity to Ω the Galois resolvent becomes reducible, a new resolvent of lower degree arises, and the degree of the affect of the equation $F(x) = 0$ increases. In this way, therefore, one comes nearer to the solution of the equation.

The Galois conception of the problem of solving an equation $F(x) = 0$ consists in this: by the successive adjunction of algebraic quantities of the simplest possible kind, the group is to be gradually diminished, or the affect raised, until it has reached its highest degree.

> The general equation of degree m has no affect in the field Ω which consists of the rational functions of the coefficients $a_1, a_2 \dots a_m$.

For the coefficients $a_1, a_2 \dots a_m$, regarded as independent variables, can also be represented as the symmetric fundamental functions of the roots $\alpha, \alpha_1, \alpha_2 \dots \alpha_{m-1}$. If then $g(t)$ is an irreducible factor of $G(t)$, expressed rationally in terms of the a, which vanishes for $t = \varrho$, then, on representing a_i and ϱ by the α, one obtains in $g(\varrho) = 0$ an identical equation. In this, however, the α may be permuted arbitrarily, whereby the a do not change, while ϱ can pass into any of the roots $\varrho, \varrho', \varrho'' \dots$ of $G(t)$. Hence $g(t)$ vanishes for all the roots of $G(t)$ and must consequently coincide with $G(t)$[32].

[32] Évariste Galois fell in a duel in the year 1832, scarcely 20 years old. The first indication of the theory that today bears his name is found in a memoir "Analyse d'un mémoire sur la résolution algébrique des équations", which appeared in 1830 in the "Bulletin des sciences mathém." of Férussac. More detailed communications are contained in the famous letter to Auguste Chevalier, written on the eve of his death, which is published in the "Revue encyclopédique" of September 1832. Only in the year 1846 did Liouville have all the already published papers of Galois, together with some investigations from his literary remains, among them the most important, "Mém. sur les conditions de résolubilité des équations par radicaux", printed in Vol. XI of his Journal. A German edition by Maser appeared in 1889 (Berlin, published by Springer). Interesting biographical communications are found in the "Revue encyclopédique" of September 1832, which Liouville did not reprint along with them and which are also lacking in the German edition. Of further writings that have contributed to the understanding or the further development of the Galois theory, let there still be mentioned: J. A. Serret ("Cours d'algèbre supérieure", II. edition, 1854, IV. 1879); Betti (Annali di scienze fisiche e matematiche [1853]); C. Jordan, "Traité des substitutions" (1870). E. Netto, "Substitutionentheorie" (Leipzig 1882).

§. 150.
Transitive and intransitive groups.

From the Galois group P we can derive a very simple criterion for whether the equation $F(x) = 0$ is reducible or irreducible.

Let us assume that $F(x)$, of degree m, is reducible, and that $f(x)$ is a factor of $F(x)$ in Ω of degree n, with $n < m$. Let the roots of $f(x)$

$$\alpha, \alpha_1, \alpha_2 \ldots \alpha_{n-1} \tag{1}$$

and the remaining roots of $F(x)$ be

$$\alpha_n, \alpha_{n+1} \ldots \alpha_{m-1}. \tag{2}$$

If α' is any one of the quantities (1), and α'' one of the quantities (2), then no permutation can occur in the group P by which α' is carried into α''; for by hypothesis

$$f(\alpha') = 0,$$

and if now a permutation occurred in P by which α' were replaced by α'', then on account of the property §. 149, a) we should also have $f(\alpha'') = 0$, which contradicts our assumption.

Thus by the permutations of P the roots (1) of $f(x)$ are permuted only among themselves.

If conversely the group P has the property that its permutations permute a part of the m roots α, such as the system (1), only among themselves, then the product

$$(x - \alpha)(x - \alpha_1) \ldots (x - \alpha_{n-1}) = f(x)$$

admits all the permutations of P, and is therefore by §. 149, b) contained in Ω; that is, $F(x)$ is reducible.

A permutation group is called transitive if it contains at least one permutation which carries an arbitrary element into any other arbitrary element; in the opposite case, that is, when the elements can be so decomposed into two or more portions that by no permutation of the group does an element of the one portion pass into an element of the other, the group is called intransitive. Accordingly we may sum up what has been proved in the theorem:

1. The equation $F(x) = 0$ is reducible or irreducible, according as its Galois group is transitive or intransitive.

If a part of the elements is so connected that by permutations from P every element of this part can pass into every other, we call the elements of this part transitively connected. The various transitively connected systems, which by all the permutations of P are permuted only among themselves, are called the systems of intransitivity.

§. 151.
Primitive and imprimitive groups.

Now let $f(x) = 0$ be an irreducible equation of degree n, so that its Galois group P is transitive. Let the roots of $f(x)$ be

$$\alpha, \alpha_1, \alpha_2 \ldots \alpha_{n-1}. \tag{1}$$

If the field $\Omega(\alpha)$ is imprimitive (§. 144) and $\Theta = \chi(\alpha)$ is an imprimitive element whose conjugate values fall into s systems of r mutually equal values each, so that

$$n = rs$$

with $r > 1$, $s < n$, then the values (1) can be decomposed into s sequences of r elements each, which we shall denote by

$$\begin{array}{l} A = \alpha, \alpha_1 \dots \alpha_{r-1} \\ B = \beta, \beta_1 \dots \beta_{r-1} \\ \dots\dots\dots\dots \\ S = \sigma, \sigma_1 \dots \sigma_{r-1} \end{array} \tag{2}$$

so that

$$\begin{array}{l} \Theta = \chi(\alpha) = \chi(\alpha_1) \dots = \chi(\alpha_{r-1}) \\ \Theta_1 = \chi(\beta) = \chi(\beta_1) \dots = \chi(\beta_{r-1}) \\ \dots\dots\dots\dots\dots\dots\dots\dots \\ \Theta_{s-1} = \chi(\sigma) = \chi(\sigma_1) \dots = \chi(\sigma_{r-1}) \end{array} \tag{3}$$

are the conjugate values of Θ. By §. 144,

$$(t - \Theta)(t - \Theta_1) \dots (t - \Theta_{s-1}) = \varphi(t) \tag{4}$$

is then an irreducible function in Ω of degree s with respect to t, whose roots are the values (3).

It now follows from (3) that the group P must be so constituted that all its permutations interchange the elements of the individual sequences $A, B \dots S$ only among themselves, and moreover permute the whole sequences $A, B \dots S$ with one another, so that two elements of one sequence are never replaced by two elements of different sequences. For if, say, a permutation π of P carried α and α_1 into β and σ, it would follow, since one may perform the permutation π [by a), §. 149] in the equation $\chi(\alpha) = \chi(\alpha_1)$, that $\chi(\beta) = \chi(\sigma)$ would also have to hold, which contra-

dicts the assumption that the values (3) are different from one another. Permutation groups P which have this property, namely that the permuted elements can be decomposed into rows $A, B \dots S$ of equally many elements in such a way that no permutation of P carries two elements of the same row into elements of different rows, are called imprimitive. The individual rows $A, B \dots S$ are called the systems of imprimitivity. Permutation groups for which such a decomposition of the elements is not possible are called primitive groups.

It may very well happen that a group is imprimitive in more than one way, that it possesses entirely different systems of imprimitivity. Thus, for example, the group of cyclic permutations of six elements $(1, 2, 3, 4, 5, 6)$ is imprimitive in a twofold manner and has the systems of imprimitivity

$$A = 1, 3, 5, \quad B = 2, 4, 6$$

and

$$A = 1, 4, \quad B = 2, 5, \quad C = 3, 6.$$

Now that the concept of primitive and imprimitive groups has thus been established, we can give the following expression to what was proved above:

1. An imprimitive field has an imprimitive group.

From the imprimitivity of the group there results an important consequence for the imprimitive fields.

We will denote by

$$\psi(x, x_1, \dots x_{r-1})$$

a rational symmetric function of the r variables $x, x_1 \dots x_{r-1}$, and set

$$\begin{array}{r} \omega = \psi(\alpha, \alpha_1 \dots \alpha_{r-1}) \\ \omega_1 = \psi(\beta, \beta_1 \dots \beta_{r-1}) \\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot\ \cdot \\ \omega_{s-1} = \psi(\sigma, \sigma_1 \dots \sigma_{r-1}); \end{array} \tag{4}$$

then it is to be proved that ω can be expressed rationally in terms of Θ, that is, is contained in the field $\Omega(\Theta)$. For we have

$$\varphi(t)\left(\frac{\omega}{t-\Theta}+\frac{\omega_1}{t-\Theta_1}+\cdots+\frac{\omega_{s-1}}{t-\Theta_{s-1}}\right)=\Phi(t) \tag{5}$$

an integral rational function of t, whose coefficients are rational functions of the $\alpha, \alpha_1 \ldots \alpha_{n-1}$, which remain unchanged when any permutation of the group P is carried out, because under these permutations the ω either remain unchanged, or are permuted among one another in the same way as the Θ. By §. 149 b) these coefficients are therefore contained in Ω. If we then substitute in (5) the value Θ for the indeterminate t, it follows, since $\varphi'(\Theta)$ is different from zero,

$$\omega=\frac{\Phi(\Theta)}{\varphi'(\Theta)}. \tag{6}$$

If we apply this to the coefficients of the product

$$(u-\alpha)(u-\alpha_1)\ldots(u-\alpha_{r-1})=\varphi(u,\Theta), \tag{7}$$

where u denotes a variable, it follows that this function of degree r, whose roots are the $\alpha, \alpha_1 \ldots \alpha_{r-1}$, can be expressed rationally in terms of Θ.

Thus the quantity α, which is originally the root of an equation of degree n in Ω, is at the same time the root of an equation of degree r, whose coefficients depend rationally on the root of an equation of degree s. An equation $f(x)=0$ whose root α has this property is called an imprimitive equation.

We can also express what has been proved thus:

2. The imprimitive field $\Omega(\alpha)$ of degree n passes, by the adjunction to Ω of the field of degree s, $\Omega'=\Omega(\Theta)$, into a field $\Omega'(\alpha)$ of degree r.

We wish further to investigate whether the imprimitivity of the group is a sufficient criterion for the imprimitivity of the field, that is, whether one can always find imprimitive elements in a field $\Omega(\alpha)$ with imprimitive group.

Let now, therefore, $f(x)=0$ be an irreducible equation of degree n with imprimitive group, and let (1) be the roots of this equation, decomposed into the sequences (2) in such a way that the elements of these sequences are not separated from one another by the permutations of the group.

We choose any symmetric function $\psi(x, x_1 \ldots x_{r-1})$ in such a way that the values

$$\begin{aligned} y &= \psi(\alpha, \alpha_1 \ldots \alpha_{r-1}) \\ y_1 &= \psi(\beta, \beta_1 \ldots \beta_{r-1}) \\ &\cdots\cdots\cdots\cdots \\ y_{s-1} &= \psi(\sigma, \sigma_1 \ldots \sigma_{r-1}) \end{aligned} \tag{8}$$

are all different from one another.

To see the possibility of this, we can, for example, set

$$\begin{aligned} \psi(t,A) &= (t-\alpha)(t-\alpha_1)\ldots(t-\alpha_{r-1}), \\ \psi(t,B) &= (t-\beta)(t-\beta_1)\ldots(t-\beta_{r-1}), \\ &\cdots\cdots\cdots\cdots \\ \psi(t,S) &= (t-\sigma)(t-\sigma_1)\ldots(t-\sigma_{r-1}) \end{aligned} \tag{9}$$

and then (by §. 143, 1.) a rational value can be found for t such that these quantities all turn out different from one another. These values can therefore be taken for the y in (8).

If we now understand by u an independent variable and set

$$\varphi(u) = (u-y)(u-y_1)\dots(u-y_{s-1}) \tag{10}$$

then $\varphi(u)$, since its coefficients remain unchanged under the permutations of the group, is a function in Ω whose roots are the quantities y. It is moreover irreducible, for from the assumed transitivity of the group it follows that if a rational function of y vanishes for one of the values (8), it must also vanish for all the others.

If ω is any symmetric function of the $\alpha, \alpha_1 \dots \alpha_{r-1}$, then from the consideration of the expression

$$\varphi(u)\left(\frac{\omega}{u-y} + \frac{\omega_1}{u-y_1} + \dots + \frac{\omega_{s-1}}{u-y_{s-1}}\right),$$

which is an integral rational function of u in Ω, we conclude, just as above, by setting $u = y$, that ω can be represented rationally in terms of y; and accordingly the function

$$\psi(t, A) = \psi(t, y), \tag{11}$$

whose roots are the quantities $\alpha, \alpha_1 \dots \alpha_{r-1}$, can also be expressed rationally in terms of y.

The equation $\psi(t, A) = 0$ is also irreducible in the field $\Omega(y)$; for if $\psi_1(t, y)$ is a rational divisor of $\psi(t, y)$, and if

$$\psi_1(\alpha, y) = 0,$$

then, on account of the transitivity of the group, we can apply to this equation a permutation by which α passes into an arbitrary quantity of the sequence A, while y remains unchanged; hence every root of $\psi(t, y)$ is at the same time a root of $\psi_1(t, y)$, and therefore the two functions are identical.

Hence the equation $f(x) = 0$ of degree n is imprimitive in the sense fixed above, i.e. a root α of this equation can be regarded as a root of an equation of degree r, whose coefficients depend on the root y of an equation of degree s.

To prove that then the field $\Omega(\alpha)$ (in the sense of §. 144) is also imprimitive, we have still to show that there are quantities in $\Omega(\alpha)$ which satisfy an irreducible equation of degree s.

We shall show that y itself is such a quantity, and hence that y can be represented rationally by α alone.

But this is very simple to conclude. By (11) we have

$$\psi(\alpha, y) = 0, \tag{12}$$

while $\psi(\alpha, y_1), \dots \psi(\alpha, y_{s-1})$ are different from zero. For if, say, $\psi(\alpha, y_1) = 0$, then α would be a root of the equation of degree r, $\psi(t, y_1) = 0$, whose roots are precisely the quantities $\beta, \beta_1 \dots \beta_{r-1}$, which are different from α. Hence the two equations

$$\psi(\alpha, u) = 0, \quad \varphi(u) = 0 \tag{13}$$

have one and only one common root $u = y$. The greatest common divisor of these two functions is therefore linear with respect to u, and it gives y rationally in terms of the coefficients of $\varphi(u)$ and $\psi(\alpha, u)$, and hence rationally in terms of α. With this the converse of 1. is proved:

3. A primitive field has a primitive group.

Fourteenth Section.

§. 152.

Effect of the Permutation Groups on Functions of Independent Variables.

For a deeper penetration into algebra, a more exact study of the permutation groups is, after the results of the preceding section, necessary. We first set up some general theorems about them, which form the foundation for what follows[33].

Let

$$u,\ u_1,\ u_2 \dots u_{m-1} \tag{1}$$

be a system of mutually independent symbols (variables), and

$$\psi(u,\ u_1,\ u_2 \dots u_{m-1}) \tag{2}$$

an integral rational function of them with coefficients from an arbitrary field.

Two such functions ψ are to be regarded as equal only when, in the expressions ordered according to powers and products of the u, corresponding terms have the same coefficients.

If we subject the indices of the variables u, or, what comes to the same thing, the u themselves to a permutation π, then the function ψ may change, or it may remain unchanged. If it remains unchanged, we say that it admits the permutation π. If it admits all $\Pi(m)$ permutations that are possible at all, then it is symmetric and can be expressed by the symmetric fundamental functions (§. 44). The other extreme case is that in which the function takes nothing but distinct values for all $\Pi(m)$ arrangements. In general, certain permutations will leave the function ψ unchanged, and others will change it. We now set up the theorem:

1. The totality of all those permutations of $u, u_1, u_2 \dots u_{m-1}$ which leave an integral function $\psi(u, u_1, u_2 \dots u_{m-1})$ unchanged is a group of permutations.

To prove it, assume that π is one of the permutations which leaves ψ unchanged. We express this by the following identity.

$$\psi = \psi_\pi$$

Now since the u are independent variables, this identical equation must remain correct when the variables are subjected to any permutation π' whatever. But when π' is applied to ψ_π, the result is the same as when the composed permutation $\pi\pi'$ is applied to ψ. It follows therefore that

$$\psi_{\pi'} = \psi_{\pi\pi'}$$

and if now

$$\psi = \psi_{\pi'}$$

then it follows that

$$\psi = \psi_{\pi\pi'},$$

i. e. if the function ψ remains unchanged under the permutations π and π', then it also remains unchanged under $\pi\pi'$, whereby, according to the definition of the group, the theorem we have stated is proved.

We have in this a means of forming groups of permutations, by taking any function of m variables and seeking out all the permutations that leave such a function unchanged. That all groups of permutations can be formed in this way will appear later.

[33] For the theory of permutations the following are especially to be noted: Cauchy, "Journal de l'École polytechn. X. cah. (1815)" (several memoirs). "C. Jordan, Traité des substitutions et des équations algébriques", Paris 1870. Netto, "Substitutionentheorie und ihre Anwendung auf die Algebra." Leipzig 1882.

2. If

$$P = \pi,\ \pi',\ \pi'',\ \pi''' \ldots$$

is a group of permutations of m elements and π_1 is any one among them, then the system

$$P_1 = \pi\pi_1,\ \pi'\pi_1,\ \pi''\pi_1,\ \pi'''\pi_1 \ldots,$$

agrees completely with P, apart from the order.

For the permutations contained in P_1, whose number is just as great as that of the permutations P, are, by the group property of P (§. 148, 3.), in any case all contained among the permutations of P. Moreover, the permutations of P_1 are (by §. 148, 2.) all different from one another, and hence P is identical with P_1, apart from the order. In the same way one can also show that

$$P_1' = \pi_1\pi,\ \pi_1\pi',\ \pi_1\pi'',\ \pi_1\pi''', \ldots$$

is identical with P.

From this the following theorems result:

3. Every permutation group contains the identical permutation.

For among the permutations of P_1, π_1 itself must also occur, and if $\pi_0\pi_1 = \pi_1$, then (by §. 148, 2.) π_0 is the identical permutation.

4. A permutation group P contains, for every permutation, also the opposite one.

For by 3. the identical permutation must occur in P_1. But if $\pi\pi_1$ is the identical permutation, then $\pi = \pi_1^{-1}$. Here π_1 may be any permutation from P.

If in place of the variables $u, u_1 \ldots u_{m-1}$ definite quantities $\alpha, \alpha_1, \alpha_2 \ldots \alpha_{m-1}$ are put, be they numerical values or quantities of any field, then Theorem 1. loses its general validity. If we take, for example, $m = 3$ and fix between the three quantities $\alpha, \alpha_1, \alpha_2$ the relation $2\alpha_2 = \alpha + \alpha_1$ while otherwise not restricting them further, then the function $\alpha_1 - \alpha_2$ remains unchanged only under the identical permutation and under the permutation

$$\begin{pmatrix} 0, & 1, & 2 \\ 1, & 2, & 0 \end{pmatrix}.$$

But these two permutations do not form a group, because the repetition of the latter,

$$\begin{pmatrix} 0, & 1, & 2 \\ 2, & 0, & 1 \end{pmatrix}$$

does not occur among them.

For these functions the following theorem now holds:

5. If P is a permutation group of m figures, and if $\alpha, \alpha_1 \ldots \alpha_{m-1}$ are arbitrary quantities different from one another, then there are rational functions of the α, even with rational coefficients, which do not change when a permutation of the group P is applied to the indices of α, and which do change when a permutation not belonging to P is applied.

To prove it, we take a function

$$\varrho = \Phi(\alpha, \alpha_1 \ldots \alpha_{m-1}),$$

such as we have already considered in §. 143, which takes $\Pi(m)$ different values when the α are permuted in all possible ways.

If we denote the values which ϱ takes through the application of the permutations of a group P by

$$\varrho, \varrho_1 \ldots \varrho_{\mu-1}, \tag{1}$$

then by Theorem 2. these quantities are only interchanged among one another when a permutation π_1 from the group P is applied to all of them at once. For the application of the permutation π_1 to all the functions (1) has the same effect as the application of the permutations contained in P_1 to ϱ. But by 2., P_1 is identical with P, apart from the order.

If, then, t denotes a variable, the function

$$\psi = (t-\varrho)(t-\varrho_1)\dots(t-\varrho_{\mu-1}) \tag{2}$$

remains unchanged when any permutation from P is applied. Now if, by a permutation not contained in P, the quantities (1) pass into

$$\varrho', \varrho_1' \dots \varrho_{\mu-1}' \tag{3}$$

then at least ϱ' certainly does not occur among the quantities (1), and consequently

$$\psi_1 = (t-\varrho')(t-\varrho_1')\dots(t-\varrho_{\mu-1}')$$

not identical with ψ. One can therefore (by §. 143, 1.) give t such a rational value that ψ is different from all functions formed after the manner of ψ_1.

It is self-evident that these considerations also hold when independent variables take the place of the α.

A function which admits all the permutations of a group P, while it changes under all permutations not belonging to P, is called a function belonging to the group P.

To the group P_0, which comprises all permutations of m figures, belong the symmetric functions. This group is therefore also called the symmetric group.

A function ϱ which belongs to the group consisting of the single identical permutation can serve as an unambiguous way of denoting the permutations; for there is only one permutation π' by which ϱ passes into ϱ'. We can therefore use for this permutation the symbol

$$\pi' = (\varrho, \varrho')$$

so that by

$$(\varrho,\varrho), \quad (\varrho,\varrho'), \quad (\varrho,\varrho'')\dots$$

all permutations of the m figures are unambiguously denoted.

§. 153.
Decomposition of Permutations into Transpositions and into Cycles.

We have already mentioned, in the second section, on the occasion of the determinants, the composition of arbitrary permutations from a sequence of transpositions, that is, the permutation of only two figures. Such transpositions we denote by placing the figures in question side by side, e.g. by $(0,1)$ the permutation of the figures 0 and 1.

A transposition, repeated twice, leads to the identity, so that every transposition is opposite to itself. Now if

$$\pi = \begin{pmatrix} 0, & 1, & 2\dots m-1 \\ a_0, & a_1, & a_2\dots a_{m-1} \end{pmatrix}$$

is an arbitrary permutation of m figures, then

$$\pi(m-1, a_{m-1}) = \pi_1$$

is a permutation which leaves $m-1$ unchanged, and which therefore by

$$\begin{pmatrix} 0, & 1, & 2 \dots m-2 \\ b_0, & b_1, & b_2 \dots b_{m-2} \end{pmatrix}$$

can be represented, and is therefore a permutation of at most $m-1$ figures. Since now also

$$\pi = \pi_1(m-1, a_{m-1})$$

it follows from this by complete induction

1. that every permutation can be decomposed (and indeed in infinitely many different ways) into transpositions.

And from this it follows further:

2. If a permutation group of m figures contains all transpositions with one fixed figure, e.g.:

$$(0,1), (0,2) \dots (0, m-1)$$

then it is identical with the symmetric group.

For such a group, as one recognises from the composition

$$(1,2) = (0,1)(0,2)(0,1)$$

contains also all the other transpositions, and by 1. all the permutations of the symmetric group can be composed from these.

A permutation π is called cyclic if the figures can be so arranged in a sequence that by π each figure passes into the following one, and the last again into the first, thus e.g.

$$\begin{pmatrix} 0, & 1, & 2 \dots m-2, & m-1 \\ 1, & 2, & 3 \dots m-1, & 0 \end{pmatrix};$$

such cyclic permutations are denoted more simply, by setting the figures of the cycle next to one another, by

$$(0, 1, 2 \dots m-1).$$

Here it is indifferent with which figure one begins; one could therefore also write

$$(1, 2 \dots m-1, 0)$$

in its place. The following theorem now holds:

3. Every permutation π can be decomposed, and indeed in only one way, into a sequence of cyclic permutations, so that no two of these cycles have a figure in common.

For if

$$\pi = \begin{pmatrix} 0, & 1, & 2 \dots m-1 \\ a_0, & a_1 & a_2 \dots a_{m-1} \end{pmatrix}, \tag{1}$$

one begins with an arbitrary figure, say with 0, and continues the sequence

$$0, \quad a_0 = b, \quad a_b = c \dots \tag{2}$$

until one strikes upon some figure for the second time. The first figure to recur must be 0, since for every figure the one preceding it in the sequence (2) is uniquely determined by (1). The figures

$$(0, b, c \dots) \tag{3}$$

then form a first cycle. If not all m figures of (1) are thereby exhausted, one picks out one of the remaining figures and proceeds in the same way, until all m figures of (1) have been accommodated

in the cycles. Since in every such cycle, for each figure, the preceding one as well as the following one is completely determined by (1), the cycles themselves are also uniquely determined. In the designation of π by the cycles, however, not only may the various cycles be arranged arbitrarily, but in each cycle one may also begin with any one of its figures. A figure that is not changed forms by itself a one-term cycle.

We choose an entirely arbitrary example, by which the procedure at once becomes clear:

$$\pi_1 = \begin{pmatrix} 0, & 1, & 2, & 3, & 4, & 5, & 6, & 7 \\ 2, & 3, & 6, & 7, & 1, & 4, & 5, & 0 \end{pmatrix} = (0,2,6,5,4,1,3,7)$$

$$\pi_2 = \begin{pmatrix} 0, & 1, & 2, & 3, & 4, & 5, & 6, & 7 \\ 4, & 2, & 1, & 3, & 5, & 7, & 0, & 6 \end{pmatrix} = (0,4,5,7,6)(1,2)(3)$$

$$\pi_3 = \begin{pmatrix} 0, & 1, & 2, & 3, & 4, & 5, & 6, & 7 \\ 7, & 2, & 3, & 6, & 5, & 0, & 1, & 4 \end{pmatrix} = (0,7,4,5)(1,2,3,6).$$

The setting of the cycles side by side is equivalent to a composition in the sense used hitherto; only it is to be remarked that permutations which contain no common figure whatever can always be interchanged in the composition.

One-term cycles, which change nothing, are also often omitted in the designation.

From the simultaneous consideration of the decomposition of permutations into cycles and into transpositions there results a proof of the division of the permutations into two classes, with which we have already become acquainted in the second section, in connection with determinants.

If τ denotes a transposition and π an arbitrary permutation, then in the compounded permutation $\tau\pi$ the number of cycles is greater by one or smaller by one than in π. The two figures that are interchanged by τ can occur either in the same cycle of π or in two different cycles. Let us assume that a cycle occurring in π is $\gamma = (1, 2 \ldots a, a+1 \ldots b)$, and that τ contains two figures that occur in γ, say $\tau = (1, a)$; then

$$\tau\gamma = (1, a+1 \ldots b)\ (2, 3 \ldots a),$$

that is, γ is decomposed by composition with τ into two cycles, while the remaining cycles are not touched by τ.

But if the two figures of τ occur in two different cycles of π, then let these two cycles be

$$\gamma = (1, 2, 3 \ldots a), \quad \gamma' = (1', 2', 3' \ldots a')$$

and $\tau = (1, 1')$. Then

$$\tau\gamma\gamma' = (1, 2', 3' \ldots a', 1', 2, 3 \ldots a),$$

that is, the two cycles γ, γ' are united by τ into a single cycle, while again the remaining cycles stay unchanged.

If, then, we have represented a permutation π of m figures, consisting of ν cycles (the one-term cycles being counted along with the rest), by μ transpositions, then by composition with these μ transpositions in reversed order it can be converted into the identical permutation, which consists of m one-term cycles. In all, therefore, $m - \nu$ cycles have been gained, and since to every transposition there corresponds a gained or a lost cycle, we must have $\mu \equiv m - \nu \pmod 2$; that is, μ is an even or an odd number according as $m - \nu$ is even or odd. But the last number depends only on the permutation π, not on the manner of the decomposition into transpositions. We have thereby the theorem:

4. The permutations of m figures fall into two kinds, of which the first is decomposable into an even, the second into an odd number of transpositions.

Each of these two kinds comprises equally many permutations, namely $\frac{1}{2}\Pi(m)$; for by the addition of a fixed transposition every permutation of the first kind passes over into one of the second kind, and conversely.

The composition of two permutations of the same kind always gives a permutation of the first kind, while a permutation of the first and one of the second kind, composed, give a permutation of the second kind, as is at once to be seen from the decomposition into transpositions. From this it follows:

5. The permutations of the first kind form a group.

If we call a function of m variables $u_0, u_1 \dots u_{m-1}$, or also of m mutually distinct quantities $\alpha_0, \alpha_1 \dots \alpha_{m-1}$, which changes its sign when two of the variables are interchanged with one another, as e. g. the product of all the differences:

$$(u_0 - u_1)(u_0 - u_2) \dots (u_0 - u_{m-1})$$

$$(u_1 - u_2) \dots (u_1 - u_{m-1})$$

$$\dots\dots\dots\dots$$

an alternating function, then such a function is not changed by the permutations of the first kind, while by a permutation of the second kind it is changed, namely transformed into the opposite value. The alternating functions therefore belong to the group of the permutations of the first kind.

As we earlier called the group to which the symmetric functions belong the symmetric group, so we call the group of the permutations of the first kind, to which the alternating functions belong, the alternating group. The group which consists of the single identical permutation we call the identical group or also the unit group.

A cyclic permutation of n terms can be decomposed into $n - 1$ transpositions, as one sees from the composition

$$(1, 2, 3 \dots n) = (1, 2)(1, 3) \dots (1, n)$$

recognises, and from this it follows that a cyclic permutation belongs to the first or to the second kind, according as the number of figures is odd or even.

Just as every permutation can be composed of transpositions, so too every permutation of the first kind can be composed of three-term cycles. It suffices if this is proved for every pair of transpositions, since every permutation of the first kind can be compounded of such pairs. But we have

$$(0, 1)(0, 2) = (0, 1, 2)$$

$$(0, 3)(1, 2) = (0, 1, 2)(0, 1, 3),$$

from which the correctness of the assertion is to be seen, since the two transpositions of a pair have either one figure in common or none. Hence:

6. All permutations of the first kind can be decomposed (in infinitely many ways) into cyclic permutations of three elements.

From 6. there follows further

7. A permutation group which contains all three-term cyclic permutations with two fixed figures must contain the whole alternating group.

For from the cycles $(0, 1, 2), (0, 1, 3), (0, 1, 4) \dots (0, 1, m - 1)$ all three-term cycles, and hence by 6. all permutations of the first kind, can be compounded, as from the compositions

$$(1, 0, 2) = (0, 1, 2)(0, 1, 2),$$

$$(0, 2, 3) = (0, 1, 2)(1, 0, 3)(1, 0, 2),$$

$$(1, 3, 4) = (1, 0, 3)(0, 1, 4)(0, 1, 3)$$

$$(2, 3, 4) = (1, 0, 2)(1, 3, 4)(0, 1, 2)$$

may be seen.

8. If m is greater than 4, then a permutation group which contains all permutations $(0,1)(2,3)$ composed of any two transpositions without a common figure is divisible by the alternating group.

For we have

$$(0,1,2) = (0,1)(3,4)(3,4)(0,2)$$

and one can therefore, if besides the four figures 0, 1, 2, 3 a fifth, 4, is present, compose all three-term cycles from pairs of transpositions of the form $(0,1)(3,4)$.

To this we append the proofs of two further important theorems relating to transitive permutation groups:

9. If a transitive permutation group of m figures contains a single transposition, then it is either the symmetric group or it is imprimitive.

Let the given group be P, and let the m figures that are permuted by it be $0,1,2\ldots m-1$, and let the transposition $(0,1)$ occur in it. Suppose that it contains altogether the transpositions

$$(0,1),(0,2),(0,3),\ldots(0,\mu-1),$$

but no other transposition with the figure 0. Now if $\mu = m$, then by 2. P is the symmetric group. But if $\mu < m$, then by the same theorem P contains all permutations of the figures

$$M = 0,1,2\ldots\mu-1.$$

From this it follows that P can contain no transposition of one of the figures $0,1,2,3\ldots\mu-1$ with another figure, say with μ. For if $(1,\mu)$ occurs in P, then $(0,\mu) = (0,1)(1,\mu)(0,1)$ also occurs in it, contrary to the hypothesis.

Now if P is transitive, there is in it a permutation π by which the figure 0 passes into a figure not contained in the system M, say into μ; and then by π none of the figures of M can pass into a figure of M; for if, e. g., 1 passes into r, then

$$\pi^{-1}(0,1)\pi = (\mu,r)$$

and (μ,r), if r belongs to M, cannot, as we have seen, occur in P. By every permutation of P, therefore, the figures of the system M are either permuted only among themselves, or they are carried over into the figures of a system M' entirely distinct from it, and in P all permutations of the figures M' also occur.

If with M and M' not all the figures are yet exhausted, there is a permutation in P by which M is carried over into a third system M'', which again can have no figure in common with M or with M'.

For if, when α denotes any one of the figures $0,1,2,\ldots n-1$, we denote by α_π and $\alpha_{\pi'}$ the figures into which α passes under π and under π', then $\alpha_{\pi\pi'}$ is the figure into which α_π passes under π', or α under $\pi\pi'$. If $\alpha,\beta,\gamma\ldots$ are the figures from the system M, then

$$\pi'^{-1}(\alpha,\beta)\pi' = (\alpha_{\pi'},\beta_{\pi'})$$
$$\pi\pi'^{-1}(\alpha,\beta)\pi'\pi^{-1} = (\alpha_{\pi'\pi^{-1}},\beta_{\pi'\pi^{-1}}) = (\gamma,\delta).$$

Now if $\alpha_{\pi'} = \gamma_\pi$ is contained in M', then $\alpha_{\pi'\pi^{-1}} = \gamma$ is contained in M, and consequently $\beta_{\pi'\pi^{-1}} = \delta$ must likewise be contained in M, since the transposition (γ,δ) occurs in P. Consequently $\alpha_{\pi'} = \gamma_\pi$ and $\beta_{\pi'} = \delta_\pi$, i.e. if one of the figures from M'' is contained in M', then M' and M'' must be identical. The totality of the figures $0,1\ldots n-1$ therefore falls apart into systems $M, M', M''\ldots$ of m figures each, which are permuted imprimitively by P; that is, the group P is imprimitive. If n is a prime, there are no imprimitive permutation groups; hence in this case a transitive group which contains a transposition must be the symmetric group. In exactly the same way we conclude from Theorem 7:

10. If a transitive permutation group of n figures contains a three-term cycle, then it contains the whole alternating group or it is imprimitive.

For if the group P contains the three-term cycles

$$(0,1,2),(0,1,3)\dots(0,1,m-1),$$

but no other with the figures $0,1$, then by 7. it contains the whole alternating group of the figures $M=0,1,2\dots m-1$. Now if $m<n$, no three-term cycle can occur in P which joins one or two of the figures of M with other figures. For if P contains e.g. $(0,m,m+1)$, then, as one sees from the compositions

$$(0,2,1)(0,m,m+1)(0,1,2)=(1,m,m+1)$$

it also contains $(1,m,m+1),\dots(m-1,m,m+1)$ and hence by 7. the whole alternating group of the figures $0,1,2\dots m+1$, which is contrary to the hypothesis. The n figures of P therefore fall apart, as before, into several systems $M,M',M''\dots$ of imprimitivity. We may add further,

11. that, when the case of imprimitivity occurs, the group P contains, in case 9. the whole symmetric group, in case 10. the whole alternating group, for each single system of imprimitivity.

Repeated composition of a permutation with itself is denoted by powers. If π is any permutation, then among the successive powers

$$\pi^0,\pi^1,\pi^2,\pi^3\dots,$$

in which $\pi^0=1$ denotes the identical permutation, the same permutation must necessarily recur. But from $\pi^u=\pi^{u+e}$ it follows that $\pi^e=1$. If e is the least positive exponent for which $\pi^e=1$, then the permutations

$$1,\pi,\pi^2\dots\pi^{e-1} \tag{4}$$

are all different from one another, while, as the sequence is continued with $\pi^e,\pi^{e+1},\dots\pi^{2e-1}\dots$, they repeat periodically in the same order. The sequence (4) is called the period of the permutation π, and e is called the degree of π. The period of π forms a group, since composing any two of its elements yields another element of the same period. If π is a cyclic permutation

$$\pi=(0,1,2\dots n-1),$$

then for even n

$$\pi^2=(0,2,4\dots n-2)(1,3\dots n-1),$$

and for odd n

$$\pi^2=(0,2,4\dots n-1,1,3\dots n-2),$$

and the degree of π is in both cases equal to n. If π is decomposed into several cycles, the degree of π is equal to the least common multiple of the number of terms of the single cycles, e. g.

$$\pi=(0,1)(2,3,4),\ \pi^2=(0)(1)(2,4,3),\ \pi^3=(0,1)(2)(3)(4),$$

$$\pi^4=(0),(1),(2,3,4),\ \pi^5=(0,1),(2,4,3),\ \pi^6=1.$$

If π occurs in any permutation group P, then P contains the whole period of π.

§. 154.

Divisors of Groups. Subsidiary Groups and conjugate Groups.

We saw in §. 152 that a function of m independent variables always determines a permutation group of m figures, but that this does not always hold when definite quantities are put in place of the independent variables. By the Galois theory, however, this difference is evened out, or at

least reduced to its kernel. Now let $F(x) = 0$ be any equation of degree m in a given field Ω with the mutually distinct roots $\alpha, \alpha_1, \alpha_2 \ldots \alpha_{m-1}$.

The normal field

$$\Omega(\varrho) = \Omega(\alpha, \alpha_1, \alpha_2 \ldots \alpha_{m-1})$$

we shall denote by N.

By the Galois group P of the field N, or of the equation $F(x) = 0$, we may understand (by §. 146, 149) either the group of substitutions of the field N, or also the permutation group of the indices of the α isomorphic with it.

Let p be the degree of P, and let us denote the operations of P (be they now substitutions of N or permutations of the α) by

$$P = \pi,\ \pi_1,\ \pi_2, \ldots \pi_{p-1} \tag{1}$$

If the system Q of the permutations contained in P

$$\varkappa,\ \varkappa_1,\ \varkappa_2, \ldots \varkappa_{q-1}. \tag{2}$$

forms a group by itself, that is, if any two of the elements of Q, compounded with one another, again yield an element of Q, then the group Q is called a divisor or factor of P[34]).

We first derive an important general theorem on the divisors of a group.

If the degree q of the divisor is smaller than the degree p of P, we take an element π_1 of P not contained in Q and form the system

$$Q\,\pi_1 = \varkappa\,\pi_1,\ \varkappa_1\,\pi_1 \ldots \varkappa_{q-1}\,\pi_1, \tag{3}$$

which we represent as the symbolic product $Q\pi_1$. The elements of (3) are distinct both from one another and from the elements of Q; for from $\varkappa_1\pi_1 = \varkappa_2\pi_1$ would follow $\varkappa_1 = \varkappa_2$, and from $\varkappa = \varkappa_1\pi_1$ would follow $\pi_1 = \varkappa_1^{-1}\varkappa$, which is not possible if π_1 does not belong to Q. The elements of $Q\pi_1$ form no group, for otherwise $\varkappa_1\pi_1\varkappa_2\pi_1 = \varkappa_3\pi_1$ or $\pi_1 = \varkappa_1^{-1}\varkappa_3\varkappa_2^{-1}$ would have to hold, which would again require that π_1 be contained in Q.

We call the system $Q\pi_1$ a subsidiary group to Q (within P). The subsidiary group remains unaltered in its totality when π_1 is replaced by any element $\varkappa\pi_1$ of $Q\pi_1$, because $Q\varkappa$ is identical with Q.

If we call every system $Q\pi$ generally a subsidiary group, so that Q itself is counted among them, then we can show that two subsidiary groups are either entirely identical, or contain no common element. For if $Q\pi_1$, $Q\pi_2$ are two subsidiary groups, and if $\varkappa_1\pi_1 = \varkappa_2\pi_2$, in which $\varkappa_1, \varkappa_2$ are two elements from Q, then, if we set $\varkappa = \varkappa_2^{-1}\varkappa_1$, it follows that $\pi_2 = \varkappa\pi$. But since $\varkappa$ belongs to Q, and consequently $Q\varkappa = Q$, it follows that

$$Q\pi_2 = Q\varkappa\pi_1 = Q\pi_1.$$

If the group P is not yet completely exhausted by (2) and (3), we can form a further subsidiary group $Q\pi_2$, and can continue in the formation of the subsidiary groups until P is decomposed into nothing but subsidiary groups. Since each of these subsidiary groups contains ν elements, there follows the important fundamental theorem of Cauchy:

1. The degree of a divisor of a group P is a divisor of the degree of P.

As a special case there is contained in this the fact that the degree of an element of the group (§. 153) is always a divisor of the degree of the group.

The decomposition of P into its subsidiary groups we can (following Galois) express very fittingly by the symbolic equation

$$P = Q + Q\pi_1 + Q\pi_2 + \cdots + Q\pi_{j-1}, \tag{4}$$

[34]Also called subgroup.

if

$$p = jq \tag{5}$$

holds. The quotient j we call the index of the divisor Q. This number expresses no peculiarity of the group Q in itself, but only a relation of Q to P.

Since, as we have already seen above, two subsidiary groups of Q are either wholly identical, or contain no common element, we can further state the theorem:

2. If π is an arbitrary permutation from P, then the system of subsidiary groups

$$Q\pi,\ Q\pi_1\pi,\ Q\pi_2\pi \ldots Q\pi_{j-1}\pi$$

is distinguished from the system

$$Q,\ Q\pi_1,\ Q\pi_2 \ldots Q\pi_{j-1}$$

only by the arrangement.

We extend the designation subsidiary groups also to systems of the form πQ, and can decompose P into such systems as well. From the decomposition (4) there follows in particular the second

$$P = Q + \pi_1^{-1}Q + \pi_2^{-1}Q + \cdots \pi_{j-1}^{-1}Q.$$

For if π_1 does not occur in Q, then π_1^{-1} does not occur in it either, and we obtain a first subsidiary group $\pi_1^{-1}Q$. If then π_2 does not occur in $Q\pi_1$, then π_2^{-1} does not occur in $\pi_1^{-1}Q$ either; for from $\pi_2^{-1} = \pi_1^{-1}\varkappa$ would follow $\pi_2 = \varkappa^{-1}\pi_1$, and so forth.

We here give the preference to the decomposition (4) on account of the significance which, as we shall see presently, it has for the algebraic problem.

We call a quantity ψ of the field N belonging to the group Q (when necessary with the addition "within P"), if ψ does not change when one of the operations of Q is applied to it, but does change when any other operation from P is applied. As one sees, this concept of belonging is somewhat extended as against the one fixed in §. 152, inasmuch as the permutations that lie outside P do not come into consideration here at all. For the case that P is the symmetric group, however, the two definitions coincide.

With this definition of belonging, the following theorems now hold generally.

3. To every divisor Q of P there belong quantities in N, and every quantity in N belongs to a determinate divisor Q of P.

The first part of the theorem is already proved in §. 152, 5. For if a function of the quantities α within the sym-

metric group belongs to a group Q, then it belongs to Q also within every other group P of which Q is a divisor.

The second part, however, follows from the fundamental properties of the Galois group.

To see this, we shall introduce the following notation. If any permutation π of the group P is applied to a function ψ of the field N, let ψ pass over into $\psi(\pi)$; if a permutation π' is applied to $\psi(\pi)$, the result is the same as if the composed permutation $\pi\pi'$ had been applied to ψ, i. e. we have

$$\psi(\pi)(\pi') = \psi(\pi\pi'). \tag{6}$$

If now ψ remains unchanged by the permutation $\varkappa$, then

$$\psi = \psi(\varkappa),$$

and to this, by §. 149, a), any permutation π from the group P may be applied. From this it follows that

$$\psi(\pi) = \psi(\varkappa\pi).$$

If therefore $\psi(\pi)$ is also equal to ψ, then $\psi(\varkappa\pi) = \psi$ as well, i. e. the permutations which leave ψ unchanged form a group, q. e. d.

If ψ is a function in N belonging to the group Q, so that

$$\psi(\varkappa) = \psi(\varkappa_1) = \cdots = \psi(\varkappa_{q-1}) = \psi, \tag{7}$$

and π_1 is any permutation of P not belonging to Q, then we can apply π_1 to the equations (7) and obtain

$$\psi(\varkappa\pi_1) = \psi(\varkappa_1\pi_1) = \cdots = \psi(\varkappa_{q-1}\pi_1) = \psi_1,$$

and ψ_1 is different from ψ. If conversely, for some π,

$$\psi_1 = \psi(\pi) = \psi(\varkappa\pi_1) \tag{8}$$

holds, then π must be contained in the sequence of substitutions

$$Q\pi_1 = \varkappa\pi_1,\ \varkappa_1\pi_1, \ldots \varkappa_{\nu-1}\pi_1 \tag{9}$$

for by applying π^{-1} to (8) it follows that $\varkappa\pi_1\pi^{-1}$ must be contained in Q, say equal to $\varkappa_1$; but from this it follows that

$$\pi = \varkappa_1^{-1}\varkappa\pi_1,$$

which is indeed contained in (9).

We can accordingly state the following theorem:

4. A function ψ belonging to Q passes, under all the permutations of a coset $Q\pi$, and

by no other operation of P into a determinate quantity ψ_1 different from ψ.

Accordingly, to the j subsidiary groups there correspond just as many functions ψ, namely

$$\psi,\ \psi_1,\ \psi_2 \ldots \psi_{j-1}, \tag{10}$$

and from Theorem 2. there results the theorem:

5. The quantities (10) undergo a permutation when one and the same operation π of P is applied to all of them simultaneously.

For we can represent the quantities (10) thus

$$\psi(\varkappa),\ \psi(\varkappa\pi_1),\ \psi(\varkappa\pi_2), \ldots \psi(\varkappa\pi_{j-1}),$$

and if now π is applied to them,

$$\psi(\varkappa\pi),\ \psi(\varkappa\pi_1\pi),\ \psi(\varkappa\pi_2\pi), \ldots \psi(\varkappa\pi_{j-1}\pi),$$

in which $\varkappa$ and π are arbitrary permutations of Q and P. But in each of the two sequences

$$\begin{aligned} &\varkappa,\ \varkappa\pi_1,\ \varkappa\pi_2 \ldots \varkappa\pi_{j-1} \\ &\varkappa\pi,\ \varkappa\pi_1\pi,\ \varkappa\pi_2\pi \ldots \varkappa\pi_{j-1}\pi \end{aligned}$$

there occurs one and only one permutation from each of the subsidiary groups (4).

The functions (10), which, as we have seen, are all different from one another, are called conjugate functions.

By what precedes we can easily form the groups to which the individual functions (10) belong. We have, e.g., $\psi_1 = \psi(\pi_1)$, and if now π is a permutation of P by which ψ_1 remains unaltered, then

$$\psi(\pi_1\pi) = \psi(\pi_1)$$

must hold. But we can apply to this equation the permutation π_1^{-1} contained in P, whereby

$$\psi(\pi_1\pi\pi_1^{-1}) = \psi$$

results; that is, $\pi_1\pi\pi_1^{-1}$ must belong to Q. If we set it equal to $\varkappa$, it follows that

$$\pi = \pi_1^{-1}\varkappa\pi_1. \tag{11}$$

If conversely π has this form, then

$$\psi_1(\pi) = \psi(\pi_1\pi) = \psi(\pi_1\pi_1^{-1}\varkappa\pi_1) = \psi(\varkappa\pi_1) = \psi_1,$$

and ψ_1 therefore admits this permutation. Accordingly we can denote the group to which ψ_1 belongs by the symbol

$$\pi_1^{-1}Q\pi_1$$

represent, which is in fact a group, as is to be seen from $\pi_1^{-1}\varkappa\pi_1\pi_1^{-1}\varkappa_1\pi_1 = \pi_1^{-1}\varkappa\varkappa_1\pi_1$. From this it follows at the same time that $\pi_1^{-1}Q\pi_1$ is isomorphic with Q (§. 149).

The groups to which the conjugate functions (10) belong, namely

$$Q, \quad \pi_1^{-1}Q\pi_1, \quad \pi_2^{-1}Q\pi_2, \ldots \pi_{j-1}^{-1}Q\pi_{j-1} \tag{12}$$

we call conjugate divisors of P, or briefly conjugate groups[35].

If π is an arbitrary permutation from P, and hence of the form $\varkappa\pi_r$, then the group $\pi^{-1}Q\pi$ is always contained among the conjugate groups (12), namely $= \pi_r^{-1}\varkappa^{-1}Q\varkappa\pi_r = \pi_r^{-1}Q\pi_r$.

The derivation of $\pi^{-1}Q\pi$ from Q is also called a transformation of the group Q by π, and $\pi^{-1}Q\pi$ a group transformed from Q.

For the formation of the permutations of the conjugate groups $\pi^{-1}Q\pi$ the following simple rule serves:

6. One obtains the permutation $\pi^{-1}\varkappa\pi$ by carrying out the permutation π in the cycles of $\varkappa$.

To prove the rule, let, decomposed into its cycles (§. 153),

$$\varkappa = (\alpha, \beta, \gamma \ldots)(\alpha', \beta', \gamma' \ldots) \ldots$$

and let

$$\pi = \begin{pmatrix} \alpha, & \beta, & \gamma & \ldots\alpha', & \beta', & \gamma' & \ldots \\ \alpha_\pi, & \beta_\pi, & \gamma_\pi & \ldots\alpha'_\pi, & \beta'_\pi, & \gamma'_\pi & \ldots \end{pmatrix}.$$

By π^{-1}, α_π passes into α; by $\varkappa$, α passes into β; and by π, β is carried into β_π. By $\pi^{-1}\varkappa\pi$, therefore, α_π passes into β_π. Since the same consideration is applicable to $\beta_\pi, \gamma_\pi \ldots$ and so forth, it follows that:

$$\pi^{-1}\varkappa\pi = (\alpha_\pi, \beta_\pi, \gamma_\pi \ldots)(\alpha'_\pi, \beta'_\pi, \gamma'_\pi \ldots),$$

by which the rule 6. is proved.

Of the decomposition into subsidiary groups we can make yet another application, to the proof of the theorem:

7. The degree of a transitive permutation group of m figures is always a multiple of m.

For all the permutations in P that leave the figure 0 unchanged form a divisor Q of P. Moreover, on account of the transitivity, there are in P permutations $\pi_1, \pi_2 \ldots \pi_{m-1}$ which carry the figure 0 into $1, 2 \ldots m-1$, and we have

$$P = Q + Q\pi_1 + Q\pi_2 + \cdots + Q\pi_{m-1},$$

hence the degree of P is equal to m times the degree of Q.

[35]The name "equally entitled subgroup" is also in use for this.

§. 155.
Reduction of the Galois resolvent by adjunction. Normal divisor of a group.

We have now to consider the gradual reduction of the group of a given equation, which can come about through the adjunction of certain algebraic quantities.

Let, as in the preceding section, P be the Galois group of degree p and Q one of its divisors of degree q and of index j; further let ψ be a function of the roots belonging to Q, and let

$$\psi, \psi_1, \psi_2 \dots \psi_{j-1} \tag{1}$$

be the conjugate quantities. We place the following theorem at the head.

1. The conjugate quantities (1) are the roots of an irreducible equation of degree j in Ω.

For if t denotes a variable, then

$$\varphi(t) = (t-\psi)(t-\psi_1)\dots(t-\psi_{j-1}) \tag{2}$$

remains unchanged (by §.154, 5.) when a permutation from P is applied; hence the coefficients of the function $\varphi(t)$, whose roots are the quantities (1), are contained in Ω [§. 149, b)].

To prove the irreducibility of $\varphi(t)$, we assume that $\Phi(t)$ is any function in Ω which vanishes for $t = \psi$, so that $\Phi(\psi) = 0$. Since all the permutations of P may be applied to this equation, it follows that $\Phi(\psi_1), \Phi(\psi_2) \dots \Phi(\psi_{j-1})$ must also be zero, and hence that $\Phi(t)$ must be divisible by $\varphi(t)$. But in this the irreducibility is contained.

To this is joined the theorem of Lagrange[36]:

2. Every quantity of the field N which admits the permutations of the group Q is contained in the field $\Omega(\psi)$, if ψ is a function belonging to Q.

A function ω which admits the permutations of Q passes, under the permutations of a subsidiary group $Q\pi_1$, into one and the same function. Hence to the conjugate values

$$\psi, \psi_1, \psi_2 \dots \psi_{j-1} \tag{3}$$

there correspond the values

$$\omega, \omega_1, \omega_2 \dots \omega_{j-1}, \tag{4}$$

which, however, are not necessarily all different from one another.

If one applies to the sequences of quantities (3), (4) one of the permutations of P, a certain permutation takes place, and indeed the same one in both sequences, since, if for example ψ_1 passes into ψ_2, then ω_1 must also pass into ω_2 (§. 154, 5.).

Let us now consider the sum

$$\varphi(t)\left(\frac{\omega}{t-\psi} + \frac{\omega_1}{t-\psi_1} + \dots + \frac{\omega_{j-1}}{t-\psi_{j-1}}\right) = \chi(t),$$

[36] Lagrange, Réflexions sur la résolution algébrique des équations. Mémoires de l'Académie de Berlin, années 1770, 1771. Oeuvres de Lagrange. Tome III. The theorem was, to be sure, given by Lagrange only in a more special form. The general formulation is due to Galois.

which is an integral function of t of degree $(j-1)$; we then find that it remains unchanged under all the permutations P and is consequently contained in Ω. If one then sets $t = \psi$ and observes that $\varphi(t)$ has no equal roots, it follows that

$$\omega = \frac{\chi(\psi)}{\varphi'(\psi)}. \tag{5}$$

Thus ω is expressed rationally in terms of ψ, and this is the content of theorem 2.

3. If we adjoin to the field Ω a function ψ belonging to Q, the group of the field N reduces to Q.

For if we denote the field $\Omega(\psi)$ by Ω', then first, every equation in Ω' between $\alpha, \alpha_1 \ldots \alpha_{m-1}$ admits the permutations of Q, because Q is contained in P, and because ψ remains

unchanged by Q; and secondly, every function that remains unchanged by Q is, by Lagrange's theorem, contained in Ω'. But these [by §. 149, a), b)] are the characteristic marks of the Galois group in the field Ω'. In order, therefore, to reduce the Galois group from degree p to degree q, one root of an auxiliary equation of degree j must be adjoined to Ω.

Theorem 3. we can also express thus:

The normal field $N = \Omega(\alpha, \alpha_1 \ldots \alpha_{m-1})$ is a field of degree p over Ω and of degree q over $\Omega' = \Omega(\psi)$.

To the lowering of the group by the adjunction of ψ there corresponds, as already follows from the general principles, a splitting of the Galois resolvent. Let us assume that (as in §. 145) $g(t) = 0$ is the Galois resolvent and ϱ one of its roots, and that by the permutations of Q, ϱ passes over into

$$\varrho, \varrho_1, \varrho_2 \ldots \varrho_{q-1},$$

and by the permutations of the subsidiary group $Q\pi_i$ into

$$\varrho_{0,i}\ \varrho_{1,i}, \varrho_{2,i} \cdots \varrho_{q-1,i},$$

then each of the factors of $g(t)$

$$g_i(t) = (t - \varrho_{0,i})(t - \varrho_{1,i}) \ldots (t - \varrho_{q-1,i})$$

is unchanged by the substitutions of Q and is therefore contained in $\Omega(\psi)$.

If we set

$$g(t, \psi) = (t - \varrho)(t - \varrho_1) \ldots (t - \varrho_{q-1}),$$

then it results, if we carry out in it a permutation from $Q\pi_i$, that

$$g_i(t) = g(t, \psi_i),$$

and $g(t)$ is thus decomposable in the following manner:

$$g(t) = g(t, \psi)\, g(t, \psi_1) \ldots g(t, \psi_{j-1}), \tag{6}$$

in which each factor is of degree q, and no two of these factors have a common divisor.

If Q and Q' are two different divisors of the group P, then Q and Q' will have certain permutations in common, among which the identical permutation is always found. It is possible that this is the only common permutation of Q and Q', and in that case these two groups are called coprime. But still more common elements may

be present. The totality R of all common permutations of Q and Q' we call the greatest common divisor, or also the intersection, of Q and Q'. This R is always a group, for if π_1 and π_2 both occur in Q as well as in Q', then $\pi_1\pi_2$ must also occur in Q and in Q', and hence also in R. The concept carries over at once to several groups $Q, Q', Q'' \ldots$

If ψ is a function belonging to Q and ψ' one belonging to Q', we can determine the rational numbers x, x' (by §. 143, 1.) so that $\omega = x\psi + x'\psi'$ becomes a function belonging to R. For ω certainly admits the permutations of R, since ψ and ψ' admit them. If then π is a permutation of P not contained in R, then certainly $\psi = \psi(\pi)$ and $\psi' = \psi'(\pi)$ do not both hold; hence we can determine x, x' so that $\omega = \omega(\pi)$ likewise does not hold. If therefore we adjoin ψ and ψ', and consequently ω, simultaneously, the group of the field N reduces by 3. to R. In the same way we may conclude for more than two groups Q, and obtain the theorem:

4. If $Q, Q', Q'' \ldots$ are divisors of P, R their intersection, $\psi, \psi', \psi'' \ldots$ functions belonging to $Q, Q', Q'' \ldots$, then by the simultaneous adjunction of $\psi, \psi', \psi'' \ldots$ the group of the field N reduces to R.

If we adjoin not merely one root ψ of the auxiliary equation $\varphi(t) = 0$, but all the quantities conjugate to ψ, namely $\psi, \psi_1, \psi_2 \ldots \psi_{j-1}$ — if, then, we derive from Ω the field

$$\Omega'' = \Omega(\psi, \psi_1, \psi_2 \ldots \psi_{j-1})$$

— then by this theorem the outcome is that the group of N in Ω'' becomes the greatest common divisor R of all the groups conjugate to Q, so that (by §. 154) R is the greatest common divisor of all the groups $\pi^{-1}Q\pi$, as π runs through the permutations of P.

Of special interest now is the case in which all the conjugate groups $\pi^{-1}Q\pi$ are identical with one another. In this case, by Theorem 2., each of the conjugate quantities $\psi, \psi_1, \psi_2 \ldots \psi_{j-1}$ is contained in $\Omega(\psi)$, the fields $\Omega(\psi), \Omega(\psi_1) \ldots \Omega(\psi_{j-1})$ and $\Omega(\psi, \psi_1 \ldots \psi_{j-1})$ are identical, and $\Omega(\psi)$ is a normal field over Ω.

We therefore call a divisor Q of the group P which has this property a normal divisor (or normal factor)[37].

But if Q is a normal divisor, then the adjunction of a single root ψ of the auxiliary equation is equivalent to the simultaneous adjunction of all these roots.

5. If Q itself is not normal, then the greatest common divisor R of all the conjugate divisors $\pi^{-1}Q\pi$ is certainly normal.

For if a permutation $\varkappa$ is contained in R, and hence in all the groups $\pi^{-1}Q\pi$, then the same holds also of every $\pi^{-1}\varkappa\pi$; that is, $\pi^{-1}R\pi$ is a divisor of R, and consequently, since it is of the same degree, $= R$.

A group which has no normal divisor other than itself and the identical group is called a simple group. The intersection of all the groups conjugate to a simple group is either the group itself or the identical group.

If Q is a normal divisor of P, and R a normal divisor of Q, then R is indeed a divisor of P, but by no means always a normal one. On the other hand, if R is a normal divisor of P, then R is also a normal divisor of every divisor Q of P of which R is a divisor.

§. 156.
The Group of the Resolvents.

The auxiliary equation $\varphi(t) = 0$, on which the determination of the function ψ depends, passes over into the Galois resolvent when the divisor Q to which ψ belongs is the identical group. For then, by the theorem of Lagrange (§. 155, 2.), every function of the field N, and hence also the roots α themselves, can be expressed rationally in terms of ψ, and N is identical with $\Omega(\psi)$.

[37] Galois speaks of the proper decomposition (décomposition propre) of a group; hence the expression "proper divisor" which is also in use; more recent writers designate the normal divisors as "distinguished or invariant subgroups".

We shall therefore call these equations $\varphi(t) = 0$, in a general sense, resolvents. But two cases are here to be distinguished.

1. If the conjugate groups $\pi^{-1}Q\pi$ are coprime, then the simultaneous adjunction of all the roots of the resolvent $\varphi(t) = 0$ is (by Theorem 4, §. 155) equivalent to the adjunction of a function belonging to the unit group, and the solution of the given equation is reduced to the complete solution of the resolvent. We have

$$N = \Omega(\psi, \psi_1 \dots \psi_{j-1}),$$

and a Galois resolvent of the equation $\varphi(t) = 0$ is at the same time a Galois resolvent of the original equation. In this case we call $\varphi(t) = 0$ a total resolvent of the given equation.

2. If the conjugate groups $\pi^{-1}Q\pi$ have a divisor R of degree r different from the unit group, which is then a normal divisor of P, then the simultaneous adjunction of all the functions conjugate to ψ is equivalent to the adjunction of a quantity belonging to R. The Galois resolvent of the given equation is not yet completely solved by this adjunction; rather it is only decomposed into factors of degree r. In this case $\varphi(t) = 0$ is called a partial resolvent.

If P is a simple group, then only total resolvents exist, whereas, if P has normal divisors, a partial resolvent can be found for every such normal divisor.

The same distinction also comes to light when we investigate the Galois group of the resolvent $\varphi(t)$. This group consists of all the permutations that are produced in the sequence of the quantities

$$\psi, \psi_1, \psi_2 \dots \psi_{j-1} \tag{1}$$

by the application of all the operations π of P; for every equation in Ω between the quantities (1) remains valid when such a permutation is carried out, since one can apply the operations π to such an equation; and if a function in Ω of the quantities (1) admits all these permutations, then it also admits all the permutations π and is therefore equal to a quantity in Ω.

The degree of this group is now to be determined. Among the operations of P, those and only those will produce no change among the quantities (1) which occur simultaneously in

the groups $\pi^{-1}Q\pi$ of all these functions, and hence also in their greatest common divisor R. Let the operations of R be denoted by σ. If then π and π' are two operations that produce the same permutation among the quantities (1), then $\pi'\pi^{-1}$ will restore the original arrangement of the ψ, and hence be equal to one of the quantities σ, or

$$\pi' = \sigma\pi.$$

We therefore have the result:

> The permutations of the subsidiary group $R\pi$, and only these, produce one and the same permutation among the quantities (1).

The degree of the Galois group of the resolvent $\varphi(t) = 0$ is therefore equal to the number of these subsidiary groups, i.e. equal to the quotient $p : r$, or the index of the divisor R of P.

If R is the identical group, so that $r = 1$, and consequently $\varphi(t) = 0$ is a total resolvent, then the degree of its group is just as high as the degree of the group of the original equation, and the two groups are moreover isomorphic. With regard to the group, then, nothing is gained. The given equation is equivalent to the resolvent, however different their degrees may be.

If, on the other hand, Q is a normal divisor, so that R is identical with Q, then $\varphi(t) = 0$ is a partial resolvent and the degree of its group is equal to the index j of the divisor Q of P. After the adjunction of a root of this resolvent, the group of the original equation reduces to Q, and hence to degree q. A splitting of the group has therefore taken place.

If one wishes to form resolvents of the lowest possible degree, one has to seek out divisors of the group P of the smallest possible index, and hence of the highest possible degree; but if at the same time a reduction of the group is to occur, then the divisors must be normal divisors.

§. 157.

Reduction of the Galois Group by Adjunction of Arbitrary Irrationalities.

According to the Galois conception, the problem of solving an algebraic equation is replaced by another,

namely, by the adjunction of algebraic quantities of the simplest possible nature, to bring about a splitting of the Galois resolvent, and thus a lowering of the degree of the group.

We now put the question thus. In the original field Ω the Galois resolvent $g(t)$ is irreducible. The field Ω is to be extended to another field Ω' in such a way that $g(t)$ becomes reducible. Here Ω' is to be an algebraic field over Ω, and so by §. 143 there must be an algebraic quantity ε such that $\Omega' = \Omega(\varepsilon)$. This quantity ε will be the root of a certain irreducible equation in Ω, which we shall denote by

$$\chi(\varepsilon) = 0 \tag{1}$$

Let us assume that in the field $\Omega(\varepsilon)$ the irreducible factor

$$g_1(t) = g_1(t, \varepsilon) \tag{2}$$

separates off from $g(t)$, having the root $t = \varrho$. The degree of $g_1(t, \varepsilon)$ we shall denote by q, and the roots by

$$\varrho, \varrho_1, \varrho_2 \dots \varrho_{q-1}; \tag{3}$$

so that

$$g_1(t, \varepsilon) = (t - \varrho)(t - \varrho_1) \dots (t - \varrho_{q-1}). \tag{4}$$

It is now first to be shown that the substitutions

$$(\varrho, \varrho), (\varrho, \varrho_1), (\varrho, \varrho_2) \dots (\varrho, \varrho_{q-1})$$

contained in the group P form a group Q. To show this, we set, as before (§. 146),

$$\varrho_i = \Theta_i(\varrho)$$

and observe that then the equation

$$g_1[\Theta_i(t), \varepsilon] = 0$$

has a root, namely $t = \varrho$, in common with $g_1(t)$, and hence, since $g_1(t)$ is irreducible, is divisible by $g_1(t)$. From this it follows that, if ϱ_1 and ϱ_2 are any two of the roots (3), then $\Theta_i(\varrho_2) = \varrho_3$ is also contained among these roots. But the substitution (ϱ, ϱ_3) is composed of (ϱ, ϱ_1) and (ϱ, ϱ_2), and with this the group property of Q is proved. We denote the index of Q by j and set

$$p = jq. \tag{5}$$

The product (4) now admits the substitutions of the group Q, and hence, if ψ, as above, denotes a function in N belonging to this group, then by §. 155, 2. the function $g_1(t, \varepsilon)$ can be expressed rationally in terms of ψ; that is, $g_1(t, \varepsilon)$ is a function of t in the field of degree j, $\Omega(\psi)$, and shall accordingly be denoted by $g(t, \psi)$.

The quantity ψ is one of the roots of an irreducible equation of degree j

$$\varphi(u) = 0, \tag{6}$$

whose remaining roots are $\psi_1, \psi_2 \dots \psi_{j-1}$, and by §. 155, (6) $g(t)$ is decomposable into the factors

$$g(t) = g(t,\psi)\, g(t,\psi_1) \ldots g(t,\psi_{j-1}). \tag{7}$$

Now

$$g(t,\psi) = g_1(t,\varepsilon),$$

and from (7) it follows that this equation does not continue to hold if, on the left-hand side, one of the other roots of (6) is put in place of ψ. One can therefore assign to t a rational value such that the equation

$$g(t,u) - g_1(t,\varepsilon) = 0 \tag{8}$$

has only the one root $u = \psi$ in common with equation (6). The greatest common divisor of (6) and (8) is then linear with respect to u, and if one sets $u = \psi$ in it, it results that ψ is rationally expressible in terms of ε. The field $\Omega(\psi)$ is therefore certainly a divisor of the field $\Omega(\varepsilon)$. The degree of the field $\Omega(\varepsilon)$, i. e. the degree of the auxiliary equation $\chi(\varepsilon) = 0$, is therefore a multiple of j and never lower than j. If the degree of the auxiliary equation is equal to j, which, for example, always occurs when the degree of χ is a prime, then $\Omega(\varepsilon)$ is identical with $\Omega(\psi)$; that is, ε is also rationally expressible in terms of ψ. In this case the conjugate values $\varepsilon, \varepsilon_1 \ldots \varepsilon_{j-1}$ are all the roots of $\chi = 0$, and one obtains them by replacing, in the rational expression of ε in terms of ψ, the function ψ by each of the conjugate functions $\psi, \psi_1 \ldots \psi_{j-1}$. Hence ε itself can be taken for ψ, and one finds a decomposition

$$g(t) = g(t,\varepsilon)\, g(t,\varepsilon_1) \ldots g(t,\varepsilon_{j-1}).$$

The quantities $\varepsilon, \varepsilon_1, \varepsilon_2 \ldots \varepsilon_{j-1}$ all belong to the field $\Omega(\varrho)$.

These results are of great importance and deserve to be specially emphasised.

Following Kronecker, if $F(x) = 0$ is the given equation and N the associated Galois field, the quantities of N are called the natural irrationalities of the equation $F(x) = 0$.

We then have the theorem:

Every possible reduction of the Galois group is brought about by the adjunction of a natural irrationality. If j is the index of the reduced group with respect to the original one, then the reduction cannot come about through the adjunction of a field of degree lower than j. But the degree can be equal to j, and then the adjoined irrationality is a natural one. If the degree of the adjoined field is greater than j, then it is a multiple of j.

§. 158.
Imprimitive Groups.

We make application of our general theorems to groups of a special nature. First we take the imprimitive groups. We have seen in §. 151 that, if $f(x) = 0$ is an irreducible equation of degree n with imprimitive group, then n can be decomposed into two factors $r.s$ in such a way that the roots split into s rows of r terms each:

$$\begin{aligned} A &= \alpha,\ \alpha_1 \ldots \alpha_{r-1}, \\ B &= \beta,\ \beta_1 \ldots \beta_{r-1}, \\ &\ldots\ldots\ldots\ldots\ldots\ldots \\ S &= \sigma,\ \sigma_1 \ldots \sigma_{r-1}, \end{aligned} \tag{1}$$

and that by the permutations of the group P the quantities of the individual rows are not separated from one another, but are only permuted among themselves, while on the other hand the rows in turn are permuted among one another.

Among the permutations of the Galois group P of $f(x)$ there will now be certain ones, whose totality we shall denote by Q,

which leave the individual rows $A, B \dots S$ in their place and only permute the indices within each row. These permutations form a group by themselves, since by the composition of two of them the rows $A, B \dots S$ cannot be interchanged. This group Q is a normal divisor of P. For if $\varkappa$ is a permutation of Q, and π one from P, then the composed permutation $\pi^{-1}\varkappa\pi$ belongs again to Q, because the permutation among the $A, B \dots S$ that is produced by π^{-1} is not altered by $\varkappa$ and is undone again by π. The group Q is intransitive, and by adjunction of a function ψ belonging to Q not only does the Galois resolvent become reducible, but the function $f(x)$ itself splits into s factors of degree r,

$$f(x) = f_\alpha(x,\psi)\, f_\beta(x,\psi) \dots f_\sigma(x,\psi). \tag{2}$$

The roots of $f_\alpha(x,\psi) = 0$ are the quantities α, and the Galois group of this equation consists of all the permutations that are produced by Q among the α.

The quantity ψ is, if j is the index of the divisor Q of P, a root of a partial resolvent of degree j, $\chi(u) = 0$, which is a normal equation and hence its own Galois resolvent (§. 155). If we denote its roots by

$$\psi,\ \psi_1,\ \psi_2 \dots \psi_{j-1}, \tag{3}$$

then the Galois group of this resolvent consists of the substitutions

$$(\psi,\psi),\ (\psi,\psi_1),\ (\psi,\psi_2) \dots (\psi,\psi_{j-1}), \tag{4}$$

which are produced by the permutations of the subsidiary groups

$$Q,\ Q\pi_1,\ Q\pi_2 \dots Q\pi_{j-1} \tag{5}$$

Each of these subsidiary groups produces a certain permutation among the rows $A, B \dots S$, and indeed these permutations are all different from one another; for if, say, the same permutation among the rows were produced by π_1 and π_2, then $\pi_2\pi_1^{-1}$ belongs to Q, and $Q\pi_1$, $Q\pi_2$ are not different subsidiary groups. The permutation group of the partial resolvent $\chi(u) = 0$ is therefore isomorphic with the group of the permutations that are produced by P among the rows $A, B \dots S$.

If we understand by Q_α the totality of the permutations of P by which the row A is not displaced, then Q_α is

certainly a group. If π_β is a permutation by which the row A passes into the row B, which by the hypothesis of transitivity must exist in P, then $Q_\alpha\pi_\beta$ is a subsidiary group of Q_α by which A passes into B. $\pi_\beta^{-1}Q_\alpha\pi_\beta = Q_\beta$ is a group conjugate to Q_α, and indeed the one by whose permutations the row B is not displaced. In this way we obtain the conjugate groups

$$Q_\alpha,\ Q_\beta \dots Q_\sigma,$$

whose greatest common divisor is the group Q of index j considered above.

A function y_α belonging to Q_α is the root of an equation of degree s in Ω, $\varphi(y) = 0$, and the remaining roots $y_\beta \dots y_\sigma$ arise from y_α by application of the permutations of the subsidiary groups

$Q_\alpha\pi_\beta, \dots Q_\alpha\pi_\sigma$. The equation $\chi(u) = 0$ of degree j found above is the Galois resolvent of the equation $\varphi(y) = 0$. We have remarked above that the group of the equation $\chi(u) = 0$ of degree j is isomorphic with the group of the permutations that are produced by P among the systems of the $A, B \dots S$. But since, on account of the transitivity of P, these systems also pass transitively into one another, j must, by the theorem of §. 154, 7., be divisible by s, and in the particular case where j is a prime, we must have $j = s$.

The simultaneous adjunction of all the roots of $\varphi(y)$ has the same effect as the adjunction of the function ψ, namely, to reduce the group of the equation $f(x) = 0$ from P to Q.

By the adjunction of y_α a rational factor $f(x, y_\alpha)$ splits off from $f(x)$, whose roots are the quantities of the row A, and by adjunction of the remaining y one obtains the decomposition

$$f(x) = f(x, y_\alpha)\, f(x, y_\beta) \dots f(x, y_\sigma), \tag{6}$$

The functions $f(x, y_\alpha), f(x, y_\beta) \dots f(x, y_\sigma)$ are precisely the factors $f_\alpha(x, \psi), f_\beta(x, \psi) \dots f_\sigma(x, \psi)$ in (2).

It is further to be remarked here that $f(x, y_\alpha)$ is irreducible in the field $\Omega(y_\alpha)$. For on account of the assumed irreducibility of $f(x)$ there are in P permutations by which α is carried into every α_i, and these permutations belong to Q_α. If therefore a rational function $\varphi(x, y_\alpha)$ vanishes for $x = \alpha$,

then each $\varphi(\alpha_i, y_\alpha)$ must be equal to zero, that is, $\varphi(x, y_\alpha)$ must be divisible by $f(x, y_\alpha)$. But it does not follow from this that $f(x, y_\alpha) = f_\alpha(x, \psi)$ is also irreducible in the field $\Omega(\psi)$.

These are the same theorems as in §. 151. It is further shown there that for y_α one can take a function of α alone, which we shall not go into again.

It is of interest, however, to follow the connection between the Galois groups of the individual factors $f(x, y_\alpha), f(x, y_\beta), \dots f(x, y_\sigma)$. We have already seen above that one obtains the group of the factor $f(x, y_\alpha)$ in the field $\Omega(\psi)$ by seeking out all the permutations of the α that are produced by Q. We shall denote this group by P_α, and likewise use the signs $P_\beta, \dots P_\sigma$ for the other factors.

We select from P any $s - 1$ permutations $\pi_\beta \dots \pi_\sigma$ of such a kind that by π_β the row A passes into B etc., by π_σ the row A into S; and for the sake of simplicity we shall now — which is evidently still entirely open to us — assume the designation of the $\beta, \dots \sigma$ so chosen that $\pi_\beta, \dots \pi_\sigma$ produce the following permutations:

$$\begin{pmatrix} \alpha_0, & \alpha_1 & \dots & \alpha_{r-1} \\ \beta_0, & \beta_1 & \dots & \beta_{r-1} \end{pmatrix}, \dots \begin{pmatrix} \alpha_0, & \alpha_1 & \dots & \alpha_{r-1} \\ \sigma_0, & \sigma_1 & \dots & \sigma_{r-1} \end{pmatrix}.$$

Now let $\varkappa$ be any element of Q which among the α effects the permutation

$$(\alpha) = \begin{pmatrix} \alpha_0, & \alpha_1 & \dots & \alpha_{r-1} \\ \alpha_{a_0}, & \alpha_{a_1} & \dots & \alpha_{a_{r-1}} \end{pmatrix}$$

then

$$\varkappa_\beta = \pi_\beta^{-1} \varkappa \pi_\beta$$

is likewise contained in Q, and by $\varkappa_\beta$ there is produced among the β the permutation

$$(\beta) = \begin{pmatrix} \beta_0, & \beta_1 & \dots & \beta_{r-1} \\ \beta_{a_0}, & \beta_{a_1} & \dots & \beta_{a_{r-1}} \end{pmatrix},$$

and hence among the indices of the β exactly the same permutation as by $\varkappa$ among the indices of α. If we take (β) and a corresponding $\varkappa_\beta$ as given, then $\pi_\beta \varkappa_\beta \pi_\beta^{-1}$ yields $\varkappa$ again, so that the permutations of the β produced by Q agree exactly with those which are produced among the α; and the same holds for the remaining rows. But with this the theorem is proved.

1. In the case of an irreducible imprimitive equation the roots in the individual rows can be so designated that the partial equations $f(x, y_\alpha) = 0, f(x, y_\beta) = 0, \dots f(x, y_\sigma) = 0$ in the field $\Omega(\psi)$ all receive the same group.

If one does not choose to designate the roots in the manner indicated, the groups will at least be isomorphic.

We further raise the question, which is in a certain measure answered by the converse of this theorem: When does a transitive group have an intransitive normal divisor?

If P is a transitive permutation group and Q an intransitive normal divisor of P, let

$$A = \alpha, \alpha_1 \dots \alpha_{r-1}$$

be one of the systems whose elements are interchanged by Q only among themselves, but in such a way that the elements of A are transitively connected by Q, that is, that by permutations from Q every α can pass into every other α. Now if by an element π from P the quantities A pass into

$$B = \beta, \beta_1 \dots \beta_{r-1}$$

then these elements must either all coincide with the α, or all be different from the α; for by the group $\pi^{-1}Q\pi$, which by hypothesis is identical with Q, the β are interchanged only among themselves, and indeed in the same manner as the α, hence transitively. If therefore only a part of the α occurred in B, then, contrary to the hypothesis, these α would be interchanged by Q with other quantities not occurring in A.

If the A do not exhaust all the elements to which the permutations of P extend, then, since P is transitive, there is a permutation π by which A is carried over into a system B entirely different from it. If $\varkappa$ is a permutation from Q, then by the permutation $\pi^{-1}\varkappa\pi$, which likewise belongs to Q, the same permutation is produced among the indices of the β as by $\varkappa$ among the indices of the α. From this it follows that by Q the β also are permuted only among themselves. If with A and B the elements are not yet exhausted, one can form in the same way a third system C, which arises from A by π' and from B by $\pi^{-1}\pi'$,

and which has no element in common with either A or B. One continues in this way until all elements are exhausted, and comes to the following theorem:

2. A transitive group has, only when it is imprimitive, an intransitive normal divisor different from the identical group. The systems of intransitivity of the normal divisor are systems of imprimitivity of the given group.

With respect to the equations, having regard to 1., we can also express theorem 2. as follows:

3. An irreducible equation $f(x) = 0$ splits, when it becomes reducible through the adjunction of all the roots of a resolvent or of one root of a normal equation of degree j, into several irreducible factors of equal degree and of equal group. The number of these factors is a divisor of the degree j and is equal to j when j is a prime.

If, then, an irreducible primitive equation is reduced by the adjunction of the roots of a resolvent, it splits into linear factors, i. e. it is completely solved. This case always occurs when the degree of the equation $f(x) = 0$ is a prime, because such an equation cannot be imprimitive.

Fifteenth Section.
Cyclic Equations.

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§. 159.
Cubic Equations.

From the standpoint now gained we shall first consider the solution of the equations of the third and fourth degree.

The group of the permutations of three figures consists of six elements, namely of the identical permutation 1, of two three-term cycles, and three transpositions:

$$(1) \qquad \begin{matrix} 1 & (0,\,1,\,2) & (0,\,2,\,1) \\ (1,\,2) & (2,\,0) & (0,\,1). \end{matrix}$$

The first three

$$1,\ (0,\,1,\,2) \quad (0,\,2,\,1)$$

form the alternating group; it consists of the powers of the cyclic permutation $\pi = (0, 1, 2)$:

$$(2) \qquad \pi^0 = 1,\ \pi,\ \pi^2$$

and is therefore called the cyclic group.

Now let $\alpha, \alpha_1, \alpha_2$ be the three roots of the cubic equation:

$$(3) \qquad f(x) = x^3 - ax^2 + bx - c = 0.$$

If we take as basis the field Ω, which consists of all rational functions of the independent variables a, b, c, then (1) is the group of this equation; but if we

$$(4) \qquad \sqrt{D} = (\alpha - \alpha_1)(\alpha - \alpha_2)(\alpha_1 - \alpha_2)$$

adjoin, in which

$$(5) \qquad D = +a^2b^2 + 18\,abc - 4a^3c - 4b^3 - 27c^2$$

is the discriminant of the equation (3), then the group reduces to (2).

In the field Ω', which arises from Ω by this adjunction, every root can be expressed rationally in terms of every other, for we have

$$\alpha_1 + \alpha_2 = a - \alpha$$
$$\alpha_1 - \alpha_2 = \frac{\sqrt{D}}{f'(\alpha)},$$

hence

$$(6) \qquad \begin{aligned} 2\alpha_1 &= a - \alpha + \frac{\sqrt{D}}{f'(\alpha)} \\ 2\alpha_2 &= a - \alpha - \frac{\sqrt{D}}{f'(\alpha)}, \end{aligned}$$

and in these the permutations π, π^2 can be carried out. The cubic equation is therefore, after adjunction of $\sqrt{D}$, its own Galois resolvent.

All this remains valid if the domain of rationality is any special field Ω in which a, b, c and $\sqrt{D}$ are contained, unless $f(x)$ is itself reducible in Ω, and so has a rational root. For besides itself and the cyclic group (2), the group (1) has only intransitive divisors, namely the unit group and three groups of the type (1, 2).

If we now wish to solve the cubic equation, that is, to reduce it to a pure cubic equation, we must seek a function v of the roots which does not itself admit the cyclic permutation π, but whose cube does. Such a function can exist, however, only if the third roots of unity, or, what is the same thing, $\sqrt{-3}$, is adjoined to the field Ω; for v must, on application of π, acquire a third root of unity as a factor.

By this adjunction no reduction of the group can occur, because the group (2) has no divisor besides the unit group, and the reduction to the unit group can take place not through a quadratic, but only through a cubic equation (§. 157). We therefore set

$$\varepsilon = \frac{-1 + \sqrt{-3}}{2}, \qquad \varepsilon^2 = \frac{-1 - \sqrt{-3}}{3}$$

and

$$v = \alpha + \varepsilon\alpha_1 + \varepsilon^2\alpha_2$$
$$v' = \alpha + \varepsilon^2\alpha_1 + \varepsilon\alpha_2; \tag{7}$$

then the effect of π is that v passes into $\varepsilon^2 v$, and v' into $\varepsilon v'$, while v^3, v'^3 and vv' remain unchanged, and hence are contained in the field Ω'. There is no difficulty in computing these quantities. If for brevity we set

$$A = \alpha^2\alpha_1 + \alpha_1^2\alpha_2 + \alpha_2^2\alpha,$$
$$A' = \alpha_1^2\alpha + \alpha_2^2\alpha_1 + \alpha^2\alpha_2,$$

then

$$A + A' = ab - 3c$$
$$A - A' = \sqrt{D},$$

and for v^3 one obtains

$$v^3 = \alpha^3 + \alpha_1^3 + \alpha_2^3 + 6\alpha\alpha_1\alpha_2 + 3\varepsilon A + 3\varepsilon^2 A',$$

hence

$$v^3 = a^3 - \frac{9}{2}ab + \frac{27}{2}c + \frac{3}{2}\sqrt{-3D}, \tag{8}$$

and likewise

$$v'^3 = a^3 - \frac{9}{2}ab + \frac{27}{2}c - \frac{3}{2}\sqrt{-3D}. \tag{9}$$

But we also have

$$vv' = \alpha^2 + \alpha_1^2 + \alpha_2^2 - \alpha\alpha_1 - \alpha\alpha_2 - \alpha_1\alpha_2 = a^2 - 3b, \tag{10}$$

and from (8), (9), (10) one obtains, in agreement with an earlier formula [§. 47, (8)],

$$27D = 4(a^2 - 3b)^3 - (2a^3 - 9ab + 27c)^2.$$

If to (7) we further add the equation

$$a = \alpha + \alpha_1 + \alpha_2,$$

then one finds, in agreement with the Cardan formula,

$$3\alpha = a + v + v'$$
$$3\alpha_1 = a + \varepsilon^2 v + \varepsilon v'$$
$$3\alpha_2 = a + \varepsilon v + \varepsilon^2 v'.$$

§. 160.
Permutation groups of four elements.

A good example of the Galois theory is offered by the equations of the fourth degree, where the circumstances are so simple

that everything can easily be surveyed, and yet the most important phenomena of group formation come to light in the process.

From four figures, 0, 1, 2, 3, one can form 24 permutations, and so high, therefore, is the degree of the Galois group of the general equation of the fourth degree in the field of the rational

functions of the coefficients (cf. the end of §. 149). We represent these permutations by their cycles in the following way, where 1 denotes the identical permutation:

$$
\begin{gathered}
P = 1, \quad (0,1), \quad (0,2), \quad (0,3), \quad (1,2), \quad (1,3), \quad (2,3) \\
(0,1)(2,3), \quad (0,2)(1,3), \quad (0,3)(1,2) \\
(0,1,2), \quad (0,1,3), \quad (0,2,3), \quad (1,2,3) \\
(0,2,1), \quad (0,3,1), \quad (0,3,2), \quad (1,3,2) \\
(0,1,2,3), \quad (0,1,3,2), \quad (0,2,3,1) \\
(0,2,1,3), \quad (0,3,1,2), \quad (0,3,2,1).
\end{gathered} \tag{1}
$$

We thus have in P, besides the identical permutation, six transpositions, eight three-term cycles, six four-term cycles, and three permutations of the form $(0,1)(2,3)$, which we shall call transposition pairs.

In P the alternating group of degree 12 is contained as a normal divisor. This group can contain no single transposition and no four-term cyclic permutation, and there thus remain the following:

$$
\begin{gathered}
Q = 1, \quad (0,1)(2,3), \quad (0,2)(1,3), \quad (0,3)(1,2) \\
(0,1,2), \quad (0,1,3), \quad (0,2,3), \quad (1,2,3) \\
(0,2,1), \quad (0,3,1), \quad (0,3,2), \quad (1,3,2).
\end{gathered} \tag{2}
$$

We now wish to seek out all the existing divisors of the group P. In doing so we leave aside the intransitive groups, which all consist either of three or of six permutations of only three elements, or of the permutations of two pairs of two elements. We therefore seek out only the transitive divisors of P. Moreover, in the case of a system of conjugate divisors we restrict ourselves to one representative, from which the others can indeed be derived by permutation of the figures (§. 154, 6.).

An essential simplification is brought about by the following two theorems:

1. If a transitive permutation group of four figures contains a three-term cycle,

it is the symmetric or the alternating group, and

2. if it contains two transpositions with a common element, it is the symmetric group.

For if the cycle $(0,1,2)$ occurs in a group, and the group is moreover transitive, then it contains a permutation π by which 0 is carried into 3. In the three-term cycle $\pi^{-1}(0,1,2)\pi$ the figure 3 therefore certainly occurs; it is thus, say, $(0,1,3)$, and accordingly we have in the group

$$
\begin{gathered}
(0,1,2), \quad (0,1,2)^2 = (0,2,1) \\
(0,1,3), \quad (0,1,3)^2 = (0,3,1) \\
(0,1,2)(0,3,1) = (1,2,3) \\
(1,2,3)^2 = (1,3,2) \\
(0,3,1)(0,1,2) = (0,3,2) \\
(0,3,2)^2 = (0,2,3),
\end{gathered}
$$

that is, all the three-term cycles that exist at all. Accordingly all permutations of the alternating group occur in the group in question (§. 153), and it is therefore either exhausted therewith, or it contains in addition a permutation of the second kind and is then the symmetric group.

But if a transitive group contains the two transpositions $(0,1)$, $(0,2)$, then it also contains

$$(0,1)(0,2) = (0,1,2),$$

and with it the whole alternating group, and since it also contains a transposition, it is the symmetric group.

If now, then, a group different from P and Q contains first a transposition $(0,2)$, then, if it is transitive, it must also contain $(1,3)$, and accordingly the whole intransitive group of the fourth degree:

$$1, \quad (0,2), \quad (1,3), \quad (0,2)(1,3). \tag{3}$$

If, then, the group is to be transitive, further permutations must be added. These can be sought only among the four-term cycles or among the pairs of transpositions. But a four-term cycle in which 0 and 2 adjoin one another cyclically cannot occur, because, combined with $(0,2)$, it would give a three-term cycle, such as:

$$(0,2)(0,2,3,1) = (0,3,1).$$

Likewise 1 and 3 cannot adjoin one another cyclically, and there remain therefore only the following:

$$(0,1,2,3), \quad (0,3,2,1), \quad (0,3)(1,2), \quad (0,1)(2,3). \tag{4}$$

But if one of these permutations is added to (3), then all the others must also be admitted, for we have

$$(0,2)(0,1,2,3) = (0,3)(1,2)$$
$$(1,3)(0,1,2,3) = (0,1)(2,3)$$
$$(0,2)(1,3)(0,1,2,3) = (0,3,2,1),$$

and one can therefore, by means of the transpositions $(0,2)$ and $(1,3)$, derive each of the permutations (4) from each of the others. From the assumption, then, that a transposition occurs at all in a transitive group different from P and Q, it follows that this must be the group of the eighth degree

$$\begin{gathered} P_1 = 1, \quad (0,2), \quad (1,3), \quad (0,2)(1,3), \quad (0,1)(2,3), \quad (0,3)(1,2) \\ (0,1,2,3), \quad (0,3,2,1) \end{gathered} \tag{5}$$

or at least a group conjugate with it.

That P_1 really is a group one easily recognises by composing any two of its permutations.

There are three distinct groups conjugate with P_1, which one obtains by starting, instead of from $(0,2)$, $(1,3)$, from the transpositions $(0,1)$, $(2,3)$ or $(0,3)$, $(1,2)$. The intersection of these three groups is a normal divisor of P and at the same time of Q, namely

$$Q_1 = 1, \quad (0,2)(1,3), \quad (0,1)(2,3), \quad (0,3)(1,2). \tag{6}$$

If a transitive group different from P and Q contains no transposition, then it either contains no four-term cycle and is then identical with (6), or it contains a four-term cycle and then contains also the whole cyclic group, such as

$$P_2 = 1, \quad (0,1,2,3), \quad (0,2)(1,3), \quad (0,3,2,1). \tag{7}$$

If such a group contains yet a second four-term cycle, say $(0,1,3,2)$, then on account of $(0,1,2,3)(0,1,3,2) = (0,3,1)$ it is the symmetric group. But if, besides P_2, it contains also one of the permutations of (6), say $(0,1)(2,3)$, then on account of

$$(0,1,2,3)(0,1)(2,3) = (1,3)$$

it is identical with P_1.

From P_2 likewise three conjugate groups arise, which however have no common divisor other than 1.

Thus, apart from P, Q, P_1, Q_1, P_2 and the divisors of P conjugate with P_1 and P_2, there are no transitive permutation groups of four elements.

Among the intransitive groups, the group (3):

$$P_3 = 1, \ (0,2), \ (1,3), \ (0,2)(1,3),$$

still deserves to be singled out, which likewise gives rise to a system of three conjugate groups.

§. 161.
Solution of the biquadratic equations.

The various methods of solution of the biquadratic equations differ from one another in the order in which the various divisors of the group P are employed.

Let α, α_1, α_2, α_3 be the roots of the biquadratic equation

$$f(x) = x^4 - a_1x^3 + a_2x^2 - a_3x + a_4 = 0. \tag{1}$$

We denote the discriminant by D and set

$$\sqrt{D} = (\alpha - \alpha_1)(\alpha - \alpha_2)(\alpha - \alpha_3)(\alpha_1 - \alpha_2)(\alpha_1 - \alpha_3)(\alpha_2 - \alpha_3). \tag{2}$$

Moreover we recall the expressions for the invariants (§. 64):

$$A = a_2^2 - 3a_1a_3 + 12a_4 \tag{3}$$

$$B = 27a_1^2a_4 + 27a_3^2 + 2a_2^3 - 72a_2a_4 - 9a_1a_2a_3, \tag{4}$$

by means of which D is expressed in the form:

$$27D = 4A^3 - B^2. \tag{5}$$

If one first adjoins $\sqrt{D}$, the group of the equation reduces to Q, and if one then seeks a function belonging to Q_1, the group is reduced by it to Q_1. Since Q_1 has index 3 within Q, such a function will depend on a cubic equation, and indeed, since Q_1 is a normal divisor of Q, on a normal equation. Let us take, for example,

$$y = (\alpha - \alpha_1)(\alpha_2 - \alpha_3) \tag{6}$$

as a function belonging to Q_1, then the two values conjugate to it are

$$\begin{aligned} y_1 &= (\alpha - \alpha_2)(\alpha_3 - \alpha_1) \\ y_2 &= (\alpha - \alpha_3)(\alpha_1 - \alpha_2). \end{aligned} \tag{7}$$

These three quantities are the roots of a cubic partial resolvent, whose coefficients are easily formed as invariants of f (cf. §. 64, 65, where the quantities y, y_1, y_2 were denoted by U, V, W). The equation becomes

$$y^3 - Ay + \sqrt{D} = 0. \tag{8}$$

Its discriminant is B^2, and for the product of the root differences the sign can easily be determined (by comparison of the terms of highest order according to the Gauss definition, §. 45, or also from §. 64). Thus one obtains:

$$(y - y_1)(y - y_2)(y_1 - y_2) = B. \tag{9}$$

The equation (8) is, as already remarked, a normal equation, and it is easy to represent y_1, y_2 actually in terms of y. If one multiplies (9) by $(y_1 - y_2)$, one finds, by making use of the formulae $(y - y_1)(y - y_2) = 3y^2 - A$, $(y_1 - y_2)^2 = y^2 - 4y_1y_2 = 4A - 3y^2$:

$$\begin{aligned} B(y_1 - y_2) &= -9y^4 + 15Ay^2 - 4A^2 \\ &= 6Ay^2 + 9y\sqrt{D} - 4A^2 \\ y_1 + y_2 &= -y, \end{aligned} \tag{10}$$

whereby y_1, y_2 are determined. If therefore one further adjoins y, the group of the biquadratic equation reduces to

$$Q_1 = 1,\ (0,2)(1,3),\ (0,1)(2,3),\ (0,3)(1,2).$$

This group is imprimitive (and indeed in three ways), and the equation can now be solved by two square roots. This is best reached in the following way. The three quantities

$$\begin{aligned} v_1 &= (\alpha + \alpha_1 - \alpha_2 - \alpha_3)^2 \\ v_2 &= (\alpha - \alpha_1 + \alpha_2 - \alpha_3)^2 \\ v_3 &= (\alpha - \alpha_1 - \alpha_2 + \alpha_3)^2 \end{aligned} \tag{11}$$

all admit the permutations of the group Q_1 and are therefore rationally representable in terms of y. One obtains this representation as follows: We have

$$\begin{aligned} v_1 &= a_1^2 - 4(\alpha + \alpha_1)(\alpha_2 + \alpha_3) \\ y_1 - y_2 &= (\alpha + \alpha_1)(\alpha_2 + \alpha_3) - 2(\alpha\alpha_1 + \alpha_2\alpha_3) \\ a_2 &= (\alpha + \alpha_1)(\alpha_2 + \alpha_3) + \alpha\alpha_1 + \alpha_2\alpha_3, \end{aligned}$$

and thus, if one forms v_2, v_3 in the same way:

$$\begin{aligned} 3v_1 &= 3a_1^2 - 8a_2 - 4y_1 + 4y_2 \\ 3v_2 &= 3a_1^2 - 8a_2 - 4y_2 + 4y \\ 3v_3 &= 3a_1^2 - 8a_2 - 4y + 4y_1. \end{aligned} \tag{12}$$

But there is yet another relation, which results from the fact that the product

$$(\alpha + \alpha_1 - \alpha_2 - \alpha_3)(\alpha - \alpha_1 + \alpha_2 - \alpha_3)(\alpha - \alpha_1 - \alpha_2 + \alpha_3)$$

is a symmetric function of the α, which one easily finds to be equal to

$$a_1^3 - 4a_1a_2 + 8a_3$$

We therefore make the signs of the three root quantities depend on one another by the relation

$$\sqrt{v_1}\sqrt{v_2}\sqrt{v_3} = a_1^3 - 4a_1a_2 + 8a_3 \tag{13}$$

and set:

$$\begin{aligned} \sqrt{v_1} &= \alpha + \alpha_1 - \alpha_2 - \alpha_3 \\ \sqrt{v_2} &= \alpha - \alpha_1 + \alpha_2 - \alpha_3 \\ \sqrt{v_3} &= \alpha - \alpha_1 - \alpha_2 + \alpha_3 \\ a_1 &= \alpha + \alpha_1 + \alpha_2 + \alpha_3. \end{aligned} \tag{14}$$

From this it then follows:

$$\begin{aligned} 4\alpha &= a_1 + \sqrt{v_1} + \sqrt{v_2} + \sqrt{v_3} \\ 4\alpha_1 &= a_1 + \sqrt{v_1} - \sqrt{v_2} - \sqrt{v_3} \\ 4\alpha_2 &= a_1 - \sqrt{v_1} + \sqrt{v_2} - \sqrt{v_3} \\ 4\alpha_3 &= a_1 - \sqrt{v_1} - \sqrt{v_2} + \sqrt{v_3}. \end{aligned} \tag{15}$$

If one gives $\sqrt{D}$ in (8) the opposite sign, and replaces y by $-y$, then by (10) one must interchange y_1, y_2 with $-y_2$, $-y_1$. Thus v_1 remains unchanged, and v_2 and v_3 are interchanged with one another. If one takes for y another root of the equation (8), say y_1, then on account of (9) y_1, y_2 must be replaced by y_2, y, and the quantities v_1, v_2, v_3 undergo a cyclic permutation, whereby the quantities α_1, α_2, α_3 in (15) are also cyclically permuted. If, finally, one chooses the signs of $\sqrt{v_1}$, $\sqrt{v_2}$, $\sqrt{v_3}$

differently, but in such a way that (13) always remains satisfied, then certain permutations likewise occur among the quantities (15).

Whatever values one may assign to the many-valued algebraic quantities that occur in the solution, the expressions (15) thus always represent the roots of our biquadratic equation in some order.

With regard to the remaining ways of solving the biquadratic equation we can be briefer.

If we first adjoin not $\sqrt{D}$ but a function belonging to the group P_1, we obtain a cubic resolvent which is not a normal equation. For this function we can choose, say, y^2, which satisfies the following cubic equation in that quantity:

$$y^6 - 2Ay^4 + A^2y^2 - D = 0 \tag{16}$$

Better still, one takes as roots of the cubic resolvent

$$z = y_1 - y_2, \quad z_1 = y_2 - y, \quad z_2 = y - y_1. \tag{17}$$

Then z belongs to the group P_1, as already follows from the second equation (10), or as may also be seen directly. From (8) and (9) one obtains for z the cubic equation

$$z^3 - 3Az + B = 0. \tag{18}$$

If we further set

$$u = \alpha\alpha_1 + \alpha_2\alpha_3 \tag{19}$$

then

$$z = a_2 - 3u$$

and hence obtain the resolvent for u:

$$u^3 - a_2u^2 + (a_1a_3 - 4a_4)u + 4a_2a_4 - a_1^2a_4 - a_3^2 = 0. \tag{20}$$

But if one adjoins not one but all the roots of equation (18) or (20), one arrives again at the group Q_1. The quantities v_1, v_2, v_3 are by (12) expressed immediately in terms of z, z_1, z_2, and one can therefore also apply the formulae (15) for the representation of the α by this procedure.

One can further arrive at the solution of the biquadratic equation by first adjoining a function of the roots belonging to the cyclic group P_2 (a cyclic function), as for example

$$w = \alpha\alpha_1^2 + \alpha_1\alpha_2^2 + \alpha_2\alpha_3^2 + \alpha_3\alpha^2. \tag{21}$$

This function is six-valued and is therefore the root of an equation of the sixth degree. But this equation of the sixth degree is, as one easily sees, imprimitive, and can be

to an equation of the third degree and to two square roots. This equation of the sixth degree is a total resolvent, and two of its roots suffice for the rational representation of the roots α. The form of these resolvents does not turn out simple.

Finally, one may also aim, by the adjunction of a function ξ belonging to the group P_3, to make the function $f(x)$ directly reducible and to decompose it into two quadratic factors. Such a function ξ is likewise the root of an equation of the sixth degree, which, however, also reduces by imprimitivity to the third and second degree. If one takes, e. g.,

$$\xi = \alpha + \alpha_2 - \tfrac{1}{2}a_1 = \tfrac{1}{2}\sqrt{v_2}, \tag{22}$$

then of the six roots two at a time are equal and opposite, so that a cubic equation for ξ^2 results. By means of one of these quantities ξ the quadratic factors of $f(x)$ can then be represented rationally.

If, to simplify the formulae, we assume $a_1 = 0$ and

$$f(x) = x^4 + ax^2 - bx + c, \tag{23}$$

then

$$(\alpha\alpha_2 - \alpha_1\alpha_3)\,\xi = b$$
$$\alpha\alpha_2 + \alpha_1\alpha_3 = \xi^2 + a,$$

hence

$$2\,\alpha\alpha_2 = \frac{b}{\xi} + \xi^2 + a, \qquad 2\,\alpha_1\alpha_3 = -\frac{b}{\xi} + \xi^2 + a, \tag{24}$$

so that the two quadratic factors of $f(x)$ become the following:

$$\begin{aligned} x^2 - \xi x + \tfrac{1}{2}\Big(\frac{b}{\xi} + \xi^2 + a\Big) &= 0 \\ x^2 + \xi x + \tfrac{1}{2}\Big(-\frac{b}{\xi} + \xi^2 + a\Big) &= 0, \end{aligned} \tag{25}$$

and for ξ^2 there results from (24), by multiplication, the cubic equation

$$4c = (\xi^2 + a)^2 - \frac{b^2}{\xi^2}$$

or

$$\xi^6 + 2a\xi^4 + (a^2 - 4c)\xi^2 - b^2 = 0. \tag{26}$$

If one takes for ξ any root of this equation of the sixth degree, then each of the equations (25) gives a pair of roots of $f(x) = 0$.

§. 162.
Abelian Equations.

The degree of a transitive permutation group of m figures is always divisible by m, and therefore never smaller than m. For if P is such a group and Q_0 is the totality of the permutations of P that leave the figure 0 unchanged, then Q_0 is also a group, and therefore a divisor of P. Now, on account of the assumed transitivity, one can find in P a system of permutations $\pi_1, \pi_2 \ldots \pi_{m-1}$ that carry 0 into 1, 0 into 2, $\ldots$ 0 into $m-1$, and then $Q\pi_1$ is the system of all those permutations of P that change 0 into 1. Accordingly

$$P = Q_0 + Q_0\pi_1 + Q_0\pi_2 + \cdots + Q_0\pi_{m-1}, \tag{1}$$

and so the degree of P is equal to the product of m and the degree of Q_0.

The Galois group of an irreducible equation is therefore never of lower degree than the equation itself.

A normal equation we have earlier defined by the stipulation that it be irreducible and that each of its roots should be rationally expressible in terms of every other (§. 145). From this it follows that the group of a normal equation must reduce to the identical group when one root is adjoined; and conversely, an equation whose group is so constituted is, provided it is at the same time irreducible, always a normal equation.

Now P reduces, by adjunction of the root α_0, to Q_0, by α_1 to $\pi_1^{-1}Q_0\pi_1$, etc., and so if P is to be the group of a normal equation, it is necessary and sufficient that Q_0 be the unit group, and hence that the group be exhausted by $\pi, \pi_1, \pi_2, \ldots \pi_{m-1}$. We therefore have:

In order that an irreducible equation be a normal equation, it is necessary and sufficient that the degree of the group agree with the degree of the equation.

We first consider here the special class of equations that includes the cyclotomic equations first solved by Gauss,

which Abel taught how to solve in general, and which we shall therefore, after him, call Abelian equations[38].

An equation of degree m, $F(x) = 0$, with the roots $\alpha, \alpha_1, \alpha_2 \ldots \alpha_{m-1}$, is called an Abelian equation if every root is rationally expressible in terms of one of them, α, and if, when

$$\alpha_1 = \Theta_1(\alpha), \quad \alpha_2 = \Theta_2(\alpha) \ldots \alpha_{m-1} = \Theta_{m-1}(\alpha) \tag{2}$$

are these rational expressions, the condition

$$\Theta_h\Theta_k(\alpha) = \Theta_k\Theta_h(\alpha) \tag{3}$$

holds for every pair of these functions.

Here the symbol $\Theta_h\Theta_k(\alpha)$ means that the function $\Theta_h(x)$ is to be formed for the argument $x = \Theta_k(\alpha)$. This whole definition refers, of course, to a definite field Ω assumed as rational. To these equations certainly belong, when Ω is the field of rational numbers, the equations by which the roots of unity are determined. For every mth root of unity is, if r is a primitive one among them, contained in the form $\Theta_h(r) = r^h$, and we have

$$\Theta_h\Theta_k(r) = \Theta_k\Theta_h(r) = r^{hk}.$$

Likewise, when the field Ω contains the mth roots of unity r, the pure equation

$$x^m - a = 0,$$

in which a belongs to the field Ω, belongs to the Abelian equations. For if α is one of its roots, then all the roots are contained in the form

$$\alpha_h = r^h\alpha$$

we have therefore $\Theta_h(x) = r^h x$, and hence

$$\Theta_h\Theta_k(x) = \Theta_k\Theta_h(x) = r^{h+k}x.$$

If the function $F(x)$ is not irreducible, then it has a definite irreducible factor $\varphi(x)$ which has the root α, and among the roots of $\varphi(x) = 0$ the relations expressed by (2) and (3) likewise hold. Hence $\varphi(x) = 0$ is, by §. 145, a Galois resolvent of

$F(x) = 0$, and when $\varphi(x) = 0$ is solved, all the roots of $F(x)$ are thereby known. It therefore suffices if we confine ourselves to the consideration of irreducible Abelian equations.

The Galois group of an Abelian equation, whether it be irreducible or not (provided only that its roots are distinct from one another), has the property that in the composition of its permutations the commutative law holds; it is therefore a commutative group.

Namely, let

$$\alpha,\ \alpha_1 = \Theta_1(\alpha),\ \alpha_2 = \Theta_2(\alpha) \ldots \alpha_{m-1} = \Theta_{m-1}(\alpha) \tag{4}$$

be the roots of $F(x)$, and

$$\alpha,\ \alpha' = \Theta'(\alpha),\ \alpha'' = \Theta''(\alpha) \ldots \tag{5}$$

the roots of the irreducible factor $\varphi(x)$ of $F(x)$ contained among them.

Since, as already remarked, $\varphi(x) = 0$ is a Galois resolvent, the group of the equation consists of the substitutions of the field $\Omega(\alpha)$, that is, of the substitutions

$$\sigma = (\alpha, \alpha),\ \sigma' = (\alpha, \alpha'),\ \sigma'' = (\alpha, \alpha'') \ldots \tag{6}$$

[38] Abel, Mémoire sur une classe d'équations résolubles algébriquement. Crelle's Journal f. Mathematik, Bd. 4, 1829. Oeuvres complètes, nouvelle édition, 1881, Bd. 1, S. 418.

But this group obeys the commutative law; for let

$$\sigma' = (\alpha, \alpha') = [\alpha, \Theta'(\alpha)]$$
$$\sigma'' = (\alpha, \alpha'') = [\alpha, \Theta''(\alpha)];$$

then by §. 147

$$\sigma'\sigma'' = [\alpha, \Theta'(\alpha)][\Theta'(\alpha), \Theta'\Theta''(\alpha)] = [\alpha, \Theta'\Theta''(\alpha)]$$
$$\sigma''\sigma' = [\alpha, \Theta''(\alpha)][\Theta''(\alpha), \Theta''\Theta'(\alpha)] = [\alpha, \Theta''\Theta'(\alpha)].$$

On account of (3) we therefore have $\sigma'\sigma'' = \sigma''\sigma'$, where σ', σ'' may be any two of the substitutions σ. Consequently the group of the substitutions of the field $\Omega(\alpha)$, and with it also the isomorphic permutation group of the equation $F(x) = 0$, is commutative.

The converse now also holds, but here the irreducibility must be presupposed.

An irreducible equation $\varphi(x) = 0$ with commutative group is an Abelian equation.

Let $\alpha, \alpha_1, \alpha_2 \ldots \alpha_{m-1}$ be the roots of $\varphi(x) = 0$ and P the group of this equation, which is transitive because of the irreducibility of $\varphi(x)$, and which we now moreover assume to be commutative. Let Q be the divisor of P that leaves the element 0

fixed. If then π_i is a permutation that carries the figure 0 into i, then $\pi_i^{-1} Q \pi_i$ is the group of the permutations in P that do not alter the figure i. But since in every composition of permutations from P the components may be interchanged, we have

$$\pi_i^{-1} Q \pi_i = \pi_i^{-1} \pi_i Q = Q,$$

that is, the group Q also leaves the figure i unchanged. On account of the transitivity of P, however, i may denote any of the figures $1, 2 \ldots m-1$; consequently Q consists of the single identical permutation, and in P there is, besides the identical one, no permutation that leaves a figure unchanged. But if one adjoins a root α, the group P reduces to the unit group, or the equation is solved; $\varphi(x) = 0$ is a normal equation and hence its own Galois resolvent. If $\alpha_k = \Theta_k(\alpha)$, then the Galois group of this equation consists of the substitutions:

$$\sigma_k = [\alpha, \Theta_k(\alpha)],$$

and we have

$$\sigma_h\sigma_k = [\alpha, \Theta_h\Theta_k(\alpha)], \quad \sigma_k\sigma_h = [\alpha, \Theta_k\Theta_h(x)].$$

Since the group is to be commutative, we must have

$$\Theta_h\Theta_k(\alpha) = \Theta_k\Theta_h(\alpha)$$

that is, $\varphi(x) = 0$ is an Abelian equation. This is the reason why the commutative groups are also called Abelian groups (§. 148). It therefore also follows from the theorem given above on the group of normal equations that in a transitive Abelian permutation group the number of permutations agrees with the number of the permuted figures.

Here we have presupposed the irreducibility of the given equation. If we wish to take into consideration also reducible equations with commutative group, we need only put in place of the equation itself one of its Galois resolvents.

It remains, finally, to remark that in a commutative group every divisor is normal, since we always have $\pi^{-1}\varkappa\pi = \varkappa$ when π and $\varkappa$ are arbitrary elements of a commutative group.

§. 163.
Reduction of Abelian equations to cyclic ones.

We have already seen above that in a transitive Abelian group no permutation other than the identical one occurs which leaves a figure unchanged. But from this there follows yet another theorem. Let us assume that a permutation π of such a group is decomposed into its cycles, and that the cycle which contains the fewest terms is an r-term one. Then π^r will leave the terms of this cycle unchanged, and must therefore be the identical permutation. But from this it follows that all the remaining cycles of π, if there are any others, must likewise consist of r terms; hence:

1. A permutation of a transitive Abelian group contains only cycles with the same number of terms.

The number r of the terms of a cycle must be a divisor of m, if m is the degree of the group, and if $m = rs$, then s is the number of the r-term cycles of which π consists.

If now P is the group of an Abelian equation $F(x) = 0$, we take any non-identical permutation π out of P and decompose it into its cycles

$$\pi = \gamma\gamma_1\gamma_2 \cdots \gamma_{s-1}, \tag{1}$$

in which each of the cycles γ relates to r roots of the equation $F(x) = 0$. We shall arrange these roots in such a way that

$$\begin{aligned} \gamma &= (\alpha, \alpha_1, \alpha_2 \ldots \alpha_{r-1}) \\ \gamma_1 &= (\beta, \beta_1, \beta_2 \ldots \beta_{r-1}) \\ &\vdots \\ \gamma_{s-1} &= (\sigma, \sigma_1, \sigma_2 \ldots \sigma_{r-1}) \end{aligned} \tag{2}$$

holds.

If π_1 is any permutation of P, then on account of the permutability

$$\pi_1^{-1}\pi\pi_1 = \pi. \tag{3}$$

By the theorem on the formation of the permutations $\pi_1^{-1}\pi\pi_1$ (§. 154, 6.), π must therefore not be changed when the permutations π_1 are carried out in the cycles of π. But since the cycles are completely determined, apart from their ar-

rangement and their initial element, it follows that by the application of any permutation π_1 from P the elements of the individual cycles γ are not separated from one another, but can only be interchanged among themselves (cyclically), and moreover the cycles can be interchanged with one another. The group is therefore, if $s > 1$, imprimitive.

A rational function of the arguments $\alpha, \alpha_1, \ldots \alpha_{r-1}$ which does not change its value when the α are subjected to the cyclic permutation γ and its repetitions is called a cyclic function of the α. If one denotes by

$$\omega = \psi(\alpha, \alpha_1 \ldots \alpha_{r-1}) \tag{4}$$

such a cyclic function, and by

$$\omega,\ \omega_1,\ \omega_2 \ldots \omega_{s-1} \tag{5}$$

the conjugate values of ω, setting thus e. g.

$$\omega_1 = \psi(\beta, \beta_1 \ldots \beta_{r-1}), \tag{6}$$

then these quantities are, by §. 158, the roots of an irreducible equation of degree s:

$$\Phi(t) = 0, \tag{7}$$

whose group one obtains by seeking out the permutations of the quantities (5) produced by P.

But one obtains the permutation effected by $\pi_1\pi_2$ among the quantities (5) by composing the two permutations produced by π_1 and π_2 singly, and the group of the permutations of (5) is therefore also commutative. Hence $\Phi(t) = 0$ is an Abelian equation of degree s.

If one adjoins ω, then $F(x)$ splits into s factors of degree r:

$$F(x) = F(x,\omega)F(x,\omega_1)\dots F(x,\omega_{s-1}),$$

of which the first, $F(x,\omega)$, has the roots $\alpha, \alpha_1 \dots \alpha_{r-1}$, and the group of the equation $F(x,\omega) = 0$ consists solely of the period of the cyclic permutation γ.

We shall call an equation whose group consists of a single cycle and its repetitions a cyclic equation, so that the cyclic equations are the simplest special case of the Abelian equations. We have then proved that the solution of every Abelian equation is reduced to the solution of an Abelian equation of lower degree and to the solution of a series

of cyclic equations. This theorem can again be applied to the auxiliary equation of degree s, and one can continue in this way until this auxiliary equation reduces to the first degree. With this, then, one obtains the result:

The solution of an Abelian equation can always be reduced to the solution of a sequence of cyclic equations whose degrees are divisors of the degree of the given equation.

It is not necessary to include irreducibility in the definition of cyclic equations. We may therefore frame the definition generally as follows:

An equation of degree m, $F(x) = 0$, with m distinct roots is called a cyclic equation in the field Ω if its roots $\alpha, \alpha_1, \alpha_2 \dots \alpha_{m-1}$ are not rational, but can be so arranged that the cyclic functions of the roots are rational in Ω.

If

$$\pi = (\alpha, \alpha_1, \alpha_2 \dots \alpha_{m-1}) \tag{7}$$

holds, then every function that admits the permutations of the cyclic group

$$C = 1, \pi, \pi^2, \pi^3 \dots \pi^{m-1} \tag{8}$$

must be contained in Ω. The Galois group of a cyclic equation is either the period C itself, and then the equation is irreducible, or it is a divisor of C; in that case, if e and f are two integer factors of m and

$$m = ef$$

it consists of the permutations

$$C_e = 1, \pi^e, \pi^{2e} \dots \pi^{(f-1)e}, \tag{9}$$

and the cyclic equation splits into e factors of degree f. For if we take a function $\psi(\alpha, \alpha_1 \dots \alpha_{m-1})$ belonging to the group C, then by hypothesis it is equal to a quantity in Ω, which we denote by a. No permutation not contained in C is then applicable to the rational equation $\psi = a$, and consequently the group of the equation can contain no substitutions other than those which occur in C. Now if π occurs in the group of the equation, then that group is identical with C, and since C is transitive, the equation is irreducible. But if π^e is the lowest power of π which occurs in the

group of the equation, then C_e is this group. But C_e is intransitive when $e < m$, and the equation is reducible.

The reduction carried out above can also be applied to the cyclic equations whose degree is not a prime.

For if $m = ef$ is any decomposition of m into two factors, the permutation π^e splits into e cycles γ of f terms each:

$$\begin{aligned} \gamma &= (\alpha, \alpha_e, \alpha_{2e} \dots \alpha_{(f-1)e}), \\ \gamma_1 &= (\alpha_1, \alpha_{e+1}, \alpha_{2e+1} \dots \alpha_{(f-1)e+1}), \\ &\dots\dots\dots\dots\dots\dots\dots \\ \gamma_{c-1} &= (\alpha_{e-1}, \alpha_{2e-1}, \alpha_{3e-1} \dots \alpha_{m-1}). \end{aligned} \tag{10}$$

Now we determine a function ψ belonging to the first of these cycles, and set

$$\begin{aligned} \eta &= \psi(\alpha, \alpha_e, \alpha_{2e} \dots \alpha_{(f-1)e}), \\ \eta_1 &= \psi(\alpha_1, \alpha_{e+1}, \alpha_{2e+1} \dots \alpha_{(f-1)e+1}), \\ &\dots\dots\dots\dots\dots\dots\dots \\ \eta_{e-1} &= \psi(\alpha_{e-1}, \alpha_{2e-1}, \alpha_{3e-1} \dots \alpha_{m-1}). \end{aligned} \tag{11}$$

These quantities are all different from one another, but under a cyclic permutation of their arguments they remain unchanged. By application of the permutation π the quantities

$$\eta, \eta_1, \eta_2 \dots \eta_{e-1}$$

pass cyclically into one another, and by the hypothesis their cyclic functions, and consequently also their symmetric functions, are therefore rational. They are thus the roots of a cyclic equation of degree e, while $F(x)$ splits into e factors of degree f

$$F(x) = F(x, \eta)\, F(x, \eta_1) \dots F(x, \eta_{e-1})$$

each of which yields a cyclic equation of degree f for the roots of one of the cycles γ.

For if we set, say,

$$F_1 = (x - \alpha)(x - \alpha_e) \dots (x - \alpha_{(f-1)e}),$$

then this function admits the permutations of the period of γ. Hence it also admits the permutations of the group which, after adjunction of η, becomes the Galois group of our equation, and which indeed can consist only of powers of $\gamma, \gamma_1 \dots \gamma_{e-1}$. Hence, by the theorem of Lagrange [§. 155, (2)], F_1 is rationally expressible in terms of η, so $F_1 = F(x, \eta)$. The equation

$F(x, \eta) = 0$ is, however, again cyclic in $\Omega(\eta)$, since the cyclic functions of its roots belong to this field.

The cyclic equations, like all Abelian equations, have the property that every root is rationally expressible in terms of every other. Here these expressions can be arranged cyclically in the following manner. If $\alpha, \alpha_1, \alpha_2 \dots \alpha_{m-1}$ are the roots of the cyclic equation $F(x) = 0$, then the polynomial of degree $(m-1)$ in x

$$F(x) \left(\frac{\alpha_1}{x - \alpha} + \frac{\alpha_2}{x - \alpha_1} + \dots + \frac{\alpha}{x - \alpha_{m-1}} \right) = \Psi(x)$$

is unchanged by the permutation π and therefore contained in Ω. If we set $x = \alpha, \alpha_1 \dots \alpha_{m-1}$ therein and introduce the sign

$$\frac{\Psi(x)}{F'(x)} = \Theta(x)$$

it follows that

$$\alpha_1 = \Theta(\alpha), \alpha_2 = \Theta(\alpha_1), \dots \alpha_{m-1} = \Theta(\alpha_{m-2}), \alpha = \Theta(\alpha_{m-1}), \tag{11}$$

and this holds, whether $F(x)$ be reducible or irreducible, provided only that $F(x)$ and $F'(x)$ have no common divisor.

The solution of the cyclic equations of degree m is hereby made dependent on the solution of cyclic equations whose degrees are the prime factors of m. If, for example, m is a power of 2, the solution is accomplished by a sequence of square roots. In this way Gauss first treated the cyclotomic equations[39].

We finally attach here the remark, important for what follows, that for a prime degree the concepts of the normal equation and of the cyclic equation coincide. For in any case a cyclic equation of prime degree, since it is irreducible, is a normal equation. And if, conversely, the degree n of a normal equation, which is at the same time the degree of the group of this equation, is a prime, and π is a non-identical permutation of this group, then the degree of π, which must indeed be a divisor of n, is equal to n, and the group of the equation is $1, \pi, \pi^2 \dots \pi^{n-1}$, hence cyclic.

§. 164.
Resolvents of Lagrange.

The method of solving cyclic equations with which we now wish to become acquainted is applicable uniformly to prime degrees and to composite degrees. For this purpose one makes use of certain expressions, known under the name of the resolvents of Lagrange[40], which are of great use in all investigations concerning the algebraic solution of equations.

Let $F(x) = 0$ be an equation with the roots $\alpha, \alpha_1, \alpha_2 \dots \alpha_{m-1}$. We denote by ε any mth root of unity and introduce the notation

$$(\varepsilon, \alpha) = \alpha + \varepsilon\alpha_1 + \varepsilon^2\alpha_2 + \cdots + \varepsilon^{m-1}\alpha_{m-1}. \tag{1}$$

The sums so defined are those which one calls the Lagrange resolvents. When these functions are known for all mth roots of unity ε, the equation itself is also solved, for by §. 133 (6) we have

$$\overset{\varepsilon}{\Sigma}\varepsilon^k = m \text{ or } = 0, \tag{2}$$

according as k is divisible by m or not, and from this it follows

$$m\alpha = \overset{\varepsilon}{\Sigma}(\varepsilon, \alpha), \tag{3}$$

in which the sums extend over all mth roots of unity ε. One can also express the other roots in the same way:

$$m\alpha_k = \overset{\varepsilon}{\Sigma}\varepsilon^{-k}(\varepsilon, \alpha), \tag{4}$$

so that in fact everything is reduced to the knowledge of ε and of the functions (ε, α).

The sums (2) and (3) may also be represented in a somewhat different manner, since all the mth roots of unity are powers of a primitive one among them. If, then, one understands by ε a fixed primitive mth root of unity, and by λ an index which runs through a complete system of residues with respect to the modulus m, say the

sequence of numbers $0, 1 \dots m-1$, then for the equations (2), (3) we can set

$$m\alpha = \overset{\lambda}{\Sigma}(\varepsilon^\lambda, \alpha), \quad m\alpha_k = \overset{\lambda}{\Sigma}\varepsilon^{-\lambda k}(\varepsilon^\lambda, \alpha).$$

We investigate these resolvents (ε, α) first as functions of the m independent variables α, and, for the sake of a simpler presentation of the formulae, we shall agree that $\alpha_m = \alpha_0 = \alpha$, and in general that $\alpha_h = \alpha_k$ shall hold whenever $h \equiv k \pmod m$. We then obtain the whole system of the

[39] Gauss, Disquisitiones arithmeticae, Sectio VII.

[40] Lagrange, Réflexions etc., see p. 508. Earlier we understood by resolvents solving equations; here they are solving functions.

variables α when in α_h we let the index run through a complete system of residues to the modulus m.

For these resolvents the following theorems now hold.

1. If one applies to the indices of the α the cyclic permutation $\pi = (0, 1, \ldots m-1)$, then (ε, α) passes into $\varepsilon^{-1}(\varepsilon, \alpha)$, and by the permutation π^k, (ε, α) passes into $\varepsilon^{-k}(\varepsilon, \alpha)$.

This is shown immediately by the definition (1) of the resolvents.

We further understand by ν an arbitrary positive exponent, and form $(\varepsilon, \alpha)^\nu$ according to the polynomial theorem. In the developed power we set $\varepsilon^m = 1$ and then arrange according to powers of ε. There then results an expression of the form

$$\begin{aligned}(\varepsilon, \alpha)^\nu &= A_0^{(\nu)} + \varepsilon A_1^{(\nu)} + \varepsilon^2 A_2^{(\nu)} + \cdots + \varepsilon^{m-1} A_{m-1}^{(\nu)} \\ &= \sum_{0, m-1}^{h} \varepsilon^h A_h^{(\nu)},\end{aligned} \tag{5}$$

in which the $A_0^{(\nu)}, A_1^{(\nu)} \ldots A_{m-1}^{(\nu)}$ are forms of degree ν with integral coefficients and the variables α, but independent of ε. Here too let $A_h = A_k$ hold whenever $h \equiv k \pmod m$. We then prove the theorem:

2. If one applies to the indices of the α the cyclic permutation π, then the indices of the coefficients of $(\varepsilon, \alpha)^\nu$ undergo the cyclic permutation π^ν, i.e. $A_h^{(\nu)}$ passes into $A_{h+\nu}^{(\nu)}$.

To prove this, we remark that the formula (5) remains correct for every arbitrary mth root of unity ε, including 1, and accordingly there follows from (2)

$$mA_k^{(\nu)} = \sum_{\varepsilon} \varepsilon^{-k} (\varepsilon, \alpha)^\nu. \tag{6}$$

If one carries out on the right-hand side of this formula the permutation π in the indices of the α, then by 1. there results

$$\sum^{\varepsilon} \varepsilon^{-k-\nu} (\varepsilon, \alpha)^\nu,$$

that is, $A_k^{(\nu)}$ passes into $A_{k+\nu}^{(\nu)}$, as is asserted in Theorem 2.

Theorem 2. is a special case of a general theorem. If we develop a product of arbitrarily many factors

$$(\varepsilon, \alpha)^\nu (\varepsilon^{\lambda_1}, \alpha)^{\nu_1} (\varepsilon^{\lambda_2}, \alpha)^{\nu_2} \ldots,$$

in which $\nu, \nu_1, \nu_2 \ldots$ are positive and $\lambda_1, \lambda_2 \ldots$ arbitrary integers, and order it, as before $(\varepsilon, \alpha)^\nu$, according to powers of ε, the result may be

$$(\varepsilon, \alpha)^\nu (\varepsilon^{\lambda_1}, \alpha)^{\nu_1} (\varepsilon^{\lambda_2}, \alpha)^{\nu_2} \cdots = \sum_{0, m-1}^{h} \varepsilon^h B_h, \tag{7}$$

in which the B_h are forms of the variables α independent of ε. Since this development also holds for all mth roots of unity ε, one can apply the formula (2) and obtains

$$B_k = \sum^{\varepsilon} \varepsilon^{-k} (\varepsilon, \alpha)^\nu (\varepsilon, \alpha^{\lambda_1})^{\nu_1} (\varepsilon, \alpha^{\lambda_2})^{\nu_2} \ldots$$

from which one can conclude, by 1., that B_k passes by the substitution π into

$$B_{k+\nu+\lambda_1\nu_1+\lambda_2\nu_2+\cdots}$$

We have therefore:

3. The permutation π, applied to the indices of the α, calls forth among the indices of the coefficients of the product, ordered according to ε,

$$(\varepsilon,\alpha)^{\nu}(\varepsilon^{\lambda_1},\alpha)^{\nu_1}(\varepsilon^{2},\alpha)^{\nu_2}\dots \tag{8}$$

the permutation $\pi^{\nu+\lambda_1\nu_1+\lambda_2\nu_2\cdots}$.

In these theorems the permutation π may be repeated; they remain correct when π is replaced by any power of π, and they hold whatever mth root of unity ε may be. But if ε is an imprimitive root of unity, certain simplifications occur, with which we must still become acquainted. If e is a divisor of m,

$$m = ef$$

and ε is any eth root of unity, then the expression (1) becomes

$$\begin{aligned}(\varepsilon,\alpha) = \alpha &+ \varepsilon\alpha_1 + \cdots + \varepsilon^{e-1}\alpha_{e-1}\\ &+ \alpha_e + \varepsilon\alpha_{e+1} + \cdots + \varepsilon^{e-1}\alpha_{2e-1}\\ &\cdots\cdots\cdots\cdots\cdots\cdots\cdots\\ &+ \alpha_{e(f-1)} + \varepsilon\alpha_{e(f-1)+1} + \cdots + \varepsilon^{e-1}\alpha_{m-1}\end{aligned}$$

We now introduce the notation:

$$\begin{aligned}\eta &= \alpha + \alpha_e + \alpha_{2e} + \cdots + \alpha_{(f-1)e}\\ \eta_1 &= \alpha_1 + \alpha_{e+1} + \alpha_{2e+1} + \cdots + \alpha_{(f-1)e+1},\\ &\cdots\cdots\cdots\cdots\cdots\cdots\cdots\\ \eta_{e-1} &= \alpha_{e-1} + \alpha_{2e-1} + \alpha_{3e-1} + \cdots + \alpha_{m-1},\end{aligned} \tag{9}$$

and call these quantities, as Gauss did in the special case of the division of the circle, the f-term periods of the quantities α. We then have

$$(\varepsilon,\alpha) = \eta + \varepsilon\eta_1 + \varepsilon^2\eta_2 + \cdots + \varepsilon^{e-1}\eta_{e-1}, \tag{10}$$

for which we may also write (ε,η).

The application of the permutation π to the indices of α brings forth, among the indices of the quantities η, which are to be taken with respect to the modulus e, the cyclic permutation

$$\gamma = (0,1,2\dots e-1) \tag{11}$$

Corresponding to formula (10), we can now also write the formulas (5) and (7) thus:

$$(\varepsilon,\eta)^{\nu} = E_0^{(\nu)} + \varepsilon E_1^{(\nu)} + \varepsilon^2 E_2^{(\nu)} + \cdots + \varepsilon^{e-1}E_{e-1}^{(\nu)}$$

$$(\varepsilon,\eta)^{\nu}(\varepsilon^{\lambda_1},\eta)^{\nu_1}(\varepsilon^{\lambda_2},\eta)^{\nu_2}\cdots = G_0 + \varepsilon G_1 + \varepsilon^2 G_2 + \cdots + \varepsilon^{e-1}G_{e-1},$$

in which then

$$\begin{aligned}E_k^{(\nu)} &= A_k^{(\nu)} + A_{e+k}^{(\nu)} + \cdots + A_{(f-1)e+k}^{(\nu)}\\ G_k &= B_k + B_{e+k} + \cdots + B_{(f-1)e+k}\end{aligned}$$

and $E_k^{(\nu)}$ and G_k are homogeneous polynomials in the α, which can also be represented as functions of the η.

The quantities η_k, $E_k^{(\nu)}$, G_k remain unchanged when the permutation π^e is applied to their indices, that is, when the index is altered by a multiple of e; and we can therefore now complete Theorems 2. and 3. as follows:

4. If ε is an arbitrary eth root of unity and e a divisor of m, then, when the permutation π is applied to the indices of α, the indices k of the coefficients $E_k^{(\nu)}$ of $(\varepsilon,\eta)^{\nu}$ undergo the permutation γ^{ν}, and the indices of the coefficients G of the product $(\varepsilon,\eta)^{\nu}(\varepsilon^{\lambda_1},\eta)^{\nu_1}(\varepsilon^{\lambda_2},\eta)^{\nu_2}\dots$ the permutation $\gamma^{\nu+\lambda_1\nu_1+\lambda_2\nu_2\dots}$

§. 165.
Solution of the Cyclic Equations.

The Lagrange resolvents, by application of the theorems now proved, lead to the solution of the cyclic equations — more precisely, to the reduction to pure equations.

We now understand by the α no longer arbitrary variables, but the roots of a cyclic equation, so that the cyclic functions of the α are to be regarded as known quantities.

By the theorem of §. 164, 2., the coefficients of $(\varepsilon, \alpha)^m$ are cyclic functions of the α.

If we understand by $a_0, a_1 \dots a_{m-1}$ quantities in Ω and set

$$\psi_\lambda = a_0 + a_1\varepsilon^\lambda + a_2\varepsilon^{2\lambda} + \cdots + a_{m-1}\varepsilon^{(m-1)\lambda}, \tag{1}$$

then from this theorem it follows that

$$(\varepsilon^\lambda, \alpha) = \sqrt[m]{\psi_\lambda}. \tag{2}$$

If ε denotes therein a primitive mth root of unity, then all the resolvents are contained in the form $(\varepsilon^\lambda, \alpha)$.

If we observe further that $(1, \alpha) = a$, as the sum of the roots, belongs to the known quantities, then by §. 164 (3) we have

$$m\alpha = a + \sqrt[m]{\psi_1} + \sqrt[m]{\psi_2} + \cdots + \sqrt[m]{\psi_{m-1}}, \tag{3}$$

and with this α is expressed by radicals of degree m, which under the root signs contain, besides the quantities that occur in Ω from the outset, also mth roots of unity.

Each of these radicals has, considered by itself, m different values, which differ from one another by mth roots of unity as factors. If we give each mth root all its values, we obtain from (3) many different values of α, among which, by §. 164, (4), all the roots $\alpha, \alpha_1, \alpha_2 \dots \alpha_{m-1}$ occur. But the number of the expressions thus derived from (3) is much greater, and it remains to distinguish those which are to be retained from those which are to be discarded. The following way leads to this most simply.

If we apply the theorem of §. 164, 3. to only two factors, it follows that

$$(\varepsilon, \alpha)^\nu (\varepsilon^\lambda, \alpha)^u$$

is a function in $\Omega(\varepsilon)$ when $\nu + \lambda\mu \equiv 0 \pmod m$. If we therefore set $\mu = 1$, $\nu = m - \lambda$, it follows, if

$$\chi_\lambda = b_0^{(\lambda)} + b_1^{(\lambda)}\varepsilon + \cdots + b_{m-1}^{(\lambda)}\varepsilon^{m-1}$$

denotes a quantity in $\Omega(\varepsilon)$, that

$$(\varepsilon, \alpha)^{m-\lambda}(\varepsilon^\lambda, \alpha) = \chi_\lambda,$$

and hence by (2)

$$\sqrt[m]{\psi_\lambda} = \frac{\chi_\lambda}{\left(\sqrt[m]{\psi_1}\right)^{m-\lambda}} = \frac{\chi_\lambda \left(\sqrt[m]{\psi_1}\right)^\lambda}{\psi_1}, \tag{4}$$

and by this, if ε denotes a fixed primitive mth root of unity, all the radicals occurring in (3) are expressed rationally in terms of one of them, $\sqrt[m]{\psi_1}$.

If one gives this one its m different values, one obtains from (3) precisely the m different values α.

There is only one exceptional case in which this procedure is not applicable, namely when $\psi_1 = 0$. But by a small modification of the procedure we can free ourselves from such an exceptional case. To this we premise the following.

Let p be a prime dividing m, and $m = pn$; as above, let ε be any fixed primitive mth root of unity. Then there is always a λ not divisible by p such that $(\varepsilon^\lambda, \alpha)$ is different from zero. For if we form the difference $\alpha_n - \alpha_0$ according to the formula §. 164, (4), we obtain

$$m(\alpha_n - \alpha_0) = \sum_{0,\, m-1}^{\lambda} (\varepsilon^{-n\lambda} - 1)(\varepsilon^\lambda, \alpha).$$

Now, however, $\varepsilon^{-n\lambda} - 1$ is always $= 0$ whenever λ is divisible by p, and if $(\varepsilon^\lambda, \alpha)$ vanishes in all the other cases, that is, where λ is not divisible by p, then $\alpha_n = \alpha_0$, contrary to the hypothesis that the α are all to be different. There is therefore at least one λ not divisible by p such that $(\varepsilon^\lambda, \alpha)$ is different from zero.

We now decompose m into its prime factors and set

$$m = p_1 p_2 \ldots,$$

in which $p_1, p_2 \ldots$ are powers of different primes. We further set

$$m = p_1 m_1 = p_2 m_2 \ldots,$$

and choose, as is always possible by what has just been proved, λ_1 relatively prime to p_1, λ_2 relatively prime to p_2 etc., so that

$$(\varepsilon^{\lambda_1}, \alpha), \quad (\varepsilon^{\lambda_2}, \alpha) \ldots$$

are different from zero. Then by the Theorem §. 164, 3.:

$$(\varepsilon^\lambda, \alpha)(\varepsilon^{\lambda_1}, \alpha)^{m_1 \nu}(\varepsilon^{\lambda_2}, \alpha)^{m_2 \nu} \ldots = \chi_\lambda \tag{5}$$

is a quantity contained in $\Omega(\varepsilon)$, if

$$\lambda \equiv -\nu(\lambda_1 m_1 + \lambda_2 m_2 + \cdots) \pmod{m}. \tag{6}$$

But $(\varepsilon^{\lambda_1}, \alpha)^{m_1}$ is a root of degree p_1 of a function in $\Omega(\varepsilon)$, and we therefore set

$$(\varepsilon^{\lambda_1}, \alpha)^{m_1} = \sqrt[p_1]{\varphi_1}, \quad (\varepsilon^{\lambda_2}, \alpha) = \sqrt[p_2]{\varphi_2} \ldots, \quad (\varepsilon^\lambda, \alpha) = \sqrt[m]{\psi_\lambda}. \tag{7}$$

Then by (5) we have

$$\sqrt[m]{\psi_\lambda} = \frac{\chi_\lambda}{(\sqrt[p_1]{\varphi_1}\, \sqrt[p_2]{\varphi_2} \ldots)^\nu}. \tag{8}$$

Now $(\lambda_1 m_1 + \lambda_2 m_2 + \cdots)$ is relatively prime to m, since $m_2 \ldots$ are divisible by p_1, while $\lambda_1 m_1$ is relatively prime to p_1; and thus from (6) one obtains for each λ a number ν completely determined with respect to the modulus m. If, then, we substitute the expressions (8) in (3) and assign to the radicals $\sqrt[p_1]{\varphi_1}$, $\sqrt[p_2]{\varphi_2} \ldots$ all their values, we obtain exactly m different values for α, and no more.

The last results we can use to derive a form of the representation of the roots α in a somewhat generalised shape, which Abel communicates in the place cited, and which relates to the case where the field Ω is real, i.e. consists of nothing but real numbers. The functions φ, ψ, χ, as we have used them above, are then composed of real numbers and of the root of unity ε, which one finds by the division of the circumference of the circle into m equal parts; one may, for instance, set

$$\varepsilon = e^{\frac{2\pi i}{m}} = \cos\frac{2\pi}{m} + i \sin\frac{2\pi}{m}$$

The function φ_1 passes over, when ε is changed into ε^{-1}, into the conjugate imaginary value, which we denote by φ_1'.

We shall choose a positive quantity ϱ_1 and an angle Θ_1 in such a way that

$$\varphi_1 = \varrho_1 e^{i\Theta_1}, \qquad \varphi_1' = \varrho_1 e^{-i\Theta_1}, \tag{9}$$

or

$$\begin{aligned} \varphi_1 &= \varrho_1(\cos\Theta_1 + i\sin\Theta_1) \\ \varphi_1' &= \varrho_1(\cos\Theta_1 - i\sin\Theta_1), \end{aligned} \tag{10}$$

from which it further follows that

$$\varrho_1^2 = \varphi_1\varphi_1'. \tag{11}$$

Now

$$(\varepsilon^{\lambda_1}, \alpha)(\varepsilon^{-\lambda_1}, \alpha) = \pm a_1 \tag{12}$$

is a quantity of the field $\Omega(\varepsilon)$ by (§. 164, 3.), and indeed, since it does not change on passing to the conjugate imaginary value, that is, on the interchange of ε and ε^{-1}, it is a real quantity. We shall determine the sign so that a_1 is positive. Hence by (11), since ϱ_1^2 is positive,

$$\varrho_1^2 = (\pm a_1)^m = a_1^m,$$

and from this it follows that for odd m the upper sign holds in any case; for even m the lower can also occur. We then have

$$\varrho_1 = \sqrt{a_1^m}, \tag{13}$$

where the square root is to be taken positive.

Further, for the same reason,

$$\frac{\varphi_1 + \varphi_1'}{2} = b_1, \qquad \frac{\varphi_1 - \varphi_1'}{2i} = c_1 \tag{14}$$

are real quantities in $\Omega(\varepsilon)$, and it follows from (10) that

$$\cos\Theta_1 = \frac{b_1}{\sqrt{a_1^m}}, \qquad \sin\Theta_1 = \frac{c_1}{\sqrt{a_1^m}}, \tag{15}$$

from which the relation further follows:

$$a_1^m = b_1^2 + c_1^2.$$

Accordingly we obtain

$$\sqrt[p_1]{\varphi_1} = \sqrt{a_1^{m_1}}\, e^{\frac{i\Theta_1}{p_1}};$$

and one now proceeds in the same way with the functions φ_2 and so on. One thus determines $\Theta_2 \ldots$ from a system of equations like (15), and, if one further puts

$$m_1\Theta_1 + m_2\Theta_2 + \cdots = \Theta \tag{16}$$

one obtains, by (8)

$$\sqrt[m]{\psi_\lambda} = \chi_\lambda \left(\sqrt{a_1^{m_1} a_2^{m_2} \ldots}\right)^{-\nu} e^{-\frac{i\Theta}{m}\nu}. \tag{17}$$

By (15) we can set

$$e^{i\Theta_1} = \frac{b_1 + ic_1}{\sqrt{a_1^m}}, \qquad e^{i\Theta_2} = \frac{b_2 + ic_2}{\sqrt{a_2^m}} \cdots,$$

and if, then, by decomposition into the real and imaginary part we obtain

$$(b_1 + ic_1)^{m_1}(b_2 + ic_2)^{m_2} \cdots = B + iC \tag{18}$$

and set

$$a_1^{m_1} a_2^{m_2} \cdots = A \tag{19}$$

it follows by (16) that

$$e^{i\Theta} = \frac{B + iC}{\sqrt{A^m}},$$

hence

$$\sqrt{A^m}\cos\Theta = B, \qquad \sqrt{A^m}\sin\Theta = C, \tag{20}$$

and

$$\sqrt[m]{\psi_\lambda} = \chi_\lambda \sqrt{A}^{-\nu}\left(\cos\frac{\Theta\nu}{m} - i\sin\frac{\Theta\nu}{m}\right). \tag{21}$$

The angle Θ is determined by (20) only up to a multiple of 2π, and if one therefore puts $\Theta + 2h\pi$ for Θ and lets h run from 0 to $m-1$, one obtains from (21) the m different values of the root quantity. The functions $\cos\frac{\Theta\nu}{m}$ and $\sin\frac{\Theta\nu}{m}$ can further be expressed rationally in terms of $\cos\frac{\Theta}{m}$, $\sin\frac{\Theta}{m}$. The solution of the cyclic equations in the real field Ω is thus reduced to the following.

One first adjoins to the field Ω the mth root of unity ε (division of the circumference of the circle into m equal parts). Thereupon A, B, C are known. One further adjoins the positive square root $\sqrt{A}$; then $\cos\Theta$ and $\sin\Theta$ are known by (20). Finally one adjoins $\cos\frac{\Theta}{m}$, $\sin\frac{\Theta}{m}$ (division of the angle Θ into m parts). Then the cyclic equation is solved by (21).

These considerations lead further to an interesting result on the reality relations of the roots of cyclic equations.

If, as in §. 163, we represent each root in the sequence of roots $\alpha, \alpha_1, \ldots \alpha_{m-1}$ rationally by the preceding one:

$$\alpha_1 = \Theta(\alpha), \quad \alpha_2 = \Theta(\alpha_1), \ldots \alpha_{m-1} = \Theta(\alpha_{m-2}), \quad \alpha = \Theta(\alpha_{m-1}),$$

in which, since Ω is here assumed to be real, $\Theta(x)$ denotes a real rational function of x, then it follows first that if one of the roots is real, all the others must be real as well. This is always the case for odd m, since a real equation of odd degree must always have at least one real root.

For even m imaginary roots may also be present, and if one root is imaginary, then all must be so, since, if a real root occurs, all the others are also real. If we denote by $\Theta^\nu(x)$ the ν-fold repetition of the function Θ, then

$$\alpha_{\nu+k} = \Theta^\nu(\alpha^k).$$

If then $\alpha = \alpha_0$ is conjugate imaginary with α_k, then for every ν the functions $\Theta^\nu(\alpha_0)$ and $\Theta^\nu(\alpha_k)$, that is, α_ν and $\alpha_{\nu+k}$, are also conjugate imaginary.

From this it follows that $2k = m$ must hold, and we conclude that in the case of imaginary roots

$$\alpha_k \quad \text{and} \quad \alpha_{k+\frac{m}{2}}$$

form, for every index k, a pair of conjugate imaginary roots.

The cubic equations become cyclic equations by the adjunction of the square root of the discriminant. When the discriminant is positive, the roots are, in agreement with this theorem, real.

§. 166.
Division of the angle.

Cyclic equations also include those on which the division of an angle into m equal parts depends; to these we have reduced the general cyclic equations in a real field.

The problem can be formulated thus:

When $\cos m\varphi$ and $\sin m\varphi$ are given, then $\cos\varphi$ and $\sin\varphi$ are to be found from them.

We suppose that the given values of $\cos m\varphi$ and $\sin m\varphi$ are less than 1 and that the sum of their squares is $= 1$.

We further adjoin mth roots of unity, with which we shall occupy ourselves more thoroughly in the next section, but which in any case depend on equations whose degree is lower than m. In order, however, to keep the field real, we shall adjoin, instead of the roots of unity themselves, the two real quantities

$$\sin\frac{2\pi}{m}, \qquad \cos\frac{2\pi}{m} \tag{1}$$

Let the field Ω, then, consist of all rational numbers and of the rational functions of $\cos m\varphi$, $\sin m\varphi$, $\cos\frac{2\pi}{m}$, $\sin\frac{2\pi}{m}$.

The equation of degree m on which $x = 2\cos\varphi$ depends we have set up in §. 137 in the form

$$2\cos m\varphi = A_m(x), \tag{2}$$

where $A_m(x)$ is a polynomial of degree m in x; but there we have also found the equation

$$\sin\varphi = \frac{\sin m\varphi}{B_m(x)} \tag{3}$$

by which $\sin\varphi$ is expressed rationally in terms of x. In these formulae the roots of unity are not yet contained.

The m roots of (2) now have the following meaning:

$$x_0 = 2\cos\varphi, \qquad x_1 = 2\cos\left(\varphi + \frac{2\pi}{m}\right)\ldots,$$

$$x_{m-1} = 2\cos\left(\varphi + \frac{2(m-1)\pi}{m}\right),$$

and with the aid of (2) and (3), and upon adjunction of the quantities (1), each of them can be represented as a rational function of another, namely thus:

$$x_1 = f(x_0), \qquad x_2 = f(x_1)\ldots, \qquad x_0 = f(x_{m-1}).$$

Any function of the $x_0, x_1 \ldots x_{m-1}$ can therefore be represented as a rational function $F(x_0)$; under a cyclic permutation the following transformations occur:

$$F(x_0) \text{ into } F(x_1),\ F(x_1) \text{ into } F(x_2)\ldots F(x_{m-1}) \text{ into } F(x_0)$$

For a cyclic function, therefore,

$$F(x_0) = F(x_1)\cdots = F(x_{m-1}) = \frac{1}{m}\left[F(x_0) + F(x_1)\cdots + F(x_{m-1})\right],$$

and consequently $F(x_0)$ is rational, since it is represented as a symmetric function of the roots.

The division of the angle thus depends on a cyclic equation.

This cyclic equation is irreducible; for if in any rational equation $\Phi(x_0, \cos m\varphi, \sin m\varphi) = 0$ one replaces φ by $\varphi + \frac{2\pi k}{m}$, it goes over into

$$\Phi(x_k, \cos m\varphi, \sin m\varphi) = 0$$

and is therefore satisfied for all roots of (2).

If we wished to adjoin only $\cos m\varphi$, and not at the same time $\sin m\varphi$, then the equation would no longer be a cyclic one, which one easily confirms by the example of the trisection, where precisely $\sin m\varphi$ becomes the square root of the discriminant (§. 112).

Sixteenth Section.
Division of the Circle.

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§. 167.
The cyclotomic periods and the period equations.

The most important among the Abelian equations are those on which the determination of the roots of unity depends, whose elementary properties we have already come to know in the twelfth section. On account of the relation to the geometric problem of dividing the circumference of the circle into a definite number of equal parts, all the equations connected with this are also called the cyclotomic equations.

The field Ω, which contains the quantities regarded as known, is here simply the field of rational numbers, which we shall call the field R.

We shall first occupy ourselves with the nth roots of unity under the assumption that n is an odd prime (the prime 2 gives rise to no further questions, since the only second roots of unity, ± 1, are contained in R).

Besides the rational nth root of unity 1 there exist $n-1$ more primitive ones, which can be represented by the transcendental expressions

$$e^{\frac{2\pi i}{n}}, \quad e^{\frac{4\pi i}{n}}, \quad e^{\frac{6\pi i}{n}} \ldots e^{\frac{2(n-1)\pi i}{n}}$$

If we denote one of them by r, then the whole system is also represented by the following list:

$$r,\ r^2,\ r^3 \ldots r^{n-1} \tag{1}$$

In the exponent of r only the residue with respect to the modulus n matters, since $r^h = r^k$ holds always and only when $h \equiv k \pmod{n}$.

As we have already seen in the twelfth section, the quantities (1) are the roots of an irreducible equation of degree $(n-1)$

$$X = x^{n-1} + x^{n-2} + x^{n-3} + \cdots + x + 1 = 0, \tag{2}$$

and we wish first to show that this equation is cyclic. This property results from the existence of primitive roots of the prime n, with which we occupied ourselves in §. 136. It was proved there that for every prime n there are certain numbers g, called primitive roots of n, which are characterised by the property that among the residues of the powers

$$1,\ g,\ g^2,\ g^3 \ldots g^{n-2},$$

upon division by n, each of the numbers

$$0,\ 1,\ 2, \ldots n-1$$

occurs once and only once. The residue of a power g^h remains the same when h is changed by a multiple of $n-1$. If

$$g^\alpha \equiv a \pmod{n},$$

then α is called the index of a ($\alpha = \operatorname{ind} a$) and a the number or the numerus. The index is taken with respect to the modulus $n-1$, the numerus with respect to the modulus n (§. 136).

If, then, we assume such a primitive root of n, we can, apart from the order, represent the quantities (1) thus:

$$r,\ r^g,\ r^{g^2} \dots r^{g^{n-2}},$$

or, if we set

$$r^{g^h} = r_h \tag{3}$$

then

$$r,\ r_1,\ r_2 \dots r_{n-2}. \tag{4}$$

Every number of the field $R(r)$, i. e. every rational function of r, can be brought into the form

$$\varphi(r) = b_0 + b_1 r + b_2 r^2 + \cdots + b_{n-2} r^{n-2}, \tag{5}$$

or, since by (2)

$$1 + r + r^2 + \cdots + r^{n-1} = 0$$

holds,

$$\varphi(r) = (b_1 - b_0)\, r + (b_2 - b_0)\, r^2 + \cdots + (b_{n-2} - b_0)\, r^{n-2} - b_0 r^{n-1}; \tag{6}$$

or, since the powers $r, r^2, \dots r^{n-1}$, apart from their order, agree with $r, r_1, r_2 \dots r_{n-2}$, into the form

$$\varphi(r) = a\, r + a_1 r_1 + a_2 r_2 + \cdots + a_{n-2} r_{n-2}, \tag{7}$$

in which the coefficients b and a are rational numbers.

I. Such a function $\varphi(r)$ can be equal to zero only when all its coefficients $a, a_1 \dots a_{n-2}$ vanish;

for the expression (5) shows that $\varphi(r)$ cannot vanish otherwise than when the $b_0, b_1 \dots b_{n-2}$ are zero. But if these are zero, then (6) shows that the coefficients a must also all be zero. Hence $\varphi(r)$ can be brought into the form (7) in only one way.

If we make the stipulation that a_h and r_h shall not change when the index h increases by a multiple of $n-1$, we can also set

$$\varphi(r) = \sum^{h} a_h r_h \tag{8}$$

and let h run through a complete system of residues with respect to the modulus $n-1$.

We now have to prove that the group of the equation $X = 0$ is no other than the period of the cyclic permutation

$$\pi = (r, r_1, r_2 \dots r_{n-2}).$$

This period or cyclic group we denote by

(9) $$C = 1, \pi, \pi^2 \dots \pi^{n-2}.$$

The equation $X = 0$ is in any case a normal equation, because it is irreducible, and because all its roots are, by (3), rationally expressible in terms of one among them; it is therefore its own Galois resolvent (§. 145). Its substitutions are

(10) $$(r, r), (r, r_1), (r, r_2) \dots (r, r_{n-2}),$$

and by (3) we have

$$(r, r_h) = (r_k, r_{k+h}).$$

From this it follows that (r, r_1) produces among the roots (4) the cyclic permutation π, and that therefore the substitution group

(10) is isomorphic with the permutation group C. Hence C is the Galois group of $X = 0$.

By §. 163 the equation $X = 0$ can be reduced to a sequence of cyclic equations of lower degree, if one seeks functions that belong to the divisors of the group C. But the divisors of C, if $n - 1$ is decomposed into two factors e, f, so that

$$n - 1 = ef$$

is set, are the cyclic groups

(11) $$C_e = 1, \pi^e, \pi^{2e} \dots \pi^{(f-1)e},$$

and the permutation π^e is therein equivalent to the substitution (r, r_e).

In order to arrive at a function of the roots belonging to C_e of the simplest possible form, we consider, following Gauss, the f-term periods

(12)
$$\begin{aligned} \eta &= r + r_e + r_{2e} + \dots + r_{(f-1)e} \\ \eta_1 &= r_1 + r_{e+1} + r_{2e+1} + \dots + r_{(f-1)e+1} \\ &\dots\dots\dots\dots\dots\dots\dots \\ \eta_{e-1} &= r_{e-1} + r_{2e-1} + r_{3e-1} + \dots + r_{n-2}, \end{aligned}$$

which arise from the first of them by the substitutions $(r, r), (r, r_1), (r, r_2) \dots (r, r_{e-1})$. We also designate the quantities (12) as a system of conjugate periods.

Each of these functions remains unchanged by the substitution (r, r_e), and they are all different from one another, since an equation $\eta_h - \eta_k = 0$, if h is different from k, could be written in the form

$$ar + a_1 r_1 + \dots + a_{n-2} r_{n-2} = 0$$

where the $a, a_1 \dots a_{n-2}$ are integers not all vanishing at once (+1, 0 or −1), which by Theorem I is not possible. Each of the periods (12) is therefore a function belonging to the group C_e, and these quantities are therefore the roots of an irreducible equation of degree e with integer coefficients. If one adjoins one of these quantities, the determination of r still depends on an equation of degree f.

By the general theorems of the fourteenth section, every function of the r that admits the substitutions of the group C_e can be represented rationally by any one of the quantities η. But here we can also set up the following theorem:

II. Every number in $R(r)$ which admits the substitution (r, r_e), and hence also every number of the field $R(\eta)$, can be represented as a homogeneous linear function of the periods $\eta, \eta_1 \dots \eta_{e-1}$.

For by I. we can put every number of $R(r)$ in one way into the form

$$\varphi(r) = \sum a_h r_h \tag{13}$$

where the coefficients a_h are rational numbers, and h runs through a complete system of residues with respect to the modulus $n-1$. But we have

$$\varphi(r_e) = \sum a_h r_{h+e} = \sum a_{h-e} r_h,$$

and therefore, if $\varphi(r) = \varphi(r_e)$ is to hold, we must have $a_{h-e} = a_h$ for every h; hence also $a_h = a_{h+e} = a_{h+2e} \dots$ Accordingly (13) can be put into the form

$$\varphi(r) = \sum_{0,\,e-1}^{h} a_h r_h + \sum_{0,\,e-1}^{h} a_h r_{h+e} + \cdots + \sum_{0,\,e-1}^{h} a_h r_{h+(f-1)e},$$

or

$$\varphi(r) = a\eta + a_1\eta_1 + \cdots + a_{e-1}\eta_{e-1}, \tag{14}$$

which was to be proved. It is further to be remarked here that, if φ has integer coefficients, the coefficients of the η_h also become integers.

Since every number of the field $R(\eta)$ can be represented in one and only one way in the linear form (14), we call the system of quantities $\eta, \eta_1, \eta_2 \dots \eta_{e-1}$ a basis of the field $R(\eta)$.

If one knows the representation of the product of each two periods $\eta_h \eta_k$ in the form (14), one can from it, by repeated application, represent all rational functions of the periods η in the same form. Likewise one can also easily form the equation whose roots are the e quantities η, if one applies the procedure to the product

$$F_e(x) = (x-\eta)(x-\eta_1)\cdots(x-\eta_{e-1}) \tag{15}$$

This equation can also be represented in determinant form.

Namely, let us put

$$\eta\eta_h = a_{0,h}\eta + a_{1,h}\eta_1 + \cdots + a_{e-1,h}\eta_{e-1}, \tag{16}$$

in which the $a_{0,h}, a_{1,h} \dots a_{e-1,h}$ are integers, about whose calculation further details will follow later. If one sets up this equation for $h = 0, 1, \dots e-1$ in the form:

$$\begin{aligned}
(a_{0,0}-\eta)\,\eta + a_{1,0}\eta_1 + \cdots \quad & a_{e-1,0}\eta_{e-1} = 0 \\
a_{0,1}\eta + (a_{1,1}-\eta)\eta_1 + \cdots \quad & a_{e-1,1}\eta_{e-1} = 0 \\
\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots & \\
a_{0,e-1} + a_{1,e-1}\eta_1 + \cdots (a_{e-1,e-1}-\eta)\eta_{e-1} & = 0,
\end{aligned}$$

one sees, since the $\eta, \eta_1 \dots \eta_{e-1}$ do not vanish, that the determinant of this system must be zero, hence

$$\begin{vmatrix}
a_{0,0}-\eta, & a_{1,0}, & \dots & a_{e-1,0} \\
a_{0,1}, & a_{1,1}-\eta, & \dots & a_{e-1,1} \\
\dots & \dots & \dots & \dots \\
a_{0,e-1}, & a_{1,e-1}, & \dots & a_{e-1,e-1}-\eta
\end{vmatrix} = 0, \tag{17}$$

which, if one writes x for η, must agree, apart from the sign, with $F_e(x)$.

The function $F_e(x)$ therefore has not merely rational, but integer coefficients.

One can now also form the equation of degree f whose roots are $r, r_e, r_{2e} \dots r_{(f-1)e}$; for, for an indeterminate r,

$$\Phi_e(x) = (x - r)(x - r_e) \cdots (x - r_{(f-1)e}) \tag{18}$$

is a function in the field $R(\eta)$. Its coefficients can easily be computed from the Newton formulae, by which the coefficients of an equation are expressed by means of the power sums of the roots (§. 42). For by (12) and (3) we have

$$r^{g^h} + r_e^{g^h} + \cdots + r_{(f-1)e}^{g^h} = \eta_h,$$

and this is thus directly equal to one of the power sums, and every power sum is equal to one of the periods.

But the equation (18) can be treated in just the same way as the equation (2); for the group of $\Phi_e(x) = 0$ is precisely C_e, and if e' is a divisor of f, say $f = e'f'$, then $C_{ee'}$ is a divisor of C_e. But to $C_{ee'}$ belongs the period

$$\eta' = r + r_{ee'} + r_{2ee'} + \cdots + r_{(f'-1)ee'},$$

which therefore depends on an equation of degree e' in the field $R(\eta)$.

If one takes the numbers $e, e' \dots$ to be primes, then the sequence of resolvents consists of a sequence of equations whose degrees are the prime factors of $n - 1$.

As for the group of the equation of degree e whose roots are the e quantities η, we find this very simply

from the remark that the substitution (r, r_h) calls forth among the η_k the substitution (η_k, η_{k+h}), when the index of η is taken modulo e. The group of $F_e(x) = 0$ therefore consists of the powers of the cyclic permutation

$$(\eta, \eta_1, \eta_2 \dots \eta_{e-1}).$$

§. 168.
The Gauss Method for the Computation of the Resolvents.

In order actually to carry out the solution of the cyclotomic equation in concrete cases, all that matters, after what we have developed in the preceding section, is to represent the products of any two conjugate periods as linear functions of these same periods. For this Gauss has given a simple procedure, which we shall now become acquainted with. To this end we must alter the notation of the periods a little.

We write any two of the periods §. 167, (12) in the form

$$\begin{aligned} \eta^{(\lambda)} &= r^{\lambda} + r^{\lambda'} + r^{\lambda''} + \cdots \\ \eta^{(\mu)} &= r^{\mu} + r^{\mu'} + r^{\mu''} + \cdots \end{aligned} \tag{1}$$

so that, if $\lambda \equiv g^h \pmod{n}$, then $\eta^{(\lambda)} = \eta_h$ or also $\eta^{(\lambda)} = \eta_{\operatorname{ind}\lambda}$; we then have

$$\begin{aligned} \lambda' &\equiv \lambda g^e, \quad \lambda'' \equiv \lambda g^{2e} \cdots \\ \mu' &\equiv \mu g^e, \quad \mu'' \equiv \mu g^{2e} \cdots \end{aligned} \pmod{n}; \tag{2}$$

if, then, we let the symbols s and t each run through a complete system of residues modulo f, we have

$$\eta^{(\lambda)} = \sum^{s} r^{\lambda g^{se}}, \qquad \eta^{(\mu)} = \sum^{t} r^{\mu g^{te}}.$$

The product of these is

$$\eta^{(\lambda)}\eta^{(\mu)} = \sum^{s}\sum^{t} r^{\lambda g^{se} + \mu g^{te}}.$$

If, in forming this sum, we first hold s fixed, and sum with respect to t, we may replace t by $t+s$, because both run simultaneously through a complete system of residues modulo f. Accordingly we obtain

$$\eta^{(\lambda)}\eta^{(\mu)} = \sum^{s}\sum^{t} r^{(\lambda+\mu g^{te})g^{se}}.$$

In this, however, the order of summation may be interchanged, so that we also have

$$\eta^{(\lambda)}\eta^{(\mu)} = \sum^{t}\sum^{s} r^{(\lambda+\mu g^{te})g^{se}} \tag{3}$$

The sum taken with respect to s,

$$\sum^{s} r^{(\lambda+\mu g^{te})g^{se}}$$

is, however, itself one of the periods, namely, according to the notation (1), the period

$$\eta^{(\lambda+\mu g^{te})}.$$

From (2) and (3) it therefore follows that

$$\eta^{(\lambda)}\eta^{(\mu)} = \eta^{(\lambda+\mu)} + \eta^{(\lambda+\mu')} + \eta^{(\lambda+\mu'')} + \cdots \tag{4}$$

The right-hand side of this expression contains f terms; among them the same period may of course occur several times; there may also occur among them the improper period $\eta^{(0)}$, which is to be set equal to the integer f. If one wishes to restore the homogeneous form of the expression, one makes use of the relation

$$\eta + \eta_1 + \eta_2 + \cdots + \eta_{e-1} = -1,$$

which is indeed only another way of writing the equation §. 167, (2).

When one has computed according to (4) the expression for a product $\eta\eta_h$, to which we have already given the form above,

$$\eta\eta_h = a_{0,h}\eta + a_{1,h}\eta_1 + \cdots + a_{e-1,h}\eta_{e-1}, \tag{5}$$

one obtains the product of any two of the periods by the substitution (η, η_k),

$$\eta_k\eta_{h+k} = a_{0,h}\eta_k + a_{1,h}\eta_{k+1} + \cdots + a_{e-1,h}\eta_{k-1}. \tag{6}$$

In order to illustrate these rules by a simple example, we take $n = 13$. For 13, 2 is a primitive root, and we can apply the index table given in §. 136:

(7)
$$\begin{array}{c|cccccccccccc} I & 0, & 1, & 2, & 3, & 4, & 5, & 6, & 7, & 8, & 9, & 10, & 11 \\ \hline N & 1, & 2, & 4, & 8, & 3, & 6, & 12, & 11, & 9, & 5, & 10, & 7 \end{array}.$$

If we first assume $e = 3$, $f = 4$, we obtain a cubic resolvent whose roots are

(8)
$$\begin{aligned} \eta &= r + r_3 + r_6 + r_9 \\ \eta_1 &= r_1 + r_4 + r_7 + r_{10} \\ \eta_2 &= r_2 + r_5 + r_8 + r_{11} \end{aligned}$$

From the table (7) this also gives

(9)
$$\begin{aligned} \eta &= r + r^{-5} + r^{-1} + r^5 = \eta^{(1)} \\ \eta_1 &= r^2 + r^3 + r^{-2} + r^{-3} = \eta^{(2)} \\ \eta_2 &= r^4 + r^6 + r^{-4} + r^{-6} = \eta^{(4)} \end{aligned}$$

If one applies formula (4), one obtains, e.g.,

$$\begin{aligned} \eta\eta &= \eta^{(2)} + \eta^{(-4)} + \eta^{(0)} + \eta^{(6)} = 4 + \eta^{(2)} + 2\eta^{(4)} \\ &= -4\eta - 3\eta_1 - 2\eta_2, \end{aligned}$$

and thus the formulae result

$$\begin{aligned} \eta^2 &= -4\eta - 3\eta_1 - 2\eta_2, \\ \eta\eta_1 &= \eta + 2\eta_1 + \eta_2, \\ \eta\eta_2 &= 2\eta + \eta_1 + \eta_2, \end{aligned}$$

from which, by §. 167, (17), the equation for η

$$\begin{vmatrix} -4 - \eta, & -3, & -2 \\ 1, & 2 - \eta, & 1 \\ 2, & 1, & 1 - \eta \end{vmatrix} = 0,$$

or

(10)
$$\eta^3 + \eta^2 - 4\eta + 1 = 0$$

follows. The discriminant of this equation is found to be equal to $169 = 13^2$, hence positive. The equation consequently has three real roots, and indeed, since the last term is positive, two positive and one negative root.

If we set $r = e^{\frac{2\pi i}{13}}$, then

$$\begin{aligned} \eta &= 2\cos\frac{2\pi}{13} + 2\cos\frac{10\pi}{13} = 2\left(\cos\frac{2\pi}{13} - \cos\frac{3\pi}{13}\right), \\ \eta_1 &= 2\cos\frac{4\pi}{13} + 2\cos\frac{6\pi}{13} = 2\left(\cos\frac{4\pi}{13} + \cos\frac{6\pi}{13}\right), \\ \eta_2 &= 2\cos\frac{8\pi}{13} + 2\cos\frac{12\pi}{13} = -2\left(\cos\frac{\pi}{13} + \cos\frac{5\pi}{13}\right), \end{aligned}$$

or also

$$\eta = 4\cos\frac{4\pi}{13}\cos\frac{6\pi}{13}, \qquad \eta_1 = 4\cos\frac{\pi}{13}\cos\frac{5\pi}{13},$$
$$\eta_2 = -4\cos\frac{2\pi}{13}\cos\frac{3\pi}{13}.$$

Thus η_2 is the negative root, η_1 the greater, and η the smaller of the two positive roots.

The equation of degree 4, whose roots are r, r^{-1}, r^5, r^{-5}, can now easily be formed, if one

$$\xi = r + r^{-1} = 2\cos\frac{2\pi}{13}, \qquad \xi' = r^5 + r^{-5} = 2\cos\frac{10\pi}{13}$$

sets. For we have

$$\xi + \xi' = \eta, \qquad \xi\xi' = \eta_2,$$

hence

$$\xi^2 - \eta\xi + \eta_2 = 0 \tag{11}$$

is the quadratic equation for ξ, from which one obtains the biquadratic equation for r,

$$r^4 - \eta r^3 + (\eta_2 + 2)r^2 - \eta r + 1 = 0. \tag{12}$$

The quadratic equation (11) has one positive and one negative root. The positive root is $2\cos\frac{2\pi}{13}$. Instead of computing r itself, one computes further $2\sin\frac{2\pi}{13}$, which one obtains as the positive square root $\sqrt{4-\xi^2}$.

In place of the three four-term periods η one could also have first considered two periods of six terms.

$$\begin{aligned} \xi &= r + r^3 + r^4 + r^{-1} + r^{-3} + r^{-4} \\ \xi_1 &= r^2 + r^5 + r^6 + r^{-2} + r^{-5} + r^{-6}. \end{aligned} \tag{13}$$

For the product $\xi\xi_1$ one finds by formula (4) the value $3(\xi + \xi_1)$ or -3, so that ξ, ξ_1 are the roots of the quadratic equation

$$\xi^2 + \xi - 3 = 0 \tag{14}$$

from which, since it is easy to see from the expression by the cosines that ξ is positive:

$$\xi = \frac{-1+\sqrt{13}}{2}, \qquad \xi_1 = \frac{-1-\sqrt{13}}{2}. \tag{15}$$

After adjunction of the values ξ, ξ_1 one can form a cubic equation for the two-term periods. For if one sets

$$\begin{aligned} \xi &= r + r^{-1}, \quad & \xi_1 &= r^2 + r^{-2}, \\ \xi_2 &= r^4 + r^{-4}, \quad & \xi_3 &= r^5 + r^{-5}, \\ \xi_4 &= r^3 + r^{-3}, \quad & \xi_5 &= r^6 + r^{-6}, \end{aligned}$$

then it follows that

$$\zeta = \xi + \xi_2 + \xi_4, \qquad \zeta_1 = \xi_1 + \xi_3 + \xi_5, \qquad \zeta + \zeta_1 = -1,$$

further

$$\xi\xi_2 = \xi_3 + \xi_4, \qquad \xi\xi_4 = \xi_1 + \xi_2, \qquad \xi_2\xi_4 = \xi + \xi_5,$$

$$\xi\xi_2\xi_4 = 2 + \zeta_1 = 1 - \zeta,$$

hence ξ, ξ_2, ξ_4 are the roots of the cubic equation

$$x^3 - \zeta x^2 - x - 1 + \zeta = 0. \tag{16}$$

§. 169.
Reduction of the cyclotomic equation to pure equations.

Gauss, already in his first exposition in the disq. arithm., reduced the cyclotomic equations directly to pure equations by the use of the resolvents[41]. To carry this out, we set

$$n - 1 = m,$$

and denote, as in the preceding section, by ε a primitive mth root of unity, and by $r, r_1, r_2 \ldots r_{m-1}$ the nth roots of unity in the order fixed above. The Lagrange resolvents are then

$$(\varepsilon^\lambda, r) = \sum_{0,\,n-2}^{h} \varepsilon^{\lambda h} r_h = \sum_{0,\,n-2}^{h} \varepsilon^{\lambda h} r^{g^h},$$

in which λ is an arbitrary exponent taken with respect to the modulus m. If we set $g^h = s$, so that $h \equiv \operatorname{ind} s \pmod m$, we may also write

$$(\varepsilon^\lambda, r) = \sum_{1,\,n-1}^{s} \varepsilon^{\lambda \operatorname{ind} s} r^s. \tag{1}$$

The basis for the further reduction of these expressions is the computation of the product of two such resolvents.

We denote by μ a second number like λ, and understand by s and t two symbols which, independently of one another, each run through a system of residues with respect to the modulus n, with the exclusion of zero. Then

$$(\varepsilon^\lambda, r)(\varepsilon^\mu, r) = \sum^{s}\sum^{t} r^{(s+t)} \varepsilon^{\lambda \operatorname{ind} s + \mu \operatorname{ind} t}.$$

If the summation with respect to t is carried out first, then st may be put in place of t, since s is different from zero, and hence st and t run at the same time through a complete system of residues with respect to the modulus n. Thus

$$(\varepsilon^\lambda, r)(\varepsilon^\mu, r) = \sum^{s}\sum^{t} r^{s(t+1)} \varepsilon^{(\lambda+\mu) \operatorname{ind} s} \varepsilon^{\mu \operatorname{ind} t}. \tag{2}$$

[41] Gauss, disq. arithm. art. 359, 360; disq. circa aequationes puras ulterior evolutio, Werke Bd. II. Lagrange, rés. des équations numériques. Jacobi, "Ueber die Kreistheilung und ihre Anwendung in der Zahlentheorie". Werke Bd. 6. Kummer, Crelle's Journal, Bd. 35 and Abhandl. d. Berl. Akademie 1856. Eisenstein and Cauchy are also to be mentioned. The theory of the division of the circle is presented in connected form, accompanied by the historical references, in the book by Bachmann, "Die Lehre von der Kreistheilung". Leipzig 1872.

We first dispose of the special case $\mu = -\lambda$, which we obtain from (2) when we carry out the summation with respect to s first:

$$(\varepsilon^{\lambda}, r)(\varepsilon^{-\lambda}, r) = \sum^{t} \varepsilon^{-\lambda \operatorname{ind} t} \sum^{s} r^{s(t+1)}.$$

Herein

$$\sum^{s} r^{s(t+1)} = -1, \quad \text{if } t = 1, 2 \ldots n-2$$
$$= n-1, \quad \text{if } t = n-1,$$

hence, since by §. 136 $\operatorname{ind}(n-1) = \frac{1}{2}(n-1)$ and

$$\varepsilon^{\frac{n-1}{2}} = -1$$

holds,

$$(\varepsilon^{\lambda}, r)(\varepsilon^{-\lambda}, r) = -\sum_{1, n-1}^{t} \varepsilon^{-\lambda \operatorname{ind} t} + (-1)^{\lambda} n.$$

In the sum taken with respect to t, $\operatorname{ind} t$ runs through a complete system of residues with respect to the modulus $n-1$, and hence, if λ is assumed not divisible by $n-1$,

$$\sum_{1, n-1}^{t} \varepsilon^{-\lambda \operatorname{ind} t} = 0,$$

and therefore

$$(\varepsilon^{\lambda}, r)(\varepsilon^{-\lambda}, r) = (-1)^{\lambda} n, \tag{3}$$

while, if λ is divisible by $n-1$, $(\varepsilon^{\lambda}, r) = (1, r) = -1$.

We now, then, treat the sum (2) further under the assumption that $\lambda + \mu$ is not divisible by $(n-1)$. Then $\sum^{s} \varepsilon^{(\lambda+\mu) \operatorname{ind} s} = 0$, and in the sum on the right-hand side of (2) the terms corresponding to the value $t = n-1$ may be omitted. One thus obtains

$$\begin{aligned}(\varepsilon^{\lambda}, r)(\varepsilon^{\mu}, r) &= \sum_{1, n-2}^{t} \varepsilon^{\mu \operatorname{ind} t} \sum_{1, n-1}^{s} \varepsilon^{(\lambda+\mu) \operatorname{ind} s} r^{s(t+1)} \\ &= \sum_{1, n-2}^{t} \varepsilon^{\mu \operatorname{ind} t - (\lambda+\mu) \operatorname{ind}(t+1)} \sum^{s} \varepsilon^{(\lambda+\mu) \operatorname{ind} s(t+1)} r^{s(t+1)}.\end{aligned} \tag{4}$$

Now, for fixed t, $s(t+1)$ runs at the same time as s through a complete system of residues with respect to the modulus n, and hence

$$\sum^{s} \varepsilon^{(\lambda+\mu) \operatorname{ind} s(t+1)} r^{s(t+1)} = \sum^{s} \varepsilon^{(\lambda+\mu) \operatorname{ind} s} r^{s} = (\varepsilon^{\lambda+\mu}, r),$$

and formula (4) yields

$$\frac{(\varepsilon^{\lambda}, r)(\varepsilon^{\mu}, r)}{(\varepsilon^{\lambda+\mu}, r)} = \psi_{\lambda,\mu}(\varepsilon), \tag{5}$$

if for abbreviation

$$\sum_{1, n-2}^{t} \varepsilon^{\mu \operatorname{ind} t - (\lambda+\mu) \operatorname{ind}(t+1)} = \psi_{\lambda,\mu}(\varepsilon) \tag{6}$$

is set, and $\lambda + \mu$ is assumed not divisible by $n - 1 = m$. These functions ψ are then composed of the $(n-1)$th roots of unity ε with integral coefficients, and are therefore known if the $(n-1)$th roots of unity are assumed to be known. They are numbers of the field $R(\varepsilon)$.

We now assume λ divisible by μ, and accordingly replace λ by $\mu\lambda$ and set

$$\alpha = \varepsilon^{\mu}, \tag{7}$$

so that α may also be a non-primitive $(n-1)$th root of unity, e.g. an eth root of unity (if, as above, $m = ef$ and μ is divisible by f, say taken $= f$), where however the case $\alpha = 1$, that is $e = 1$, is to be excluded. Then we set

$$\sum_{1,\,n-2}^{t} \alpha^{\operatorname{ind} t - (\lambda+1)\operatorname{ind}(t+1)} = \psi_\lambda(\alpha); \tag{8}$$

here $\psi_\lambda(\alpha)$ is a number of the field $R(\alpha)$, which can easily be computed with the aid of an index table; λ can here be taken with respect to the modulus e. Formula (5) gives for this assumption

$$(\alpha^\lambda, r)(\alpha, r) = (\alpha^{\lambda+1}, r)\psi_\lambda(\alpha), \tag{9}$$

and this formula holds so long as $\mu(\lambda+1)$ is not divisible by $n-1$; thus, if μ is coprime to m, and hence α is a primitive mth root of unity, for $\lambda = 1, 2 \ldots n-3$; and if f is the greatest common divisor of μ and m and $m = ef$, for $\lambda = 1, 2 \ldots e-2$. Formula (3) now gives, if α^λ is not $= 1$,

$$(\alpha^\lambda, r)(\alpha^{-\lambda}, r) = (-1)^{\mu\lambda} n. \tag{10}$$

Formulae (9), (10) suffice to determine not only the r themselves, but also all the periods η, by radicals whose degrees are the primes dividing m, and among which the functions ψ_λ occur as the known quantities.

Since the f-term periods (for $f = 1$) include the quantities r among them, we shall pass at once to the determination of these periods.

We shall therefore understand by α a primitive eth root of unity and set $\mu = f$. Then

$$(\alpha^\lambda, r) = r + \alpha^\lambda r_1 + \alpha^{2\lambda} r_2 + \cdots + \alpha^{(n-2)\lambda} r_{n-2},$$

or by §. 167, (12)

$$(\alpha^\lambda, r) = (\alpha^\lambda, \eta) = \eta + \alpha^\lambda \eta_1 + \alpha^{2\lambda}\eta_2 + \cdots + \alpha^{(e-1)\lambda}\eta_{e-1}, \tag{11}$$

and the formulae (9) and (10) yield

$$(\alpha^\lambda, \eta)(\alpha, \eta) = (\alpha^{\lambda+1}, \eta)\psi_\lambda(\alpha) \qquad \lambda = 1, 2 \ldots (e-2), \tag{12}$$

$$(\alpha^\lambda, \eta)(\alpha^{-\lambda}, \eta) = (-1)^{f\lambda} n. \tag{13}$$

Equation (13) shows that none of the numbers (α^λ, η) vanishes, and by (12) the $\psi_\lambda(\alpha)$ also are different from zero; and if we form equation (12) for $\lambda = 1, 2 \ldots \lambda - 1$, it follows that

$$\begin{aligned}
(\alpha, \eta)(\alpha, \eta) &= (\alpha^2, \eta)\psi_1(\alpha) \\
(\alpha^2, \eta)(\alpha, \eta) &= (\alpha^3, \eta)\psi_2(\alpha) \\
&\cdots\cdots\cdots\cdots\cdots\cdots\cdots\cdots \\
(\alpha^{\lambda-1}, \eta)(\alpha, \eta) &= (\alpha^\lambda, \eta)\psi_{\lambda-1}(\alpha).
\end{aligned}$$

From this we obtain, by multiplication and cancellation of the factor $(\alpha^2, \eta) \ldots (\alpha^{\lambda-1}, \eta)$,

$$(\alpha, \eta)^\lambda = (\alpha^\lambda, \eta)\psi_1(\alpha)\psi_2(\alpha)\ldots\psi_{\lambda-1}(\alpha). \tag{14}$$

By this, (α^λ, η) is rationally expressed in terms of (α, η). But if we set $\lambda = e - 1$ in (14), there results

$$(\alpha, \eta)^{e-1} = (\alpha^{-1}, \eta)\psi_1(\alpha)\psi_2(\alpha)\ldots\psi_{e-2}(\alpha),$$

and, by multiplication with (α, η), if (13) is used in the form

$$(\alpha, \eta)(\alpha^{-1}, \eta) = (-1)^f n$$

we obtain

$$(\alpha, \eta)^e = (-1)^f n\, \psi_1(\alpha)\psi_2(\alpha)\dots\psi_{e-2}(\alpha), \tag{15}$$

whereby the formation of (α, η) is reduced to an eth root. If we set $e = m$, we obtain (ε, r).

We shall apply this procedure to the interesting case $n = 17$, which leads to the geometrically remarkable result, first found by Gauss, that the division of the circumference of the circle into 17 parts depends on a sequence of quadratic equations, and that therefore the regular seventeen-gon can be constructed with compass and ruler.

For the prime 17, 3 is a primitive root, and one finds the index table

I	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
N	1	3	9	10	13	5	15	11	16	14	8	7	4	12	2	6

.

The two-term periods are, if we set $r = e^{\frac{2\pi i}{17}}$,

$$\begin{aligned}
\eta &= 2\cos\frac{2\pi}{17}, & \eta_1 &= 2\cos\frac{6\pi}{17}\\
\eta_2 &= 2\cos\frac{18\pi}{17} = -2\cos\frac{\pi}{17}, & \eta_3 &= 2\cos\frac{20\pi}{17} = -2\cos\frac{3\pi}{17},\\
\eta_4 &= 2\cos\frac{26\pi}{17} = 2\cos\frac{8\pi}{17}, & \eta_5 &= 2\cos\frac{10\pi}{17} = -2\cos\frac{7\pi}{17},\\
\eta_6 &= 2\cos\frac{30\pi}{17} = 2\cos\frac{4\pi}{17}, & \eta_7 &= 2\cos\frac{22\pi}{17} = -2\cos\frac{5\pi}{17}.
\end{aligned} \tag{16}$$

For the computation of the functions ψ one most simply applies the following table:

t	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$\operatorname{ind} t$	0	14	1	12	5	15	11	10	2	3	7	13	4	9	6
$\operatorname{ind}(t+1)$	14	1	12	5	15	11	10	2	3	7	13	4	9	6	8

,

from which, if α denotes an 8th root of unity, one finds by formula (8):

$$\begin{aligned}
\psi_1(\alpha) = \psi_6(\alpha) &= 2\alpha + 2\alpha^2 + 3\alpha^4 + 4\alpha^5 + 2\alpha^6 + 2\alpha^7\\
\psi_2(\alpha) = \psi_5(\alpha) &= 2 + 3\alpha + \alpha^3 + \alpha^4 + 3\alpha^5 + 4\alpha^6 + \alpha^7\\
\psi_3(\alpha) = \psi_4(\alpha) &= 3 + 3\alpha + 2\alpha^2 + 3\alpha^3 + \alpha^5 + 2\alpha^6 + \alpha^7.
\end{aligned} \tag{17}$$

If one takes

$$\alpha = \frac{1+i}{\sqrt{2}}, \qquad \alpha^2 = i, \qquad \alpha^4 = -1,$$

there results

$$\begin{aligned}
\psi_1(\alpha) = \psi_6(\alpha) &= -3 - i\sqrt{8}\\
\psi_2(\alpha) = \psi_5(\alpha) &= 1 - 4i\\
\psi_3(\alpha) = \psi_4(\alpha) &= 3 + i\sqrt{8}\\
\psi_1(i) &= -1 + 4i,
\end{aligned} \tag{18}$$

hence by (15)

$$(\alpha,\eta)^8 = 17\,(3+i\sqrt{8})^4\,(1-4i)^2. \tag{19}$$

But from (13) and (14) one obtains

$$\begin{aligned}(-1,\eta)^2 &= 17\\ (i,\eta)^2 &= (-1,\eta)(-1+4i)\\ (\alpha,\eta)^2 &= -(i,\eta)(3+i\sqrt{8});\end{aligned} \tag{20}$$

in addition one has $(1,\eta) = -1$,

$$\begin{aligned}(i,\eta)(-i,\eta) &= 17\\ (\alpha,\eta)(\alpha^{-1},\eta) &= 17\\ (\alpha^3,\eta)(\alpha^{-3},\eta) &= 17\end{aligned} \tag{21}$$

and by (9)

$$(\alpha,\eta)(\alpha^3,\eta) = (-1,\eta)(3+i\sqrt{8}), \tag{22}$$

whereby $(1,\eta)$, $(-1,\eta)$, $(\pm i,\eta)$, (α,η), (α^{-1},η), (α^{-3},η) are determined by square roots.

With the determination that we have made concerning r and α, it follows easily from a consideration of the values of the cosines (16) that $(-1,\eta)$ is positive, and hence equal to the positive square root $\sqrt{17}$, and further that (i,η) and (α,η) have positive real parts, whereby these quantities too are completely determined. We have therefore

$$\begin{aligned}(-1,\eta) &= \sqrt{17}\\ (i,\eta) &= \sqrt[4]{17}\left\{\sqrt{\frac{\sqrt{17}-1}{2}} + i\sqrt{\frac{\sqrt{17}+1}{2}}\right\},\end{aligned}$$

if all the roots are taken positive; and by the last equation (20), (α,η) can be represented, by a formula which is, to be sure, somewhat long, through square roots of real quantities. We shall not enter into the geometric constructions that attach themselves here[42].

The determination of signs employed is obtained simply if, following (16), one sets

$$\frac{1}{2}(-1,\eta) = \cos\frac{2\pi}{17} - \cos\frac{\pi}{17} + \cos\frac{8\pi}{17} + \cos\frac{4\pi}{17} - \cos\frac{6\pi}{17} + \cos\frac{3\pi}{17} + \cos\frac{7\pi}{17} + \cos\frac{5\pi}{17}.$$

But by a well-known trigonometric formula,

$$\begin{aligned}\cos\frac{2\pi}{17} - \cos\frac{\pi}{17} &= -2\sin\frac{\pi}{34}\sin\frac{3\pi}{34}\\ -\cos\frac{6\pi}{17} + \cos\frac{5\pi}{17} &= 2\sin\frac{\pi}{34}\sin\frac{11\pi}{34},\end{aligned}$$

and since $\sin\frac{11\pi}{34}$ is greater than $\sin\frac{3\pi}{34}$, the sum of these two expressions, and with it the whole sum $(-1,\eta)$, is positive

[42] Cf. on this v. Staudt, Construction des regulären Siebzehnecks. Crelle's Journ. Bd. 24, 1842.

There results further, for the real part of (i, η),

$$\eta - \eta_2 + \eta_4 - \eta_6 = 2\cos\frac{2\pi}{17} + 2\cos\frac{\pi}{17} + 2\cos\frac{8\pi}{17} - 2\cos\frac{4\pi}{17},$$

which is immediately recognised as positive; finally, for the real part of (α, η),

$$\eta - \eta_4 + \frac{1}{\sqrt{2}}(\eta_1 - \eta_3 - \eta_5 + \eta_7) = 2\cos\frac{2\pi}{17} - 2\cos\frac{8\pi}{17} + \sqrt{2}\left(\cos\frac{6\pi}{17} + \cos\frac{3\pi}{17} + \cos\frac{7\pi}{17} - \cos\frac{5\pi}{17}\right),$$

which likewise proves to be positive.

§. 170.
Properties of the Numbers ψ.

The formulae for the resolvents derived in the preceding section admit a wealth of applications, even without an exact determination of the prime n, for special assumptions about the number e; they thus give number-theoretic theorems for whole categories of primes.

For algebra they provide arbitrarily many solutions of cyclic equations of prescribed degree.

We precede the derivation of the most important of these applications for algebra with some general considerations on the functions ψ which we defined in the preceding section.

For the computation of the numbers $\psi_{\lambda,\mu}(\varepsilon)$ [§. 169, (5)] one makes use, as we have seen in the case $n = 17$, of an index table. But between these functions there exist relations by which they can be reduced to a smaller number, on which Jacobi has made remarkable investigations[43].

The function $\psi_{\lambda,\mu}(\varepsilon)$ does not change, according to its definition, when λ, μ are interchanged, although its expression

[§. 169, (6)] thereby takes on another form. We thus obtain the same function $\psi_\lambda(\alpha)$ when we replace λ, μ by $\lambda f, f$ or by $f, \lambda f$, hence [§. 169, (8)]

$$\psi_\lambda(\alpha) = \sum^{t} \alpha^{\operatorname{ind} t - (\lambda+1)\operatorname{ind}(t+1)} = \sum^{t} \alpha^{\lambda \operatorname{ind} t - (\lambda+1)\operatorname{ind}(t+1)}.$$

If, then, two numbers λ, λ' are connected with one another by the congruence

$$\lambda\lambda' \equiv 1 \pmod{e} \tag{1}$$

it follows that

$$\psi_\lambda(\alpha^{\lambda'}) = \sum^{t} \alpha^{\lambda\lambda' \operatorname{ind} t - (\lambda\lambda'+\lambda')\operatorname{ind}(t+1)} = \sum^{t} \alpha^{\operatorname{ind} t - (\lambda'+1)\operatorname{ind}(t+1)},$$

hence the formula

$$\psi_\lambda(\alpha^{\lambda'}) = \psi_{\lambda'}(\alpha). \tag{2}$$

If we then put $-\lambda - \mu$ for λ in §. 169, (5), and hence $-\lambda$ for $\lambda + \mu$ (where it must then be assumed that λ is not divisible by m), it follows that

$$\frac{(\varepsilon^{-\lambda-\mu}, r)(\varepsilon^{\mu}, r)}{(\varepsilon^{-\lambda}, r)} = \psi_{-\lambda-\mu,\mu}(\varepsilon).$$

[43] Cf. Kronecker, Zur Theorie der Abel'schen Gleichungen. Journ. f. Mathem. Bd. 93.

We multiply the left side by

$$\frac{(\varepsilon^{\lambda+\mu}, r)(\varepsilon^{\lambda}, r)}{(\varepsilon^{\lambda+\mu}, r)(\varepsilon^{\lambda}, r)} = 1,$$

and apply in the numerator and denominator the formula

$$(\varepsilon^{\lambda}, r)(\varepsilon^{-\lambda}, r) = (-1)^{\lambda} n$$

[§. 169, (3)]; then it follows that

$$(-1)^{\mu} \frac{(\varepsilon^{\lambda}, r)(\varepsilon^{\mu}, r)}{(\varepsilon^{\lambda+\mu}, r)} = \psi_{-\lambda-\mu,\mu}(\varepsilon),$$

or

$$\psi_{-\lambda-\mu,\mu}(\varepsilon) = (-1)^{\mu} \psi_{\lambda,\mu}(\varepsilon),$$

and if we replace λ, μ by $\lambda f, f$ and determine λ'' from the congruence

$$\lambda + \lambda'' + 1 \equiv 0 \pmod{e} \tag{3}$$

we obtain

$$\psi_{\lambda}(\alpha) = (-1)^{f} \psi_{\lambda''}(\alpha). \tag{4}$$

This formula can then in turn be combined with formula (2), and the number of functions ψ_{λ} to be computed for a given e, whose number amounts to $e - 2$, thereby be reduced to about the sixth part.

A third formula results when one puts ε^{-1} for ε in §. 169, (5) and then multiplies, thus

$$\psi_{\lambda,\mu}(\varepsilon)\, \psi_{\lambda,\mu}(\varepsilon^{-1}) = \frac{(\varepsilon^{\lambda},\, r)(\varepsilon^{-\lambda},\, r)(\varepsilon^{\mu},\, r)(\varepsilon^{-\mu},\, r)}{(\varepsilon^{\lambda+\mu},\, r)(\varepsilon^{-\lambda-\mu},\, r)}$$

is formed. And with the help of §. 169, (3) one thus finds the formula

$$\psi_{\lambda,\mu}(\varepsilon)\, \psi_{\lambda,\mu}(\varepsilon^{-1}) = n, \tag{5}$$

which is now again specialised by the substitution $\lambda f, f$ for λ, μ:

$$\psi_{\lambda}(\alpha)\, \psi_{\lambda}(\alpha^{-1}) = n. \tag{6}$$

Finally we wish to mention one further property of the functions ψ, which can also be read off immediately from their definition §. 169, (5). According to this, namely, if λ, μ, ν are three numbers such that $\lambda + \mu$, $\lambda + \mu + \nu$ is not divisible by m,

$$\psi_{\lambda,\mu}(\varepsilon)\, \psi_{\lambda+\mu,\nu}(\varepsilon) = \frac{(\varepsilon^{\lambda},\, r)(\varepsilon^{\mu},\, r)(\varepsilon^{\nu},\, r)}{(\varepsilon^{\lambda+\mu+\nu},\, r)}, \tag{7}$$

i. e. the product $\psi_{\lambda,\mu}(\varepsilon)\, \psi_{\lambda+\mu,\nu}(\varepsilon)$ remains unchanged when the three numbers λ, μ, ν are permuted arbitrarily (provided that $\lambda + \nu$ and $\mu + \nu$ are also not divisible by m). From this we obtain, e. g.,

$$\psi_{\lambda,\mu}(\varepsilon)\, \psi_{\lambda+\mu,\nu}(\varepsilon) = \psi_{\mu,\nu}(\varepsilon)\, \psi_{\mu+\nu,\lambda}(\varepsilon),$$

and if one replaces λ, μ, ν in it by $2\lambda f, f, f$, and observes that $\psi_{2f,2\lambda f}(\varepsilon) = \psi_{\lambda}(\alpha^2)$,

$$\psi_{2\lambda}(\alpha)\, \psi_{2\lambda+1}(\alpha) = \psi_{1}(\alpha)\, \psi_{\lambda}(\alpha^2), \tag{8}$$

provided that $2, 2\lambda+1, 2\lambda+2$ are not divisible by e.

Jacobi made the attempt, with the help of these formulae, to express, for a given e assumed to be a prime, all the $e-2$ functions $\psi_\lambda(\alpha)$ by the conjugate values of a single one, i. e. by $\psi_1(\alpha), \psi_1(\alpha^2), \psi_1(\alpha^3)\ldots$, and carried the computation through as far as $e=23$.

His conjecture, however, that such a representation is possible in general, is, according to the investigations of Kronecker, probably not correct.

We wish to carry through the case $e=7$ as an example. For this case

$$\lambda = 1,2,3,4,5$$
$$\lambda' = 1,4,5,2,3$$
$$\lambda'' = 5,4,3,2,1,$$

so the formulae (2) and (4) yield

$$\psi_4(\alpha) = \psi_2(\alpha^4), \quad \psi_5(\alpha) = \psi_3(\alpha^5)$$
$$\psi_1(\alpha) = \psi_5(\alpha), \quad \psi_2(\alpha) = \psi_4(\alpha),$$

and from this further

$$\psi_1(\alpha) = \psi_3(\alpha^5)$$
$$\psi_3(\alpha) = \psi_1(\alpha^3),$$

and if in (8) one sets $\lambda = 1$

$$\psi_2(\alpha)\,\psi_3(\alpha) = \psi_1(\alpha)\,\psi_1(\alpha^2);$$

finally, by (6)

$$n = \psi_3(\alpha)\,\psi_3(\alpha^6) = \psi_3(\alpha)\,\psi_1(\alpha^4).$$

If one now computes accordingly the formula §. 169, (15)

$$(\alpha,\eta)^7 = n\,\psi_1(\alpha)\,\psi_2(\alpha)\,\psi_3(\alpha)\,\psi_4(\alpha)\,\psi_5(\alpha),$$

there results

$$(\alpha,\eta)^7 = \psi_1(\alpha)^4\,\psi_1(\alpha^2)^2\,\psi_1(\alpha^4). \tag{9}$$

By this formula the relation from §. 169, (14)

$$(\alpha,\eta)^{14} = (\alpha^2,\eta)^7\,\psi_1(\alpha)^7$$

can be verified immediately.

We shall conclude these considerations on the functions ψ with the proof of a theorem that will be useful to us later.

If we determine a positive integer $\nu < m$ from the congruence

$$\lambda+\mu+\nu \equiv 0 \pmod m, \tag{10}$$

then we can represent the function $\psi_{\lambda,\mu}(\varepsilon)$, as it is defined by §. 169, (6), in the following way

$$\psi_{\lambda,\mu}(\varepsilon) = \sum_{1,\,n-2}^{t} \varepsilon^{\mu \operatorname{ind} t + \nu \operatorname{ind}(t+1)}. \tag{11}$$

In this we now wish to replace the mth root of unity ε by the primitive congruence root g of the prime n on which the index system is based, i. e. we wish to form the integer

$$\psi_{\lambda,\mu}(g) = \sum_{1,\,n-2}^{t} g^{\mu \operatorname{ind} t + \nu \operatorname{ind}(t+1)} \equiv \sum_{1,\,n-2}^{t} t^\mu (t+1)^\nu \pmod n \tag{12}$$

Considering this number with respect to the modulus n, we can extend the summation to $t = n-1$, since for this value $(t+1)$ is congruent to zero. We thus obtain

$$\psi_{\lambda,\mu}(g) \equiv \sum_{1,\,n-1}^{t} t^{\mu}(t+1)^{\nu} \pmod{n}. \tag{13}$$

To the power $(t+1)^{\nu}$ we can apply the binomial theorem, and we obtain, if, as before, $B_h^{(\nu)}$ denote the binomial coefficients,

$$\psi_{\lambda,\mu}(g) \equiv \sum_{0,\,\nu}^{h} B_h^{(\nu)} \sum_{1,\,n-1}^{t} t^{\mu+h} \pmod{n}. \tag{14}$$

If we set $t \equiv g^s \pmod{n}$, then s runs through a system of residues with respect to the modulus m, and we obtain

$$\sum_{1,\,n-1}^{t} t^{\mu+h} \equiv \sum_{0,\,m-1}^{s} g^{s(\mu+h)} \pmod{n}.$$

But by the summation formula for the geometric series

$$(g^{\mu+h}-1) \sum_{0,\,m-1}^{s} g^{s(\mu+h)} = g^{m(\mu+h)} - 1 \equiv 0 \pmod{n},$$

and consequently

$$\sum_{0,\,m-1}^{s} g^{s(\mu+h)} \equiv 0 \pmod{n}, \tag{15}$$

except when $\mu + h \equiv 0 \pmod{m}$, and in this case

$$\sum_{0,\,m-1}^{s} g^{s(\mu+h)} \equiv m \equiv -1 \pmod{n}. \tag{16}$$

If we take λ, μ between 0 and m, then

1) when $\lambda + \mu < m$, $\quad \nu = m - \lambda - \mu$,
2) when $\lambda + \mu > m$, $\quad \nu = 2m - \lambda - \mu$.

The exponent $\mu + h$ in the sum (14) therefore runs, in case 1), through the values $\mu + h = \mu, \mu+1, \ldots m-\lambda$, and in case 2) through $\mu, \mu+1, \ldots 2m-\lambda$.

In case 1), therefore, the case $\mu + h \equiv 0 \pmod{m}$ assumed in (16) does not occur at all; in case 2) it occurs once, for $h = m - \mu$.

Accordingly, from (14), having regard to (15) and (16), if for the binomial coefficient $B_h^{(\nu)}$ the value $\dfrac{\Pi(\nu)}{\Pi(h)\,\Pi(\nu-h)}$ is put, we have

$$\begin{aligned} &1) \quad \psi_{\lambda,\mu}(g) \equiv 0, \quad \lambda + \mu < m \\ &2) \quad \psi_{\lambda,\mu}(g) \equiv -\frac{\Pi(2m-\lambda-\mu)}{\Pi(m-\lambda)\,\Pi(m-\mu)}, \quad m < \lambda + \mu < 2m. \end{aligned} \pmod{n} \tag{17}$$

§. 171.
The Gauss Sums.

In §. 169 it is shown that, if $n-1=ef$, the periods $\eta, \eta_1, \dots \eta_{e-1}$ of an equation of degree e with integral coefficients

satisfy. This equation has only real roots if f is even, because then $\frac{1}{2}(n-1)$ is a multiple of e, and consequently r and r^{-1} occur in the same period. But if f is odd, all the roots are imaginary.

It is now of the highest interest to investigate this equation of degree e for individual particular values of e, without any special assumption about the prime n beyond the requirement that $n-1$ shall be divisible by e.

We consider the first special cases, and assume to begin with $e=2$, which is admissible for every odd prime n; here η, η_1 are thus the two periods of $\frac{1}{2}(n-1)$ terms, which we now wish to denote by A, B, so that

$$A = r + r_2 + r_4 + \cdots + r_{n-3}$$
$$B = r_1 + r_3 + r_5 + \cdots + r_{n-2}.$$

The $r, r_2, r_4 \dots$ have the exponents $g^0, g^2 \dots g^{n-3}$, i. e. the exponents of g are even numbers. The exponents of r are therefore the quadratic residues of n (§. 138). Likewise the exponents of r in the sum B are the non-residues. If, then, we denote the residues by a and the non-residues by b, we have

$$A = \Sigma r^a, \quad B = \Sigma r^b, \tag{1}$$

and $(-1, r) = A - B$. These expressions A, B, are called the Gauss sums.

If in the formula §. 169, (10) we make the assumption

$$e = 2, \quad \mu = f = \frac{n-1}{2}, \quad \lambda = 1, \quad \alpha = -1,$$

there results

$$A - B = \pm\sqrt{(-1)^{\frac{n-1}{2}}\, n}, \tag{2}$$

while on the other hand

$$A + B = -1,$$

hence

$$2A = -1 \pm \sqrt{(-1)^{\frac{n-1}{2}}\, n}, \quad 2B = -1 \mp \sqrt{(-1)^{\frac{n-1}{2}}\, n}. \tag{3}$$

The sign that is to be given to the root depends on the choice of r. But once r has been fixed upon, the sign is completely determined. Its ascertainment presents peculiar difficulties, which gave occasion to a separate memoir by Gauss[1]).

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[1]) Summatio quarundam serierum singularium. Gauss Werke, Bd. II. In another way, by application of the higher analysis, the sign

Let μ and ν be two positive integers and x a variable. We define a rational function of x by the equation

$$(\mu, \nu) = \frac{(1-x^\mu)(1-x^{\mu-1})\dots(1-x^{\mu-\nu+1})}{(1-x)(1-x^2)\dots(1-x^\nu)}. \tag{4}$$

If $\nu = \mu + 1$, this function vanishes identically, on account of the factor $1 - x^{\mu-\nu+1}$, and likewise when ν becomes still greater.

We set up for these functions some recursion formulae, which follow immediately from an inspection of formula (4):

(5) $$(1 - x^{\mu-\nu+1})(\mu+1, \nu) = (1 - x^{\mu+1})(\mu, \nu)$$

(6) $$(1 - x^{\nu+1})\ (\mu, \nu+1) = (1 - x^{\mu-\nu})(\mu, \nu),$$

and by the combination of (5) and (6), after μ has been replaced by $\mu + 1$ in (6),

(7) $$(1 - x^{\nu+1})(\mu+1, \nu+1) = (1 - x^{\mu+1})(\mu, \nu).$$

If one subtracts (6) from (7) and divides by the non-vanishing $1 - x^{\nu+1}$, it follows further

(8) $$(\mu+1, \nu+1) = (\mu, \nu+1) + x^{\mu-\nu}(\mu, \nu).$$

If we set, by way of definition, $(\mu, 0) = 1$, and $(\mu, \nu) = 0$ for negative ν, then these formulae all hold generally for every integer ν.

From the last formula we conclude by complete induction that (μ, ν) is an integral function of x. For if this property is proved for any one μ, it follows for the next greater μ from formula (8). But, for example, $(1, \nu)$ is equal to 1 or equal to 0, and so the theorem is generally correct.

If we put $\mu + 1$ in place of μ in (8), we can with its help transform formula (5) thus,

(9) $$(1 - x^{\mu+1})(\mu, \nu) = (\mu+1, \nu) + (\mu+1, \nu+1) - (\mu+2, \nu+1).$$

We now consider an integral function

(10) $$f(x, \mu) = 1 - (\mu, 1) + (\mu, 2) - (\mu, 3) + \cdots = \overset{\nu}{\Sigma}(-1)^{\nu}(\mu, \nu),$$

determined by Dirichlet, Cauchy, Kronecker. The path that Gauss takes, which we follow in the text, uses only algebraic means. Kronecker has given in Liouville's Journal Ser. 2, Bd. 1 a way of determining the sign which, with the use of a remark of Dedekind (Schlömilch's Ztschr. Bd. 15, Literaturzeitung S. 21), likewise becomes a purely algebraic one.

in which the summation over ν extends from 0 to μ, but may also be extended arbitrarily further, since (μ, ν) vanishes when ν is negative or greater than μ. Accordingly we also have

$$f(x, \mu) = -\overset{\nu}{\Sigma}(-1)^{\nu}(\mu, \nu+1),$$

and if we therefore multiply formula (9) by $(-1)^{\nu}$ and sum with respect to ν, there results

(11) $$(1 - x^{\mu+1})f(x, \mu) = f(x, \mu+2).$$

From this it follows that $f(x, \mu)$ vanishes for every odd μ, since $f(x, 1) = 0$. On the other hand, there results directly from (10)

$$f(x, 2) = 1 - x,$$

and therefore from (11), for every even μ,

(12) $$f(x,\mu) = (1-x)(1-x^3)(1-x^5)\dots(1-x^{\mu-1}).$$

For if this formula is correct for μ, its correctness for $\mu+2$ follows from (11).

We now apply this identical transformation to our problem, by setting $\mu = m = n-1$ and $x = r$. Then

$$\frac{1-x^m}{1-x} = \frac{1-r^{-1}}{1-r} = -r^{-1}, \quad \frac{1-x^{m-1}}{1-x^2} = \frac{1-r^{-2}}{1-r^2} = -r^{-2}\cdots,$$

hence, so long as $0 \leqq \nu \leqq m$,

$$(m,\nu) = (-1)^\nu r^{-(1+2+\cdots+\nu)} = (-1)^\nu r^{-\frac{\nu(\nu+1)}{2}},$$

or, replacing r by r^{-2},

$$(m,\nu) = (-1)^\nu r^{\nu^2+\nu}.$$

Now

$$\nu^2+\nu \equiv \nu^2-(n-1)\nu \equiv \left(\nu-\frac{n-1}{2}\right)^2 - \left(\frac{n-1}{2}\right)^2 \pmod n,$$

hence

$$(m,\nu) = (-1)^\nu r^{-\frac{m^2}{4}} r^{\left(\nu-\frac{m}{2}\right)^2},$$

and it therefore becomes, by the defining equation (10),

$$f(r^{-2},m) = r^{-\frac{m^2}{4}} \Sigma r^{\left(\nu-\frac{m}{2}\right)^2},$$

in which ν runs through a complete system of residues with respect to the modulus n. Now simultaneously with ν, $\nu - {}^1/_2 m$ also runs through such a system of residues, so that we may also write

$$f(r^{-2},m) = r^{-\frac{m^2}{4}} \Sigma r^{\nu^2}$$

can set. Now since in the sequence of numbers ν^2 the number 0 occurs, and moreover every quadratic residue occurs twice [for ν^2 and $(-\nu)^2$], we have

$$r^{\frac{m^2}{4}} f(r^{-2},m) = 1+2A = A-B,$$

and from formula (12) it follows that

$$A-B = r^{\frac{m^2}{4}}(1-r^{-2})(1-r^{-6})\dots(1-r^{-2n+4}).$$

Now

$$r^{\frac{m^2}{4}} = r\, r^3 r^5 \dots r^{n-2},$$

and so we finally obtain the formula

(13) $$A-B = (r-r^{-1})(r^3-r^{-3})\dots(r^{n-2}-r^{-n+2}).$$

If ν denotes an odd and μ an even number between 0 and $^1/_2 n$, this product consists of the factors

$$r^\nu - r^{-\nu}, \quad r^{n-\mu} - r^{-n+\mu} = -(r^\mu - r^{-\mu}).$$

The number of factors of the second category is as great as the number of even numbers between 0 and $^1/_2 n$, or equal to the number of positive integers that are smaller than $^1/_4 n$. If therefore, understanding by l an integer, we set $n = 4l+1$ or $4l+3$, then l is this number, and we can set

$$A - B = (-1)^l \overset{\nu}{\Pi}(r^\nu - r^{-\nu}) \overset{\mu}{\Pi}(r^\mu - r^{-\mu}),$$

for which, if h runs through the sequence of numbers from 1 to $^1/_2(n-1)$, we may also set

$$A - B = (-1)^l \underset{1, \frac{n-1}{2}}{\overset{h}{\Pi}}(r^h - r^{-h}).$$

If we now set

$$r = e^{\frac{2\pi i k}{n}}, \quad r^h - r^{-h} = 2i \sin \frac{2\pi k h}{n},$$

then

$$A - B = (-1)^l (2i)^{\frac{n-1}{2}} \overset{h}{\Pi} \sin \frac{2\pi h k}{n}.$$

The product that still occurs here we have already determined earlier [§. 138, (4)]:

$$2^{\frac{n-1}{2}} \overset{h}{\Pi} \sin \frac{2\pi k h}{n} = \left(\frac{k}{n}\right) \sqrt{n},$$

wherein the $\sqrt{n}$ is to be taken positive.

If we further set $^1/_2(n-1) = 2l$ or $= 2l+1$, according as $n = 4l+1$ or $4l+3$, we now obtain, with exact determination of sign,

$$(14) \qquad \begin{aligned} A - B &= \left(\frac{k}{n}\right) \sqrt{n}, \qquad n \equiv 1 \pmod 4 \\ &= i \left(\frac{k}{n}\right) \sqrt{n}, \quad n \equiv 3 \pmod 4. \end{aligned}$$

§. 172.

The periods of $^1/_3(n-1)$ and $^1/_4(n-1)$ terms.

We now pass to the case $e = 3$, in which $n-1$ must be assumed divisible by 3, so that $n = 7, 13, 19, 31, 37, 43 \ldots$ We denote by ϱ an imaginary third root of unity:

$$\varrho = \frac{-1 + \sqrt{-3}}{2},$$

and determine the three periods η, η_1, η_2 of $^1/_3(n-1)$ terms each, which, as we have seen in §. 168, are the roots of a cyclic cubic equation. By §. 169 we have

$$\psi_1(\varrho) = \underset{1, n-2}{\overset{t}{\Sigma}} \varrho^{\operatorname{ind} t - 2 \operatorname{ind}(t+1)},$$

for which, since $-2 \equiv 1 \pmod 3$, we may also set

(1) $$\psi_1(\varrho) = \sum_{1,\,n-2}^{t} \varrho^{\mathrm{ind}(t^2+t)}$$

This number $\psi_1(\varrho)$ can be represented in one of the two forms

(2) $$\psi_1(\varrho) = a + b\varrho = \frac{A + b\sqrt{-3}}{2}$$

in which a, b, A are integers and $A = 2a - b$.
We then have

(3) $$\psi_1(\varrho^2) = a + b\varrho^2 = \frac{A - b\sqrt{-3}}{2},$$

and the formulae (10), (15) of §. 169 yield

(4) $$\eta + \eta_1 + \eta_2 = -1$$

(5) $$(\varrho, \eta)^3 = n\psi_1(\varrho), \quad (\varrho^2, \eta)^3 = n\psi_1(\varrho^2)$$

(6) $$(\varrho, \eta)(\varrho^2, \eta) = n, \quad \psi_1(\varrho)\psi_1(\varrho^2) = n,$$

in which

(7) $$(\varrho, \eta) = \eta + \varrho\eta_1 + \varrho^2\eta_2, \quad (\varrho^2, \eta) = \eta + \varrho^2\eta_1 + \varrho\eta_2.$$

For example, for $n = 7, 13$, if one takes the primitive roots 3, 2 as basis and applies the index tables, one obtains:

$n = 7$

N	1	2	3	4	5	6
I	0	2	1	4	5	3

$n = 13$

1	2	3	4	5	6	7	8	9	10	11	12
0	1	4	2	9	5	11	3	8	10	7	6

for $n = 7$, $\quad a + b\varrho = -(1 + 3\varrho)$,

for $n = 13$, $\quad a + b\varrho = -(4 + 3\varrho)$.

From these formulae we can easily derive the cubic equation whose roots are the η, η_1, η_2. On account of (4) it has the form

(8) $$\eta^3 + \eta^2 + \beta\eta + \gamma = 0,$$

and the integers β, γ are to be determined.

If we carry out the multiplication in (6), there results

$$\begin{aligned} n &= \eta^2 + \eta_1^2 + \eta_2^2 - \eta\eta_1 - \eta\eta_2 - \eta_1\eta_2 \\ &= (\eta + \eta_1 + \eta_2)^2 - 3\beta, \end{aligned}$$

hence

$$3\beta = -(n-1). \tag{9}$$

Carrying out the cubes in (5) yields, if for the moment we set

$$s_3 = \eta^3 + \eta_1^3 + \eta_2^3, \quad s = \eta^2\eta_1 + \eta_1^2\eta_2 + \eta_2^2\eta,$$

$$s' = \eta_1^2\eta + \eta_2^2\eta_1 + \eta^2\eta_2$$

the relation

$$n\psi_1(\varrho) = s_3 - 6\gamma + 3s\varrho + 3s'\varrho^2.$$

But from (8) and (9) one obtains $s_3 = -n - 3\gamma$, hence

$$n[\psi_1(\varrho) + 1] = -9\gamma + 3s\varrho + 3s'\varrho^2, \tag{10}$$

and likewise

$$n[\psi_1(\varrho^2) + 1] = -9\gamma + 3s\varrho^2 + 3s'\varrho, \tag{11}$$

to which is added, if one raises (4) to the cube,

$$n - 1 = -9\gamma + 3s + 3s'$$

as well. If one adds the last three equations, it follows that

$$n[\psi_1(\varrho) + \psi_1(\varrho^2)] + 3n - 1 = -27\gamma,$$

or finally, by (2) and (3),

$$-27\gamma = nA + 3n - 1. \tag{12}$$

From this relation it follows, since γ is an integer,

$$A \equiv 1 \pmod 3 \tag{13}$$

and from this $A^3 \equiv 1 \pmod 9$. Now by (2), (3) and (6)

$$n = a^2 - ab + b^2 \tag{14}$$

or

$$4n = A^2 + 3b^2 \tag{15}$$

and if we substitute this value in (12),

$$-4 \cdot 27\gamma = A^3 + 3Ab^2 + 3A^2 + 9b^2 - 4,$$

and from this

$$3b^2 \equiv 0 \pmod 9.$$

Thus b is divisible by 3, and if we set $b = 3B$, it follows from (15) that

$$4n = A^2 + 27B^2. \tag{16}$$

We can also easily form the square root of the discriminant, namely

$$\sqrt{D} = (\eta - \eta_1)(\eta - \eta_2)(\eta_1 - \eta_2) = s - s', \tag{17}$$

if we subtract (10) and (11) from one another. From this one obtains

$$\sqrt{D} = nB. \tag{18}$$

The cubic equation whose roots are the three quantities η therefore has, as indeed already follows from what was remarked at the beginning of §. 171, three real roots. It is simplified if one sets

$$3\eta + 1 = \xi$$

and then yields

$$\xi^3 - 3n\xi - nA = 0. \tag{19}$$

If one substitutes on the left-hand side for ξ in succession

$$-2\sqrt{n}, \quad -\sqrt{n}, \quad +\sqrt{n}, \quad +2\sqrt{n}, \tag{20}$$

there result the values

$$-n(2\sqrt{n} + A), \quad +n(2\sqrt{n} - A), \quad -n(2\sqrt{n} + A),$$
$$+n(2\sqrt{n} - A),$$

which, because by (16) A is absolutely smaller than $2\sqrt{n}$, have alternating signs. Hence in each of the three intervals bounded by the values (20) there lies one root of equation (19).

On the question which of the three roots of the cubic equation (19) is to be set for $\xi = 3\eta + 1$, which for $3\eta_1 + 1$ or for $3\eta_2 + 1$, this much may be said in general: the sign of the product (17) must agree with the sign of (18), which is determined by the definition (1), (2), though it still depends on the choice of the primitive root g. By this, three of the six possible assignments are excluded. Which assignment among the three remaining is to be made depends on the choice of r, and would give occasion to investigations similar to those carried out in the previous section for the determination of the sign of the square root[44].

We emphasise further the theorem proved by the formulae (14), (16):

If n is a prime of the form $3k+1$, then n is representable by the form $a^2 - ab + b^2$ and $4n$ by the form $A^2 + 27B^2$, where a, b, A, B are integers.

From this it follows further that n is also representable in the form $x^2 + 3y^2$. For if of the two numbers a, b one, say a, is even, then

$$n = \left(\frac{a}{2} - b\right)^2 + \tfrac{3}{4}a^2,$$

and if a and b are both odd, then

$$n = \tfrac{1}{4}(a + b)^2 + \tfrac{3}{4}(a - b)^3.$$

The two-term periods of the 9th roots of unity also satisfy a cubic equation, and because this equation plays an important role in the general theory of the cubic cyclotomic fields to be treated later, we shall consider it here for the sake of completeness, although it properly belongs to a more general region, in which the degrees of the roots of unity are no longer primes.

A 9th root of unity r satisfies the equation of degree 6

$$r^6 + r^3 + 1 = 0, \tag{21}$$

[44] Kummer, Journ. f. Mathematik, Bd. 32.

and we have three conjugate two-term periods

$$\eta = r + r^{-1}, \quad \eta' = r^2 + r^{-2}, \quad \eta'' = r^4 + r^{-4}, \tag{22}$$

which are the roots of the cubic equation

$$\eta^3 - 3\eta + 1 = 0 \tag{23}$$

are. If ϱ is a third root of unity, we can set $\varrho = r^3$, and obtain the resolvents

$$(\varrho, \eta) = \eta + \varrho\eta' + \varrho^2\eta'', \quad (\varrho^{-1}, \eta) = \eta + \varrho^{-1}\eta' + \varrho^{-2}\eta'', \tag{24}$$

for which, by (22), making use of (21), one obtains

$$(\varrho, r) = 3r, \quad (\varrho^{-1}, r) = 3r^{-1},$$

hence

$$(\varrho, r)^3 = 27\varrho, \tag{25}$$

$$(\varrho, r)(\varrho^{-1}, r) = 9, \tag{26}$$

which is entirely analogous to the formulae (5) and (6).

We enter briefly also on the case $e = 4$, which occurs for the primes n of the form $4f + 1$ (f an integer), thus for $n = 5, 13, 17, 29, 37, 41 \ldots$

Let the four periods of f terms each again be $\eta, \eta_1, \eta_2, \eta_3$.

For α in §. 169 (8), we have to set i, and thus obtain, since $-2 \equiv +2 \pmod 4$,

$$\begin{aligned} \psi_1(i) &= \sum i^{\operatorname{ind} t + 2\operatorname{ind}(t+1)} \\ \psi_2(i) &= \sum i^{\operatorname{ind}(t^2+t)} = a + bi, \end{aligned} \tag{27}$$

wherein a and b are integers. $\psi_1(i)$ has the same form as $\psi_2(i)$. But we shall presently reduce the one function to the other.

By the results of §. 171 we have

$$(-1, \eta) = \eta + \eta_2 - \eta_1 - \eta_3 = \sqrt{n}, \tag{28}$$

and hence

$$2(\eta + \eta_2) = -1 + \sqrt{n}, \quad 2(\eta_1 + \eta_3) = -1 - \sqrt{n}, \tag{29}$$

where the sign of $\sqrt{n}$ is to be determined according to §. 171 and may, on a suitable assumption about r, be taken positive.

We have further, by §. 169, (14), (15)

$$\begin{aligned} (i, \eta)(-i, \eta) &= (-1)^f n \\ (i, \eta)^2 &= (-1, \eta)\psi_1(i), \end{aligned} \tag{30}$$

hence by (28)

$$(i, \eta)^2 = \psi_1(i)\sqrt{n}, \quad (-i, \eta)^2 = \psi_1(-i)\sqrt{n} \tag{31}$$

$$(i, \eta)^4 = (-1)^f n\, \psi_1(i)\psi_2(i) = n\, \psi_1(i)^2, \tag{32}$$

from which it follows

$$\psi_1(i) = (-1)^f \psi_2(i) = (-1)^f (a + bi), \tag{33}$$

and from this by (30)

$$\psi_1(i)\psi_1(-i) = \psi_2(i)\psi_2(-i) = a^2 + b^2 = n. \tag{34}$$

To form the biquadratic equation whose roots are η, η_1, η_2, η_3, we first seek the quadratic equation with the roots η, η_2, and obtain it from

$$2(\eta + \eta_2) = -1 + \sqrt{n}$$
$$2(\eta - \eta_2) = (i, \eta) + (-i, \eta).$$

If we square these two equations and subtract the second from the first, then by (30), (31) and (33) it follows that

$$16\,\eta\,\eta_2 = 1 + n - 2\sqrt{n} - 2n(-1)^f - 2a(-1)^f\sqrt{n}. \tag{35}$$

But by (29) and (35) the coefficients of the quadratic equation sought are determined. It reads:

$$\eta^2 + \frac{\eta}{2} + \frac{1 + n - 2n(-1)^f}{16} = \left(\frac{\eta}{2} + \frac{a(-1)^f + 1}{8}\right)\sqrt{n},$$

and the biquadratic equation for the four periods is obtained by squaring both sides:

$$\left(\eta^2 + \frac{\eta}{2} + \frac{1 + n - 2n(-1)^f}{16}\right)^2 - n\left(\frac{\eta}{2} + \frac{a(-1)^f + 1}{8}\right)^2 = 0. \tag{36}$$

If we seek the coefficient of the first power of η, which must be an integer, we find for it

$$\frac{1 + n - 2n(-1)^f}{16} - n\,\frac{a(-1)^f + 1}{8}; \tag{37}$$

f is even when $n \equiv 1$, odd when $n \equiv 5 \pmod 8$, and accordingly

$$1 + n - 2n(-1)^f \equiv 0 \pmod 8.$$

But from this it follows by (37) that $a(-1)^f + 1$ must be divisible by 4, hence

$$a \equiv -(-1)^f \pmod 4, \tag{38}$$

and from this it is further to be concluded, by (34), that b must be even, hence

$$b \equiv 0 \pmod 2. \tag{39}$$

The biquadratic equation takes a simpler form under the substitution

$$4\eta + 1 = \xi \tag{40}$$

and by (36) becomes:

$$[\xi^2 + n(1 - 2(-1)^f)]^2 - 4n[\xi + (-1)^f a]^2 = 0, \tag{41}$$

or, if we separate the two cases $n \equiv 1$ or $\equiv 5 \pmod 8$:

$$\begin{aligned}(\xi^2 - n)^2 - 4n(\xi + a)^2 &= 0\\ (\xi^2 + 3n)^2 - 4n(\xi - a)^2 &= 0.\end{aligned} \tag{42}$$

We wish here also to emphasise the theorem expressed in formula (34):

Every prime of the form $4f+1$ can be decomposed into the sum of two squares.

§. 173.

The complex numbers of Gauss.

The theorem last proved forms the foundation for the theory of the number field which arises from the field of rational numbers by adjunction of the imaginary unit $i = \sqrt{-1}$, and which by our stipulation we have to denote by $R(i)$[1]).

We have here to distinguish the primes of the form $4f + 1$ from those of the form $4f + 3$, and for the sake of brevity we wish to combine the former with the prime 2 and denote them by p, the latter by q; thus we set

$$p = 4f + 1, 2; \quad q = 4f + 3$$

The following theorems then hold:

1. For every p there are two integers a and b such that

$$p = a^2 + b^2$$

holds.

For the primes of the form $4f + 1$ this has been proved in the concluding theorem of the last section, and for $p = 2$ it is immediately evident from $2 = 1^2 + 1^2$.

Against this stands a second theorem:

2. A prime q is never representable in the form $a^2 + b^2$, or the still more general one:

3. The sum of two squares of integers $a^2 + b^2$ is divisible by a prime q only when a and b are divisible by q.

Theorem 2. follows simply from the fact that, when $a^2 + b^2$ is odd, one of the two numbers a, b is even, and hence its square divisible by 4, while the other is odd, and hence its square

[1]) Gauss, Theoria residuorum biquadraticorum, commentatio secunda. Werke, Bd. II.

$\equiv 1 \pmod 4$, and hence $a^2 + b^2 \equiv 1 \pmod 4$. Theorem 3., which moreover includes the second, is proved as follows.

Suppose that $a^2 + b^2$, but not b, is divisible by q; we then determine b' from the congruence $bb' \equiv 1 \pmod q$ and obtain from $a^2 + b^2 \equiv 0 \pmod q$

$$(ab')^2 \equiv -1 \pmod q.$$

But this is impossible, since by §. 138, 4. the number -1 is a quadratic non-residue for every prime q.

The field $R(i)$ is the totality of all numbers of the form $x + yi$, where x, y are integral or fractional rational numbers.

The numbers $a+bi$, in which a and b are integers in R, are called the integers of the field $R(i)$.

The product of two conjugate numbers

$$\xi = x + yi, \quad \xi' = x - yi,$$

that is

$$\xi\xi' = x^2 + y^2$$

is called the norm of ξ and is denoted by $N(\xi)$. The norm of a non-vanishing number ξ is a positive rational number, and the norm of an integer is an integer. The norm of a product or of a quotient is equal to the product or the quotient of the norms.

An integer whose norm is equal to 1 is called a unit.

Since $a^2 + b^2$ (for integers a, b) can be $= 1$ only when $a = \pm 1$, $b = 0$ or $a = 0$, $b = \pm 1$, there are in $R(i)$ only the four units

$$+1, \quad -1, \quad +i, \quad -i.$$

The reciprocal value of a unit is also a unit.

Two numbers, of which one arises from the other by multiplication with a unit, are called associated numbers.

Every complex number belongs to a system of four associated numbers

$$a + bi, \quad -a - bi, \quad -b + ai, \quad b - ai.$$

The sum, difference and product of two integers are again integers.

An integer α is called divisible by an integer β if a third integer γ exists, such that $\alpha = \beta\gamma$.

If α is divisible by β, then $N(\alpha)$ is divisible by $N(\beta)$. For from $\alpha = \beta\gamma$ it follows that $N(\alpha) = N(\beta)N(\gamma)$.

If α, β, γ are integers in $R(i)$ and α is divisible by γ, then $\alpha\beta$ is also divisible by $\gamma\beta$; and if α and β are divisible by γ, then $\alpha \pm \beta$ is also divisible by γ.

Every integer is divisible by every unit.

The units, however, are divisible only by units; for if $\alpha\beta$ is a unit, then $N(\alpha)N(\beta) = 1$, hence $N(\alpha)$ and $N(\beta) = 1$, i. e. α and β are units.

Associated numbers are mutually divisible by one another, and when two numbers are mutually divisible, they are associated. For if $\alpha : \beta$ and $\beta : \alpha$ are both integers, then the product of these two numbers is $= 1$; hence both are units.

For the numbers of the field $R(i)$ the same laws hold concerning the decomposition into prime factors as for the real integers.

An integer α of the field $R(i)$ which is not a unit is called composite if it can be decomposed into several integer factors, none of which is a unit.

If it cannot be so decomposed, it shall be called a prime in the field $R(i)$.

If $\xi = x + yi$ is a fractional number of the field $R(i)$, then an integer $\mu = m + ni$ can be determined so that the norm of the difference $\xi - \mu$, that is $(x-m)^2 + (y-n)^2$, is smaller than, or at least not greater than, $1/2$; for one need only choose the integral rational numbers m, n so that $x - m$ and $y - n$, taken absolutely, are not greater than $1/2$, and hence their squares not greater than $1/4$.

If, then, α and α_1 are two integers in $R(i)$, and α_1 is different from zero, one can accordingly determine the integers μ and α_2, so that

$$\frac{\alpha}{\alpha_1} - \mu = \frac{\alpha_2}{\alpha_1}, \quad \frac{N(\alpha_2)}{N(\alpha_1)} \leqq \frac{1}{2},$$

hence

$$\alpha = \mu\alpha_1 + \alpha_2, \quad N(\alpha_2) < N(\alpha_1).$$

If α_2 is not zero, one can proceed in the same way with the numbers α_1, α_2 and obtains

$$\alpha_1 = \mu_1\alpha_2 + \alpha_3, \quad N(\alpha_3) < N(\alpha_2),$$

and in this way one determines a sequence of integers $\alpha, \alpha_1, \alpha_2, \alpha_3 \ldots$ with steadily decreasing norm, which can be continued so long as zero does not occur among them. Since these norms are positive rational integers which always decrease, zero must appear after a finite number of steps, and one thus obtains a system of equations

$$\begin{aligned} \alpha &= \mu\alpha_1 + \alpha_2, \\ \alpha_1 &= \mu_1\alpha_2 + \alpha_3, \\ &\cdots\cdots\cdots\cdots \\ \alpha_{h-2} &= \mu_{h-2}\alpha_{h-1} + \alpha_h, \\ \alpha_{h-1} &= \mu_{h-1}\alpha_h. \end{aligned} \tag{1}$$

From this one concludes that the whole sequence of numbers $\alpha, \alpha_1, \alpha_2 \ldots$, and thus in particular also α and α_1, are divisible by α_h, and conversely, that every common divisor of α and α_1 is a divisor of all the following α, and hence also of α_h. Every other number that has these two properties must be associated with α_h, and one therefore calls α_h (and every number associated with α_h) the greatest common divisor of α and α_1.

If from the algorithm (1) we eliminate the intermediate α, by substituting α_2 from the first equation into the second, then α_3 from the second into the third, there results a result of the form $\alpha\varkappa + \alpha_1\lambda = \alpha_h$, and we can therefore, if instead of $\alpha, \alpha_1, \alpha_h$ we write α, β, δ, state the theorem:

4. If α, β are any two integers in $R(i)$, and δ their greatest common divisor, then the integers $\varkappa, \lambda$ can be so determined that

$$\alpha\varkappa + \beta\lambda = \delta \tag{2}$$

holds.

If the greatest common divisor δ of two numbers α, β is a unit, then α and β are called relatively prime, and in this case we can satisfy the equation

$$\alpha\varkappa + \beta\lambda = 1 \tag{3}$$

by integral $\varkappa, \lambda$.

If β is a prime, then either α is divisible by β, or α and β are relatively prime. In the latter case equation (3) holds. If then γ is another number in $R(i)$, it follows from (3) that

$$\alpha\gamma\varkappa + \beta\gamma\lambda = \gamma.$$

Accordingly, if $\alpha\gamma$ is divisible by β, then γ too must be divisible by β. We therefore have the following fundamental theorem:

5. A product of two or more integers in $R(i)$ is divisible by a prime in $R(i)$ only if at least one of its factors is divisible by this prime.

If one now further makes the remark that a prime can be divisible by another only if the quotient is a unit, that is, if the two are associated, and regards associated primes as not essentially different, it follows:

6. An integer of the field $R(i)$ can always and in only one way be decomposed into a product of primes.

For let α be any integer in $R(i)$. If α is not itself a prime, it is decomposable, say into $\gamma\beta$; but the norms of γ and of β are smaller than the norm of α. If β is not yet a prime, it is again decomposable, say into $\delta\varepsilon$; the norms of these factors, which are all positive integers, decrease continually, and one must therefore necessarily at some point come upon a factor which is a prime. Hence every number α is certainly divisible by a prime π. If accordingly $\alpha = \pi\alpha'$, then the same holds of α', and we have, say, $\alpha' = \pi'\alpha'' \ldots$ Since the norms of $\alpha, \alpha', \alpha'' \ldots$ again all decrease, one must, in continuing this sequence, finally come upon a unit. Hence α is decomposable into a finite number of prime factors, thus

$$\alpha = \pi\pi'\pi'' \ldots,$$

where a unit factor remaining over at the end can be united with one of the primes π. If now $\varkappa, \varkappa', \varkappa'' \ldots$ are likewise primes and $\alpha = \varkappa\varkappa'\varkappa'' \ldots$, then it follows from

$$\varkappa\varkappa'\varkappa'' \cdots = \pi\pi'\pi'' \ldots$$

by 5. that one of the primes π, e.g. the first, is divisible by $\varkappa$, and hence is not essentially different from $\varkappa$. If one cancels on both sides by $\varkappa = \pi$, one can repeat the same inference with $\varkappa'$, and so forth, and thus finds that each of the primes $\varkappa$ must also occur among the π, and if among the $\varkappa$ a prime occurs several times, it must occur at least as often among the π. But the same inference can also be made in the reverse direction, and from this it follows that the totality

the $\varkappa$, apart from unit factors, must agree with the totality of the π.

It now remains to actually determine the primes of the field $R(i)$.

Every prime is a divisor of infinitely many rational integers, e.g. of the norm and all its multiples. Among the rational positive integers that are divisible by a prime π there is one, which we shall denote by n, that is the smallest, and this must be a prime in the field R. For if it were decomposable in R, then by 5. one of its factors, and hence a still smaller number, would have to be divisible by π. Likewise, conversely, every real prime n is also divisible by at least one prime π. Let, then, $n = \pi\alpha$, from which, by forming the norm, $n^2 = N(\pi)N(\alpha)$ follows. Since n is a prime and $N(\pi)N(\alpha)$ integers, two cases are possible:

$$\begin{aligned} &1. \quad N(\pi) = n, \quad N(\alpha) = n, \\ &2. \quad N(\pi) = n^2, \quad N(\alpha) = 1. \end{aligned}$$

In the first case, if π' denotes the number conjugate to π and $\pi = a + bi$ is set, then $n = \pi\pi' = a^2 + b^2$, $\alpha = \pi'$, and one sees that this case occurs only when n belongs to the primes p that are decomposable into the sum of two squares. Conversely, every such number p can be decomposed into two conjugate factors $\pi\pi'$, of which neither is a unit. The primes p are therefore not primes in the field $R(i)$.

In the second case, which accordingly occurs with the primes q, α is a unit, hence n is associated with π; that is, the real primes q are also primes in the field $R(i)$.

The prime 2 belongs, as already remarked, to the first kind, and $2 = (1 + i)(1 - i)$. But it occupies a special position, because $1 - i = -i(1 + i)$, and hence $2 = -i(1 + i)^2$. The real prime 2 is therefore (apart from the factor $-i$) the square of a prime in the field $R(i)$. This case, as one easily sees, occurs with none of the remaining primes p.

The totality of the primes of the field $R(i)$ thus consists of the real primes q and of the factors of the primes p. If p is decomposed into the sum of two squares, $a^2 + b^2$, then one also knows the conjugate factors $a \pm bi$ of p. To this

come also the associated numbers. From 5. it then further follows that a prime p can be decomposed in only one way into the sum of two squares. By this decomposition, however, the numbers a, b are not yet completely determined; rather, they can still be interchanged with one another and furnished with two signs. This comes down to saying that one can replace π by any of

the four associated numbers $a+bi, -a-bi, -b+ai, b-ai$. If p is odd, then of the two numbers a, b one must be even, the other odd. If we choose, say, for b the even one of the two numbers, and determine the sign so that $a \equiv 1 \pmod 4$, then among the four associated numbers a definite one is selected, which one may call the primary one[45].

This definition of the primary numbers can be carried over to all integers of the field $R(i)$ whose norm is odd, and one then has the law that the product of two primary numbers again yields a primary number.

The system of four associated numbers in the field $R(i)$ is analogous to the pair of opposite numbers in the field R. In R one regards the positive numbers as the primary ones.

By the formula (27) of the preceding section, for any given prime p one of the factors $a+bi$ is derived from the division of the circle. By the formulae §. 172, (38), (39) this number is primary if $p \equiv 5 \pmod 8$, but opposite to the primary one if $p \equiv 1 \pmod 8$. But as to which of the two conjugate numbers $a \pm bi$ is represented by these formulae, we have no general criterion.

For example, the complex primes in $R(i)$ whose norms are less than 200 are:

$$1+i,\ 1+2i,\ 3+2i,\ 1+4i,\ 5+2i,\ 1+6i,\ 5+4i,$$
$$7+2i,\ 5+6i,\ 3+8i,\ 5+8i,\ 9+4i,\ 1+10i,\ 3+10i,$$
$$7+8i,\ 11+4i,\ 7+10i,\ 11+6i,\ 13+2i,\ 9+10i,\ 7+12i,$$
$$1+14i.$$

§. 174.
The Field of the Third Roots of Unity.

The principal theorem of the preceding section, that every integer of the field $R(i)$ can be decomposed in only one way into indecomposable factors, and that in consequence the concept of the indecomposable number coincides with that of the prime, rests, as one sees, in the main on the algorithm of the greatest common divisor contained in the formulae §. 173, (1); and thus if in another field one has such an algorithm, terminating after finitely many steps, the same conclusions can be drawn.

This is the case for the field which can be derived from R by means of the third root of unity $\varrho = \dfrac{-1+\sqrt{-3}}{2}$, and which may be denoted by $R(\varrho)$ or also by $R(\sqrt{-3})$. This field consists of the totality of all numbers of the form $\xi = x + \varrho y$, in which x, y are rational numbers, and the norm of such a number is

$$N(\xi) = (x+\varrho y)(x+\varrho^2 y) = x^2 - xy + y^2$$
$$= \frac{2x-y+y\sqrt{-3}}{2} \cdot \frac{2x-y-y\sqrt{-3}}{2} = \frac{(2x-y)^2+3y^2}{4}.$$

A number $\alpha = a + b\varrho$ is called an integer in $R(\varrho)$ if a, b are rational integers. The units in this field result from

$$N(\alpha) = a^2 - ab + b^2 = 1,$$

or

$$(2a-b)^2 + 3b^2 = 4,$$

which is satisfied only for $b=0$, $a=\pm 1$ or $b=\pm 1$, $a=\pm 1$, so that here one has six units

$$\pm 1, \quad \pm\varrho, \quad \pm\varrho^2.$$

[45] Gauss gives at the place mentioned two different determinations of the primary numbers to choose from, of which this is the first. He afterwards retains the second.

A system of associated numbers is here

$$\pm(a+b\varrho), \quad \pm(a\varrho+b\varrho^2) = \pm[-b+(a-b)\varrho]$$
$$\pm(a\varrho^2+b) = \pm(b-a-a\varrho).$$

If ξ is an arbitrary fractional number in $R(\varrho)$, one can determine the integer μ so that in $\xi-\mu = x+\varrho y$ the

components x and y are absolutely not greater than $1/2$, and hence that

$$N(\xi-\mu) = x^2 - xy + y^2 \leqq \frac{3}{4}.$$

On the basis of this property one can, just as in the preceding section, determine the greatest common divisor of two numbers α, β, and thereby prove that every integer in $R(\varrho)$ can be decomposed into primes of this field in only one way, if associated numbers are not regarded as essentially different.

As in the preceding section one concludes that one obtains all the primes of the field $R(\varrho)$ by decomposing all real primes, and that a real prime either splits into two conjugate prime factors in $R(\varrho)$, $p = \pi\pi'$, or is a prime in the field $R(\varrho)$ as well.

The prime 3 splits into the two factors $(1-\varrho)$, $(1-\varrho^2)$; but these are associated, and so $3 = -\varrho^2(1-\varrho)^2$ is associated with the square of a prime in $R(\varrho)$. One may choose $\sqrt{-3}$ for this prime. All primes of the form $3f+1$ can, by §. 172, be decomposed into two conjugate factors in $R(\varrho)$, while the primes of the form $3f+2$ are never decomposable, because an expression of the form a^2-ab+b^2 cannot be $\equiv 2 \pmod 3$. Thus one obtains all the primes in $R(\varrho)$ by seeking out the real primes of the form $3f+2$ and the conjugate factors of the remaining real primes.

If $a+b\varrho$ is a complex integer whose norm a^2-ab+b^2 is not divisible by 3, then of the two numbers a, b either one must be divisible by 3 and the other not, or we must have $a \equiv b$ and consequently $a^2-ab+b^2 \equiv 1 \pmod 3$. Among the associated numbers

$$a+b\varrho, \quad b+a\varrho^2, \quad b-(a-b)\varrho,$$

there is therefore always one and only one in which the coefficient of ϱ (or of ϱ^2) is divisible by 3. Let us assume that $b = 3B$ is divisible by 3, and set $N(a+b\varrho) = p$; then it follows, if in addition $2a-b = A$ is set,

$$4p = A^2 + 27B^2,$$

in agreement with §. 172. If p is a real prime, and $4p$ has been represented in this form, one obtains from it the complex factors of p

$$p = \left(\frac{A+3B}{2} + 3\varrho B\right)\left(\frac{A-3B}{2} - 3\varrho B\right).$$

The complex primes whose norm is under 200 are here $1-\varrho$, $1+3\varrho$, $4+3\varrho$, $5+3\varrho$, $5+6\varrho$, $7+3\varrho$, $7+6\varrho$, $5+9\varrho$, $7+9\varrho$, $8+9\varrho$, $10+3\varrho$, $11+3\varrho$, $11+9\varrho$, $7+12\varrho$, $13+6\varrho$, $13+3\varrho$, $14+9\varrho$, $13+12\varrho$, $14+3\varrho$, $11+15\varrho$, $16+9\varrho$, $13+15\varrho$.

Seventeenth Section.
Algebraic Solution of Equations.

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§. 175.
Reduction of the Group by Pure Equations.

One of the oldest questions, in connection with which the more recent algebra has chiefly developed, is that of the so-called algebraic solution of equations, by which one understands a representation of the roots of an equation by a series of radicals, or their computation by a finite chain of root extractions. On this question the theory of groups sheds the brightest light.

If we first make precise the question at issue, it is evidently this: whether and how one can extend the field Ω by successive adjunction of root quantities (radicals) in such a way that either all or at least a part of the roots are contained in the extended field. A root quantity, however, is one which is not itself contained in Ω, but of which some integral power is; that is, if a is a quantity in Ω, it is the root of an equation of the form

$$y^m - a = 0,$$

that is, of a pure equation.

If an irreducible equation is to be algebraically solvable, or at least to have one or several algebraically representable roots, then after a successive adjunction of roots of pure equations in finite number the given equation must become reducible, since a part of its roots is to be contained in the extended field. Since the initial group P is transitive, this group must finally become intransitive, or

reduce to the unit group. Hence the case must at all events occur at some point that the group P is reduced by the adjunction of a root of a pure equation.

The investigation of this question is extraordinarily simplified if one transforms it somewhat further [46].

We have seen in §. 162 that the pure equations belong to the Abelian ones, that all Abelian equations are solvable through a chain of cyclic equations of prime degree, and these latter by radicals.

We therefore replace the question of solvability by radicals by the equivalent one of solvability by roots of Abelian equations. If we replace the Abelian equation by a chain of cyclic equations of prime degree, then the first reduction of the group will occur through the adjunction of the root of one such equation, and so we stand first of all before the question:

Under what conditions is the group P of an equation of degree n, $f(x) = 0$, reduced by the adjunction of a root of a cyclic equation of prime degree m?

We do not restrict ourselves here to irreducible equations $f(x) = 0$, but discuss the question in general, always under the self-evident assumption that $f(x)$ has no multiple roots.

It is conceivable that it may be necessary first to adjoin to the field Ω various roots of cyclic equations (e. g. roots of unity) which do not alter the group P. But since we are now concerned only with ascertaining the necessary properties of the group P, we assume that all the necessary preparations have been made, and the field Ω so constituted that by the adjunction of a root of the cyclic equation $\varphi(x) = 0$, of prime degree m, the group P reduces to one of its divisors, Q.

If we denote the roots of $\varphi(x) = 0$ by

$$\varepsilon,\ \varepsilon_1,\ \varepsilon_2 \ldots \varepsilon_{m-1},$$

then all these quantities are expressible rationally (in Ω) through an arbitrary one among them, and if P is the group of $f(x) = 0$

in Ω, then Q is the group of the same equation in $\Omega(\varepsilon)$, or, what is the same thing, in $\Omega(\varepsilon_1), \Omega(\varepsilon_2) \ldots \Omega(\varepsilon_{m-1})$. But now we can apply the concluding theorem in §. 157. By this theorem the index j of the divisor Q of P must be a divisor of m, and since m is assumed to be a prime, we have $m = j$. Moreover, by the same theorem, ε is rationally representable by the roots of the equation $f(x) = 0$:

[46] This form of the question was first pointed out by C. Jordan (Traité des substitutions p. 386).

$$\varepsilon = \psi(x_0, x_1 \dots x_{m-1}).$$

This function belongs to the group Q, and if we apply to it all the permutations of the group P, we obtain the functions $\varepsilon, \varepsilon_1, \varepsilon_2 \dots \varepsilon_{m-1}$ and no others. These functions belong to the conjugate groups $\pi^{-1}Q\pi$. But since each of these functions is rationally expressible by every other, they must all belong to the same group, that is, Q is a normal divisor of P.

We have thus proved the first theorem:

I. If the group P of an equation is reduced by the adjunction of the roots of an Abelian equation, then P has a normal divisor Q of prime index.

This theorem can also be reversed.

For if the group P has a normal divisor Q of index m, then we can choose a function ψ belonging to Q, and the functions conjugate with it, $\psi, \psi_1, \psi_2 \dots \psi_{m-1}$, all belong to the same group. The field $\Omega(\psi)$ is a normal field and ψ the root of a normal equation. In the field $\Omega(\psi)$, Q is the group of $f(x) = 0$ (§. 155). But if m is a prime, then by §. 163 ψ is the root of a cyclic equation, and thus it is proved:

II. If the group P of $f(x) = 0$ possesses a normal divisor Q of prime index m, then by the adjunction of the root of a cyclic equation of degree m the group P is reduced to Q.

§. 176.
Metacyclic Equations.

We shall call an equation whose complete solution can be reduced to a chain of cyclic equations call a metacyclic equation. The cyclic equations themselves are contained among them as a special case, and by what was remarked in §. 175 the metacyclic equations are the same as the equations solvable by radicals. If P is the group of such an equation, then by the preceding section it must possess a normal divisor of prime index j_1, which we now wish to denote by P_1. If P_1 consists of the single identical permutation, then P is itself cyclic and of prime degree. If P_1 is not the unit group, then P_1 must in turn contain a normal divisor P_2 of prime index j_2, and so on, until we finally arrive at the unit group.

That this composition of its group P is also the sufficient condition for a metacyclic equation follows from the preceding section. We therefore pronounce the theorem:

III. The necessary and sufficient condition for a metacyclic equation is that there is a sequence of groups

$$P, P_1, P_2, P_3 \dots$$

whose first is the Galois group of the equation, whose last is the unit group, and of which each following one is a normal divisor of the one next preceding, of prime index.

Accordingly we call a permutation group P which has this property, for which therefore a chain of groups

$$P, P_1, P_2 \dots P_{\mu-1}, 1$$

can be so determined that each term is a normal divisor of the preceding, of prime index, a metacyclic group[47].

We have here obtained the condition for the complete solvability of an equation by a chain of cyclic equations. But it is still a question of the further inquiry,

whether one or several of the roots can be represented in this way, while others admit no such representation. This question is justified only for irreducible equations, since for reducible equations all conceivable combinations can occur; and here the following theorem now holds:

> IV. If a root of an irreducible equation is determinable by the solution of cyclic equations, then the equation is metacyclic.

If an irreducible equation of degree n, $f(x) = 0$, has even one root which becomes rational through successive adjunction of roots of cyclic equations, then it must necessarily become reducible through this adjunction, since in the end a linear factor must separate off. Let P then be the group of our equation, after everything necessary has been adjoined so far that $f(x)$ is not yet reducible, but splits into factors through the next adjunction of the root ε of a cyclic equation of prime degree m. Then by §. 175, P must have a normal divisor Q of index m, to which the group of the equation reduces after the adjunction of ε, and the permutation group Q must be intransitive. If Q is the unit group, then $f(x) = 0$ is completely solved by the adjunction of ε. But if Q is still different from the unit group, then the theorems of §. 158 come into force. According to these, $f(x)$ splits after the adjunction of ε — since the index of Q is here a prime — into m factors, $\varphi(x, \varepsilon), \varphi_1(x, \varepsilon), \ldots \varphi_{m-1}(x, \varepsilon)$, of the same degree μ, and we have $\mu m = n$.

If, for example, n is a prime, then $\mu = 1$ must hold; the functions φ are linear, and the equation $f(x) = 0$ is completely solved; hence Theorem IV is correct for equations of prime degree. In the general case, one of the factors of degree μ, $\varphi(x, \varepsilon), \varphi_1(x, \varepsilon), \ldots \varphi_{m-1}(x, \varepsilon)$, say $\varphi(x, \varepsilon)$, must have a root which becomes rational through the adjunction of the roots of cyclic equations; and if we therefore assume that our theorem is already proved for equations of degree μ, it follows that $\varphi(x, \varepsilon) = 0$ itself, and hence also its group, is metacyclic.

But since by §. 158 the different equations $\varphi = 0$, $\varphi_1 = 0, \ldots \varphi_{m-1} = 0$ have the same group, they are all, and consequently also $f(x) = 0$, metacyclic.

Under the assumption, then, that Theorem IV is correct for equations of degree μ, its correctness follows for equations of degree μm; and since it holds for equations of prime degree, it is proved in general.

§. 177.
Simplicity of the alternating group.

We have seen earlier (§. 149) that if we regard the coefficients of an equation of degree n as independent variables, and the field of all rational functions of these coefficients as the domain of rationality, the Galois group of the equation is the symmetric group. In the symmetric group there is always contained a normal divisor of index 2, the alternating group, to which the group of the equation reduces when the square root of the discriminant is adjoined. For four figures the alternating group, which consists of the identical permutation, eight three-term cycles, and three pairs of transpositions, has the normal divisor of index 3:

$$1, \quad (0, 1)(2, 3), \quad (0, 2)(1, 3), \quad (0, 3)(1, 2).$$

This group has three different normal divisors of index 2, of which we prefer one, 1, $(0, 1)(2, 3)$, of which in turn the unit group is a normal divisor of index 2. The group of the 24 permutations of

[47] The expression "metacyclic groups" was first used by Kronecker, though in a more restricted sense. I should like here to propose this slight generalisation of an already known expression in place of the expression "solvable groups" used by Frobenius and Hölder.

four figures is therefore metacyclic, and on this rests every method of solution of the biquadratic equation (§. 160, 161).

We now wish to prove that if n is greater than 4, the alternating group has no normal divisors at all besides the unit group, or, in the designation introduced earlier, is simple. From this it then follows that the condition which we found to be necessary for the algebraic solvability of an equation is not fulfilled for equations of higher than the fourth degree whose group is the symmetric or the alternating group, and that therefore equations of higher than the fourth degree, so long as the coefficients are independent variables, are no longer algebraically solvable.

The proof that the alternating group is simple may be conducted as follows:

Let A be the alternating group of the permutations of n figures $0, 1, 2 \dots n-1$, and Q a normal divisor of A. We have seen in §. 153, 6. and §. 154, 6. that all permutations of A can be composed of three-term cycles, and that, if $\varkappa$ and π are any permutations whatever, one obtains the transformed permutation of $\varkappa$, $\pi^{-1}\varkappa\pi$, by carrying out in the cycles of $\varkappa$ the permutations π. Now if $\varkappa$ is a three-term cycle, say (0, 1, 2), then π can be chosen from A so that $\pi^{-1}\varkappa\pi$ represents any arbitrary three-term cycle of the n figures; for in

$$\pi = \begin{pmatrix} 0,\ 1,\ 2,\ 3 \dots n-1 \\ a_0,\ a_1,\ a_2,\ a_3 \dots a_{n-1} \end{pmatrix}$$

one may choose the first three figures a_0, a_1, a_2 arbitrarily, and, if it is necessary in order that π belong to A, further interchange a_1 and a_2. In this way there arises from $\varkappa$ one of the two cycles $(a_0,\ a_1,\ a_2)$, $(a_0,\ a_2,\ a_1)$, of which each is the second power of the other. Now if Q is a normal divisor of A, and $\varkappa$ a permutation. from Q, then $\pi^{-1}\varkappa\pi$ is also contained in Q whenever π is contained in A, and from this it follows that, if a three-term cycle occurs in Q, then Q is identical with A.

Our proof now rests on the fact that, if $\varkappa$ is any permutation in Q, then $\pi^{-1}\varkappa\pi$, and consequently also

$$\lambda = \varkappa^{-1}\pi^{-1}\varkappa\pi \tag{1}$$

must occur in Q; and it is then to be shown that, if $\varkappa$ is any non-identical permutation, π can always be so chosen from A that the permutation λ becomes a three-term cycle, and consequently Q identical with A.

For this purpose we suppose both $\varkappa$ and π decomposed into their cycles, and remark that in the formation of λ one need not take into account at all those cycles of $\varkappa$ whose figures remain unchanged by π, because they cancel each other in $\varkappa^{-1}$ and $\varkappa$. We must now consider the various possible forms of $\varkappa$ one by one.

1. Let $\varkappa$ contain a cycle of more than three figures, say $(1,\ 2,\ 3 \dots m)$; we take $\pi = (1,\ 2,\ 3)$ and obtain

$$\lambda = \varkappa^{-1}\pi^{-1}\varkappa\pi = (1,\ m,\ m-1 \dots 2)(2,\ 3,\ 1,\ 4 \dots m) = (1,\ 2,\ 4)$$

Thus a three-term cycle occurs in Q.

2. Let $\varkappa$ contain two three-term cycles $(1,\ 2,\ 3)(4,\ 5,\ 6)$. We take $\pi = (1,\ 3,\ 4)$ and obtain

$$\lambda = (1,\ 3,\ 2)(4,\ 6,\ 5)(3,\ 2,\ 4)(1,\ 5,\ 6) = (1,\ 2,\ 5,\ 3,\ 4).$$

This permutation λ, which is contained in Q, falls, however, under case 1.

3. Let $\varkappa$ contain a three-term and a two-term cycle $(1,\ 2,\ 3)(4,\ 5)$. (That in $\varkappa$, if it belongs to A, a second cycle of even number of terms must also occur, is immaterial here). For $\pi = (1,\ 2,\ 4)$ we get

$$\lambda = (1, 3, 2)(4, 5)(2, 4, 3)(1, 5) = (1, 2, 5, 3, 4),$$

which again falls under case 1.

4. Let $\varkappa$ contain three transpositions, $(1, 2)(3, 4)(5, 6)$. For $\pi = (1, 3, 5)$ it follows that

$$\lambda = (1, 2)(3, 4)(5, 6)(3, 2)(5, 4)(1, 6) = (1, 3, 5)(2, 6, 4),$$

which falls under case 2.

5. Let $\varkappa$ contain two transpositions and one unchanged element $(1, 2)(3, 4)(5)$. One sets $\pi = (1, 2, 5)$ and obtains

$$\lambda = (1, 2)(3, 4)(5)(2, 5)(3, 4)(1) = (1, 5, 2).$$

With this all cases are exhausted when $n > 4$. For $n = 4$ there still remains the one case in which $\varkappa$ consists of two transpositions, and it is just this that brings about the special behaviour at $n = 4$, by which the algebraic solution of the equation of degree 4 is made possible[48].

It follows further from this theorem that the symmetric group has no normal divisors other than itself, the alternating group and the unit group. For if S is the symmetric group, A the alternating group, and Q a normal divisor of S, then the greatest common divisor Q' of A and Q is a normal divisor of A, and is therefore equal to A or equal to 1.

If Q' is equal to A, then Q is either likewise equal to A or equal to S. But if Q' is $= 1$, then Q contains no permutation of the first kind besides the unit. If, then, $\varkappa$ and λ are two permutations of Q, different from one another and different from the unit, then $\varkappa^2$ and $\varkappa\lambda$, being of the first kind, must be $= 1$; that is, λ must be $= \varkappa$. Thus Q can contain at most one permutation $\varkappa$ different from the identical one. But since Q is to be a normal divisor of S, for every permutation π of the symmetric group we must have $\pi^{-1}\varkappa\pi = \varkappa$; that is, $\varkappa$ may not change when any permutation whatever is carried out in its cycles. But this is possible only when there are only two figures 1, 2 present at all, and $\varkappa = (1, 2)$. But then $1, (1, 2)$ is the whole group S.

§. 178.

Non-metacyclic equations in the field of the rational numbers.

By the theorem of the preceding section the proof has been given that an equation of degree n, when n is greater than 4, can no longer be solved algebraically if the coefficients are regarded as independent variables. But of still greater interest is the question whether there are, in the field of the rational numbers, equations of degree n that are not algebraically solvable. The question can be put somewhat more generally, namely thus: whether there are equations with integral coefficients whose group is the symmetric group, and which therefore, according to a mode of expression explained earlier, have no affect.

If one forms the Galois resolvent $G(t) = 0$ of degree $\Pi(n)$ of a general equation of degree n, $f(x) = 0$, with indeterminate coefficients a, then $G(t)$ is an integral function of the variables t and a which cannot be decomposed into factors that are again rational functions of t and a. If one substitutes for the variables a quantities of any field Ω, and $G(t)$ then becomes reducible in this field, the equation $f(x) = 0$ has an affect in the domain of rationality Ω. Our question thus comes down to whether one can put for the variables a such rational numbers in $G(t)$ that $G(t)$ remains irreducible in the field of the rational numbers.

[48]The first complete proof that the general equation of degree higher than the 4th is not solvable by radicals is due to Abel (Crelle's Journal, Bd I, 1826). On the older attempts at a proof by Ruffini (1799 to 1806) and their relation to the Abel proof, compare the memoir of Burkhardt, "Die Anfänge der Gruppentheorie und Paolo Ruffini" (Abhandlungen zur Geschichte der Mathematik VI. Supplement zu Schlömilch's Zeitschrift. Leipzig 1892).

This question has found a quite general answer in a memoir by Hilbert[49], in which the general theorem is proved that in an irreducible function of arbitrarily many variables one can substitute for an arbitrary part of the variables rational numbers such that an irreducible function of the remaining variables results. We cannot enter upon this general theorem here.

We shall, however, answer the question posed much more simply, though by far not so generally, by showing that equations without affect can be found for every prime degree.

We have proved in §. 153, 9. that a transitive permutation group of n figures which is not the symmetric group, if n is a prime, can contain no single transposition.

By an equation with an affect we have understood one whose group is not the symmetric group. If, then, the irreducible equation $f(x) = 0$ has an affect and its degree is a prime, its group P must be transitive, and it can contain no transposition of two roots. If therefore we adjoin any $n-2$ of the roots to the domain of rationality, the group must reduce to the unit group, since besides this only the permutation of the two last roots could still remain, which does not occur in P. From this it follows, then, that the two last roots are contained in the extended field Ω, or the theorem:

1. If an irreducible equation whose degree n is a prime has an affect, then any two of its roots can be expressed rationally in terms of the remaining ones.

From this follows as a corollary:

2. If the field Ω contains only real numbers, then an irreducible equation of prime degree n with an affect in Ω cannot have two imaginary and $n-2$ real roots.

Now there are, however, innumerable equations of every arbi-

trary degree n, with real coefficients, which have two conjugate imaginary and $n-2$ real roots; for in

$$\begin{aligned} f(x) &= (x-\alpha_1)(x-\alpha_2)\dots(x-\alpha_n) \\ &= x^n + a_1x^{n-1} + a_2x^{n-2} + \dots + a_n \end{aligned}$$

one may take α_1, α_2 conjugate imaginary at will and the remaining α real. But the number of real and imaginary roots of $f(x)$ does not change when the coefficients are varied continuously within certain finite limits (cf. the seventh section), and consequently there are also such equations whose coefficients are rational numbers, since one can always find rational numbers in arbitrary nearness to any given real numbers.

It remains, then, only to show that these rational coefficients can be so chosen that $f(x)$ becomes irreducible. But this follows from the following theorem:

3. If p is a prime, $c_0, c_1, c_2 \dots c_n$ a sequence of integers, of which c_0 is not divisible by p, while $c_1, c_2 \dots c_n$ are divisible by p, but c_n is not divisible by p^2, then the function

$$\varphi(x) = c_0x^n + c_1x^{n-1} + \dots + c_n$$

is irreducible.

For if $\varphi(x)$ splits into two factors with rational coefficients, then (by §. 2) the factors may also be assumed to have integral coefficients. So let

$$\varphi(x) = (\alpha_0x^h + \alpha_1x^{h-1} + \dots + \alpha_h)(\beta_0x^k + \beta_1x^{k-1} + \dots + \beta_k),$$

h and k both greater than zero and their sum equal to n.

Since $\alpha_h\beta_k = c_n$, one of the two factors α_h, β_k must be divisible by p, and the other not. Suppose that β_k is divisible by p and α_h not divisible; and since not all the β can be divisible by p, because otherwise c_0 too would be divisible by p, let β_ν be not divisible by p, and $\beta_{\nu+1}, \beta_{\nu+2} \dots \beta_k$ divisible by p. The coefficient of $x^{k-\nu}$ in the product of the two factors is then $\alpha_h\beta_\nu + \alpha_{h-1}\beta_{\nu+1} + \dots$, and

[49] Ueber die Irreducibilität ganzer rationaler Functionen mit ganzzahligen Coëfficienten. Journal für Mathematik, Bd. 110.

hence not divisible by p. It would therefore have to be that $k - \nu = n$, which is not possible, since already $k < n$ must hold.

If, then, one sets for the coefficients of $f(x)$:

$$a_1 = \frac{c_1}{c_0}, \quad a_2 = \frac{c_2}{c_0} \cdots a_n = \frac{c_n}{c_0},$$

then $f(x)$ is irreducible, and one still has freedom enough in the choice of the integers c to bring the rational fractions $a_1, a_2 \ldots a_n$ arbitrarily near to any arbitrarily given system of values.

But with this the theorem is proved:

4. Of every arbitrary prime degree there are infinitely many equations with rational coefficients without affect.

The proof that has been given here shows that such affect-free equations can be found whose coefficients fill a finite region everywhere densely — namely the region in which the real coefficients of the equations with only two conjugate imaginary roots lie. The proof is therefore, apart from the fact that it is restricted to prime degrees, also not exhaustive in this respect, that it gives us no information about the remaining regions, in which in all probability the matter stands just the same.

§. 179.
Solution by real radicals.

In the solution of the cubic equations with real coefficients, the two cases in which the discriminant is negative or positive have been distinguished from ancient times. In the second case, although the equation then has precisely three real roots, one has, in the application of the Cardan formula, to extract the cube root of an imaginary expression, and the attempt to represent the roots in real form leads always back to a cubic equation of the same character. For this reason this case has been called the casus irreducibilis. The cubic equations of the casus irreducibilis are a special case of the cyclic equations with real roots, which we came to know in §. 165, and in whose solution likewise roots of imaginary quantities, or what comes to the same thing, divisions of angles, occur. That these roots of imaginary quantities or divisions of angles can in no way be avoided, we can now prove.[50]

We assume a real domain of rationality Ω, and take in it a normal equation $g(t) = 0$, i. e. an irreducible equation whose roots are all rationally expressible in terms of an arbitrary one of them, ϱ. If one of the roots is real, then all the others must be real as well, and $g(t)$ thus has either only real or only pairwise conjugate imaginary roots. $g(t) = 0$ can be the Galois resolvent of any given equation $F(x) = 0$, whether irreducible or reducible, and if therefore $g(t)$ has only real roots, then $F(x)$ too can have only real roots, because the roots of $F(x)$ are rationally representable in terms of a root of $g(t)$. If among the roots of $F(x)$ there are imaginary ones, then $g(t)$ has only imaginary roots. But if $F(x)$ has only real roots, then the roots of $g(t)$ are also real, because they can indeed be expressed rationally in terms of the roots of $F(x)$.

Now if $g(t)$ is reduced by the adjunction of a root ε of an irreducible equation $\chi = 0$ whose degree is a prime, then, as we have seen in §. 157, all the roots of χ are contained in $\Omega(\varrho)$, and if therefore ϱ is real, then all the roots of χ are real.

1. A normal equation with real roots can therefore be reduced only by such irreducible equations of prime degree as have only real roots.

We now ask whether a reduction of the normal equation can be effected by the adjunction of a real radical $\sqrt[p]{a}$. We may here assume that the degree p of the radical is a prime, because every radical can be reduced to a series of extractions of roots of prime degree. We can also presuppose that a is not the pth power of a rational quantity, because otherwise the real root $\sqrt[p]{a}$ would be rational.

[50] Cf. Hölder, Mathematische Annalen, Bd. 38; Kneser, ebendas., Bd. 41.

Under this assumption, as we now wish to prove, $x^p - a$ is irreducible. For if we denote one, e. g. the real value $\sqrt[p]{a}$, by α, and an imaginary pth root of unity by ε, then

$$\alpha,\ \varepsilon\alpha,\ \varepsilon^2\alpha \ldots \varepsilon^{p-1}\alpha \tag{1}$$

the roots of the equation

$$x^p - a = 0. \tag{2}$$

If this equation is reducible, then

$$x^p - a = f_1(x) f_2(x), \tag{3}$$

and f_1, f_2 are functions in Ω of degree lower than p. One part of the roots (1) will fall on $f_1 = 0$, another part on $f_2 = 0$. If then μ is the degree of f_1, then for some exponent λ

$$\varepsilon^\lambda \alpha^\mu = b$$

must be a quantity in Ω [the term in $f_1(x)$ independent of x]; hence, if one raises to the pth power,

$$a^\mu = b^p. \tag{4}$$

Now $0 < \mu < p$, and therefore μ and p are relatively prime, so that two integers h and k can be determined from the equation

$$\mu h + pk = 1$$

Then by (4)

$$a = a^{\mu h} a^{pk} = (b^h a^k)^p;$$

thus, contrary to the hypothesis, a would be the pth power of a quantity in Ω.

This theorem, as one sees, is independent of the assumption that Ω be a real field. But if we take it to be real, and p is not equal to 2, then the equation (2) has imaginary roots, and by 1. cannot reduce a normal equation $g(t) = 0$. But if $p = 2$, then $g(t)$ can be reduced by this equation (2) only if a is positive and the degree of $g(t)$ is an even number; hence:

2. A normal equation of odd degree cannot be reduced by the adjunction of a real radical.

This case occurs in the casus irreducibilis of the cubic equation, when one has adjoined the square root of the discriminant to the domain of rationality of the coefficients. For since the discriminant is positive, the domain of rationality remains real, and it also remains real however many real radicals one adjoins. If the equation is thus to be solvable by real radicals, it must at last split under such adjunctions, which by 2. is impossible.

In the same case are found the cyclic and in general all irreducible Abelian equations of odd degree, which therefore can never be solved by real radicals.

Accordingly we can, for example, distinguish the cubic equations in the field of the rational numbers into three or four kinds, which all occur in certain geometric problems famous from antiquity. We here leave the reducible equations out of account. The degree of the Galois group must then always be divisible by 3 (§. 154, 7.) and is therefore either equal to 3 or equal to 6. By §. 157 the group can therefore never be reduced by adjunction of mere square roots to a lower than the third degree; hence the equation also cannot be solved by square roots. The corresponding geometric problems cannot be solved with compass and ruler.

If the group is of degree 3, we have a cyclic equation. Here belong the equations arising from the division of the circle, e.g. the one on which the construction of the regular heptagon depends. The three roots of such an equation are real and cannot be expressed by real radicals. Among the cubic equations with a group of degree 6 we must distinguish those with positive and those with negative discriminant.

The first belong to the casus irreducibilis and can be reduced to the equation on which the trisection of an arbitrary angle depends. To the equations with negative discriminant belong also, as special cases, the pure cubic equations $x^3 = a$, when a is not a cube number. For $a = 2$ there results the equation on which the Delian problem of the duplication of the cube depends.

§. 180.

Metacyclic equations of prime degree.

The general conditions which we found in §. 176 are not yet simple enough to permit an immediate application to the finding of metacyclic equations or to the decision on the solvability of a proposed

equation to admit solution by radicals. We therefore derive, at first under the assumption that the degree n of the equation is a prime, another criterion first set up by Galois.

Now let $f(x) = 0$ be an irreducible equation of degree n, and n a prime greater than 2. If $f(x)$ is to be algebraically solvable, then by §. 176 its group P must be metacyclic, i.e. there must be a chain of groups

$$P,\ P_1,\ P_2 \ldots P_{\mu-1},\ 1 \tag{1}$$

each of which is a normal divisor of the one immediately preceding, with prime index. By successive adjunction of roots of cyclic equations of prime degree the group of the equation is reduced from P to $P_1, P_2 \ldots P_{\mu-1}, 1$.

The function $f(x)$ itself, as was already shown in §. 176, cannot be reduced before the last adjunction has been made, whereupon it splits into n linear factors. The degree of the last cyclic equation, and hence the degree of the next-to-last group $P_{\mu-1}$, must therefore by §. 158, 3. be equal to n.

The degree of an element of a group is always a divisor of the degree of the group, and hence $P_{\mu-1}$ can contain, besides the identical one, only permutations of order n. It is therefore identical with the period $1, \pi, \pi^2 \ldots \pi^{n-1}$ of one of its elements. From this it follows by §. 153 that π must be a cyclic permutation of all n figures, and we can choose the designation of the roots of $f(x)$ so that

$$\pi = (0,\ 1,\ 2 \ldots n-1)$$

holds. If we denote the roots by x_z, and stipulate that $x_z = x_{z'}$ shall hold when $z \equiv z' \pmod{n}$, then by π every z passes into $z+1$, by π^2 into $z+2$, and so on, and we can therefore say that the group $P_{\mu-1}$ consists of the substitutions for z

$$(z,\ z+b), \qquad b = 0,\ 1,\ 2 \ldots n-1 \tag{2}$$

Every irreducible metacyclic equation of prime degree can therefore be transformed by adjunction of radicals into a cyclic equation.

The substitutions $(z, z+b)$ form a special case of the more general linear substitution

$$(z,\ az+b), \tag{3}$$

in which a and b are fixed numbers, which may also be reduced according to the modulus n, but where for a the value 0 of course

is to be excluded, because otherwise by (3) all the different z would pass into the same b, and thus no permutation of the quantities x_z would be expressed at all. But if a is different from 0 (mod n), then a permutation will always be represented by (3), because then $az + b \equiv az' + b$ can hold only when $z \equiv z'$. We call (3) a substitution for z, which produces a permutation among the x. The number of the different linear substitutions of the form (3) is $n(n-1)$. To them correspond just as many different permutations among the x; for it is only then that, for every z,

$$az + b \equiv a'z + b' \pmod{n}$$

can hold, when $b \equiv b'$ (to be concluded from $z = 0$) and $a \equiv a'$ (to be concluded from $z = 1$).

The totality of the permutations that are represented by (3) forms a group; for let

$$\lambda = (z,\ az + b), \qquad \lambda' = (z,\ a'z + b')$$

be two of these permutations; then

$$\lambda\lambda' = \begin{pmatrix} x_z \\ x_{az+b} \end{pmatrix} \begin{pmatrix} x_z \\ x_{a'z+b'} \end{pmatrix} = \begin{pmatrix} x_z \\ x_{a'(az+b)+b'} \end{pmatrix},$$

hence

$$\lambda\lambda' = \lambda'' = (z,\ aa'z + a'b + b'), \tag{4}$$

which is again of the form $(z, a''z + b'')$.

The group formed by all substitutions of the form (3), which is a generalisation of the cyclic group (2), we shall call a linear group[51]; the permutations contained in it shall be called linear permutations.

In the linear group various divisors are contained. Among them we find a group of degree $n - 1$, namely (z, az), which may itself in turn have divisors.

The composition rule (4) shows how we can form all divisors of the linear group. If $\lambda = (z, az+b)$ is a linear substitution, then by (4), through repetition, we obtain for every exponent h

$$\lambda^h = [z,\ a^h z + (1 + a + \cdots + a^{h-1})b],$$

and for $a = 1$

$$\lambda^h = (z,\ z + hb).$$

If, then, in a group L there occurs a substitution λ in which $a = 1$ and b is different from zero, then by the above L contains the whole cyclic group $(z, z + b)$ $(b = 0, 1, \ldots n - 1)$.

If a is not equal to 1, we set

$$1 + a + \cdots + a^{h-1} = \frac{a^h - 1}{a - 1},$$

and conclude from this that, if e is the least positive exponent for which $a^e \equiv 1 \pmod{n}$, then λ is of degree e.

The period of the substitution λ is, unless $a = 1$, an intransitive group. There is a figure z which is not altered by λ and its repetitions, which is determined from the congruence $z \equiv az + b \pmod{n}$, and can be denoted by

$$z_0 \equiv \frac{b}{a - b} \pmod{n}$$

if by $\frac{\mu}{\nu} \pmod{n}$ is understood an integer which, by multiplication with the factor ν, becomes congruent to μ with respect to the modulus n.

The cyclic group $(z, z + b)$ is transitive and of degree n. It can be shown that every transitive linear group (mod n) must contain the cyclic group.

Let g be a primitive root of n and α the index of a, i. e. $g^\alpha \equiv a \pmod{n}$ (§. 136). For brevity we call α, for the moment, also the index of the substitution $\lambda = (z, az + b)$. The index may be assumed to be less than $n - 1$ and equal to or greater than zero. If it is equal to zero, then λ is either the identical substitution, or it belongs to the cyclic group. By (4) the rule holds that the index of a compounded substitution is equal to the sum of the indices of the components.

[51] Kronecker calls only this group metacyclic.

From this it follows that all the indices of the substitutions of a linear group L are multiples of the least positive one among them, and that consequently, if α_0 is this least positive index and $a^{\alpha_0} = a_0$ is set, all the substitutions of L can be represented in the form

$$\lambda = (z, a_0^h z + b)$$

in which a_0 is held fixed, and only h and b run through certain sequences of numbers.

A substitution of the form $\lambda_0 = (z, a_0 z + b_0)$ must occur in the group L.

If we now form, according to (4), the composition

$$\lambda\lambda_0^{-h} = \left(z,\, z + a_0^{-h} b + \frac{a_0^{-h} - 1}{a_0 - 1} b_0\right),$$

then we obtain the identical substitution when, and only when,

$$\frac{b_0}{a_0 - 1} \equiv \frac{b}{a_0^h - 1} \pmod{n}.$$

If this congruence is satisfied for all substitutions λ of L, then the element $\dfrac{b_0}{a_0 - 1}$ remains unchanged by the whole group, and L is intransitive. Consequently, in a transitive group L there is always an element λ for which $\lambda\lambda_0^{-h}$ is not the identical substitution, although the index is zero, and L therefore contains the whole cyclic group.

If one forms the various powers of λ_0, it turns out that the exponent h in the substitutions λ of the group L can take every value. If e is the least positive exponent that satisfies the condition $a_0^e \equiv 1 \pmod{n}$, then the values $h = 0, 1, 2 \ldots e - 1$ therefore occur, and here e is a divisor of $n - 1$. From the composed substitution

$$(z, a_0^h z + b)(z, z + 1) = (z, a_0^h z + b + 1),$$

which after all must also occur in L, one sees further that with every value of h each of the values $b = 0, 1, 2, \ldots n - 1$ occurs in combination, and the whole group L therefore consists of the substitutions

$$\lambda = (z, a_0^h z + b), \qquad \begin{matrix} h = 0, 1 \ldots e - 1 \\ b = 0, 1 \ldots n - 1' \end{matrix} \tag{5}$$

and is of degree en. Conversely, every system of substitutions of this form constitutes a group.

Of the linear groups the following theorem now holds:

I. If a transitive linear group L is a normal divisor of another permutation group P of the same n figures, then P is also a linear group.

The theorem can be proved as follows: If π is an arbitrary permutation:

$$\pi = \begin{pmatrix} 0, & 1, & 2 \ldots n - 1 \\ a_0, & a_1, & a_2 \ldots a_{n-1} \end{pmatrix},$$

which can also be represented by (z, a_z), then one can determine an integral function $\varphi(z)$ of z with rational coefficients, whose degree is not higher than $n - 1$, so that $a_z = \varphi(z)$ can be set. One need only determine the n coefficients in $\varphi(z)$ from the n linear equations

$$a_0 = \varphi(0), \quad a_1 = \varphi(1), \ldots a_{n-1} = \varphi(n - 1)$$

For this one may use the Lagrange interpolation formula derived in §. 29, which, however, can be given a still simpler form, since here only congruences with respect to the modulus n come into question. Namely, if one sets

$$\psi(z) = z(z - 1)(z - 2) \ldots (z - n + 1),$$

the interpolation formula just mentioned gives

$$\varphi(z) = \psi(z)\left(\frac{a_0}{\psi'(0)z} + \frac{a_1}{\psi'(1)(z-1)} + \cdots + \frac{a_{n-1}}{\psi'(n-1)(z-n+1)}\right).$$

Now, as we have seen in §. 136, the congruence

$$\psi(z) \equiv z^n - z \pmod{n}$$

is identical; hence also

$$\psi'(z) \equiv nz^{n-1} - 1 \equiv -1 \pmod{n},$$

and therefore we can represent $\varphi(z)$ with integral coefficients as follows:

$$\varphi(z) \equiv -a_0\frac{\psi(z)}{z} - a_1\frac{\psi(z)}{z-1} - \cdots - a_{n-1}\frac{\psi(z)}{z-n+1} \pmod{n}. \tag{6}$$

If, as in our problem, the numbers $a_0, a_1 \ldots a_{n-1}$ agree, apart from the order, with the numbers $0, 1, 2 \ldots n-1$, then, if $n > 2$, we have $a_0 + a_1 + \cdots + a_{n-1} \equiv 0 \pmod{n}$, and hence $\varphi(z)$ is at most of degree $n-2$, which, however, is of no significance for our proof.

Accordingly, then, every arbitrary permutation π can be represented by a substitution $[z, \varphi(z)]$.

Now if L is a transitive linear group and at the same time a normal divisor of another group P, and if λ is an arbitrary permutation of L, and π a likewise arbitrary permutation of P, then $\pi^{-1}\lambda\pi = \lambda'$ is contained in L, and $\lambda\pi = \pi\lambda'$. If, by what has just been proved, we set $\pi = [z, \varphi(z)]$ and choose for λ the cyclic substitution $(z, z+1)$, which by the hypothesis of transitivity occurs in L, then a' and a must admit of being determined so that

$$(z, z+1)[z, \varphi(z)] = [z, \varphi(z)](z, a'z + a),$$

or, if one carries out the composition,

$$\varphi(z+1) \equiv a'\varphi(z) + a \pmod{n} \tag{7}$$

for every integer z. Since $\varphi(z)$ does not reach the degree n, the coefficients of the individual powers of z on the two sides of (7) must (§. 136) be congruent, and from the comparison of the coefficients of the highest powers of z it follows that $a' \equiv 1 \pmod{n}$, and hence

$$\varphi(z+1) \equiv \varphi(z) + a \pmod{n}.$$

If one here puts $z+1, z+2 \ldots z+h-1$ for z, it follows that

$$\varphi(z+h) = \varphi(z) + ah,$$

where h can be congruent to any arbitrary integer with respect to the modulus n. In this one may now set $z = 0$, and obtains, on writing z again for h and setting $\varphi(0) = b$,

$$\varphi(z) = az + b. \tag{8}$$

The group P therefore consists entirely of linear permutations, and our theorem is proved.

We add further the remark, which follows immediately from an inspection of the formulae, that the cyclic group is a normal divisor of every linear group in which it is contained at all.

If we apply the theorem just proved to the group of the metacyclic equation, we obtain the theorem of Galois:

II. The group of a metacyclic equation of prime degree is linear.

For if we return to the chain of groups (1), we have seen that $P_{\mu-1}$ is the cyclic group. If $P_{\mu-1}$ is not identical with P, then $P_{\mu-1}$ is a normal divisor of $P_{\mu-2}$, and hence $P_{\mu-2}$ is linear. $P_{\mu-2}$ is in

turn a normal divisor of $P_{\mu-3}$, hence $P_{\mu-3}$ too is linear, and so on, until we reach the conclusion that P itself must also be linear. At the same time it follows that $P_{\mu-1}$ is a normal divisor of all the preceding groups, and hence also of P itself.

The converse theorem also holds:

III. Every irreducible equation of prime degree whose group is linear is metacyclic.

To prove it, it suffices to show that every transitive linear group L has a normal divisor L' whose index

a prime, and which itself, if it is not the unit group, is a transitive linear group.

But this is shown immediately by the representation of the substitutions of the group L by the formula (5).

If in it $e = 1$, then L is the cyclic group of degree n with the normal divisor 1. But if $e > 1$, let p be a prime dividing e, and $e = pe'$. The linear group L' of degree ne', which consists of the substitutions:

$$\lambda' = (z, a_0^{ph} z + b), \quad \begin{matrix} h = 0, 1, 2 \dots e' - 1 \\ b = 0, 1, 2 \dots n - 1, \end{matrix}$$

is certainly a divisor of L of index p. But it is also normal, as one recognises without difficulty from the rule of composition (4).

The concepts of the transitive linear and of the metacyclic groups thus coincide completely for the equations of prime degree, and we can therefore in what follows also use the two expressions synonymously.

If we wish to pass from a group P to a conjugate group $P' = \pi^{-1} P \pi$, this transition is effected by carrying out the permutation π in the cycles of all the permutations of P. P' will therefore agree with P up to the designation of the roots, and is therefore, if P is linear, also to be called linear, even though a change in the numbering of the roots is necessary in order to represent it by linear substitutions.

To simplify the application, let it further be observed that one can derive the full linear group by repetition and composition of the two substitutions

$$s = (z, z + 1), \quad t = (z, gz), \tag{9}$$

and every divisor L of the full linear group likewise from

$$s = (z, z + 1), \quad t^{\alpha_0} = (z, a_0 z) \tag{10}$$

One therefore calls the two substitutions (9) or (10) the generating substitutions of these groups.

If we set $a_0 = g^2$, we obtain a frequently occurring linear group $(z, az + b)$, in which a runs only through the quadratic residues of n, and which has been called by Kronecker the half-metacyclic group.

This representation by the generating substitutions admits a simple inference concerning the relation of the meta-

cyclic groups to the symmetric and the alternating group.

The permutation arising from s consists of a single cycle with an odd number of terms $(0, 1, 2 \dots n - 1)$ and therefore belongs to the alternating group. The substitution t leaves the index 0 unchanged and yields for the remaining figures again a single cycle. For by t, 1 passes into g, g into g^2, g^2 into g^3, etc., and since the powers $1, g, g^2 \dots g^{n-2}$, apart from their order, coincide

with the figures $1, 2 \ldots n-1$, t corresponds to the cycle $(1, g, g^2 \ldots g^{n-2})$, which consists of an even number of terms and consequently belongs to the permutations of the second kind, and is therefore not contained in the alternating group. On the other hand t^2 is again contained in it.

From this the result follows:

IV. The full metacyclic group is not a divisor of the alternating group. The greatest common divisor of the two groups is the half-metacyclic group.

Every transitive divisor of the metacyclic group is itself metacyclic.

The condition for the metacyclic equations can be given various forms, which can be derived from what has gone before.

The full linear group is contained in the symmetric group as a divisor of index $\nu = 1.2.3 \ldots (n-2)$. A function y of the n variables $x_0, x_1 \ldots x_{n-1}$ belonging to the full linear group, which we may call a metacyclic function, therefore satisfies a resolvent $F(y) = 0$ of degree ν, whose coefficients are symmetric functions of the x.

If now for x one substitutes the roots of a metacyclic equation $f(x) = 0$, then y becomes rational.

If conversely the function y becomes rational by the substitution of the roots of an irreducible equation $f(x) = 0$ for the x, while $F'(y)$ remains different from zero, then $f(x)$ is metacyclic; for then the group of $f(x) = 0$ is certainly a divisor of the full metacyclic group, and therefore, since $f(x)$ is irreducible, itself metacyclic. But it also suffices for the algebraic solubility of $f(x) = 0$ if the resolvent $F(y) = 0$ has any rational root at all, which

is not a double root. For the different roots of this resolvent belong to conjugate groups, and if therefore one of these roots is rational, then one of the conjugate groups is metacyclic. We therefore have the theorem:

V. The necessary and sufficient condition for the solvability of the equation $f(x) = 0$ by radicals is that the resolvent of degree ν, $F(y) = 0$, which one has arranged, by a suitable choice of the function y, so that it receives no double roots, has a rational root.

Another form of this condition results in the following way.

Among the permutations of a linear group, only the identical one leaves any two figures unchanged. For if $(z, az + b)$ does not alter two figures, then the congruence $az + b \equiv z \pmod{n}$ must have two distinct solutions. But this is possible only if $a \equiv 1$, $b \equiv 0 \pmod{n}$.

If, then, the group P of $f(x)$ is metacyclic, it reduces itself, by the adjunction of two arbitrary roots, to the unit, and consequently all the roots are rationally expressible in terms of any two among them. Hence:

VI. If an irreducible equation of prime degree is metacyclic, then all the roots are rationally expressible in terms of any two among them.

But this theorem also holds conversely, which we can prove simply in the following manner.

Suppose an irreducible equation of prime degree n has the property that all the roots are rationally expressible in terms of any two among them. If, say, $x_\nu = \psi(x_0, x_1)$ is such a representation, then all the permutations of the group P of our equation can be applied to this relation, and if therefore one of these permutations leaves x_0 and x_1 unchanged, then it also leaves all the x_ν unchanged, i.e. it is the identical permutation. Hence P contains, besides the identical permutation, none that leaves two figures unaltered.

If now one of the permutations $\pi = cc_1 \ldots$ of P contains two or more different cycles $c, c_1 \ldots$, and c is of degree h, but c_1 of a higher degree, then $\pi^h = c_1 \ldots$, and π^h leaves

the figures of c unchanged, while it is nevertheless not the identical substitution, because c_1^h does not leave all figures unchanged. But this is not possible when $h > 1$. Hence either π must leave one figure unchanged, or it must consist of cycles of equal degree. But since n is a prime, π can in this case form only a single cycle of degree n.

Thus P contains, besides the identical permutation, only cyclic permutations of degree n, which we shall denote by γ, and permutations $\varkappa$ which leave one figure unchanged and moreover consist

of cycles of equal degree.

Each of the permutations $\varkappa$ by which the figure 0 is left unchanged we shall denote by $\varkappa_0$. Likewise $\varkappa_1, \varkappa_2 \ldots \varkappa_{n-1}$ denote the permutations $\varkappa$ which leave the figures 1, $2 \ldots n-1$ unchanged. Since P is transitive, there must be just as many $\varkappa_0$ as $\varkappa_1$, as $\varkappa_2$, etc.

For if π is a permutation which carries 0 into 1, then every $\pi^{-1}\varkappa_0\pi$ is a $\varkappa_1$, and conversely every $\pi\varkappa_1\pi^{-1}$ is a $\varkappa_0$, and likewise for the remaining figures. If then μ is the number of the $\varkappa_0$, ν the number of the γ, m the degree of P, then, since the identical permutation is also to be added, we have

$$m = \mu n + \nu + 1. \tag{11}$$

Now the $\varkappa_0$, together with the identical permutation, form a group Q of degree $\mu + 1$ and a divisor of P, namely the totality of all permutations of P which leave 0 unchanged. If then $\pi_1, \pi_2 \ldots \pi_{n-1}$ are permutations in P which carry 0 into $1, 2 \ldots n-1$, then $Q\pi_1$ is the totality of all those permutations of P which carry 0 into 1, and so forth.

We thus obtain the decomposition of P into the subsidiary groups

$$P = Q + Q\pi_1 + Q\pi_2 + \cdots + Q\pi_{n-1},$$

from which follows

$$m = n(\mu + 1), \tag{12}$$

and hence from (11) and (12) $\nu = n - 1$.

There are therefore $n-1$ and no more cyclic permutations of degree n in P, and these consequently form, again together with the unit, a cyclic group

$$C = 1, \gamma, \gamma^2 \ldots \gamma^{n-1},$$

since with γ all powers of γ likewise occur in P.

Now if γ is a cyclic permutation of degree n, then every $\pi^{-1}\gamma^h\pi$ is also cyclic, and hence, if π belongs to P

belongs, must also be contained in C. C is therefore a normal divisor of P, and consequently, by Theorem I., the group P must be linear.

VII. An irreducible equation of prime degree, in which all the roots are rationally expressible in terms of any two of them, is metacyclic.

From these theorems we further infer a remarkable property of metacyclic equations, first noticed by Kronecker, for the case of a real domain of rationality[52]. If in such an equation two roots are real, then it follows from VI that all its roots are real. But such an equation, since it is of odd degree, must always have one real root. Hence follows

VIII. An irreducible metacyclic equation of odd prime degree with real coefficients has either all its roots real or only one.

If all the roots are real, the discriminant, as a product of nothing but squares of real quantities $(x_h - x_k)^2$, is positive. If only one root is real, then to every complex factor of the discriminant $(x_h - x_k)^2$ there corresponds a conjugate one, and their product is positive. Only when x_h and x_k are conjugate imaginary, so that $x_h - x_k$ is purely imaginary, is $(x_h - x_k)^2$ negative, and the discriminant thus receives the factor -1 for every pair of conjugate imaginary roots. The sign of the discriminant is therefore $(-1)^{\frac{n-1}{2}}$. From this follows:

[52] Ueber algebraisch auflösbare Gleichungen. Monatsbericht der Berliner Akademie, 14. April 1856.

IX. When $n \equiv 1 \pmod 4$, the discriminant is always positive; but if $n \equiv 3 \pmod 4$, the sign of the discriminant decides which of the two cases of Theorem VIII. occurs.

§. 181.
Application to the metacyclic equations of degree 5.

We make one further application of the general theorems to the equations of degree 5. If $n = 5$, the cyclic group has 5, the half-metacyclic 10, and the full metacyclic group 20 permutations. For a general equation of degree 5, a metacyclic function satisfies a resolvent of degree 6, and if this equation of degree 6 has a rational root, then the given equation of degree 5 is metacyclic. The half-metacyclic functions, i. e. the functions that admit the permutations of the half-metacyclic group, satisfy a resolvent of degree 12, which however, after adjunction of the square root of the discriminant, splits into two factors of degree 6.

If one applies the generating substitutions for the modulus 5

$$s = (z, z+1), \quad t = (z, 2z), \quad t^2 = (z, 4z)$$

to the figures $0, 1, 2, 3, 4$, there result the permutations

$$\begin{array}{cccccc} & 0, & 1, & 2, & 3, & 4 \\ (s) & 1, & 2, & 3, & 4, & 0 \\ (t) & 0, & 2, & 4, & 1, & 3 \\ (t^2) & 0, & 4, & 3, & 2, & 1. \end{array} \tag{1}$$

If we apply the substitutions s, t^2, t to the pairs of figures $(0,1), (1,2), (2,3), (3,4), (4,0)$, it follows that

$$\begin{array}{ccccccc} & (0,1), & (1,2), & (2,3), & (3,4), & (4,0) \\ (s) & (1,2), & (2,3), & (3,4), & (4,0), & (0,1) \\ (t^2) & (0,4), & (4,3), & (3,2), & (2,1), & (1,0); \end{array} \tag{2}$$

these pairs therefore remain unchanged, in their totality, by s and by t^2, and consequently by the half-metacyclic group, and a symmetric function of the corresponding pairs of roots is half-metacyclic. The substitution t effects the permutations

$$\begin{array}{ccccccc} & (0,1), & (1,2), & (2,3), & (3,4), & (4,0) \\ (t) & (0,2), & (2,4), & (4,1), & (1,3), & (3,0), \end{array} \tag{3}$$

thus carries the five pairs over into five others; and since there are only ten pairs of five things, all pairs occur in the two sequences (3).

One can now form half-metacyclic functions in very different ways. The simplest is

$$u = x_0x_1 + x_1x_2 + x_2x_3 + x_3x_4 + x_4x_0, \tag{4}$$

which by t passes into

$$u' = x_0x_2 + x_2x_4 + x_4x_1 + x_1x_3 + x_3x_0 \tag{5}$$

and u' itself belongs to the half-metacyclic functions.

If

$$f(x) = x^5 - ax^4 + bx^3 - cx^2 + dx - e = 0 \tag{6}$$

is the equation whose roots are x_0, x_1, x_2, x_3, x_4, then accordingly

$$u + u' = b. \tag{7}$$

The function

$$y = u - u' \tag{8}$$

is likewise half-metacyclic, but the square y^2 is full metacyclic and therefore the root of a resolvent of degree 6, as Lagrange already found.

If $\sqrt{\Delta}$ is the product of the ten root differences $(x_0 - x_1)$, $(x_0 - x_2)\ldots$, so that Δ is the discriminant of $f(x)$, then

$$Y = \frac{u - u'}{\sqrt{\Delta}} \tag{9}$$

is likewise full metacyclic and the root of an equation of degree 6 [1].

In order to form the conjugate functions to u, we denote by M the full metacyclic, by N the half-metacyclic, by A the alternating group, and decompose A into the subsidiary groups:

$$\begin{aligned} A = {} & N + N(1,2)(3,4) + Nt(0,1) + Nt(0,2) \\ & + Nt(0,3) + Nt(0,4). \end{aligned} \tag{10}$$

These subsidiary groups are in fact all different from one another. For if, e. g., $N = N(1,2)(3,4)$, then $(1,2)(3,4)$ would have to occur in N, and hence

$$(1,2)(3,4)t = (1,2)(3,4)(1,2,4,3) = (1,4)$$

would have to occur in M, and this is not possible, because in M no permutation appears that leaves two figures unchanged; and in

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[1] Jacobi, "observatiunculae ad theoriam aequationum pertinentes", Crelle's Journ. Bd. 13. Jacobi's Werke, Bd. 3. Cayley, philos. transactions 1861, Collected math. papers, Vol. IV, p. 309.

in a similar way one shows that no two others of these subsidiary groups are equal to one another.

According to this, the values of u conjugate within A are obtained by applying the permutations

$$(1,\,2)(3,\,4), \quad t(0,1) = (0,1,2,4,3), \quad t(0,2) = (0,2,4,3,1)$$
$$t(0,3) = (0,3,1,2,4), \quad t(0,4) = (0,4,3,1,2):$$

$$\begin{aligned} u_1 &= x_0x_1 + x_1x_2 + x_2x_3 + x_3x_4 + x_4x_0, \\ u_2 &= x_0x_2 + x_2x_1 + x_1x_4 + x_4x_3 + x_3x_0, \\ u_3 &= x_1x_2 + x_2x_4 + x_4x_0 + x_0x_3 + x_3x_1, \\ u_4 &= x_2x_0 + x_0x_4 + x_4x_1 + x_1x_3 + x_3x_2, \\ u_5 &= x_3x_2 + x_2x_4 + x_4x_1 + x_1x_0 + x_0x_3, \\ u_6 &= x_4x_2 + x_2x_0 + x_0x_1 + x_1x_3 + x_3x_4, \end{aligned} \tag{11}$$

and the corresponding $u'_1, u'_2, u'_3, u'_4, u'_5, u'_6$ are found by applying the permutation t to the u_1, and then proceeding with u'_1 just as was done with u_1, or also simply by uniting the pairs missing from each u into the corresponding u', so that for all these u the condition (7) is satisfied.

$$
\begin{aligned}
u_1' &= x_0x_2 + x_0x_3 + x_1x_3 + x_1x_4 + x_2x_4,\\
u_2' &= x_0x_1 + x_0x_4 + x_1x_3 + x_2x_3 + x_2x_4,\\
u_3' &= x_0x_1 + x_0x_2 + x_2x_3 + x_3x_4 + x_1x_4,\\
u_4' &= x_0x_1 + x_0x_3 + x_1x_2 + x_2x_4 + x_3x_4,\\
u_5' &= x_0x_2 + x_0x_4 + x_1x_2 + x_1x_3 + x_3x_4,\\
u_6' &= x_0x_3 + x_0x_4 + x_1x_2 + x_1x_4 + x_2x_3.
\end{aligned} \tag{12}
$$

The six quantities (8): $y_1, y_2, y_3, y_4, y_5, y_6$ are the roots of an equation of the sixth degree, whose coefficients depend rationally on a, b, c, d, e and $\sqrt{\Delta}$. Since the y all change their sign when $\sqrt{\Delta}$ changes its sign, this equation has the form

$$
y^6 + a_2y^4 + a_4y^2 + a_6 - \sqrt{\Delta}(a_1y^5 + a_3y^3 + a_5y) = 0, \tag{12}
$$

in which the a are rational functions of the coefficients of $f(x)$. But they must also be integral functions of these coefficients; for $a_2, a_4, \alpha_6, a_1\sqrt{\Delta}, a_3\sqrt{\Delta}, a_5\sqrt{\Delta}$ are integral functions of the x, which can indeed be regarded here as independent variables, and therefore the three last must be divisible by the difference product $\sqrt{\Delta}$, and then $a_2, a_4, a_6, a_1, a_3, a_5$ must prove to be integral symmetric functions of the x. If one determines the degrees with respect to the x, then, since the y are of the second degree, there result for

	$a_1\sqrt{\Delta}$,	a_2,	$a_3\sqrt{\Delta}$,	a_4,	$a_5\sqrt{\Delta}$,	a_6
the degrees	2	4	6	8	10	12.

But $\sqrt{\Delta}$ is of degree 10 in the x, and consequently $a_1 = 0$, $a_3 = 0$ must hold, and a_5 must be a number.

One can compute the coefficients a_2, a_4, a_5, a_6 by actual formation of the expressions according to the rules for the representation of the symmetric functions, which has been done by Cayley. But the calculation and the formulae are very lengthy.

We shall content ourselves here with forming the resolvent for a special case. The value found in doing so for the number a_5 is then of course generally valid.

Let the equation of degree 5 have the Bring-Jerrard form[53]

$$
x^5 + \alpha x + \beta = 0. \tag{13}
$$

Then a_2, a_4, a_6 are integral rational functions of α and β. Of the coefficients of the general equation (6), d is of the fourth, e of the fifth order in the x. From this it follows that in a_2, a_4, a_6 the coefficient e can occur only multiplied by one of the coefficients a, b, c, and hence that, if a, b, c are set to zero, e must drop out of these expressions. For equation (13), therefore, a_2, a_4, a_6 cannot depend on β, and, having regard to the degree, one obtains, if m, m_1, m_2, m_3 denote numbers,

$$
a_2 = m_1\alpha, \quad a_4 = m_2\alpha^2, \quad a_6 = m_3\alpha^3, \quad a_5 = m. \tag{14}
$$

The numbers m, m_1, m_2, m_3 can be determined from a special assumption. If we set $\beta = 0$, then

$$
\begin{aligned}
x_0 = 0, \quad x_1 &= \sqrt[4]{-\alpha}, \quad x_2 = i\sqrt[4]{-\alpha},\\
x_3 &= -\sqrt[4]{-\alpha}, \quad x_4 = -i\sqrt[4]{-\alpha},
\end{aligned} \tag{15}
$$

[53] Runge, Acta mathematica, Bd. 7.

and from this there results

$$\begin{aligned}\sqrt{\Delta} &= (x_0-x_1)(x_0-x_2)(x_0-x_3)(x_0-x_4)(x_1-x_2)\\ &\quad (x_1-x_3)(x_1-x_4)(x_2-x_3)(x_2-x_4)(x_3-x_4)\\ &= 16i\sqrt{-\alpha}^5 = -16\sqrt{\alpha}^5\\ \Delta &= 256\alpha^5.\end{aligned} \tag{16}$$

Since here $b=0$, by (7) and (8) we have $y=2u$, and from (11) and (15) the values of the y result:

$$\begin{aligned}y_1 = y_2 = y_3 = y_6 &= -2\sqrt{\alpha}\\ y_4 = (4-2i)\sqrt{\alpha}, &\quad y_5 = (4+2i)\sqrt{\alpha}.\end{aligned}$$

From this one obtains for $\beta=0$ the following equation, identical in y:

$$\begin{aligned}&y^6 + a_2y^4 + a_4y^2 + a_6 - a_5\sqrt{\Delta}\,y\\ &= (y+2\sqrt{\alpha})^4(y^2 - 8y\sqrt{\alpha} + 20\alpha)\\ &= y^6 - 20\alpha y^4 + 240\alpha^2y^2 + 512\sqrt{\alpha}^5y + 320\alpha^3.\end{aligned} \tag{17}$$

But from this one finds the general resolvent for the equation (13), if according to (16) one puts $\sqrt{\Delta}$ for $-16\sqrt{\alpha}^5$, thus

$$y^6 - 20\alpha y^4 + 240\alpha^2y^2 - 32\sqrt{\Delta}\,y + 320\alpha^3 = 0. \tag{18}$$

The discriminant Δ here has, as follows from formula (3) in §. 74, the expression

$$\Delta = 5^5\beta^4 + 2^8\alpha^5. \tag{19}$$

Simpler still is the equation for u, namely

$$u^6 - 5\alpha u^4 + 15\alpha^2u^2 - \sqrt{\Delta}\,u + 5\alpha^3 = 0, \tag{20}$$

and if one puts $u^2 = v$, there results for v the rational equation of degree 6

$$(v^3 - 5\alpha v^2 + 15\alpha^2 v + 5\alpha^3)^2 = \Delta v. \tag{21}$$

Another form of this equation is obtained from (17), if one multiplies the second of the three expressions by the one obtained from it by changing $\sqrt{\alpha}$ into $-\sqrt{\alpha}$. In this way one obtains

$$(v-\alpha)^4(v^2 + 6\alpha v + 25\alpha^2) = 0,$$

and (21) must pass over into this when $\beta=0$ is set. But since in (21) the part depending on β is determined by (19), the sought form of the resolvent follows:

$$(v-\alpha)^4(v^2 + 6\alpha v + 25\alpha^2) = 5^5\beta^4 v. \tag{22}$$

One may further raise the question whether the function v introduced here is always really metacyclic, whether it might not perhaps, for special equations, remain unchanged under still other permutations.

If this case should occur, then the equation (21) or (22) would have to have equal roots, or (20) equal or opposite roots.

The value $u=0$ or $v=0$ occurs only when $\alpha=0$. In that case we have in fact in (13) the well-known metacyclic equation $x^5+\beta=0$; this case, then, we now set aside. The equation (20) can only have two opposite roots when $\Delta=0$, that is, when the equation (13) has equal roots. This too is to be excluded, since we have assumed (13) to be irreducible.

There remains, then, only the question whether (20) can have equal roots. In that case, together with (20), the derived equation

$$6u^5 - 20\alpha u^3 + 30\alpha^2 u - \sqrt{\Delta} = 0$$

would have to be satisfied, and if we eliminate $\sqrt{\Delta}$ from this and from (20),

$$5u^6 - 15\alpha u^4 + 15\alpha^2 u^2 - 5\alpha^3 = 5(u^2 - \alpha)^3 = 0.$$

We should therefore have to have $v = \alpha$, and hence by (22) $\beta = 0$. But then the equation of degree 5 would again be reducible, since it has the factor x.

If one wishes, then, to test a given equation of the form (13) with regard to its solvability, one has first to establish the irreducibility, and then to investigate whether (21) or (22) has a rational root.

If α and β are integers, then a rational v must also be an integer, which is to be sought among the factors of $25\alpha^6$.

If, e. g., $\alpha = 5$, $\beta = 5t$, then, if t is an integer not divisible by 5, $x^5 + \alpha x + \beta$ is irreducible by §. 178, 3. For v one has to substitute powers of 5, with positive or negative sign, but for no such number can (22) be satisfied, because on the left-hand side the power of 5 does not become high enough.

Thus no equation of the form $x^5 + 5x + 5t = 0$ is metacyclic.

If one wishes to determine metacyclic equations, one sets in (22)

$$\beta = \alpha\mu, \quad v = \alpha\lambda,$$

whereby one obtains:

$$\begin{aligned} \alpha &= \frac{5^5\mu^4\lambda}{(\lambda-1)^4(\lambda^2+6\lambda+25)} \\ \beta &= \frac{5^5\mu^5\lambda}{(\lambda-1)^4(\lambda^2+6\lambda+25)}, \end{aligned} \tag{23}$$

and if one here puts rational quantities for λ and μ (from an arbitrary domain of rationality, e.g. also from the field of rational functions of λ and μ), then the equation of degree 5

$$x^5 + \alpha x + \beta = 0 \tag{24}$$

is algebraically solvable, since indeed the case that this equation becomes reducible also leads to algebraically solvable equations.

Conversely we can say that no irreducible equation of degree 5 of the form (24) is algebraically solvable in which the coefficients α, β are not representable in the form (23).

If we set, e.g., $\lambda = -1$, $\mu = 1$, we obtain $64\alpha = -5^4$, $64\beta = -5^4$. The equation (24) is simplified, if we introduce for x a new unknown ξ, by the equation $x\xi = 5$. We thus obtain the equation

$$\xi^5 + 5\xi^4 - 5 \cdot 64 = 0$$

as an example of an algebraically solvable equation which, moreover, is irreducible by the theorem mentioned above.

§. 182.
The Group of the Resolvent.

The theorems that we have derived in the preceding section admit a remarkable conclusion about the equations of degree 6.

We now again take the x_0, x_1, x_2, x_3, x_4 of the preceding section as independent variables. Then the six quantities

$$(1) \qquad \begin{aligned} v_1 &= (u_1 - u_1')^2, \quad v_2 = (u_2 - u_2')^2, \quad v_3 = (u_3 - u_3')^2 \\ v_4 &= (u_4 - u_4')^2, \quad v_5 = (u_5 - u_5')^2, \quad v_6 = (u_6 - u_6')^2 \end{aligned}$$

are the roots of an equation of degree 6, $F(v) = 0$, whose coefficients are rationally expressible in terms of the symmetric fundamental functions a, b, c, d, e of the x, and which is irreducible in the field of the rational functions of these quantities. This equation is a resolvent of the equation of degree 5, $f(x) = 0$, whose roots are the x.

If M is the full metacyclic group of the x, then the groups $\pi^{-1}M\pi$ conjugate to M have no other common

common divisor than the identical permutation. For every common divisor of all these groups would have to be a normal divisor of the symmetric, and consequently also of the alternating group; but since it cannot be the alternating group itself, it must by §. 177 reduce to the unit group.

The resolvent $F(v) = 0$ is therefore, according to the designation introduced in §. 156, a total resolvent of $f(x) = 0$, and its group must be of the same degree as the group of $f(x) = 0$, whose degree is $1.2.3.4.5 = 120$. Since $F(z)$ is also irreducible, the permutation group among the indices of the six quantities v is transitive. It is of degree 120, while the degree of the symmetric group of the permutations of six figures is 6.120. With this we have the theorem:

The symmetric permutation group of six figures has a transitive divisor of index 6.

In order to find this remarkable group, which we shall denote by C, we have only to investigate the influence which all the 120 permutations of the x exert on the v, or on the quantities (11) of the preceding section.

Now we have seen earlier (§. 153, 2.) that the transpositions $(0,1)$, $(0,2)$, $(0,3)$, $(0,4)$ generate, by repeated composition, the whole symmetric group of the permutations of five figures. It therefore suffices to determine the influence of these four transpositions on the indices of the v. By the application of a transposition $(0,1)$ each of the expressions (11), §. 181 passes into one of the expressions (12) and conversely, and if one determines this change for the four transpositions named, one obtains the following four generating permutations $\pi_1, \pi_2, \pi_3, \pi_4$ of the group C

$$\begin{aligned} (0,1): &\quad \pi_1 = (1,3)(2,5)(4,6) \\ (0,2): &\quad \pi_2 = (1,4)(2,3)(5,6) \\ (0,3): &\quad \pi_3 = (1,5)(2,6)(3,4) \\ (0,4): &\quad \pi_4 = (1,6)(2,4)(3,5), \end{aligned}$$

where the transpositions $(1,3)\ldots$ occurring in the π refer to the indices of the u or of the v.

Among the 120 permutations of the group C there occur, among others, also

$$\begin{aligned} \pi_1\pi_2 &= (1,2,6)(3,4,5) \\ \pi_1\pi_2\pi_3 &= (1,6,5,4) \\ \pi_1\pi_2\pi_3\pi_4 &= (2,4,6,3,5) \\ (\pi_1\pi_2\pi_3\pi_4)^3 &= (2,3,4,5,6). \end{aligned}$$

According to this, a function of the six quantities v belonging to the group C can easily be formed.

A function such as

$$v_1v_3 + v_2v_4 + v_5v_6,$$

remains unchanged by π_1 and by $\pi_1\pi_2$; but if we apply to it the powers of the cyclic permutation

$(2, 3, 4, 5, 6)$, five functions arise from it, which we shall denote as follows:

$$
\begin{aligned}
w_0 &= v_1v_2 + v_4'v_5 + v_3v_6 \\
w_1 &= v_1v_4 + v_2v_6 + v_3v_5 \\
w_2 &= v_1v_6 + v_2v_5 + v_3v_4 \\
w_3 &= v_1v_3 + v_2v_4 + v_5v_6 \\
w_4 &= v_1v_5 + v_2v_3 + v_4v_6.
\end{aligned} \tag{2}
$$

If we apply the permutations $\pi_1, \pi_2, \pi_3, \pi_4$ to the functions (2), these functions merely pass into one another, and indeed in the following way:

$$\pi_1 = (w_0, w_1), \quad \pi_2 = (w_0, w_2), \quad \pi_3 = (w_0, w_3), \quad \pi_4 = (w_0, w_4),$$

i.e. the $\pi_1, \pi_2, \pi_3, \pi_4$ correspond to the transpositions $(0, 1)$, $(0, 2)$, $(0, 3)$, $(0, 4)$ among the indices of the w, and thus by §. 153, 2. the symmetric group of the indices of the w corresponds to the whole group C. A symmetric function of the five quantities w which is not at the same time a symmetric function of the v, such as the sum of the w, is therefore a function belonging to C.

One can, for example, take for this the sum of the squares

$$W = w_0^2 + w_1^2 + w_2^2 + w_3^2 + w_4^2,$$

or a function

$$(\lambda - w_0)(\lambda - w_1)(\lambda - w_2)(\lambda - w_3)(\lambda - w_4)$$

for an arbitrary rational λ.

Such a quantity W is the root of an irreducible equation of degree 6, whose coefficients are symmetric functions of the six quantities v.

Eighteenth Section.
Roots of Metacyclic Equations.

§. 183.
Statement of the Problem. Lemma.

In the preceding section we have found general criteria by which one can decide whether a given equation of prime degree is metacyclic or not. But with this the subject is by no means exhausted. Rather, according to the form of the statement of the problem as it derives from Abel, the task is to give a procedure by which one can find all metacyclic equations. In this form the problem, on which Abel left behind only brief indications without proofs, was taken up by Kronecker, and completely solved in such a way that at first not the equations themselves, but their roots, are found[54]. Thus Kronecker has given, for every prime n, an expression which is derived from a given field Ω by repeated extraction of roots, and which has the double

property that every root of an irreducible metacyclic equation of degree n in Ω is contained in this expression, and that conversely every such expression also satisfies an irreducible equation of degree n in Ω.

[54] Abel, Sur la résolution algébrique des équations (Oeuvres complètes, ed. Sylow, tome II, p. 217), letter to Crelle of 14 March 1826 (Oeuvres tome II, p. 266). — Kronecker, Ueber die algebraisch auflösbaren Gleichungen. Monatsberichte der Berl. Akademie 1853, 1856. — H. Weber, Ueber algebraisch auflösbare Gleichungen von Primzahlgrad. Sitzungsberichte der Gesellschaft zur Beförderung der Naturwissenschaften zu Marburg 1892. The following presentation is simplified as against the one given there, and corrected in one point.

The foundation for this investigation is furnished by the resolvents of Lagrange.

Now let n be a prime, let $x_0, x_1, x_2 \dots x_{n-1}$ be a system of independent variables, and let the notation be so chosen that $x_n = x_0$, $x_{n+1} = x_1 \dots$ Finally, let ε be an nth root of unity. We then set

$$(1) \qquad (\varepsilon, x) = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \cdots + \varepsilon^{n-1} x_{n-1} = \sum_{0,n-1}^{h} \varepsilon^h x_h.$$

These expressions we have already come to know in the fourteenth section under the name of the resolvents of Lagrange, and we have there (§. 164) investigated their behaviour with respect to the cyclic permutations.

It is now a matter of determining the influence of linear permutations, of the kind considered in §. 180, on these functions.

We first send ahead a lemma, which we really need only reproduce from what has gone before. The imaginary nth roots of unity ε are the roots of an equation $X = 0$, if

$$(2) \qquad X = \xi^{n-1} + \xi^{n-2} + \cdots + \xi + 1$$

is set, and ξ denotes a variable.

The function X is, as we have seen earlier (§. 134), irreducible in the field R of the rational numbers.

If we take as basis any field Ω in which X is irreducible, and derive from it the field $\Omega(\varepsilon)$, which consists of all rational functions of ε with coefficients in Ω, then it follows by §. 142 that every quantity of this field can be put in one and only one way into the form

$$(3) \qquad c_0 + c_1\varepsilon + c_2\varepsilon^2 + \cdots + c_{n-2}\varepsilon^{n-2}$$

in which the c are quantities in Ω.

For brevity we shall call this form of the quantities in $\Omega(\varepsilon)$ the normal form, and shall accordingly state the theorem:

1. If Ω is a field in which the cyclotomic equation $X = 0$ is irreducible, then every

quantity of the field $\Omega(\varepsilon)$ can be brought in only one way into the normal form

$$\Phi(\varepsilon) = c_0 + c_1\varepsilon + c_2\varepsilon^2 + \cdots + c_{n-2}\varepsilon^{n-2}$$

in which the coefficients c are contained in Ω.

In this theorem we can take for Ω the field that arises from R by the adjunction of n independent variables $x_0, x_1, x_2 \dots x_{n-1}$; for if X were reducible in this field, so that

$$X = X_1 X_2,$$

where X_1, X_2 are integral functions of ε and rational functions of the x, then one could put for the variables $x_0, x_1 \dots x_{n-1}$ such rational numbers that X_1, X_2, without changing their degree with respect to ε, pass into functions in R (§. 143), and this contradicts the irreducibility of X in R.

This field Ω has various divisors in which X is likewise irreducible, and which can therefore all be taken for Ω in Theorem 1. Such divisors are obtained if one fixes any permutation group P of the n figures $0, 1, 2 \dots n-1$ and considers the totality of all rational functions of $x_0, x_1 \dots x_{n-1}$ in R that admit the permutations of this group.

The totality of all these functions evidently forms a field, and thus to every permutation group of the n figures one obtains a definite associated field Ω. The quantities of such a field can be represented rationally by the symmetric functions of the x and by a function belonging to the group P.

We call the field Ω so determined the field belonging to the group P. Every such field can take the place of Ω in 1. (Cf. §. 152.)

We can therefore state the second theorem:

2. If a rational function $\Phi(x_0, x_1 \dots x_{n-1}, \varepsilon)$ of the field $\Omega(\varepsilon)$ remains unchanged when the permutations of a group P are applied to the indices of the x, then the coefficients $c_0, c_1, c_2 \dots c_{n-2}$ of the normal form of Φ also admit the permutations of P.

If $\Phi(\varepsilon)$ is any element of the field $\Omega(\varepsilon)$, one obtains the conjugate quantities by putting for ε any other

root of $X = 0$, that is, carries out the substitution $(\varepsilon, \varepsilon^h)$ for $h = 1, 2 \dots n-1$. All these fields are again contained in $\Omega(\varepsilon)$; they are therefore identical with one another, i.e. $\Omega(\varepsilon)$ is a normal field. If a quantity $\Phi(\varepsilon)$ has the property of remaining unchanged when ε is replaced by all the ε^h, then it is itself contained in Ω. This already follows from the general theorems of §. 144. But we see its correctness at once if, under the hypothesis $\Phi(\varepsilon^h) = \Phi(\varepsilon)$, we set

$$\Phi(\varepsilon) = \frac{1}{n-1} \sum_{1, n-1}^{h} \Phi(\varepsilon^h) = c_0 - \frac{c_1 + c_2 + \dots + c_{n-2}}{n-1}$$

from which it is then to be concluded, by 2., that $c_1, c_2 \dots c_{n-2}$ are equal to zero. Hence:

3. If a quantity in $\Omega(\varepsilon)$ admits all the substitutions $(\varepsilon, \varepsilon^h)$, then it is contained in Ω.

§. 184.
Theorems on the Resolvents.

We now consider the Lagrange resolvents (ε, x) under the hypothesis that ε is an imaginary nth root of unity and the x are independent variables, and apply to them the theorems of the preceding section.

We first investigate the influence of the linear permutations on (ε, x); for this it suffices to determine the influence of the two generating substitutions

$$s = (h, h+1), \quad t = (h, gh) \tag{1}$$

where h, taken modulo n, runs through the indices of x, and g denotes a primitive root of the prime n (§. 180).

But if we apply to

$$(\varepsilon, x) = x_0 + \varepsilon x_1 + \varepsilon^2 x_2 + \dots + \varepsilon^{n-1} x_{n-1} \tag{2}$$

the substitution s, this function goes over into

$$x_1 + \varepsilon x_2 + \varepsilon^2 x_3 + \dots + \varepsilon^{n-1} x_0 = \varepsilon^{-1}(\varepsilon, x).$$

The effect of the substitution s, then, is that

$$(\varepsilon, x) \quad \text{goes over into} \quad \varepsilon^{-1}(\varepsilon, x)$$

We now pass to the substitution t, and in

$$(\varepsilon, x) = x_0 + \sum_{1, n-1}^{h} \varepsilon^h x_h \tag{2}$$

put x_{gh} for x_h; this function then goes over into

$$x_0 + \sum_{1, n}^{h} \varepsilon^h x_{gh}. \tag{3}$$

In this sum h may run through an arbitrary system of residues with respect to the modulus n, with the exclusion of 0, and we can therefore replace h by $g^{-1}h$; thereby the sum (3) becomes

$$x_0 + \sum^{h} \varepsilon^{g^{-1}h} x_h = (\varepsilon^{g^{-1}}, x).$$

If one applies the permutations s and t repeatedly, there results the influence of s^λ, t^λ, which consists in the permutations

$$(\varepsilon, x), \varepsilon^{-\lambda}(\varepsilon, x)$$
$$(\varepsilon, x), (\varepsilon^{g^{-\lambda}}, x)$$

and so in particular, for $\lambda = -1$,

$$(\varepsilon, x), \varepsilon(\varepsilon, x)$$
$$(\varepsilon, x), (\varepsilon^{g}, x),$$

and we obtain the following theorem:

4. The generating permutations of the metacyclic group, s, t, effect the permutations

$$(s)\colon (\varepsilon, x), \varepsilon^{-1}(\varepsilon, x); \qquad (s^{-1})\colon (\varepsilon, x), \varepsilon(\varepsilon, x)$$
$$(t)\colon (\varepsilon, x), (\varepsilon^{g^{-1}}, x); \qquad (t^{-1})\colon (\varepsilon, x), (\varepsilon^{g}, x).$$

But from 2. and 4., because s is the generating permutation of the cyclic group, there results the consequence:

5. If one represents the functions

$$(\varepsilon, x)^n = \Phi(\varepsilon), \quad (\varepsilon^\lambda, x)(\varepsilon, x)^{-\lambda} = F(\varepsilon) \tag{4}$$

in the normal form [§. 183, (3)], then the coefficients are cyclic functions of $x_0, x_1 \ldots x_{n-1}$ (§. 163).

Here λ may denote any arbitrary integer that is not divisible by n.

From the function $F(\varepsilon)$ there arise, if we take $\lambda = g$ and put for ε its $n-1$ different values, $n-1$ functions, which all remain unchanged under s, and which we shall denote as follows:

$$\begin{aligned} (\varepsilon^g, x)(\varepsilon, x)^{-g} &= f_0 \\ (\varepsilon^{g^2}, x)(\varepsilon^g, x)^{-g} &= f_1 \\ \cdots\cdots\cdots\cdots & \\ (\varepsilon^{g^{n-1}}, x)(\varepsilon^{g^{n-2}}, x)^{-g} &= f_{n-2}. \end{aligned} \tag{5}$$

so that, if $(\varepsilon^g, x)(\varepsilon, x)^{-g} = f(\varepsilon)$ is set, we have

$$f_h = f(\varepsilon^{g^h}) \tag{6}$$

If one applies the permutation t^{-1} to (5), then, as is to be seen from Theorem 4., the functions

$$f_0, f_1, f_2 \ldots f_{n-2}$$

undergo a cyclic permutation $(0, 1, 2 \ldots n-2)$.

Moreover, so long as the x are variable, the functions f_ν are, as one sees from (5), all different from one another since they are decomposed into different linear factors.

If, then, we form a cyclic function of these $n-1$ quantities

$$C(f_0, f_1, f_2 \ldots f_{n-2}), \tag{7}$$

this remains unchanged both by s and by t^{-1}, hence by t and the whole metacyclic group, and we obtain the theorem:

6. If a cyclic function of the quantities $f_0, f_1 \dots f_{n-1}$ is represented in the normal form, then the coefficients are metacyclic functions of the variables $x_0, x_1 \dots x_{n-1}$.

Here a metacyclic function is to be understood as any function that remains unchanged by the permutations of the linear group.

But the functions $f_0, f_1 \dots f_{n-2}$ also pass cyclically into one another if one leaves the x unchanged and makes the substitution $(\varepsilon, \varepsilon^g)$ for ε. If, then, the cyclic function C of the f does not contain ε in its coefficients, then C remains unchanged by all the substitutions $(\varepsilon, \varepsilon^h)$, and we obtain from 3. the theorem:

7. If $C(f_0, f_1 \dots f_{n-2})$ is a cyclic function of the quantities $f_0, f_1 \dots f_{n-2}$ with rational coefficients, then C is a metacyclic function of the variables $x_0, x_1 \dots x_{n-1}$ with rational coefficients.

If we raise the functions $f_0, f_1 \dots f_{n-2}$, as they are given by (5), in order to the powers $g^{n-2}, g^{n-3} \dots 1$

and then multiply everything, and if we further observe that $\varepsilon^{g^{n-1}} = \varepsilon$, then in the product of the left-hand sides of the equations (5) all the resolvents except (ε, x) cancel out, and it follows that

$$(\varepsilon, x)^{1-g^{n-1}} = f_0^{g^{n-2}} f_1^{g^{n-3}} \cdots f_{n-2}. \tag{8}$$

If we now denote by λ a still undetermined integer exponent, we derive from (8) the following relation:

$$(\varepsilon^\lambda, x)^n = \left[(\varepsilon, x)^{\frac{g^{n-1}-1}{n}} (\varepsilon^\lambda, x)\right]^n f_0^{g^{n-2}} f_1^{g^{n-3}} \cdots f_{n-2}. \tag{9}$$

If we understand by g_1 any fixed primitive root of n, we can set $g = g_1 + ln$, where l denotes an arbitrary integer, since the primitive roots are defined only up to multiples of n. Then

$$g^{n-1} \equiv g_1^{n-1} + n(n-1) l g_1^{n-2} \pmod{n^2},$$

from which

$$\frac{g^{n-1}-1}{n} \equiv \frac{g_1^{n-1}-1}{n} - l g^{n-2} \pmod{n},$$

and l can be so determined that

$$\frac{g^{n-1}-1}{n} \equiv 1 \pmod{n}$$

holds.

In (9) we now set, under this assumption,

$$\lambda = \frac{g^{n-1}-1}{n} \equiv 1 \pmod{n}, \tag{10}$$

and, as in (4),

$$F(\varepsilon) = (\varepsilon^\lambda, x)(\varepsilon, x)^{-\lambda} = (\varepsilon, x)^{nk}, \tag{11}$$

where $\varepsilon^\lambda = \varepsilon$, and $1-\lambda$, which is divisible by n, is set $= nk$. From this we obtain

$$(\varepsilon, x)^n = [F(\varepsilon)]^n f_0^{g^{n-2}} f_1^{g^{n-3}} \cdots f_{n-2}. \tag{12}$$

These formulae can be generalised if one observes that the functions $f_0, f_1 \ldots f_{n-2}$ undergo a cyclic permutation when the substitution $(\varepsilon, \varepsilon^g)$ is carried out. If we therefore stipulate that $f_h = f_k$ is to hold when $h \equiv k \pmod{n-1}$, and set

$$F(\varepsilon^{g^\nu}) = F_\nu = (\varepsilon^{g^\nu}, x)^{nk}, \tag{13}$$

then the substitution $(\varepsilon, \varepsilon^{g^\nu})$ in (12) yields

$$(\varepsilon^{g^\nu}, x)^n = F_\nu^n f_\nu^{g^{n-2}} f_{\nu+1}^{g^{n-3}} \cdots f_{\nu+n-2}. \tag{14}$$

These formulae hold for every ν, if we further stipulate with regard to the F_ν that $F_h = F_k$ shall hold whenever $h \equiv k \pmod{n-1}$. It then follows from 4. that the functions $F_0, F_1 \ldots F_{n-2}$ remain unchanged by the substitution s, and undergo a cyclic permutation by t^{-1}.

If we wish here to reduce the exponents to their least positive residues with respect to the modulus n, we set

$$g^\nu = nq_\nu + r_\nu, \quad 0 < r_\nu < n, \quad r_0 = 1, \tag{15}$$

and the numbers r_ν coincide, in some order, with the numbers $1, 2 \ldots n-1$, so that r_0 is always $= 1$.

Then by (5) we have

$$f_{\nu-1} = (\varepsilon^{r_\nu}, x)(\varepsilon^{r_{\nu-1}}, x)^{-g}, \tag{16}$$

and we further set

$$\Phi_\nu = F_\nu f_\nu^{q_{n-2}} f_{\nu+1}^{q_{n-3}} \cdots f_{\nu+n-3}, \tag{17}$$

and obtain from (14)

$$(\varepsilon^{r_\nu}, x)^n = \Phi_\nu^n f_\nu^{r_{n-2}} f_{\nu+1}^{r_{n-3}} \cdots f_{\nu+n-2}^{r_0}. \tag{18}$$

The exponents r_ν remain unchanged when g is replaced by $g - ln$, and are therefore independent of the condition (10) and can be derived from an arbitrary primitive root g.

The characteristic properties of the functions f_ν, F_ν, Φ_ν occurring in these formulae, whose totality we shall denote by ω_ν, are, to repeat them once more, the following:

α) By the cyclic permutation $s = (h, h+1)$ among the indices of x the functions ω_ν are not changed.

β) By the linear permutation $t^{-1} = (h, g^{-1}h)$ among the indices of the x the cyclic permutation $(0, 1, 2 \ldots n-2)$ is produced among the indices of the ω.

γ) By the substitution $\sigma = (\varepsilon, \varepsilon^g)$ the same cyclic permutation $(0, 1, 2 \ldots n-2)$ is effected among the indices of ω.

δ) The cyclic functions of ω_ν are metacyclic functions of the x and free of ε.

By the theorem of Lagrange (§.155) every function Θ_ν possessing the four properties α), β), γ), δ) can be expressed rationally, in the field Ω of the metacyclic functions of x, by one of them, ω_ν, provided only that the $\omega_0, \omega_1 \ldots \omega_{n-2}$ are different from one another; one may therefore choose for these functions the f_ν.

For if u denotes a variable, and if

$$(u-\omega_0)(u-\omega_1)\cdots(u-\omega_{n-2}) = \varphi(u), \tag{19}$$

then the function $\varphi(u)$ admits the permutations s, t and also the substitution $\sigma = (\varepsilon, \varepsilon^g)$; and hence, by Theorem 3. of §.183, it is an integral function of u with coefficients metacyclic in regard to x, and independent of ε.

If now $\Theta_0, \Theta_1 \ldots \Theta_{n-2}$ is a system of functions possessing the properties $\alpha)\beta)\gamma)\delta)$, then the sum

$$\sum_{0,\,n-2}^{\nu} \frac{\Theta_\nu\, \varphi(u)}{u-\omega_\nu} = \chi(u) \tag{20}$$

is an integral function of degree $(n-2)$ of u, which likewise admits the substitutions s, t, σ and which therefore has coefficients in Ω. If one sets

$$\frac{\chi(u)}{\varphi'(u)} = \Theta(u) \tag{21}$$

and then sets $u = \omega_\nu$ in (20), it follows that

$$\Theta_\nu = \Theta(\omega_\nu),$$

where Θ denotes a rational function whose coefficients are metacyclic functions of the x, and no longer contain ε.

§. 185.
Roots of metacyclic equations.

In the formulae of the preceding section, the roots $\xi_0, \xi_1 \ldots \xi_{n-1}$ of a metacyclic equation of degree n, irreducible in some field $\mathfrak{K}$, are now to be introduced in place of the variables $x_0, x_1 \ldots x_{n-1}$, and indeed in such an order that the metacyclic functions of the ξ are rational (in $\mathfrak{K}$), which by the theorems of the preceding section is always possible. But in doing so we first make two further restrictive as-

sumptions, from which we shall afterwards free the result again. These assumptions are:

1. that by this substitution of the ξ for the x none of the resolvents (ε, ξ), in which ε is an imaginary nth root of unity, vanishes.

By (13), (16), (17), §.184 the functions f_ν, F_ν, Φ_ν then receive definite values different from zero, and we make the second assumption,

2. that by the same substitution no two of the functions $f_0, f_1 \ldots f_{n-1}$ become equal to one another.

We assume that by the substitution of the ξ for the x

$$f_0, f_1 \ldots f_{n-2}$$

pass into

$$k_0, k_1 \dots k_{n-2},$$

so that the $k_0, k_1 \dots k_{n-2}$ are therefore different from one another, and

$$\Phi_0, \Phi_1 \dots \Phi_{n-2}$$

into

$$K_0, K_1 \dots K_{n-2}$$

where, however, among the K equal ones may also occur.

On account of the property $\delta)$ of the functions f_ν, the quantities $k_0, k_1 \dots k_{n-2}$ are the roots of a cyclic equation of degree $(n-1)$ (hence even), $\psi(u) = 0$, so that, if u denotes a variable,

$$\psi(u) = (u - k_0)(u - k_1) \cdots (u - k_{n-2}) \tag{1}$$

is a function of degree $(n-1)$ in u with rational coefficients. The quantities $k_0, k_1 \dots k_{n-2}$ can be expressed rationally in terms of one another in the form

$$k_1 = \Theta(k_0), \quad k_2 = \Theta(k_1), \dots k_{n-1} = \Theta(k_{n-2}), \quad k_0 = \Theta(k_{n-1}) \tag{2}$$

(§. 163).

Likewise the quantities K_ν can, by the concluding theorem of §.184, be expressed in terms of the k_ν, and indeed in the form

$$K_0 = \Phi(k_0), \quad K_1 = \Phi(k_1), \dots K_{n-2} = \Phi(k_{n-2}), \tag{3}$$

in which Φ denotes a rational function (in $\mathfrak{R}$).

Accordingly the formula (18), §.184 yields us the following result: We put, as an abbreviation,

$$\tau_\nu = \sqrt[n]{k_\nu}, \tag{4}$$

and obtain

$$(\varepsilon^{r_\nu}, \xi) = K_\nu \tau_\nu^{r_{n-2}} \tau_{\nu+1}^{r_{n-3}} \dots \tau_{\nu+n-2}^{r_0}, \tag{5}$$

and if one adds all these formulae, and further sets the rational quantity $(1, x) = A$:

$$n\xi_0 = A + \sum_{0, n-2}^{\nu} K_\nu \tau_\nu^{r_{n-2}} \tau_{\nu+1}^{r_{n-3}} \dots \tau_{\nu+n-2}^{r_0}. \tag{6}$$

By §. 184, (16) and (17) the quantities K_ν and k_ν are all different from zero.

By (6) one of the roots ξ is represented.

If we set, for abbreviation,

$$R_\nu = k_\nu^{r_{n-2}} k_{\nu+1}^{r_{n-3}} \dots k_{\nu+n-2}$$

and write (6) in the form

$$n\xi_0 = A + K_0 \sqrt[n]{R_0} + K_1 \sqrt[n]{R_1} + \cdots + K_{n-2} \sqrt[n]{R_{n-2}}, \tag{7}$$

then ξ_0 is thus expressed by $n-1$ radicals of degree n, each of which can in itself have n different values. But these values cannot be independent of one another, since otherwise the number of values of ξ that result from (7) would be too great. One obtains in fact from (5), if by §. 184, (16) one sets

$$(\varepsilon^{r_\nu}, \xi)(\varepsilon^{r_{\nu-1}}, \xi)^{-g} = k_{\nu-1}$$

$$K_\nu \sqrt[n]{R_\nu} = K_{\nu-1}^g k_{\nu-1} \left(\sqrt[n]{R_{\nu-1}} \right)^g, \tag{8}$$

and can accordingly express all these radicals rationally by one of them and by k_ν.

But the expression (6) has over (7) the great advantage that, however one may determine the $n-1$ radicals $\sqrt[n]{k_\nu}$, it nevertheless represents only n different values, namely the n roots ξ.

To prove this, we denote by $\varepsilon_0, \varepsilon_1 \ldots \varepsilon_{n-2}$ any arbitrary system of nth roots of unity, and replace in (5)

$$\sqrt[n]{k_0}, \quad \sqrt[n]{k_1} \ldots \sqrt[n]{k_{n-2}}$$

by

$$\varepsilon_0 \sqrt[n]{k_0},\ \varepsilon_1 \sqrt[n]{k_1} \ldots \varepsilon_{n-2} \sqrt[n]{k_{n-2}}\,:$$

Then (5) passes into another form, which we represent thus

$$n\xi = A + \sum_{0,\,n-2}^{\nu} E_\nu K_\nu \tau_\nu^{r_{n-2}} \tau_{\nu+1}^{r_{n-3}} \ldots \tau_{\nu+n-2}^{r_0}, \tag{9}$$

in which E_ν is an nth root of unity, which is expressed by means of ε as follows:

$$E_\nu = \varepsilon_\nu^{r_{n-2}} \varepsilon_{\nu+1}^{r_{n-3}} \ldots \varepsilon_{\nu+n-2}^{r_0}. \tag{10}$$

Now, by the definition of r_ν [§. 184, (15)]

$$r_\nu \equiv g r_{\nu-1} \pmod{n}, \tag{11}$$

and by (10), since the index of r is to be taken with respect to the modulus $n-1$,

$$E_{\nu-1} = \varepsilon_{\nu-1}^{r_{n-2}} \varepsilon_\nu^{r_{n-3}} \ldots \varepsilon_{\nu+n-3}^{r_0}$$
$$E_{\nu-1}^g = \varepsilon_\nu^{r_{n-2}} \varepsilon_{\nu+1}^{r_{n-3}} \ldots \varepsilon_{\nu-1}^{r_0},$$

hence

$$E_\nu = E_{\nu-1}^g = E_0^{r_\nu}. \tag{12}$$

If, then, the ε are determined in any manner whatever, we obtain

$$E_0 = \varepsilon_0^{r_{n-2}} \varepsilon_1^{r_{n-3}} \dots \varepsilon_{n-2}^{r_0}, \tag{13}$$

and thereby, by (12), the remaining E_ν are completely determined. Thus in fact only n values result from (6), which one can obtain, for example, by assigning to one of the radicals τ_ν its various values. (Compare with this the Cayley solution of the cubic equations, §. 36.)

If one wishes to determine the quantities $\xi_0, \xi_1 \dots \xi_{n-1}$ in the order which underlies the formation of the cyclic and metacyclic group, one must, in (5), multiply by ε^{-hr_ν} before the summation, and finds (§. 133)

$$n\xi_h = A + \sum_{0,n-2}^{\nu} \varepsilon^{-hr_\nu} K_\nu \tau_\nu^{r_{n-2}} \tau_{\nu+1}^{r_{n-3}} \dots \tau_{\nu+n-2}^{r_0}, \tag{14}$$

in which an altered determination of the radicals occasions only a cyclic permutation of the ξ_h.

§. 186.
Removal of the Restrictive Assumptions.

It would be an essential restriction of these investigations on solvable equations if the assumptions 1., 2. of the preceding section had to be maintained.

But this is not necessary, as we now wish to show.

Now let $\eta_0, \eta_1 \dots \eta_{n-1}$ be the n roots of any irreducible metacyclic equation of degree n.

We introduce the new unknowns $\xi_0, \xi_1 \dots \xi_{n-1}$ by a system of equations

$$\xi_0 = \psi(\eta_0), \quad \xi_1 = \psi(\eta_1), \dots \xi_{n-1} = \psi(\eta_{n-1}) \tag{1}$$

in which $\psi(y)$ is to denote an integral function of degree $(n-1)$

$$\psi(y) = a_0 + a_1 y + a_2 y^2 + \dots + a_{n-1} y^{n-1} \tag{2}$$

with indeterminate coefficients from the field $\mathfrak{K}$. We assume, however, that the n quantities $\xi_0, \xi_1 \dots \xi_{n-1}$ are distinct from one another; then the ξ are likewise the roots of an irreducible metacyclic equation, and by (1) a Tschirnhausen transformation is expressed (§. 52); ξ_0 is a primitive element of the field $\mathfrak{K}(\eta_0)$; consequently the fields $\mathfrak{K}(\xi_0)$ and $\mathfrak{K}(\eta_0)$, and likewise the conjugate fields $\mathfrak{K}(\xi_r)$ and $\mathfrak{K}(\eta_r)$, are identical with one another. Hence also, if we set

$$\chi(x) = b_0 + b_1 x + b_2 x^2 + \dots + b_{n-1} x^{n-1} \tag{3}$$

in which the coefficients b are likewise quantities in $\mathfrak{K}$,

$$\eta_0 = \chi(\xi_0), \quad \eta_1 = \chi(\xi_1) \dots \eta_{n-1} = \chi(\xi_{n-1}). \tag{4}$$

It is now easy to see that one can dispose of the coefficients $a_0, a_1 \dots a_{n-1}$ in (1) in such a way that the hypotheses 1. and 2. of the preceding section are fulfilled for the ξ. For if we regard the a as independent variables, then the ξ determined by (1) likewise become variables, which are

connected with the a by a linear substitution with non-vanishing determinant. The determinant of this substitution is namely

$$\begin{vmatrix} 1, & \eta_0, & \eta_0^2 & \dots & \eta_0^{n-1} \\ 1, & \eta_1, & \eta_1^2 & \dots & \eta_1^{n-1} \\ \cdot & \cdot & \cdot & \cdot & \cdot \\ 1, & \eta_{n-1}, & \eta_{n-1}^2 & \dots & \eta_{n-1}^{n-1} \end{vmatrix},$$

that is, equal to the difference product of the η, which by hypothesis is different from zero.

The functions (ε, x) and the differences $f_\alpha - f_\beta$, which by §. 184 are non-vanishing functions of the variables $x_0, x_1 \dots x_{n-1}$

therefore, when the substitution $x_h = \psi(\eta_h)$ is made for the x, they pass over into functions of the variables a which likewise cannot vanish identically. By the theorem of §. 143, 1. one can therefore set for the quantities $a_0, a_1 \dots a_{n-1}$ such rational numbers that the functions (ε, x) and the differences $f_\alpha - f_\beta$, which then pass over into (ε, ξ) and into $k_\alpha - k_\beta$, become different from zero, and that the ξ_h also remain different from one another, which was to be proved.

Having thus established this, we introduce the system of variables $x_0, x_1 \dots x_{n-1}$ and a system of variables y depending on it,

$$y_0 = \chi(x_0), \quad y_1 = \chi(x_1), \dots y_{n-1} = \chi(x_{n-1}) \tag{5}$$

in which χ denotes the function (3), and set

$$\frac{(\varepsilon^{r_\nu}, y)}{(\varepsilon^{r_\nu}, x)} = \Theta_\nu. \tag{6}$$

If any permutation is applied to the indices of the x, the indices of the y undergo the same permutation. If, then, we apply to (6) the cyclic permutation s, Θ_ν does not change (by §. 184, 4.).

If, then, we represent Θ_ν in the normal form of §. 183, its coefficients are cyclic functions of $x_0, x_1 \dots x_{n-1}$.

If one applies to Θ_ν the permutation $t^{-1} = (h, g^{-1}h)$, then Θ_ν passes over, by §. 184, 4., into $\Theta_{\nu+1}$, or the Θ_ν undergo a cyclic permutation.

But the substitution $\sigma = (\varepsilon, \varepsilon^g)$ has the same effect as well, and therefore the system of functions $\Theta_0, \Theta_1 \dots \Theta_{n-2}$ has the properties $\alpha)$ $\beta)$ $\gamma)$ $\delta)$ (§. 184).

By the theorem proved at the end of §. 184, therefore, when we replace the x by the ξ and consequently the y by the η, Θ_ν passes over into a rational function of k_ν

$$Q_\nu = Q(k_\nu) \tag{7}$$

and from (6) it follows that

$$(\varepsilon^{r_\nu}, \eta) = Q_\nu(\varepsilon^{r_\nu}, \xi). \tag{8}$$

If one substitutes herein for $(\varepsilon^{r_\nu}, \xi)$ the expression (5), §. 185, one obtains for $(\varepsilon^{r_\nu}, \eta)$ an expression of entirely the same form, only with the difference that in place of K_ν there has entered $Q_\nu K_\nu$. The functions $Q_\nu K_\nu$ have essentially the same properties as the function K_ν, only that they also

may in part be zero. By the same alteration, §. 185 (6) then yields the value of η_0.

With this, then, the restrictive assumptions of §. 185 are removed, and we have the general theorem:

I. Every root ξ of a metacyclic equation of prime degree n can be represented in the form

$$\xi = A + \sum_{0,\,n-2}^{\nu} K_\nu \tau_\nu^{r_{n-2}} \tau_{\nu+1}^{r_{n-3}} \cdots \tau_{\nu+n-2}^{r_0}, \tag{9}$$

where A is a rational quantity, $k_0, k_1 \ldots k_{n-2}$ are the roots, different from one another and from zero, of a cyclic equation of degree $(n-1)$, and K_ν is a rational function of k_ν, whose form is the same for all ν. The exponents $r_0, r_1 \ldots r_{n-2}$ are the least positive residues of the numbers $1, g, g^2 \ldots g^{n-2}$, where g is a primitive root of n. The n values that one obtains from (9) when one assigns to the radicals $\tau_\nu = \sqrt[n]{k_\nu}$ their various values are the n roots of one and the same rational equation.

The formula (9) results from §. 185 (6) when A and K_ν are replaced by nA and nK_ν, which has been done for simplification.

But the converse of Theorem I can now also be stated.

To prove this, we denote by ε any imaginary nth root of unity and set

$$\begin{aligned} \xi_h &= A + \sum_{0,\,n-2}^{\nu} \varepsilon^{hr_\nu} K_\nu \tau_\nu^{r_{n-2}} \tau_{\nu+1}^{r_{n-3}} \cdots \tau_{\nu+n-2}^{r_0} \\ h &= 0, 1, 2 \ldots n-1, \end{aligned} \tag{10}$$

or, abbreviated,

$$\xi_h = A + \sum_{0,\,n-2}^{\nu} \varepsilon^{hr_\nu} K_\nu \sqrt[n]{R_\nu}, \tag{11}$$

where, as before,

$$\sqrt[n]{R_\nu} = \tau_\nu^{r_{n-2}} \tau_{\nu+1}^{r_{n-3}} \cdots \tau_{\nu+n-2}^{r_0} \tag{12}$$

is set. We now have to investigate the changes that result for ξ_h when we

1. determine one of the radicals $\tau_\nu = \sqrt[n]{k_\nu}$ differently, and
2. permute the $k_0, k_1 \ldots k_{n-2}$ cyclically.

We wish to give one of the radicals, say τ_α, a different sign, that is, to make the permutation

$$(\tau_\alpha,\ \varepsilon^\beta \tau_\alpha) \tag{13}$$

where β can be an arbitrary exponent, and α is one of the indices $0, 1 \ldots n-2$.

In (12) the radical τ_α has the exponent $r_{n+\nu-\alpha-2} \equiv r_{\nu-\alpha-1}$, and hence to the permutation (13) there corresponds the permutation

$$\left(\sqrt[n]{R_\nu},\ \varepsilon^{\beta r_{\nu-\alpha-1}} \sqrt[n]{R_\nu} \right). \tag{14}$$

Now, by the meaning of the numbers r, we have generally $r_{\alpha+\beta} \equiv r_\alpha r_\beta \pmod{n}$, and if therefore one introduces the permutation (13) into (11), h passes over into $h + \beta r_{-\alpha-1}$.

1. Thus the permutation (13) calls forth among the indices of ξ the cyclic permutation

$$(h,\ h + \beta r_{-\alpha-1})$$

If, secondly, we make the cyclic permutation

(15) $$(\tau_0,\ \tau_1, \dots \tau_{n-2}),$$

then this corresponds to the permutations

(16) $$\left(\sqrt[n]{R_\nu},\ \sqrt[n]{R_{\nu+1}}\right), \quad (K_\nu, K_{\nu+1}), \quad \nu = 0, 1, 2 \dots n-2,$$

and ξ_h passes over into

$$A + \sum^{\nu} \varepsilon^{hr_\nu} K_{\nu+1} \sqrt[n]{R_{\nu+1}} = A + \sum \varepsilon^{hr_{\nu-1}} K_\nu \sqrt[n]{R_\nu}$$
$$= A + \sum \varepsilon^{hg^{-1}r_\nu} K_\nu \sqrt[n]{R_\nu}.$$

2. Accordingly the indices of ξ suffer, through the permutation (15), the permutation

$$(h,\ g^{-1}h).$$

If we now consider any rational symmetric (or even merely metacyclic) function of the ξ:

$$S(\xi_0,\ \xi_1 \dots \xi_{n-1}),$$

then, when we introduce the values (10), we obtain from it a rational function of the radicals

$$\tau_0,\ \tau_1, \dots \tau_{n-2}.$$

But this function does not change when one of these radicals is given another of its n values, and consequently the function must depend rationally on $k_0, k_1 \dots k_{n-2}$. On account of

2. but neither does this function change when the cyclic permutation $(k_0, k_1 \dots k_{n-2})$ is performed among the k_ν, and since now the quantities k_ν are the roots of a cyclic equation in the field $\mathfrak{K}$, the function $S(\xi_0, \xi_1 \dots \xi_{n-2})$ is a quantity in $\mathfrak{K}$, i.e. rational. If one applies this to the coefficients of the equation whose roots are the quantities (10), it follows that these quantities are the roots of an equation of degree n in $\mathfrak{K}$.

As for the irreducibility of this equation, the following is to be remarked on that point:

The radicals $\sqrt[n]{R_\nu}$ are, if one of them is rational, all rational, and if one is irrational, so are the others [§. 185, (8)]. Which of the two cases occurs therefore depends, by §. 179, on whether the function $x^n - R_0$ is reducible or irreducible in $\mathfrak{K}$. If it is reducible, then one of the quantities ξ_h is rational in the field $\mathfrak{K}$, and the equation for ξ is reducible. All the quantities ξ are contained in the field $\mathfrak{K}(\varepsilon)$, whose degree is at most equal to $n-1$.

But if $\sqrt[n]{R_0}$ is irrational, then $x^n - R_0$ is irreducible in $\mathfrak{K}$. Every equation $\Phi(\xi_0) = 0$ passes, on substitution of the value of ξ_0, into an equation $\Psi(\sqrt[n]{R_0}) = 0$ whose coefficients are contained in $\mathfrak{K}$, and which must therefore continue to hold when $\sqrt[n]{R_0}$ is replaced by any one of its n values.

But thereby ξ_0 passes into each of the values $\xi_0, \xi_1 \dots \xi_{n-1}$, i.e. the equation of degree n whose roots are the n quantities ξ is irreducible in $\mathfrak{K}$.

We can therefore state the converse of Theorem I as follows:

II. Every quantity ξ contained in the form (9) is the root of an equation of degree n in $\mathfrak{K}$, and this equation is irreducible if one leaves aside the special case that one of its roots is rational and the others are contained in $\mathfrak{K}(\varepsilon)$.

§. 187.
Reality Relations.

If the field $\mathfrak{K}$ is a real one, then, as we have seen in §. 180, there are two kinds of metacyclic equations of prime degree n: those with one real and $n-1$ imaginary roots, and those with all real roots. Likewise we have seen in §. 165 that there are two kinds of cyclic equations of an even degree, namely those with all real and those with all imaginary roots.

If now the cyclic equation whose roots are the quantities $k_0, k_1 \ldots k_{n-1}$ belongs to the first kind, that is, if $k_0, k_1 \ldots k_{n-1}$ are real and the radicals $\tau_0, \tau_1, \ldots \tau_{n-1}$ are also taken to be real, then formula (10) of §. 186 shows that ξ_0 is real, and that ξ_h and ξ_{-h} are conjugate imaginary.

If on the other hand the k are imaginary, then by §. 165 k_ν and $k_{\nu+\frac{n-1}{2}}$ are conjugate imaginary, and by formula (8) in §. 185 we can set, if Φ denotes a rational function,

$$\sqrt[n]{R_{\nu+\frac{n-1}{2}}} = \Phi(k_\nu)\left(\sqrt[n]{R_\nu}\right)^{g^{\frac{n-1}{2}}},$$

or also

$$\sqrt[n]{R_{\nu+\frac{n-1}{2}}}\ \sqrt[n]{R_\nu} = \Phi(k_\nu)\left(\sqrt[n]{R_\nu}\right)^{g^{\frac{n-1}{2}}+1}.$$

Now because g is a primitive root of n, $g^{\frac{n-1}{2}}+1$ is divisible by n, and the right side therefore becomes rational. If then Ψ denotes a rational function, we have

$$\sqrt[n]{R_{\nu+\frac{n-1}{2}}}\ \sqrt[n]{R_\nu} = \Psi(k_\nu),$$

and this formula shows that

$$\Psi(k_\nu) = \Psi\left(k_{\nu+\frac{n-1}{2}}\right),$$

hence that $\Psi(k_\nu)$ is real. From this it follows that

$$\sqrt[n]{R_{\nu+\frac{n-1}{2}}} \quad \text{and} \quad \sqrt[n]{R_\nu}$$

are conjugate imaginary; for, first, their nth powers are conjugate imaginary, and hence they themselves are so, at first up to an nth root of unity as factor. But since their product is real, this nth root of unity must be $= 1$. Since, moreover,

$$r_\nu \equiv -r_{\nu+\frac{n-1}{2}} \pmod{n}$$

formula (11) shows that in this case the roots ξ_z are all real.

With this, then, the theorem is proved for a real field $\mathfrak{K}$:

If the cyclic equation whose roots are the k_0, $k_1 \ldots k_{n-2}$ has real roots, then of the roots ξ one is real and the rest imaginary; if this cyclic equation has imaginary roots, then all the ξ are real.

§. 188.
Metacyclic Equations of the Fifth Degree.

Through the theorems of the preceding section the roots of a metacyclic equation of prime degree have been brought into a generally valid form, which permits all these quantities to be actually formed, if one presupposes a knowledge of the roots of cyclic equations. Whole systems of cyclic equations of every degree are furnished to us, e.g., by the theory of the division of the circle, whose roots are the cyclotomic periods. Thus we can, for example, form arbitrarily many metacyclic equations for the field of rational numbers. That all these equations in the field of rational numbers are contained among them is a very remarkable theorem due to Kronecker, which we shall come

to know in the second volume. Here, as an application and illustration of what precedes, we wish further to treat the special problem of finding, in an arbitrary field $\mathfrak{K}$, the roots of all metacyclic equations of the fifth degree.

For this it is necessary and sufficient that we determine generally the roots k_0, k_1, k_2, k_3 of a cyclic equation of the fourth degree, and indeed under the assumption that k_0, k_1, k_2, k_3 are different from one another and none of them is zero.

As a foundation we here use the function

$$w = (k_0 - k_2)(k_1 - k_3), \tag{1}$$

which must be different from zero, and which for a real field $\mathfrak{K}$ is always real, since either k_0, k_1, k_2, k_3 are real or k_0, k_2 and k_1, k_3 form two conjugate imaginary pairs.

Under the cyclic permutation $(0, 1, 2, 3)$, w changes its sign. Hence the square of w is a cyclic function, which by hypothesis is to be known. We therefore set, denoting by the small latin letters $a, b, c \ldots$ quantities in $\mathfrak{K}$, that is, rational quantities,

$$w = 4\sqrt{c}, \tag{2}$$

and remark further that every function of the k which changes its sign under the cyclic permutation $(0, 1, 2, 3)$ is the product of a rational quantity with $\sqrt{c}$.

It is then only a matter of forming cyclic functions in sufficient number, so that the four quantities k_0, k_1, k_2, k_3 can be computed from them and expressed algebraically in terms of mutually independent rational quantities.

Now two further cyclic functions are

$$(k_0 - k_2)^2 + (k_1 - k_3)^2, \qquad \frac{(k_0 - k_2)^2 - (k_1 - k_3)^2}{(k_0 - k_2)(k_1 - k_3)},$$

and we thus obtain, if a, b are rational quantities,

$$\begin{aligned} (k_0 - k_2)^2 + (k_1 - k_3)^2 &= 8\,b \\ (k_0 - k_2)^2 - (k_1 - k_3)^2 &= 8\,a\sqrt{c}. \end{aligned} \tag{3}$$

The three rational quantities a, b, c are, however, not independent of one another; rather, by (2) and (3) there subsists between them the relation

$$b^2 = c\,(1 + a^2). \tag{4}$$

From (3) it follows further that

$$\begin{aligned} k_0 - k_2 &= 2\sqrt{b + a\sqrt{c}} \\ k_1 - k_3 &= 2\sqrt{b - a\sqrt{c}}, \end{aligned} \tag{5}$$

and between the two square roots there subsists the relation

$$\sqrt{c} = \sqrt{b + a\sqrt{c}}\,\sqrt{b - a\sqrt{c}}. \tag{6}$$

If we further denote by C and B again two rational quantities, then

$$\begin{aligned} k_0 + k_1 + k_2 + k_3 &= 4\,C, \\ k_0 - k_1 + k_2 - k_3 &= 4\,B\sqrt{c}, \end{aligned} \tag{7}$$

hence

$$\begin{aligned} k_0 + k_2 &= 2\,(C + B\sqrt{c}), \\ k_1 + k_3 &= 2\,(C - B\sqrt{c}), \end{aligned} \tag{8}$$

from which by (5) it follows:

$$
\begin{aligned}
k_0 &= C + B\sqrt{c} + \sqrt{b + a\sqrt{c}} \\
k_1 &= C - B\sqrt{c} + \sqrt{b - a\sqrt{c}} \\
k_2 &= C + B\sqrt{c} - \sqrt{b + a\sqrt{c}} \\
k_3 &= C - B\sqrt{c} - \sqrt{b - a\sqrt{c}}.
\end{aligned} \tag{9}
$$

Conversely, it is also easy to show that these quantities are the roots of a biquadratic cyclic equation. For if we set, as an abbreviation,

$$
r = \sqrt{c}, \quad \varrho = \sqrt{b + a\sqrt{c}}, \quad \varrho' = \sqrt{b - a\sqrt{c}}, \tag{10}
$$

and hence by (6)

$$
r = \varrho\varrho', \quad \varrho^2 = b + ar, \quad \varrho'^2 = b - ar,
$$

then we can represent every rational function of k_0, k_1, k_2, k_3 as a linear function with rational coefficients of the six radicals

$$
1, \quad r, \quad \varrho, \quad \varrho', \quad r\varrho, \quad r\varrho'
$$

The cyclic permutation (k_0, k_1, k_2, k_3) corresponds to the permutation

$$
\begin{array}{llll}
r, & \varrho, & \varrho', \\
-r, & \varrho', & -\varrho,
\end{array}
$$

and if one repeats this permutation, one sees that a cyclic function of the k cannot change if, in the linear representation by these six quantities, the following permutations are made:

$$
\begin{array}{llllll}
1, & r, & \varrho, & \varrho', & r\varrho, & r\varrho' \\
1, & -r, & \varrho', & -\varrho, & -r\varrho', & r\varrho \\
1, & r, & -\varrho, & -\varrho', & -r\varrho, & -r\varrho' \\
1, & -r, & -\varrho', & \varrho, & r\varrho', & -r\varrho,
\end{array}
$$

and if one adds the four expressions thus obtained, one obtains a rational expression for the cyclic function of the k.

If one wishes to form the biquadratic equation whose roots are the k, it is best to introduce $k - C = x$ as the unknown

in. One obtains in this way, by a simple calculation, the equation

$$
x^4 - 2\,(B^2c + b)\,x^2 - 4\,B\,a\,c\,x + B^4c^2 - 2\,B^2bc + c = 0. \tag{11}
$$

The representation of the k by the formulae (9) is, however, not yet completely satisfactory, because among the rational quantities a, b, c occurring in them the relation (4) still holds, and we seek a representation by independent quantities.

One special case we must first dispose of, namely $b = 0$, which has $a = i$ as a consequence. Then the formulae (9) give

$$
\begin{aligned}
k_0 &= C + B\sqrt{c} + \frac{1+i}{\sqrt{2}}\sqrt[4]{c} \\
k_1 &= C - B\sqrt{c} + \frac{1-i}{\sqrt{2}}\sqrt[4]{c} \\
k_2 &= C + B\sqrt{c} - \frac{1+i}{\sqrt{2}}\sqrt[4]{c} \\
k_3 &= C - B\sqrt{c} - \frac{1-i}{\sqrt{2}}\sqrt[4]{c},
\end{aligned}
$$

and the biquadratic equation (11) becomes

$$x^4 - 2\,B^2 c\,x^2 - 4\,i\,B\,c\,x + B^4 c^2 + c = 0.$$

But this case belongs here only when i is contained in the field $\mathfrak{R}$, and hence never for real fields.

If, however, b does not vanish, we introduce a new rational quantity h by setting

$$b = h\,(1 + a^2), \quad c = h^2\,(1 + a^2)$$

whereby the relation (4) is then identically satisfied, and the formulae (9), if one replaces Bh by B, give

$$\begin{aligned}
k_0 &= C + B\sqrt{1+a^2} + \sqrt{h\,(1 + a^2 + a\sqrt{1+a^2})} \\
k_1 &= C - B\sqrt{1+a^2} + \sqrt{h\,(1 + a^2 - a\sqrt{1+a^2})} \\
k_2 &= C + B\sqrt{1+a^2} - \sqrt{h\,(1 + a^2 + a\sqrt{1+a^2})} \\
k_3 &= C - B\sqrt{1+a^2} - \sqrt{h\,(1 + a^2 - a\sqrt{1+a^2})}
\end{aligned} \tag{12}$$

This expression for k_0 has moreover the advantage that, however one may determine the signs of the square roots occurring in it, it represents only four distinct values.

In A b e l a somewhat different expression is found for k_0, namely

$$k_0 = C + B\sqrt{1+e^2} + \sqrt{h\left(1 + e^2 + \sqrt{1+e^2}\right)},$$

which arises from (12) when one sets $a = 1 : e$ and then replaces B and h by Be and he^2. The expression (12) is thus more general in so far as it also embraces the special case $a = 0$, in which the biquadratic equation splits into two quadratic equations.

Now, in order to represent the root of a metacyclic equation of degree 5, these expressions for k_0, k_1, k_2, k_3 are to be substituted into the formula (9) of §. 186. If we assume $g = 2$, the exponents r_0, r_1, r_2, r_3 become congruent in turn with 1, 2, 4, 8, hence equal to 1, 2, 4, 3, and there results, when, as in §. 185,

$$\begin{aligned}
\tau_0 &= \sqrt[5]{k_0}, \quad \tau_1 = \sqrt[5]{k_1} \\
\tau_2 &= \sqrt[5]{k_2}, \quad \tau_3 = \sqrt[5]{k_3}
\end{aligned}$$

is set,

$$\begin{aligned}
\xi = A &+ K_0\,\tau_0^3\tau_1^4\tau_2^2\tau_3 + K_1\,\tau_1^3\tau_2^4\tau_3^2\tau_0 \\
&+ K_2\,\tau_2^3\tau_3^4\tau_0^2\tau_1 + K_3\,\tau_3^3\tau_0^4\tau_1^2\tau_2.
\end{aligned} \tag{13}$$

The coefficients in this expression, K_0, K_1, K_2, K_3, are rational functions of the k_0, k_1, k_2, k_3, which are themselves cyclically permuted by cyclic permutation of the k. According to A b e l one is to set

$$K_0 = A_1 + A_2 k_0 + A_3 k_2 + A_4 k_0 k_2$$

wherein A_1, A_2, A_3, A_4 are rational; but this is not general enough, because between $k_0 k_2$, k_0 and k_2 there subsists a linear relation easily derivable from (12). Most simply one expresses the k in terms of the three radicals

$$\begin{aligned}
r &= \sqrt{1+a^2}, & \varrho &= \sqrt{h\left(1 + a^2 + a\sqrt{1+a^2}\right)} \\
\varrho' &= \sqrt{h\left(1 + a^2 - a\sqrt{1+a^2}\right)}, & \varrho\varrho' &= hr
\end{aligned}$$

One can express the radical ϱ' linearly in terms of $r\varrho$ and ϱ, as one sees from the formula

$$hr^2\varrho' = r\varrho\varrho'^2$$

when one substitutes on the right for ϱ'^2 its expression in terms of r and bears in mind that r^2 is rational. Thus one obtains, when instead of the cyclic permutation of the k one takes the permutation

$$\begin{pmatrix} r, & \varrho, & \varrho' \\ -r, & \varrho', & -\varrho \end{pmatrix}$$

applies, and denotes by A_1, A_2, A_3, A_4 rational quantities:

$$\begin{aligned} K_0 &= A_1 + A_2 r + A_3 \varrho + A_4 r \varrho \\ K_1 &= A_1 - A_2 r + A_3 \varrho' - A_4 r \varrho' \\ K_2 &= A_1 + A_2 r - A_3 \varrho + A_4 r \varrho \\ K_3 &= A_1 - A_2 r - A_2 \varrho' - A_4 r \varrho'. \end{aligned} \tag{14}$$

These quantities K_0, K_1, K_2, K_3 are themselves again the roots of a cyclic biquadratic equation. They have only apparently a more general form than the k; for if one puts K_0 into the form

$$K_0 = A_1 + A_2 r + \sqrt{\varrho^2 (A_3 + A_4 r)^2},$$

one recognises the form of k_0 in (9) again, naturally with altered a, b, c.

Corrections.

Page 182, in the formula, line 12, read x_m instead of x_n.
Page 347, in the formula, line 24, read $(2x^2 + 1)^2$ instead of $(2x^2 - 1)^2$.
